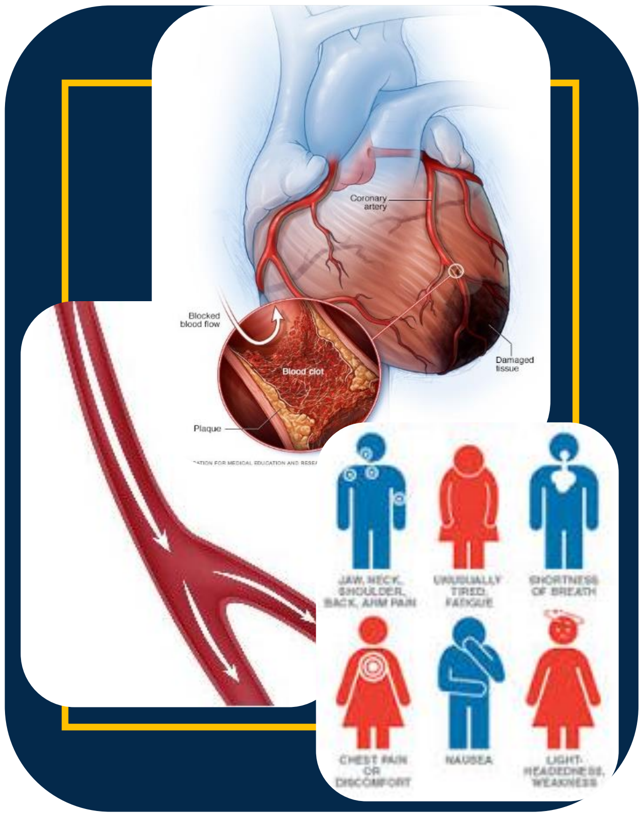
PATHOPHYSIOLOGY NOTES

(1st Edition)



A Comprehensive Resource for Students Enrolled in LUSL2107

Contributors

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Amber Doyle (4th-year BScN)

Conner Smith (3rd-Year BScN)

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Preface

Welcome to pathophysiology, a new learning experience that is critical to your profession! Pathophysiology is the science of disease. It is all about studying the causes of and alterations in cells, tissues and organs beyond physiology. It details the processes and mechanisms that explain clinical manifestations which guide patient care.

This resource is composed of my own lecture notes developed over the last 5 years, fortified and enriched by student experiences and perspectives. This collaborative effort places your needs – as a student – at the forefront, and students like you helped to bring this resource into existence by sharing their vision for how best to enrich student learning and success.

There are two pillars underlying the philosophy behind generating this document. First, is building on what you already know to strengthen and expand your basic science foundation about the mechanisms of disease. Second, is incorporating students' input to enhance the efficiency of the document as a study resource.

Throughout this document, we will discuss important pathophysiological concepts, as well as selected diseases related to the various body systems, with focus on etiology and abnormalities of cells, tissues and organs that explain patients' symptoms and signs. The document is complemented by illustrations of the general principles of pathophysiology and explanatory diagrams.

Acknowledgements

I truly value the contribution of each member of the team. They have all been my students, have successfully completed this course, and have been the top of their class. Their dedication in bringing this document to light have been exemplary. I acknowledge Conner Smith, now a third year BScN student, for putting together the nucleus of my original set of notes and adding his own notes from in-class and other resources. I acknowledge Amber Tamara Doyle for completing the second round of revisions and for enriching the document with her cumulative experience as a fourth year BScN student. Finally Kathryn Osborne, a 2016 graduate from the SLC-BScN program and now RN, who not only completed the third round of revision, enriched the content with practical experience and enhanced the scientific quality of the document, but also planned and led the whole project over the past year and kept everyone on track. Additionally, I am so grateful for the precise and special touches of Kseniia Reidman, an SLC Office Administration Student, in formatting the document and getting it ready for print.

This is only the first edition and I would certainly welcome your feedback towards refining this document to best suit your learning. As you move through the document, I encourage you to make note of areas of strength, as well as areas for improvement, and participate in future opportunities for further development of the resource. On behalf of the team, I wish you success and hope you find this document both useful and enjoyable.

Maha Othman

Contributor Bios

Dr. Maha Othman (MD, MSc, PhD)

Dr. Othman is Physician, Scientist, Educator, Mentor and Leader with over 20 years of experience in Health Care, Research and Academia. She obtained her MD and MSc in Clinical Pathology in Mansoura University, Egypt. She then completed a PhD in Pathology, with a specialization in Haemostasis, from Southampton University, UK, and then a post-doctoral research training at Queen's University, Canada.

Her strong passion for teaching and learning led her to supervise and mentor numerous trainees at both the postgraduate and undergraduate levels. Dr. Othman is currently a professor at St Lawrence College teaching several courses in foundational science and research within the Bachelor of Sciences in Nursing program where she strives to create new learning opportunities for her students. She is also an adjunct associate professor at the Queen's University School of Medicine, where she maintains an active research lab. Dr. Othman has over 75 publications in peer-reviewed journals and is an internationally recognized expert in platelet-type von Willebrand's disease. She is an Associate Editor for Seminars in Thrombosis and Haemostasis Journal, a reviewer for multiple Haemostasis journals, and a member of several scientific organizing committees and advisory boards for international haemostasis conferences. She is the current chairman of the International Society on Thrombosis and Haemostasis's (ISTH's) Scientific and Standardization Committee (SSC) for Women's Health Issues in thrombosis and Haemostasis.

Dr. Othman recently completed the Executive Women in Leadership Certificate from Cornell University and is a decisive and reliable leader who values transparency, effective communication and collaboration. She enjoys new challenges, works hard to break obstacles and strives to make a difference and promote the team she leads. She led the development of departmental scholarship and research strategic plan and participated actively in shaping up the international SSC's strategic plan.

Dr. Othman volunteers extensively in her community. She is a professional mentor at Kingston Keys Job Centre and is a member of the Board of Directors of Kingston Community Health Centre. She supports Indigo Girls' (non-for-profit- organization) mission to Educate, Inspire and Empower women.

In her spare time, Dr. Othman enjoys travel, music and intercultural dining.

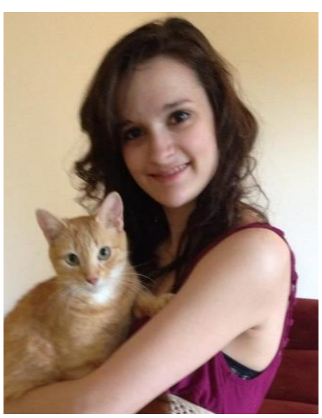
Kathryn Osborne (BScN, RN)

Kathryn is a 2018 graduate of the SLC-BScN program, and now practicing RN, with additional courses in mammalian physiology, biology, and chemistry from the University of Ottawa. She is currently an emergency room nurse in a rural hospital with additional training in perianesthesia nursing.

Kathryn has been committed to this project from its inception when she created and led tutorials in pathophysiology for nurses (LUSL 2107, the course for which this resource was developed) using a Peer Assisted Learning approach. With the expertise, guidance, and support of Dr. Othman, the she co-authored her first peer-reviewed article on the 3-year project published in 2018 in the Journal of Nursing Education and Practice. This experience has created a passion not only for enriching nursing and undergraduate education, but also research as a means of bringing forward new and creative ideas. She envisions further nursing education with more student-focused learning and leadership initiatives that engage and enrich the undergraduate nursing student experience and foster an enthusiasm for learning. In practice, teaching and education remain forefront for Kathryn, with patient education as a pillar to health, well-being, and autonomy.

Outside of work, academia, and family, Kathryn enjoys reading a wide variety of books from history and philosophy to genre fiction, listening to music, visiting museums and attending the occasional Theatre production (of which the Stratford Festival is a particular favourite).

Amber Doyle (4^{th}-year BScN)



A 4th year student nurse who developed a passion for harm reduction when working at Milhaven Institution and HIV/AIDS Regional Services. She also developed an aptitude for teaching while tutoring peers throughout her academic career.

She hopes to begin her career working with marginalized populations as a Street Health Nurse. Eventually, she envisions drawing from her various experiences in order to educate and inspire the next generation of nurses.

Conner Smith (3rd-year BScN)

Conner is a 3rd year BScN student on route to graduate in the 2021 academic year. He demonstrated great initiative by creating an original set of resourceful study notes that eventually developed into a collaborative team project for future pathophysiology students. He hopes students will benefit from this comprehensive learning tool as well as contribute to the tool's readability and overall dynamic. Conner enjoys helping fellow nursing students academically by tutoring multiple science courses including Pathophysiology, where he utilizes a comprehensible student-centered approach. This project has helped spark Conner's interest as a future educator in nursing where his qualities can make an impact.

When not completing assignments or studying, Conner enjoys spending time with classmates and family. He has taken up skiing on available weekends and continues to play various sports throughout the weeknights. Conner has recently made time every night to watch The Office with his pet gecko. This resource is to help guide your learning from a past student perspective; the contributors wish you best of luck in the course and semester!

Kseniia Reidman (2nd-year Business Administration and Accounting)

Kseniia Reidman is a second-year Business Administration-Accounting student at St. Lawrence College, where she also works part-time in the role of student Office Assistant for the Health Sciences office.

Kseniia had come to Canada 3 years ago and decided to build a new career. Kseniia holds an equivalent of a master's degree in education and has vast experience working in different fields such as education, financial sector and customer service. Kseniia is an enthusiastic, organized and self-driven professional with excellent leadership and social skills.

Pathophysiology ("Patho") Notes

1: Introduction to Pathophysiology / Altered cell and tissue biology

Learning Outcomes

Introduction to Pathophysiology

- Define pathophysiology and demonstrate an understanding of the scope of science of pathophysiology.
- Define and use appropriately the various pathophysiological terms.

Altered Cellular and Tissue Biology

- Describe cellular structure, components, organelles and their functions (review).
- Define and distinguish cellular adaptation versus cellular injury.
- Describe the cellular adaptations made in each of the following processes: atrophy, hypertrophy, hyperplasia, dysplasia (atypical hyperplasia), and metaplasia.
- Discuss physiologic versus pathogenic cellular adaptations.
- Describe the mechanisms of cellular injury that can occur as a result of the following causes: hypoxia, chemicals, free radicals, infectious agents, asphyxial injuries, immunologic and inflammatory responses, genetic factors, nutritional imbalances, and physical trauma.
- Describe the cellular damage incurred in reperfusion (reoxygenation) injury.
- Describe the characteristics of the following intentional and unintentional injuries: blunt force injuries, sharp force injuries, gunshot wounds, and asphyxial injuries.
- Define the following terms associated with blunt force injuries: contusion, hematoma, abrasion, laceration, and fracture.
- Define the following terms association with sharp force injuries: incised wound, stab wound, puncture wound, and chopping wound.
- Define the following terms associated with gunshot and asphyxial injuries: entrance wound, exit wound, suffocation, strangulation, drowning, and chemical asphyxiation.
- Define necrosis.
- Identify and describe the mechanism and resulting damage of coagulative, liquifactive, caseous, fat, and gangrenous necrosis.
- Compare and contrast cellular necrosis with apoptosis.
- Describe the cellular mechanisms of normal degenerative changes of aging.
- Describe the clinical manifestations of somatic death

Definitions: Altered Cellular and Tissue Biology

Term	Definition
Actin Filament	- Twisted protein fibers that are responsible for cell movement.
Acute Disease	- Symptoms/signs of the disease developing quickly, within short period.
Anaplasia	- Undifferentiated cells, with variable nuclear and cellular structures. No structure.
Anoxia	- Lack of oxygen supply to tissues.
Atrophy	- Decreases in cell size resulting in reduced tissue mass. - Examples: 1. Decrease or stoppage of exercise leading to smaller muscle mass (as occurs in bed-bound patients) 2. Smaller breast size after menopause
Autopsy (post-mortem examination)	- Examination of the body/organs after death. Performed to determine the exact cause of death.
Biopsy	- Excision of a part of the living tissue. - Ex. taking a piece of a tumor to examine.
Centriole	- Complex assembly of microtubules that occurs in pairs.
Chronic Disease	- Symptoms/signs of the disease developing gradually, persisting for longer time.
Complications	- New, secondary or additional problems.
Cytoskeleton	- Supports organelles and cell shape and plays a role in cell motion.
Cytoplasm	- Semifluid matrix that contains the nucleus and other organelles.
Dysplasia	- Cells vary in size and shape within a tissue - No obvious order or structure;
Etiology	- The cause of a disease. - Broad categories of etiology include:

	A. Identifiable (80%): infection, hereditary, immune, malnutrition. B. Idiopathic: No known cause. C. Iatrogenic: drug/intervention induced. D. Nosocomial: Hospital related.
Exacerbation	Periods when symptoms become worse and more severe.
Hyperplasia	- Increase in number of cells resulting in increased tissues mass. - Example: Mammary glands in lactation.
Hypertrophy	- Increase in cell size resulting in increased cell mass. - Example: Skeletal and heart muscles of weight lifters.
Hypoxemia	- Lack of oxygen due to reduced blood flow (more specifically arterial blood)
Hypoxia	- Insufficient oxygen delivery to cells resulting in tissue injury. - May be related to decreased oxygen content in inspired air, decreased hemoglobin for oxygen binding, or due to cardiovascular or respiratory disease.
Intermediate Filaments	- Intertwined proteins fibers that provide support and strength
Ischemia	- Insufficient blood flow to tissues. - May lead to subsequent cell injury or death.
Latent state	- No apparent clinical symptoms or signs. - Can flare up at any time. - Example: Shingles (Herpes-Zoster) caused by Varicella-Zoster virus (VZV).
Metaplasia	- Mature cell type is replaced by a different mature cell type which can lead to cancer - Example: stratified squamous epithelium replaces pseudostratified columnar epithelium in trachea in response to chronic smoking.

Microtubule	- Tubule of protein molecules present in cytoplasm, centrioles, cilia and flagella
Mitochondria	- Cell organelle in which energy is generated during oxidative phosphorylation. “Power house of the cell”
Neoplasia	- "New growth" - commonly called a tumor.
Nuclear Envelope	- A double membrane that surrounds the nucleus in the cell. - Separates nucleus from cytoplasm.
Nuclear Pore	- Openings embedded with proteins that regulate passage into and out of the nucleus.
Nucleolus	- Centre of nucleus and site of ribosome synthesis.
Nucleus	- Command center of cell.
Pathophysiology	- Science that provides understanding of the mechanisms of disease (impaired physiology). - Functional changes in cells, tissues and organs by disease and/or injury that lead to the signs and symptoms of the disease.
Pathogenesis	- Pattern of changes associated with the development of diseases (natural course of the disease from exposure to recovery).
Pathology	- Science of diseases. Abnormal states or structural alterations in cells, tissues and organs.
Macroscopic (Gross) Pathology	- Investigation at the organ or system level.
Microscopic Pathology	- Investigation at the cellular or tissue level.
Peroxisomes	- Vesicles that contain enzymes that detoxify potentially harmful chemicals.
Plasma Membrane	- Phospholipid bilayer surrounding the cell, in which proteins are embedded.
Precipitating Factors	- Elements that trigger actual development of a disease. Example: Poor diet and exercise can precipitate diabetes.
Predisposing Factors	- Tendencies that promote a disease. Example: Genetic factor leading to diabetes.

Prognosis	- The expected outcome of the disease, such as recovery, chronic state, survival rate. - Prognosis affects treatment decisions for patients and health practitioners.
Remissions	- Manifestations of the disease subsiding or are absent. Example: Leukemia following successful chemotherapy.
Ribosomes	- Cell organelles; small complexes of RNA and protein for protein synthesis.
Rough ER	- Cell organelle; internal membranes studded with ribosomes that carry out protein synthesis.
Secretory Vesicle	- Vesicle fusing with the plasma membrane, releasing materials to be secreted from the cell (performs exocytosis).
Sequelae	- Residual outcomes of primary condition. Example: a scar following a burn.
Smooth ER	- Cell organelle; a system of internal membranes that aid in the manufacturing of carbs and lipids.
Subclinical state	- Pathologic changes with no obvious manifestations.
Syndrome	- Collection of signs and symptoms that affect more than one organ/system.

Altered Cellular and Tissue Biology

Types/Causes of Cellular injury

1. **Oxygen deficit**: Hypoxia and ischemia
2. Free Radicals
3. Chemical Injury
4. Physical Injury
5. Nutritional Injury
6. Mechanical Injury
7. Infection
8. Immunological/inflammatory injury
9. Genetic defects

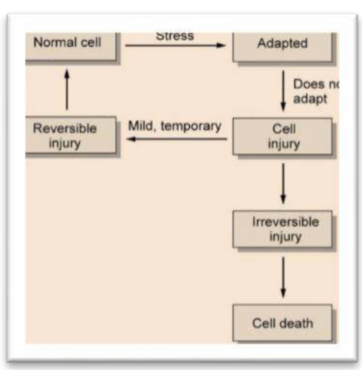


Figure 1: Cellular response to adaptation

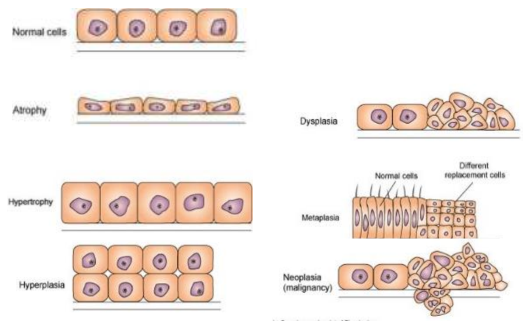


Figure 2: Cellular adaptation

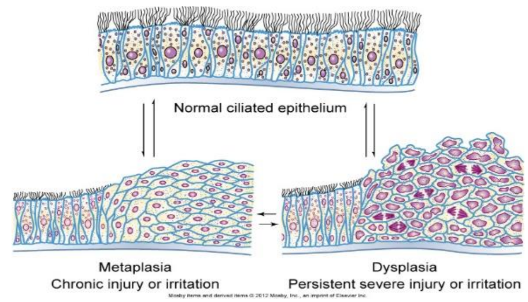


Figure 3: Cellular response in epithelium (e.g. trachea) to chronic, persistent injury or irritation (e.g. smoking)

1. Hypoxic Injury

- *Definitions: Hypoxia, Anoxia, Hypoxemia, Ischemia*
- Gas exchange takes place in the alveoli (between air and blood) and at tissues (between blood and cells)
- Hypoxic injury can occur as a result of:
  - **Lack of oxygen in the air**
    - E.g. High altitude, being trapped in an enclosed space such as an elevator or underground.
  - **Problems with oxygen transport**:
    - E.g. Conditions affecting RBC formation and structure, and subsequent oxygen carrying capacity.
    - E.g. decreased RBCs production (erythropoiesis).
  - **Disease of the respiratory and/or cardiovascular system**:
    - Blood is not getting pumped/circulated adequately.
    - Poor gas (oxygen) exchange in the lungs and/or delivery to tissues.
  - **Narrowing of an artery or vein**
    - E.g. atherosclerosis.
  - **Obstruction of an artery or vein**
    - E.g. blood clot (thrombosis).

Myocardial Infarction (MI)

- Acute obstruction of coronary artery leading to myocardial cell death.
- There is hypoxia, which progresses to anoxia if blood flow is not restored.
  - In response to hypoxia, the heart modifies its method of oxygen uptake and utilizes myoglobin stores [*Remember:* Myoglobin = hemoglobin in heart]
  - Myoglobin stores are limited and eventually run out.

- The heart becomes anoxic and cardiac cells decrease their metabolism to conserve energy until eventually no ATP is produced.
- The Na-K pump fails due to lack of ATP.
- Na and water leak into cell causing it to swell and burst.

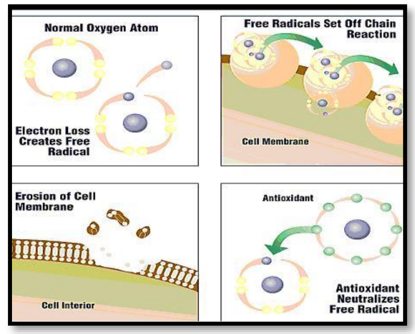
▪ **Diagnostic Testing**
  - Troponin and cardiac enzyme test, ECG
    - Elevated troponin I is a marker for MI, however, depending on the timing in relation to onset of chest pain, levels may not be elevated. For this reason, Troponin is repeated in 4 hours for patients with suspected MI.
    - Elevated Troponin I after 4 hours is indicative of MI
  - ECGs are critical to determine treatment approach.
    - Do an ECG for any suspected MI*
    - *In practice, ECGs are completed on most adults presenting with sudden onset chest pain, especially with cardiac features, to rule out MI
    - Extra consideration should be taken for persistent, generalized, radiating pain in women with no apparent cause of onset as women with MI often present atypically from males
      - This includes generalized abdominal pain, back pain, neck, that is not reproducible with specific movements or is related to a specific diagnosis or injury

▪ *Reperfusion*:
  - Restoration of blood flow to an obstructed area.
  - This occurs via collateral circulation.

▪ **Reperfusion Injury**:
  - Cellular damage caused by restoration of blood flow to obstructed area
  - Occurs due to an increase in intracellular calcium and formation of free radicals:
    - Calcium damages mitochondria → Therefore no ATP production
    - Calcium increases cellular enzyme activities causing membrane damage, nucleus damage, and further decrease in ATP production.
    - Free radical causes further cell membrane damage and release of calcium stores contributing to mitochondrial calcium overload.
  - Cells attempt to detoxify radicals but they may fail:
    - Toxic free radicals can be stabilized by donating or accepting an electron from another molecule (antioxidants).
    - With significant free radical formation, there are not enough antioxidant molecules to stabilize all the unstable oxygen species.

2. Free radicals

- Molecules with unpaired, negatively charged electron in their outer orbit.
- Free radicals come from many sources such as:
  - Aging, respiration, metabolism
  - Infection, cancer
  - Reperfusion, inflammation
  - Drugs, chemicals, radiation.



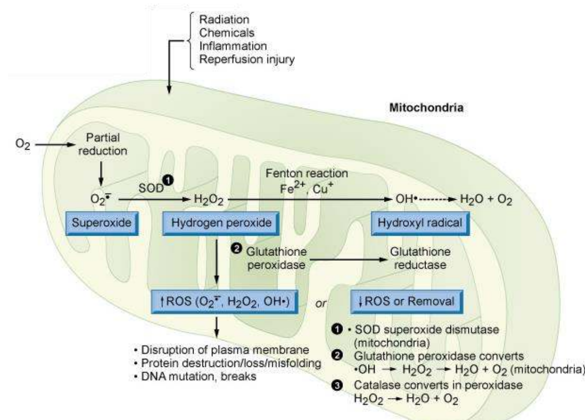


Figure 4: Formation and detoxification of free radicals

- Unstable and highly reactive, therefore called reactive oxygen species (ROS).
- Free radical injury has many effects, including:
  - Lipid peroxidation.
  - Alteration to proteins → impaired enzymatic activity and folding
- Free radicals must accept or donate an electron from another molecule to become stable, and they do this by binding to other molecules
  - This binding can be injurious if these ROS bind to molecules such as proteins, lipids, and carbohydrates, as well as those key to survival such as molecules of cellular membranes and nucleic acids (DNA)
  - ROS set of a chain reaction as they bind to stable molecules, in turn making these previously stable molecules unstable free radicals.
- Peroxisome is the cell that detoxifies the free radical by adding something to make them safe and "stable".
- Antioxidants can also act as binding molecules to stabilize free radicals and prevent their binding to other body molecules.
  - Free radicals can be eliminated by Antioxidants (vitamins A,C,E), spontaneously, or via enzymes. These enzymes include:
    - Superoxide dismutase (in the *mitochondria*) $O_2 + 2H \rightarrow H_2O_2$
    - Catalase (in *peroxisomes*): $2\ H_2O \rightarrow 2\ H_2O + O_2$
    - Glutathione peroxidase (in *cytoplasm*): H_2O_2 or $2\ OH + 2\ GSH \rightarrow 2\ H_2O + GSSH$
- **Antioxidants stabilize, peroxisomes detoxify**

- *Oxidative Stress*: When excess ROS production overwhelms endogenous antioxidant systems leading to cellular injury.

3. Chemical Injury

- Chemicals can cause cellular injury by direct (binding to molecular components of the cell) and indirect (lipid peroxidation or formation of free radicals).
- **Examples**:
  - Alcohol, ethylene glycol, lead, carbon tetrachloride (CCL_4), cyanide and carbon monoxide (CO).

a) Alcohol toxicity

- **Ethanol (alcohol)** turns into acetaldehyde in the liver by alcohol dehydrogenase.
  - This leads to the formation of free radicals and acetic acid by the liver
    - Creates a state of acidosis
  - Acetic acid leads to fatty change
  - Acetate can further breakdown to acetyl CoA and enter the Krebs cycle
    - Leads to build-up of ketone bodies

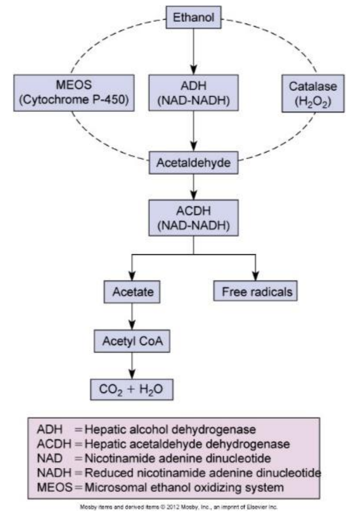


Figure 5: Metabolism and detoxification of ethanol (alcohol)

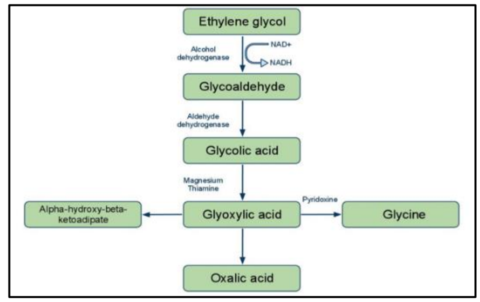
- **Manifestations**:
  - CNS depression, metabolic acidosis
  - Inhibition of gluconeogenesis
  - Fetal alcohol syndrome (FAS)
    - Ethanol can cross the placental barrier and affect the developing fetus
    - Children are affected both physically and cognitively
  - **Acute toxicity manifests mainly by CNS depression, chronic toxicity manifests mainly by liver damage**.

Figure 6: Fetal alcohol syndrome

b) *Ethylene Glycol Toxicity*

- Found in antifreeze, toxicity is usually via ingestion.
- Metabolized to glycolic acid leading to severe acidosis and acute renal injury.
- Direct effect on CNS with altered mental status
    - The person will feel drowsy for the first hour following intoxication

Figure 7: Ethylene glycol (antifreeze) toxicity



- Eventually oxalic acid can be made which is a toxic metabolite
    - This leads to the formation of calcium oxalate crystal in urine (renal stones).
- Serum levels of ethylene glycol peak within 1-4 hours
    - CNS depression occurs during the 1^{st} hour
- Patients develop heart failure or pulmonary edema within 12 hours
    - Renal tubular necrosis within 24-72 hours (late stage).

c) *Lead Toxicity*

- Lead is a heavy metal, absorbed by inhalation or ingestion (paint, dust, occupational)
    - When in the body, lead mimics other metal cofactors (calcium, iron, zinc) in enzymatic reactions.
    - As a result, it inhibits 2 enzymes involved in heme synthesis: delta-ALAD (δ-aminiolevulinic acid dehydratase) and ferrochelatase.
    - Lead also interferes with neurotransmitter activity.
- Toxicity can be acute or chronic.
    - Not super common anymore with lead free toys, pencils, paints and car gas.
- **Manifestations include**:
    - **Anemia (Iron Deficiency):** Lead can replace iron in hemoglobin and inhibits the synthesis of RBCs
    - Lead can also replace calcium in bone (teeth), blood, and impair nerve conduction, muscle contraction etc.

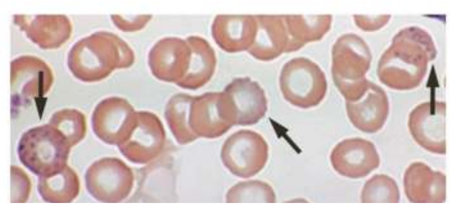


Figure 8: Blood smear of person with lead toxicity

- Calcium is a co-factor for blood coagulation so risk for bleeding is expected.
- In the cell, calcium is in the cisternae of the endoplasmic reticulum
- Intracellular free calcium stimulates contraction of muscles. Lead replaces this and those mechanisms are impaired.

- Blue gums (lead lines), dense metaphyseal lines on x-ray.
- Peripheral neuropathy (nerve conduction inhibited)
  - Manifested as wrist/foot drop in adults, poor hand-eye coordination in children.
- Lead crosses the placenta which can lead to miscarriages, stillbirths, low birth weight, and cognitive impairments.

Figure 9: Blue gum line in lead toxicity

d) Carbon monoxide

- Produced from incomplete combustion of organic materials
- Binds to Hb and forms carboxyhemoglobin (COHb)
- This is known as the silent killer (so is hypertension). *No taste or smell* but it is fatal. The carbon is tightly bound to Hb, so does not release oxygen in the tissues. Tissues get hypoxic and die.

e) Cyanide

- Binds/ inactivates cytochrome oxidase in mitochondria.
- Disrupts mitochondrial cellular respiration.
- Causes respiratory failure.

4. Physical injury

- **Thermal:**
  - *Hyperthermia (>38.5°C):* High heat damages protein and stops enzyme activity (eg. Heat stroke)
  - *Hypothermia (<35°C):* Excessive exposure to cold, alcohol intoxication. (cooling of the surgical field to decrease circulation and reduce bleeding, if overdone → hypoxic tissue.
  - *Burns*
- **Radiation:** UV
  - Exposure creates thymine dimers leading to faults in DNA replication when the cell divides and subsequent mutations.
  - A method for killing bacteria.

- **Pressure:**
  - Air pressure
    - Explosion e.g. Blast injuries: complex physical trauma.
    - Water pressure.
- **Surgery:**
  - Deliberate tissue injury.
  - Potential damage to the cells as a post-operative complication.

5. Nutritional Injury

- **Nutritional Deficits**
  - *Malnutrition*: not enough intake of any nutrient (CHO, Fat, Protein, Minerals)
  - *Dehydration*: Water loss (may also result in electrolyte imbalance)
  - *Hypoalbuminemia*: Albumin = 60% of plasma proteins; therefore, it is largely responsible for the oncotic pressure, so low albumin leads to edema/ascites since fluid is not retained in blood vessels.
  - *Hypoglycemia*: Low blood glucose level → low glucose inside cells →reduce energy production.
  - *Vitamin deficiencies* (Vitamins ADEK and folic acid).
- **Nutritional Excess**
  - Excess nutrients can also cause injury e.g. Hypervitaminosis D, hypercalcemia.

6. Mechanical Injury

a. Blunt force:

- Application of mechanical energy to the body resulting in the tearing, shearing, or crushing of tissues e.g. MVAs
- **Manifestations**:
  - **Contusion / hematoma** = bruise.
  - Stages:
    - I. Trauma to the site = rupture of BV → blood (RBCs; low oxygen) in subcutaneous tissue (hematoma) → blue/purple skin.
    - II. Lysis of the escaped RBCs → free Hb which is decomposed further → hemosiderin (brownish skin).
    - III. Hemosiderin turns into biliverdin → (yellowish skin)
  - **Abrasion**, **laceration**, **fractures**.

b. Penetrating injuries

- Stab wounds, gunshot wounds. If gunshot went into body it is damaging mechanically, can rupture BV→ internal hemorrhage.

- Ruptured spleen = almost fatal (storage site for blood)

c. Asphyxial Injuries

- Effects due to hypoxia
    - Suffocation (choking).
    - Strangulation (hanging, ligature, manual strangulating).
    - Chemical asphyxiants (cyanide and hydrogen sulfide).
    - Drowning.

d. Gunshot wounds

- Effects
    - Tissue disruption.
    - Bleeding, hypovolemic shock.
    - Pneumothorax.
- $KE = \frac{1}{2} MV^2$
    - Wounds produced by missiles of higher mass (M) and/or velocity (V) produce greater tissue disruption
        - I.e. faster & heavier = more damage

7. Infectious Injury

- *Note: Details of infectious agents in Microbiology course*
- **Bacteria**
    - Produce toxins (endotoxin or exotoxin).
- **Viruses**
    - Decrease the ability to synthesize proteins.
    - Change the cell's antigenic properties.
- **Other microbes**

8. Immunologic and Inflammatory Injury (Details in Microbiology course)

- Induced by:
    - **Cells:** phagocytic cells (neutrophils, macrophages, monocytes)
        - Alterations in cell membrane
        - Changes in membrane permeability
            - These changes cause unintentional injuries and problems that will need to be repaired
    - **Mediators:** substances release by immune inflammatory cells
        - Histamine, cytokines, antibodies, complement, and proteases

- Histamine from basophils circulates in body leading to itching, edema

- **Remember Inflammation = tissue response to ANY injury

9. Genetic Injury

- Genetics play a substantial role in cellular structure and function including:
  - Cell shape
  - Protein structure and functions
  - Receptors and recognition
  - Transport mechanisms
- **Examples**
  - *Sickle cell* disease (missense mutation in globin protein; abnormal Hb S; red cell hemolysis; anemia)
  - *G6PD* (X-linked; enzyme deficiency; red cell hemolysis; anemia)
  - *CML (chronic myeloid lymphoma); Philadelphia chromosome*
    - Chromosome 22 to chromosome 9 translocation.

Pathological themes - Cellular injury

a) Mitochondrial damage
  - O_2 depletion
  - Impaired oxidative metabolism → decreased ATP production
  - ROS = Reactive oxygen species (free radicals)

b) Membrane damage:
  - *Plasma membrane damage*: failure of Na-K pump; increases cellular water
  - *Lysosomal membrane damage*: enzyme release, activation & digestion of cell components
  - *Mitochondrial membrane damage* → decreased ATP

c) Protein synthesis impairment:
  - Ribosomes separate from the swollen endoplasmic reticulum → decreases protein synthesis

d) Lipid peroxidation:
  - Causing damage to phospholipid cell membranes

Manifestations of Cellular injury - Accumulations

a) Water:

- Necessary for life, but when there is too much, this can lead to flooding
- Water accumulation = hydropic degeneration

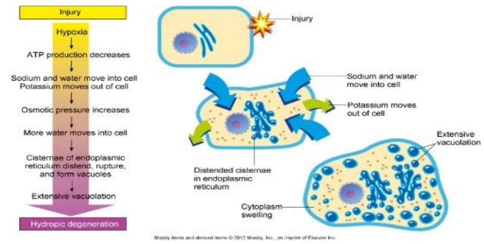


Figure 10: Cellular injury from water accumulation (hydropic degeneration)

b) Lipids and carbs

c) Glycogen

d) Proteins

e) Pigments

- Melanin, hemosiderin, bilirubin

f) Calcium

- Calcium is stored inside Cisternae of ER (inactive state) and is only released when needed (e.g. muscle contraction, cofactor in the activation of enzymes)
- Free Ca^{2+} activates enzymes → hormonal action, calcium mediated, kinases etc

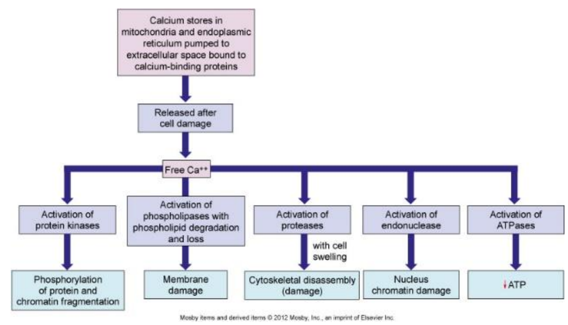


Figure 11: Cellular damage by excess free calcium

- Therefore, it is essential for the regulation of body processes to have calcium remain stored in the ER because if calcium were to be free it would continuously activate body processes (above)

g) Urate

Cell Death

- Happens when you can't reverse the changes through adaptation to injury/stress

Apoptosis

- Programmed cell death.
- Normal occurrence in the body:
  - Aging/senescence of cells e.g. RBCs killed every 120 days
  - Tissue involution following hormonal withdrawal
  - Elimination of self-reactive lymphocytes (immune tolerance) in Thymus
  - Breaking self-tolerance → autoimmune disease

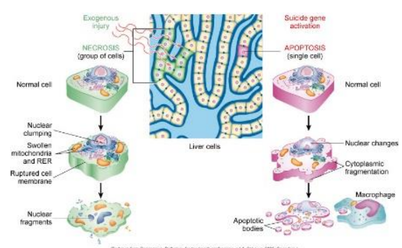


Figure 12: Cell death - Necrosis vs apoptosis

Autophagy

- Self-eating membrane extension, fusion with lysosomes
- Survival mechanism in deprivation and starvation
- Degenerative CNS disease and cancer

Necrosis

- Un-programmed cell death
- Cell changes include:
    - **Karyolysis**: Nuclear dissolution and chromatin lysis
    - **Pyknosis**: Clumping of the nucleus
    - **Karyorrhexis**: Fragmentation of the nucleus

Types of necrosis:

1. *Liquefaction necrosis*
    - Tissue transforms into a liquid viscous mass due to release of hydrolytic enzymes
        - Common in brain, liver and lungs
        - Due to bacterial infection

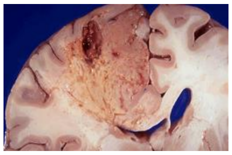


Figure 13: Liquifaction necrosis

2. *Coagulative necrosis*
    - Denaturation of structural proteins (coagulation) including lysosomal enzymes
    - Like an egg on a hot pan

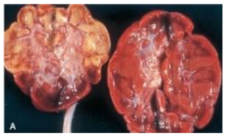


Figure 14: Coagulative necrosis

3. *Fat necrosis*
    - Tissue broken down into fatty acids by action of lipases
        - Liver, breast, pancreas and other abdominal organs
        - Common in liver. Lipases break down hepatocytes into fatty acids

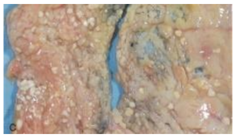


Figure 15: Fat necrosis

4. *Caseous necrosis*
    - Combination of coagulative and liquefactive necrosis
    - Thick, yellowish "cheesy" substance
        - Tuberculosis pulmonary infection

Figure 16: Caseous necrosis

5. *Infarction*
    - Area of dead tissues as a result of oxygen deprivation
    - E.g Myocardial infarction

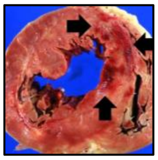


Figure 17: Infarction

6. Gangrene

- Area of dead tissues due to severe and persistent hypoxic injury
    - **Dry gangrene** (arterial occlusion; feet and toes; DM)
    - **Wet gangrene** (venous occlusion; most tissues/organs strangulated hernia)

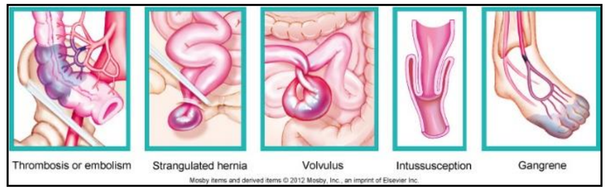


Figure 18: Types of gangrene

    - **Gas gangrene:** hypoxia + gas production (Clostridium perfringens)

Aging

- Humans have normal life span and life expectancy
- **Aging vs disease**
    - Aging is a physiological process (not a disease). Atrophy, decline in number and size of cells, low blood supply, poor healing, poor skin elasticity…etc
- **Cellular aging**
    - Atrophy, decreased function/renewal, and organelle damage
- **Tissue and systemic aging**
    - Progressive stiffness and rigidity
    - Sarcopenia (muscle loss; decreased body mass with aging)
- **Frailty**
    - Complex clinical **syndrome**, common in elderly
    - Affects: mobility, balance, muscle strength, motor activity, cognition, nutrition, endurance and bone density
    - Person vulnerable to falls, fractures, functional decline

Somatic Death

- Death of the entire person

Manifestations:

- Post-mortem change is diffuse and does not involve the inflammatory response
    - Cessations of respiration and circulation
    - Pupil dilation
    - Gradual lowering of body temp
    - Loss of elasticity and transparency of skin
    - Muscle stiffening (rigor mortis)
    - Skin discoloration

Types of post-mortem changes:

- *Algor mortis*
    - Reduction in body temp
- *Livor mortis*
    - Purplish red discoloration of the skin
- *Rigor mortis*
    - Stiffness of body parts, visible within 2-4 hours, fully developed 6-12 hours after death
- **Post-mortem decomposition**
    - *Autolysis*
    - *Putrefaction*: obvious around 24-48 hours after death. Rotting or decaying of the body.
    - *Maceration*: aseptic autolysis of dead fetus in utero within amniotic fluid.

Helpful Hint

Think Al Gor(e) and his focus on temperature change to remember the relationship of algor mortis and reduction in body temperature

2: Mechanisms of Self-Defence

Learning Outcomes

- Review basis, types and mechanisms of immune response (innate and adaptive)
- Explain defects in mechanisms of defense and review common infections
- Explain various mechanisms of alterations of immune response and provide examples
- Define inflammation and describe in details its mechanisms including the roles of blood vessels, blood cells, coagulation system, complement system, inflammatory mediators, platelets and endothelium
- Explain pathology of acute vs chronic inflammation
- Describe mechanisms and steps of wound healing and explain disorders of these mechanisms
- How do we defend against the 9 ways of cellular destruction? For example:
    - Oxygen deprivation leads to impaired organelles
    - Free radicals are generated all the time from processes such as catabolism and inflammation

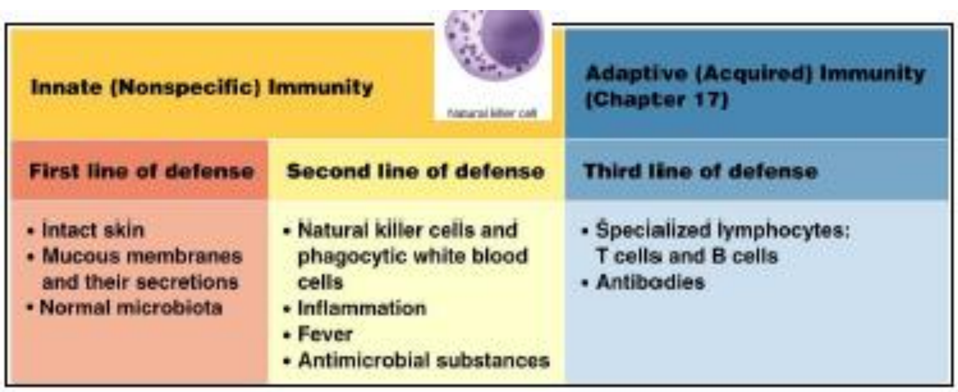


Figure 1: Overview of immunity - Innate vs adaptive

Innate immunity (natural; native)

First line

- **Physical barriers**
    - Skin integrity
    - Mucous membranes lining the GI and resp tracts
    - Cilia
- **Chemical barriers**
    - **Lysozymes in** Saliva, tears, sweat, ear wax and low pH
    - Natural antimicrobial substances

- **Normal flora**

Second line:

- **Cellular response of innate immunity**
  - Natural killer (NK) cells (from lymphocyte)
  - *Phagocytes*: Cells that engulf and digest (phagocytosis) other debris, cells and organisms
    - Macrophages and neutrophils are phagocytes
    - Monocytes are precursors for macrophages found in tissue
- **Fever**
- **Inflammation**

Inflammation

- The first response to injury
- A biological response of vascular tissue (blood vessels, cells in blood, endothelium) to harmful stimuli
- **Causes**
  - Any of the 9 causes of injury previously discussed
  - Ischemia, infection, mechanical damage, nutrient deprivation, extremes of temperature, radiation, etc
- **Goals**
  - Prevent and limit injury and further damage
  - Control the response the inflammatory response
  - Initiate adaptive immune response (preparation for 3rd line of defense)
  - Initiate healing

Pathophysiology of Inflammation

- *Blood vessel dilation*:
  - Increases vascular permeability/leakage
  - This is the role of the endothelium which acts to leak mediators to tissue
- *White blood cell (WBC) response*:
  - WBCs leave the blood vessel and migrate to the site of injury and release mediators of inflammation
  - WBCs exit the blood vessel through the processes of:
    1. **Margination ("Pavementing"):** WBCs align along the walls of the blood vessels
    2. **Adherence** of WBCs to the inner wall of the blood vessel (endothelium)

3. **Emigration** of cells from vessel through endothelial junctions (diapedesis)
    - At sites of endothelial cell retraction (opening from dilation), the neutrophils exit the vessel by means of diapedesis (blood to tissue)

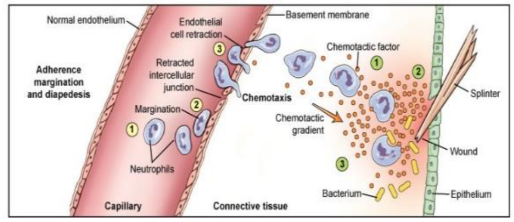


Figure 2: Processes of white blood response to inflammation (exiting of blood vessel and migration to site by chemotaxis)

- *Chemotaxis*:
    - Chemotaxis is when the neutrophils detect chemotactic factors (released in response to injury) through cell surface receptors.
    - This process guides neutrophils to migrate toward the site of injury, after which they become immobile
- *Opsonization* (by Antibodies and C3b from complement system)
    - Increase adherence of phagocyte to the target cell, making it more readily and more efficiently engulfed by phagocytes
    - Once bound, the target cell is digested by phagocytosis

- *Release of mediators from inflammatory cells* (mast cell = tissue bound basophil).
- *Plasma protein release*

Plasma Protein Systems

1. **Complement system**
2. **Coagulation system**
3. **Kinin system**

- Each system is composed of a group of proteins with a specific task
- Each system has a different task within inflammation

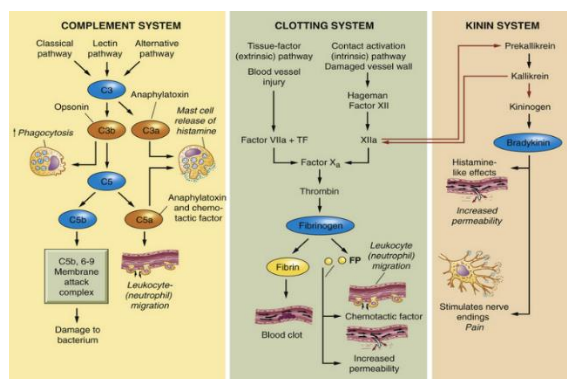


Figure 3: Plasma protein systems (overview)

Common to all 3 systems

- **Inactive enzymes** (proenzymes or zymogen)
- **Sequential activation**
    - First proenzyme is converted to an active enzyme
    - Substrate of the activated enzyme becomes the next component in the cascade.

Complement System

- A group of circulating proteins that when activated, promotes phagocytosis, activates chemical mediators
- Ultimately forms a membrane attack complex (MAC – C5b, 6-9) that will bind to and kill bacteria

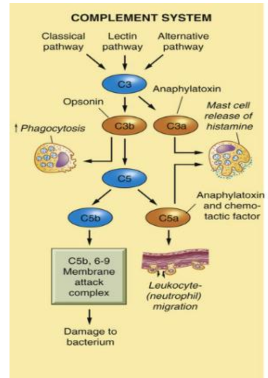


Figure 3a: The complement system

- The MAC can also activate other inflammatory components
- A major component of inflammation
- Can be activated by 3 pathways:

I. Classical:
   - Activated by antibodies. They make C1 which can eventually make C3 and C5

II. Lectin:
   - Activated by plasma proteins, especially MBL (mannose-binding lectin = recognizes a sugar on bacteria and activates complement system)

III. Alternative:
   - Activated by substances on the surface of infectious organisms. Uses factor B, factor D and properdin to form a complex that makes C3

- All 3 pathways lead to C3 activation and fragmentation
   - **C3a**: Potent anaphylatoxin (induces degranulation of mast cells)
   - **C3b**: Serve as opsonin for phagocytosis or proteolytically activate C5.
   - **C5a**: Potent anaphylatoxin and chemotactic factor
   - **C5b**: Activates MAC, which damages bacteria
- Complement proteins can be measured in serum. We usually look for C3a and C5a (active forms)
- **ANTIBODIES ARE NOT NEEDED TO START THE COMPLEMENT SYSTEM. SO, IT IS A PART OF BOTH <u>INNATE</u> AND <u>ADAPTIVE</u> IMMUNITY.

Coagulation (Clotting) System

- Forms fibrin meshwork at an injured or inflamed site.
- Fibrin strands contain platelets and trap RBCs and other cells
- **Goal of fibrin**: To stop bleeding and stop spread of pathogens (trap microbes)
   - Fibrin provides a framework for repair/ healing

Formation of Fibrin

- A series of sequential events leading eventually to two main reactions to build fibrin:
   1. Prothrombin (inactive) → thrombin
   2. Thrombin activated fibrinogen → fibrin

Activation of the Coagulation System

- Two methods of activation:
    1. Classical System, and
    2. Cell Model Theory

Classical system

- **Extrinsic and intrinsic activation**
    - *Extrinsic system* activated by reaction of Factor VII with tissue factor release from a damaged blood vessel
    - *Intrinsic system* activated when Factor XII reacts with negatively charged subendothelial substances (exposed from irritation and damage within the blood vessel
- Both extrinsic and intrinsic systems converge at Factor X activating the common pathway, leading to fibrin formation (as explained above)

Extrinsic system:	**Intrinsic system**:
- Activated by TF - Tested via PT, INR	- Activated by contact - Tested via PTT

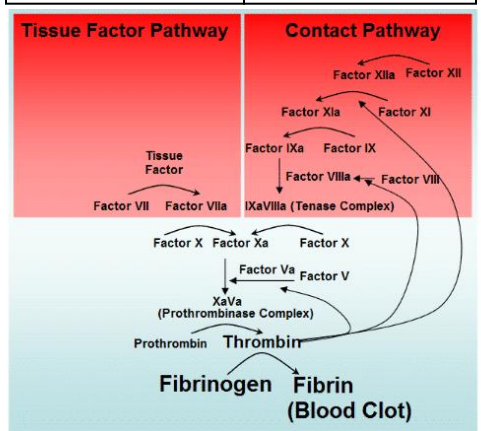


Figure 4: Coagulation system - Extrinsic vs intrinsic pathway activation of classical system

Diagnostic testing for classical system:

- **Prothrombin (PT; INR)**: Tests EXTRINSIC system activated by TISSUE DAMAGE (tissue factor)
- **Thromboplastic times (PTT):** Tests INTRINSIC system activated by CONTACT (endothelial damage)
- **Clotting factor assays**

Cell model theory

- Initiation, activation, amplification taking place on the surface of any TF expressing cell

Kinin-Kallikrein System

- **Roles of kinin-kallikrein system**:
    - Inflammation, blood pressure control, coagulation, and pain
- Kininogens are activated by kallikrein

Bradykinin

- Primary kinin
    - Causes vasodilation
    - Acts with prostaglandins to induce pain
    - Causes smooth muscle contraction

- Increases vascular permeability

- Bradykinin functions similarly to histamine and increases vascular permeability, it also stimulates nerve endings to cause pain
- **Effects:**
  - Mediates pain
  - Increases permeability (vasodilator)
  - Smooth muscle contractions
  - Stimulates leukocyte chemotaxis
- **Relationship of Kinin-Kallikrein system to coagulation (clotting) system**:
  - *Factor XII* (of clotting system) stimulates *Prekallikrein* and kallikrein
    - Hence, another name for Factor XIIa is *Prekallikrein activator*
  - The *XIIa* produced by the clotting system can also be *activated by Kallikrein* of kinin system
  - *Prekallikrein* is enzymatically converted to kininogen, which activates bradykinin

Cellular Mediators of Inflammation

Cell types

> **Helpful Hint**
>
> Granulocytes can be remembered as the "phils" because they are "philled"

- *Granulocytes*: Contain proteins – some are mediators of inflammation
  - **Neutrophils**: phagocytic WBCs and the predominant cell which kill bacteria in acute inflammation
  - **Eosinophils**: help regulate the inflammatory response (limit and control)
  - **Basophils**: release histamine in allergy and parasitic infestation
- *Monocytes/macrophages*
  - **Monocytes**: phagocytic WBCs
    - Monocytes are produced in the bone marrow, enter the circulation
    - Can migrate to the inflammatory site where they phagocytose microbes
    - Can develop into tissue macrophages
  - **Macrophages**: mature form in the tissue
  - Both monocytes and macrophages can:
    - Promote angiogenesis
    - Release cytokines and growth factors that promote epithelial cell division
    - Activate fibroblasts
    - Promote the synthesis of extracellular matrix and collagen formation (healing)

- *Lymphocytes*:
  - T cells mediate cell mediated immunity and B mediate antibody mediated immunity during inflmamatin. T cells support the proliferative phase of wound healing
- *Mast cells* (located in connective tissue):
  - A basophil in the tissue
- *Platelets*: Coagulation cascade takes place on its surface

Cell Products (cytokines)

- *Cytokines* are small proteins (5-20kDa)
  - Known as primary inflammatory mediators
  - Released by inflammatory cells and affect the behaviour of other cells
  - Important in cell signalling
  - Can be proinflammatory and anti-inflammatory
- **Lymphokines** = cytokines from lymphocytes
- **Monokines** = cytokines from monocytes
- **Chemokines** = chemotactic and primarily attract leukocytes to sites of inflammation
  - Can be made from macrophages, fibroblasts and endothelial cells
  - Most act locally over a short distance, but some such as the IL-1 has systemic induction (long distance), form an endogenous pyrogen

Types of Cytokines

1. Interleukins (IL)
2. Interferons (IFN)
3. Tumor necrosis factor alpha (TNFa)
4. Chemokines (much smaller; 8-10 kDa)

1. Interleukins

- Produced by macrophages and lymphocytes
- Produced in response to a pathogen or stimulation by other products of inflammation.
- Many types (1-36), and the list keeps growing
- They manage inflammation in different ways:

a) IL-1: pro-inflammatory
  - Produced in 2 forms: alpha and beta (mainly from macrophages)
  - Activates monocytes, macrophages and lymphocytes, which enhances both innate and acquired immunity, and acts as a growth factor for many cells
  - Increases:

- The number of neutrophils in circulation (proliferation)
- Neutrophils attraction to inflammation site (chemotaxis)
- Cellular respiration and lysosomal enzyme activity (cellular killing)

- An endogenous pyrogen (causes fever) that reacts with hypothalamus to raise temp

b) IL-10: anti-inflammatory

- Primarily produced by lymphocytes
- Suppresses the growth of other lymphocytes and the production of proinflammatory cytokines by macrophages, leading to anti-inflammatory effects

2. Interferons

- Related to interference with viral infections
- Produced by:
  - Virally infected cells
  - Other cells in response to bacterial, parasites and tumors
- Activate immune cells; NK and macrophages
- **Types:**

a) IFN alpha/beta:

- Type 1 interferon
- Induce production of antiviral proteins
- These are produced and released by virally infected cells
- DO NOT kill the virus directly but instead and induce antiviral proteins and protect neighboring healthy cells

b) IFN gamma:

- Type 2 interferon
- Produced by lymphocytes and activates macrophages
- Kills infectious agents (viruses and bacteria) and enhances the development of acquired immune response against viruses

3. Tumor Necrosis Factor-Alpha

- Secreted by macrophages (mainly) in response to microbe recognition
- **Secreted by other cells:**
  - $CD4^+$ lymphocytes
  - NK cells
  - Neutrophils
  - Mast cells

  - Endothelium
  - Synoviocytes
  - Hepatocytes
- Plays a key role in the regulation of inflammatory/immune cell function
- **Major effects:**
  - Induces fever by acting as endogenous pyrogens
  - Regulates gene expression
  - Stimulates synthesis of inflammatory mediators
  - Inhibits tumorigenesis and viral replication
  - Induces apoptosis
  - Causes muscle wasting (cachexia)
  - Stimulates intravascular thrombosis

4. Chemokines

- Small cell signalling cytokines (8-10 kDa) – just small cytokines
- SECONDARY inflammatory mediators
- **Synthesised and released by:**
  - Macrophages
  - Fibroblasts
  - Endothelial cells
- **Function:**
  - Guide the migration to the worst site of inflammation
  - Recruit inflammatory cell (e.g. Leukocytes) to the site of inflammation
  - Direct lymphocytes to lymph node

> **Helpful Hint**
>
> Think of chemokines as the "Traffic Cops" for processes of the immune system

Mast Cell Mediators

- **Roles**:
  - Mast cells are the most important cellular activator of the inflammatory response
  - Filled with granules and located in the loose connective tissues close to blood vessels near the body's outer surface
- **Major Effects:**
  - Cause temporary, rapid constriction of smooth muscle and dilation of the postcapillary venules → Increased blood flow into the microcirculation
  - Also causes increased vascular permeability resulting from retraction of endothelial cells lining the capillaries and increased adherence of leukocytes to the endothelium

a) Histamine

- Vasoactive amine that causes dilation of post-capillary venules
- **Binds specific receptors** on the other cells
- DEGRANULATION PRODUCT

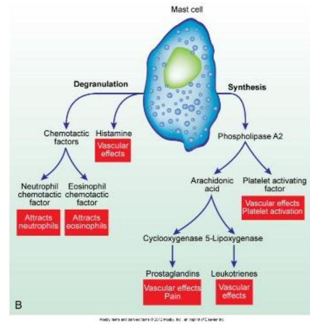


Figure 5: Mast cell processes in inflammation

b) Chemotactic factors

- Neutrophil chemotactic factor
- Eosinophil chemotactic factor
- DEGRANULATION PRODUCT

c) Prostaglandins and leukotrienes

- Products of arachidonic acid pathway
- Induce pain, inhibited by ASA/NSAIDS
- Similar effect to histamine
- Can also act as hormones (chem)
- SYNTHESIS PRODUCT

d) Platelet-activating factor

- Similar effect to leukotrienes + platelet activation
- SYNTHESIS PRODUCT

Endothelium

- Lines blood vessels
- Attached to underlying connective tissue matrix
- Critical in pathophysiology of inflammation
- Interact with circulating cells, platelets, plasma proteins
- Endothelial injury initiates platelet adherence
- Release/regulate inflammatory mediators
- Simple squamous cells lining blood vessel; have tight junction which can be loosened

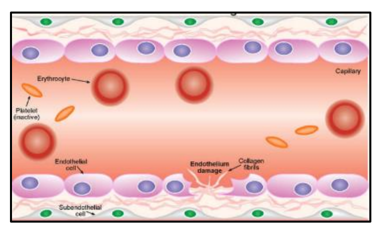


Figure 6: Endothelium in inflammation

Platelets

- Activated by rough endothelium, tissue destruction and inflammation
- Activation leads to interaction with coagulation factors and activates the cascade
    - The cascade is only activated when there is tissue factor release
- They have no nucleus
- Mother cell is megakaryocyte in bone marrow

- **Normal count**: 150,000-450,000 /L
- **Life span**: 8-11 day

Pathology of Acute Inflammation

- Vascular changes
    - Vessels become more porous and leak fluid
- "Acute" refers to the increase in neutrophils as part of the inflammatory response

Types of Exudative Fluids

- In acute inflammation there is a leakage of circulating fluid into tissue (exudative fluids)
- **The four types of exudative fluids are**:

a) *Serous:*
- watery. Indicates early inflammation

b) *Fibrinous:*
- Thick, clotted. Indicates advanced inflammation.

c) *Purulent (suppurate):*
- Pus. Indicates bacterial infection

d) *Hemorrhagic:*
- Bloody; Indicates bleeding


Figure 7: Serous exudate


Figure 8: Fibrinous exudate


Figure 9: Purulent (suppurate) exudate


Figure 10: Hemorrhagic exudate

Manifestations of Acute Inflammation

Local

- Heat, swelling, redness, pain, loss of function

Systemic

- **Fever**
    - Caused by exogenous and endogenous pyrogens
    - Pyrogens act directly on the hypothalamus
- **Leukocytosis**
    - Increased numbers of circulating leukocytes
    - Normal count: 4-11 x 10^9/L
- **General lab markers inflammation**: ESR, CRP
    - **ESR:** erythrocyte sedimentation rate. Basic non-specific test. Examines the sedimentation of red cell mass vertically over time
    - **CRP:** C-reactive protein measured in blood. Pathology of Chronic inflammation, acute phase reactant

- Acute inflammation lasting ~2 weeks (more neutrophils infiltration) and chronic lasts longer (more lymphocytes infiltration)

Chronic Inflammation

Causes of Chronic Inflammation:

- Unresolved acute inflammation
- Microbe surviving inside macrophage
- Persistence of stimulus
  - Toxins, chemicals, particulate matter, or physical irritants

Pathophysiological features of chronic inflammation

- **Dense infiltration of lymphocytes, fibroblasts and macrophages**
  - If macrophages are unable to protect the host from tissue damage, the body will attempt to wall off and isolate the infected area, thus forming a granuloma.
  - TNF-alpha primarily drives the formation of a granuloma.

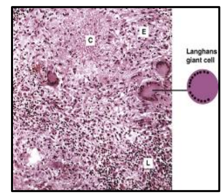


Figure 11: Langerhans giant cell of chronic inflammation

- **Giant cell infiltration and granuloma formation**
  - When macrophages form, they can make a giant cell. Which is active phagocyte that can engulf very large particles.
  - The giant cell and epithelioid cell form the centre of the granuloma and are surrounded by wall of lymphocytes.
- **Simultaneous destruction and healing of tissues**

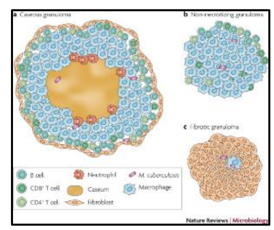


Figure 12: Giant cell and granuloma formation in chronic inflammation

Examples of chronic inflammatory conditions:

- TB
- Rheumatoid arthritis

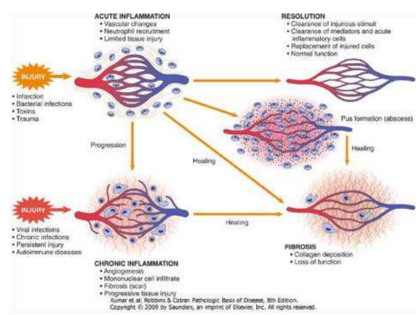


Figure 13: Acute vs chronic inflammation

2: Wound Healing

- The process where the skin (or organ/tissue) repair themselves after injury
- Requires healthy blood vessels and connective tissue

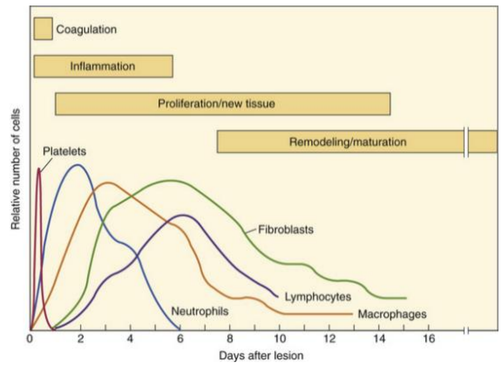


Figure 14: 3 phases of wound healing

Three phases of wound healing:

I. Inflammation (including coagulation) (1-2 days)

- Infiltration of cells of inflammatory response, phagocytosis of debris
- Release of cytokines and inflammatory mediators

II. Proliferation and new tissue formation (3-4 days)

- **Angiogenesis** (new blood vessel formation)
- Fibroblasts grow and form new connective tissue (granulation tissue)
- **Re-epithelialization**: epithelium crawl to provide a cover for new tissue

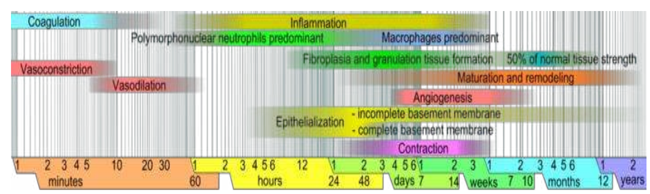


Figure 15: Timescale (minutes-years) for phases of wound healing

III. Remodeling and maturation (few weeks – 2 years)

- Collagen is remodeled and realigned along tension line
- Myofibroblasts decrease wound size (wound contraction)
- Un-necessary cells undergo apoptosis

Types of wound healing:

Primary intention

- Minimal tissue loss
- E.g. Most surgical wounds
- Form clean scar

Secondary intention

- Lots of tissue loss
- May need pack with gauze/drain
- Healing is slower
- Form broader scar
- Examples: tooth extraction, burns, pressure ulcers

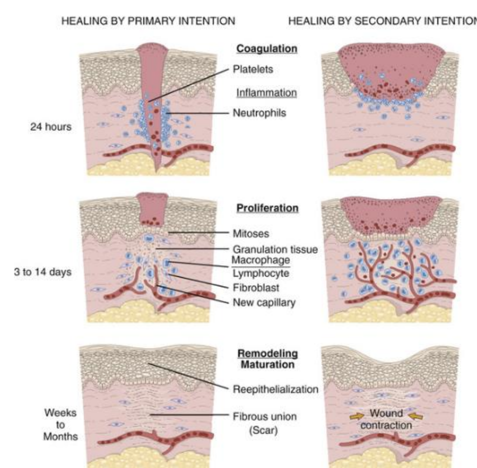


Figure 16: Healing by primary vs secondary intention

Dysfunctional/Complications of Wound Healing

Dysfunctional inflammation

- **Caused by**:
  - Bleeding, hypovolemia, poor inflammatory response
- **Results in**:
  - Fibrous adhesion/infection

Deficient scar:

- Caused by:
  - Inadequate granulation tissue
- Results in:
  - Wound dehiscence, rupture, increase risk of infection

Excessive scar:

- Caused by:
  - Impaired collagen matrix remodeling
- Results in:
  - Keloid/hypertrophic scar

Impaired epithelialization

- Caused by:
  - Steroids, hypoxemia, nutritional deficiencies
- Results in:
  - Incomplete healing

Impaired contraction

- Deficient: decreases engraftment in skin grafts
- Excessive (contracture): impairs joint function post burns

Others:

- Calcification
- Pigmentation
- Pain
- Incisional hernia

Pathophysiological conditions of impaired wound healing

Diabetes Mellitus

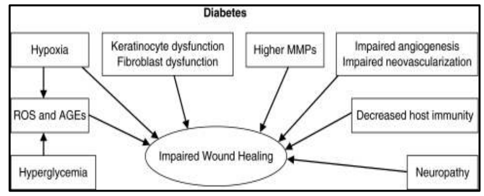


Figure 17: Diabetes and impaired wound healing

- Worst of chronic conditions that can impair wound healing
- Impaired due to:
    - Neuropathy
    - Vascularization (angiopathy)
    - Impaired coagulation
    - Hyperglycemia impairs cell function
- Stress
- Extremes of age
- Disease involving hypoxemia
- Obesity
- Alcoholism
- Poor nutrition
- Smoking
- Medications
    - Steroids
    - Chemotherapy

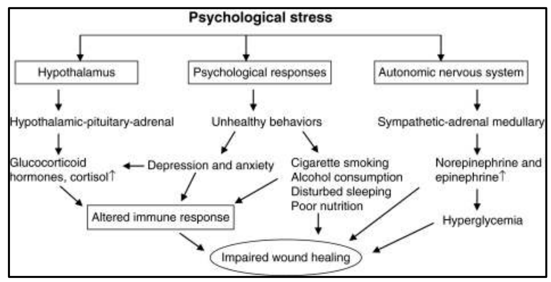


Figure 18: Psychological conditions and impaired wound healing

Wound Healing and Extremes of Age

Neonates

- Transient depression of inflammatory and immune function
- Inefficient neutrophils chemotaxis
- Complement deficiency
- Increased susceptibility to sepsis

Elderly:

- Impaired function of innate immune cells (eg. Phagocytes)
- Increased susceptibility to infections
- **Impaired inflammation likely due to**:
    - Associated chronic illness: DM, CVD, etc.
    - Polypharmacy
- Impaired healing due to loss of regenerative ability of skin

2: Adaptive immunity (3rd line of defence)

Overview of adaptive immunity

Goals

- Destruction of foreign organisms that are resistant to innate immunity and inflammation
- Long term, highly effective protection against future exposure

Characteristics of the adaptive immune response

Inducible

- Requires stimulus
- Diverse and specific

End products:

- Specific immunoglobulins
- Active lymphocytes (T cells, B cells)
- Memory cells

Components

- **Humoral** (B cell mediated), leading to Ig (antibodies)
  - Bind to antigens on bacteria and viruses
- **Cellular** (T cell mediated), leading to generation of effector cells
  - Kill cell directly

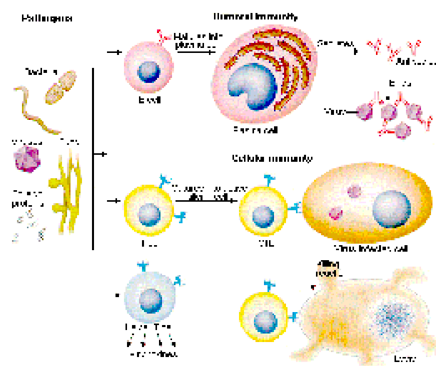


Figure 19: Adaptive immunity (3rd line defense)

Types

- **Active**:
  - Exposure to antigen
  - Immunization
- **Passive**:
  - Via pre-formed antibodies (Ig)

Antigen recognition, processing and presentation

- Antigen processed, then presented by APCs (dendritic cells, macrophages)
- **Recognition occurs**:
  - By means of clusters of differentiation (CD)
  - In conjunction with MHC
- **Results**: (B and T cell activation)
  - B cells differentiate into plasma cells (Ig producing cells)

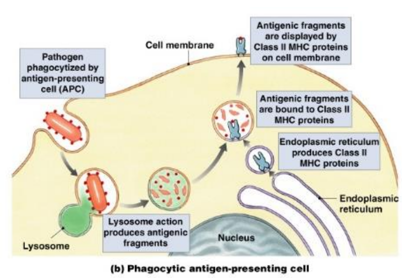


Figure 20: Processes of antigen recognition, processing and presentation

- T cells differentiate into effector cells; cytotoxic T cells (killer cells)

Antibodies:

Antibody (Ig) Structure

- **4 polypeptide chains**
  - Light chains (2) and heavy chains (2)
- **Antigen-binding fragment (Fab)**
  - Recognition sites (receptors) for antigenic determinants
- **Crystalline fragment (fc)**
  - Responsible for biologic function
- **Hinge region**
- **Variable region (V)**
  - Variation in sequence of AA
  - Supports endless diversity of Ig generation
- **Constant region**
  - Similar sequence of AA
  - Responsible for common features of Ig class

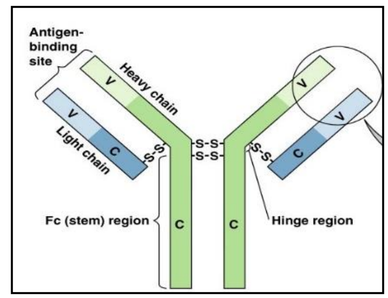


Figure 21: Antibody structure

Antibody classes

- **IgG**- most abundant, crosses placenta, secondary response to infection
- **IgM**- pentamer, first response to infection (primary response)
- **IgA**- dimer, in body secretions
- **IgD**- on B cells, initiate immune response
- **IgE-** Allergic reactions, lysis of parasitic worms

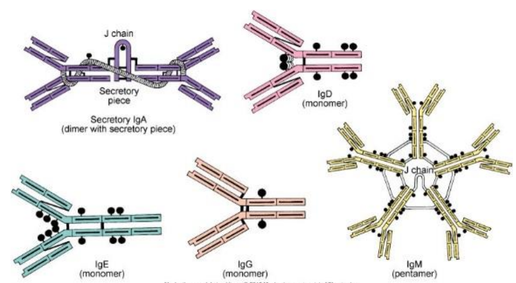


Figure 22: Antibody classes

Antibody functions

- **Destroys antigens**:
  - Directly
    - Neutralization
    - Agglutination/precipitation
  - Indirectly
    - Stimulate phagocytosis
    - Stimulate complement

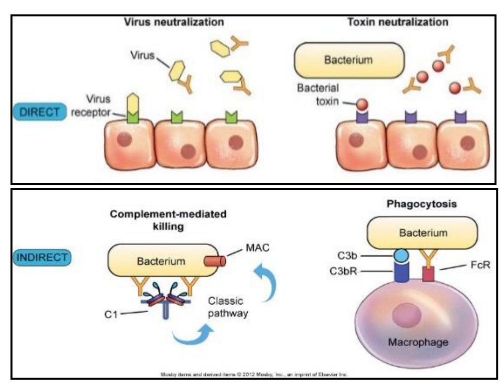


Figure 23: Antibody functions

Cells of Immune Response

Phagocytic cells:

- Macrophages, neutrophils, monocytes

Antigen presenting cells:

- Dendritic cells, macrophages

Specific cells of immune response

- B cells (originates in bone marrow)
- T cells (originates in thymus)
    - *T helper – CD4*
        - Require that antigens be presented in conjunction with *MHC class II* molecules
        - **TH1**: Activate cells related to cell-mediated immunity
        - **TH2**: Activate B cells to produce Ab
    - *Cytotoxic T – CD8*
        - Require that antigens be presented in conjunction with MHC class I molecules
        - Recognize Ag + *MHC I*
        - Induce apoptosis in target cell
    - *T regulatory – CD25 (& some CD4)*
        - Regulate other T cells
        - Regulate T cells response against self
        - Previously named T supressor **(Ts)**

Helpful Hint

To remember origins for B and T-cells, think:

B = Bone marrow

T = Thymus

Primary and Secondary Immune responses

Primary response

- Initial exposure
- Lag phase
- Mediated by IgM
- IgM detected 5-7 days

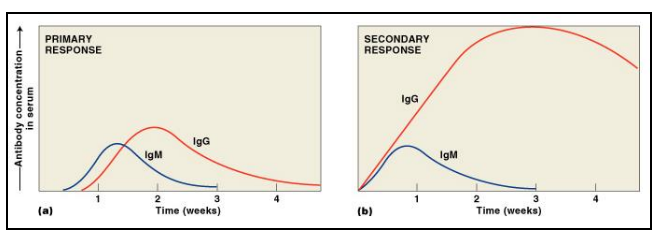


Figure 24: Primary vs secondary immune response

Secondary response

- Second exposure
- More rapid response and larger amounts of Ig produced
- Facilitated by memory cells
- IgM is produced in similar quantities to the primary response, but IgG is produced in considerably greater numbers.

Aging and Immune function

- Decreased T cell activity differentiation
- Decreased antibody response to antigens
- Increase in circulating antigen-antibody complexes
- Increase in circulating autoantibodies
- Decrease in circulating memory B cells

Infections and defects in mechanisms of defense

- Inadequate response = deficiency
- Over/ Inappropriate response = hypersensitivity and autoimmunity

Immune deficiencies

- Failure of immune mechanisms of self-defence
    - **Primary** (congenital) immunodeficiency
    - **Secondary** (acquired) immunodeficiency
        - Due to another illness, drug, or environmental exposure
        - More common

Primary Immune Deficiencies (with selected examples)

Most are the result of a <u>single gene defect</u>

Clinical presentation:

- Recurrent or persistent infections
- Developmental delay as a result of infection

Five groups:

1. *B lymphocyte deficiencies*
    - **Hypogammaglobulinemia** or **agammaglobulionemia**
2. *T lymphocyte deficiencies*
    - **DiGeorge syndrome**: Partial or complete absence of T cell immunity
        - Congenital thymic aplasia or hypoplasia and diminished parathyroid gland development.
        - It is caused by the lack or partial lack of the thymus, greatly resulting decreased T cell numbers and function.
        - Leads to:
            - Neural tube defect
            - Abnormal development of facial features (low set ears, fish shaped mouth, wide eyes)

        - Calcium deficiency

3. *Combined T and B cell deficiencies*
    - Severe combined immunodeficiencies **(SCID)**
        - Low T cells, very low IgM and IgA from underdeveloped thymus
    - **Wiskott-Aldrich syndrome (WAS)**
        - X linked recessive disorder.
        - IgM antibodies are depressed.
        - Very susceptible to bacterial infections and bleeding
    - **Ataxia-telangiectasia (AT)**
4. *Complement defects*
    - **C3 deficiency**
        - Very severe. Lose C3 in the compliment cascade. Results in recurrent life-threatening infections with encapsulated bacteria
5. *Phagocyte defects:*
    - **Chediak-Higashi syndrome** (lysosomal problem)
    - **Myeoloperoxidase deficiency** (chronic granulomatous disease)
        - Defect in myeloperoxidase–hydrogen peroxide system
        - Causes deficient production of hydrogen peroxide needed in phagocytic killing
        - Results in recurrent pneumonia; tumour-like granulomata in lungs, skin bones; and other infections

Secondary Immune deficiencies

- Also known as, acquired deficiencies
- More common than primary.

Causes:

- Psychologic stress
- Dietary insufficiencies
- Malignancies
- Liver disease
- Physical trauma
- Drugs
- **Viral**:
    - E.g. Acquired immunodeficiency syndrome AIDS

Evaluation of immunity

- **Complete blood count (CBC)** with a differential
  - Subpopulations of lymphocytes, granulocytes
- Plasma levels of **immunoglobulins (Ig)**
  - Subpopulations of immunoglobulins
- Bone marrow examination
- Serum complement levels (C3a, C5a)
- Skin tests
- Evaluation of body systems to identify causes

Hypersensitivity

- Exaggerated immunologic response to an antigen that results in disease or host damage

Classification: (based on immune mechanism)

- **Type I**
  - Anaphylaxis (IgE mediated)
- **Type II**
  - Tissue specific reactions
- **Type III**
  - Immune complex mediated
- **Type IV**
  - Delayed cell mediated immunity

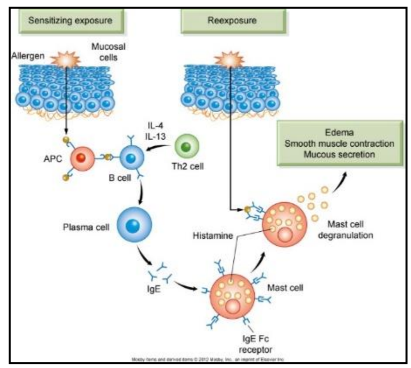


Figure 25: Type-I hypersensitivity

Type I

- **Immediate hypersensitivity reactions**: occur within mins to a few hrs after exposure
- Also known as *allergy* or *anaphylaxis*
- Response to environmental antigens (allergens):
  - Pollens, molds, fungi, foods, animals
- **IgE mediated**
  - IgE binds via its Fc receptor on surface of mast cells
  - Degranulation and histamine release

Examples

- Bronchial asthma
- Bee sting

Clinical manifestations

- Itching/urticaria (hives)

- Conjunctivitis
- Rhinitis
- Hypotension
- Bronchospasms
- Dysrhythmias
- GI cramps and malabsorption

Genetic predisposition

- Most often

Tests

- Food challenges
- Skin tests
- Lab tests

Desensitization

- With caution

Type II

- ****Tissue specific**
- Immune response is directed to a specific cell/tissue/organ

Mechanism

- Cells/tissues are destroyed by:
  - Antibodies
  - Complement system
  - Phagocytosis
- **Result:** cell/organ malfunctions

Examples

- **Grave's disease** (antibody against thyroid tissue)
- **Pernicious anemia**
- Some types of **glomerulonephritis** (Berger's Disease)

Type III

- ****Immune complex mediated**

Mechanism

  - Antigen-antibody complexes are formed in circulation and are later deposited in vessel walls or extravascular tissues

- **Not organ specific** (unlike type II)

Examples

- **Serum sickness; *Raynaud phenomenon*** **Figure 26: Integrated processes of antigen recognition and cellular destruction**
    - A condition caused by the temperature-dependent position deposition of immune complexes in the capillary beds of the peripheral circulation. Certain immune complexes precipitate at temperatures below normal body temp, particularly in the nose, toes, tips of fingers and are called cryoglobulins. The precipitates block circulation and cause localized pallor and numbness, followed by cyanosis.
- **SLE (also considered autoimmune disease)**
- Post-streptococcal glomerulonephritis

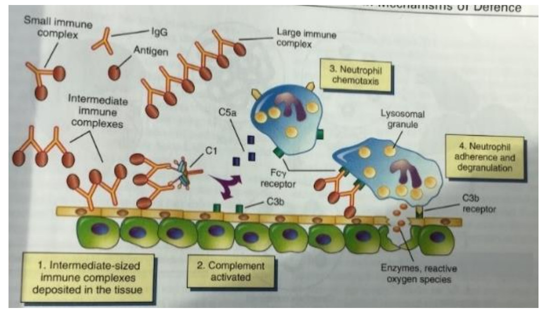


Figure 26: Type-III hypersensitivity

Type IV

- **Cell-mediate
- Do not involve antibody
- Mediated by delayed cytotoxic T lymphocytes response
    - Direct killing of cellular targets by Tc or recruitment of phagocytic cells by TH1 cells
- **Examples**:
    - Rheumatoid arthritis
    - Graft rejection
    - Tuberculosis
    - Contact dermatitis

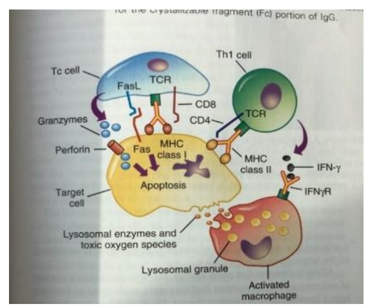


Figure 27: Type-IV Cell-mediated immune response

Autoimmunity

- Breakdown of self-tolerance
    - Body recognizes self-antigens as foreign and react to it
- Genetic and environmental factors

Examples:

- *Antigenic mimicry*:
    - *Rheumatic fever*: reactions that occur within minutes to a few hours after exposure to an antigen
    - *Post infectious glomerulonephritis*
- *Neoantigen*:
    - Newly acquired and expressed antigen
    - *Cancer immunity*

- *Multisystem autoimmunity*:
    - *Systemic Lupus Erythematosus (SLE)*

Systemic Lupus Erythematous (SLE)

- Chronic multisystem inflammatory/autoimmune disease

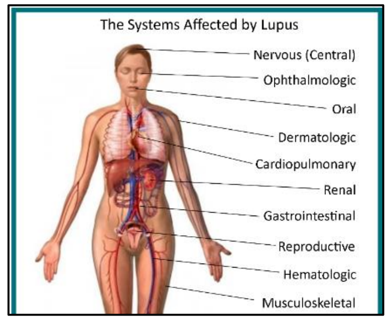


Figure 28: Systemic lupus erythematous (SLE)

Causes:

- Genetics
- Drug induced: hydralazine, procainamide and isoniazid

Pathophysiology

- Development of autoantibodies against:
    - Nucleic acids
    - Erythrocytes
    - Coagulation proteins
    - Phospholipids
    - Lymphocytes
    - Platelets

No	Criteria
1	Malar rash
2	Discoid rash
3	Photosensitivity
4	Oral ulcers
5	Arthritis
6	Serositis
7	Renal disease a. > 0.5 g/d proteinuria, or b. ≥ 3+ dipstick proteinuria, or c. Cellular casts
8	Neurologic Disorder a. Seizures, or b. Psychosis (without other cause)
9	Hematologic disorders a. Hemolytic anemia, or b. Leukopenia (< 4000/mL), or c. Lymphopenia (< 1500/mL), or d. Thrombocytopenia (< 100,000/mL)
10	Immunologic abnormalities a. Positive LE cell preparation, or b. Antibody to native DNA, or c. Antibody to Sm. or d. False-positive serologic test for syphilis
11	Positive antinuclear antibody (ANA)

Figure 29: Table - Criteria for diagnosis of SLE

Common features

- Facial rash (malar rash)
- Discoid rash
- Photosensitivity
- Oral or nasopharyngeal ulcers
- Non-erosive arthritis
- Serositis
- Renal manifestations
- Neurologic manifestations
- Hematologic manifestations
- Immunologic manifestations
- Positive antinuclear antibodies (ANA)

Figure 30: Malar rash of SLE

Criteria for Dx:

- **Patient must have at least 4 out of the 11 symptoms = Positive SLE

Alloimmunity

- Reacting to allo-antigens (external Ag from other species)
- **Includes**:

- Transplant/graft rejection
- Transfusion reactions

Transplant rejection

- Classified according to time
- **Hyper acute (rare)**
  - Immediate reaction (white-graft)
  - Pre-existing antibody to the antigens within the graft (type II reaction to HLA antigens on the vascular endothelial cells in the grafted tissue)
- **Acute**
  - Rapid; weeks to months
  - Cell-mediated immune response against major HLA antigens
  - Infiltration of lymphocytes and macrophages (type IV reaction)
- **Chronic**
  - Months to years
  - A weak cell-mediated response against minor HLA antigens
  - Slow progressive organ failure

Graft-Versus-Host-Disease

- T cells in the graft are mature and capable of cell-mediated destruction of tissues of the recipient
  - Occurs in immunocompromised people
- People with SCID are at high risk of GVHD
  - GVHD targets skin, liver, mouth, eyes and GI tract
- Risk can be diminished by removing mature T cells from the tissue used to treat individuals with immune deficiencies

Transfusion reactions

- ***ABO blood groups (mismatched transfusion):**
  - Reactions that occur within minutes to a few hours after exposure to an antigen
  - Platelets alloimmunization (Platelets are covered in proteins and some people can develop ABs against platelets that are not their own.)

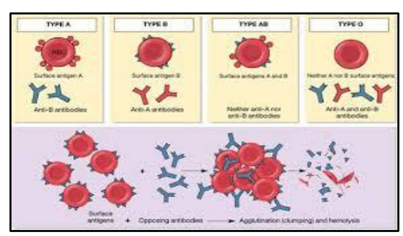


Figure 31: Transfusion reaction - ABO blood groups

- **Rh incompatibility in pregnancy and "Hemolytic disease of newborn"**
  - Caused by IgG (anti-D alloantibodies) produced inside the body of Rh-negative mothers against erythrocytes of Rh-positive fetuses (passed via placenta)

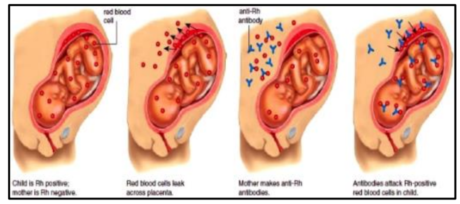


Figure 32: Transfusion reaction - Rh incompatibility in pregnancy

Review: Bacterial vs Viral infection

Bacterial:

- Antibody mediated
- Antigen presentation activates
  - Th cells
  - Tc cells
- B cell activation and differentiation into plasma cell
- Plasma cells secrete antibodies

Viral:

- Antibody + cell mediated
- Similar to above bacterial infection (for ab mediated portion)
- For cell mediated
  - Tc and NK cells activated by contact with virally infected cells
  - Kill (lyse) virally infected cells

3: Cellular Proliferation and Cancer

Learning Outcomes

Cellular Proliferation and Cancer

- Define the terms **tumor**, **neoplasia**, **cancer**, **benign**, **malignant**, and **carcinoma in situ**.
- Identify and differentiate types of cancers based on cell type, such as adenocarcinoma, lymphoma, or sarcoma.
- Define the following concepts: **autonomy**, **transformation**, **anchorage-independence**, **immortal**, **anaplasia**, **pleomorphic**, **stem cells, tumor markers, clonal proliferation**.
- Describe how the following mechanisms promote tumor growth: autocrine stimulation, increase in growth factor receptors, mutated cell-surface receptor, mutation in ras intracellular protein, and inactivation of Rb tumor suppressor.
- Define and relate **angiogenesis** to cancer growth.
- Describe the role of telomeres and telomerase in tumor development and compare and contrast the gene products of **proto-oncogenes**, **oncogenes**, and **tumor suppressor genes**.
- Describe how the following genetic events can activate oncogenes: point mutations, chromosome translocations, gene amplification, tumor suppressor gene mutation and loss of heterozygosity.
- Define **mutagen** and **carcinogen.**
- Describe the role of chronic inflammation and cancer cell development and list bacteria and viruses that can cause of cancer.
- Describe cancer cell growth and behaviour including steps for metastatic spread and mechanisms that favour or inhibit local spread of cancerous cells.
- Describe the mechanisms of the following environmental factors in cancer development: tobacco use, ionizing radiation, ultraviolet radiation, alcohol consumption, sexual and reproductive behaviour, physical activity, occupational hazards and exposure, air pollution, electromagnetic fields, and diet.
- Describe the common clinical manifestations of cancer: anemia, leukopenia, thrombocytopenia, infection, **paraneoplastic syndrome**, fatigue, and pain and the syndrome of cachexia.
- Describe methods for diagnosis of cancer including biopsy types.
- Compare and contrast the modalities for the treatment of cancer: chemotherapy, surgery, radiation, and immunotherapy
- Describe common side effects associated with cancer treatment in the following systems: GI tract, bone marrow, integument, and reproductive tract

Definitions: Cellular Proliferation and Cancer

Term	Definition
Cancer	-Derived from the word for crab (it can spread); karkinoma (carcinoma)
Carcinoma in Situ	-High grade local dysplasia; hasn't invaded the basement membrane
Neoplasia	-Tumour
Tumor	-Abnormal cell growth
Benign	- See table 1
Malignant	- See table 1
Definitions Related to Cancer Growth	
Anaplasia	-Poorly differentiated cells
Anchorage-independence	- Growth does not need adhesion to surface
Angiogenesis	- New blood vessel formation
Autonomy	- No dependency; self-growth
Clonal proliferation	- Rapid cellular proliferation that leads to same clone
Immortal	- No death
Pleomorphic	- Variable shape
Transformation	- Change into a different form
The Syndrome of Cachexia	- A syndrome of progressive weight loss, anorexia, and persistent erosion of body cell mass in response to a malignant growth. - Seen in 80% of cancer patients at death
Oncogenes	- Abnormal gene that **promotes** tumors.
Paraneoplastic Syndrome	- Symptoms caused by the biologic activity of a tumor marker - E.g. Catecholamines secreted by a benign tumor of the adrenal medulla (pheochromocytoma) leading to tachycardia, hypertension, etc.
Proto-oncogenes	- Normal gene that directs protein synthesis and cellular growth
Stem Cells	- Cell capable of self-renewal and differentiation

	- Exist locally in each tissue to help repair - Limitation in nervous and muscle tissue
Telomerase	- Enzyme that adds new DNA repeats to telomeres allowing cells to divide indefinitely
Telomeres	- Protective cap at the end of chromosomes - Shortens with each cell division
Tumor Markers	- Substances produced by cancer cells - Can be measured in blood, CSF or in urine
Tumor Suppressor Genes	- Genes that encodes proteins which in their normal state negatively regulate cell proliferation or promote apoptosis

Review

- The following refers to areas of previous knowledge that applies to the current unit and will help to understand and apply the concepts during the lecture.
- It is recommended that previous lectures and resources are used to clarify details in areas as determined by the needs of you, the student.

Anatomy and Physiology:

- **There are 4 types of body tissues:**
    1. Epithelium
    2. Connective tissue
    3. Muscle
    4. Neural
- Epithelium is classified into epithelia and glands.
    - Epithelia can be further divided by cell structure, each one having distinct properties related to the locations in which they are found.
    - Epithelia can be classified by cell shape (squamous, columnar) and density (simple, stratified).
        - The general exception is pseudostratified epithelium lining the respiratory tract.
    - There are 2 types of glandular tissue:
        1. Exocrine
        2. Endocrine
- **There are 3 types of connective tissue (CT):**

1. CT Proper
2. Fluid CT
3. Supportive CT

Clinical Chemistry Review Topics:

- DNA structure, transcription, and gene expression
- Terms: gene, chromosome, chromatin, histone

Pathophysiology Review Topics:

- Pathological cellular adaptation

Cancer

- *Cancer*: Derived from the word for crab (it can spread), karkinoma
- Cancer is a leading cause of death in Canada and is responsible for 30% of all deaths. Cancer and then heart disease. 1 in 3 people in Canada gets cancer.
- **Most common cancer types in Canada:**
  - Breast → Prostate → Colorectal → Lung

Tumor

- Also called a *neoplasm* = new growth
- Classified as either *benign* or *malignant* (see below comparison)

Comparing Benign and Malignant Tumors (Table)

Benign	Malignant
Grow slowly	Grow rapidly
Well-defined capsule	Not encapsulated
Not invasive	Invasive
Well-differentiated	Poorly differentiated
Low mitotic index	High mitotic index
Do not metastasize	Can metastasize

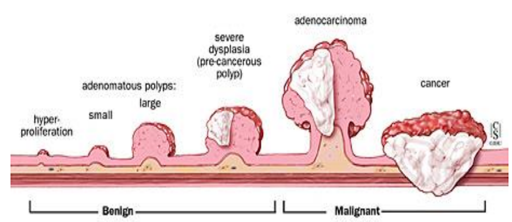


Figure 1: Benign vs malignant tumors

Cancer Cells in the Laboratory

Characteristics of Cancer Growth

- Less need for growth factors
- Lack of contact inhibition
- *Anchorage independence*
  - Do not require attachment to a surface and therefore will continue to grow when suspended in soft agar gel
- *Immortality* – do not die

Human cancer cell lines:

- Can grow independently
  - A piece can be dissected from a tumor, cells teased and frozen/preserved in tubes and use to study later.
  - These cells can live forever and are used for research (drugs, animals, immunity).
- Can create tumors and metastasize in mice with no immune system
- **Examples of human cancer cell lines** (used for research)
  - HELA cells (cervical cancer)
  - MDA (breast cancer)
  - DU145 (prostate cancer)

Tumor Classification and Nomenclature

Benign Tumors

- Naming according to the tissue of origin
- Mostly include the suffix "-oma"
- **Common examples**:
  - *Glioma* = brain
  - *Chondroma* = cartilage
  - *Leiomyoma* = uterine fibroid
  - *Lipoma* = fat tissue
  - *Benign melanoma* = skin
- However, use a cautionary approach to benign tumors
  - Not all tumors ending in "oma" are benign, SOME CAN BE MALIGNANT (e.g. melanoma and lymphoma).
  - *A benign tumor can progress to become malignant and requires frequent monitoring and/or treatment depending on stage, type, risk, and patient preference.
  - *Benign does not mean asymptomatic. An otherwise benign tumor may cause other symptoms depending on the location and relationship to other organs.
    - For example, compression of blood vessels and organs (especially those that do not adapt to invasive growth such as the brain)

Malignant Tumors

- Key characteristics of malignant tumors:
  - *Anaplasia* = poorly differentiated
  - *Pleomorphic* = vary in size/shape
  - Metastasize – the most deadly characteristic of malignant

- Malignant tumors are named according to their tissue of origin (if known), followed by organs/organ systems of metastasis.
    - **Epithelial tissue (carcinoma)**
        - Adenocarcinoma (from or form ductal/glandular structures)
        - Squamous cell carcinoma
        - Anaplastic carcinoma (heterogeneous/undifferentiated)
    - **Connective tissue (sarcoma)** = CT, muscle or bone
        - 90% diagnosed in children and young adults
        - Fibrosarcoma = fibrous tissue
        - Osteosarcoma = bone
        - Rhabdomyosarcoma = muscle
        - Leukemia (blood origin, still CT)
        - Lymphoma = lymph node

Other Nomenclature

- **Carcinoma in situ (CIS)**
    - High grade dysplasia
    - Local (pre-invasive) epithelial or glandular origin
    - "local" – hasn't invaded the basement membrane. Usually intraductal, confined to the duct, cannot leave
- **Cancer of unknown primary origin (CUP)**
    - Metastasized but do not know where from
    - If you find cells and tissues outside of where they are supposed to be = CUP
    - Example: squamous epithelial cell in the lung

Tumor Microenvironment

- Cancer cells are surrounded by a complex microenvironment:
    - Comprised of numerous different cells:
        - Endothelial cells of blood and lymphatics
        - Stromal fibroblasts
        - Immune/ inflammatory cells
    - Often hypoxic → angiogenesis
- This is known as tumour microenvironment and determines the tumour behaviour and response -or lack of- to therapy
- The tumour cell makeup (microenvironment) contributes to tumor heterogeneity:

- Every tumor matures in its own way, the composition of cells is different, making a unique microenvironment.
  - It is like an ID
- This is the reason why some tumors but not others of the same type resist chemotherapy (subject of research)

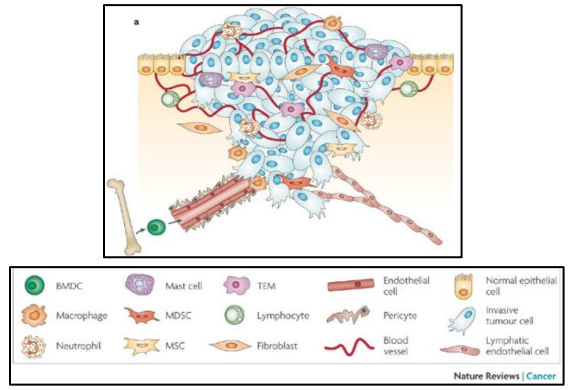


Figure 2: The tumor microenvironment (note significant cell heterogeneity)

Cancer Metabolism

- Cancer cells have distinct nutritional requirements compared to normal cells and therefore must develop mechanisms to meet these requirements. This includes:
  - Extract nutrients from blood stream
  - Use aerobic glycolysis (glycolysis (no kreb's cycle) even in presence of O_2) to support growth and proliferation
    - Glycolysis makes 2 pyruvates (2 ATP) and is anaerobic process (typically)
    - Because cancer cells use glycolysis even in aerobic conditions (i.e. with oxygen present) they never use oxygen and will always have energy to divide.
    - This is unlike obligate aerobes (like human cells) that will stop dividing and eventually die in the absence of oxygen.

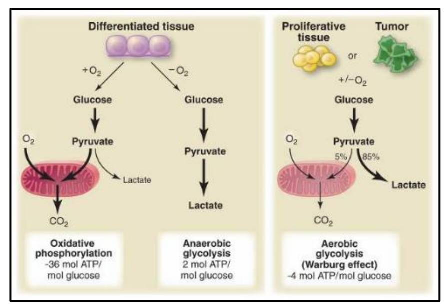


Figure 3: Energy production in normal vs. proliferative (tumor) tissue

  - Divide even in hypoxic or acidic states
    - The tumor environment becomes acidic from glycolytic energy production as a result of the production of ketones from acetyl CoA and lactate.

Cancer and Angiogenesis

- *Angiogenesis*: Growth of new vessels
- **Advanced cancer can secrete angiogenic factors**
  - Vascular endothelial growth factor (VEGF)
  - Platelet derived GF
  - Basic fibroblast GF
- *Why do tumors need new BV given they don't need oxygen??*
  - THEY NEED THE NUTRIENTS

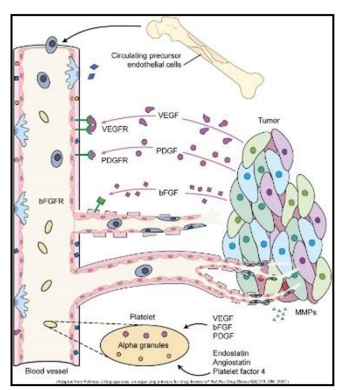


Figure 4: Angiogenesis and the presence of growth factors

Genes, DNA and Cancer

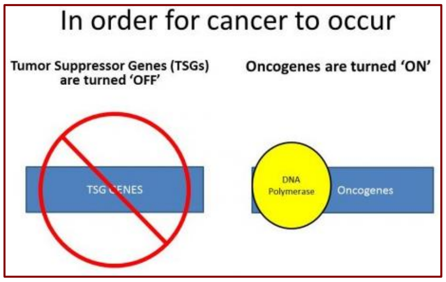
Telomeres and immortality

- Normal somatic cells cannot divide indefinitely
- *Telomeres*:
    - Protective caps in chromosomes' ends
        - E.g. TTAGGG is a sequence that caps the end of a chromosome in somatic cells
    - Become shorter with each cell division
        - As the cells divide telomeres become closer together until eventually, they are lost
- *Telomerase* adds new DNA repeats to telomeres allowing cells to divide indefinitely.
    - Stem cells express telomerase
    - Cancer cells activate telomerase
        - This allows unlimited division and proliferation
- CANCER CELLS ARE IMMORTAL BECAUSE OF THEIR TELOMERES

Oncogenes and Tumor-Suppressor Genes

- *Oncogene*:
    - Abnormal gene that **promotes** tumors.
- *Proto-oncogene:*
    - Normal gene that directs protein synthesis and cellular growth.
    - **Can turn into an oncogene due to**:
        - Mutation
            - Point mutation in RAS gene (pancreatic and colorectal cancer)
            - Point mutation of EGF receptor tyrosine kinase (lung cancer)
        - Chromosomal translocation
        - Activation/increased expression
            - Gene amplification is one example that can result in increased expression of gene.
                - E.g. N-MYC oncogene is amplified in 25% of childhood neuroblastomas.
                - The HER2 gene is amplified in 20% of breast cancers.
- *Tumor-suppressor genes (antioncogenes)*
    - Gene that encodes proteins which in their normal state **negatively regulate cell proliferation or promote apoptosis.**

- Prevent cancer
- These genes must be inactivated for cancer to occur
- Both tumor suppressor genes from mom and dad must be inactivated for an oncogene to be activated



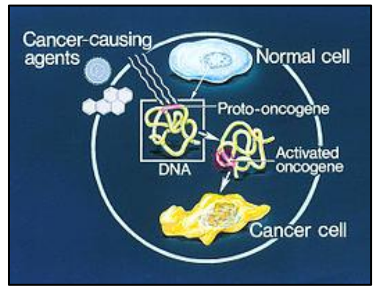


Figure 5: Oncogenes and tumor-suppressor genes in cancer development

Genetic changes in Cancer – Examples

Point mutations in proto-oncogenes

- RAS gene mutations: pancreas and colorectal cancers
- EGF receptor tyrosine kinase: lung cancer

Chromosomal translocation

- T(8;14) results in activation of MYC proto-oncogenes → **Burkitt lymphoma**
  - MYC gene is usually on chromosome 8 and is regulated at low levels of producing lymphocytes.
  - The chromosomes translocate, the MYC gene is now on chromosome 14 under B cell IgG control.
  - MYC genes are now produced at high levels, hence the lymphoma (high lymphocytes).

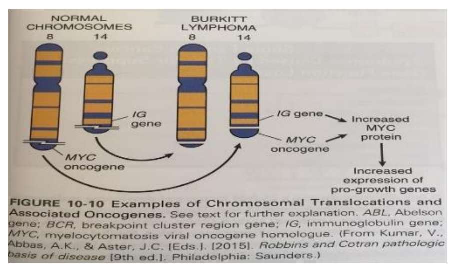


Figure 6: Burkitt lymphoma chromosomal translocation

- T(9;22) results in BCR-ABL fusion genes → chronic myeloid leukemia (**CML)**
  - "Philadelphia chromosome" (name referring to high levels of CML in Philadelphia in the 60s)
  - BCR is on chromosome 9, and ABL is on chromosome 22. The 2 genes are then fused together as a fusion gene ; new chromosome 22.

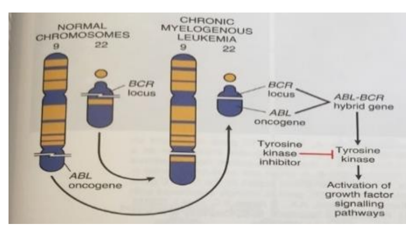


Figure 7: CML chromosomal translocation

- BCR-ABL fusion gene codes for unregulated protein tyrosine kinase that promotes growth of myeloid cells. CML = chronic myeloid (myelogenous) leukemia

Gene amplification

- Duplication of a gene that results in multiple copies
- N-MYC oncogene is amplified by 25% in childhood neuroblastoma cases
- HER2 gene is amplified in 20% of breast cancers

Loss of function of tumor suppressor genes

- Retinoblastoma (RB1 gene)
- Neurofibromatosis (NF1 gene)
- Breast cancer (BRCA1 gene)
    - Use it to test family members as well → could be predisposed to cancer

Tumor markers

- Substances produced by cancer cells
- Present on cell membranes, in blood, CSF or in urine
- **Used to**:
    - Screen and ID individuals at high risk for cancer
    - Diagnose specific types of tumors
    - Observe clinical course of cancer and treatment effectivenss
- The problem = specificity, there are a lot of **false positives**
    - Often the presence of a marker cannot tell where the tumor is or what type
    - *Example*: CA125 is present in many cancer (lung, gut, pancreas, breast, ovaries, endothelium)

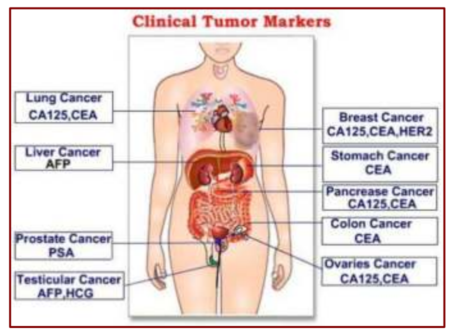


Figure 8: Clinical tumor markers

Marker name	Type of tumor
Alpha fetoprotein (AFP)	Liver cancer
Prostatic specific antigen (PSA)	Prostate cancer
Bence Jones protein	Multiple myeloma
Homovanilic/ Vanilylmandelic acid HVA/VMA)	Neuroblastoma Pheochromocytoma

Figure 9: Table - Tumor markers

Cancer stem cells

- Stem cells have the ability of self-renewal and differentiation
    - Stem cell division can create new stem cells
    - **Stem cells are pluripotent**:
        - Have the ability to differentiate into multiple types of cells
- Some cancer cells possess characteristics of stem cells

- Chemotherapy fails because some cancer cells behave like stem cells

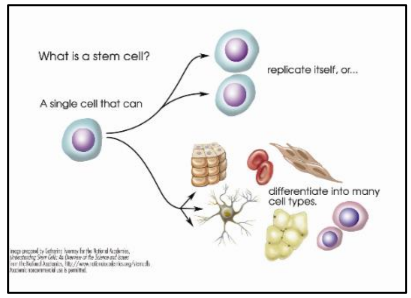


Figure 10: Stem cells

Inflammation and Cancer

- Chronic inflammation is critical for cancer development
  - Cytokine release from inflammatory cells
  - Free radicals
  - Promoting mutation
  - Decreased response to DNA damage
- *Inflammation generates free radicals (toxic), leads to cancer.
  - The continual irritation of cells is what causes the cancer.
  - Cytokines can be released, can mediate cancer.
    - These cytokines can affect DNA and mutate the chromosomes, oncogenes could be expressed.
- Examples:
  - Smoking → Chronic bronchitis → lung cancer
  - Inflammatory bowel disease (Crohn's or Ulcerative colitis) → colorectal carcinoma
  - Reflux esophagitis → irritation → esophageal carcinoma
  - Skin inflammation → melanoma

Immune system and cancer

- Normal immune system protects against cancer:
  - NK cells mediate natural anticancer protection (immune surveillance)
  - Antitumor effect of inflammatory cytokines: TNFα, IL-2,11,12
- Immunosuppressive drugs can promote cancer:
  - Non-Hodgkin lymphoma (10x)
- Immunodeficiency diseases can promote cancer:
  - Kaposi sarcoma in AIDS (1000x)
- Tumors stimulate inflammatory cells to secrete cytokines that promote their survival
- Tumors learn to escape adaptive immunity
  - In tumor microenvironment → disable cytotoxic T cells, inhibit then from activation into Killer T (cytotoxic CD8 lymphocyte)

Infection and Cancer

- Viruses:
  - Hepatitis B and C viruses → Hepatocellular carcinoma
  - Epstein-Barr virus (EBV) → Burkitt's Lymphomas (t8,14)
  - Kaposi sarcoma herpes virus (KSHV) → Kaposi sarcoma
  - Human papillomavirus (HPV) → cervical cancer
  - HTLV virus → adult T-cell leukemia/lymphoma
- Bacteria
  - Helicobacter pylori → stomach carcinoma, mucosa-associated lymphoid tissue (**MALT**) lymphoma
  - Chronic bacterial cholecystitis → gall bladder cancer
- Helminths
  - Schistosomiasis → Bladder, liver, rectal cancer

Chemicals and Cancer

- Cigarette smoke is the most common world-wide carcinogen:
  - Lung (squamous cell carcinoma, adenocarcinoma, small cell carcinoma)
  - Larynx /esophagus (squamous cell carcinoma)
  - Pancreas (adenocarcinoma)
  - Kidney (renal cell carcinoma)
  - Bladder (transitional and squamous cell carcinoma)
  - Cervix (cervical carcinoma)
- Ethanol is associated with:
  - Hepatocellular carcinoma
  - Squamous cell carcinoma of the esophagus (especially in conjugation with cigarette smoke)
- Arsenic; in cigarette smoke is associated with:
  - Angiosarcoma of the liver
  - Lung cancer
- Asbestos; old buildings, is associated with
  - Bronchogenic carcinoma and mesothelioma
    - Asbestos used to be an insulator. Famous now for causing mesothelioma (pleural tumor).

Cancer Spread

Local spread

- Spread of cancer from primary site of origin to a nearby site directly connect with it. They do not leave the organ.
- **Mechanisms**:
    - Mechanical pressure
    - Release of lytic enzymes – kill tissue around them
    - Decreased cell-to-cell adhesion
    - Increased motility – they can push through other cells to grow

Cancer Invasion

Metastasis

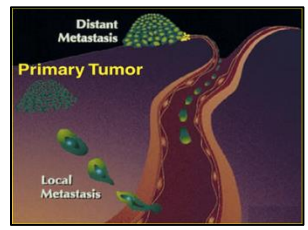


Figure 11: Distant versus local metastasis

- Spread of cancer from primary site of origin to a distant site (not directly connected with it)
- **Steps**:
    - Direct or continuous extension
    - Penetration into lymphatics, BV or body cavities
    - Travel via lymph or blood
- Only few cancerous cells are required for metastasis.
- Tumor will start growing in a secondary site.
- Organ tropism
    - Preferential growth of cancer cells in certain organs
        - Growth factors, chemokines, hormones and chemotactic factors
        - E.g. Bone metastasis in prostate cancer

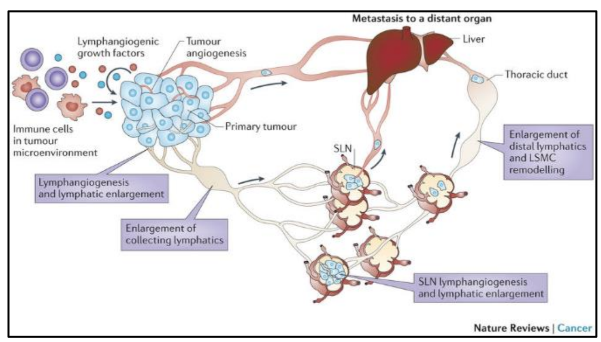


Figure 12: Mechanisms of cancer metastasis

*Local mass though blood or lymph. No idea where it will end up. They have some preferences though: **prostate likes to go to bone, liver likes to go to lung.** *

Diagnosis

- Tumors can be discovered on:
    - Screening
    - Routine investigation
    - Investigation of symptoms

- Diagnosis is based on
  - Tumor biopsy and clinical staging
  - Tumor markers
  - Immunohistochemistry
  - Cell phenotyping
  - Genetic analysis

Tissue biopsy

Types:

- *Excisional/incisional biopsy*:
  - Removal of the entire tumor = excisional
  - Removal of small part of tumor = incisional

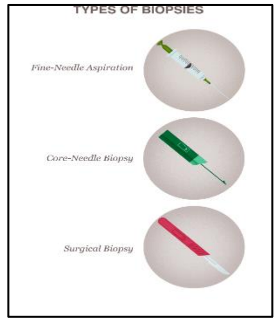


Figure 13: Types of biopsies

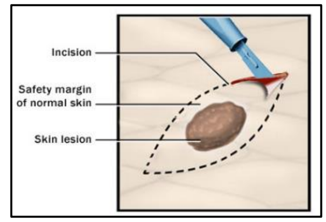


Figure 14: Surgical biopsy of skin lesion

- *Needle biopsy*:
  - Fine need aspiration (FNA): removal of fluid and very small pieces of tissue from the tumor
  - Core needle biopsy: small cylinder of tissue
- *Exfoliative cytology*:
  - Gentle scraping or brushing some cells from the organ or tissue being tested. E.g. Pap smear

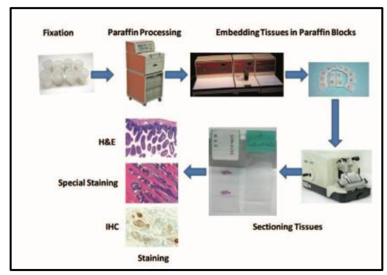


Figure 15: Processing and histopathology of biopsies

Staging

- **Microscopic analysis**
  - Stage 0: high grade dysplasia
  - Stage 1: local growth (cancer only in the origin site)
  - Stage 2: local invasion (cancer that is locally invasive)
  - Stage 3: spread to regional structures (regional spread such as lymph nodes)
  - Stage 4: distant metastasis (distant spread)

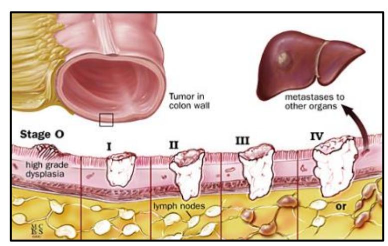


Figure 16: Tumor staging - Microscopic analysis

- **World Health Organization's TNM system**
  - T for tumor spread
  - N for node involvement
  - M for the presence of distant metastasis

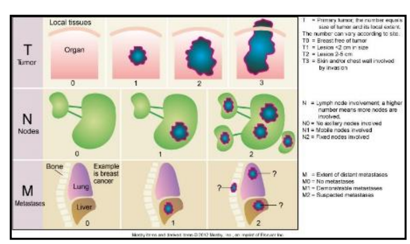


Figure 17: TNM classification (example, breast cancer)

Clinical manifestations of Cancer

- **Manifestations are based on site, tumor size, effects:**

- Physical pressure/obstruction
- Loss of normal function
- Distant/complex symptoms (paraneoplastic syndrome)
    - Due to secretion of bioactive compounds
    - Example: pancreatic tumor secreting ACTH → Cushing Syndrome

- **Manifestations include:**
    - Pain (from the pressure of the tumor, or psychological fear)
    - Fatigue = most common symptom
    - Symptoms of cachexia
    - Symptoms of anemia, leukopenia, thrombocytopenia
    - Infection
    - GIT manifestations
    - Hair and skin manifestations
    - Mass

Pain

- Little or no pain in the early stages of malignancy
- Influenced by fear, anxiety, sleep loss, fatigue, and overall physical deterioration
- **Mechanisms**
    - Pressure
    - Obstruction
    - Invasion of sensitive structures
    - Stretching of visceral surfaces
    - Tissue destruction
    - Inflammation
- Pain leads to fatigue!

Fatigue

- Most frequently reported symptom of cancer and cancer treatment
- Subjective clinical manifestation
- Tiredness, weakness, lack of energy, exhaustion, lethargy, inability to concentrate, depression, sleepiness, and lack of motivation
- **Suggested causes**:
    - Sleep disturbances

- Biochemical changes from circulating cytokines secondary to disease and treatment
- Psychosocial factors
- Nutritional factors

Cachexia

- Syndrome of cachexia
  - A syndrome of progressive weight loss, anorexia, and persistent erosion of body cell mass in response to a malignant growth.
  - Most severe form of malnutrition
  - Seen in 80% of cancer patients at death
  - **Includes**:
    - Anorexia, early satiety, weight loss
    - Anemia, asthenia, taste alterations
    - Altered protein, lipid and carbohydrate metabolism
    - ***Most of these are from the decreased amount of muscle and fat available.**

Figure 18: Syndrome of cachexia

Anemia

- A reduction of hemoglobin/red cell mass in the blood
- **Mechanism**:
  - Chronic bleeding resulting in iron deficiency
  - Severe malnutrition
  - Malignancy in blood-forming organs
    - E.g. Bone marrow
  - Medication

Leukopenia and thrombocytopenia

- Direct tumor invasion of bone marrow is a main cause
- Toxic effect of chemotherapy on bone marrow is another
  - Chemo affects rapid growing cells (such as bone marrow cells)

Infection

- Risk increases when the absolute neutrophil and lymphocyte counts fall
- Leading cause of complications in persons with malignant diseases
- The advanced malignancies are immunosuppressive, and so are the radiotherapy and chemotherapy options

Gastrointestinal Symptoms

- Mostly due to cancer treatment (i.e. chemotherapy and radiation)
- **Includes**:
    - Stomatitis (oral ulcers) from decreased cell turnover
    - Malabsorption and diarrhea
    - Nausea and vomiting
    - Patient may need enteral or parenteral nutrition from malabsorption

Hair and Skin

- Mostly due to cancer treatment (i.e. chemotherapy and radiation)
    - Therapy injures rapid-growing cells of the body including hair and skin
    - These are some of the most visible symptoms which can contribute to the psychological effects of the disease
- **Includes**:
    - Alopecia (Hair loss)
    - Skin erythema, breakdown

Cancer treatment

- Chemotherapy
- Radiotherapy
- Hormonal therapy
- Surgery

Chemotherapy

- Cytotoxic drugs that target vital cellular machinery or metabolic pathways critical to cell growth and replication
    - They target the DNA of cancer cells.
- Nonselective
- Goal of therapy is to reduce cancer cells so the body’s defence can eradicate the remainder
- Single or combination chemotherapy
- *Regimen*:
    - **Induction**: initial, for shrinkage, or disappearance of tumor
    - **Adjuvant**: after surgical removal, to eliminate micro-metastases
    - **Neo-adjuvant**: before surgery or radiation to reduce size. Make it easier for the surgeon

Ionizing radiation

- **Goals**:
  - Eradicate cancer without excessive toxicity
  - Avoid damage to normal structure
- **Effect**: damage of cancer cell's DNA

Hormone therapy

- Receptor activation or blockage
- Interferes with cellular growth and signalling
- **Example**:
  - Androgen deprivation therapy (ADT)/castration for prostate cancer
  - Anti-estrogens in breast cancer (Breast tissue growth is dependent on estrogen)

Surgery

- Prevention of cancer by the removal of pre-cancerous lesions
  - E.g. Colon polyps
- Removal of cancer mass
- Biopsy for diagnosis
- Lymph node sampling for staging
- Debulking surgery (in huge/invasive cancer mass)
- Palliative surgery

Childhood Cancer

- **Incidence**
  - Overall, rare but leading cause of death from diseases in children
  - In 2004, mortality rate was 2.4% per 100,000 cases
  - Survival rates have dramatically improved over past 30 years
  - 5-year survival rate is 83% → not too bad
- **Origin**
  - Most originate from the **mesodermal** germ layer
  - Mostly sarcomas in kids
  - Carcinomas in adults
  - Diagnosed during peak growth periods
  - Fast growing and without early signs

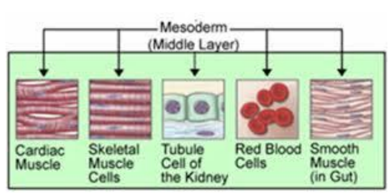


Figure 19: Cell types originating from the mesoderm germ layer

Common Types

Leukemia

- Most common malignancy in children
    - E.g. Acute lymphocytic leukemia (ALL)
- Children with “Down Syndrome” have a 10-20 X greater risk
    - Therefore, always monitor for ALL

Sarcoma

- Bone tumors: E.g. Osteosarcoma and Ewing Sarcoma
    - Both bone tumors

Figure 20: Bone tumors - Osteosarcoma (left) and Ewing sarcoma (right)

Embryonic tumors

- Originate during intrauterine life
- Immature embryonic tissue unable to mature or differentiate into fully developed cells
- Diagnosed early in life, E.g. Glioma (brain cancer), craniopharyngioma (pituitary gland derived tumor)

Other common cancers

- **Neuroblastoma** (nerve tissue cancer – tends to start in adrenal glands)
- **Wilms tumor** (nephroblastoma) = kidney cancer mass
- **Rhabdomyosarcoma** = muscle cancer
- **Retinoblastoma** = cancer in the retina

Figure 21: Wilms tumor (two images)

Etiology of Childhood Cancers

Genetic factors

- **Chromosomal abnormalities**
    - Aneuploidy, amplifications, deletions, translocations
    - Certain congenital syndromes and cancers may occur together
        - Urogenital abnormalities and Wilms tumor
        - Down syndrome and leukemia
- **Oncogenes and tumor-suppressor genes**
    - *E.g. Fanconi anemia*

- Autosomal recessive condition that results in impaired DNA repair and child is at risk for development of acute leukemia.
- Can cause:
    - Acute myelogenous leukemia, myelodysplastic syndrome, hepatic tumors
    - Child will have abnormal number of thumbs, hyperpigmentation, short stature.

Environmental factors

- **Prenatal exposure**
    - Drugs and ionizing radiation
- **Childhood exposure**
    - Second-hand smoke
    - Ionizing radiation
    - *Drugs*: anabolic steroids, cytotoxic agents, immunosuppressive agents
    - *Viral infections*: Epstein-Barr virus, AIDS

Prognosis

- More than 70% of children tumors can be cured
- Survival rates higher in children under 15 years
- Younger age more likely to. Be enrolled in clinical trials
- Survivors have increased risk of cancer later in life
- Residual and long-term effects of treatment
- Psychologic sequelae

Cancer Epidemiology

Genetics, Epigenetics, environment in Cancer

- **Genetics**: Heredity, DNA composition
- **Epigenetics**: change in genetic expression (phenotype) without DNA mutation
    - Involves factors that silence active genes or activate silent ones
    - E.g. DNA methylation
- Epigenetic processes can be influenced by environment/lifestyle including in utero
- Two-thirds of cancers are caused by environmental/lifestyle factors interacting with genes

Environmental Risk factors

Tobacco

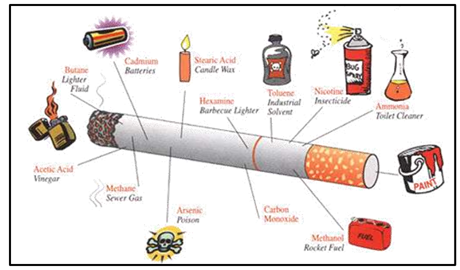


Figure 22: Tobacco and cancer risk

- Multipotent carcinogenic mixture
- Linked to many cancers
  - Oropharynx, larynx, esophagus, lung, stomach, liver, pancreas, kidney
  - Acute myeloid leukemia
  - Cervix, bladder colorectal
- Second-hand smoke contains many toxic chemicals (7000)
- Cigar and pipe smoking are equally harmful

Diet

- ~30% of the overall risk factors for cancer
- Mutagens found in human diet as a result of industrial/environmental contamination (pesticide/water)
- Aflatoxins and liver cancer
  - Aspergillus flavus; fungus infecting peanuts, tree nuts and grains. This can produce animal food-products that contain aflatoxins. Women can pass in aflatoxin to infants through breast milk
- Carcinogenic chemicals. In food
- May influence epigenetics

Obesity

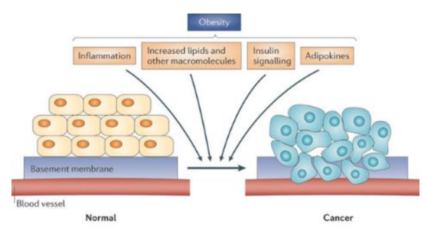


Figure 23: Obesity and cancer risk

- Based on BMI
- Related to increased incidence of several cancers
  - *Examples*: esophagus, gastric, colorectal, liver, gallbladder, pancreatic, breast, uterine, cervical, ovarian and kidney
- Casual relationship to cancer is unclear
- Gender differences in obesity-related cancers
- Abdominal obesity can be associated with insulin resistance, hyperinsulinemeia and increased steroids, all of which increase risk for cancer development

Alcohol

- Risk factor for oral cavity, pharynx, hypopharynx, larynx, esophagus, and liver cancers

- Substantial alcohol consumption (e.g. 3 or more drinks per day) has been associated with high risk
- Cigarette/alcohol combinations increases a person's risk for cancer
- Genetic factors involved

Ionizing radiation

- X-rays, radioisotopes and other radioactive sources
- Exposure causes cell death, mutations, chromosome aberrations (disorder)
- Mutations in germ cells are heritable
- Increased use of diagnostic testing is of concern
    - CT scans provide 50x more radiation to stomach compaired to an x-ray.

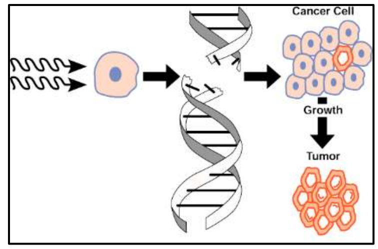


Figure 24: Ionizing radiation and cancer risk

UV radiation

- Principle source is sunlight (ultraviolet)
- Causes basal cell carcinoma, squamous cell carcinoma and melanoma (increased incidence)
- Produces ROS and other free radicals
- Promotes skin inflammation
- UVC rays do not enter the earth's ozone (shortest wavelength = 100-290 nm)
- UVB rays reach the skins surface – cause surface tanning, burns, signs of aging (medium wavelength = 290-320 nm)
- UVC rays penetrate deep into the skin's layers and release free radicals and cause DNA changes that can result in skin cancers. Wavelength is the longest (320-400 nm)

Electromagnetic radiation (EMR)

- Non-ionizing, low-frequency radiation
- Microwaves, cell phones, and power frequency radiation associated with electricity and radio waves, fluorescent lights, computers and other electrical equipment
- USA recommended that low-frequency EMFs be classified as possible carcinogens

Physical activity (30 mins a day)

- Reduces cancer risk (post-menopausal breast, colorectal, endometrial)
    - Decreases insulin and insulin-0like growth factors
    - Decreased obesity
    - Decreased inflammatory mediators and free radicals
    - Increased gut motility

- Less exposure of gut to dietary carcinogens

Sexual/reproductive behavior

- Carcinogenic types of human papillomavirus
  - 99.7% of women with cervical cancer
  - HPV vaccination

Occupational hazards

- Upper respiratory passages, lung, bladder, peritoneum
- Substantial number of occupational carcinogenic agents:
  - Asbestos (mesothelioma and lung cancer)
  - Dyes, rubber, paint, explosives, rubber cement, heavy metals, air pollution

Air pollution

- **Outdoor**
  - Heavy metal and aromatic hydrocarbon emissions from industry
- **Indoor**
  - Environmental tobacco smoke (ETS)
  - Radon gas
  - Heating and cooking combustion sources
  - Asbestos
  - Inorganic arsenic

Nutrition

- **Foods that decrease cancer risk**
  - Fruits/vegetables; fiber; foods containing vitamins A, B6, C, D, E and folate; whole grains; legumes/nuts
- **Foods that increase cancer risk**
  - Fat (especially omega-6 fatty acids), high glycemic index carbs; high preservatives; alcohol; grilled/blackened foods; fried foods, high calcium (>2000mg)

4: Alterations of the Endocrine System

Learning Outcomes

Alterations of the Endocrine System

- Describe names, location and hormones secreted by endocrine glands and their function (review).
- Describe three ways target cells fail to respond to hormones, creating hormonal dysfunction.
- Compare the syndrome of inappropriate antidiuretic hormone secretion (SIADH) and diabetes insipidus in regards to: causative factors, pathophysiology, manifestations, treatment, and prognosis.
- Describe the causes and clinical manifestations of hyper- and hypopituitarism.
- Describe the manifestations and consequences of pituitary adenomas and prolactinomas.
- Describe the progression of hyperthyroidism through Graves disease and thyroid storm in relation to cellular changes, manifestations, treatments and complications.
- Describe the causes, and outcomes for disorders that produce hypothyroidism.
- Differentiate between primary and secondary hyperparathyroidism.
- Describe the similarities and differences in the onset, etiology and pathophysiology of type 1 and type 2 diabetes mellitus.
- Describe the acute and chronic complications of diabetes mellitus.
- Compare hypercortical function (Cushing disease and syndrome) and hypocortical function (Addison disease) including causative factors, pathophysiology, manifestations, treatment and prognosis.
- Describe common tumors of the adrenal medulla.

The Reproductive System

- Describe the normal structure and function of the reproductive system (review).
- Define alterations of sexual maturation.
- Describe types, manifestations of menstrual alterations.
- Describe pelvic relaxation disorders.
- Describe benign growths and proliferative conditions of reproductive tract.
- Discuss main features and impact of infertility.
- Describe selected disorders of the prostate gland.
- Describe briefly male and female sexual dysfunction.
- Describe briefly benign and malignant breast lesions.

Definitions: Alterations of the Endocrine System

Term	Definition
Hyperfunction	- Increased function of an endocrine gland
Hypofunction	- Decreased function of an endocrine gland
Primary endocrine disorders	- Endocrine defect in the primary gland
Secondary endocrine disorders	- Endocrine defect outside the primary gland
Hypopituitarism	- Pituitary hypofunction
Myxedema	- Untreated adult hypothyroidism
Cretinism	- Untreated congenital hypothyroidism
RAS	- Renin-Angiotensin-System
Juxta glomerular apparatus	- A specialized structure formed by distal convoluted tubule and the glomerular afferent arteriole, located near the vascular pole of the glomerulus - Secretes renin and erythropoietin
Conn's syndrome	- Adrenal glands secreting excess aldosterone
Cushing’s syndrome	- Adrenal gland secreting excess steroids
Pheochromocytoma	- Tumour of adrenal medulla
Addison’s disease	- Adrenal gland hypofunction

Endocrine System: Anatomy and Physiology Review

Endocrine glands

- **Hypothalamus**
  - Secretes releasing hormones (RH) which in turn stimulates hormone release from the anterior pituitary
  - Hormones secreted by the hypothalamus: GHRH, TRH, CTH, GnRH, Dopamine (PIH), MSH IH
    - Note: Dopamine and MSH IH are inhibiting hormones
- **Pituitary**
  - Anterior and posterior lobes

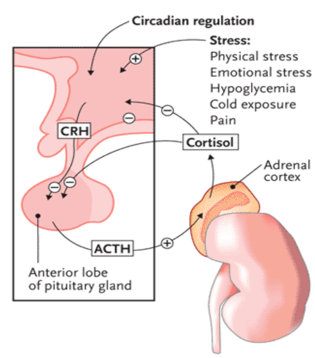


Figure 1: Hypothalamus and pituitary regulation of cortisol secretion in response to stress

- Anterior lobe secretes hormones in response to the releasing hormones from hypothalamus
    - **Anterior hormones**: GH, TSH, ACTH, FSH, LH, Prolactin, MSH
- Posterior lobe secretes hormones in direct response to neuronal signals from the hypothalamus. *No releasing hormone*
    - **Posterior pituitary hormones**: ADH, Oxytocin

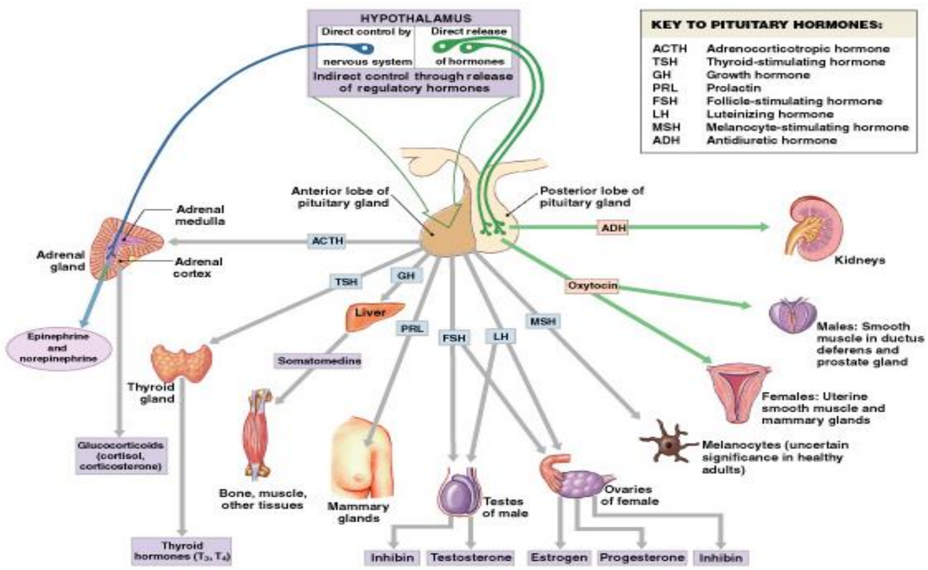


Figure 2: Hypothalamus and Pituitary Hormones, Targets, and Function

- **Thymus** = immune tolerance
- **Thyroid**
- **Adrenal gland**
- **Pancreas** = endocrine and exocrine
- **Gonads**
- *Note*: There are some endocrine cells in the kidney as part of the juxtaglomerular apparatus (JGA). These cells release renin and erythropoietin hormones
    - Renin stimulates mechanisms to regulate blood pressure
    - Erythropoietin stimulates red blood cell formation

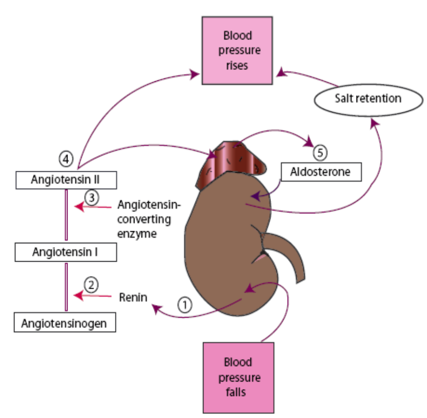


Figure 3: Renin-Angiotensin-Aldosterone-System (RAAS) of kidneys

Regulation of Calcium

- There are 3 hormones responsible for the regulation of calcium
  - **Parathyroid hormone**
    - Breaks down bone to increase blood calcium
  - **Calcitriol** (active Vit D = D_3 = 1,25 cholecalciferol)
    - Stimulated calcium absorption in intestine
  - **Calcitonin**
    - Inhibits bone resorption to decrease blood calcium

> **Helpful Hint**
>
> To remember the functions of Calcitriol and PTH, use the "R"s and think:
>
> "calcit**R**iol and pa**R**athy**R**oid hormone **Raises** calcium levels

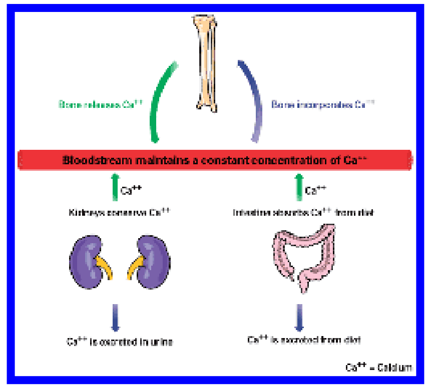


Figure 4: Mechanisms of Calcium Regulation

Urine Analysis

- Urine may contain small amounts of glucose, urobilirubin and protein
- Urine has NO bacteria, yeast, ketones, nitrates, leukocytes, bilirubin
- Urine may have crystals occasionally
- If hypovolemic, the osmolarity (amount of solutes per solvent) increases
  - This may be observed as more concentrated, darker urine
  - The person may also have decreased output relative to normal

Normal urine analysis

- Normal values are as follows:
- Color – Yellow (light/pale to dark/deep amber)
- Clarity/turbidity – Clear or cloudy
- pH – 4.5-8
- Specific gravity – 1.005-1.025
- Glucose - ≤130 mg/d
- Crystals – Occasionally
- Bacteria – None
- Yeast - None
- Casts – 0-5 hyaline casts/lpf
- Ketones – None
- Nitrates – Negative
- Leukocyte esterase – Negative
- Bilirubin – Negative
- Urobilirubin – Small amount (0.5-1 mg/dL)
- Blood - ≤3 RBCs
- Protein - ≤150 mg/d
- RBCs - ≤3RBCs/hpf
- WBCs - ≤2-5 WBCs/hpf
- Squamous epithelial cells - ≤15-20 squamous epithelial cells/hpf

Figure 5: Expected results for normal urine analysis

Volume status	Urine osmolality (mOsm/kg)	Urine sodium (mEq/L)	Etiology
Hypovolemic	↑	≤20	Nonrenal sodium loss
		>20	Renal sodium loss
Hypervolemic	↑	≤20	Edematous states
		>20	Kidney failure
Euvolemic	↑		SIADH
	Intermediate	>20	Potomania
	↓		Primary polydipsia
	Variable		Excess fluid intake

↑= >300; **intermediate**: 150-300; ↓: ≤100-150

Figure 6: Volume status, urine osmolality and sodium

Etiology of endocrine disorders

Primary:

- Pathologies within endocrine organs
- Congenital, inflammatory, metabolic, immune, malignancy (tumor)

Secondary:

- Pathologies outside the primary endocrine organs (in the controlling gland)
- Secondary pathologies can stem from the following mechanisms:
    - Hypothalamus or pituitary gland (these are the controlling glands)
    - Target cell/tissue
        - Non-responsive (receptor associated issues)
            - Low number of receptors
            - Impaired receptor function
            - Antibodies against receptors

Endocrine Disorders -Basics

Types

- *Hyperfunction*: elevated hormone levels
- *Hypofunction*: decreased hormone levels

Etiology/malfunction

- Reduced or excess synthesis
- Failure of feedback mechanisms
- Excessive degradation
- Inactivation or hyperstimulation by antibodies
- Ectopic hormone secretion

Alterations of the Hypothalamic-Pituitary axis

- Note: RH = releasing hormone; IH/IF = inhibiting hormone
- Dopamine (referred to in *Figure 7* as PIH [prolactin inhibiting hormone]) and MIH (melanocyte inhibiting hormone) act to decrease hormone production opposite to releasing hormones

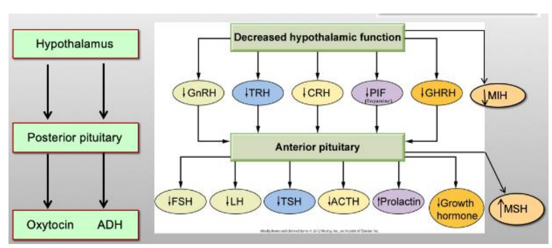


Figure 7: Hypopituitary-axis and hormonal regulation

- Therefore, decreased production of inhibiting hormone would increase hormone production/release from the pituitary
- *Example: Dopamine as an Inhibiting Hormone*
    - Decreased production of dopamine (PIH) leads to increased prolactin levels

Hypopituitarism

- Decreased function of the pituitary
- Can be from inadequate levels of hypothalamic hormones
- Defects in pituitary gland itself

Causes

- Invasive, space occupying lesion (**Tumor or aneurysm**)
- Infections: (TB, syphilis, meningitis)
    - TB encephalitis is rare, but it can occur to the pituitary
- Trauma (severe head trauma)
- Autoimmune disease
    - Leads to destruction of the gland
- Sheehan syndrome (postpartum pituitary infarction that leads to necrosis)
    - Postpartum bleeding leads to ischemia and can lead to subsequent ischemic damage to the pituitary

Pathophysiology

- Inflammation, ischemic necrosis, infarction, fibrosis
- Edema and swelling of the pituitary within the sella turcica further impede blood supply to the gland causing further shrinkage and fibrosis

Clinical manifestations

- Could see symptoms related to specific hormone deficiency (e.g. FSH, LH, ACTH)

Diagnosis

- Diagnosis is often challenging
- Triple bolus test = insulin + TRH + GnRH and then assess. You should see hypoglycemia, elevated TSH and elevated FSH and LH. If you don't see this = pituitary issue
- CT scan can view the gland for trauma or tumor
- Laboratory results need to be interpreted in conjunction with/or in light of clinical manifestations

Prolactinoma (hyperpituitarism)

- The most common pituitary tumor

- Benign, hyperfunctioning, slow-growing pituitary adenoma

Pathophysiology:

- Mass secreting prolactin
- Sustained elevation of serum prolactin
- Compression effects from tumor mass (e.g. visual disturbances)

Clinical manifestations

- Compression effects: optic chiasm; visual impairment, headache
- *Functional effects*:
    - Female: amenorrhea, galactorrhea, hirsutism, infertility, osteopenia
    - Male: gynecomastia, erectile dysfunction

Treatment

Medication: Dopamine agonists

- Since the tumor secretes prolactin, dopamine (which has the same chemical structure as PIH) can be used to decrease and maintain prolactin levels PIH is the same as dopamine in structure
- Dopamine drugs can be used to inhibit synthesis of prolactin in remaining tumor mass following surgery

Surgical approach: removal of the tumor

Diabetes Insipidus (DI)

- The inability to retain water leading to excessive diuresis evidence by urination

Central (Neurogenic) DI

- No ADH secretion
- Excessive diuresis due to failed ADH synthesis/release
- Potential causes include:
    - Idiopathic, head trauma, tumors, vascular, autoimmune, infection, drugs and surgery

Nephrogenic DI

- Renal resistance to ADH
    - ADH is made, but is not working in the kidney
- May be related to disease or drugs

Pathophysiology:

- Failure of ADH action on the distal convoluted tubule (DCT) and collecting ducts
- Inability to concentrate urine

Clinical manifestations

- Polyuria, polydipsia, Nocturia

- Inappropriate diluted urine
- Urine osmolarity is less than serum osmolarity (pee is has less solutes than blood)

Diagnostic Tests:

- Water deprivation test
  - Patient is not given water
  - The ADH level should increase with a decrease the frequency and amount of urination
- DDAVP (synthetic ADH) test
  - Patient gets synthetic ADH
  - If there is no change in urine concentration, then it confirms a nephrogenic DI

Syndrome of Inappropriate ADH secretion (SIADH)

- Water retention due to ADH hypersecretion
  - Decreased urine output and/or concentrated urine

Etiology:

- Ectopic (paraneoplastic syndrome; GI, bladder, prostate)
  - Usually a tumor somewhere that is secreting ADH
- CNS disorders: stroke, infection, tumors

Pathophysiology

- Enhance renal water retention
- Dilutional hyponatremia, plasma hypo-osmolarity
- Concentrated urine

Clinical manifestations

- Patient will present with symptoms of hyponatremia. E.g. Confusion, delirium, muscle weakness, seizures, coma

Disorders of the Thyroid and Parathyroid

Thyroid Hormone Review

- Iodide (I-) uptake
- Iodination of tyrosine
- Synthesis of thyroid hormones (thyroid peroxidase)
- Thyroid hormones produced:
  - Tetraiodothyronine (thyroxine)(T4)
  - Triiodothyronine (T3): from T4

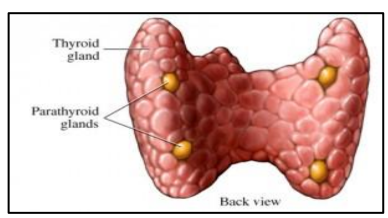


Figure 8: Anatomy of the thyroid gland

- Thyroglobulin
  - Carrier molecule for thyroid hormones
  - Without this molecule synthesized hormones cannot exit the follicle

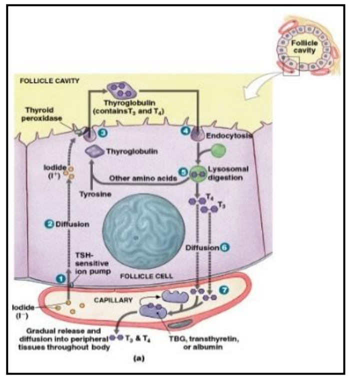


Figure 9: Production and release of thyroid hormones with a follicle

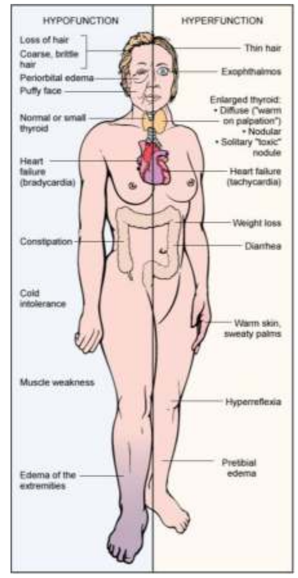


Figure 10: Comparison of hyperfunction and hypofunction of the thyroid gland

Hyperthyroid

Primary:

- Graves disease
- Multinodular goitre
- Solitary hyperfunctioning (toxic) nodule
- Thyroid cancer

Secondary (less common)

- TSH secreting pituitary adenoma
- Ectopic secretion (paraneoplastic syndrome)

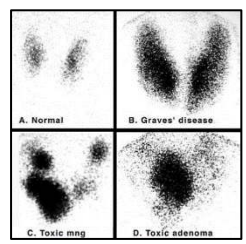


Figure 11: Radioactivity in Graves' versus other multimodule goitre and thyroid cancer (adenoma)

Pathophysiology

- **Graves**: Type II hypersensitivity; autoantibodies against TSH receptor known as TSI (thyroid stimulating Ig)
- Elevated TH → hypermetabolic state (insomnia, infertility, hair loss, irritability, lose weight)

Clinical manifestations

- Manifestations of hypermetabolic state
- Enlargements of thyroid gland (goitre)
- **Ophthalmopathy** (big eyes)
- **Dermopathy** (pretibial myxedema)
  - Due to Ig reacting with TSH receptors in connective tissue
  - Seen in Graves
- *Thyrotoxic crisis* (storm):
  - Acute severe elevation of TSH →
    - Severe tachycardia, hyperthermia, CHF
    - Diarrhea
    - Agitation, alteration in mental status

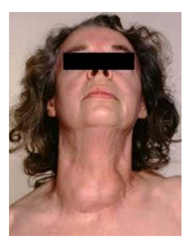


Figure 12: Goitre typically seen in Grave's disease

Figure 13: Example of pretibial myxedema

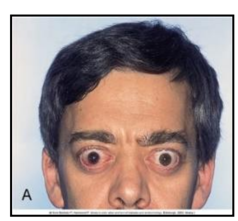


Figure 14: Example of exopthalmos

- **Precipitating factors**
    - Infection
    - Trauma
    - Pre-eclampsia

Hypothyroidism

More common than hyperthyroidism

Primary

- Congenital
- Autoimmune thyroiditis (**Hashimoto's thyroiditis**)
    - **Most common cause for hypothyroidism**
    - Slow destruction and inflammation of thyroid tissues by autoreactive T cells and peroxidase antibodies in blood
- Iatrogenic loss of thyroid tissue (surgery, radiation, medication)
- Viral thyroiditis (late phase)
- Iodine deficiency

Secondary

- Pituitary failure of TSH release due to: tumor, trauma, infarction, hemorrhage

Clinical manifestations

- Manifestations of hypometabolic state
    - Sluggish, slow, cold intolerance
- *Myxedema* in adults and *cretinism* in infants
    - Cretinism = congenital hypothyroidism
        - No thyroid or little thyroid tissue.
        - The child will be cognitively be disabled without T
        - The child can survive on maternal T4 for approximately 3-4 months before requiring levothyroxine
        - If there is no early screening, the condition may remain unknown as signs may not manifest until 4 months old.
        - Signs and symptoms include:
            - Protruding tongue
            - Hoarse cry
            - Hypotonic muscles
            - Abdominal protrusion

- Eventually the child will be dwarf, have slow milestones
  - Eg dentition, cognitive delay
- *Myxedema coma*: Acute, severe hypofunction of the thyroid gland

Alterations of Parathyroid Gland

- The parathyroid gland is responsible for the regulation of calcium (see *Figure 4*)

Hyperparathyroidism

- Primary hyperparathyroidism
  - Excess PTH secretion due to disease in one or more parathyroid glands. Makes too much.
- Secondary hyperparathyroidism
  - Chronic hypocalcemia E.g. Renal disease. stimulates calcium release

Clinical manifestations

- Hypercalcemia
- Hypercalciuria; renal stones (calcification)
- Hypophosphatemia (opposite of calcium)

Hypoparathyroidism

- Abnormally low PTH levels
  - Most common reason is iatrogenic: surgical removal of thyroid (parathyroid is also removed as it is anatomically right at posterior aspect of the thyroid gland)

Figure 15: Tetany in the hand

Manifestations:

- Hypocalcemia E.g. Tetany
- Hyperphosphatemia

Pancreatic Dysfunction

Anatomical definitions relating to the pancreas: Head, body, tail, duct

Ampulla of Vater = pancreatic duct opens into duodenum

Exocrine = acini; secreting digestove enzymes

Islets of Langerhans: endocrine cells that secreting glucagon (alpha cells) and insulin (beta cells)

Diabetes Mellitus (DM)

- Multisystem disease characterized by a group of insulin related metabolic alterations
- The state of DM in Canada
  - Today ¼ Canadians have DM or are Pre-DM
  - By 2020, it will be 1/3 people

- In 2015, diabetes cost 15 billion in health care fees

DM – Types and Causes

Primary

- **Type 1 DM**
  - Insulin deficiency: destruction of beta cells of islets of Langerhans
  - Referred to as "insulin dependent DM"
- **Type 2 DM**
  - Begins with insulin resistance (cells fail to respond to insulin). With disease progresses a lack of insulin develop
  - Referred to as "non insulin dependent DM"

Secondary DM

- Pancreatic causes
- Endocrine causes E.g. Cushing's
- Drugs: steroids, thiazide diuretics

Gestational

- High blood glucose level during pregnancy
- 2-10% of pregnancies

Manifestation

- Hyperglycemia
- Polydipsia
- Polyuria
- Polyphagia
- Weight loss
- Fatigue
- Obesity, dyslipidemia (type 2)
- Symptoms of complications
- Retinopathy

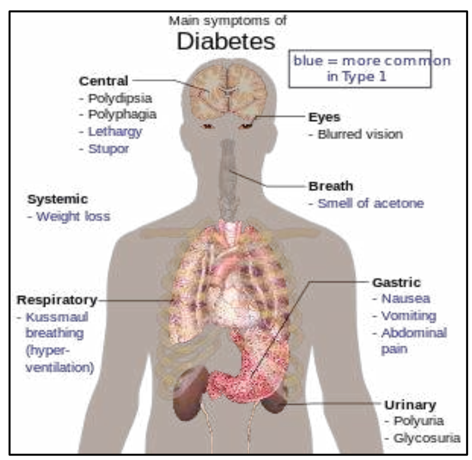


Figure 16: Most common, featured symptoms of Diabetes mellitus

Figure 17: Table-Complications of Diabetes Mellitus

Acute	Chronic
▪ Hypoglycemia - More common in Type I	▪ Hyperglycemia and non-enzymatic glycosylation
▪ Hyperglycemia - Diabetic ketoacidosis (Type I) - Hyperosmolar hyperglycemic nonketotic syndrome (HHNKS) (Type II)	▪ Microangiopathy - Retinopathy - Nephropathy - Neuropathy
	▪ Macroangiopathy - Coronary artery disease (CAD) - Cardiovascular accident (CVA; Stroke) - Peripheral artery disease (PAD)
	▪ Repeated infections
	▪ Skin complications

Hypoglycemia

Causes:

- Over dose of medications
- Missing meals
- Poor blood glucose monitoring

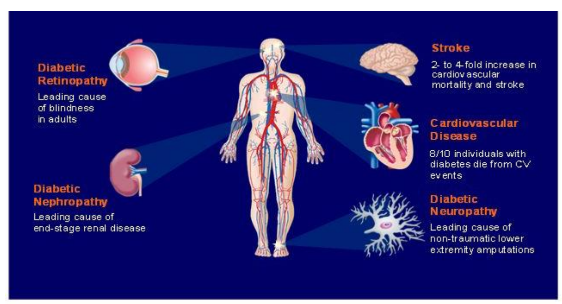


Figure 18: Complications of Diabetes mellitus (angiopathies)

Manifestations

- Pallor, tremor, tachycardia, palpitation, headache, dizziness, irritability, poor judgement, confusion, fatigue, hunger, visual disturbances, seizures, coma
- * Brain cannot have low glucose for 6-7 minutes

Complications of Diabetes Mellitus: DKA and HHNK

Diabetic Ketoacidosis

- Emergency life-threatening complication
- Seen with type 1 DM and could be first symptom of undiagnosed DM

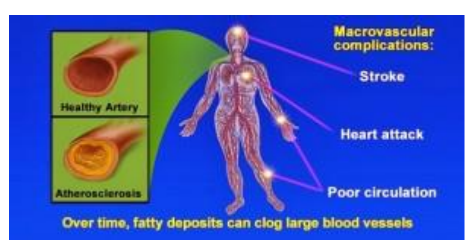


Figure 19: Macrovascular complications of Diabetes mellitus

Pathophysiology

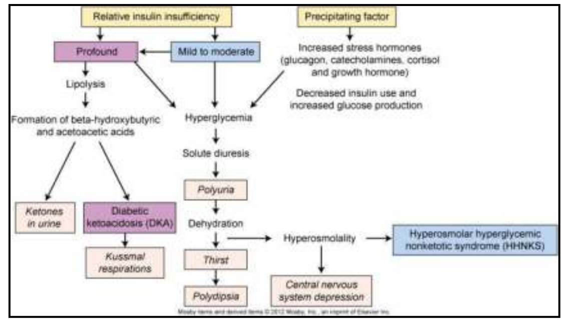


Figure 20: Pathophysiology of DKA and HHNK

- Insulin deficiency (facilitated by other anti-insulin hormones)
- Lipolysis leads to the formation of ketone bodies
    - Results in ketonemia and ketonuria (ketones in the blood and urine)
    - Osmotic diuresis
- Acidosis
- Hyperkalemia
    - Shift of potassium from ICF to ECF
    - Hydrogen exchanges to try to reduce the acidosis leaving excess potassium
    - A common issue with Insulin treatment (electrolytes monitoring is a must during DKA treatment)

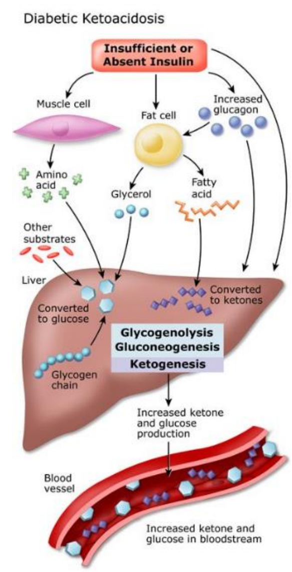


Figure 21: Pathophysiology of diabetic ketoacidosis

Clinical manifestations

- Kussmaul's respiration (rapid, deep)
- Acetone smell (from the production of ketones)
- Dehydration
- Abdominal pain, vomiting
- Impaired consciousness
    - May lead to coma and death
- Ketones present in blood and urine
    - The main ketone acids are acetoacetate and hydroxybutyrate

Hyperosmolar hyperglycemic nonketotic syndrome (HHNK)

- Hyperglycemic crisis characterized by hyperosmolarity and dehydration without ketosis
- In type 2, usually precipitated by infection/acute illness

Pathogenesis:

- Very high blood glucose: (usually >600mg/dl (30mmol/L
- Increase in blood osmolarity (>320 mOsm)
- Osmotic diuresis
- No ketosis (some insulin inhibits fat tissue breakdown)

Clinical manifestations

- Older non-compliant patient

- Dehydration and thirst
- Impaired consciousness

Disorders of the adrenal cortex

Hyperfunction:

- *Hyperaldosteronism*: excess aldosterone
  - Primary = Conn's disease
  - Secondary = RAS hyperactivity
- ***Cushing** syndrome*: Excess cortisol
- *Adrenal hyperplasia*: excess sex hormones

Hypofunction:

- *Addison's disease*

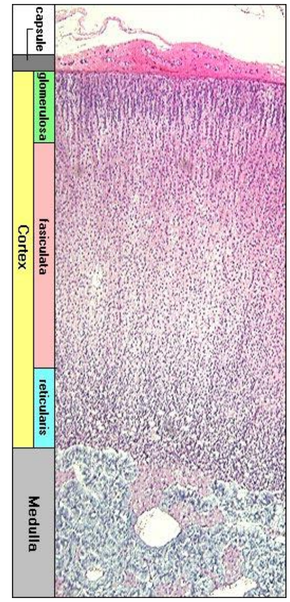


Figure 22: Cellular layers of the adrenal gland

Disorders of the Adrenal medulla

- Hyperfunction
  - Tumors of chromaffin cells (**pheochromocytoma**)
  - Secrete catecholamines (continuous or episodic)

Conn's Disease

- Increased aldosterone secretion in the absence of activation of the renin-angiotensin-aldosterone-system (RAS)

Causes:

- Adrenal hyperplasia
- Tumors: adenoma, carcinoma

Pathophysiology

- Salt and water retention
- Hypertension
- Hypokalemia
- Alkalosis
  - Hydrogen is excreted with potassium

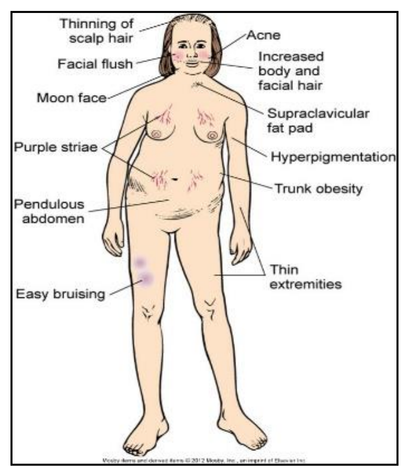


Figure 23: Manifestations of Cushing's syndrome

Cushing's syndrome

- Increased cortisol secretion

Causes:

- Iatrogenic (most common)
  - Long-term steroid and/or ACTH therapy
- Adrenal adenoma or carcinoma

- Cushing's disease
  - Increased pituitary ACTH
- Ectopic ACTH secretion (parneoplastic syndrome)

Manifestations

- Obesity
  - Notably in the face ('moon face') and upper back (buffalo hump)
  - Truncal obesity (most commonly from steroid therapy)
- Infection:
  - Excess steroids impair the function of WBCs and therefore have anti-inflammatory/ anti-immune effects
  - On the long-term the depressed immune system lead to increased susceptibility to infection
  - Important to educate on steroid therapy; caution and monitoring if long term
- Hypertension
- Peripheral edema (from hypervolemia)
- Hyperglycemia; secondary DM on the long term
- Osteoporosis
- Recurrent infections
- Psychosis
- Thin and atrophic skin with bruises

Addison's Disease

- Destruction of all zones of adrenal cortex
  - Includes glucocorticoids, mineralocorticoids, androgens

Figure 24: Hyperpigmentation of skin and mucous membranes seen in Addison's disease

Causes

- **Primary (adrenal gland disorder)**
  - Usually autoimmune-mediated destruction of the adrenal glands (represents 80% of cases)
  - Chronic infection, such as Tuberculosis (TB adrenalitis) (approximately 20% of cases)
- **Secondary (pituitary gland disorder)**
  - Low ACTH secretion
  - Sudden withdrawal of prolonged steroid therapy
  - Destruction of the pituitary gland (e.g. from trauma or a tumor)

Manifestations

- Fatigue and muscle weakness
- Hypotension
- Anorexia
- Nausea and vomiting
- Abdominal pain
- Weight loss
- Hyperpigmentation
  - High ACTH can bind to melanocyte in dermal areas which darkens skin
- Mental depression
- Acute severe conditions (Addisonian Crisis)
  - Sudden drop in BP from hypovolemia (shock). Can occur from sudden withdrawal in steroid therapy
  - Can be fatal

Pheochromocytoma

- Chromaffin cell tumors
- Neuroendocrine tumor of adrenal medulla secretes catecholamines leading to overstimulation of the SNS.

Clinical manifestations

- Hypertension
  - Severe $\pm$ paroxysmal
  - Resistant to treatment
- Headache
- Palpitations and tachycardia
- Sweating
- Anxiety and tremors
- Hyperglycemia

Complications:

- Pheo crisis: in response to anesthesia, stress, drugs
- Cardiomyopathy

Lab features

- 24-hour urinary catecholamines and metanephrines (most specific)
- **Urine vanillylmandelic acid (VMA)**

- **Tumor marker for pheochromocytoma**

- Plasma metanephrine

4: Alterations of the Reproductive system

Selected Disorders of the Female Reproductive System

- Hormonal and menstrual alterations
    - Amenorrhea (cessation of menstruation)
    - Anovulation (failure of mid cycle ovulation)
    - Dysmenorrhea and Premenstrual tension syndrome (painful menstruation)
    - Dysfunctional uterine bleeding (uterine bleeding of undetermined cause)
- *Polycystic ovarian syndrome*
- Alterations of sexual maturation
    - Delayed puberty
    - Precocious puberty
- Sexual dysfunction
    - Disorders of sexual desire (decreased or increased)
    - Anorgasmia (orgasmic dysfunction)
    - Dyspareunia (painful intercourse)
- Disorders of fertility
- Pelvic relaxation disorders
    - Cystocele (bladder hernia), rectocele (rectum protrusion/drop), urethrocele (prolapse of the female urethra)
    - Vaginal prolapse
    - Uterine prolapse
- Growths and proliferative conditions
    - *Endometriosis*
    - Benign ovarian cysts
    - Endometrial polyps
    - Leiomyomas
    - Cancers: cervical cancer, vaginal cancer, endometrial, vulvar cancer, ovarian

Selected Disorders of the Male Reproductive System

- Disorders of the urethra
    - Urethritis
    - Urethral strictures

- Disorders of the penis
  - Congenital anomalies
  - Penile cancer
  - Balanitis (Inflammation of the head of the penis)
- Disorders of the scrotum and testes
  - Varicocele (varicose veins in the scrotum; rarely painful but can → infertility)
  - Hydrocele (fluid in the scrotum; a painless swelling of one or both testicles)
  - Spermatocele (painless fluid filled cyst in the epididymis)
  - Testicular torsion (Twisting of the spermatic cord → severe testicular pain)
  - Epididymitis (inflammation of epididymis → severe testicular pain)
- Disorders of fertility
- Disorders of the prostate
  - Benign prostatic hyperplasia
  - Prostate cancer
  - Prostatitis
- Erectile dysfunction
- Disorders of the breast
  - Gynecomastia
  - Breast cancer

Endometriosis

- A major cause of chronic pelvic pain in females
- Significant social and psychological impact
- Ectopic endometrial cells
  - Endometrium grows outside the uterus

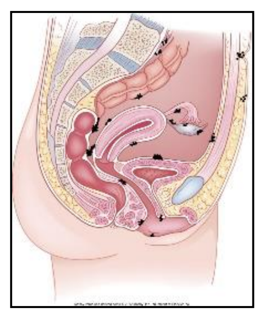


Figure 25: Sites of ectopic endometrial cells

Cause and Pathophysiology

- Currently cause is unknown
  - May have genetic influences
  - Multifactorial
- Most accepted theory: “retrograde menstruation”
  - Menstrual fluids move through the fallopian tubes and into the pelvic cavity; ectopic tissue settles in the peritoneum.
- Ectopic tissue is influenced by hormonal changes like cells found inside the uterus

Clinical manifestations:

- Pelvic pain
- Infertility

Polycystic Ovary Syndrome (PCOS)

- Hypothalamic pituitary disease resulting in enlarged polycystic ovaries and chronic anovulation
- Common in young women

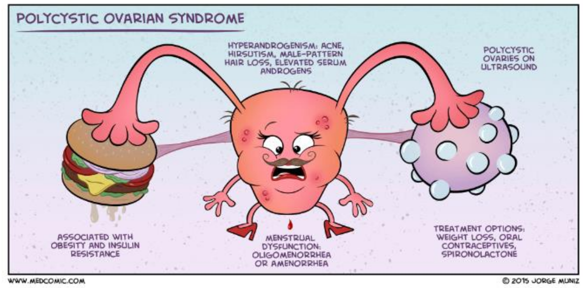


Figure 26: Polycystic ovarian syndrome (comic representation)

Pathophysiology:

- Impaired ovarian follicle development leading to increased androgen production
- High GnRh pulses leads to increased pituitary LH and ovarian androgen synthesis
- Increased androgen conversion to estrogen in adipose tissue
- **Insulin resistance**
  - Insulin stimulates ovarian follicle to secrete more androgens
- **Ovaries have 12 or more cysts**
  - Cysts may form ovarian follicles which then secretes androgens
  - Androgens are converted into estrogens in adipose tissue

Clinical manifestation:

- Weight gain and obesity
- Oligomenorrhea, Amenorrhoea, hirsutism, acne
  - Features of increased androgen production
- No ovulation leads to infertility
- Insulin resistance
  - Increased risk for DM in the future when combined with obesity

Diagnosis

- Two of the following are required for the diagnosis of PCOS
  - Few or no ovulatory cycles
  - Elevated androgen levels
  - Polycystic ovaries (do not have to be present, and their presence does not constitute PCOS)

Benign Prostatic Hyperplasia → increased 5AR

- Prostatic enlargement due to:
  - Hyperplasia (more common)
  - Hypertrophy

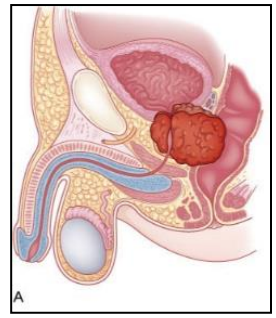


Figure 27: Anatomy of benign prostatic hyperplasia

Pathophysiology

- Person is usually >50 years-old
- Normal growth/function of prostate depends on reduction of testosterone to Dihydrotestosterone (DHT) by 5-alpha reductase enzyme (5AR).
  - 5-AR converts testosterone from the testicles into DHT
  - Elevated 5-AR causes over growth of prostate.
- Prostatic over growth leads to ureteral compression and bladder outflow obstruction

Clinical manifestations

- Of bladder outflow obstruction: Urinary retention, dribbling, hesitancy

5: Blood (Disorders of the Hematological System)

Learning Outcomes

- Describe the cellular and fluid components of blood and functions of blood cells (review).
- Define *anemia* and describe the various methods of classifying the anemias.
- Describe the manifestations of *anemia* and the pathophysiology underlying the clinical manifestations.
- Compare and contrast the pathophysiology underlying *iron deficiency, pernicious* and *folate deficiency anemias.*
- Describe the *normocytic-normochromic anemias*.
- Define *polycythemia vera* and describe its causes and multiple system manifestations related to the increased viscosity and volume of blood.
- Describe the different types of alterations in leukocyte function.
- Define *agranulocytosis* and describe its clinical manifestations.
- Describe the manifestations of *infectious mononucleosis* and *its* complications beyond the immune system.
- Classify *leukemias* as it relates to the maturity of the cells and appearance of the total leukocyte count and differential.
- Differentiate the *leukemias* by manifestations, treatment options and prognosis.
- Describe *Hodgkin* and *non-Hodgkin lymphomas*, focusing on differential diagnosis, manifestations, treatment and prognosis.
- Describe the pathophysiology, clinical manifestations, and treatment of *multiple myeloma*.
- Describe the causes of *splenomegaly*.
- Describe the causes of *thrombocytopenia*.
- Describe the various causes of *impaired hemostasis*.
- Describe the pathophysiology of *disseminated intravascular coagulation*.
- Describe the conditions that predispose an individual to the development of *thrombosis*.

Definitions: Blood (Disorders of the Hematological System)

Term	Definition
Anemia	- -Reduction in number and/or function of erythrocytes, hemoglobin, hematocrit, or any combination of these
Ecchymosis	- Subcutaneous spot of bleeding with diameter larger than 1 centimetre (more than 5mm)
Sideroblastic Anemia	- Anemia related to inefficient iron uptake and subsequent abnormal hemoglobin synthesis - Characterized by ringed sideroblasts in bone marrow
Sideroblasts	- RBCs containing iron granules that have not been synthesized into hemoglobin and form a ring around the nucleus
Leukemia	- Abnormal increase of WBC count
Pallor sign in blood smear	- Pale central regions of the red cells seen on blood smear, indicating low Hb.
Petechiae	- Red/purple discolored skin spots - Do not blanch on applying pressure - Caused by subcutaneous bleeding - Secondary to platelet, vascular or, coagulation disorders - Measure less than 3 mm
Plasma Oncotic Pressure	- Pressure exerted by proteins in plasma that helps contain the fluid within the blood vessel
Polycythemia	- Increased red cell count above normal
Polycythemia Vera	- Myeloproliferative disorder of BM leading to excess production of red cells
Purpura	- Same as petechiae but measure less 3–10 mm
Heme	- A complex formed of an iron atom surrounded by protoporphyrin ring. This complex structure is located in the centre of each of the 4 globin chains making the Hb

Review: Anatomy and Physiology of the Hematological System

Composition of Blood

- **Critical facts at a glance**:
- Albumin is the most abundant plasma protein
  - *Plasma oncotic pressure*
    - Solutes within blood that holds the fluid within the vessel
    - Loss of albumin (or other plasma proteins) causes fluid to leak into the interstitial space
      - Manifested as edema

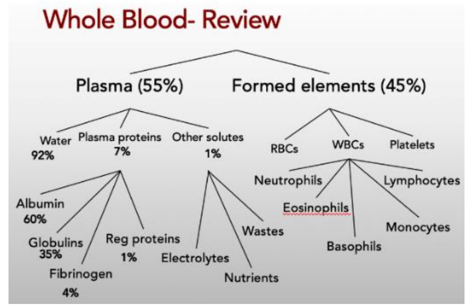


Figure 1: Whole Blood Composition (Review)

- *Leukemia* = immature cells in blood circulation
- Bone marrow contains multiple stages of cellular development and maturation but only mature cells are released into the blood
- There are 5-6 million RBC, 9-11 thousand of WBCs and 150-450 thousand of platelets per cubic millimeter

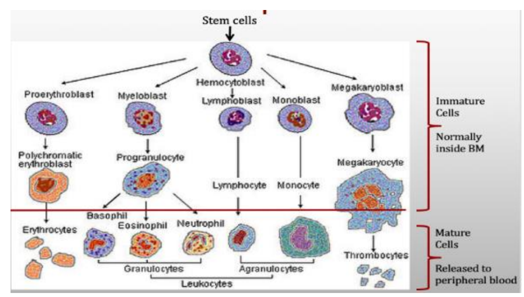


Figure 2: Maturation of white blood cells

- Vacutainers have different colored caps, indicating the type of anticoagulant added to the container to prevent clotting of the blood sample
  - If no anticoagulant were added, the sample would begin to clot once collected

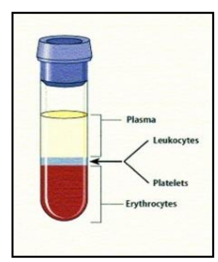


Figure 3: Separated blood components

- Hemoglobin is a quaternary protein composed of heme + globin (peptide chains).
  - Four globin chains; 2 alpha and 2 beta, each chain has *heme* (iron surrounded by porphyrin ring) in its centre
  - RBC has a 120-day life span
  - Platelets have 10 day life span
  - Some WBCs live for years

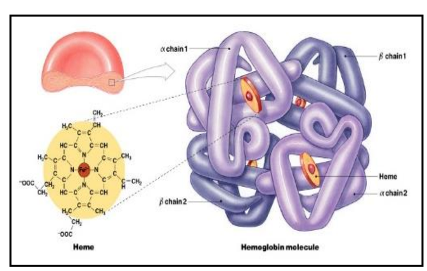


Figure 4: RBC Molecular Structure

- Life Cycle of a Red Blood Cell
  - Hemoglobin is broken into heme and globin
  - Globin is recycled to amino acids
  - Heme is broken into iron and biliverdin
  - Iron is reused as it goes back the bone marrow through blood circulation via transferrin
  - Biliverdin (green) turns into bilirubin (yellow) that goes to the liver
  - Bilirubin has three possible outcomes:
    1. Some goes with bile to the intestine to help fat digestion; part of which then gets reabsorbed into circulation
    2. Some turns into urobilin and is excreted in stool.
    3. Some goes straight to the kidney from the liver (via circulation) to be excreted in urine.

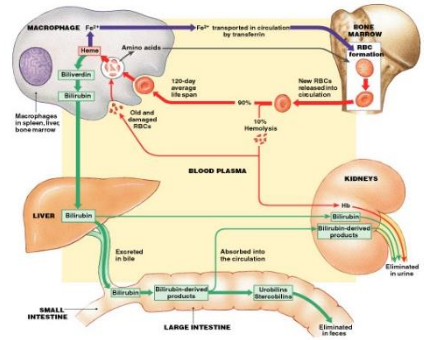


Figure 5: Life Cycle of a Red Blood Cell

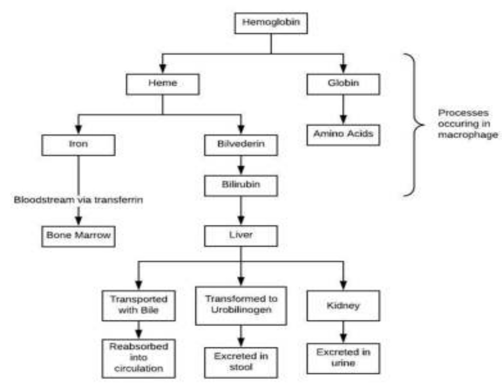


Figure 6: Life Cycle of an RBC

Disorders of the Hematological System

Anemia

- Reduced number and/or function of erythrocytes (red blood cells), hemoglobin, hematocrit, or any combination of these
- Anemia is a very common pathophysiology worldwide

Etiology

- Impaired erythrocyte production
- Increased erythrocyte destruction
  - Or a combination of both impaired production and increased destruction
- Blood loss (acute or chronic)
  - Leads to overall decrease in erythrocytes
- Anemia may also result with:
  - Increased erythrocyte destruction in the spleen
  - Impaired bone marrow production (either in producing too few, or poor quality)
  - Nutrient deficiencies (i.e. Iron, B12, Folic Acid)
- There is a relationship between RBC structure and anemia
  - RBCs are 7um biconcave discs, this is the size/shape is required to enable them to stack and move through capillaries

- Low hemoglobin is visible microscopically in a blood smear as a, "*pallor sign*", the RBC looks empty, no redness in the middle

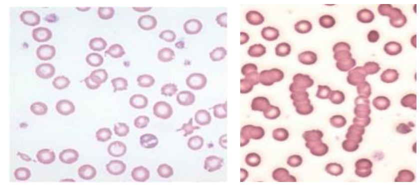


Figure 7: The "pallor sign" of anemia

Table: Abbreviations Related to Diagnostic Studies in Anemia	
Abbreviation	**Full Term**
CBC	Complete blood count (gives full picture of blood)
Hb (g/dL)	Hemoglobin
RBC count	Number of RBCs
RDW	Red cell distribution width (represents variation in size among red cell population)
Hct (%)	Hemotocrit (also known as PCV) Normal value 40-50% depending on gender (males have higher Hct)
PCV	Packed cell volume Another term for hematocrit referring to the length of sedimented RBCs column following centrifugation in capillary tubes
Three Blood Indices	
MCV	Mean corpuscular volume. A measure of the average volume of a red blood cell (fL)
MCH	Mean corpuscular hemoglobin. Amount of Hb in an average red cell (pg)
MCHC	The concentration of haemoglobin in a given volume of packed blood (%)
Other Cellular Components of Blood	
WBC count	White blood cell count. Includes both the total WBCs and differential which gives percentages of each type of WBC
Platelet count	Platelet count. Normal = 150,000-450,000 / mm^3
MPV	Mean platelet volume. Represents platelet size

Table: Normal Values for Complete Blood Count (CBC)	
Component	**Normal values**
Total erythrocytes	4.2-6.1 x 10^{12}/L
Erythrocyte size (RDW)	11.5-14.5%
Hb (total Hb in the blood)	13-17 g/dL
Hematocrit (PCV)	40-45%
MCV (mean corpuscle volume)	84-96 femtolitres
MCH (mean corpuscle HB)	28-34 picograms
MCHC (mean corpuscle hemoglobin concentration)	32-36%
Total leukocytes	5-10 x 10^{9}/L
Lymphocyte	20-40% of leukocyte differential
Monocyte/Macrophage	5-10% of leukocyte differential
Eosinophil	1-4% of leukocyte differential
Neutrophil	55-70% of leukocyte differential
Basophil	0.5-1% of leukocyte differential
Platelet count	150-400 x 10^{9}/L
MPV (mean platelet volume)	7.4-10.4 femtolitres

Classification of Anemia

- Anemia is classified according to:
-
    - Red cell volume (MCV)
    - Hb content (MHC, MCHC)

- Microcytic hypochromic anemia
    - Low indices
    - RBC are small and low Hb
- Normocytic normochromic anemia
    - Normal indices
    - Normal cell and normal Hb but reduced number/ impaired function
- Macrocytic anemia
    - High MCV
    - MCHC is never high (RBCs can't be overpacked)

Pathophysiology & clinical manifestations of anemia

- Pathophysiology
  - Reduced blood oxygen-carrying capacity
  - Hypoxemia leading to hypoxia which impair functions of al body cells (all organs can be affected)
  - Manifestations vary according to severity and the ability of the body to compensate
- Classic symptoms
  - Fatigue
  - Weakness
  - Dyspnea
  - **Pallor**
    - Key sign of anemia
    - *Always check the conjunctiva

Figure 8: Pallor in the hand (right) compared to normal coloration

Microcytic-Hypochromic Anemias

- Red cells are small and contain less hemoglobin (MCV <80 fL)

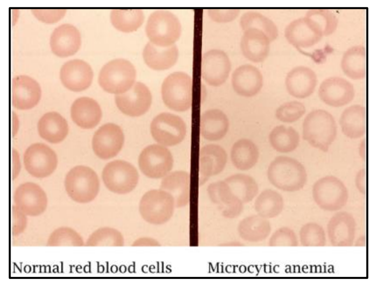


Figure 9: Microscopic comparison of normal RBCs (left) to Microcytic-hypochromic RBCs (right)

General causes:

- Iron deficiency
- Impaired iron uptake, metabolism, absorption, transport
- Impaired porphyrin formation (ring around heme)
- Impaired globin synthesis (the structure of the protein that makes hemoglobin; hemoglobinopathies)
- Will details 3 examples of microcytic-hypochromic anemias:
  - Iron deficiency anemia
  - Sideroblastic anemia
  - Thalassemia (form of Hemoglobinopathies)

Iron deficiency anemia

- The most common type of anemia worldwide

Etiology:

- *Chronic blood loss*. Potential causes include:
  - Women with heavy menses or chronic spotting related to menses
  - Bleeding gastric or duodenal ulcer

- Medications:
  - E.g. Excessive use of Ibuprofen inducing gastritis or ulcer
- *Defective iron utilization*
  - The body is unable to absorb, uptake and use the iron effectively
    - E.g. Surgical procedures such as gastric bypass that decrease stomach acidity, intestinal transit time, and/or absorption
    - Chronic disease eg renal failure
- *Nutritional*
  - Low dietary iron

Pathophysiology

- Impaired function of mitochondrial enzymes (dependent on iron as cofactor) leads to impaired metabolism and ATP generation. This explains fatigue
- Decreased iron stores (ferritin) combined with increase need for hemoglobin synthesis to compensate for blood loss
- Eventually low serum iron

Clinical Manifestations

- *Early*:
  - Fatigue, weakness, shortness of breath, pallor
- *Advanced*:
  - Brittle, thin, coarsely rigid, and spoon shaped nails (koilonychia)
  - Angular stomatitis
    - Inflamed and dry mouth, lips, tongue
    - Atrophied tongue papillae leading to soreness, redness and burning
    - Typically resolves within 1-2 weeks of iron replacement
- Other manifestations related to additional physiological processes requiring iron:
  - Gastritis
  - Neuromuscular changes
  - Irritability, headache
  - Numbness, tingling, vasomotor disturbances
  - Pica: cravings for non-nutritional substances such as clay, dirt and chalk

Main lab data:

- Low serum iron
- Low ferritin (iron storage protein)

Sideroblastic anemia

- A group of disorders characterized by anemia of varying severity related to inefficient iron uptake leading to abnormal hemoglobin synthesis, as well as the presence of ringed sideroblasts in the bone marrow
  - *Sideroblasts* = erythrocytes containing iron granules that have not been incorporated into hemoglobin

Causes:

- May be hereditary or acquired
- **Hereditary**
  - Rare and almost exclusively in males
  - Present in infancy or childhood but may not manifest until midlife when other conditions occurring from iron overload (e.g. diabetes, cardiac failure) begin to manifest
- **Acquired**:
  - Myelodysplastic syndromes (MDS)
    - Usually all stem cell lineage are abnormal; however, in some cases only the erythrocytic line of stem cells is affected
    - If all stem cell lineages, impaired platelets = bleeding, and impaired granulocytes = infection will be observed
  - Patients are treated with blood transfusions with the risk of iron overload
  - Of those who survive, 40% develop Acute myeloblastic leukemia (AML)
- **Reversible**:
  - Acquired SA associated with secondary conditions that can be corrected once treatment is initiated to treat underlying cause. Includes:
    - Excessive alcohol use resulting in folate deficiencies
  - Drug reactions
    - E.g. Anti-tuberculous agents which can interfere with B12 metabolism or directly injure mitochondria
  - Copper deficiency
  - Hypothermia
  - Pyridoxine deficiency (vitamin B6)
  - Lead poisoning

Pathophysiology

- Ineffective iron uptake dysfunctional hemoglobin synthesis
- Cell injury due to inflammatory reaction and altered mitochondrial activity

Clinical and lab manifestations

- Common signs/symptoms of anemia
- Ringed Sideroblasts in bone marrow are diagnostic
  - Test with blue stain to see the Sideroblasts
- Manifestation of siderosis (depositing excess iron in body)
- Can be treated with pyroxidine (B6) if that is the issue

Treatment:

- Hereditary SA can be usually treated with pyroxidine (B6) therapy at 50-200 mg/day
- Death related to SA is rare and usually occurs from secondary condition

Hemosiderosis (iron overload)

- Increased plasma iron leading to deposition in various body tissues

Causes:

- *Congenital*:
  - Genetic defect increasing iron absorption
  - Known as Hemochromatosis
  - Iron is absorbed but not utilized, so level rises
- *Acquired* - Severe chronic hemolysis of any cause
  - Iron is released from Hb
  - Multiple frequent blood transfusions (thalassemias, sickle cell, MDS)
  - Excess parenteral iron supplements (iron poisoning)

Clinical manifestations

- Due to deposition of iron in tissues: mainly liver, heart, pancreas, spleen
- *Hemosiderosis* focal iron deposition related to excess iron within an organ that does not result cause no tissue damage
- *Hemochromatosis* is typically a systemic process in which iron deposition can cause tissue damage

Treatment

- Phlebotomy

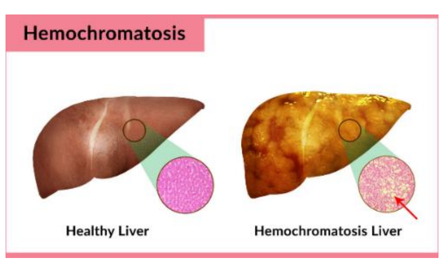


Figure 10: Comparison of healthy liver to hemochromatosis liver

Figure 11: Organs affected in iron overload

Haemoglobinopathies

- Hemoglobinopathies are a set of inherited disorders affecting the protein structure and thus function of the hemoglobin molecule

Pathophysiology

- Increased RBC destruction due to altered structure
- Impaired function of RBCs will affect oxygen carrying capacity and distribution to tissues
- Increased destruction combined with altered function leads to manifestations of anemia

Thalassemias

- The most common hemoglobinopathy
- Hereditary disorder primarily affecting those of African, Middle Eastern, and Mediterranean descent
- Defective hemoglobin synthesis due to defects in the alpha or beta chains of the globin molecule
    - Results in low intracellular hemoglobin (hypochromic); and
    - Relative excess of the unaffected chain

Pathophysiology and symptoms:

- **Bone marrow hyperplasia**
    - BM increase synthesis of RBCs to compensate for low levels of Hb
    - This hyperactivity leads to abnormal bone (face and skull) structure
- **Splenomegaly**
    - Due to increased destruction of old/damaged cells
    - Worsens the anemia
- **Iron overload**
    - If multiple transfusions are required

Normocytic-Normochromic Anemia

- Red cells are relatively normal in size and Hb content but low in number

Causes and classifications:

- *Bone marrow failure* (aplastic anemias)
    - Disorder affecting hemopoetic stem cell precursors (located in the bone marrow) and therefore affecting all blood cell lines
    - Leads to pancytopenia
    - Bone marrow biopsy is required to determine whether cause is pure red cell aplasia or hypoplasia

- *Post-hemorrhagic anemia*:
- *Hemolytic anemia*:
    - Increased hemolysis of cells stemming from hereditary abnormalities in RBCs or acquired disease conditions
    - Trauma to RBCs within blood vessels from shear forces against vessel wall cause RBC fragmentation and intravascular hemolysis
        - This can occur with prosthetic heart valves
        - Conditions with both hemolysis and deposition of fibrin and platelets within the blood vessel that leads to microvascular narrowing (e.g. disseminated intravascular coagulation (DIC), hemolytic uremic syndrome (HUS) – both discussed in later chapters)
- *Immune destruction of RBCs*:
    - Antibodies bind to RBCs leading to premature destruction
    - Two types:
        - Warm antibody disease mediated by IgG
        - Cold antibody disease mediated by IgM
- *Drug induced hemolysis*
- *Chronic inflammation or infection*
    - e.g. AIDS, rheumatoid arthritis, SLE
    - May occur in malignancies
        - Decreased RBC lifespan, or a failure of compensatory mechanisms related to progression of disease
- *Hemoglobinopathies associated with hemolysis*
    - Sickle cell anemia
        - Globin chain deficiency leading to sickle (fragile) RBCs
        - Sufficient amounts of hemoglobin is produced, however there is increased destruction of defective (sickled; fragile) cells
        - The remaining RBCs being normal in size and structure (hence, normocytic-normochromic classification)

Pathophysiology and manifestations

- The pathophysiology and manifestations vary according to disease conditions
- Underlying diseases causing hemolysis determines the pathophysiology and subsequent symptoms
    - For example, a person who has suffered acute hemorrhage and a person with a bone marrow or autoimmune condition will both have normocytic normochromic anemia, however one results from severe blood loss while the other results from an impaired erythropoiesis

- Generally, conditions with increased hemolysis will lead to splenomegaly and bone marrow hyperplasia (if the bone marrow is able to continue to produce cells)

Macrocytic (megaloblasic) Anemia

- MCV >100 + hyper-segmented neutrophils
  - Neutrophils are normally bi-lobed, but in macrocytic anemia are multi-lobed due to B12 and B9 deficiency

Cause:

Vitamin B12 (Cobalamin) deficiency

- Synthetic form of Vitamin B12 is Cyanocobalamin
- Can be caused by low levels of intrinsic factor (IF)
  1) IF is produced by parietal cells of the stomach lining
  2) Cause of Pernicious Anemia (see below)
- Other potential causes include:
  1) Insufficient intake (primarily seen in vegans)
  2) Malabsorption related to decrease IF follow gastric bypass surgery, autoimmune condition, or other gastric conditions that impact acidity
- The manifestations take a long time to develop and are bad once developed
- B12 deficiency leads to demyelination of nerves resulting in memory loss, ataxia, dementia
  - The relationship between B12 and dementia has led to increased attention to B12 supplements in the treatment of Alzheimer's disease

Folic acid (B9) deficiency

- Vitamin B9 = Folic acid (folate)
- More common than B12 deficiency
- This deficiency causes impaired DNA and RNA synthesis in bone marrow (erythropoiesis)
- Symptoms show once stores are depleted (3-4 months) and include:
  1) Cheilosis (dry, cracked mouth)
  2) Stomatitis (swelling of mouth)
  3) Can cause GI disorders
  4) Causes no neuropsychiatric symptoms.
- Treatment for deficiency is supplements

Pathophysiology

- Vitamins B12 and B9 are coenzymes required for DNA synthesis

- Impaired metabolic reactions involving homocysteine and methylmalonate, as they require B12 and B9 to complete the reactions
  - Levels of homocysteine and methylmalonate increase
- There is impaired DNA synthesis and nuclear maturation
  - Inadequate DNA synthesis leads to defective nuclear maturation and rapidly proliferating cells
  - The RBC overproduces hemoglobin during a delayed division leading to macrocytosis
    - (1) Macrocytosis: Disproportionate growth of nucleus & cytoplasm
    - (2) This results in a large cell with a small nucleus
- Timing of manifestations resulting from B12 and/or B9 deficiency are dependent on body stores.
- It is when body stores are depleted that symptoms begin to manifest
  - B12 stores last 3-4 years
  - B9 stores last 3-4 months
  - It is deficiency of both resulting in impaired DNA.

Pernicious Anemia

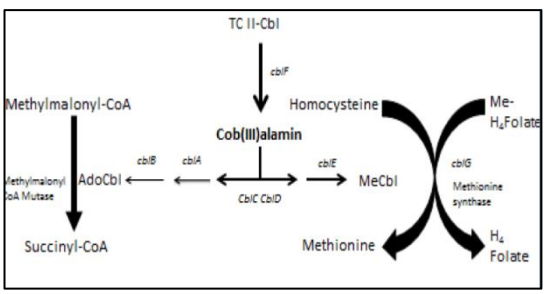


Figure 12: Metabolism of Vitamin B12 (cobalamin)

- Autoimmune vitamin B_{12} deficiency (seen with autoimmune thyroiditis, type 1 DM and vitiligo)

Etiology/Patho:

- Lack of intrinsic factor required for B12 absorption (anti-intrinsic factor antibody)

Manifestations:

- General manifestations of anemia
  - Dyspnea, weakness, fatigue
- Neurologic manifestations related to demyelination of nerves

Table: Summary of B_{12}-deficiency and B_9-Deficiency Anemias

	B_{12}-deficiency Anemia	**Folic Acid (B_9) deficiency Anemia**
Body stores	3-4 years	3-4 months
Lab features	- Elevated homocysteine - Elevated methylmalonate - Low B_{12} - Normal folate	- Elevated homocysteine - Low folate - Normal B_{12}
Clinical features	- *Neuropsychiatric symptoms*: - Memory loss - Dementia - Irritability	- No neuropsychiatric symptoms

  - Loss of vibration sense
  - Ataxia
  - Spasticity
- **Others**:
  - Loss of appetite
  - Abdominal pain
  - Beefy red tongue (atrophic glossitis)
  - Icterus (jaundice)
  - Splenic enlargement = due to wanting to save RBC

Polycythemia

- Overproduction of RBCs
- Can be classified into:

1. Absolute polycythemia
   - **Primary** (Polycythemia Rubra Vera; PRV)
     - Abnormality of stem cells in the bone marrow. High number of RBCs = sticky blood; increase blood viscosity inducing clotting tendency
   - **Secondary**
     - Condition of chronic hypoxia:
       - Leading to an increase in erythropoietin (EPO) hormone
       - Increased EPO leads to increased production of RBCs by the bone marrow
       - The body increases RBC production in response to need
     - Potential causes:
       - High altitude
       - Hypoxia associated with respiratory or cardiovascular diseases
       - EPO secreting tumor
2. Relative polycythemia
   - Results from dehydration
     - Clinically manifests with an increase in blood viscosity

Leukemia

- Malignant disorder of the blood-forming organs (cancer of white blood cells)
- Neoplasms with widespread involvement of the bone marrow and often, though not always, peripheral blood

Etiology:

- Unknown, chemo, radiotherapy, exposure to chemicals (benzene), genetic (E.g. Down's Syndrome associated with increased risk for AML)

Pathophysiology:

- Replacement of bone marrow (BM) and/or lymphoid organs with leukemic (immature) cells
- There is a depression of normal bone marrow function; red cells, white cells, platelets
- Organomegaly (organs enlarge) of organs related to the immune system
  - E.g. Lymph nodes and spleen

Clinical manifestations

- Anemia (reduced RBCs)
- Increased susceptibility to infection (abnormal WBCs)
- Bleeding manifestations (low platelets)
  - Petechiae
  - Purpura
  - Ecchymosis
  - Hemorrhage eg epistaxis, GIT bleeding
- Weight loss
- Lymphadenopathy, hepatosplenomegaly
  - Results from leukemic cells multiplying in organs
- Bony aches
  - The BM hyper activity puts pressure on bones.
- Elevated uric acid
  - As the abnormal cells and their DNA are destroyed, uric acid builds up.
  - Manifests as renal stones, uric acid crystals in urine

- *LEUKEMIC TRIAD = **ANEMIA + INFECTION + BLEEDING**

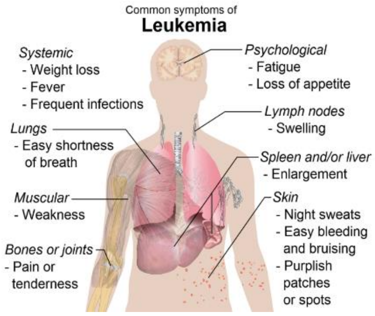


Figure 13: Classic symptoms of leukemia

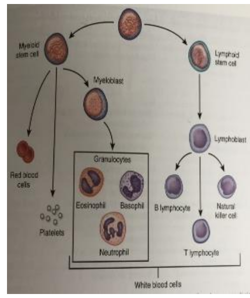


Figure 14: Lymphoid and myeloid stem cell maturation

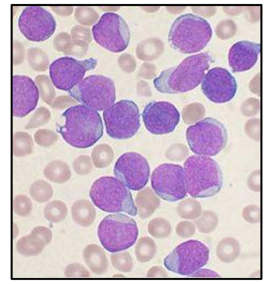


Figure 15: Lymphoblasts in blood smear

Classification/Types of Leukemia

Acute Leukemia

- Immature cells (blast cells)
  - Blasts leak from the bone marrow to cause acute disease
- *Acute Lymphoid leukemia (ALL)*
  - More common in children
- *Acute Myeloid leukemia (AML)*
  - More common in adults
  - Also, more serious in adults too

Table: Summary Classification of Leukemias

	Acute	**Chronic**
Lymphoid Stem Cell Precursor	ALL	CLL
Myeloid Stem Cell Precursor	AML	CML

Chronic Leukemia

- Mature cells, but abnormal function
  - Cells are more differentiated and cause disease over a longer period of time
- *Chronic lymphoid leukemia (CLL)*
  - More common in adults
- *Chronic myeloid leukemia (CML)*
  - More common in adults

Stem Cell Classification

- ALL and CLL come from the lymphoid stem cell
- AML and CML come from the myeloid stem cell

ALL

- Most common childhood leukemia

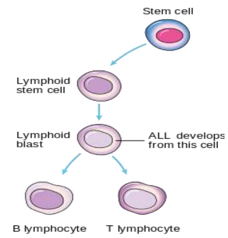


Figure 16: Development of ALL from lymphoid stem cell

Pathological features:

- Marked leukocytosis
- Lymphoblasts in peripheral blood and bone marrow

Cytogenetics:

- T(4;11) (q21;q23)
- T(8;14)(q24;q32)
- T(11;14)(p13;q11)
- Note: When reading translocation mutations:
  - E.g. T(**4**;11)(**q21**;q23)
  - **chromosome 4**, **band 21** is translocated with chromosome 11's band 23"

AML

- Most common adulthood acute leukemia
- Down syndrome and Fanconi anemia are known to increase the risk for AML

Pathological features:

- Marked leukocytosis
- Myeloblasts in peripheral blood and bone marrow

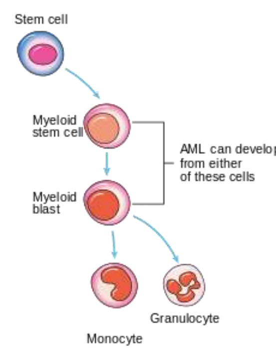


Figure 17: Development of AML from myeloid stem cell

Cytogenetics

- T(8;21)(q22;q22)
- Inv (16)(p13.1;q22)
- T(16;16)(p13.1;q22)

CML

Most common adulthood chronic leukemia

Pathological features:

- Marked leukocytosis
  - Present in terminal phase of CML
- Myeloblasts in the peripheral blood and bone marrow
  - Peripheral blood look like bone marrow!!
- Basophilia
  - Basophilia present in terminal, final stage of CML
- Splenomegaly
- Blastic crisis

- Occurs in stage 2, the acute/ accelerative phase
- Very painful

- Lymphadenopathy
  - Usually only found in the acute phase of the disease

Cytogenic

- Philadelphia chromosome:
  - T(9;22)(q34;q11) that creates the BCR-ABL fusion gene (oncogene)

Lymphomas

- Malignant transformation and proliferation of lymphocytes and their precursors/derivatives in lymphoid tissues
  - A cancer of both the blood (lymphocytes) and lymphatic system (lymphoma)
  - Lymphomas primarily affect the lymph nodes
  - The patient usually presents with a mass

2 major categories of lymphoma:

- *Hodgkin lymphoma*
  - Neoplasm of Reed-Sternberg cells
- *Non-Hodgkin lymphoma*
  - Includes 4 cell-types as classified by WHO
    - Precursor (immature) B-cell neoplasms
    - Peripheral (mature) B-cell neoplasms
    - Precursor (immature) T-cell neoplasms
    - Peripheral (mature) T-cell neoplasms

Pathophysiology & manifestations (for both Hodgkin and Non-Hodgkin Lymphoma):

- Main pathophysiological feature is lymphadenopathy
- Weight loss and night sweats
- Fever, mediastinal mass, splenomegaly, abdominal mass, pruritus

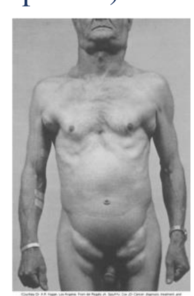


Figure 18: Lymphadenopathy (visible in groin and axilla)

Diagnostic Testing and Staging

- Anemia, leukocytosis, eosinophilia
- Elevated ESR and ALP
- LN biopsy and staging

- Histopathology: Reed-Sternberg cells; essential for diagnosis
- **Ann-Arbor staging (see Table below)**
    - Staging classification used for Hodgkin lymphoma

Hodgkin Lymphoma

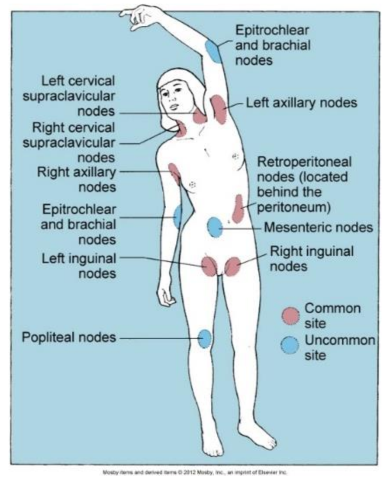


Figure 19: Body lymph node sites

- HL affects people aged 20-40 and 60-80
- Typical cases associated with infectious mononucleosis (EBV is oncogenic virus)
- Presence of abnormal B cells called *Reed-Sternberg cells (RS)*
    - RS cells are usually infected with EBV which is likely linked to the cause
    - The cells are large and binucleate, they are necessary for diagnosis but are not specific to HL
- Curable disease

Table 4. Ann Arbor Staging Classification for Hodgkin Lymphoma[a]

Stage	Description
I	Involvement of a single lymphatic site (i.e., nodal region, Waldeyer's ring, thymus, or spleen) (I); or localized involvement of a single extralymphatic organ or site in the absence of any lymph node involvement (IE).
II	Involvement of two or more lymph node regions on the same side of the diaphragm (II); or localized involvement of a single extralymphatic organ or site in association with regional lymph node involvement with or without involvement of other lymph node regions on the same side of the diaphragm (IIE).
III	Involvement of lymph node regions on both sides of the diaphragm (III), which also may be accompanied by extralymphatic extension in association with adjacent lymph node involvement (IIIE) or by involvement of the spleen (IIIS) or both (IIIE,S).
IV	Diffuse or disseminated involvement of one or more extralymphatic organs, with or without associated lymph node involvement; or isolated extralymphatic organ involvement in the absence of adjacent regional lymph node involvement, but in conjunction with disease in distant site(s). Stage IV includes any involvement of the liver or bone marrow, lungs (other than by direct extension from another site), or cerebrospinal fluid.
Designations applicable to any stage	
A	No symptoms.
B	Fever (temperature >38.0°C), drenching night sweats, unexplained loss of >10% of body weight within the preceding 6 months.
E	Involvement of a single extranodal site that is contiguous or proximal to the known nodal site.
S	Splenic involvement.

[a]Reprinted with permission from AJCC: Hodgkin and non-Hodgkin lymphomas. In: Edge SB, Byrd DR, Compton CC, et al., eds.: AJCC Cancer Staging Manual. 7th ed. New York, NY: Springer, 2010, pp 607-11.[21]

From: Childhood Hodgkin Lymphoma Treatment (PDQ®)

(National Institute of Health, 2019)

Non-Hodgkin Lymphoma

- Lymphoma from either B cell, T cell and/or NK cell neoplasms
  - Proliferation occurs through lymphoid tissue with differing patterns and responses to treatment
- Able to spread throughout body
  - In contrast HL is usually localized, affecting a certain chain of lymph nodes
- Median age for diagnosis is 67 years

Multiple Myeloma (MM)

- Malignant proliferation of plasma cells

Pathophysiology:

- Proliferation of plasma cells
  - These cells infiltrate bone marrow and aggregate creating tumor masses in bone
- Myeloma cells produce monoclonal immunoglobulins (antibodies) which are proteins
  - High levels of these proteins in the blood leads to increased blood viscosity

Clinical and lab manifestations:

- Anemia
  - Depressed bone marrow so can't make RBCs
- Bone lesions
  - Lead to bone aches
  - Prone to bone fractures
- Bence-Jones protein
  - The plasma cells can make a "free immunoglobulin light chain" (the Bence-Jones protein)
  - Circulate in blood and kindey, can cause renal failure
- Hypercalcemia
  - From bone break down
  - Leads to reduced bone density
- Renal failure
  - BJ protein buildup can block kidney tubules

Disorders Related to Alterations of the Clotting System (Bleeding Defects)

- Disorders that lead to an inability to form fibrin at an injured site

Pathophysiology:

- 3 potential causes leading to development of clotting system disorder:
  - Clotting factor defects
  - Platelet defects
  - Vascular (blood vessel) issues
    - Impaired myogenic effect
    - Impaired endothelium (vasculitis)

Diagnostics Used for Diagnosis and Treatment

- *Should know the normal CBC*
- PT (Prothrombin time)/ INR
  - The length of time it takes make prothrombin (a clot)
  - Normal range is 10-14 seconds
  - Longer times indicate bleeding issues
    - Potential cause from vitamin K deficiency
    - Oral anticoagulants (warfarin); titrated according to INR
  - Shorter times indicate clotting issues
- aPTT (Activated partial thromboplastin time)
  - A substance is added to the blood sample to make it clot faster
  - Normal range is 30-40 seconds
  - If you take heparin, your result would be higher.
    - Heparin is titrated according to aPTT

Selected Disorders Related to Clotting factors

Hereditary:

- Haemophilia A (deficiency of coagulation protein VIII)
- Von Willebrand disease (vWD) (vWF protein defects)
  - The most common mild bleeding disorder
  - Impaired VWF dependent platelet functions
- Both Haemophilia and Von Willebrand disease are very common

Acquired:

- Vitamin K deficiency

- Liver disease
  - Due to defective synthesis of coagulation factors
  - Impaired synthesis of vitamin k dependent factors (II,VII,IX,X) and proteins C and S (natural anticoagulants)

Platelet Disorders

Thrombocytopenia

- Platelet count <150,000/mm3

Selected Causes for Thrombocytopenia:

- Hypersplenism
  - Overactive spleen that destroys RBCs and platelets
- DIC (disseminated intravascular coagulation)
  - Excessive generation of thrombin and fibrin, usually from exposure to tissue factor in the circulation, leading to activation of the coagulation cascade
  - Platelet aggregation combined with consumption of coagulation factors leads to a combination of clot formation and bleeding
  - Thrombocytopenia resulting from platelet consumption in formation of clots
- HIT (heparin induced thrombocytopenia)
  - Within 7-10 days of heparin use
  - Abnormal antibodies against heparin; bound to a protein called platelet factor 4
  - Bleeding due to low platelets
  - Thrombosis may result from activated platelet
- ITP (idiopathic/immune thrombocytopenia purpura)
  - Mainly children
  - IgG against platelets
  - Antibody-coated platelets are sequestered and removed from the circulation
  - Leads to splenomegaly

Manifestations

- Petechiae, purpura
- Major bleeds

Thrombocytosis

- Elevated platelet production
- Platelet count >500,000/mm3

Selected causes:

- Essential thrombocythemia

- Myeloproliferative disorder affecting all lineage or just platelet precursor cells

Manifestations

- Microvascular thrombosis (blockages in small blood vessels)
    - Symptoms depend on vessels affected
    - Primarily occurs in hands and feet, lead to:
        - Redness
        - Warm to touch
        - Feel burning sensations in affected digits and/or limbs
        - Standing and warmth make it worse and relief from elevation and cooling
        - May develop erythromyalgia secondary to microvascular thrombosis
            - This is the paroxysmal vasodilation of small arteries (most often in hands and feet but can include face, ears and knees)
            - Causes burning pain, redness, and localized warmth
    - Microvascular thrombosis in other areas with associated symptoms include:
        - Eye (ocular migraines)
        - CNS (transient ischemic attacks)
        - Chest (pulmonary embolism)
- Bleeding events associated with thrombocytosis include easy bruising, GI bleeds, and epistaxis

Thrombophilia conditions

- Hypercoaguability of the blood
- Causes:
    - AT-III (antithrombin III) deficiency
    - Protein C deficiency and factor V Leiden (protein C resistance)
    - Protein S deficiency
    - Prothrombin gene mutations
    - Antiphospholipid syndrome

Thrombosis

- Blood clot forms inside vessel
- 3 conditions (*Virchow's triad*) together increase risk/ can lead to thrombosis
    - Hypercoagubility of blood
    - Vessel wall injury
    - Stasis of blood

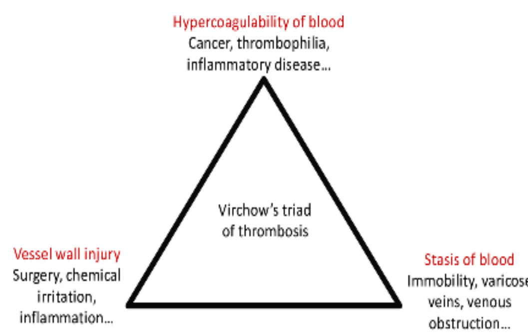


Figure 20: Virchow's Triad of thrombosis

6: Alterations of the Cardiovascular System

Learning Outcomes

Alterations of the Cardiovascular System

- Describe the normal structure and function of the cardiovascular systems (review).
- Describe the alterations in vascular flow (including thrombus formation, emboli, traumatic injury, atherosclerotic plaques, vasospastic disease and varicosities) that result in outcomes such as deep venous thrombosis (DVT), stasis ulcers, chronic insufficiencies and superior vena cava syndrome.
- Define and characterize primary, secondary, isolated, complicated, malignant and orthostatic *hypertension*.
- Describe the differences between true and false *aneurysms*.
- Distinguish between a *thrombosis*, *embolus*, *thromboembolism*, and describe conditions contributing to this pathology.
- Distinguish between *arteriosclerosis* and *atherosclerosis* and explain the pathophysiology and complications of *atherosclerosis.*
- Describe *peripheral artery disease* (PAD).
- Describe the cardiovascular risk factors and pathology of acute myocardial infarction.
- Describe the progression of *coronary artery disease* from ischemia to infarction, including clinical symptoms, diagnostic evaluation of myocardial infarction and critical timing for intervention.
- Distinguish between *myocardial ischemia* and *myocardial infarction*.
- Describe the association between *dyslipidemia* and coronary artery disease.
- Describe the clinical symptoms, evaluation and pathophysiology of *pericardial disease*.
- Define and differentiate the disorders of the myocardium: myocarditis and cardiomyopathy.
- Compare and contrast *valvular stenosis* and *regurgitation*.
- Describe the pathophysiology and manifestations of *rheumatic heart disease* including *infective endocarditis*.
- Define heart failure and compare the pathophysiology and manifestations of *right-sided and left-sided heart failure*.
- Define and classify *dysrhythmia* and describe common causes, types, hemodynamic effects and clinical presentations.

Definitions: Alterations of the Cardiovascular System

Term	Definition
Aneurysm	- Localized dilation or outpouching of a vessel wall or cardiac chamber - Caused by weakening of vessel wall
Embolism	-Detached fragment of a thrombus
True and False aneurysm	- A true aneurysm is one that involves all three layers of the wall of an artery (intima, media and adventitia) - A false aneurysm (pseudoaneurysm) is a - collection of blood leaking completely out of an artery or vein, but confined next to the vessel by the surrounding tissue
Hypertension	- Elevated blood pressure above normal range
Pericardial effusion	Accumulation of fluid in the pericardial cavity Transudate: serous/clear fluid Exudate: infected fluid
Regurgitation	'Leaky' valves lead to backflow of blood Eg mitral regurgitation, would increase workload (preload) on the left ventricle and decrease ejection of blood into aorta and body circulation
Stenosis	Narrowing of the valves; restricts blood flow through the heart
Thromboembolism	- A condition in which thrombosis develops followed by embolism eg DVT followed by PE
Thrombus	- Blood clot inside blood vessels

Review

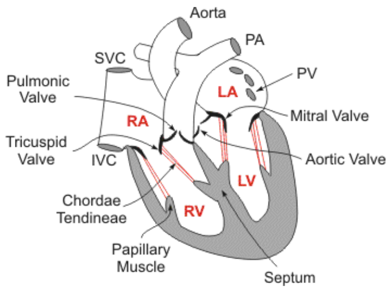


Figure 1: Heart anatomy

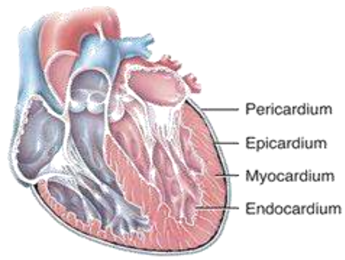


Figure 2: Layers covering heart

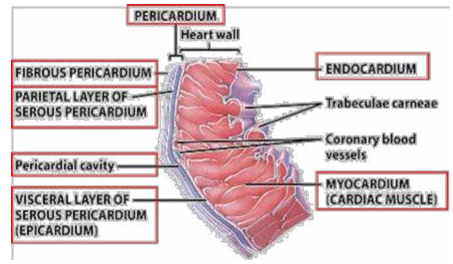


Figure 3: Heart layers showing pericardial cavity

- Chambers, valves and great vessels (review)
- Pulmonary and systemic circulation (review)
- **Layers of the heart**: pericardium, myocardium, endocardium
- Endocardium continues with endothelium lining blood vessels
- There is a space between visceral and parietal pericardium that contains fluid. This is where pericardial effusion occur
- Electrical conduction system of the heart is responsible for stimulating contraction of the muscle
    - Impulse begins in the SA node (also termed the 'pacemaker' of the heart') located in the right atrium
    - The impulse progress through the RA, then left-atrium (LA) to reach AV node where there is a slight delay
    - The impulse pass through the septum to the left and right bundle branches (left and right ventricles)

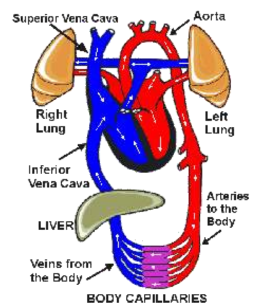


Figure 4: Circulation of blood via the heart

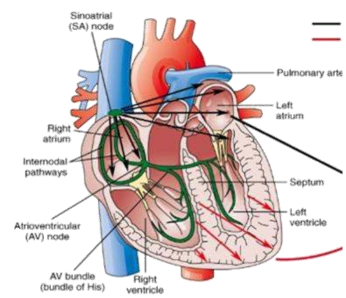


Figure 5: Conduction system of the heart

    - The last phase of the impulse is passing via Purkinje fibers, which are small branches that distribute the impulse throughout the thickness ventricular walls
- Oxygen deprivation to the heart muscle may cause conduction issues and subsequent dysrhythmias.

Disorders of the Cardiovascular System

Arteriosclerosis:

- Stiffness, loss of elasticity, and thickening of the arterial wall
- Leads to narrowing and subsequent obstruction of the lumen resulting in decreased blood flow

Atherosclerosis:

- Most common form of arteriosclerosis
- Can affect any vessel
- A chronic inflammatory response to lipid accumulation in the intimal layer of the arterial walls
  - Over time there is progressive hardening of the artery, loss of elasticity, and plaque formation (*atheromas*)
  - Narrowing of the artery leads to low blood supply
  - The process starts from birth and continues throughout the lifespan
  - Lifestyle (i.e. diet, activity, substance use) impacts the rate and severity of plaque build-up

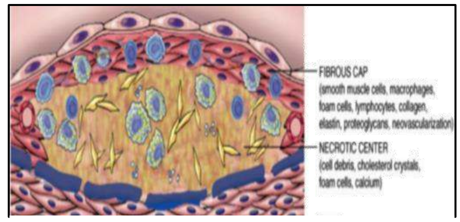


Figure 6: Formation of atheroma within a blood vessel

Manifestations:

- When atherosclerosis develops in a coronary artery this leads to coronary artery disease (CAD), which can progress to angina and MI

Non-modifiable Risk Factors

Age

Gender

Family history

Modifiable Risk Factors

- Lifestyle
- Obesity
- Physical inactivity
- Smoking

Diseases that Increase Risk for Atherosclerosis

- Type 2 DM (can be reversed)
  - Those with poor diabetes management are at increased risk
- Hyperlipidemia
- Hypertension

Atherosclerosis Pathophysiology:

- **Endothelial damage**
    - Multiple causes for endothelial cell damage (e.g. smoking, HTN, diabetes, high LDL, inactivity, etc.)
    - Damaged endothelium lose its protective antithrombic vasodilating mechanisms
    - Damaged endothelium evokes inflammation with attraction of inflammatory cells including macrophages and generation of oxygen free radicals
    - Oxidized LDL in the intima recruits monocytes that differentiate into macrophages
        - These engulf oxidized LDL to form foam cells

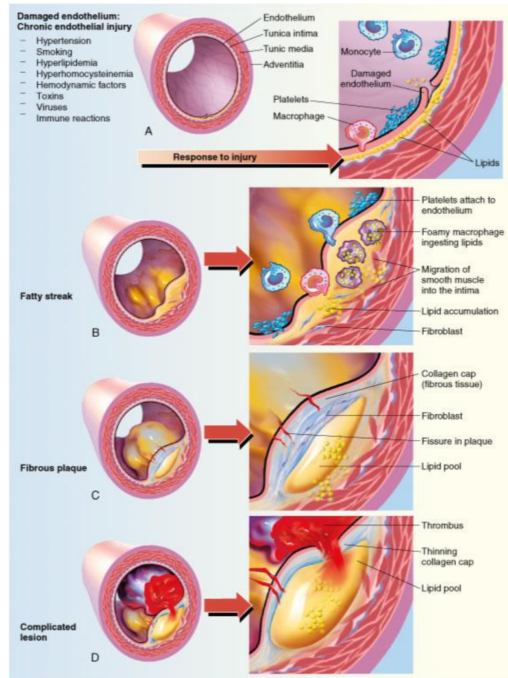


Figure 7: Pathophysiology of atherosclerosis

- **Fatty streaks/plaque**
    - Lots of foam cells creates what's known as a fatty streak
    - Fatty streaks produce more free radicals and inflammatory mediators
    - Macrophages also release growth factors that stimulate smooth muscle proliferation
- **Fibrous plaque**
    - Collagen grows over the streak to make a fibrous cap fibrous plaque
    - Can calcify and then protrude into the lumen to obstruct blood flow
- *Complicated plaque* and thrombus formation
    - Some plaques can become "unstable" and can rupture
    - A plaque that ruptures is a 'complicated plaque'
    - Exposure of the underlying tissue
        - Results in platelet adhesion, initiation of the clotting cascade, and thrombus formation
    - Thrombus can occlude a vessel
        - Partial occlusion leads to ischemia
        - Complete occlusion leads to infarction

- **Summary of atherosclerosis pathophysiology:**
  - *Stage 1*: Scratch and damage to the endothelium
  - *Stage 2*: Inflammation, collection of macrophages and other inflammatory cells
  - *Stage 3*: This makes it easy to generate a thrombus on top of plaque or exposed suburothelium

Coronary Artery Disease (CAD)

- Atherosclerosis of the coronary arteries
- Narrowing or blocking of coronary arteries → decreased blood flow→ low O2 supply to the heart → low tissue perfusion→ myocardial ischemia.

Clinical manifestations:

- Asymptomatic
- Angina pectoris
- Myocardial infarction

Cardiac complications

- Dysrhythmia
- Heart failure
- Sudden death

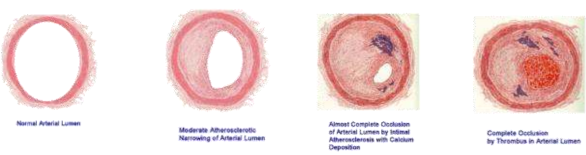


Figure 8: Progression of atherosclerosis within an artery; From normal, healthy artery (left) to occlusion (right)

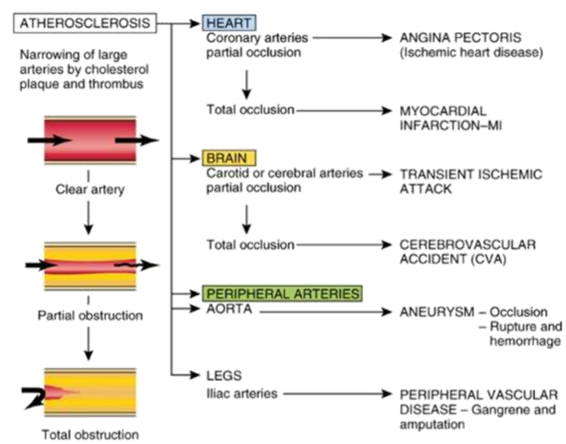


Figure 9: Progression and associated outcomes of coronary artery disease

Angina

- Sudden onset of chest pain due to myocardial infarction
  - Chest pain is a symptom of lactic acidosis (or the build-up of lactic acid as a by-product of anerobic metabolism in the absence of oxygen)
- The factors affecting angina pain are complex and multifactorial
- Precipitating factors:
  - Exertion
  - Stress
  - Heavy meals
  - Cold weather

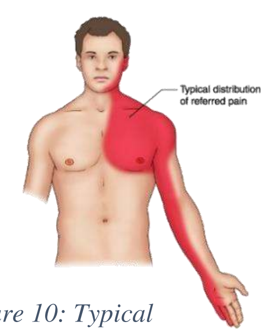


Figure 10: Typical distribution of referred pain in angina

- Relieving factors:
    - Rest
    - Nitrates
        - For angina related to insufficient blood supply and narrowed arteries pain should be relieved with medications
        - If angina is due to an obstruction of the artery pain medications will not relieve pain as they do not fix the block (medical intervention is required)

Types of Angina

Typical (exertional)

- Chest pain with exertion
- Due to coronary obstruction
- *2 patterns*
    1. **Stable**
        - Predictable and more easily controlled with used of medication by the patient
    2. **Unstable**
        - Prolonged pain
        - May happen at rest
        - More dangerous (MI)

Variant (Prinzmetals's)

- Chest pain at rest
- Due to coronary vasospasms
- Unpredictable

- Both typical and variant angina are treated with nitrates
    - Nitrous oxide is naturally occurring in the endothelium
    - Controls the vasomotor physiology (i.e. dilation and constriction of the blood vessel)
    - Giving the patient more of the drug will cause arteries to vasodilate
    - It is also an anticoagulant

Myocardial Infarction

- Ischemic necrosis of part of the heart muscle due to total occlusion of a branch of coronary artery
- Left ventricle (left main coronary artery) is more commonly affected

- There is the leakage of enzymes and other proteins
  - These are cardiac markers that can be measured in the blood to diagnose as well as guide and evaluate treatment
  - Cardiac markers include:
    - CK-MB (specific to the heart muscle)
    - Troponin
      - Specific to the heart and essential for diagnosis of MI
    - Myoglobin
      - Earliest marker visible in the blood, however less specific

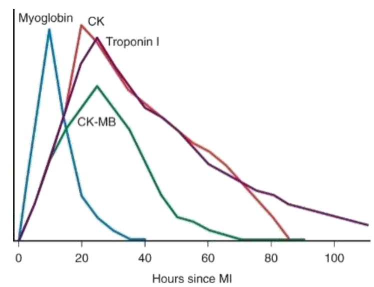


Figure 11: Changes in cardiac enzymes over time following cardiac injury

- Healing creates a fibrous scar that can weaken and rupture
- Role of collateral circulation
  - Vasculature that bypasses an occlusion
- Risk of ischemia reperfusion injury
  - Return of oxygen to the heart leads to free radicals which can lead to further damage

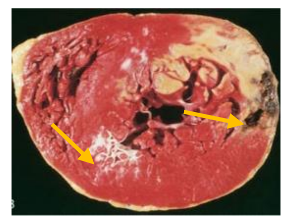


Figure 12: Fibrous tissue scars (arrows) following cardiac injury/infarction

Manifestations

- Size and location of infarction determines clinical manifestations and severity and also intervention and outcomes
- **Pain**:
  - Similar to angina but more severe, prolonged, and does not respond to rest and vasodilators
- Pallor, sweating, dyspnea
- Hypotension
- Fever (low grade)
  - Whenever a person presents pale with sweating, chest pain (or left arm, shoulder, jaw, back pain), and hypotension (verified with blood pressure) always suspect heart attack
  - For women, symptoms may be low grade and prolonged over time (denial is common)
- Administering 2 chewable ASA (160mg) remains the standard with chest pain or suspicion of myocardial infarction

ECG Changes in MI

- ECG waves represent each part of the cardiac cycle:
    - P wave = atrial contraction
    - QRS complex = ventricular contraction and atrial relaxation
    - T wave = ventricular relaxation

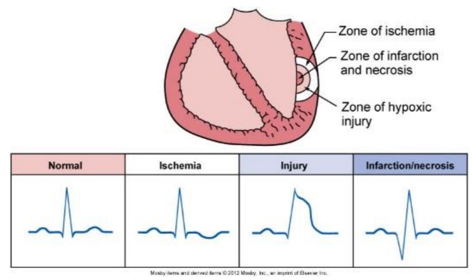


Figure 13: ECG changes related to cardiac damage

- ECG changes associated with myocardial injury:
    - **Ischemia** = T wave inversion (NSTEMI) – partial thickness
    - **Injury** = ST segment elevation (STEMI) – full thickness, all the myocardium in a section is dead.
    - **Infarction/necrosis** = PQ depression

Complications

- Cardiac arrest
- Myocardial rupture whereby a piece of necrotic tissue can break off
- Left ventricle dysfunction
    - May be a precursor to future heart failure and long-term need for nitrates to manage angina

Rheumatic fever

- Acute systemic immune/inflammatory condition that affects the heart as well as other organs
    - Abnormal humoral cell-mediated response to group-A streptococcus (GAS) cell membrane antigens called M-proteins

Etiology

- Develops 2-3 weeks after an untreated infection by beta hemolytic streptococcus group A (strep pyogenes)
    - E.g. Tonsillitis, pharyngitis
- In certain percentage of normal population ~ 50 cases per 100,000 children

Pathophysiology

- Antibodies formed against streptococci cross react with similar self antigens in normal tissue of the heart, joints, skin, muscle, and brain leading to a diffuse autoimmune inflammatory response
    - Leads to cardiac and extracardiac lesions
- Repeated attacks can lead to chronic proliferative changes and scarring

Cardiac lesions

- Develop from the 'inside out' beginning with valves and endocarditis followed by myocarditis and finally affecting the pericardium (pericarditis)
- *Pericarditis*
  - When paired with effusion may impair cardiac filling
    - As the thin pericardial space around the heart fills with fluid it limits the pumping action of the heart
  - Often indicated by pericardial friction rub audible on auscultation
- *Myocarditis:*
  - Inflammation of the myocardium
    - *Aschoff bodies* are localized areas of inflammation in connective tissue of the heart
      - Granulomas of fibrous tissue and lymphocytic, macrophage infiltrations
        - Localized fibrin deposits surrounding necrotic centres
      - These structures are slowly replaced a fibrous scar
      - They can also merge to form multinucleated cells
      - Formation of masses that impair the pumping action of the heart by decreasing elasticity
      - Also leads to conduction disturbances
        - Potential ECG changes include varying forms of heart block

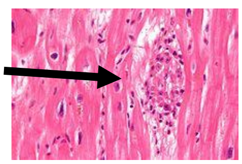


Figure 14: Aschoff body

- *Endocarditis*
  - Inflammation of the endocardium (inner lining of the heart including valves)
  - Primarily manifests as valvular lesions
    - Edema of valve cusps with secondary erosion along lines of leaflet contact where they may rub against each other
    - *Verrucae*: small, wart-like clumps of vegetation composed of fibrin and platelets are deposited along eroded areas of leaflets, as well as the chordae tendinae
      - Leads to fibrosis, scarring, and *stenosis* or *regurgitation*
    - Mitral valve is most commonly affected
  - Future risk of infective endocarditis, particularly with repeated infection
    - Another attack of strep worsens the condition of heart
    - It is important to remmebr it is not the infection, but the antibody in response to the microbe that targets the heart and joint tissue

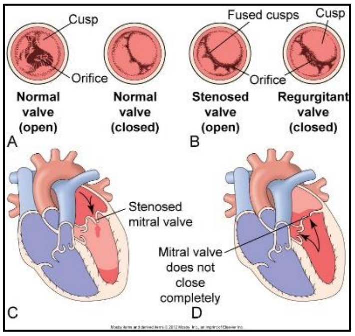


Figure 15: Valvular changes occurring in rheumatic fever

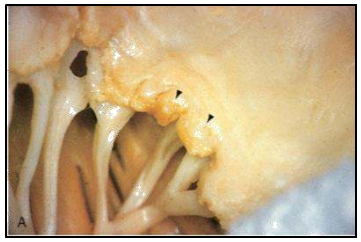


Figure 16: Verrucae along the edge of a valve leaflet; Also thickening of the chordae tendinae visible

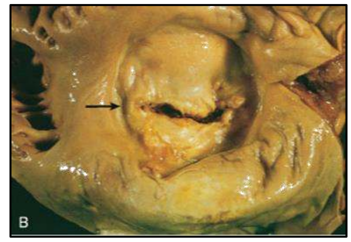


Figure 17: Thickening of valve leaflets and subsequent valve stenosis

Extracardiac lesions

- *Joints*: Polyarthritis
  - Primarily in large joints (knee and ankle)
  - Fleeting effect (one knee hurts and then the other)
  - No permanent damage
  - "ARF licks the joints but bites the heart"
- *Skin*
  - *Erythema marginatum*
    - Red nonpruitic macular rash with pale center
  - *Subcutaneous nodules:*
    - Small, non-tender bumps located on extensor surfaces of the elbow, wrist, knees, ankles


Figure 18: Erythema marginatum


Figure 19: Subcutaneous nodules

- *Brain*
  - Basal nuclei lesion → Sydenham's chorea
  - Manifests with jerky movements

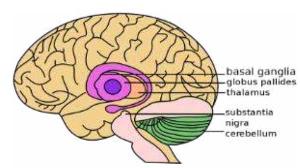


Figure 20: Locations of basal nuclei lesions

Rheumatic Fever – Diagnosis

- Diagnosis is made using the "*Modified Jones Criteria for Initial Diagnosis of Acute Rheumatic Fever* ARF" whereby symptoms are categorized into 'Major' and 'Minor'
- **Major Manifestations**
  - Carditis
  - Polyarthritis
  - Erythema marginatum
  - Subcutaneous nodules
  - Chorea

- **Minor Manifestations**
  - Fever
  - Arthralgia
  - History of rheumatic fever
  - Elevation of acute phase reactants
    - C-Reactive Protein (CRP) and erythrocyte sedimentation rate (ESR) are two acute phase reactants that when elevated indicate infection/inflammation processes
      - These should return to normal over 3 months as an exacerbation or infection resolves
      - Can be used to monitor for effective treatment and progression
    - They are non-specific for ARF, but when included with other manifestations may be used for diagnosis
- For diagnosis the person must have:
  a) 2 major manifestations
  b) 1 major manifestation + 2 minor + history of strep infection

Rheumatic Fever - Investigations

- **Lab**:
  - Elevated ASO (antistreptolysin O) antibodies titer
  - Inflammatory marker for strep
  - Elevated CRP (C-reactive protein)
  - Elevated ESR
  - Leukocytosis
  - Anemia
  - Joint aspiration
    - Excludes other causes of arthritis
- **Radiology**
  - Echocardiography
    - Used to detect carditis, particularly if there it is subacute
  - ECG
    - When combined with cardiac markers, can be used to rule out myocarditis

Infective endocarditis

- Inflammation of the endocardium including valves caused by infection

Causes:

- There are usually two precipitating factors leading to IE:
    1. Predisposing abnormality of the endocardium (may be acquired)
    2. Microorganisms in the bloodstream (bacteremia)
        - **Bacteria** (most common)
            - Subacute: Non-virulent bacteria
                - E.g. Strep viridans
            - Acute: Highly virulent bacteria
                - E.g. Staph aureus
                - Manifestations are sudden, more aggressive, and cause more damage
        - **Others**:
            - E.g. fungal, viral, rickettsial

Risk Factors

- IVDU (IV drug use)
- Needles used for tattoos and body piercing
- Prosthetic valves
- Congenital heart disease

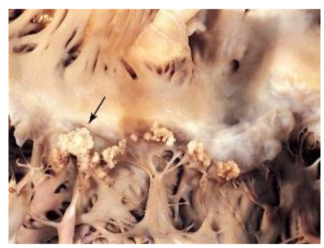


Figure 21: Vegetative growths on endocardium

Pathophysiology

- Three stages of disease progression:
    1. **Bacteremia**
        - The microorganism is present in the blood
    2. **Adhesion**
        - The microorganisms adhere to abnormal or damaged endothelium by way of surface adhesions
    3. **Colonization**
        - Proliferation of the organism with inflammation
        - Leads to mature vegetations on the valve cusps
        - Vegetations are large fragile masses composed of fibrin strands, platelets, other blood cells, microorganisms
        - Cause more damage to the valves and interfere with normal function
        - May break away leading to septic emboli and distant infection

Clinical Manifestations

- Fever, malaise, fatigue, anorexia
- New murmur (or a change in character)
    - Audible on auscultation
- *Roth's spots*
    - Retinal hemorrhages
    - Red spots visible when using ophthalmoscope
- *Osler's nodes*
    - Painful, raised red/brown nodules
    - Note the contrast to painless subcutaneous nodules of Rheumatic fever
- *Janeway Lesions*
    - Painless and flat lesions on the sole of foot
- Splinter hemorrhage under nails
- Splenomegaly

Figure 22: Osler's nodes

Figure 23: Splinter hemorrhages

Complications:

- Septic emboli
    - Leads to abscesses from localization in different organs or to organ infarctions from blockage of blood vessels
    - Myocardial abscesses with tissue destruction and potential conduction abnormalities
- CHF
    - From sudden, severe valve regurgitation
- Arthritis
- Glomerulonephritis
    - Occurs when emboli from vegetation in heart tissue reaches the kidney
- Septic shock
- Stroke / TIA
- Can ultimately affect any organ leading to failure

Treatment

- IV antibiotics
- Valve debridement and/or repair, replacement
- Remove potential sources for bacteria

Pericarditis

- Inflammation of the pericardium
    - Pericardial membranes become roughened and can lead to *pericardial effusion*
    - Effusion may be serous, purulent or fibrinous

Etiology:

- Any infection can lead to pericarditis
    - E.g. Post viral, Tuberculosis infection
- Rheumatic fever
- Malignancy
- Trauma

Pathophysiology:

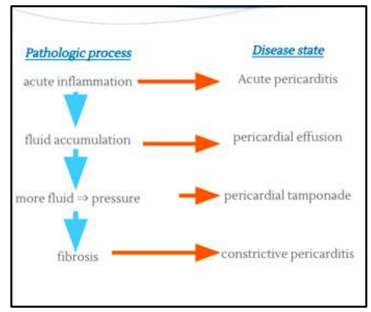


Figure 24: Pathological development of pericarditis (acute to constrictive)

- *Acute pericarditis* is most often idiopathic or a post-viral complication
    - Other causes include:
        - Myocardial infarction
        - Trauma
        - Malignancy
        - Connective tissue disorders (e.g. systemic lupus erythematous)
    - It is also the most common cardiac complication associated with HIV
- *Chronic pericarditis*
    - Recurrent, long-standing pericarditis □ constrictive pericarditis
    - Leads to fibrous scarring with occasional calcification of the pericardium causing the visceral and parietal layers to adhere and obliterating the pericardial cavity
    - Constrictive pericarditis develops as fibrotic lesions eventually encase the heart in a hard shell
        - This compresses (constricts) the heart reducing cardiac output

Manifestations:

- **Dry pericarditis**
    - No effusion
    - Pericardial friction rub on auscultation
    - Pleuritic chest pain
- **Wet pericarditis**
    - Pericardial effusion
        - May be serous, purulent, fibrinous, or hemorrhagic

- Tachycardia and palpitation
- Dyspnea and cough
- Distended neck veins
    - There is backflow of blood from right-atrium into the neck veins due to decreased pumping ability of the heart
- Distant heart sounds
    - A result of increased fluid accumulation surrounding heart
- **Cardiac tamponade (Pulsus paradoxus)**
    - Life threatening complication of pericardial effusion
    - Rapid fluid accumulation can create enough pressure to compress the heart
        - The danger is that this pressure will be enough to equal diastolic pressures and interfere with atrial filling during diastole causing increased systemic venous pressures and symptoms of right heart failure
        - Decreased atrial filling can eventually lead to decreased ventricular filling and subsequent circulatory collapse
    - *Pulsus paradoxus*: Arterial pressure during expiration exceeds arterial pressure of inspiration by more than 10mmHg
        - In the context of pericardial effusion this indicates tamponade and decreased ventricular filling combined with reduced blood volume in all four cardiac chambers
- **Effusion**
    - Rapid accumulation of fluid in pericardium obstructing heart →Prevents ventricular filling
    - The pericardium can stretch to accommodate gradual accumulation of fluid without compressing the heart
    - Transudate indicative of fluid overload while exudate indicates inflammation seen in acute pericarditis
    - *Cardiocentesis*: Removal of fluid from the pericardial space
- Decrease in SBP by 10 mmHg during inspiration

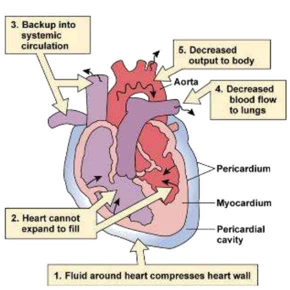


Figure 25: Pericardial effusion

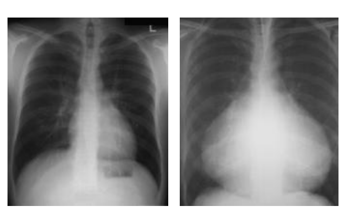


Figure 26: Comparison of pericardial effusion to normal heart on x-ray

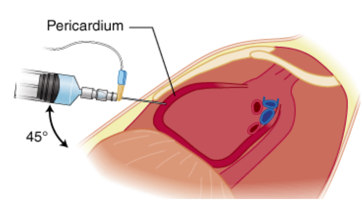


Figure 27: Cardiocentesis to remove pericardial fluid

Myocarditis and Cardiomyopathy

- Disease of the myocardium
- Most result from remodelling caused by neurohumoral responses of the heart muscle to ischemic heart disease and/or hypertension

Etiology

- Infection
  - Most commonly viral
- Hypertension
- Toxins
  - E.g. Alcohol, chemotherapy, radiation
- Immunological
  - E.g. SLE
- Infiltrative or proliferative disorders
- Nutritional deficiencies

TABLE 23-6 CLINICAL MANIFESTATIONS OF VALVULAR STENOSIS AND REGURGITATION

MANIFESTATION	AORTIC STENOSIS	MITRAL STENOSIS	AORTIC REGURGITATION	MITRAL REGURGITATION	TRICUSPID REGURGITATION
Most common cause	Congenital bicuspid valve, degenerative (calcific) changes with aging, rheumatic heart disease	Rheumatic heart disease	Infective endocarditis; aortic root disease (connective tissue diseases, Marfan syndrome); dilation of aortic root from hypertension and aging	Myxomatous degeneration (mitral valve prolapse)	Congenital
Cardiovascular outcome (untreated)	Left ventricular hypertrophy followed by left heart failure; decreased coronary blood flow with myocardial ischemia	Left atrial hypertrophy and dilation with fibrillation, followed by right ventricular failure	Left ventricular hypertrophy and dilation, followed by left heart failure	Left atrial hypertrophy and dilation, followed by left heart failure	Right heart failure
Pulmonary effects	Pulmonary edema: dyspnea on exertion	Pulmonary edema: dyspnea on exertion, orthopnea, paroxysmal nocturnal dyspnea, predisposition to respiratory tract infections, hemoptysis, pulmonary hypertension	Pulmonary edema with dyspnea on exertion	Pulmonary edema with dyspnea on exertion	Dyspnea
Central nervous system effects	Syncope, especially on exertion	Neural deficits only associated with emboli (e.g., hemiparesis)	Syncope	None	None
Pain	Angina pectoris	Atypical chest pain	Angina pectoris	Atypical chest pain	Palpitations
Heart sounds	Systolic murmur heard best at right parasternal second intercostal space and radiating to neck	Low rumbling diastolic murmur heard best at apex and radiating to axilla, accentuated first heart sound, opening snap	Diastolic murmur heard best at right parasternal second intercostal space and radiating to neck	Murmur throughout systole heard best at apex and radiating to axilla	Murmur throughout systole heard best at left lower sternal border

With data from Braunwald E, editor: *Heart disease: a textbook of cardiovascular medicine*, ed 5, Philadelphia, 1997, Saunders; Carabello BA, Paulus WJ: Valvular heart disease. In Crawford MH, DiMarco JP, editors: *Cardiology*, London, 2001, Mosby.

Figure 28: Summary Table Manifestations of Valvular Stenosis and regurgitation (taken from Huether & McCance, 2012)

Pathophysiology

- Dependent on type of cardiomyopathy
- *Hypertrophic*
  - Thickening of the myocardium whereby cells become pathologically enlarged
  - Divided into two major categories:
- **Hypertrophic Obstructive Cardiomyopathy**
  - Most common inherited cardiac disorder
  - Thickening of the septal wall can obstruct ventricular outflow when heart rate increases, and intravascular pressure decreases
    - There is increased risk for ventricular arrhythmia
    - Most commonly seen in response to exercise, which explain why this is the most common cause of sudden death in young athletes
- **Hypertensive or Valvular Hypertrophic Cardiomyopathy**
  - Most commonly seen in individuals with hypertension or valvular stenosis
  - Increased resistance to ventricular ejection leads to hypertrophy of myocytes to compensate for increased myocardial workload

- Myocardium is weak and inflamed
- The heart can contract but cannot relax leading to a high ejection fraction (it is unable to relax)

- *Dilated*
  - Most often due to ischemia, alcohol, or drug toxicity
    - Other causes include peripartum complications, genetic disorder, or infection
  - Characterized by impaired systolic function leading to increased intracardiac volume, ventricular dilation, and systolic heart failure
    - The heart can relax but is unable to contract leading to a low ejection fraction
    - Myocardium is damaged from toxic agents (including hypoxia)
      - Damage to the lumen means the muscle can no longer perform the pumping action required for systole
- *Restrictive*
  - The heart is unable to move due to stiffening of the vessel, usually from scarring
    - Loss of elasticity (stretch)
    - The heart becomes like a stiff box and is unable to accommodate changes in pressure that occur with filling
  - Inability to properly contract and relax leads to poor pumping action and decreased ejection fraction
  - There is decreased filling of heart chamber, increased diastolic pressure in one or both ventricles, and systolic function remains near normal or normal
  - Potential causes include:
    - Sarcoidosis
    - Amyloidosis
    - Hemochromatosis
    - Fibrosis

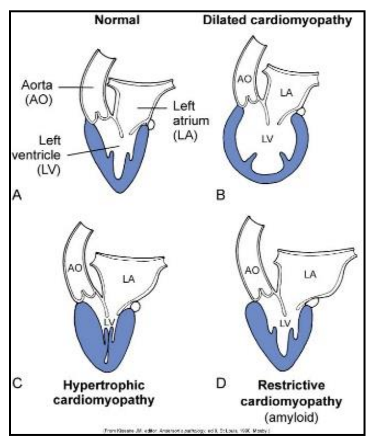


Figure 29: Comparison cardiomyopathies (dilated, hypertrophic, restrictive) to normal heart

Cardiac Dysrhythmia (Arrhythmia)

- Abnormal heart rate or rhythm
- May be congenital or acquired and range in severity from asymptomatic to severe and life threatening

Causes:

- Abnormal rate of impulse generation from the SA node
  - May be related to environmental sources such as neurohumoral stimulation

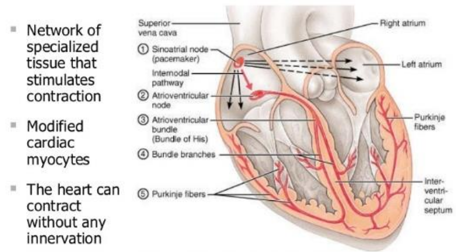


Figure 30: The conduction system of the heart

- Damage to the conduction system
- *Electrolyte imbalance*
  - Sodium (hypo/ hypernatremia)
  - Potassium (hypo/ hyperkalemia)
  - Calcium (hypo/ hypercalcemia)
  - Magnesium (hypo/ hypermagnesia)
- *Exaggerated autonomic effect*
  - E.g. Activation of the sympathetic nervous system related to release of catecholamines or other neuroendocrine/ neurohumoral activation of the nervous system
- Fever, hypoxia, infection, ischemia
- Anemia
- Thyrotoxicosis
- Drug induced
  - E.g. Digitalis

Clinical manifestations:

- Asymptomatic
- Palpitations
- Dizziness, syncope (temporary loss of consciousness)
- Hypotension
- Cardiac arrest

Classifications of Dysrhythmias

- *SA node abnormalities*
  - Sinus tachycardia = >100 and regular
  - Sinus bradycardia = <60 and regular
- *Atrial conduction abnormalities*
  - Atrial flutter = 160-350 bpm
    - Slower ventricular rate (AV nodal delay) – regular
  - Atrial fibrillation (AF) = >350 bpm
    - No P wave

  - Slower ventricular rate – irregular
  - Risk of thrombosis
- *AV Node Abnormalities*
  - Supraventricular tachycardia (SVT)
  - Heart block
- *Ventricular abnormalities*
  - Ventricular arrhythmias are usually severe with high mortality rate requiring immediate intervention through defibrillation, cardioversion, and/or medication
  - VT (Ventricular tachycardia):
    - Dangerous: low ventricular filling → low SV → low CO
  - VF (Ventricular fibrillation):
    - Muscle fibers contract independently & rapidly
    - Zero CO → severe hypoxia
    - If not treated immediately → cardiac arrest

Selected Dysrhythmias - Atrial fibrillation (AF)

- AF is the most common cardiac arrhythmia and commonly undiagnosed
  - People may live with the condition in a stable state for many years
  - When it is an unstable AF people will begin to notice symptoms related to decreased cardiac output
    - E.g. Weakness, fatigue, intermittent palpitations
    - Because the symptoms are often intermittent and mild at first, people may ignore them until they interfere with daily function
- **Pathophysiology**;
  - Ectopic atrial foci fire in a chaotic manner preventing normal coordinated atrial contraction
    - Multiple impulses are sent to the ventricle leading to ineffective pumping and ejection of blood into circulation
    - Blood stagnates in the atria increasing risk for clotting
    - Increased stroke risk
      - Patient requires increased anticoagulation to prevent stagnation and clot formation
      - PT/INR for AF patients is higher than average (2.0-3.0)
  - Mitral stenosis is the single most important risk factor

- AF can be presenting symptom of hyperthyroidism (and therefore should be considered when obtaining patient history)
- AV node only conducts some of the atrial impulses to ventricles, resulting in the classic irregularly irregular rhythm

▪ **Clinical manifestations:**
  - Rapid and irregular heartbeat, which causes palpitations and exercise intolerance
    - Atrial rate > 500 bpm, ventricular rate is typically 120-180 bpm
  - Blood pressure varies beat to beat
  - Venous stasis leading to:
    - Congestive symptoms such as shortness of breath and edema
    - Thrombosis

▪ **Diagnostic Findings**
  - *ECG*: Irregularly irregular rhythm with a disorganized baseline electrical activity
    - No P-Waves and a narrow QRS
  - When taking peripheral pulse with finger probe, the pulse will seemingly jump all over the place indicating an irregular rhythm
    - This should always be confirmed with manual palpation
    - Suspect AF and, if previously undiagnosed, obtain ECG and inform Physician or most responsible provider

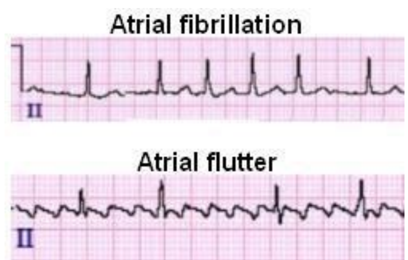


Figure 31: ECG comparison atrial fibrillation to atrial flutter

Cardiac arrest

▪ Cessation of all activities of the heart

Classification:

▪ **Shockable**:
  - VF
  - Pulsless VT

▪ **Non- shockable:**
  - Pulseless electrical activity
  - Asystole

Heart Failure

- Inability of the heart to pump sufficient quantities of blood to meet body's metabolic needs

Precipitating factors

- Precipitating factors increase risk and worsens the condition; however, they do not indicate a cause for the disorder
    - Ischemia
    - Drugs
    - Diet
    - Emotional / physical stress
    - Sudden AF
    - Infection
    - Anemia
    - Thyroid disease
    - PE (pulmonary embolism)

Etiology

- **Cardiac**
    - CAD – MI
        - Ischemic heart disease and hypertension and two of the most significant risk factors
    - Cardiomyopathy
    - Valvular heart disease
    - Congenital heart defects
    - Myocarditis
- **Extracardiac**
    - Hypertension
    - Pulmonary disease
    - Renal disease
    - Hyperthyroidism
    - Severe anemia
    - DM
        - Resulting from micro and macro angiopathy

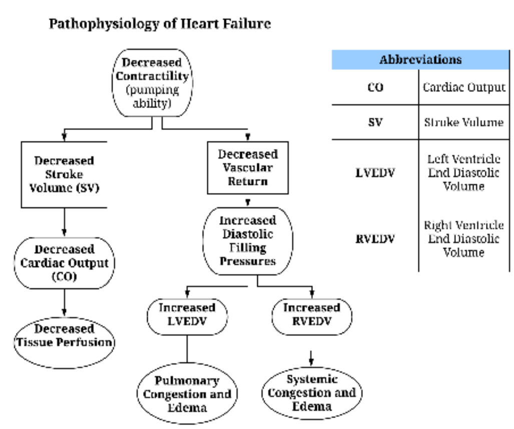


Figure 32: Pathophysiology of heart failure (simplified)

Pathophysiology

- Decreased pumping capability (or contractility) of the heart, leads to:
  - Decreased stroke volume
  - Decreased cardiac output
  - Decreased tissue perfusion

Compensatory mechanisms:

- To compensate for decreased cardiac output, the body works to increase blood volume and heart rate by:
  - RAS activation leads to vasoconstriction and increased blood volume
  - SNS activity causes increased heart rate and vasoconstriction
- These mechanisms increase cardiac workload and thereby lead to further progression of disease
- Ventricular remodelling with chamber dilatation and cardiomegaly (hallmark of heart failure)
  - There is an increase in myocardial oxygen demand, further decrease in contractility leading to further tissue ischemia

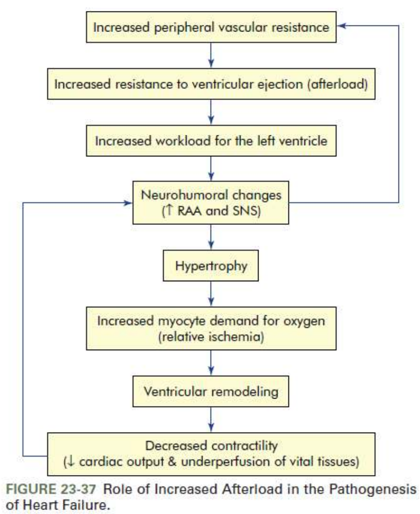


Figure 33: Compensation and impact of afterload on progression of heart failure (Huether & McCance, 2012)

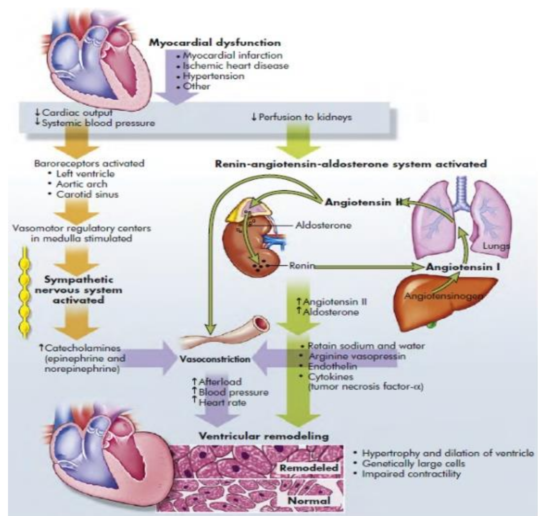


Figure 34: Ventricular remodelling and the development of heart failure (Huether & McCance, 2012)

Manifestations from hypoxia

- May be general and/or exertional (manifest or worsen with activity)
  - Dyspnea
  - Fatigue
  - Cough
  - Organ failure due to prolonged ischemia

Manifestations due to compensation

- Tachycardia
- Pallor
- Oliguria (minimal urinary output)
  - From decreased renal perfusion
  - Renal failure is common and may require dialysis
- Left and/or right-sided CHF

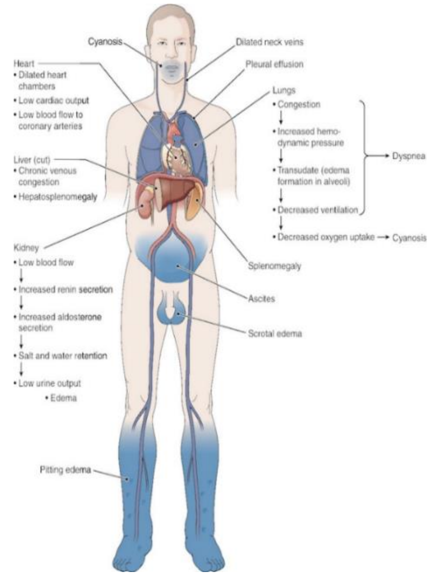


Figure 35: Manifestations of heart failure

Left sided CHF Manifestations

- Pulmonary congestion
- Dyspnea
- Orthopnea
- Paroxysmal nocturnal dyspnea (PND)
- Cough
  - May be accompanied with hemoptysis (coughing blood)
- Wheezes and rales audible on auscultation
- Pleural effusion
- Acute left sided failure will manifest with pulmonary edema
  - Related to rapid accumulation of fluid and increased ventricular pressures

Right sided CHF Manifestations

- Systemic congestion
- Dependent edema
- Digestive disturbances
- Hepatomegaly / Splenomegaly
- Distended jugular veins
- Ascites (can get hernia)

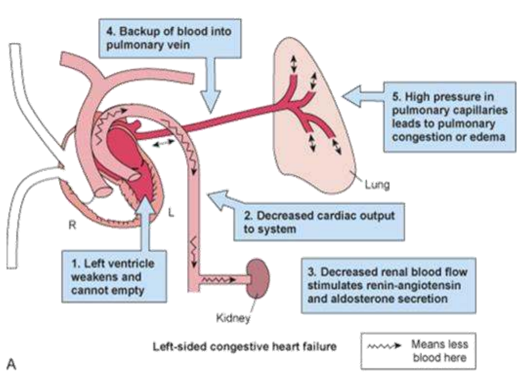


Figure 36: Symptom pathophysiology for L-sided heart failure

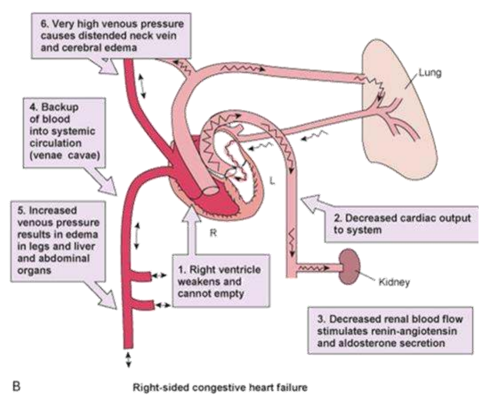


Figure 37: Pathophysiology of manifestations for R-side heart failure

Vascular disorders

Arterial disorders:

- Hypertension
- Aneurysm
- Peripheral vascular disease

Venous disorders:

- *Thromboembolism* (mainly DVT and PE)
- Varicose veins

Hypertension

- Consistent elevation of systemic arterial blood pressure
  - Generally, BP >140/90 is considered mild hypertension for the average person
  - For people with DM, BP > 130/80
  - For those older than 80 years, BP >150/90
    - Due to the stiffening of blood vessels that occurs naturally with aging

BP	Mild	Moderate	Severe
SBP	140-159	160-179	≥ 180
DBP	90-99	100-109	≥ 110

Figure 38: Categories of hypertension

- Isolated systolic hypertension (ISH) is consistently elevated SBP >140mmHg with diastolic (DBP) remaining within normal range (<90mmHg)
  - ISH is rising within every age group and is strongly associated with increased risk of cardiovascular and cerebrovascular events

Types of Hypertension

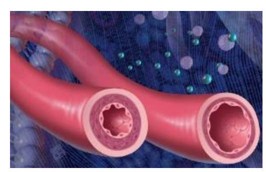


Figure 39: Narrowing of vascular lumen contributing to hypertension

Primary hypertension (90%)

- Also called essential or idiopathic
- Most common type
  - Gradual onset
  - Most likely related to combination of aging and modifiable environmental and lifestyle factors

Secondary hypertension (10%)

- Develops in response to other disease processes including:
  - Renal disorders
    - E.g. Glomerulonephritis, pyelonephritis, polycystic kidney
  - Endocrine disorders
    - E.g. Primary hyperaldosteronism, pheochromocytoma, Cushing’s syndrome, hyperthyroidism

- Vascular
    - Renal artery stenosis
- Drug induced
    - E.g. Steroids, estrogens
- Pregnancy

- Note that many of the conditions related to the development of HTN are chronic in nature indicating a relationship between these processes and the chronic inflammation as well as physiologic/pathologic adaptations that occurs

Risk factors:

- **Nonmodifiable**:
    - Age
    - Male gender
    - Race
        - E.g. African Americans have higher than average blood pressure
    - Family history
        - Genetic predisposition bis believed to be polygenic and related to defects associated with renal sodium excretion, SNS and RAAS activity, as well as cell membrane sodium and calcium transport
- **Modifiable**
    - Obesity
        - Related to hormones secreted in obesity and insulin resistance that may develop
    - High salt intake
        - Leads to increased vascular volume from decreased salt excretion by the kidneys
    - Physical activity
    - Stress
    - Smoking
    - Alcohol consumption
    - Drugs
        - E.g. Oral contraceptives

Pathophysiology:

- Sustained increase in peripheral resistance (arteriolar vasoconstriction), increase in blood volume, or both

- There is damage to the arterial wall leading to sclerosis (hardening and thickening) and narrowing of the arteries
- Weakening of the arterial wall can lead to an aneurysm

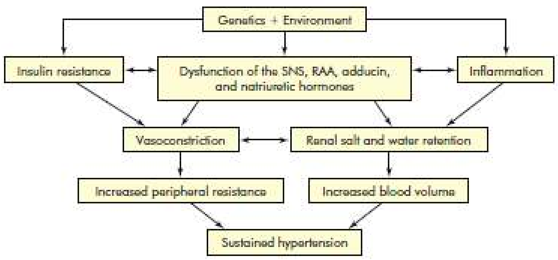


Figure 40: Pathophysiology of hypertension (from Huether & McCance, 2012)

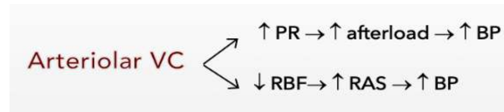


Figure 41: Relationship of arteriolar vasoconstriction on elevation in BP

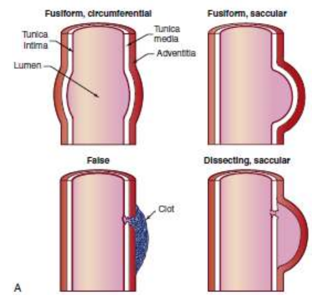


Figure 42: Types of aneurysms (Huether & McCance, 2012)

  - *Aneurysms* are often asymptomatic and may go undetected
  - Continued growth may lead to rupture and hemorrhage
- Decreased perfusion leads to organ ischemia and subsequent end-organ complications

Clinical manifestations:

- Termed the silent killer as it is usually asymptomatic
  - Diagnosis based on regular BP measurement
    - Some may have physiologic/anatomic damage despite normal BP measurements
      - If these changes remain undetected and untreated they may become established early in adulthood and begin accelerating in progression leading to increased risk for complications starting in middle decades of life
  - Potential symptoms include:
    - Morning occipital headache
    - Manifestations of end organ failure
- Clinical manifestations are usually related to tissue and/or organ damage outside of the vascular system and are therefore dependent on the system affected
- These clinical manifestations include:
  - **Cardiovascular**
    - Liver hypertrophy
    - Heart failure

- CAD
- PAD
- **Cerebrovascular**
  - TIA
  - Cerebral stroke
  - Encephalopathy
- **Visual System**
  - Retinopathy
  - Papilledema
  - Loss of vision
- **Renal System**
  - Renovascular disease
  - Renal failure

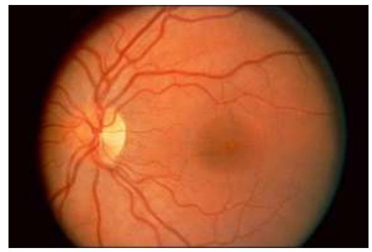


Figure 43: Hypertensive retinopathy; Papilledema; Loss of vision

Resistant and Malignant Hypertension

- *Resistant HTN*:
  - Persistently elevated BP despite medical treatment
  - Common risk factors: old age, obesity, renal disease
  - Requires more aggressive control, often with 3 or more drugs
- *Malignant HTN*:
  - Rapidly progressive hypertension with DBP usually >140mmHg
  - Hypertensive emergency
  - Cerebral arterioles are unable to regulate blood flow to cerebral capillary beds leading to:
    - Cerebral edema and cerebral dysfunction
    - Papilledema
    - Cardiac failure
    - Uremia
    - Retinopathy
    - Cerebrovascular accident (CVA; stroke)
  - Severe HTN + acute target organ damage

Hypotension

- BP < 90/60 and symptomatic

Causes:

- Orthostatic hypotension
  - Decreased BP upon standing
    - Diagnostically, a decrease in SBP >20mmHg or decrease in DBP >10mmHg within 3 minutes of standing
  - Due to lack of BP compensation in response to gravity
  - Most common type of hypotension
- Hypovolemia
- Excessive vasodilation (usually peripherally)
- Drug induced
- Shock
  - Hypotension is common and may require treatment with vasopressors to prevent complications from ischemia
- Potential secondary disorders leading to hypotension include:
  - Endocrine disorders
  - Metabolic disorders
  - Diseases of the central or peripheral nervous system

Manifestations

Dizziness

Blurred vision or vision loss

Syncope

Venous Thromboembolism (VTE)

- Occlusion within venous bed by: clot (thrombus, embolus)

Pathophysiology:

- **Virchow triad** (Vascular damage, Hypercoagulability, Circulatory stasis)
  - Two key states of circulatory stasis are post-operative patients (especially orthopedic surgery) and atrial fibrillation

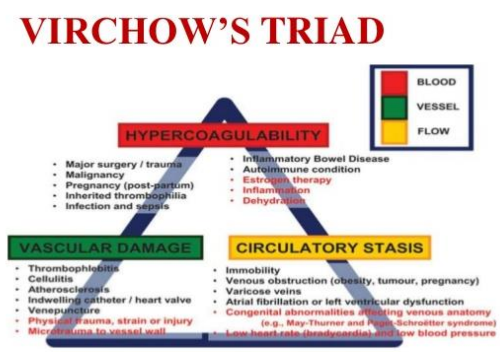


Figure 44: Virchow's Triad

  - Hypercoagulability
    - Inherited abnormalities that increase risk in the absence of other usual risk factors include ATIII deficiency, prothrombin mutations, as well as deficiencies in proteins C, S
    - Increased blood viscosity caused by eg. Polycythemia rubra vera

- *Main conditions associated with VTE:*
  - Deep vein thrombosis (DVT)
  - Pulmonary embolism (PE)

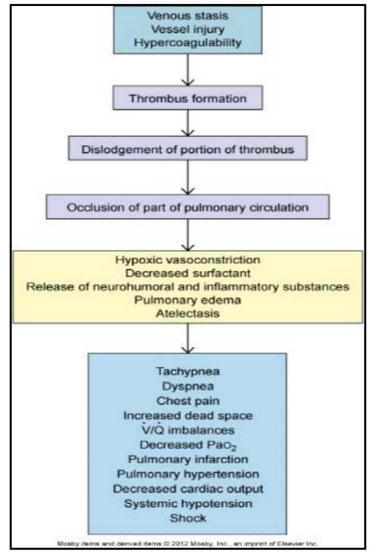


Figure 45: Pathway of pulmonary embolism

Clinical manifestations DVT:

- Swelling and pain in calf
- Homan's sign
  - Lift the knee and dorsiflex foot elicits pain in affected extremity

Clinical manifestations of PE:

  - Dyspnea, chest pain on inspiration
  - Tachycardia, cyanosis, hypotension

Diagnostic Findings:

- **Radiology**
  - Doppler Ultrasound for DVT
  - CT and pulmonary angiography, V/Q scan for PE
- **Laboratory Testing**
  - Elevated D-dimer
    - Rule-out criteria
    - A negative D-Dimer rules out PE, however, a positive D-Dimer does not confirm PE but instead indicates further testing

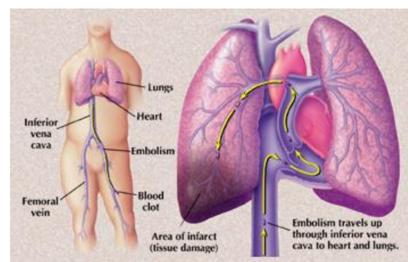


Figure 46: Pulmonary embolism

Varicose Veins

- Distended and tortuous (twisted and turning) veins leading to pooling of blood

Pathophysiology

- Valve damage caused by:
  - Trauma to saphenous veins damaging one or more valves
  - Gradual venous distention
    - Caused by the action of gravity on blood in the legs while standing
    - When a valve is damaged, a section of the vein is subject to increased pressure from a larger volume of blood leading to:
      - Engorged vein
      - Increased hydrostatic pressure and subsequent edema

Risk factors:

- Age
- Female gender
- Family Hx
- Obesity
- Pregnancy
- DVT
- Prior leg injury

7: Alterations of the Nervous System

Learning Objectives

- Describe the anatomy of the nervous system: Somatic and Autonomic divisions (review).
- Describe the sensory and motor functions of the peripheral nervous system (review).
- Explain the pathophysiology of traumatic brain injury
- Explain the pathophysiology of various spinal cord injuries and degenerative diseases of the spine (spinal shock, autonomic hyperreflexia, degenerative disk disease; spondylolysis, spondylolisthesis, and spinal stenosis)
- Explain the pathophysiology and types of CVA and aneurysms. Compare and contrast the types of CVA: *thrombotic stroke, transient ischemic attack (TIA), stroke-in-evolution, completed stroke, embolic stroke, and hemorrhagic stroke.*
- Explain the pathophysiology and types of headache
- Explain the pathophysiology of common CNS infections
- Explain the pathophysiology and clinical manifestations of multiple sclerosis
- Explain the pathophysiology and clinical manifestations Guillain-Barré Syndrome and Myasthenia gravis
- Explain the physiology of consciousness, arousal and pathophysiological impairment, describe the patterns of clinical manifestations (level of consciousness, pattern of breathing, vomiting, pupillary changes, oculomotor responses, and motor responses) that are used to evaluate the level of arousal and the differences between brain death and cerebral death
- Explain briefly the pathophysiology and types of data processing deficits, define and describe the following terms used to characterize specific data processing disorders: *agnosia*, *dysphasia*, aphasia, acute confusional states, and dementias.
- Explain pathophysiological basis and types of seizures, differentiate among the different types of seizures and seizure syndromes
- Explain the pathophysiology of Dementia and Alzheimer ‘s disease
- Characterize cerebral hemodynamics and stages of increased intracranial pressure and describe the four types of cerebral edema: vasogenic, cytotoxic, ischemic, and interstitial. Describe hydrocephalus
- Review nervous system infections and distinguish bacterial meningitis, viral meningitis, fungal meningitis and tubercular meningitis and central nervous system abscesses. Compare and contrast *meningitis* and *encephalitis*
- Describe extrapyramidal motor disorders: basal ganglia and cerebellar impairments.
- Explain the pathophysiology and clinical manifestations of Huntington's disease

- Explain the pathophysiology of Parkinson's disease and differentiate between the types of hypokinetic disorders: akinesia, bradykinesia, and loss of associated movement.
- Describe the alterations in movement & related terminology and compare upper and lower motor neuron lesions
- Explain briefly the pathophysiology of pain, temperature and sleep disorders
- Describe *nociception* (perception of pain), *nociceptors*, *pathways of nociception,* and *neuromodulation of pain, d*ifferentiate among *acute pain, chronic pain, somatic pain, visceral pain, referred pain, neuropathic pain, peripheral pain* and *central pain and* define *pain threshold* and *pain tolerance*
- Describe the process of normal thermoregulation and discuss briefly impairments of thermal regulation

Definitions: Alterations of the Nervous System

Term	Definition
Pain transmission via δ (delta) and C fibers	Pain transmitted along A-delta nerve fibres is sharp and well localized Pain transmitted along C nerve fibers is dull, aching, and poorly localized
ACUTE PAIN	
Acute somatic	Pain due to organic cause or pathology related to external structure
Acute visceral	Pain due to organic cause or pathology related to internal structure
DEGENERATIVE DISEASES OF THE SPINE	
Spondylolisthesis	Spinal vertebra slipping out of place
Spondylolysis	Structural defect in lamina or neural arch of vertebra (in pars interarticularis)
Herniated intervertebral disc	Displacement of nucleus pulposus or annulus fibrosus of vertebral disc
DISORDERS OF EXPRESSION	
Hypermimesis	Pathologic laughter or crying
Hypomimesis	Loss of emotion in language (arhapsody)
Dyspraxias/ Apraxias	Difficulty planning and executing coordinated motor movements
DISORDERS OF GAIT	

Basal ganglia/Senile gait	Stooped, hyperflexed posture with narrow base and short-stepped gait with decreased arm swing
Cerebellar gait	Wide base with feet often turned inward or outward for stability
Scissors gait	Legs adducted so they touch and then swing around each other
Spastic gait	Shuffling gait with one leg extended and the other held stiff, which can often lead to it being dragged
DISORDERS OF POSTURE/STANCE	
Dystonia	Maintenance of abnormal postures through muscular contractions
Decerebrate posturing	Extension outwards from body
Decorticate posturing	Flexion inwards towards 'core' of body
DYSSOMNIAS	
Insomnia	Poor or lack of ability to sleep
Obstructive sleep apnea (OSA)	Most common sleep disorder with complex pathology including hypoxia and oxidative stress
FRACTURES	
Simple	Single break in a vertebra that usually affects transverse or spinous process
Comminuted	Vertebral body shattered into several fragment
Compression	Vertebral body compressed anteriorly
Dislocation	Vertebral body slides on another
Nociception	Perception of pain
Nociceptors	Receptors for pain
PARASOMNIAS	
Somnambulism	Sleep walking
PROCESSES OF NOCICEPTION	
Transduction	Activation of nociceptors
Transmission	Conduction to dorsal horn and up via spinal cord

Perception	Awareness of pain
Modulation	Facilitation or inhibition of transmission before during, or after perception
Pain dominance	Pain at one location may cause an increase in the threshold in another location
Pain threshold	Point at which a stimulus is perceived as pain
Pain tolerance	The intensity or duration of pain that a person will endure before initiation of pain responses
Referred pain	Pain in an area distant from its point of origin

Review: Nervous System Anatomy and Physiology

Quick and critical facts:

- Sensation does NOT mean awareness
  - For a person to be aware of a sensation it must reach the cortex
- There is typically no awareness of visceral (those from organs) sensations
- There is awareness of somatic sensations
  - I.e. those coming from environment. General (pain, temp, touch). Special (vision, smell, sound, equilibrium)
- The somatic nervous system is voluntary and consists of nerves and skeletal muscles
- The *autonomic nervous system* (ANS) is involuntary and consists of autonomic nerves and:
  - Smooth muscle (walls of viscera and glands)
  - Cardiac muscle

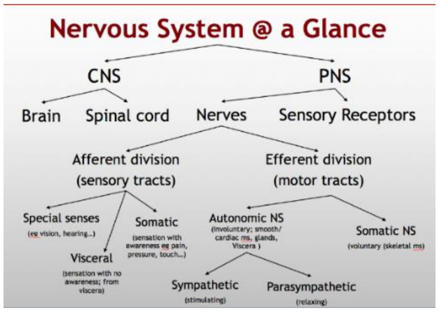


Figure 1: Divisions and structure of the nervous system

- The ANS is further divided into sympathetic and parasympathetic branches
  - The sympathetic branch stimulates the body in preparation for action
    - Also known as the "*fight or flight*" response
  - The parasympathetic branch relaxes muscles to promote repair and restoration
    - Also known as "*rest and digest*"

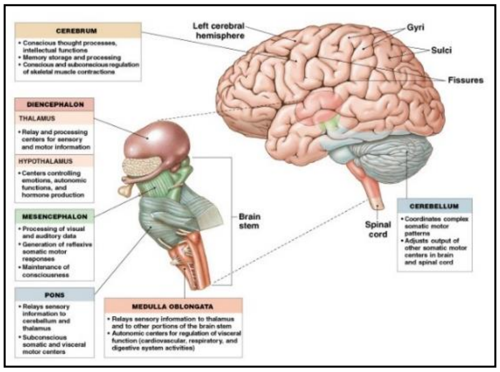


Figure 2: Anatomy of the brain and brainstem

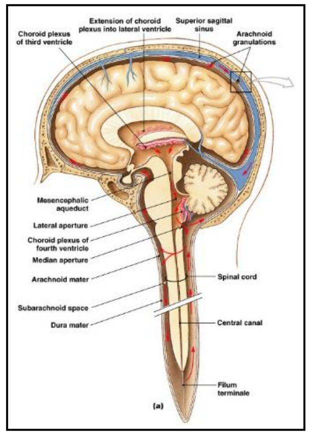


Figure 3: Anatomy of brain and brainstem

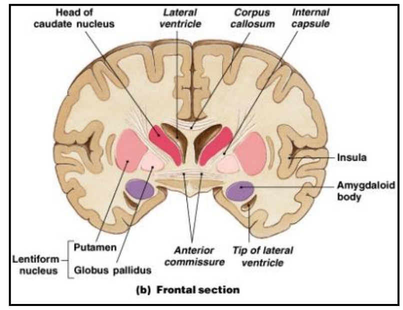


Figure 4: Frontal cross of brain

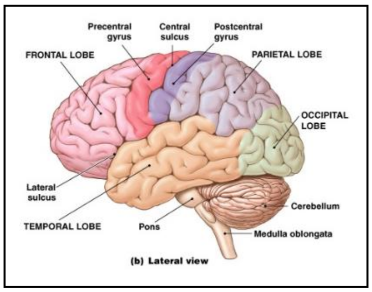


Figure 5: Surface anatomy of the brain

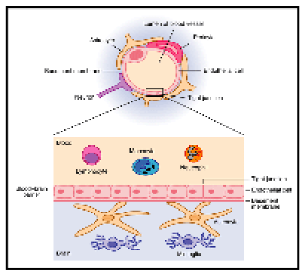


Figure 6: Cells of the nervous system and blood brain barrier

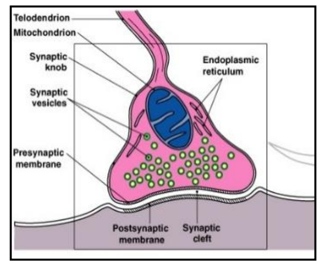


Figure 7: Neuronal synapse

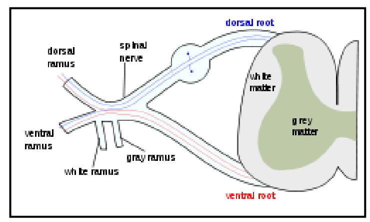


Figure 8: Peripheral nerves

- The nervous system is composed of gray and white matter in different configurations depending on the location within the nervous system
    - Within the brain, grey matter is on the outside and white matter on the inside, while in the spinal cord this is the reverse (see Figures 4, 11)

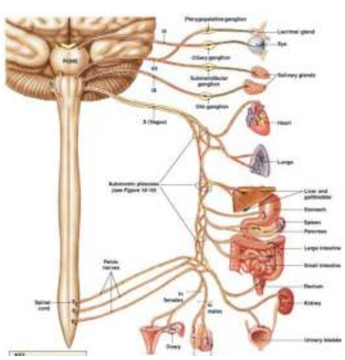


Figure 9: Craniosacral regions of the autonomic nervous system

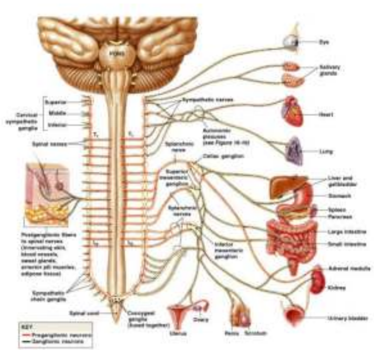


Figure 10: Thoracolumbar region of the autonomic nervous system

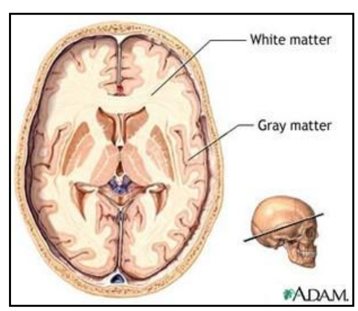


Figure 11: Brain cross section showing white and grey matter

Table: Configurations of Grey and White Matter of the Nervous System by Location		
Region of Nervous System	**White Matter (myelinated)**	**Grey Matter (unmyelinated)**
Brain	*Inside*	*Outside*
Spinal Cord	*Outside*	*Inside*

- SAME DAVE is the mneumonic that can be used for remembering the relationships and locations between nerves and the spinal cord
    - **SAME**: Sensory-Afferent; Motor-Efferent
    - **DAVE**: Dorsal-Afferent; Ventral-Efferent
- Ependymal cells line the ventricles and the choroid plexus
    - Responsible for cerebrospinal fluid (CSF), approximately 150mL, in circulation

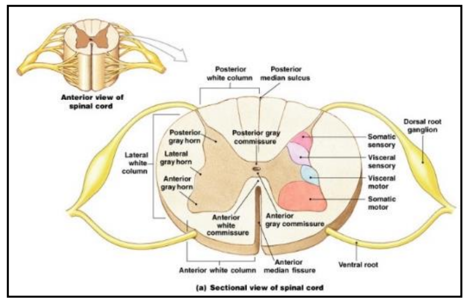


Figure 12: Relationship between central and peripheral nervous system in the spinal cord

Disorders of the Central and Peripheral Nervous Systems and the Neuromuscular Junction

Traumatic Brain Injury (TBI)

- Brain injury is **leading cause of death** and disability for Canadians under the age of 40
- ~1.5 M Canadians live with the effects of an acquired brain injury
- The annual incidence of ABI is > that od MS, SCI, HIV/AIDS and Breast Cancer combined!

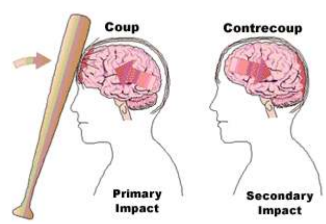


Figure 13: Coup versus contrecoup impact in TBI

Etiology

Caused by any accident in which head trauma occurs

- Damage can be at the site of impact (coup) or opposite (contrecoup) from recoil, and often both occur as the brain impacts the interior of the cranium
- Common examples include: Transportation (vehicle and pedestrian collisions), falls (particularly in older adults), sports-related, violence

Types of trauma:

- Closed (blunt, non-missile) trauma

- Open (penetrating, missile) trauma

Pathophysiology:

- Brain hematoma develops following trauma
- May be located in one of 3 spaces between meningeal layers
  - Epidural
  - Subdural
  - Intracerebral

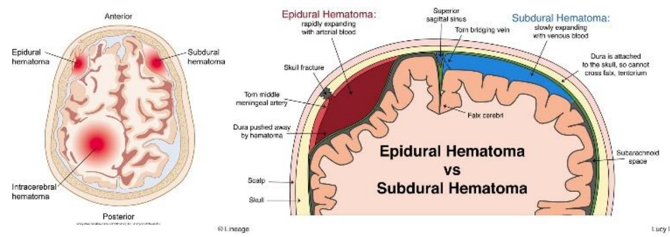


Figure 14: Types of hematoma by location (epidural, subdural, intracerebral)

- Focal or diffuse neuronal injury occurs
  - As blood accumulates, there is increased pressure within the cranium which damages neurons to produce clinical symptoms
  - Clinical symptoms are dependent on the location of injury
- *Focal injury* is localized to specific brain regions area
- *Diffuse neuronal (axonal) injury (DAI)* occurs when there is a twisting of the brain within the cranium producing shearing forces that effectively shred the axons themselves
  - DAI generally produces more severe symptoms and prognosis

Concussion

- Mild TBI causing alteration in brain functions +/- loss of consciousness
- Common causes include (blunt head trauma, car crash, physical assault, falls)

Clinical manifestations:

- **Mild concussion:**
  - Immediate but transient clinical manifestations; 1 to several minutes, possibly with amnesia
- **Classic cerebral concussion**
  - Loss of consciousness < 6 hours
  - Amnesia with confusional state lasting hours to days

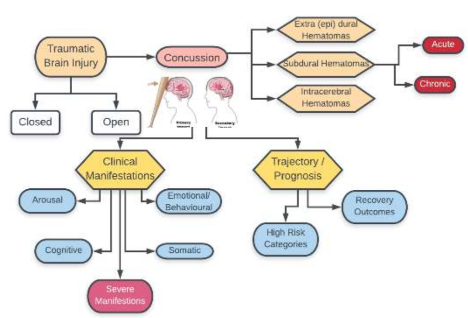


Figure 15: Classifying TBI and manifestations

- **General manifestations of concussion**
  - Headache
  - Sleep disturbance
  - Nausea and/or vomiting
  - Blurred vision
  - Attention impairment

- Reduced processing speed
- Drowsiness
- Emotion/behavior changes
- Posttraumatic seizure
    - If the patient is going to have a seizure it will usually occur within the first 24h following the traumatic event

Prognosis

- Symptoms are usually transient, with peak of symptoms occurring in the first 18-36 hours
- May experience full recovery or a range of residual impairments from mild (i.e. increased sensitivity to sound or light) to severe (i.e. permanent cognitive or physical impairment)

Fractures of the Spine

Types:

- *Simple fracture*
    - Single break usually affects transverse or spinous process
- *Compression fracture*
    - Vertebral body compressed anteriorly
- *Comminuted (burst) fracture*
    - Vertebral body shattered into several fragment
- *Dislocation*
    - Vertebral body slides on another

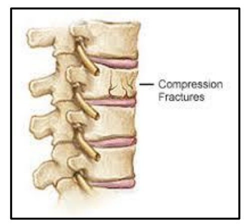


Figure 16: Compression fracture of the spine

Location:

- Most common fracture locations are:
    - Cervical (1, 2, 4-7)
    - T1-L2

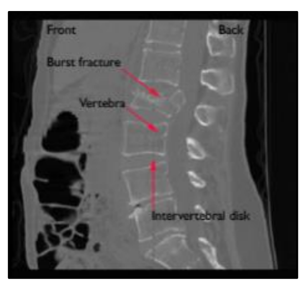


Figure 17: Comminuted (burst) fracture on x-ray

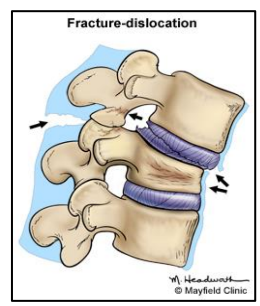


Figure 18: Dislocation fracture of the spine

Pathology and complications:

- Spinal cord injuries due to compression of spinal cord parenchyma, central canal and vascular structures

Spinal Cord Injuries (SCI)

Causes:

- Hyperextension injury
- Flexion injury
- Axial compression injury

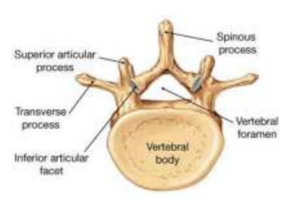


Figure 19: Anatomy of a vertebra

- Flexion-rotation injury

Pathophysiology:

- Hemorrhage in grey matter and pia-arachnoid region of the meninges
    - Increases in size until entire grey matter is hemorrhagic and necrotic
- Edema in white matter
    - Microcirculation block that reduces vascular perfusion to the region leading to ischemia and necrosis
    - Maximum pathology from hemorrhages and edema occur at the level of injury + 2 segments above and below
- It takes 24 h regain circulation in white matter and longer to gray matter
- Inflammation and healing start 36-48 h
- Collagen replacement/repair in 3-4 weeks

Spinal shock:

- Stopping of spinal cord activities at and below the level of injury
- Complete loss of reflex function (skeletal, bladder, bowel, thermal control, and autonomic control) below level of lesion
    - May last from a few days to 3 months with an average of 7-20 days
    - Ends with reappearance of reflex activity, hyperreflexia, spasticity, and reflex emptying of the bladder

Complications of spinal cord injuries

Paraplegia:

- Lesion in thoracic, lumbar, or sacral regions
- Impairment in motor, sensory functions of the lower limbs

Quadriplegia:

- Lesion at high level C1–C7
- Impairment in motor or sensory function of all limbs and torso

Degenerative Disorders of the Spine

Degenerative Disc Disease (DDD)

- Progressive disease common in individuals over the age of 30 related to structural defects from loss of cartilage in the spine leading to vertebral compression
- Both part of the normal aging process as well as having genetic component involving genes that code for cartilage in the spine
    - **Potential environmental causes include:**

- Decrease/ loss of blood supply
- Biochemical factors (e.g. inflammatory mediators)
- Biomechanical factors of the intervertebral disc tissue (e.g. mechanical loading and compression of vertebra)

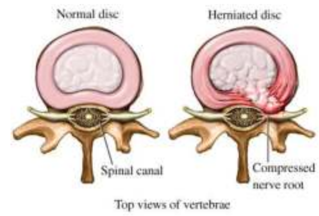


Figure 20: Comparison of normal and herniated intervertebral disc

- Pathological findings include spondylolysis, spondylolisthesis, disc protrusion (herniation), and spinal stenosis

Spondylolysis:

- Structural defect in lamina or neural arch
- Defect in pars interarticularis (small segment of bone connecting facet joints in the spine)
- Sports related, common in L5
- Cause pain, reduced mobility
- Can progress to *spondylolisthesis* (vertebra slipping out of place)

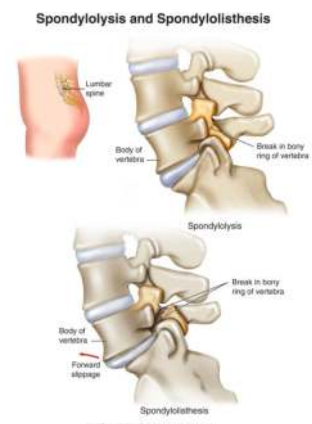


Figure 21: Spondylolysis and spondylolisthesis of the lower spine

Herniated intervertebral disc:

- Displacement of nucleus pulposus or annulus fibrosus
- **Clinical manifestations**:
  - Pain, paresthesia

Spinal stenosis:

- **Causes**:
  - Herniated disk, trauma, tumor
- **Pathophysiology**:
  - Abnormal narrowing of spinal canal
  - Mostly cervical and lumbar
- **Clinical manifestations:**
  - Could be asymptomatic
  - Symptoms are typically bilateral
  - Discomfort/pain of the neck/ lower back
  - Numbness, tingling
  - Weakness of (upper/ lower limb)

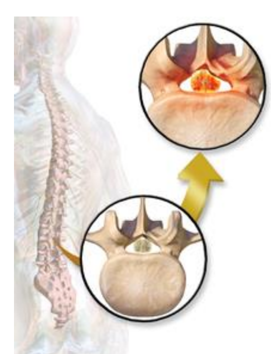


Figure 22: Spinal stenosis

Cerebrovascular Accident (CVA)

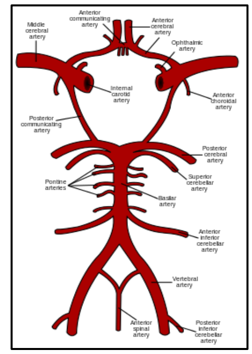


Figure 23: Cerebral vasculature

- Also known as stroke
- Leading cause of death and disability worldwide
- Loss of brain function due to impaired blood supply
- Brain vasculature = Circle of Willis (review)

Causes

- Majority caused by cerebral ischemia
  - *Thrombotic stroke*
    - Block by *thrombus* in a cerebral vessel directly leading to the brain
    - *Middle cerebral artery* is a major site
  - *Embolic stroke*
    - Block by *embolus* (e.g. thrombus fragments, air, fat) from other locations in the body's vasculature that migrates to the brain and block a cerebral vessel

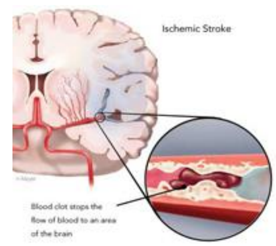


Figure 24: Ischemic stroke

- Another, less common, cause is cerebral hemorrhage (*hemorrhagic stroke*):
  - Due to trauma or ruptured aneurysm
  - A medical emergency

Risk factors:

- Hypertension, DM, high cholesterol, smoking, atrial fibrillation, previous stroke (or transient ischemic attack)
- Consistent with risk factors for myocardial infarction

Pathophysiology:

- *Review Atherosclerosis - lecture 6*

Clinical manifestations:

- Depends on site and extent of occlusion. potential symptoms include:
- Transient ischemic attacks (TIAs)
- Altered level of consciousness, confusion
- Unilateral motor, sensory impairment
- Aphasia (failure to understand or formulate speech)
- Vision impairment of one side of the visual field
- Kernig sign
- Brudzinski sign

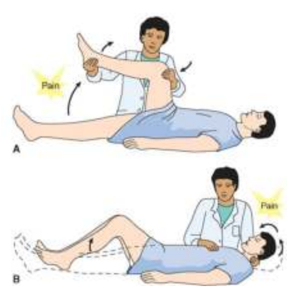


Figure 25: Kernig (A) and Brudinski (B) Signs

Intracranial Aneurysm

- Localized, blood-filled bulge in the blood vessel wall

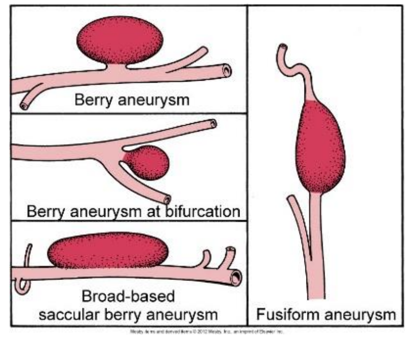


Figure 26: Types of aneurysms

Cause:

- Weakened vessel wall
    - May be congenital or acquired
    - Any blood vessel can be affected

Examples:

- Common sites for intracranial aneurysms within the Circle of Willis:
    - Anterior communicating artery (35%)
    - Internal carotid artery (30%-including the carotid artery itself, the posterior communicating artery, and the ophthalmic artery)
    - Middle cerebral artery (22%)
    - Posterior circulation sites, most commonly the basilar artery tip.
- Outside the brain, Aorta is a common site

Complications:

- Bulge may increase in size leading to rupture
    - Characterized by a "thunder clap" headache
- Continuous bleeding into subarachnoid space leading to hemorrhage
    - Person may get sentinel 'warning' headaches related to increase in the size of the bulge and/or leakage of blood into subarachnoid space
    - May experience subsequent hypovolemic shock, death

Headache

- Pain anywhere in the region of the head or neck
- Non-specific symptom with many causes (mild benign & serious)

Pathophysiology:

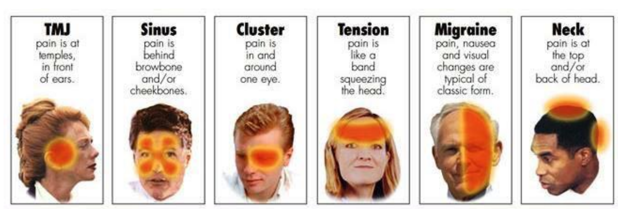


Figure 27: Types of Headache

- Many theories! Traction/irritation of meninges and spasm/ dilation of blood vessels, stimulating nociceptors
- Brain tissue lacks pain receptors, is insensitive to pain
- Pain originates from nearby pain-sensitive structures: periosteum, muscles, subcutaneous tissues, etc.

Headache - Migraine

- Familial, episodic disorder with headache as a marker
- **Characteristic features:**
  - Repeated, episodic headache lasting 4 to 72 hours
  - Usually women 25 - 55 y (More than men)
- **Causes**:
  - Genetic/environmental, neuronal dysfunction
- **Diagnosis**
  - Unilateral, throbbing pain, moderate or severe, worsened by movement
  - Associated with any one of the following:
  - Nausea, vomiting
  - Photophobia or phonophobia

Figure 28: Unilateral location of migraine headache

- **Triggers**:
  - Altered sleep patterns
  - Overexertion
  - Weather change
  - Stress or relaxation from stress
  - Hormonal changes (menstrual periods)
  - Excess afferent stimulation (bright lights, strong smells)
  - Skipping meals
  - Chemicals (alcohol or nitrates)

Headache - Cluster

- Chronic, in clusters (minutes to hours) for a period of days, followed by a long period of spontaneous remission
- Usually men 20 - 50 y
- **Cause**
  - Trigeminal activation and autonomic dysfunction
- **Associated clinical features**:
  - Tearing on affected side, ptosis of the ipsilateral eye, stuffy nose

Headache - Tension

- Most common
- **Cause**:

- Stress and fatigue (most common causes)
- Muscular tension

- **Features**:
  - Average onset 2nd decade
  - Mild to moderate bilateral headache with a sensation of a tight band or pressure around the head with gradual onset of pain
  - Occurs in episodes and may last for several hours or several days

*Infection/Inflammation of the CNS

Meningitis

- Inflammation of meninges
  - Bacterial meningitis
  - Non- bacterial, or aseptic (viral, non-purulent) meningitis

Encephalitis

- Inflammation of brain tissues
- Commonly caused by arthropod-borne viruses and herpes simplex virus

Meningo-encephalitis

- Inflammation of both the meninges and brain tissues

Infection/Inflammation of the CNS - *Abscess*

- Localized collection of pus within the parenchyma
  - Divided into three classifications:
    1. **Extradural**: Associated with osteomyelitis of the cranial bone
    2. **Intracerebral**: Arising from a vascular source
    3. **Spinal**: Located within the spinal cord
  - Pus may collect anywhere within the nervous system and lead to manifestations as it compresses nearby nerves
- **Causes**:
  - Open trauma
  - Post neurosurgery
  - Spread from nearby septic foci
    - Primarily the middle ear, mastoid air cells, nasal cavity, nasal sinuses
  - Hematogenous spread from distant foci

Clinical Manifestations – Brain abscess

- Early:
  - Low-grade fever

- Headache (most common symptom)
- Neck pain/ stiffness (more often in extradural abscesses)
- Confusion
- Drowsiness
- Sensory/communication deficits

▪ Late:
  - Distractibility
  - Memory deficits
  - Visual impairment
  - Ataxia
  - Dementia

Spinal cord abscess

▪ Severe pain, spasms of the back muscles and limited movement
▪ Progressive compression symptoms
▪ Paralysis

DEMYELINATING DEGENERATING DISORDERS

Multiple Sclerosis (MS)

▪ Demyelinating disorder of the **CNS**
▪ Acquired, progressive, autoimmune-inflammatory disorder
▪ Cause not totally understood, antibodies that destroy myelin sheath

Pathophysiology:

▪ Inflammation/autoimmune response targeting myelin sheaths of neurons
  - A previous viral illness in a susceptible individual, leads to inflammation
  - Early inflammation and demyelination lead to irreversible damage and scarring (sclerosis)

▪ *MS Lesions*: Scar-like plaques that develop throughout the CNS
  - They can appear in wherever there is localized demyelination (plaques)
  - Can be focal or diffuse
  - Can occur anywhere within in the brain, spinal cord, white, and/or grey matter

▪ There is thinning, then complete loss of myelin and breakdown of axons
  - Leads to impaired conduction of electrical signals

▪ Re-myelination takes place but becomes successively less effective with time, aggravating symptoms

Clinical Manifestations

- Onset usually between 20-40 years
- More common in women, however, males often have a more aggressive course
- Phenotypes (based on the clinical course):
    - Relapsing-remitting (90% initial representation)
        - Without treatment these persons will progress to the progressive types
    - Primary progressive (PPMS)
    - Secondary progressive (SPMS)
    - Progressive relapsing
- Symptoms include:
    - Paresthesia
    - Weakness
    - Impaired gait
    - Visual disturbances
    - Urinary incontinence

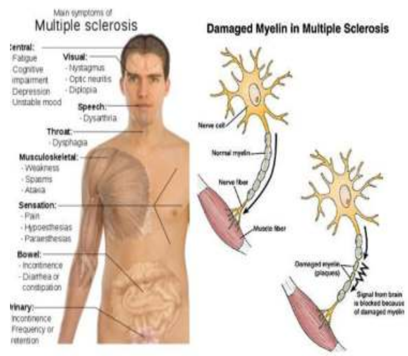


Figure 29: Pathophysiology and manifestations of multiple sclerosis

Guillain-Barré Syndrome

- Demyelinating disorder of the peripheral nervous system (LMN)

Pathophysiology:

- Days or weeks after flu-like illness
    - Most often respiratory or gastrointestinal viral illness
    - E.g. *Campylobacter jujini*
- Acute, autoimmune inflammatory polyneuropathy
    - Axonal demyelination as antibodies form to attack the myelin covering peripheral nerves
    - Acute onset of motor, sensory, or autonomic symptoms

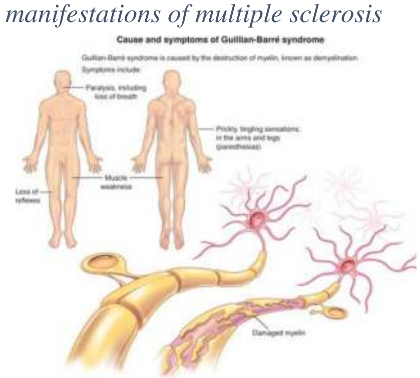


Figure 30: Causes, pathophysiology, and manifestations in Guillain-Barre syndrome

Clinical manifestations:

- Symptoms are related to antibody subtypes
- Most common form is ascending motor paralysis:
    - Rapid-onset of muscle weakness starts at lower limbs
    - Demyelination works upwards and can affect the lungs leading to cessation of breathing
    - It can affect respiratory muscles causing respiratory failure (~25% of cases)

- Loss of deep tendon reflex
- Blood pressure fluctuations and irregularities in the heartbeat

Myasthenia Gravis

- A disorder of the neuromuscular junction (NMJ) whereby autoantibodies damage/destroy acetylcholine (Ach) receptors affecting transmission of nerve impulses to the muscle
- Acquired chronic autoimmune disease

Pathophysiology:

- Circulating IgG against Ach receptors
  - Inhibit the excitatory effects of Ach on nicotinic receptors (post synaptic) at NMJ.
  - No nerve impulse transmission and subsequently excitation of the muscle
- Targets the head and neck
- Usually unilateral

Clinical Manifestations:

- Progressive weakness and fatigue
- Affects muscles of eyes & throat causing diplopia and difficulty chewing, talking, swallowing
- Manifestations of other associated autoimmune diseases

8: Alterations in Cognitive Systems, Cerebral Hemodynamics and Motor Function

Consciousness

- State of awareness of oneself and the surrounding environment

Clinical assessment:

- **Simple procedure**:
    - Checking ability to move and react to physical stimuli
    - Response in a meaningful way to questions and commands
    - Orientation to person, place, time
        - Identify name, current location, and current day and time
    - Check Level of consciousness (LOC)
- **Complex procedure**:
    - Glasgow Coma Scale, based on:
        - Eye response
        - Verbal response
        - Motor response
    - Gives cumulative score ranging from 3-15 with 3 being unconscious and 15 normal cognitive function

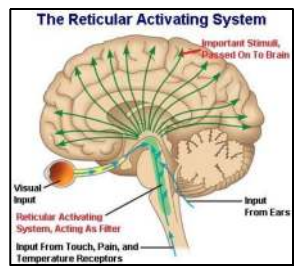


Figure 31: Reticular activating system (RAS)

Alterations in Cognitive Systems - Arousal & Awareness

Arousal: being awake or reactive to stimuli

Awareness: encompasses all cognitive functions

Clinical assessment & impairment

- Five categories for assessment:
    1. Level of consciousness
        - Awake = intact respiration, endocrine, RAS (The reticular activating system) which works with the limbic system (emotions) to coordinate the brain with the environment
    2. Pattern of breathing
    3. Pupillary changes & eye responses
    4. Oculomotor responses
    5. Motor responses
        - Vomiting, yawning, hiccups

Acute Confusional States (ACS) - *Delirium*

- Transient disorder of awareness that result from cerebral dysfunction

Causes:

- Secondary to drug intoxication, metabolic disorder, or nervous system disease
- Common in post-op patients
- Increased in elderly population

Types

- Hyperkinetic – They pull IV lines, get out of their hospital bed
- Hypokinetic – Flaccid, no response to anything

Brain Death

- Brain can no longer maintain internal homeostasis
  - Absence of functioning in brain including the brainstem and cerebellum
  - Condition observed for an extended period of time

Brain death criteria:

- Established etiology capable of causing neurological death in the absence of reversible conditions capable of mimicking neurological death
- Unresponsive coma with bilateral absence of motor responses, excluding spinal reflexes
- No brainstem function
- No spontaneous respiration (apnea)
- Absence confounding factors
  - Occurs because of known structural defect or metabolic disease

Cerebral Death

- Cerebral death (irreversible coma) is death of the cerebral hemispheres **exclusive of the brainstem and cerebellum**
- No behavioural or environmental responses
- The brain can continue to maintain internal homeostasis
- Survivors of cerebral death:
  - Remain in coma
  - Emerge into a persistent vegetative state
  - Progress into a minimally conscious state (MCS)
  - *The patient will never regain behavioural responses to stimuli*

Seizures

- Sudden (definitive feature), transient, excessive electric activity within the brain (neuron firing)
- Can be motor, sensory, autonomic, or psychic in nature
- Can be partial (focal) or generalized

Etiology

- Etiologic factors are wide and diverse in range
- *Epilepsy*:
  - A specific chronic CNS disease of abnormal brain activities characterized by ≥ 2 unprovoked seizures to make a diagnosis
  - Seizure activity with no underlying, diagnosable cause
  - Affects both males and females of all races, ethnic backgrounds and ages.
  - Symptoms can vary widely from simply stare blankly for a few seconds during a seizure, repeated twitches of arms or legs or severe lasting ones
- **CNS lesions, including:**
  - Meningitis
  - Multiple sclerosis (MS)
  - Tetanus
  - Trauma
- **Biochemical disorders including:**
  - Elevated bilirubin in children
  - Fever
  - Encephalitis
  - Electrolyte imbalance
  - Hypoglycemia
  - Lead poisoning

Manifestations

- Loss of consciousness
- Sudden loss of autonomic control
  - Incontinence is common and can be embarrassing for the person when they recover
- *Convulsions*: Jerky (rapid and repeated) contract-relax body movements
  - Also called *tonic-clonic* seizures

- *Aura*: Perceptions experienced a few seconds-hours before seizure (or migraine) begins
    - **Examples**: strange light, unpleasant smell or confusing thoughts
- A seizure is **NOT** a disease, it is a manifestation within a larger disorder

Data Processing Deficits

- The brain is responsible for receiving, processing, and interpreting data from sensors throughout the body
    - Brodmann's cytoarchitectural map identifies, generally, specialized areas of the brain used for receiving and processing this data

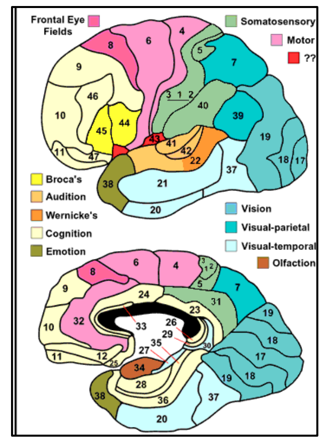


Figure 32: Brodmann's cytoarchitectural map

- *Agnosia*: Inability to process sensory information and recognize form and nature of objects
    - Poor recognition
    - May be tactile, visual, auditory, etc., or a combination depending on the areas of the brain impacted
- *Dysphasia*: Impairment of production or comprehension of language
    - Reduced ability to speak, read, or write
    - Classified both anatomically and functionally including:
        - Expressive dysphasia
        - Receptive dysphasia
        - Transcortical dysphasia
            - Person is able to repeat (echolalia) and recite with fluent speech, however comprehension is impaired, and speech makes no sense
            - They are unable to read or write in their own language
    - *Aphasia:* Severe form of dysphasia whereby the person is unable to communicate
        - Example: A person can see the glass in their hands, they know what it is, but they cannot verbally tell you they are holding a glass

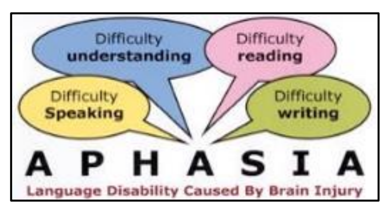


Figure 33: Types of aphasia

Dementia

- **Progressive** failure of cerebral functions that is not caused by an impaired level of consciousness
- Often includes decreased functioning and losses in:
    - Orientation
    - Memory
    - Language

   - Judgment
   - Decision making
- As intellectual function declines, the person will exhibit behaviour changes becoming more repetitive, obsessive, and decline in social functioning

Causes:

- Alzheimer disease
   - Most famous (see below)
- Vascular dementia
- Dementia with Lewy bodies (DLB)

Mechanisms:

- Neuron degeneration
- Brain tissue compression
- Atherosclerosis
- Brain trauma
- CNS infections
   - Such as HIV and slow-growing viruses associated with Creutzfeldt-Jakob disease
- Genetic predisposition associated with neurodegenerative diseases

Alzheimer Disease (AD)

- Structural changes in the brain leading to dramatic decline in intellectual functions.

Etiology

- Unknown, however there is a genetic link to Chromosome 21 in those with early onset AD and chromosome 19 in those with late onset

Pathophysiology

- Premature neuron death and cellular changes, characterized by:
   - *Amyloid plaques* accumulates in extracellular spaces
   - *Neurofibrillary tangles* (twisted bundles of filaments) intracellularly
   - Intracellular vacuoles leading cellular swelling and death with shrinkage of brain matter

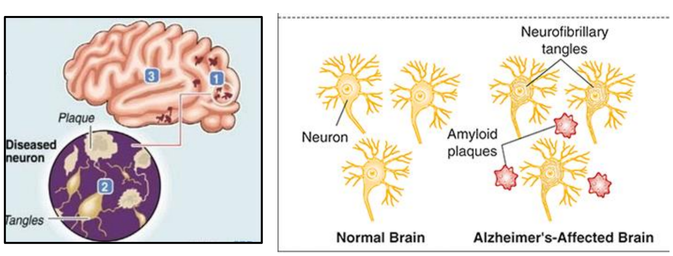


Figure 34: Plaques and tangles implicated in Alzheimer's disease

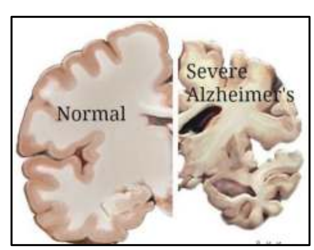


Figure 35: Comparison of normal brain to that of person with severe Alzheimer's

Theories

- Mutation in genes coding amyloid precursor protein
- Alteration in apolipoprotein E (ApoE)
- Impairment in choline acetyltransferase

Clinical manifestations:

- Forgetfulness
- Emotional upset
- Disorientation
- Confusion
- Lack of concentration
- Decline in abstraction, problem solving, and judgment
- Rule-out diagnosis
    - Diagnosis is made by ruling out other causes of dementia
    - The only definitive diagnosis can be made on autopsy

Cerebral Hemodynamics

Basics

- *Cerebral blood flow* (CBF): Amount of blood flow to the brain
    - Maintained at a rate required to meet metabolic needs
    - Constitutes 15% of cardiac output
    - Calculated by the formula CBF = CPP/CVR
- *Cerebral perfusion pressure* (CPP): Pressure required to perfuse cells of brain
    - Calculated by the formula **CPP =MAP-ICP**
- *Intracranial pressure* (ICP)
    - Normally 5 to 15 mmHg

Increased Intracranial Pressure (IICP)

- Caused by increase in intracranial content
    - For example, tumor, edema, excessive CSF, or hemorrhage

Stages:

- **Stage 1**
    - Minimal increase in ICP
    - Usually compensated
- **Stage 2**

- Systemic arteriolar vasoconstriction to increase MAP and CPP and maintain neuronal oxygenation

- **Stage 3**
  - Sustained increased ICP approaching MAP

Cerebral Edema

- Increase in the fluid (intracellular or extracellular) within the brain

Causes:

- Brain trauma
- Non traumatic causes
  - Ischemic stroke
  - Cancer
  - Inflammation (meningitis/encephalitis)

Pathophysiology and Types:

Vasogenic:

- Disruption of blood brain barrier (BBB)
- Intravascular proteins and fluid escape to brain parenchyma

Cytotoxic:

- BBB remains intact, toxins impair cellular metabolism

Interstitial:

- Disruption of Blood-CSF barrier
- CSF flows to interstitial space

Hydrocephalus

- Abnormal accumulation of CSF in the ventricles, subarachnoid space within the brain

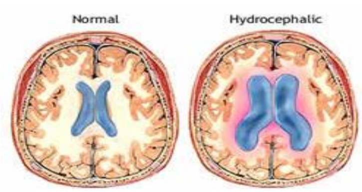


Figure 36: Comparison normal brain (right) to brain with hydrocephalus (left)

Pathophysiology and features:

- Interference in CSF flow
- Increased intracranial pressure
- Head enlargement
- Impairment of cognitive functions
  - Related to increased compression on neuronal axons and decreased oxygen perfusion and ICP increases

Treatment

- Shunt inserted to drain fluid

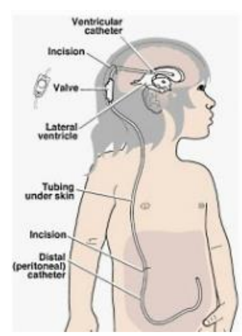


Figure 37: Shunt used to drain fluid in hydrocephalus

8: Alterations in Movement

Alterations in Movement - Terminology

Upper motor neuron syndromes:

- *Hemiparesis* or *hemiplegia*
- *Diplegia*
- *Paraparesis* or *paraplegia*
- *Quadriparesis* or *quadriplegia*

Lower motor neuron syndromes:

- *Flaccid paresis* or *flaccid paralysis*
- *Hypertonia* and *hypotonia*
- *Hyperkinesia*
    - Excessive, purposeless movement
- *Paroxysmal dyskinesias*
    - Abnormal, involuntary movements that occur as spasms
- *Tardive dyskinesia*
    - Involuntary movement of face, lip, tongue, trunk, and extremities
    - Usually a side effect of antipsychotic medication
- *Hypokinesia*, *Akinesia*, Bradykinesia
    - Slowed or absent movement

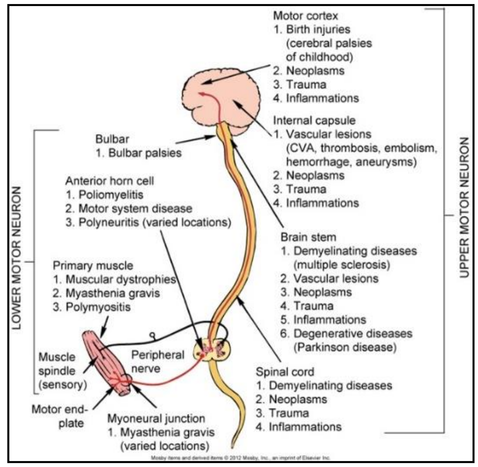


Figure 38: Motor function syndromes

Huntington Disease

- Autosomal dominant degenerative disorder

Pathophysiology:

- CAG repeats in *Huntington gene* located on chromosome 4
- There is depletion of gamma-aminobutyric acid (GABA) (an inhibitory transmitter)
    - Leads to increase in involuntary movement
- Severe degeneration of the basal ganglia (caudate nucleus) and cerebral cortex

Clinical manifestations:

- ***Chorea*** **(characteristic manifestation)**
    - Non-repetitive muscular contractions, usually in the extremities of the face
    - Random, irregular pattern of rapid, involuntary muscular contractions
    - Decreases with sleep and rest
    - Increases with emotional stress and attempted voluntary movement

- Related to changes occurring at the level of the basal ganglia
- General restlessness, lack of coordination, involuntary eye movements
- Subtle changes in personality, emotion, cognition abilities

Parkinson Disease

- A common chronic and progressive neurodegenerative disorder

Pathophysiology:

- Severe degeneration of the basal ganglia
- Diminished dopamine released by substantia nigra
- Thalamus and globus palladium become overactive resulting in motor impairments

Clinical Manifestations:

- Parkinsonian tremors, rigidity, bradykinesia
- Postural disturbances
- Autonomic and neuroendocrine symptoms
- Cognitive-affective symptoms

Disorders of Posture, Gait & Expression - Terminology

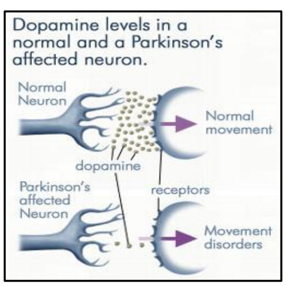


Figure 39: Dopamine levels in normal neurons versus those of one with Parkinson's disease

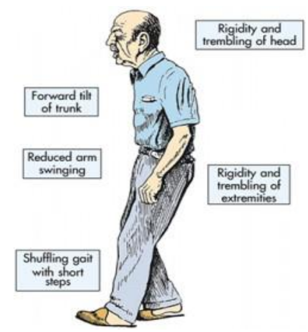


Figure 40: Classic Parkinsonian gait and stance

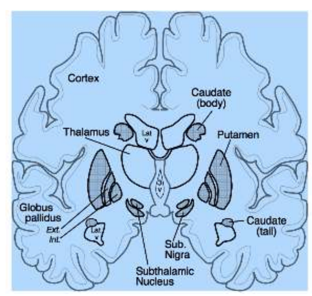


Figure 41: Brain structures impacted in Parkinson Disease

Disorders of Posture (Stance):

- *Dystonia*:
  - Maintenance of abnormal postures through muscular contractions
  - Dystonic and associated movements
    - o Dystonic movements last seconds while dystonic postures are held for longer periods
- *Decorticate*:
  - Flexion inwards towards 'core' of body
- *Decerebrate*:
  - Extension outwards from body

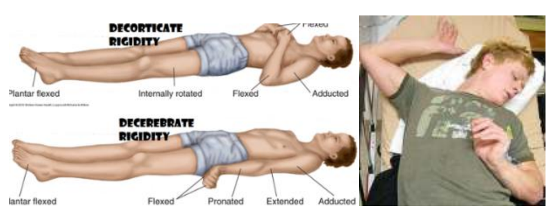


Figure 42: Decorticate and decerebrate posturing

- *Basal ganglion*
- *Senile*:
    - Increased flexed posture similar to basal ganglion

Disorders of Gait

- *Spastic gait*:
    - Shuffling gait with one leg extended and the other held stiff, which can often lead to it being dragged
    - Associated with unilateral injury
- *Scissors gait*:
    - Legs adducted so they touch and then swing around each other
    - Associated with bilateral injury and spasticity
- *Cerebellar gait*:
    - Wide base with feet often turned inward or outward for stability

Figure 43: Disorders of gait (spastic, scissors)

- *Basal ganglion gait*:
    - Stooped, hyperflexed posture with narrow base and short-stepped gait with decreased arm swing
- *Senile gait*:
    - Similar to the basal ganglion gait

Disorders of Expression:

- *Hypermimesis:*
    - Pathologic laughter or crying
- *Hypomimesis:*
    - Loss of emotion in language (arhapsody)
    - Also, receptive arhapsody where the person is unable to understand emotional speech and facial expressions
- *Dyspraxias and apraxias:*
    - Difficulty planning and executing coordinated motor movements
        - Examples of tasks include speaking, writing, using tools/utensils, playing sports, following instructions, and the ability to focus
    - May be developmental or associated with vascular disorders (common in stroke), trauma, infection, tumors, degenerative disorders

8: Pain, Temperature, Sleep, and Sensory Function

Pain

- *An an unpleasant sensory and emotional experience associated with actual or potential tissue damage or described in terms of such damage

Neuroanatomy of Pain

- ***Nociceptors***
  - Receptors for pain
  - Free nerve endings in skin, muscle, joints, arteries, viscera
  - Can also detect chemical, mechanical, thermal stimuli
- **Pathways of nociception**:
  - Located in the spinothalamic tract
  - **Processes required for nociception**:
    - *Transduction*: Activation of nociceptors
    - *Transmission*: Conduction to dorsal horn and up via spinal cord
    - *Perception*: Awareness of pain
    - *Modulation*: Facilitation or inhibition of transmission before during, or after perception
      - **Excitatory neuromodulators**
        - Substance P, glutamate
      - **Inhibitory neuromodulators**
        - GABA, glycine, serotonin, norepinephrine, *endorphin*

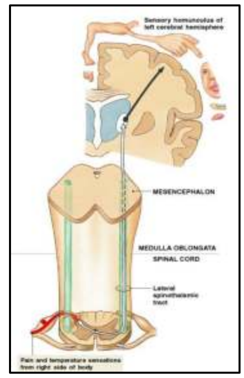


Figure 44: Pathways of nociception in the spinothalamic tract

Neuromodulation of Pain- *Endogenous Opioids*

- Short sequences of amino acids that bind to opioid receptors
  - Produce similar effect to opiates such as morphine
- **Types**:
  - *Endorphins*:
    - Block transmission of pain signals
    - Produce a feeling of euphoria
  - Enkephalins
  - Dynorphins
  - Endomorphins

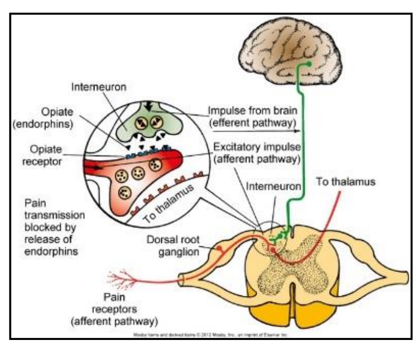


Figure 45: Endogenous opioids in the brain

Clinical Description of Pain

- *Pain threshold*:
  - Point at which a stimulus is perceived as pain
- *Perceptual dominance*:
  - Pain at one location may cause an increase in the threshold in another location
- *Pain tolerance*:
  - Duration of time or the intensity of pain that a person will endure before initiation of pain responses

Types/ Categories of Pain

Psychogenic pain

- Physical pain that is caused, increased, or prolonged by mental, emotional, or behavioral factors

Referred pain:

- Pain in an area distant from its point of origin
- Area of referred pain is supplied by the same spinal segment as the actual site eg myocardial infarction pain

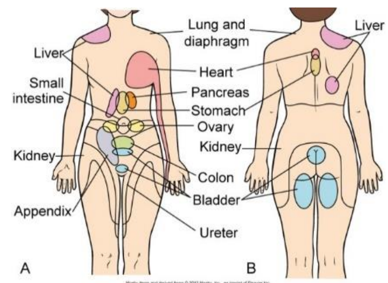


Figure 46: Referred pain locations

Acute Pain:

- Protective mechanism
- Alerts an individual to a condition or experience that is harmful to the body
- **Acute somatic**
  - Arises from skin, joints, and muscles
  - Pain is transmitted along A-delta and C-fibres
    - *A delta fibres*: pain is sharp and well localized
    - *C fibres*: dull, aching, and poorly localized
- **Acute visceral**
  - Pain in the internal organs and lining of body cavities
  - Transmitted by C fibres:
    - Poorly localized with an aching, gnawing, throbbing, or intermittent cramping quality
  - Often radiates or is referred

Chronic Pain

- May be persistent or intermittent
- Usually defined as lasting at least 3 to 6 months

- Response patterns vary
- Produces significant behavior and psychologic changes
- Usually no physiological signs as in acute
- **Common types**:
    - *Myofascial pain syndromes*:
        - Injury to the muscle and fascia with spasm, tenderness, and stiffness
    - *Chronic postoperative pain*:
        - Pain persisting beyond expected recovery time
    - *Cancer pain*:
        - Pain occurring from the tumor, procedure, treatment, infection
        - Requires different treatment approach than other types of chronic pain
        - Opioids were initially designed for treatment of cancer pain
    - *Neuropathic pain*: resulting from trauma or nerve disease (non-nociceptors)
        - **Peripheral**: e.g. diabetic neuropathy
        - **Central**: e.g. phantom limb (following amputation)

Temperature Regulation

- *Homeostasis*: Maintenance of a ‘steady state’ within the body
    - The body regulates temperature to maintain homeostasis (review these mechanisms from previous lectures)
- Age is a key factor affecting temperature regulation, particularly in the very young and old
    - **Pediatrics**:
        - Children produce sufficient body heat, but are unable to conserve heat produced due to:
            - Small body size and high body surface to weight ratio
            - Thin subcutaneous layer
        - Brown-fat contributes in increased heat production in cold temperatures
    - **Aging**
        - Slow blood circulation, vasoconstrictive response, and decreased metabolic rate
        - Decreased sweating, shivering, and perception of heat and cold

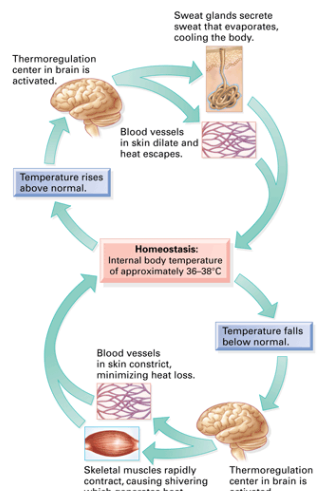


Figure 47: Homeostatic processes in temperature regulation

Mechanisms of Temperature Regulation

- **Heat Production and Conservation**
    - Processes regulated by the hypothalamus and include:
        - Chemical reactions of metabolism
        - Skeletal muscle contraction
        - Chemical thermogenesis
        - Vasoconstriction
        - Voluntary mechanisms
- **Mechanisms of Heat Loss:**
    - Radiation
    - Conduction
    - Convection
    - Vasodilation
    - Decreased muscle tone
    - Evaporation
    - Increased respirations
    - Voluntary measures
    - Adaptation to warmer climates

TABLE 13-5 MECHANISMS OF HEAT PRODUCTION AND HEAT LOSS

CONDITION	DESCRIPTION
Heat Production	
Chemical reactions of metabolism	Occur during ingestion and metabolism of food and while maintaining body at rest (basal metabolism); occur in body core (e.g., liver)
Skeletal muscle contraction	Gradual increase in muscle tone or rapid muscle oscillations (shivering)
Chemical thermogenesis	Epinephrine is released and produces rapid, transient increase in heat production by raising basal metabolic rate; quick, brief effect that counters heat lost through conduction and convection; involves brown adipose tissue, which decreases markedly in older adults; thyroid hormone increases metabolism
Heat Loss	
Radiation	Heat loss through electromagnetic waves emanating from surfaces with temperature higher than surrounding air
Conduction	Heat loss by direct molecule-to-molecule transfer from one surface to another, so that warmer surface loses heat to cooler surface
Convection	Transfer of heat through currents of gases or liquids; exchanges warmer air at body's surface with cooler air in surrounding space
Vasodilation	Diverts core-warmed blood to surface of body, with heat transferred by conduction to skin surface and from there to surrounding environment; occurs in response to autonomic stimulation under control of hypothalamus
Evaporation	Body water evaporates from surface of skin and linings of mucous membranes; major source of heat reduction connected with increased sweating in warmer surroundings
Decreased muscle tone	Exhausted feeling caused by moderately reduced muscle tone and curtailed voluntary muscle activity
Increased respiration	Air is exchanged with environment through normal process; minimal effect
Voluntary mechanisms	"Stretching out" and "slowing down" in response to high body temperatures; increasing body surface area available for heat loss; dressing in light-colored, loose-fitting garments
Adaptation to warmer climates	Gradual process beginning with lassitude, weakness, and faintness; proceeding through increased sweating, lowered sodium content, decreased heart rate, and increased stroke volume and extracellular fluid volume; and terminating in improved warm weather functioning and decreased symptoms of heat intolerance (work output, endurance, and coordination increase; subjective feelings of discomfort decrease)

Figure 48: Mechanisms of heat loss and production

Fever

- Resetting of the hypothalamic thermostat
    - Elevation occurs in response to exogenous/ endogenous *pyrogens*
- *Fever of unknown origin* (FUO)
    - Temperature >38.3C that remains undiagnosed after 3 days in hospital or ≥ 2 outpatient visits
- **Benefits of Fever**:
    - Kills many microorganisms
    - Promotes lysosomal breakdown and auto-destruction of cells
    - Increases lymphocytic transformation and phagocyte motility
    - Augments antiviral interferon production and phagocytosis

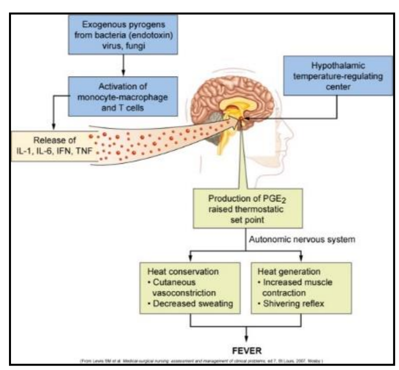


Figure 49: Pathophysiology of fever

Hyperthermia

- Not mediated by pyrogens & no resetting of hypothalamic set point. MORE serious than fever.
    - **41º C (105.8º F):** nerve damage produces convulsions
    - **43º C (109.4º F):** death results
- **Forms of hyperthermia include:**
    - Heat cramps
    - Heat exhaustion
    - Heatstroke

Malignant hyperthermia:

- Complication of inherited muscular disorders
- Can happen after surgery
- Precipitated by inhaled anesthetics, neuromuscular blocking agents

Pathophysiology:

- Impaired calcium release/uptake during muscle contraction
- Increased oxygen consumption and lactic acid production

Clinical features:

- Sustained muscle contractions
- Absent reflexes, fixed pupils, apnea, **flat ECG**

Heat Cramps

- Severe spasmodic cramps in abdomen and extremities following exercise in hot weather
- *Causes:*
    - Common in individuals not accustomed to heat
    - Performing strenuous work in warm climates
- *Pathophysiology:*
    - Prolonged sweating and associated sodium loss
- *Clinical manifestations:*
    - Fever
    - Rapid pulse
    - Increased blood pressure often accompanies the cramps

Heat Exhaustion

- A result of prolonged high core or environmental temperatures
- *Pathophysiology:*
    - Prolonged vasodilation and profuse sweating

- *Clinical manifestations:*
  - Dehydration, depressed plasma volumes, hypotension, decreased cardiac output, tachycardia
  - Dizziness, weakness, nausea, confusion, and syncope

Heatstroke ("Sun stroke")

- Potentially lethal condition, results of an overstressed thermoregulatory center
- Children are more susceptible:
  - They produce more metabolic heat when exercising
  - There is a greater surface area to mass ratio
  - Sweating capacity is less than in adults
- *Pathophysiology:*
  - Cardiovascular and thermoregulatory centers may cease functioning
  - Brain cannot tolerate temperatures >40.5° C (104.9° F)
- *Clinical manifestations:*
  - Cerebral edema
  - Degeneration of the CNS
  - Swollen dendrites
  - Renal tubular necrosis
  - Death

Hypothermia

- Body temperature less than 35° C. Can do this on purpose (operations to decrease BF)

Pathophysiology:

- CNS and respiratory depression
- Vasoconstriction & slow flow in the microcirculation
- Ice crystals form inside the cells, causing them to rupture and die
- Slow rate of cellular metabolism
- Increased blood viscosity
- Coagulation, and ischemic tissue damage

Types:

- **Accidental hypothermia**
  - Commonly the result of sudden immersion in cold water or prolonged exposure to cold
- **Therapeutic hypothermia**

  - Used to slow metabolism and preserve ischemic tissue during surgery or limb re-implantation
  - May lead to ventricular fibrillation and cardiac arrest

Trauma and Temperature Change

- CNS trauma leads to central fever
- Inflammation, increased ICP, intracranial bleeding
- **Examples**:
  - Accidental injuries
  - Major surgery
  - Thermal burns
- Temp monitoring is part of routine vital sign assessments

Sleep

- Active, multiphase process
- Hypothalamus is the major sleep center
- **Sleep occurs in two phases**:
  1. *Rapid eye movement (REM) sleep*:
     - 20% to 25% of sleep time
     - Also known as paradoxic sleep
     - Occurs every 90 minutes beginning after 1 to 2 hours of sleep
  2. *Non-rapid eye movement (NREM) sleep*:
     - 75% to 80% of sleep time
     - *Four stages evaluated by EEG*:
       - Stage I
       - Stage II
       - Stage III
       - Stage IV

Sleep Disorders

Primary sleep disorders

- *Dyssomnias*:
  - *Insomnia*: inability to fall asleep
  - *Primary and secondary hypersomnia*:
    - Primary: occurs with no other medical conditions present. The only symptom is excessive fatigue.

- Secondary hypersomnia is due to other medical conditions. Examples include:
  - Parkinson's disease, kidney failure, and chronic fatigue syndrome

- *Parasomnias*:
  - *Somnambulism* (sleep walking)
  - *Night terrors* (child is asleep and has extreme fear, once awakened they forget the terror).
  - Restless legs syndrome:
    - Uncontrollable urge to move your legs, usually because of an uncomfortable sensation
    - Runs in families
    - No known cause. likely an imbalance of the brain chemical dopamine

Secondary sleep disorders

- **Alterations in the quality and/or quantity of sleep caused by primary diseases**
  - Depression
  - Pain
  - *Obstructive sleep apnea*: Most common sleep disorder with complex pathology including hypoxia and oxidative stress
  - Alterations in thyroid hormone secretion
- **Sleep-provoked disorders**
  - Sleep stage alterations produced in certain disease states

9: Alterations of Pulmonary Functions

Learning objectives

- Explain the pathophysiology of obstructive and restrictive lung diseases
- Discuss the pathophysiology and the various clinical presentations of COPD, chronic bronchitis/emphysema, bronchial asthma, bronchiectasis, emphysema
- Discuss sleep breathing disorder (eg.OSA), aspiration, atelectasis, pulmonary fibrosis, inhalation disorders, pulmonary edema, ARDS
- Explain the pathophysiology and clinical presentations pulmonary hypertension
- Explain the pathophysiology and clinical presentation of deep vein thrombosis and pulmonary embolism
- Explain the pathophysiology and clinical presentation of selected childhood pathologies: cystic fibrosis, upper airway obstruction and infections

Definitions: Alterations of Pulmonary Functions

Term	Definition
Chest wall disorders	Disorders that compromise chest wall and restrict respiration
Dyspnea	Uncomfortable breathing/shortness of breath
Empyema	Pus in the pleural space
Flail chest	Multiple adjacent ribs are broken in multiple places Segment of ribs are separated
Hemoptysis	Coughing up blood
INHALATION DISORDERS	- Entry of a substance with air inhalation into respiratory tract
Acute Respiratory Distress Syndrome (ARDS)	- Fulminant form of respiratory failure characterized by diffuse alveolo-capillary injury
Atelectasis	- Lung collapse resulting in reduced or absent gas exchange - May affect part or all of one lung - It can be a disease, or it can be caused by a disease
Pulmonary Edema	- Fluid inside the lungs
Pulmonary Fibrosis	- Scarring of lung tissue

OBSTRUCTIVE LUNG DISORDERS	- Pulmonary diseases characterized by airway obstruction that is worse with expiration
Bronchial asthma	- Acute/chronic inflammatory/immune disorder of the airways
Chronic bronchitis	- Chronic inflammatory response within bronchi: - Hyper-secretion of mucus, chronic productive cough at least 3 months/year & at least 2 consecutive years
Bronchiectasis	- Permanent enlargement of bronchioles most commonly secondary to infection
Emphysema	- Permanent enlargement of alveoli accompanied by destruction of alveolar walls without obvious fibrosis
Respiratory Tract Infections	
Pneumonia	- Inflammation of the lung tissue
Acute Bronchitis	- Acute inflammation of the bronchi - Commonly follows a viral infection of upper resp tract - No pulmonary consolidation or chest infiltrates
Obstructive sleep apnea (OSA)	- Partial or complete upper airway obstruction during sleep
Orthopnea	- Dyspnea when a person is lying down
Paroxysmal nocturnal dyspnea (PND)	- Attacks of severe shortness of breath/cough at night during sleep
Pneumothorax	- Abnormal collection of air in pleural cavity
Pleural Effusion	- Excess fluid accumulation in the pleural cavity
PULMONARY FUNCTION DISORDERS IN CHILDREN	
Cystic fibrosis (CF)	- Autosomal recessive genetic disorder - Thick tenacious secretions affecting multiple systems (pulmonary and GI)
Croup	- URT infection causing barking like cough
Acute Epiglottitis	- Severe, rapidly progressing, life-threatening infection of epiglottis and surrounding area
PULMONARY VASCULAR DISORDERS	

Pulmonary embolism (PE)	- Occlusion within pulmonary vascular bed by: blood clot, fat/air bubble
Pulmonary Hypertension	- Elevation of pressure in pulmonary vasculature - Pressure > 20 mm or elevation by 5-10 mm Hg above normal
RESTRICTIVE LUNG DISEASES	- Diseases characterized by restriction of full expansion of lungs
Lung Aspiration	- Entry of material from oropharynx or GI into respiratory tract - Types of aspirate: pharyngeal secretions, food/drink, gastric content
Stridor	- High-pitched breath sound due to turbulent air flow in larynx or lower resp tract

Review: Nervous System Anatomy and Physiology

- **Respiratory system is composed of two parts**:
  - Upper and lower respiratory tracts
    - Everything after larynx = Lower Respiratory
- Respiratory system can be divided into *conducting* and *respiratory* components

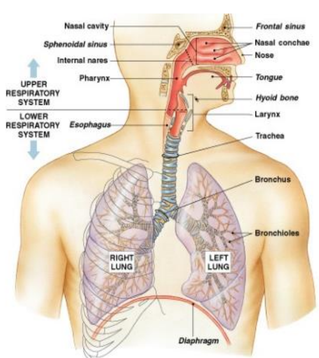


Figure 1: Anatomy of the upper and lower respiratory tracts

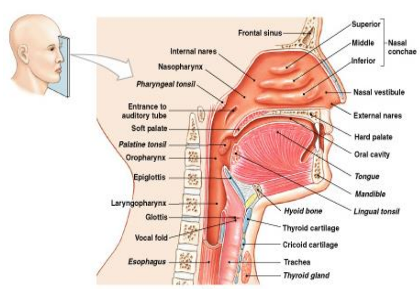


Figure 2: Anatomy of the upper respiratory tract

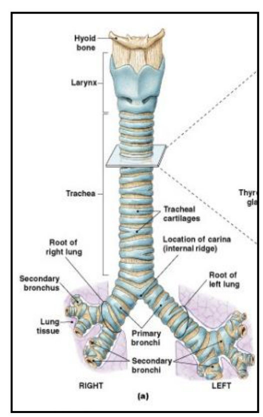


Figure 3: Conducting components of the lower respiratory tract

- Structure of lung alveoli and physiology of gas exchange

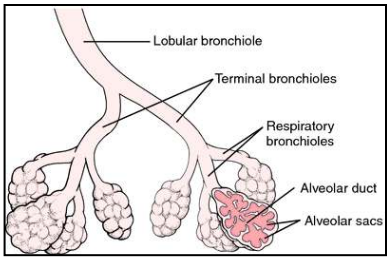


Figure 4: Cross section of alveoli of the lower respiratory tract

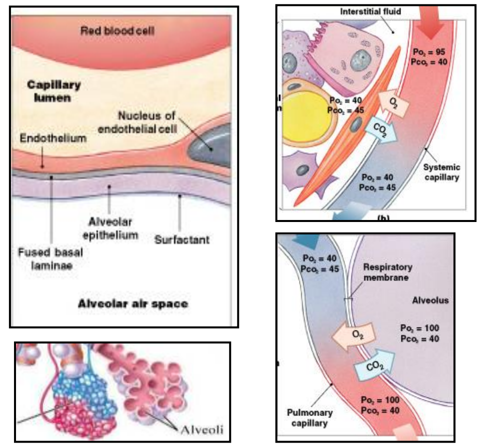


Figure 5: Sites and surfaces of gas exchange

- Structure and function of the pleural fluid
- Pulmonary & systemic circuits
- **Respiratory volumes (4) and capacities (4):**
    - Inspiratory reserve volume
    - Tidal volume
    - Expiratory reserve volume
    - Residual volume
    - Inspiratory capacity
    - Vital capacity
    - Functional residual capacity
    - Total lung capacity

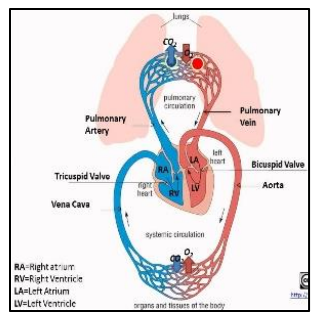


Figure 6: Pulmonary and systemic circuits

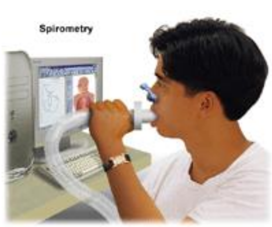


Figure 7: Spirometry measurement of respiratory volumes and capacities

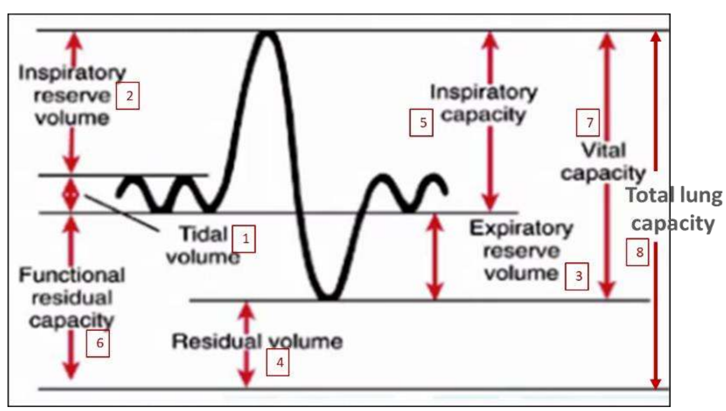


Figure 8: Respiratory volumes (#1-4) and capacities (#5-8)

Manifestations of Pulmonary Diseases

- *Dyspnea:*
  - Uncomfortable breathing/shortness of breath
- *Orthopnea:*
  - Dyspnea when a person is lying down
- *Paroxysmal nocturnal dyspnea* (PND):
  - Attacks of severe shortness of breath/cough at night during sleep
- Cough:
  - Protective reflex
  - Acute/chronic
- Pain
- *Stridor:*
  - high-pitched breath sound due to turbulent air flow in larynx or lower
- Wheezes & crackles:
  - **Video**: https://www.youtube.com/watch?v=xnubmmeDWrw&list=PLLKSXV1ibO86qgE2y9cMqNFmh6LfOa8RM

- **Abnormal respiratory rate/ pattern**:
  - Tachypnea/ bradypnea
  - Hyperpnea/hypopnea/apnea
  - Kussmaul respiration – DKA (or any acidosis)
  - Cheyne-Stokes respiration
- Abnormal sputum
  - Can be colored or clear, can increase or decrease in amount and frequency
- *Hemoptysis:*
  - Coughing up blood (cancer or TB)
- *Empyema:*
  - Pus in the pleural space, not a disease but a manifestation

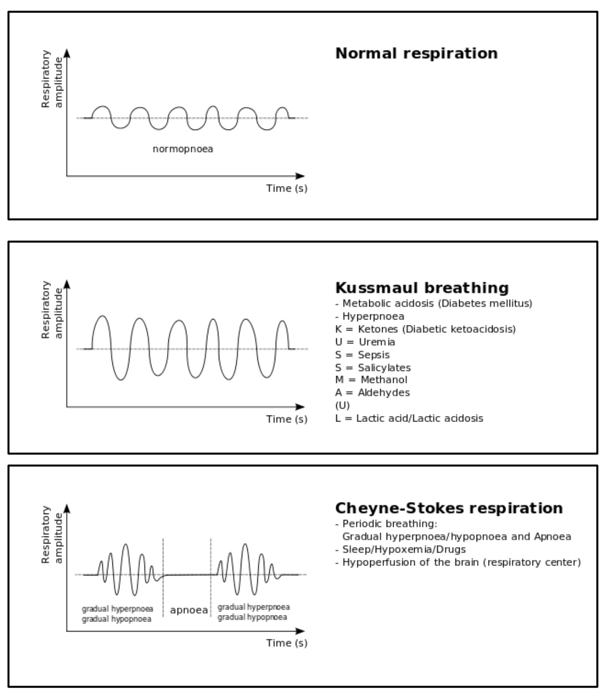


Figure 9: Respiratory rates and patterns.

- Hypoventilation
    - Leads to Hypercapnia
- Hyperventilation
    - Hypocapnia
- Cyanosis
- Clubbing

Disorders of the Respiratory System

Chest Wall Disorders

- Disorders that compromise chest wall and restrict respiration

Causes:

- Deformities
- Musculoskeletal disorders
- Immobilization
- Obesity

Pathophysiology:

- Mechanical restrictions of inspiration and expiration

Clinical manifestations:

- Breathing difficulty
- Of the underlying condition

Flail Chest

- Multiple adjacent ribs are broken in multiple places, separating a segment

Etiology:

- Severe trauma, falls

Pathophysiology:

- Segment of chest wall moves independently in the opposite direction of the lung's intention
- Due to difference in pressure, flail segment goes in while the rest of the chest is moving out (vice versa) → *paradoxical breathing*

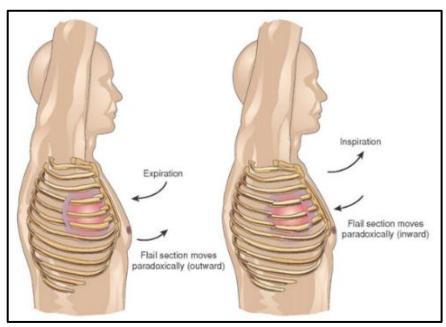


Figure 10. Paradoxical breathing

Clinical manifestations:

- Chest pain
- Dyspnea

Pneumothorax

- Abnormal collection of air in pleural space

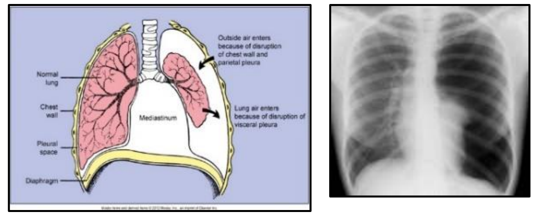


Figure 11. Pneumothorax

Etiology:

- Physical trauma
- Complication of medical/surgical intervention

Pathophysiology:

- Uncoupling of the lung from the chest wall, air interferes with normal respiration
- Increased pressure inside > atmospheric pressure,
    - Compresses the lung
    - Lung collapse (atelectasis)
- Pressure displaces the mediastinum and cause cardiopulmonary impairment
- Xray: black = air, tissue and fluid = white

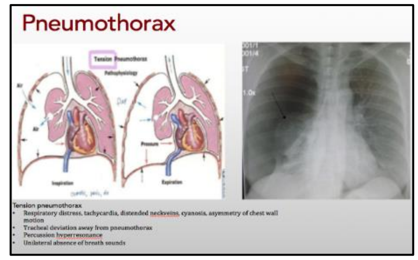


Figure 12. Tension pneumothorax with mediastinal deviation

Types:

- **Spontaneous pneumothorax**:
    - Rupture of a subpleural bleb
- **Secondary pneumothorax:**
    - Due to lung pathology, e.g. COPD
- **Open pneumothorax:**
    - Stab or bullet wounds
- **Tension pneumothorax:**
    - One-way valve allowing air and pressure to increase progressively inside pleural space
        - Significant impairment of respiration and/or circulation

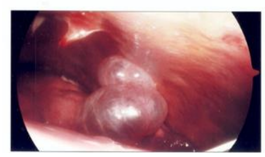


Figure 13. Subpleural bleb

Clinical manifestations & diagnosis:

- Chest pain, resp. distress
- Diminished breath sounds
- Hyper-resonant on percussion
- X-ray

Pleural Effusion

- Excess fluid accumulation in the pleural cavity

Figure 14. Pleural effusion

Causes/Types

- **Transudative effusion**
  - Congestive heart failure
  - Liver cirrhosis
  - Nephrotic syndrome
- **Exudative effusion**
  - Infection
  - Malignancy
  - Pulmonary embolism, infarction
  - Pancreatitis
  - Hemothorax
  - Trauma

Pathophysiology:

- Fluid limits expansion of lungs; impair breathing
- Increased hydrostatic pressure
- Injuries to the pleural lining
- >300 ml = detectable clinical signs
- Xray: tissue and fluid = white

Clinical manifestations & diagnosis:

- Chest pain (sharp pain worsen by cough or deep breaths)
- Impaired chest movement on the affected side
- Dullness to percussion over the fluid
- Diminished breath sounds on the affected side
- Pleural friction rub
- X ray confirms condition

9: Pulmonary Disorders

Restrictive lung diseases

- Pulmonary, extra-pulmonary, diseases characterized by <u>restriction of full expansion of lungs</u>
- **Pathology**: ↓ lung volume, ↓ ventilation/oxygenation, ↑effort in breathing

Main manifestation:

- Dyspnea

Common diseases/conditions:

- Aspiration
- Atelectasis
- Pulmonary fibrosis
- Inhalation disorders
- Pulmonary edema
- ARDS

Obstructive lung diseases

- Pulmonary diseases characterized by airway obstruction that is worse with expiration
- **Pathology**:
    - Inflamed and easily collapsible airways, frequent medical visits/ hospitalizations

Main manifestations:

- Cough, dyspnea, and wheezing

Common diseases/conditions:

- COPD
- Chronic bronchitis/Emphysema
- Bronchial Asthma
- Bronchiectasis
- Sleep breathing disorder (E.g. OSA)

Restrictive Lung disorders

Lung Aspiration

- Entry of material from oropharynx or GI into resp. tract
- Types of aspirate: pharyngeal secretions, food/drink, gastric content

Etiology:

- Medical procedure; during positive pressure ventilation
- Meconium aspiration syndrome

Risk factors:

- Impaired consciousness (TBI), alcohol or drug overdose, general anesthesia
- Tracheal intubation, presence of gastric tube

Pathophysiology:

- Depends on particle size, volume, chemical composition of aspirated material, and underlying patient health status
- Air way obstruction
- Damage of lung tissue (acidity, microbes)

Clinical manifestations:

- Range from no effect, to pneumonia, to death within minutes from asphyxiation

9: Inhalation Disorders

- Entry of a substance with air inhalation into resp. tract

Types and Causes:

- Toxic gases: e.g. CO, SO_2
- Pneumoconiosis (occupational lung disease)
    - Silica
    - Asbestos
    - Coal

Pathophysiology & clinical manifestations:

- Depend on dose inhaled, particle size, underlying patients' health status

Atelectasis

- Lung collapse resulting in reduced or absent gas exchange
- May affect part or all of one lung
- It can be a disease, or it can be caused by a disease

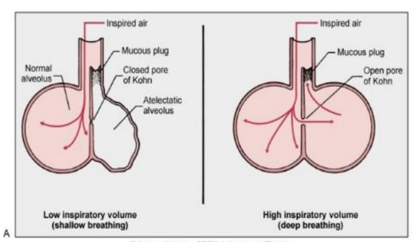


Figure 15: Atelectasis

Etiology and Pathophysiology:

- Post-surgical (most common cause)
- Airway block (foreign body, mucus plug, tumor, lymph node, tubercle)
- Poor surfactant

Clinical manifestations:

- Breathing difficulty (rapid and shallow)
- Low oxygen saturation
- Cyanosis
- Pleural effusion (transudate type)

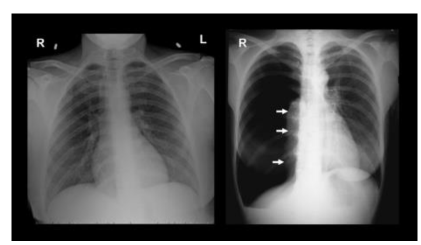


Figure 16: Atelectasis on an x-ray

Pulmonary Fibrosis

- Scarring of lung tissue (alveoli)

Etiology:

- Idiopathic
- Inhalation, e.g. asbestos
- Inflammation, e.g. pneumonitis
- Autoimmune, e.g. SLE

Pathophysiology:

- Replacement of normal lung parenchyma with fibrous tissue
- Reduction of lung compliance
- Defective perfusion

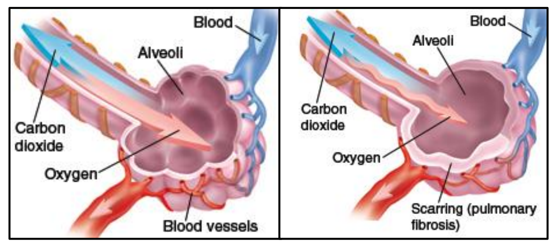


Figure 17: Replacement of normal lung tissue with fibrous tissue

Clinical manifestations & evaluation:

- Progressive dyspnea with exertion
- X-ray to show fibrosis

Pulmonary Edema

- Fluid inside the lungs

Etiology and diseases associated with lung edema:

- Left side valvular dysfunction
- Left ventricular failure
- Injury to capillary endothelium
- Blockage of lymphatic vessel

Pathophysiology:

- Increase in atrial pressures
- Increase in capillary permeability
- Accumulation of fluid in interstitial space
- Increased pulmonary capillary hydrostatic pressure

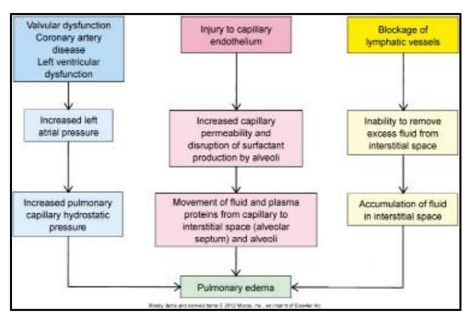


Figure 18: Pathophysiology of Pulmonary Edema

Clinical manifestations:

- **Dyspnea**, orthopnea, PND
- Cough (pink, frothy sputum)
- Excessive sweating, anxiety, pale skin
- **Anxious**

Acute Respiratory Distress Syndrome (ARDS)

- Fulminant form of respiratory failure characterized by diffuse alveolo-capillary injury

Causes:

- Sepsis **(main reason)**
- Multiple trauma
- Pneumonia
- Near drowning
- Drug overdose

Pathophysiology:

- Injury to the pulmonary capillary endothelium
- Pulmonary edema
- Inflammation and platelet activation
- Surfactant inactivation/loss

Clinical manifestations & diagnosis:

- Dyspnea and hypoxemia
- Hypoventilation, increased Pco2
- Respiratory acidosis
- Hypotension

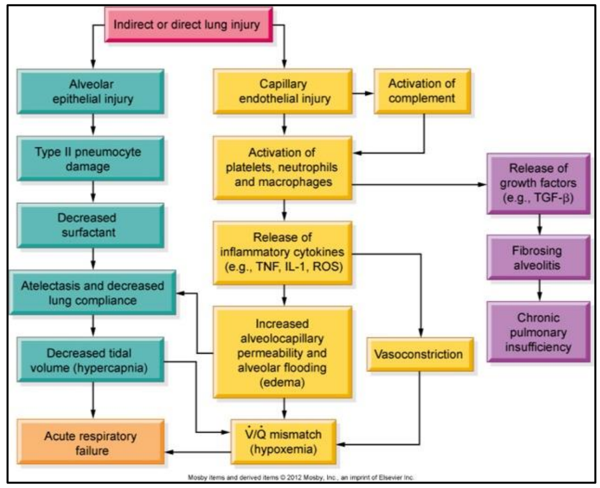


Figure 19: Pathophysiology of ARDS

9: Obstructive Lung disorders

Bronchial Asthma

- **The number 1 respiratory disease**
- Acute/chronic inflammatory/immune disorder of the airways

Etiology:

- Complex environmental and genetic factors

Pathophysiology:

- Children and adults
- Hyper-responsive airways
- Bronchospasm, inflammation, vascular congestion, increased mucous secretion
- IgE mediated degranulation of mast cells
- May lead to status asthmaticus

Precipitating factors:

- Allergens, drugs, others

Clinical manifestations/ diagnosis:

- *Expiratory wheezing, dyspnea, tachypnea, cough
- Spirometry, family history, symptoms, etc.

*Seen everywhere in the world. The terminal airways are blocked from mucus and inflammation of the wall. This is an inflammatory disorder that is IgE mediated from mast cell degeneration. The smooth muscle wall responds quickly to antigen in exposed air. This disease can be acute or chronic and can run in a family. Status asthmaticus = severe acute asthmatic attack.

Chronic Bronchitis

- Chronic inflammatory response within bronchi:
    - Hyper-secretion of mucus, chronic productive cough at least 3 months/year & at least 2 consecutive years

Cause:

- **Smoking** (main reason)
- Air pollution
- Long term exposure to irritants (occupation related)
- Genetic factors

Pathophysiology:

- Constant irritation will mucus production, size & number of mucous glands
- Mucus is thicker than normal
- Secondary bacterial infection adds to this inflammatory state

Clinical manifestations & evaluation:

- Chronic, worsening productive cough
- Diffuse, bilateral wheezes & rhonchi
- Spirometry can measure the amount of airflow obstruction
- Usually complicated by emphysema
- Colored sputum = secondary bacterial infection

Bronchiectasis

- Permanent enlargement of bronchioles most commonly secondary to infection.
    - "ectasis" = dilation. You lose wall elasticity

Etiology:

- Cystic fibrosis (50%), COPD
- Infection; pneumonia, tuberculosis
- Immune disorders

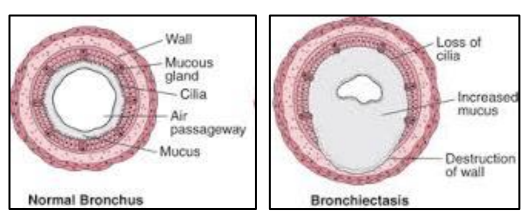


Figure 20: Pathophysiology of bronchiectasis compared to normal bronchus

Pathophysiology:

- Chronic inflammation

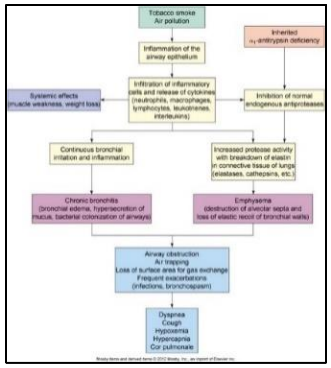
- Accumulation of/inability to clear secretions
- Progressive destruction of the normal lung architecture

Clinical manifestations:

- Chronic cough, green/yellow sputum, dyspnea
- Breath indicative of active infection
- Crepitation/crackles base of the lung

Complications:

- Emphysema
- Secondary amyloidosis



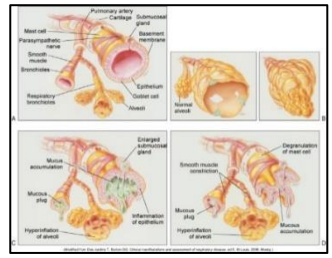


Figure 21(a)(b): Pathophysiology of obstructive pulmonary disease

Emphysema

- Permanent enlargement of alveoli accompanied by destruction of alveolar walls without obvious fibrosis

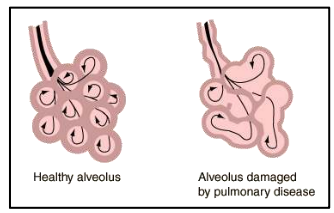


Figure 22: Damaged alveolus of pulmonary disease

Etiology:

- COPD **(main reason)**

Pathophysiology:

- Loss of elastic recoil
- Dilation and combining air spaces reduces surface area for oxygen exchange
- Ruptured alevoli

Clinical manifestations

- As in other COPD
- Labored breathing
- **Barrel chest**

*Respiratory Tract Infections

Pneumonia:

- Discussed in detail in LUSL2036
- **Pathophysiology**:
  - Acute inflammation; accumulation of neutrophils & macrophages, inflammatory mediators, congestion, exudate release, etc.
  - Consolidation (inflamed, thickened and scarred airway)

Tuberculosis

- Discussed in detail in LUSL2036
- **Pathophysiology**:

- Chronic inflammation, tubercle formation, caseous necrosis

Acute bronchitis

- Acute inflammation of the bronchi
- Commonly follows a viral infection of upper resp tract
- Similar symptoms to pneumonia but does not demonstrate pulmonary consolidation and chest infiltrates

Obstructive Sleep Apnea

- Partial or complete upper airway obstruction during sleep

Etiology:

- Adenotonsillar hypertrophy is the most common cause.
- Pharyngeal web = hyperplasia tissue that blocks the nasopharynx.
- At night the nasopharynx relaxes → high CO_2 so patient wakens abruptly to temporarily correct breathing pattern (fix apnea).

Pathophysiology:

- Mild, moderate, severe based on AHI (apnea-hypopnea index)
- Disruption of normal ventilation and sleep patterns
- Oxidative stress
    - The cells get tired. The endothelium releases NO to dilate the vessels to help with O_2
- Hypercoagulability and CV complications

Clinical Manifestations and Diagnosis:

- Snoring and labored breathing during sleep
- Daytime sleepiness
- Chronic mouth breathing
- Sleep studies (Polysomnography)
- **Sleep questionnaires**:
    - Berlin and *STOP-Bang

9: Pulmonary Vascular Disorders

Pulmonary Embolism (PE)

- Occlusion within pulmonary vascular bed by clot (thrombus, embolus), fat/air bubble, amniotic fluid droplet

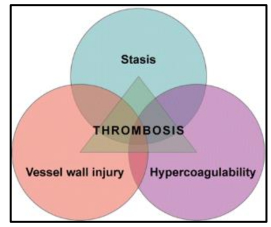
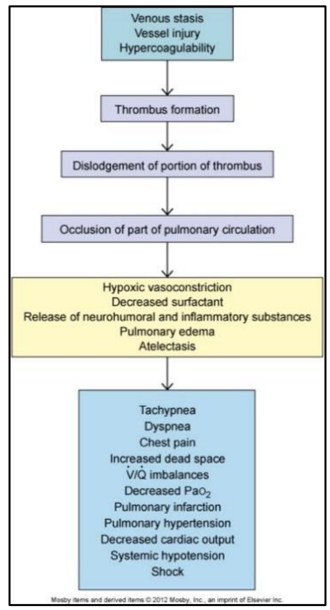


Figure 23: Virchow's triad

Etiology and Pathophysiology:

- Deep vein thrombosis (DVT) in the lower limb
- Virchow triad
- Hypoxemia, low oxygen saturation
- Perfusion is affected, not ventilation



Clinical manifestations and evaluation:

- Dyspnea, chest pain on inspiration
- Tachycardia, cyanosis, hypotension

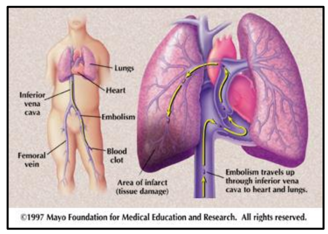


Figure 24: Pathophysiology of pulmonary embolism

Lab and radiology:

- Elevated D-dimer
- CT & pulmonary angiography
- Doppler US

Pulmonary Hypertension

- Elevation of pressure in pulmonary vasculature
- Pressure > 20 mm or elevation by 5-10 mm Hg above normal

Etiology & classifications:

- Primary pulmonary hypertension (Idiopathic)
- Pulmonary hypertension due to left heart disease
  - E.g. mitral valve
- Pulmonary hypertension due to lung disease/hypoxia
  - E.g. COPD
- Pulmonary hypertension due thrombo-embolic disease
  - E.g. lung clots

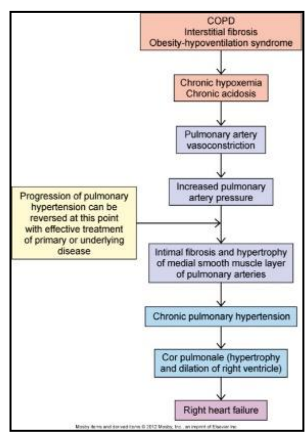


Figure 25: Pathophysiology of pulmonary hypertension

Clinical manifestations:

- Dyspnea
- Fatigue
- Dizziness

Complications:

- RT ventricular failure
- **Cor Pulmonale:**
  - RV enlargement/failure secondary to pulmonary hypertension

9: Malignancies of the Respiratory Tract:

Lung (bronchogenic) cancer

Etiology & risk factors:

- Cigarette smoking: most common cause (heavy smokers 20 x > risk than nonsmokers)
- Environmental or occupational risk factors e.g. asbestos, radon

Types:

- **Non-small cell carcinoma:**
  - Squamous cell carcinoma
  - Adenocarcinoma
  - Large cell carcinoma
- **Small cell carcinomas:**
  - Highly malignant
  - Can arise outside lungs (prostate, cervix, LN)
  - Can be hormone secreting (paraneoplastic)

9: Selected Pulmonary Function Disorders in Children - Chapter 27

Cystic Fibrosis

Etiology & Pathophysiology:

- Autosomal recessive- mutations in transmembrane conductance regulator ***(CFTR) gene***
  - Damages the chloride channels that helps make the mucus, this reduces the water, so the mucus is now thick
- Multisystem disease
- Exocrine or mucus-producing glands secrete abnormally thick mucus
- In the lungs, thick secretions obstruct the bronchioles and predispose to chronic lung infections
- Chronic inflammation leads to hyperplasia of goblet cells, bronchiectasis, pneumonia, hypoxia, fibrosis, etc.

Upper Airway Obstruction with Croup

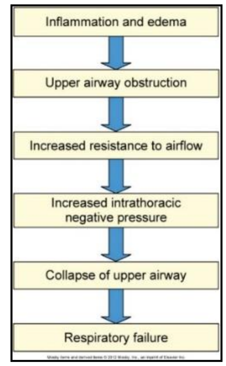


Figure 26: Pathophysiology of Croup

- *Croup*:
  - URT infection causing barking like cough
- One cause of stridor (noisy breathing) (others are fb.)
- Acute laryngotracheobronchitis:
  - Children from 6 months to 5 years
  - Commonly caused by a virus (influenza, or RSV)
  - Usually occurs after an episode of rhinorrhea, sore throat, low-grade fever
  - URT inflammation → obstruction → stridor (~70% narrowing)
  - Life-threatening

*Acute Epiglottitis

- Severe, rapidly progressive, life-threatening infection of the epiglottis and surrounding area
- Historically caused by *Haemophilus influenzae* type B
  - 80%-90% decreased incidence due to HIB vaccination

Manifestations:

- High fever
- Irritability
- Sore throat
- Inspiratory stridor/ muffled voice
- Severe respiratory distress

Aspiration Disorders

Etiology and pathophysiology:

- Aspiration of foreign bodies
  - In children, ages of 1-3
  - Any foreign substance: food, meconium (neonates), secretions (salivary or gastric), or environmental
  - <u>Inflammation</u> of the lung tissue (aspiration pneumonitis)
- Lung damage depends on volume and pH of aspirate
- Leading cause of death in children specially, neurologically compromised

Clinical Manifestations:

- Coughing
- Choking

- Gagging
- Wheezing
- Symptoms depend on size/ nature of the foreign body

Respiratory distress syndrome (RDS) of newborn

Etiology & Pathophysiology:

- Prematurity & lack of adequate surfactant
- Primarily a disease of preterm infants
- Widespread atelectasis, respiratory distress, and pulmonary hypertension

Clinical manifestations:

- Tachypnea
- Expiratory grunting
- Nasal flaring
- Dusky skin

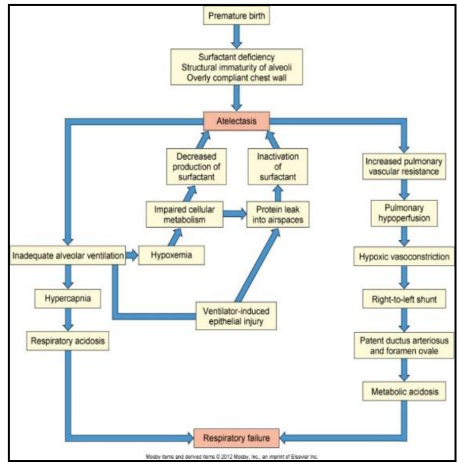


Figure 27: Pathophysiology of RDS of the newborn

10: Alterations in Renal and Urologic Systems / Water, Acid-Base and Electrolyte Imbalance

Learning Objectives: Alterations in Renal System and Urinary Tract

- Describe the normal structure and function of the renal system.
- Describe the anatomic causes of resistance to urine flow, causes and effects of *obstruction* in various locations within the urinary tract and signs of *urinary obstruction.*
- Describe the two most common tumors of the renal and urologic systems: *renal carcinoma* and *bladder tumors.*
- Describe the pathophysiology and clinical manifestations of *kidney stone formation.*
- Characterize the following lower urinary tract obstructions: *bladder neck dyssynergia, prostate enlargement, urethral structure, pelvic organ prolapse.*
- Describe what is meant by *neurogenic bladder* and *overactive bladder syndrome.*
- Describe the etiology, infectious agents, manifestations, treatments and complication of urinary tract infections: *acute cystitis, interstitial cystitis, acute pyelonephritis* and *chronic pyelonepthritis.*
- Identify the causes, types, classifications of *glomerulonephritis* and the resulting changes in glomerular structure and function.
- Describe the features and progression of *nephrotic syndrome* from causation through complications.
- Differentiate among *prerenal, intrarenal* and *postrenal* causes and the clinical manifestations, outcomes, and complications of *acute renal failure.*
- Describe the pathophysiology of *acute tubular necrosis* (ATN).
- Describe the clinical manifestations of *chronic renal failure* and the *multiple system outcomes* (including electrolyte and acid-base alterations) of chronic renal failure.

Learning Objectives: Water/Acid/Base/Electrolyte Imbalance

- Describe the roles of *water* and the *principal electrolytes* in maintaining homeostasis including two functional fluid compartments of the body (review).
- Describe water movement between plasma and interstitial fluid (review).
- Describe water movement between the intracellular fluid compartment and the extracellular fluid compartment (review).
- Describe the causation, pathophysiologic process, and clinical manifestations of *edema*.
- Describe the regulatory processes for *sodium* and *water* balance in the body, including the role of *antidiuretic hormone, renin-angiotensin-aldosterone,* and *atrial natriuretic hormone*.
- Define *hypotonic, isotonic,* and *hypertonic* alterations in water balance and give an example of each.

- Describe the basic causes and clinical manifestations of *hypernatremia* and *hyponatremia, hyperchloremia* and *hypochloremia*
- Describe the clinical manifestations of *water deficit*.
- Describe the clinical manifestations of the *syndrome of inappropriate secretion of ADH* (SIADH).
- Describe the distribution, function and regulation of *potassium* in the body.
- Describe the basic causes and clinical manifestations of *hyperkalemia* and *hypokalemia.*
- Describe the role of *hydrogen ion* concentration in cellular function and dysfunction.
- Describe how the *plasma buffering systems* help prevent significant fluctuations in pH.
- Describe how the *lungs* and the *kidneys* regulate *acid-base balance*.
- Differentiate between *respiratory* and *metabolic acid-base disorders* by causes and mechanisms of compensation.

Definitions: Alterations of the Renal and Urologic Systems

Term	Definition
Afferent and efferent arterioles	Arterioles taking blood in and out of the glomerulus
Excretion	Moving solutes from tubular cell to the tubular lumen then into the urine flow
Glomerular Filtrate/ Glomerular fluid	Material passing through the glomerular filter to the bowman's capsule
Filtration	Passage of material from blood through the glomerular filter to the bowman's capsule
Hydronephrosis	Dilation/ swelling of kidney back-up of urine
Hydroureter	Dilation/ swelling of ureter due to back-up of urine
Interstitial fluid	Fluid in the interstitial space (space between cells of CT tissue)
Plasma	Fluid component of blood
Reabsorption	Regaining back material from tubular lumen to tubular cell then to peritubular blood vessel and thus back to circulation
Secretion	Moving solutes from peritubular blood vessel to peritubular space then to tubular cell

Tubular fluid	Fluid inside lumen of kidney tubules
Urine	- Fluid running in tubular fluid eventually into renal pelvis, ureter and urethra to be excreted to the outside - Has normal composition (review)
Thrombotic microangiopathies (TMA)	- Group of disorders characterised by a thrombotic triad: Hemolytic anemia, Renal damage, Thrombosis. Each has specific features - Includes: HUS, TTP, HSP
- **Henoch-Schonlein Purpura (HSP)**	- A form of TMA common in children - Characterized by vasculitis in skin, intestine and kidney - Main symptoms: purpuric rash with small bruises over the legs or buttocks, abdominal, joint pains, renal damage (latter is rare)

Review: Anatomy and Physiology of the Renal System and Urinary Tract

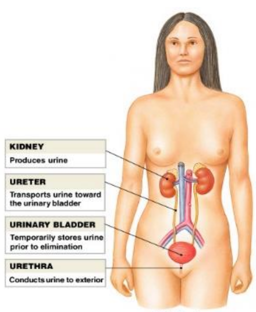


Figure 1: Urinary tract (anterior view)

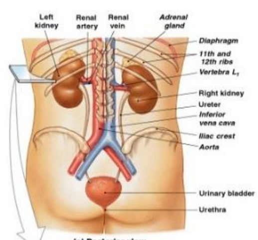


Figure 2: Urinary tract (posterior view)

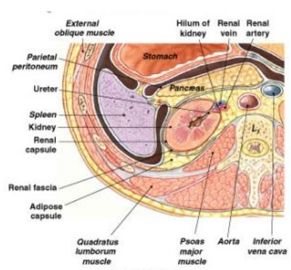


Figure 3: Urinary tract (superior view)

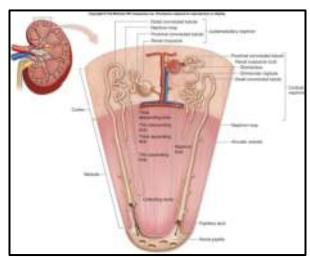


Figure 4: Gross and microscopic anatomy of renal pyramid

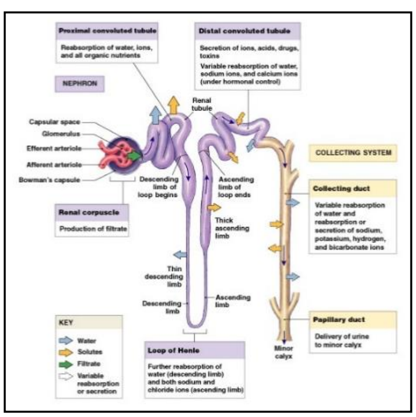


Figure 5: Structure and function of a nephron

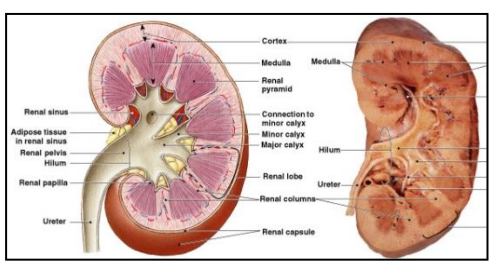


Figure 6: Sectional anatomy of the kidney

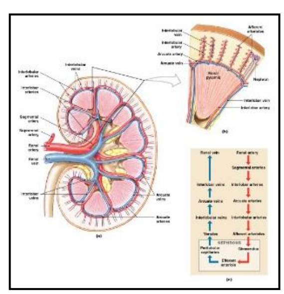


Figure 7: Blood supply of the kidney

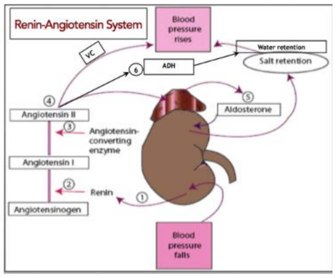


Figure 8: Renin-angiotensin-aldosterone-system (RAAS)

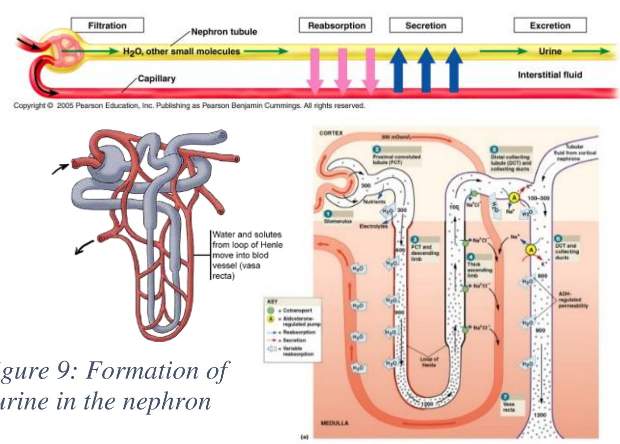


Figure 9: Formation of urine in the nephron

Normal Values (NV) for Renal Functions	
Analyte	**NV**
Serum creatinine	
Serum urea (BUN)	
Total serum protein	
Serum albumin	
Plasma osmolality	
Urine osmolality	
GFR	
Serum Electrolytes (ECF)	
Sodium (Na)	
Potassium (K)	
Calcium (Ca)	
Chloride (Cl)	
Bicarbonate (HCO3)	

Renal Diseases

Classifying Renal Diseases

Pre-renal:

- Any process that decreases renal blood flow

Renal:

- *Glomerular disease*:

Spectrum ranging from glomerulonephritis to nephrotic syndrome

- *Tubular-interstitial disease*:
  - **Acute tubular necrosis (ATN):**
    - Ischemic

        - Nephrotoxic insult
    - **Acute interstitial nephritis**
- *Vascular disease*:
    - **Vascular occlusion** - renal artery/vein thrombosis, thrombotic microangiopathies (TTP, *HUS*, *HSP*)
    - **Intrarenal vasculitis** – e.g. Wegener granulomatosis

Post-renal:

- Any process that obstructs urine outflow

Urinary Tract Obstruction

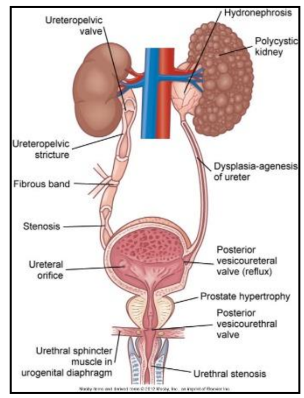


Figure 10: Anatomy of the urologic system

- Interference with urine flow within the urinary tract
- Also known as "obstructive uropathy"

Etiology:

- May be anatomic or functional obstruction, including:
    - Compression by tumors
        - E.g. kidneys, bladder, uterus, colon
    - Compression by enlarged prostate, or prolapsed pelvic organ
    - Pelvic adhesions (retroperitoneal fibrosis)
    - Stones
    - Strictures due to inflammation/scarring of walls
- **Sites of obstructions:**
    - Obstruction can occur anywhere in the pathway of urine flow
    - Effects depend on the site, cause and degree of obstruction

Pathophysiology & Complications:

- Dilation proximal to site of obstruction
    - *Hydronephrosis*
        - Dilation of renal pelvis & calyces
    - *Hydroureter*
        - Dilation of ureter
    - Compensatory hypertrophy & urine accumulation
    - Tubular and later glomerular damage if obstruction is not relieved

Clinical manifestations:

- Difficulty to void, decreased urine output

- Slowed stream, "dribble"
- Urgency especially at night
- False sense of fullness
- Blood in the urine
- Severity of manifestations will depend on:
  - Location
  - Degree
  - Unilateral versus bilateral
  - Upper versus lower urinary tract
  - Duration and cause of obstruction

Kidney Stones

- Masses of crystals, protein, or other substances that form within the kidney and may obstruct the urine flow

Etiology & risk factors:

- Gender & race
- Dietary pattern
- Fluid intake
- Geographic location; temp, humidity, rainfall
- Seasonal factors
- Occupation

Pathophysiology:

- Salt supersaturation
- Precipitation of a salt from liquid to solid state
- **Crystallization**: growth from a nucleus into a stone
- *Factors affecting stone formation and crystallization*:
  - Temperature
  - pH
  - Organic material
  - Influence of additional substances:
    - Tamm-Horsfall protein (a crystal growth-inhibiting substance)

Classification of stones:

- Calcium oxalates/phosphate are most common (70-80%)
- Magnesium/ammonium phosphate

- Uric acid stones
- Stones < 5mm will likely be excreted via urine, while those > 1cm will not

Clinical manifestations & Diagnosis

- *Renal colic*:
  - Moderate to severe spasms of pain originating in flank, radiating to groin
- May have associated nausea, vomiting
- Urinary urgency and frequency
- **Diagnostic testing** includes:
  - Urine analysis
    - Hematuria is an indicator of stones
  - **Imaging** (x-ray, ultrasound, and/or computed tomography)
    - Kidney, ureter, bladder

Neurogenic Bladder

- Bladder dysfunction caused by neurogenic disorders (brain, spinal cord, peripheral nerve) interfering with micturition

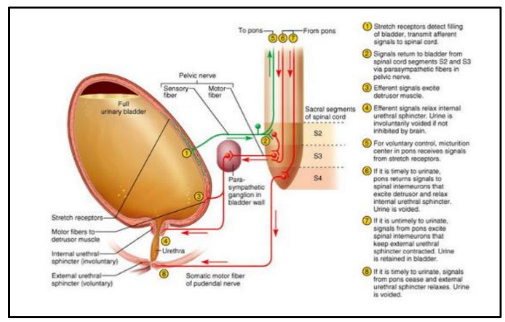


Figure 11: Process of micturition

Etiology:

- Multiple of neurologic disorders including:
  - Parkinson's disease
  - Multiple sclerosis (MS)
  - Infection of the brain or spinal cord
  - CVA
  - Peripheral neuropathies
  - Pelvic surgery
  - Congenital (I.e. spina bifida)

Pathophysiology:

- Lesions in brain centers or the spinal cord
- Upper or lower motor neuron lesions impacting the ability to control micturition thus leading to overactive or underactive bladder syndrome

Clinical manifestations:

- Frequency, urgency, nocturia
- Retention with overflow
- Stress incontinence

Renal Tumors- Renal Cell Carcinoma

- Malignant tumor (adenocarcinoma) originating from tubular epithelium
- Most common in males > 60 years
- 25% present with metastasis, including to lung, liver, bone, and lymph nodes

Clinical manifestations:

- Usually presents in advanced cases
- Dull aching flank pain
- Mass
- Hematuria

Renal Tumors- Wilm's Tumor

- A common childhood cancer
    - Prevalence in children 2-4 years old
- Mostly unilateral

Clinical Presentation:

- Abdominal swelling/ mass
- Pain
- Hematuria

Urinary Tract Infection (UTI)

- Inflammation of urinary epithelium mostly due to infection
- **Most common pathogen**:
    - *Escherichia coli*

Lower UTI (cystitis/urethritis):

- Inflammation of the bladder (cystitis) and/or urethra (urethritis)

- **Clinical manifestations**:
    - Frequency
    - Dysuria
    - Urgency
    - Lower abdominal and/or suprapubic pain

Upper UTI (pyelonephritis):

- Infection of kidney and renal pelvis

- **Risk/ predisposing factors**:
    - Vesicoureteral reflux
    - Urinary tract obstruction

- Previous pyelonephritis
- Immunocompromised state

▪ **Common pathogens**:
  - *E. coli, Proteus, Pseudomonas*

▪ **Pathophysiology**:
  - Ascending infection from lower UTI
  - Persistent or recurring episodes of acute pyelonephritis
  - Scarring, further kidney damage & renal failure

▪ **Additional considerations**
  - What is the overall clinical condition or associated comorbidity?
  - Is a lab diagnosis available?
  - Is the person pregnant and are there complications?

Glomerulonephritis

▪ Inflammation/damage of the glomeruli

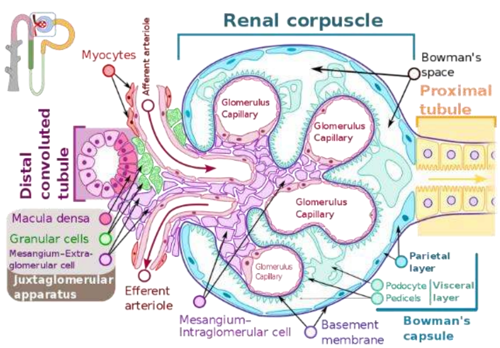


Figure 12: Anatomy of a kidney glomerulus (4)

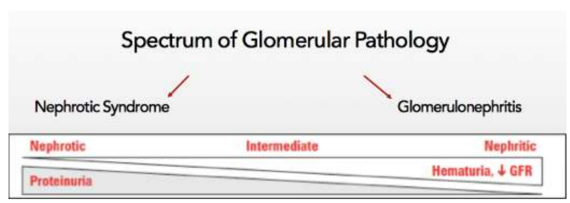


Figure 13: Spectrum of glomerular pathology

Etiology/Forms:

▪ **Immune injury (most common):**
  - *Post-streptococcal glomerulonephritis*:
    - Most common cause of hematuria in children
    - Several weeks following infection with group A *Streptococci*
    - **Immune complex deposition in** glomeruli (Type III hypersensitivity)
  - *IgA nephropathy (Berger's disease)*:
    - A common cause/type of glomerulonephritis world wide
    - Common between late teens and late 30s and runs in families
    - **IgA** deposition in glomeruli (Type II hypersensitivity)

- *Good Pasture syndrome*:
    - Develops one week following upper respiratory tract infection (URTI)
    - Pathological Triad:
        1. Anti-GBM antibodies
        2. Rapidly progressive glomerulonephritis
        3. Pulmonary hemorrhage
- **Non-immune injury**:
    - Drugs
    - Toxins
    - Ischemia
        - Vasculitis (E.g. Wegener granulomatosis)
- **Secondary to systemic diseases**:
    - SLE (Lupus nephritis)
    - DM (Kimmelstiel–Wilson nodules is a histopathological finding)
    - Amyloid
    - HTN

Pathophysiology:

- Glomerular inflammation (review pathology of inflammation)
- Rapidly progressive

Clinical manifestations:

- Hematuria with red cell casts
- Edema
- Oliguria
- Hypertension
- Low grade fever, flank pain

Diagnosis

- Clinical presentation
- Renal biopsy (invasive technique) but confirms pathological form and guides treatment

Complications:

- Can proceed to nephrotic syndrome

Nephrotic Syndrome

- A syndrome whereby a glomerular lesion leads to massive protein loss

Etiology:

- Diabetes mellitus and hypertension are the most common causes for nephrotic syndrome
- Glomerulonephritis
- Others:
    - Amyloidosis
    - Cancer
    - Drugs
    - SLE
    - HIV

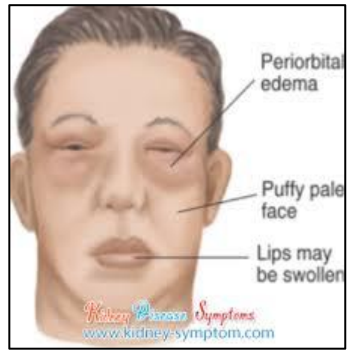


Figure 14: Manifestations of nephrotic syndrome

Types (histo-pathophysiological):

- **Minimal change disease**
    - Mainly children, NSAIDs
- **Membranous glomerulonephritis**
    - Adults, NSAIDs, cancer
- **Focal segmental glomerulosclerosis**
    - HIV patients

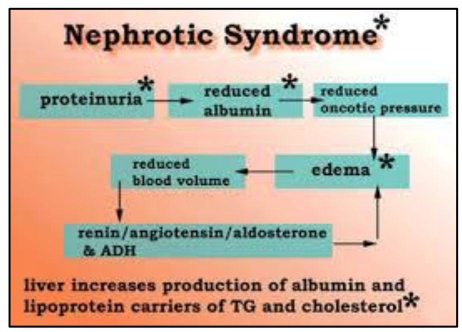


Figure 15: Pathophysiology of nephrotic syndrome

Pathophysiology and clinical manifestations:

- Glomerular injury that leads to protein leakage
- **Proteinuria**: ≥3.5 g of protein in urine/ day
- **Hypoalbuminemia**: <3 gm/dL
- **Edema**
    - May be periorbital, generalized and/or pitting
- **Hyperlipidemia**, **lipiduria**
    - Hyperlipidemia occurs from decreased protein levels
    - Proteins are required in the form of lipoproteins for the uptake of lipids into tissue
    - Absence of proteins leads to build-up of lipids in the blood
- **Hypogammaglobulinemia**
- Vitamin D deficiency
- Hypercoagulable state
- Manifestations of RAAS stimulation

Acute Kidney Injury (AKI)

- Sudden decline of kidney function
  - Occurs on a spectrum ranging from minimal changes in kidney function to kidney failure requiring dialysis and kidney transplant
- Failure of kidney function means:
  - Reduction of glomerular filtration and creatinine clearance →
  - Accumulation of nitrogenous waste in blood →
  - Encephalopathy and impairment of brain function (uremia and high creatinine).

Stages of AKI:

Renal insufficiency:

- Decline kidney functions by 25% of normal (GFR~ 25-30mL/min)
- Mild elevation of serum creatinine

Renal failure:

- Significant loss of kidney function, requiring dialysis

End stage renal failure (ESRF; uremia):

- Renal function is less than 10%, requiring dialysis

Etiology & pathophysiology:

- AKI can originate from causes before (pre-renal), within (intrarenal), or after (post-renal) the kidney
- *Pre-renal* causes are occurred before the kidney and are due to impaired renal perfusion
  - The most common causes for AKI, including:
    - **Hypovolemia**:
      1. Hemorrhage (E.g. trauma, GI bleeding, childbirth)
      2. Loss of plasma volume (E.g. Burns)
      3. Loss of water/electrolytes (E.g. Severe vomiting, diarrhea, intestinal obstruction, diuretic overuse)
    - **Hypotension/ Systemic vasodilation** (E.g. Shock, CHF, massive pulmonary embolism)
    - **Cirrhosis** through the process of ↓ albumin → ↓ colloid oncotic pressure → ↓ intravascular volume
- *Intra-renal*: Causes from within the kidney and include renal parenchymal and interstitial disorders
  - *Acute tubular necrosis (ATN)* is the most common
    - Causes for ATN include:
      1. **Ischemic**: E.g. Post-operative ischemia, sepsis

2. **Toxic**: E.g. Drugs, antibiotics eg gentamicin
3. **Crystals/pigments**: E.g. Uric acid, oxalates, myoglobin, hemoglobin

- *Post-renal*: AKI originating after the kidney, and the most rare
    - *Obstructive uropathy*: E.g. Enlarged prostate, bladder stones, uretheral, ureteric (bilateral)

Clinical manifestations:

- Oliguria in various levels
- Manifestation of underlying pathology

Pathogenesis & Stages:

1. **Initiation phase** (first 36h)
    - Kidney injury is evolving with slight decrease in urine output and increase in blood urea nitrogen (BUN)
    - Prevention of injury is possible
2. **Maintenance phase**
    - Established kidney injury and dysfunction
    - Sustained oliguria (40-400mL/day)
    - Increase in ECF volume → can be associated with edema, pulmonary congestion
    - Hyperkalemia may be present leading to EKG changes and increased risk of sudden death
    - Metabolic acidosis
3. **Recovery phase** (2-3 weeks)
    - Brisk diuresis (up to 3L/day) leading to urinary loss of electrolytes Ca, K, PO4
    - Hypokalemia (one of most serious complications of recovery) leading to EKG changes and presenting a risk for cardiac arrhythmias
    - Renal function eventually recovers with BUN and serum creatinine returning to baseline

Chronic Kidney Disease (CKD)

- Chronic progressive loss of renal function
- CKD is defined as glomerular filtration rate (GFR) < 60 mL/ min/1.73 m^2, for 3 months or more irrespective of cause

Etiology:

- **Systemic diseases**:
    - E.g. Hypertension and Diabetes mellitus
- **Intrinsic kidney disease**:

- Glomerular or tubular
- Glomerulonephritis
- Pyelonephritis
- Obstructive uropathy

Stages:

- **Normal Kidney Function**: GFR >90mL/min
- **Mild**: GFR 60-89mL/min
- **Moderate**: GFR 30-59mL/min
- **Severe**: GFR 15-29mL/min
- **End stage**: GFR <15mL/min (regular dialysis or transplantation)

Pathophysiology:

- Continued loss of functioning nephrons
- Glomerular hypertrophy and adaptive hyperfiltration
- Proteinuria and glomerulosclerosis
- RAAS activation and increased Angiotensin II leading to glomerular and systemic hypertension
- Erythropoietin deficiency
- Kidney unable to excrete potassium (K), Phosphate (PO_4), Magnesium (Mg)
- Tubulointerstitial inflammation and fibrosis

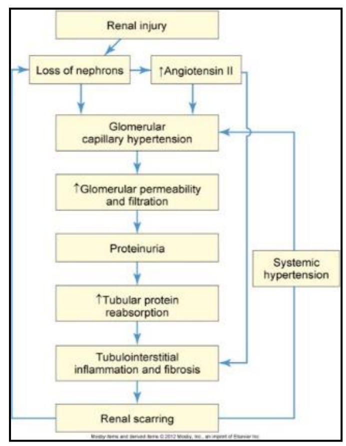


Figure 16: Pathophysiology in acute kidney failure

Clinical manifestations:

- Manifestations of uremia
- All systems are affected:
  - Reduced GFR, increased plasma creatinine, BUN
  - Increased sodium and water retention leading to edema and hypertension
  - Metabolic acidosis when GFR 30-40%
  - Bone change and subsequent fractures
  - Proteinuria and impaired fat and carbohydrate metabolism
  - Hypertension, CHF, edema
  - Pulmonary edema, Kussmaul respiration
  - Anemia, hypercoagulability & bleeding
  - Increased risk of infection

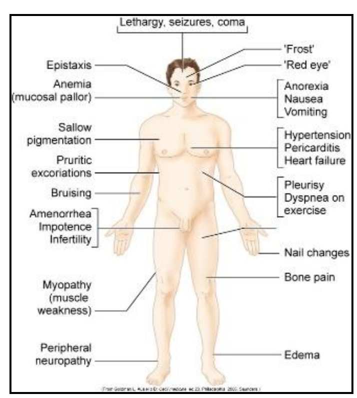


Figure 17: Signs and symptoms of kidney failure

- Uremic encephalopathy
- Peripheral neuropathy
- Impaired sexual/reproductive functions (anovulation, erectile dysfunction)
- Pruritus, sallow skin color

Thrombotic microangiopathies

- Vascular kidney lesions where thrombosis is common
- Characterized by a " thrombotic" triad of symptoms:
    1. Thrombocytopenia & Purpura
    2. Hemolytic anemia
    3. Renal damage / failure
- A common cause of acute renal failure
- Two common conditions, each exhibit additional different pathologies:
    1. **Hemolytic Uremic Syndrome**
    2. **Thrombotic Thrombocytopenic**

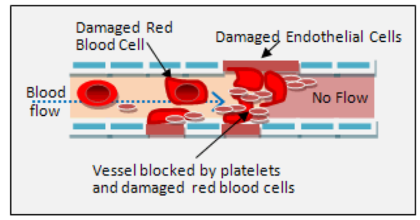


Figure 18: Vessel occlusion related to platelet accumulation and/or damaged red blood cells

Hemolytic-Uremic Syndrome (HUS)

Etiology:

- **Infections**
    i. *Bacterial*:
        - *E coli*; O:157 H:7
        - *Shiga* toxins (secreted by this *E coli* strain; names after *Shigella shiga* toxin as it has similar effect) causing damage to the blood vessel (endothelium/RBCs)

    ii. Viral infection

Pathophysiology

- **Thrombotic Triad**
- **Glomerular damage**
    - Cell swelling
    - Blood vessel occlusion with fibrin clots
- Acute hemolytic anemia (**schistocytes seen in peripheral blood smear**)
- Splenomegaly
- Thrombi in microcirculation including renal vasculature

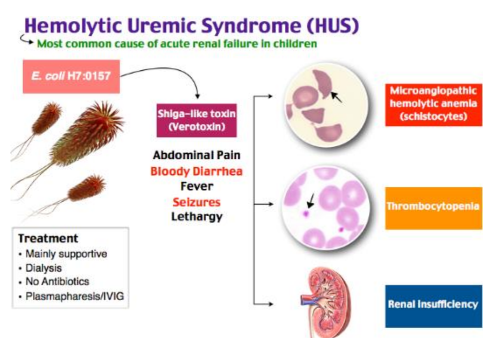


Figure 19: Hemolytic uremic syndrome (overview) [5]

Clinical manifestations

- Sudden onset of pallor, bruising or purpura, irritability, and oliguria
- Slight fever
- Anorexia
- Vomiting
- Diarrhea
  - Stool is watery and blood stained
- Abdominal pain, mild jaundice (due to hemolysis)
- Renal failure is apparent within the first few days, causing:
  - Metabolic acidosis
  - Hyperkalemia
  - Hypertension

Thrombotic Thrombocytopenic Purpura (TTP)

Etiology & Pathophysiology:

- **Thrombotic Triad**
- **Deficiency in ADAMTS13 enzyme**
  - ADAMTS13 is responsible for cleavage of large multimers of VWF (von Willebrand factor) protein

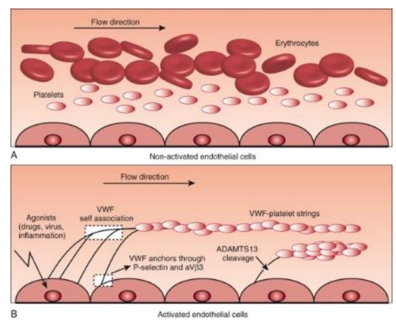


Figure 20: ADAMTS13 and VWF protein in TTP [6]

  - VWF is protein, which has two critical roles in coagulation cascade and clot formation at the site of injury
    - VWF is a sticky multimeric protein which binds platelet and injured subendothelium to create thrombus at site of injury.
  - VWF is a carrier for FVIII and protects it from proteolytic cleavage
  - The deficiency in ADAMTS13 enzyme → elevated levels of VWF large multimers → thrombosis
- Genetic mutations

Clinical manifestations:

- Differentiating features from HUS would be neurologic complications:
  - Recurrent symptomatic episodes including seizures and vision loss
- **Lab features:**
  - Serum ADMTS 13
  - Serum VWF

10: Fluids and Electrolytes, Acids and Bases

Water Balance

Total body water (TBW)

- **Intracellular fluid**
- **Extracellular fluid**:
  - Interstitial fluid
  - Intravascular fluid
  - Lymph, synovial, intestinal, CSF, sweat, urine, pleural, peritoneal, pericardial, and intraocular fluids

Age-Related Considerations

- **Children**
  - More susceptible to changes in body fluids
  - Greater body surface area
- **Elderly**
  - Decreased % total body water
  - Increased adipose tissue and decreased muscle mass
  - Diminished thirst perception
  - Decline in renal function

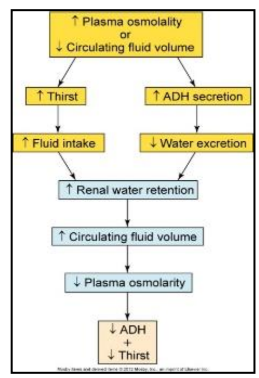


Figure 21: Mechanisms to maintain water balance in the human body

Water Balance- Regulation

- **Mechanisms**
  - [ADH]
    - Increases when person is dehydrated leading to water retention
  - Thirst sensation
  - Osmoreceptors (hypothalamus)
    - Stimulated by hyperosmolality and plasma volume depletion
  - Baroreceptors (blood vessels):
    - Stimulated by high blood pressure (hypervolemia)

Water Movement Between Fluid Compartments

- **Forces favoring filtration**:
  - Capillary hydrostatic pressure (blood pressure)
  - Interstitial oncotic pressure (water-pulling)

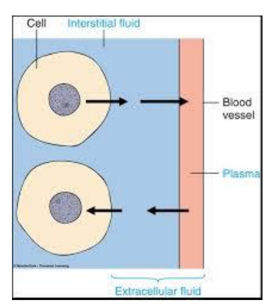


Figure 22: Movement of water between fluid compartments

- **Forces favoring reabsorption**:
    - Plasma oncotic pressure (water-pulling)
    - Interstitial hydrostatic pressure
- **Starling hypothesis**
    - **Net filtration** = [Forces favoring filtration] – [Forces opposing filtration]

Edema

- Accumulation of fluid within the interstitial spaces
- May be generalized OR organ specific

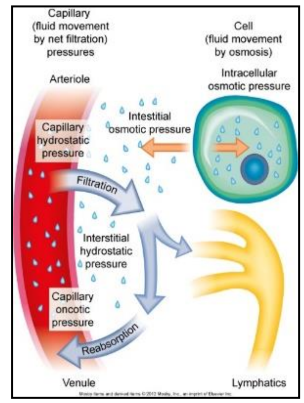


Figure 23: Water movement between fluid compartments

Causes:

- Increase in capillary hydrostatic pressure
- Decrease in plasma oncotic pressure
- Increase in capillary permeability
- Lymph obstruction (lymphedema)

Types

- **Pitting**:
    - Pregnancy
    - Heart failure
- **Non pitting edema**:
    - Lymphedema (due to lymphatic obstruction)
    - Note: Pretibial myxedema is not edema!

Sodium and Chloride Balance

Sodium

- Primary ECF cation
- Regulates osmotic forces, thus water
- **Functions**:
    - **N**euromuscular excitability
    - Acid-base balance
    - Cellular chemical reactions
    - Membrane transport

Chloride

- Primary ECF anion
- **Functions**
    - Electroneutrality

Regulation:

- Renin-angiotensin-aldosterone system (RAAS)
    - Aldosterone leads to:
        - Sodium and water reabsorption
        - Potassium and hydrogen secretion (loss in urine)
- Natriuretic peptides

Hyponatremia

- Serum sodium level <135 mEq/L
- Cause plasma hypo-osmolality and cellular swelling
    - Decreases the ECF osmotic pressure, so water moves into the cell via osmosis

Causes:

- Pure sodium loss
    - Occurs through body processes such as sweat
- Low intake
- Dilutional hyponatremia

Effects:

- Lethargy, confusion, decreased reflexes, seizures, and coma
- Hypovolemia, hypotension, tachycardia
- Dilutional hyponatremia is associated with weight gain, edema, ascites, and jugular vein distention

Hypernatremia

- Serum sodium >147 mmol/L
- Related to sodium gain or water loss

Causes:

- Water movement from the ICF to the ECF
- Administration of hyperosmolar solutions

Effects:

- Intracellular dehydration, convulsions
- Pulmonary edema
- Hypertension

Hypochloremia

- Usually the result of hyponatremia or elevated bicarbonate concentration
- Develops as a result of vomiting and the loss of HCl

- Observed in cystic fibrosis
- Alters acid-base balance

Hypokalemia

- Potassium level <3.5 mmol/L

Causes:

- **Shift of K+ from ECF into ICF**:
    - Insulin
        - E.g. Hyperglycemic patients receiving Insulin require fluid containing potassium to accommodate for the shift that occurs as potassium moves into cells
    - B2 agonists
- **Increased renal excretion**:
    - E.g. hyperaldosteronism

Effects:

- Skeletal muscle weakness
- Smooth muscle atony
- ECG changes

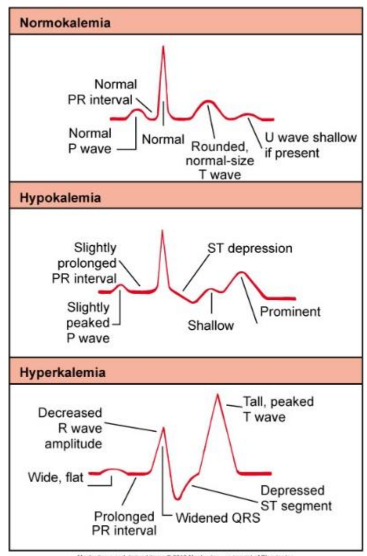


Figure 24: ECG changes related to potassium level

Hyperkalemia

- Potassium level >5.5 mmol/L
- Rare because of efficient renal excretion

Causes:

- **Shift of K+ from ICF into ECF**:
    - Insulin deficiency
        - E.g. diabetic ketoacidosis
    - Acidosis
    - Massive cell destruction (tumor lysis syndrome)
    - Decreased renal excretion
- Tissue trauma, crush injury
- Potassium-sparing diuretics

Effects:

- Dysrhythmias
- Cardiac arrest during diastole

Calcium & Phosphate

Calcium:

- 99% of stores are located in bone and 10% in serum
    - Of the 10% in serum, 50% is ionized and 50% is complexed with albumin (40%) or inorganic elements (10%), such as phosphate
- **Functions**:
    - Bone & teeth integrity
    - Blood clotting
    - Hormone secretion
    - Cell receptor function
    - Plasma membrane stability
    - Transmission of nerve impulses
    - Muscle contraction

Phosphate:

- Similar to calcium, most phosphate is also located in the bone
- **Functions**:
    - Bone integrity
    - High-energy bonds located in ATP
    - Creatine phosphate
    - Acts as an anion buffer

Regulation

- Calcium and phosphate concentrations are rigidly controlled
    - $Ca^{+2} \times HPO_4^{-} = K_{(constant)}$
    - If the concentration of one increases the other decreases;
        - I.e. If phosphate level increases, calcium absorption from gut decreases
- **Serum Ca^{+2}** : 8.8-10.5 mg/dL
- **Serum PO_4^{-}** : 2.5 to 4.5 mg/dL

Hypercalcemia

Causes

- Hyperparathyroidism
- Malignancy
- Sarcoidosis
- Excess vitamin D

Effects

- **Nonspecific symptoms including**:
    - Fatigue
    - Weakness
    - Lethargy
    - Anorexia
    - Nausea and/or vomiting
- Renal stones, polyuria, renal failure
- Bone pain
- Dysrhythmias, with potential for cardiac arrest during systole

Hypocalcemia

Causes

- PTH deficiency
- Vitamin D deficiency
- Inadequate intestinal absorption
- Bone/ tissue deposition of Ca

Effects

- Increased neuromuscular excitability
- Tingling and/or muscle spasm occurring in the hands, feet, and/or facial muscles
    - Muscle spasms may also manifest as intestinal cramping, hyperactive bowel sounds
- *Tetany*:
    - Carpopedal spams

Figure 25: Tetany of the hand

Acid-Base Imbalances

- **Normal arterial blood pH**
    - 7.35 to 7.45
    - Obtained by arterial blood gas (ABG) sampling
        - Venous blood gas (VBG) sampling is also used, however, the reference ranges differ
        - VBGs are often used for initial testing to rule out respiratory conditions and complications (e.g. in the emergency departments), as both registered nurses and laboratory technicians can collect venous blood samples when a Physician is unavailable

- **Acidosis**
  - Systemic increase in H^+ concentration or decrease in bicarbonate
- **Alkalosis**
  - Systemic decrease in H^+ concentration or increase in bicarbonate

Acid-Base Balance: Basic facts

- **Acid-base balance**: Careful regulation to maintain a normal pH
- **pH**: inverse logarithm of the H^+ concentration
  - When H^+ is high in number, pH is low (acidic);
  - When H^+ is low, pH is high (alkaline)
- **pH scale**: Logarithmic scale ranging from 0 -14 whereby 0 is very acidic, 14 is very alkaline
  - Each number represents a factor of 10
  - I.e. If a solution moves from a pH of 6 to a pH of 5, the number H^+ have increased 10 times (the solution is 10 times more acidic)
- **Sources of acids**
  - End products of protein, carbohydrate, and fat metabolism
- To maintain normal pH (7.35-7.45), excess H^+ is neutralized or excreted
- *Two major organs are involved in the regulation of acid and base balance*:
  1. **Lungs**
  2. **Kidneys**
- *Body acids exist in two forms*:
  1. **Volatile**
     - H_2CO_3 (can be eliminated as CO_2 gas)
  2. **Nonvolatile**
     - Sulfuric, phosphoric, and other organic acids
     - Eliminated by the renal tubules with the regulation of HCO_3^-

Buffering Systems

- A buffer is a chemical that can bind excessive H^+ or OH^- without a significant change in pH
- A buffering pair consists of a weak acid and its conjugate base
- The most important plasma buffering system is the carbonic acid-bicarbonate system
  - $CO_2 + H_2O \leftrightarrow H_2CO_3 \leftrightarrow H^+ + HCO_3^-$

Carbonic Acid-Bicarbonate System

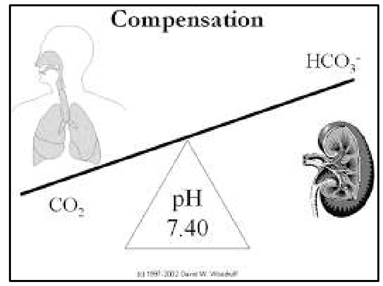


Figure 26: Respiratory (lungs) and metabolic (kidneys) compensation in carbonic-acid-bicarbonate system

- Operates in lungs and kidneys
- The greater the PCO_2, the more H_2CO_3 is formed
  - At pH of 7.4 (normal), the ratio of HCO_3 to H_2CO_3 is 20:1
  - HCO_3 and H_2CO_3 can increase or decrease, but ***ratio*** must be maintained to maintain normal pH required for body function
- For example, if HCO_3 decreases, pH decreases → (acidosis)
  - pH can be returned to normal if carbonic acid also decreases
  - This type of pH adjustment is referred to as compensation
- Respiratory system compensates by increasing ventilation to expire CO_2 or by decreasing ventilation to retain CO_2
- Renal system compensates by producing acidic or alkaline urine

Other Buffering Systems

Protein buffering (hemoglobin)

- Proteins have negative charges, so can serve as buffers for H^+

Renal buffering

- Secretion of H^+ in the urine and reabsorption of HCO_3^-

Ion exchange (between ICF and ECF)

- Exchange of K^+ for H^+ (through Na/K balance, Na/H balance) in acidosis and alkalosis

Acidosis and Alkalosis

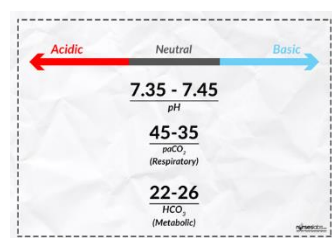


Figure 27: Normal ranges for determining acidosis and alkalosis

- Four categories of acid-base imbalances:
  1. **Respiratory acidosis**
     - Elevation of pCO_2 as a result of hypoventilation
  2. **Respiratory alkalosis**
     - Depression of pCO_2 as a result of hyperventilation
  3. **Metabolic acidosis**
     - Depression of HCO_3^- or an increase in non-carbonic acids
  4. **Metabolic alkalosis**
     - Elevation of HCO_3^- usually caused by an excessive loss of metabolic acids

Anion Gap

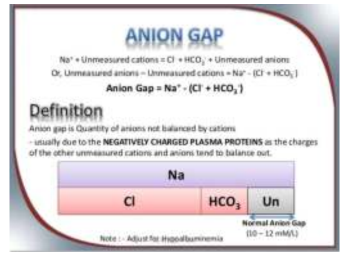


Figure 28: Definition of the anion gap

- DifferenceCalculated measure of the difference between cations and anions in plasma or urine
- A calculated parameter
- Used cautiously to distinguish different types of **metabolic acidosis**
- **Normal anion gap** = <12 mmol/L
  - By rule, the concentration of anions (–) should equal the concentration of cations (+), but not all anions are routinely measured
  -
- The anion gap increases in acidosis
  - Acidotic conditions can be remembered by the acronym MUD PILES CAT

Table - Conditions Leading to Acidosis (MUD PILES CAT)

M	methanol
U	uremia
D	diabetic / starvation ketoacidosis
P	phenformin/ paraldehyde
I	isoniazid, iron, ibuprofen
L	lactate
E	ethylene glycol
S	salicylates
C	cyanide, carbon dioxide
A	alcoholic ketoacidosis
T	toluene, theophylline

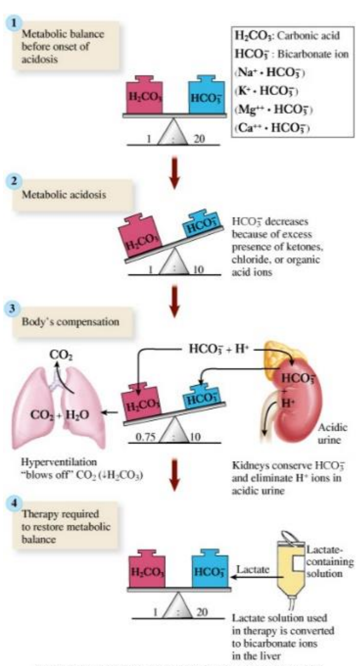


Figure 29: Metabolic acidosis and compensation

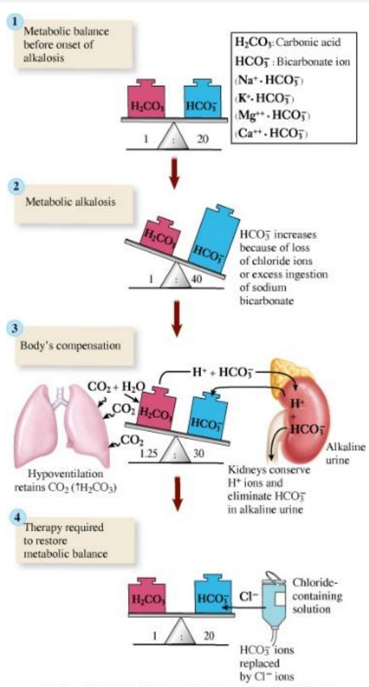


Figure 30: Metabolic alkalosis and compensation

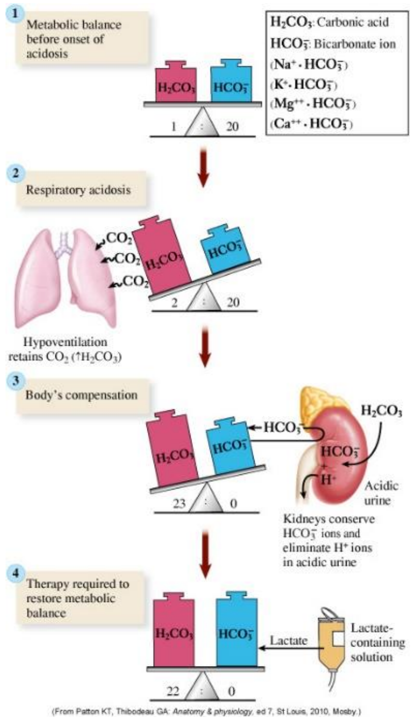


Figure 31: Respiratory acidosis and compensation

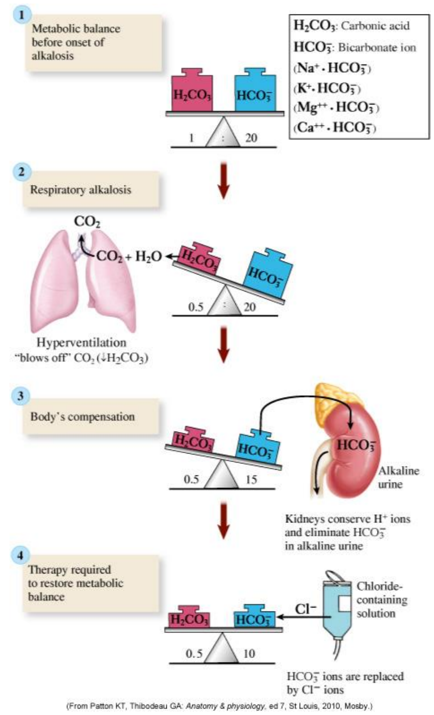


Figure 32: Respiratory alkalosis and compensation

11: Alterations of the Gastrointestinal Tract

Learning Objectives

- Review clinical manifestations of gastrointestinal dysfunction
- Explain the pathophysiology and clinical presentation of *achalasia*, hiatus hernia and *GERD*. Distinguish symptoms among those conditions
- Explain the pathophysiology of *pyloric* and *intestinal obstruction*
- Explain the classification of *peptic ulcer disease*
- Explain the pathophysiology and types of *gastrointestinal bleeding*
- Explain the pathophysiology of *appendicitis*
- Explain the pathophysiology of *inflammatory bowel disease (IBD)*; *Crohn's disease*, *ulcerative colitis (UC)*, *inflammatory bowel syndrome (IBS)*
- Explain the pathophysiology of *malabsorption syndrome*; various types eg. *pancreatitis*, *celiac disease*, *gall bladder diseases*
- Explain the pathophysiology of *diverticular diseases*
- Explain the pathophysiology of *liver cirrhosis* and *portal hypertension*
- Classify the different types of *jaundice* and describe causes and the clinicopathological features of each
- Explain the pathophysiology of *viral hepatitis* and *diagnostic markers*
- Explain briefly the pathophysiology of *obesity* and *malnutrition*

Clinical Manifestations of Gastrointestinal Dysfunction

- **Anorexia**
  - Lack of desire to eat despite physiologic stimuli that would normally produce hunger

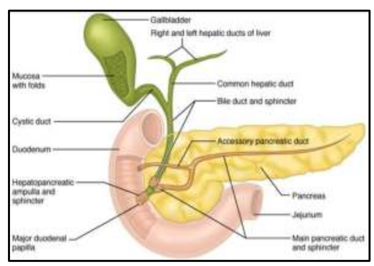


Figure 1: Accessory organs of the gastrointestinal tract

- **Nausea**
  - Subjective experience associated with a number of conditions
- **Vomiting**
- **Projectile vomiting**
  - Spontaneous vomiting that does not follow nausea
- **Retching**
  - Nonproductive vomiting
- **Constipation**
  - Infrequent or difficult defecation
    - Due to:

- Low-residue diet
- Sedentary lifestyle
- Changes in bowel habits
- Disease conditions:
  - GI diseases
  - Neurogenic disorders
- Medications

- **Diarrhea**
  - Increased frequency of bowel movements
  - Alterations in volume, fluidity, weight of stools…etc
    - Due to:
      - Infection
      - Malabsorption syndromes
      - Autoimmune disease
      - Medications
- **Abdominal pain**
  - Symptom of a number of gastrointestinal disorders
  - Occurs in many forms, including:
    - Parietal pain
    - Visceral pain
    - Referred pain
  - Some GI diseases have distinctive pain patterns, such as rebound tenderness in appendicitis, however, often abdominal pain is a vague symptom that requires other diagnostic and clinical investigations to determine the cause
- **Gastrointestinal bleeding**
  - Upper GI
    - Esophagus, stomach, or duodenum
  - Lower GI
    - Jejunum, ileum, colon, or rectum
  - Forms of bleeding:
    - *Hematemesis*: Vomiting of blood
    - *Melena*: Passage of black, tarry stools (digested blood)
    - *Hematochezia*: Passage of fresh blood through the anus usually with stools

        - Indicates colonic origin
    - *Occult bleeding*: Hidden blood in stool
        - Detected using special techniques in the laboratory
    - *Bleeding per rectum*
        - Passage of fresh blood through the anus due to fissure or hard stools
- **Dysphagia/ odynophagia**: Difficulty/ pain in swallowing
    - Mechanical obstructions
    - Functional obstructions

Achalasia

- Failure of relaxation of smooth muscle fibers (SMF) of the lower esophageal sphincter (LES)

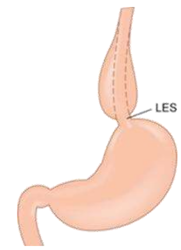


Figure 2: Location of the lower esophageal sphincter implicated in achalasia

Etiology & Pathophysiology:

- A normal functioning LES maintains a zone of high pressure to prevent chyme reflux
- Angle of His acts as a valve to prevent backflow
- A balance of excitatory/ inhibitory impulses to SMF regulates muscular coordination required in the esophagus and LES
    - The GI tract is stimulated (excited) by the neurotransmitter acetylcholine and inhibited by nitric oxide and vasoactive intestinal peptide
    - Imbalance in these impulses result in failure of relaxation and subsequent obstruction causing food to build-up in the esophagus behind the obstruction

Clinical manifestations/diagnosis:

- Dysphagia to both solids and liquids
- Regurgitation
- Chest pain
- Obstruction appear in esophageal manometry, barium studies
    - On x-ray the esophagus appears massively dilated
- Similar conditions: GERD, hiatus hernia, psychosomatic disorders

Figure 3: Massively dilated esophagus seen in achalasia

Gastroesophageal Reflux Disease (GERD)

- Reflux of chyme from the stomach to the esophagus
- Leads to inflammation (reflux esophagitis)

Etiology & Pathophysiology:

- Hiatal hernia due to mechanical and motility factors

- Impaired esophageal motility
  - Eg. Multiple sclerosis
- Increased gastric acidity:
  - Eg. Zollinger-Ellinson syndrome
- Increased abdominal pressure
- Obesity increases the severity

Clinical manifestations:

- Heartburn
- Regurgitation of chyme
- Mid-epigastric pain within 1 hour of eating
- **Others**:
  - Increased salivation
  - Chest pain
    - This can be very intense and present similar to a myocardial infarction
    - Patients presenting with central, epigastric pain require an ECG to rule out cardiac involvement, however, in the absence of other risk factors acid reflux should be considered as part of the differential diagnosis and the administration of an antacid to alleviate the pain

Figure 4: Barrett's esophagus

Complications:

- Strictures and narrowing (reflux-induced inflammation), Barrett's esophagus (**Figure 4**)

Hiatal hernia

- Protrusion (or herniation) of the upper part of the stomach into the thorax through a tear or weakness in the diaphragm

Etiology & Pathophysiology:

- Defect/ weakness in the diaphragm
- Increased intra-abdominal pressure
  - Eg. Straining, severe violent cough

Types:

- Sliding hiatal hernia (Figure 5-A)
- Para-esophageal (rolling) hiatal hernia (Figure 5-B)

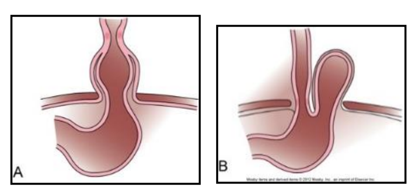


Figure 5: Sliding hiatal hernia (A) and para-esophageal (rolling) hiatal hernia (B)

Clinical manifestations:

- No symptoms
- Chest pain
- Symptoms of acid reflux

- Shortness of breath (hernia's pressure on the diaphragm)
- Palpitations (from irritation of the vagus nerve (cranial nerve X)

Pyloric Obstruction

- Blocking or narrowing of the opening between stomach & duodenum

Etiology & Pathophysiology:

- Can be acquired or congenital (pyloric atresia)
- Obstruction impairs emptying of gastric contents into the duodenum
- Persistent vomiting leads to hypochloremia and subsequent metabolic alkalosis
    - Compensatory respiratory acidosis → hypoventilation (pCO_2)
- Hypovolemia (decreased fluid volume) leads to dehydration and electrolyte disturbance, and secondary hyperaldosteronism

Manifestations:

- Epigastric pain and fullness
- Nausea
- Vomiting
- Succussion splash (gastric splash)
- Malnutrition, dehydration, and extreme debilitation (with prolonged obstruction)

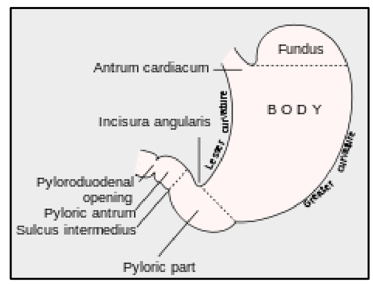


Figure 6: Anatomy of the stomach (focus on pylorus)

Intestinal Obstruction

- Mechanical or functional obstruction of the intestine, preventing the normal transit of products of digestion

Etiology:

- Adhesions post abdominal surgery (most common cause)
- Hernias containing bowel
- Inflammatory strictures
    - E.g. Crohn's disease
- Intussusception (telescoping bowel into distal segment causing ischemia and necrosis)
- Foreign bodies
- Intestinal atresia
- Neoplasms
- Diverticulitis / Diverticulosis

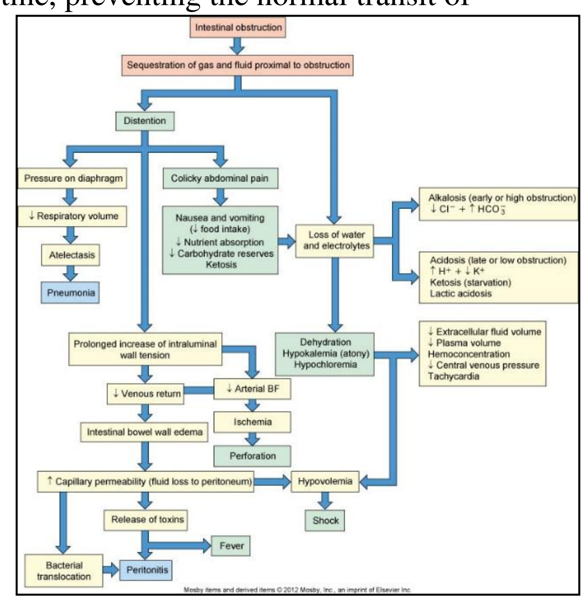


Figure 7: Intestinal obstruction pathophysiology and clinical manifestations

- Fecal impaction
- Functional obstruction
    - E.g. Paralytic ileus:
        - Impairment of normal propulsive ability of the GI tract
        - A common post-surgery side effect

Pathophysiology and Clinical Manifestations:

- Can occur at any level distal to the duodenum
    - Mechanical obstruction of chyme flow through the intestinal lumen
    - Functional obstruction related to poor intestinal motility without an obstructing lesion
- A medical emergency
- Distension and colicky pain
- Nausea and vomiting
- Hypovolemia, dehydration, shock
- Electrolyte imbalance
- Silent abdomen (in complete obstruction)

Diagnostic Testing and Evaluation

- Radiology
    - Abdominal imaging by x-ray, computed tomography, and ultrasound can assist in determining presence and location of obstruction
    - Endoscopy can confirm

Gastritis

- Inflammatory disorder of the gastric mucosa

Etiology:

- Peptic ulcer/*H pylori* and NSAIDs are the most common
- Others:
    - Alcohol, severe illness, pernicious anemia

Pathophysiology:

- Can be acute or chronic (*review pathophysiology of inflammation)*
- NSAIDs (*consider mechanism of action)*
    - Eg. NSAIDs inhibit prostaglandins which stimulate mucus production to protect the lining of the stomach

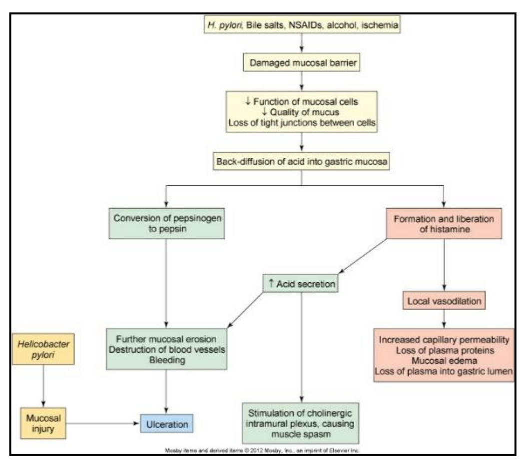


Figure 8: Pathophysiology of gastric ulcers

- Sites: fundal or antral

Clinical manifestations:

  - *Abdominal pain*: central, dull aching
  - Gastric ulcer and its complications (see **Figure 8**)

Peptic Ulcer Disease

- A break or ulceration in the protective mucosal lining of stomach, or duodenum

Etiology and risk factors:

- **Main pathology**: increased gastric acid secretion
- High gastrin levels
- Use of NSAIDs
- Cigarette smoking
- Zollinger-Ellison syndrome
  - Excess secretion of the hormone gastrin from one or more tumors located in the pancreas and/or duodenum
  - Causes increase in gastric acid secretion
- *H pylori* toxins promote inflammation and ulceration

Pathophysiology:

- Duodenal ulcers are the most common
  - Gastric ulcers tend to develop in antral region
- Acute or chronic
- Superficial erosions or deep true ulcers

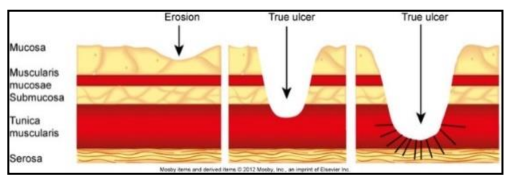


Figure 9: Degrees of erosion in stomach mucosa from superficial erosions to "true ulcer"

Clinical manifestations:

- Pain
- Manifestation of hyperacidity
- *Stress ulcer*: A peptic ulcer that is related to severe illness, sepsis, neural injury, or systemic trauma, eg. TBI, burns

Pathophysiology:

- Physiologic stress of a serious illness
- Ischemia and hypoxic injury

Gastrointestinal Bleeding

Etiology:

- **Upper GI bleeding** (more common):
  - Peptic ulcer disease (associated with *H. pylori* infection)
  - Esophagitis and esophageal varices due to liver cirrhosis
  - Medications (eg. NSAIDs)
  - Others: cancer, angiodysplasia
- **Lower GI Bleeding**:
  - Hemorrhoids
    - Ulcerative colitis and Crohn's disease
  - Others: cancer, angiodysplasia

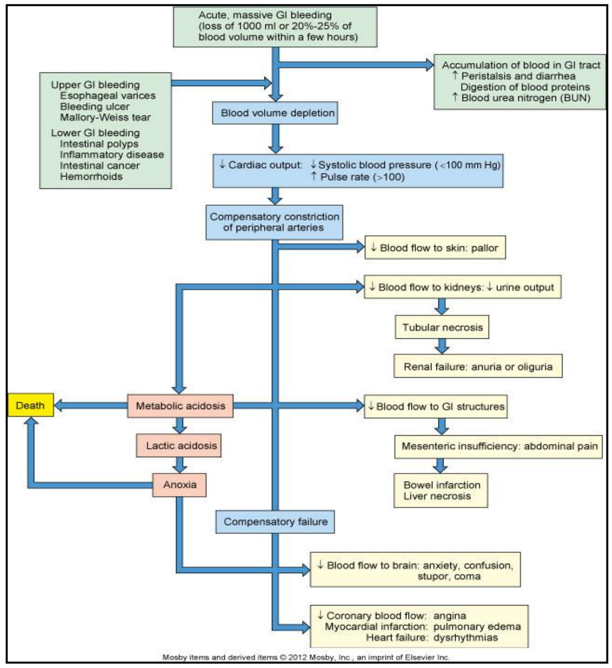


Figure 10: Pathophysiology, clinical manifestations and complications of gastrointestinal bleeding

Appendicitis

- Inflammation of the vermiform appendix

Etiology:

- Lymphoid hyperplasia
- Obstruction of lumen: Fecaliths, FB
- Infection

Pathophysiology:

- Mechanical blockage of the appendix leads to increased intraluminal pressure, decreased blood flow, inflammation
- Bacterial growth increases inflammation (swelling, tissue injury)

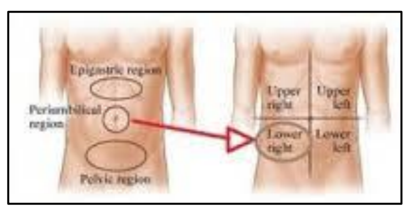


Figure 11: Locations of pain in appendicitis

Clinical manifestations

- Epigastric and RLQ pain
- Rebound tenderness
- Peritonitis (most serious complication)

Inflammatory Bowel Diseases (IBD) -KNOW

- A group of disorders characterized by chronic and relapsing inflammation of the bowel
- 1 /150 Canadians is living with Crohn’s or Colitis (the highest rate worldwide)
- A leading cause for colorectal cancer

- Colorectal cancer is 3rd most common cancer in Canada
- Anyone > 50 y should be screened for colon cancer

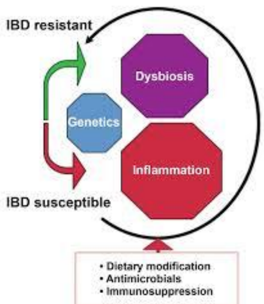


Figure 12: Therapeutic modifications in management of IBD

Figure 13: Ulcerative colitis and Crohn's disease

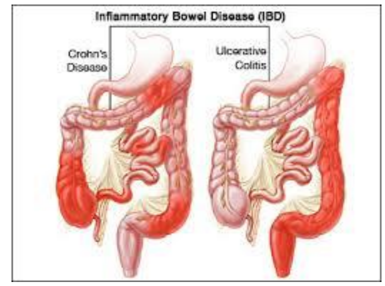


Figure 14: Sites of lesions in UC versus Crohn's disease

	Crohn's Disease	Ulcerative Colitis
Affected sites 	Any part of GI from mouth to anus Commonly terminal ileum	Colon , rectum
Lesion Spread	Discontinuous or " skip lesions"	Continuous
Extent of inflammation	Deep (transmural)	Superficial (mucosa, sub-mucosa)
Intestinal manifestations	Abdominal pain, cramping bloating Severe diarrhea, bloody stools Weight loss Mouth ulcers	
Pathophysiology	Inflammation Focal neutrophil infiltration Granuloma formation Mouth ← Ulcerations → lower colon/ rectum	
Extra intestinal manifestations	Uncommon	Common (eg arthritis, spondylitis)
Cancer risk	1-3%	Up to 25%

Table - Comparison of Crohn's disease and Ulcerative Colitis

Irritable Bowel Syndrome (IBS)

- A **functional** gastrointestinal disorder with no specific structural or biochemical alterations

Clinical manifestations:

- Recurrent abdominal pain and discomfort associated with altered bowel habits
  - Can be diarrhea or constipation-predominant or alternating
- Gas, bloating, and nausea

- Symptoms are usually relieved with defecation
- Associated with anxiety, depression, chronic fatigue syndrome

	Irritable Bowel Syndrome	Ulcerative Colitis	Crohn's Disease
Symptoms	Bloating Cramping Diarrhea Nausea Constipation Abdominal Pain Gas Incomplete Bowel Movements Fatigue	Bloating Cramping Nausea Vomitting Severe Diarrhea Bloodly Stool Abdominal pain Weight Loss Fatigue	Bloating Cramping Nausea Vomitting Severe Diarrhea Bloodly Stool Mucus in Stool Abdominal pain Weight Loss Higher Likelihood of Fistulae Fatigue
Diagnosis	Elimination of other diseases. Colonoscopy & biopsy does not show signs of disease.	Ulcer formation on interior wall of large intestine are visible via colonoscopy. Biopsy will also show signs of disease.	Attacks the interior wall & inner layers of the intestinal wall. Ulceration is less likely, but intestine will show inflammation. Biopsy will show signs of disease.
Classification	Functional Disorder	Inflammatory Bowel Disease	Inflammatory Bowel Disease
Possible Causes	Usually develops after trauma to the gut such as food poisoning or stomach flu. A recent study suggests micro-inflammation causing overproduction of histamines may be the root cause of IBS. Also considered are: bacterial overgrowth, nervous system disorders or psychosomatic causes.	Autoimmune system of the body attacks the gut, causing inflammation & ulcers on interior wall of large intestine. Genetics, bacterial infection or overuse of antibiotics may also play a part.	Autoimmune system of the body attacks the gut, causing deep inflammation in the large intestine and possibly small intestine. Genetics, bacterial infection and/or overuse of antibiotics may also play a part.
Common Treatments	Diet Probiotics Fiber Antispasmodics Laxatives Antidepressants	Diet Antibiotics Anti-inflammatories Corticosteriods Immunosuppressive Drugs Surgery	Diet Antibiotics Anti-inflammatories Corticosteriods Immunosuppressive Drugs Surgery

Figure 15: Table - Comparison IBS, Ulcerative Colitis and Crohn's disease (for review)

Pancreatitis

- Inflammation of the pancreas

Etiology & Pathophysiology:

- Acute or chronic
- Injury/ inflammation leads to leakage of pancreatic enzymes
- Autodigestion of pancreatic tissue and other organs
- Can be associated with several other clinical disorders

Clinical manifestations and evaluation:

- Mid epigastric pain radiating to the back
- Fever and leukocytosis
- Hypotension and hypovolemia
  - Enzymes increase vascular permeability
- Elevated serum lipase & amylase

Malabsorption Syndromes

- Pathological interference with normal physiological sequence of digestion, absorption and transport of nutrients
  - *Maldigestion*: failure of chemical processes of digestion
  - *Malabsorption*: failure of intestinal mucosa to absorb digested nutrients
  - Combined

Etiology (selected):

- **Infections**: traveler diarrhea, intestinal parasites, intestinal TB, HIV
- **Surgical**: post gastrectomy, vagotomy, bariatric surgery
- Celiac disease
- Enzyme deficiencies:
  - Intestinal disaccharidase deficiency
  - Eg. lactase intolerance
- Pancreatic disease
  - Eg. cystic fibrosis
- Terminal ileal disease such as Crohn's disease, surgical resection
- Obstructive jaundice
- **Endocrine diseases**: thyroid, adrenal
- **Diet**: fiber deficiency
- Malnutrition

Malabsorption Syndromes- Selected

Pancreatic insufficiency

- Insufficient pancreatic enzyme production (lipase, amylase, trypsin, chymotrypsin)
- **Causes**:
  - Pancreatitis
  - Pancreatic carcinoma
  - Pancreatic resection
  - Cystic fibrosis
- Fat maldigestion → steatorrhea and weight loss

Lactase deficiency

- No breakdown of lactose into monosaccharides → reduced absorption of monosaccharides
- Fermentation of lactose by bacteria → gas, cramping pain, flatulence, and osmotic diarrhea

Bile salt deficiency

- **Cause**: liver disease and bile obstruction
- **Pathophysiology**:
    - Bile salts are synthesized from cholesterol/ bile acids inside liver
    - Conjugated bile salts are needed to emulsify/absorb fat
    - Poor intestinal absorption of lipids causes steatorrhea, diarrhea, and poor absorption of fat-soluble vitamins (A, D, E, K)

Fat-soluble vitamin deficiencies and their effects:

Vitamin A

- Night blindness

Vitamin D

- Decreased calcium absorption
- Bone pain
- Osteoporosis
- Fractures

Vitamin K

- Prolonged prothrombin time (PT)
- Bleeding tendencies

Diverticulitis & Diverticulosis

Pathophysiology:

- Acute inflammation of a **diverticulum** (mucosal outpouching of colon wall), especially the sigmoid colon
    - Results from increased pressure within the lumen as diameter shrinks in response to consumption of low residue, refined diets
- Obstruction by fecal matter and undigested food particles
- Increased colonic intraluminal pressure
- Hypertrophy, thickened mucosal folds

Clinical manifestations:

- Asymptomatic (Diverticulosis, absence of inflammation)
- Left lower quadrant abdominal pain
- Fever, leukocytosis, melena

Complications:

- Abscess, perforation, fistula

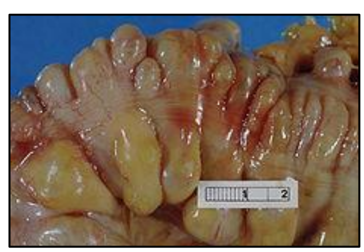


Figure 16: Diverticular disease

Jaundice (icterus)

- Occurs in unconjugated and conjugated hyperbilirubinemia

Table – Etiologies of Hyperbilirubinemia (Jaundice; icterus)		
Etiology	**Unconjugated**	**Conjugated**
Overproduction	- Hemolysis - Increased ineffective hematopoiesis	- **Hepatocellular disease**: cirrhosis, hepatitis - Sepsis
Decreased hepatic uptake	- Gilbert's syndrome - **Drugs**: eg. rifampin	
Decreased hepatic conjugation	- Prematurity/ neonatal jaundice - Crigler-Najjar	- Biliary obstruction - Gall bladder disease (stone, inflammation, Mg) - Pancreatic disease (inflammation, cancer)

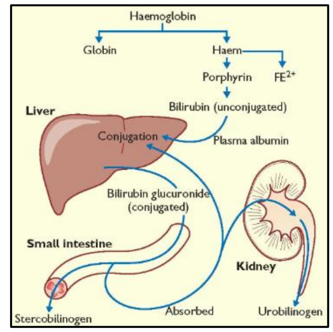


Figure 17: Etiology of hyperbilirubinemia (jaundice; icterus)

Viral Hepatitis

- Systemic viral disease that primarily affects the liver
- Acute or chronic or fulminant hepatitis

Causes:

	Genetics	Transmission	Chronic disease?	Carrier Sate?	Liver cancer?	*Vaccine?
Hepatitis A	RNA	Fecal-oral	No	No	No	Yes inactivated
Hepatitis B	**DNA**	Parenteral, Sexual	Yes (10%)	Yes	Yes	Yes recombinant
Hepatitis C	RNA	Parenteral	Yes (60-80%)	Yes	Yes	No
Hepatitis D	RNA	Parenteral, HBV coinfection	Yes	Yes	Yes	Yes HBV vaccine
Hepatitis E	RNA	Fecal-oral	No	No	No	No

HBV Serological Diagnosis and Interpretation					
	HBsAg*	**HBeAg**	**HBc Ab (IgM)**	**HBc Ab (IgG)**	**HBsAb**
Acute Infection	+ < 6mo	+	+	-	-
Chronic Infection	+ > 6mo	+	-	+	-
Immunized	-	-	-	-	+

Liver Cirrhosis

- Irreversible damage due to chronic/advanced liver disease
- At least 70- 80% of liver replaced by fibrous tissue

Etiology:

- Chronic alcoholism
- Chronic hepatitis (viral hepatitis B, C)
- Primary biliary cirrhosis (autoimmune)
- Haemochromatosis
- Alpha-1 antitrypsin deficiency,

Pathophysiology and Clinical Manifestations

- Low albumin, ascites, peripheral edema
- Portal hypertension, esophageal varices, splenomegaly, caput medusa
- Increased bilirubin, Jaundice
- Bleeding tendencies
- **Manifestations of increased estrogens**:
    - Gynecomastia
    - Impotence
    - Infertility
    - Loss of sexual drive
    - Testicular atrophy
- Fetor hepaticus
- **Skin manifestations**:
    - Spider nevi, palmar erythema
- **Hepatorenal syndrome**:
    - Renal failure due to advanced liver disease

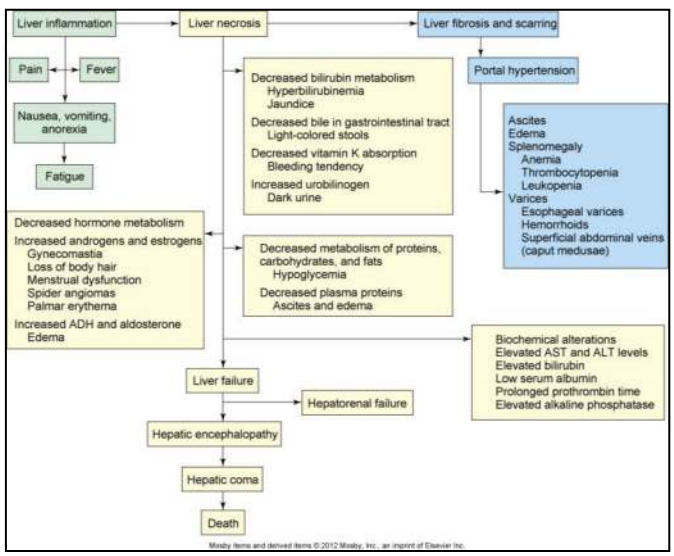


Figure 18: Pathophysiology and clinical manifestations of liver cirrhosis

- **Hepatic encephalopathy**:
  - Neurologic manifestations of impaired cognitive function
  - Flapping tremor (results from ammonia crossing blood brain barrier of the brain)

Portal hypertension (review anatomy of liver and portal circulation)

- ↑pressure in portal system due to resistance to portal blood flow

Etiology:

- **Pre-hepatic**: portal vein thrombosis or congenital atresia
- **Intrahepatic**: *liver cirrhosis*, hepatic fibrosis, massive fatty change, military TB
- **Post-hepatic**: obstruction at any level between liver and heart; thrombosis in hepatic vein or IVC, constrictive pericarditis

Pathophysiology:

- Open portosystemic collaterals
- Systemic venous congestion
- Reduced blood flow to liver
- Toxic substance pass directly to systemic circulation

Clinical manifestations/complications:

- Bleeding varices in lower esophagus, stomach
- Splenomegaly
- Ascites
- Hepatic encephalopathy
- Hepatorenal syndrome

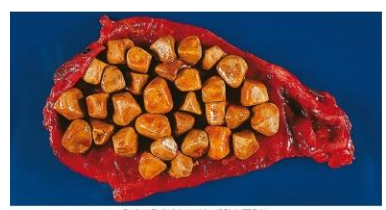


Figure 19: Gallstones

Gallbladder Diseases

- **Obstruction**
- **Inflammation (cholecystitis)** and the most common
- **Cholelithiasis**: gallstone formation

Gall Bladder Stones

Risk factors:

- Obesity
- Middle age
- Female
- Native American ancestry

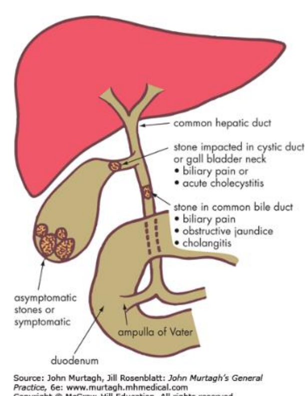


Figure 20: Pathophysiology of cholelithiasis (gallstones)

Pathophysiology:

- Most common is the cholesterol and pigmented stones
- **Theories**:
    - Enzyme defect increases cholesterol synthesis
    - Decreased secretion of bile acids to emulsify fats
    - Decreased reabsorption of bile acids from ileum
    - Gallbladder hypomotility and stasis
    - Genetic predisposition

Obesity at a Glance

- Body mass index (*BMI) > 30
- **Regulation of body weight**:
    - Hypothalamus
    - Hormones controlling appetite/weight:
        - Insulin
        - Leptin
        - Ghrelin
        - Adiponectin
- A major cause of morbidity, mortality and increased health care costs

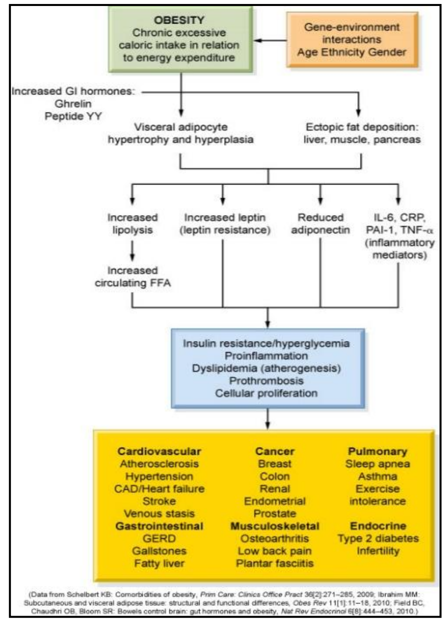


Figure 21: Pathophysiology of obesity

*Starvation & Malnutrition

- Decreased caloric intake leading to weight loss and eventually cachexia
- Short-term starvation can be compensated by:
    - Glycogenolysis
    - Gluconeogenesis
- Long-term starvation leads to malnutrition disorders:
    - **Marasmus
    - **Kwashiorkor

Cancers of the Gastrointestinal Tract

- Esophagus
- Stomach
- Liver
- Gallbladder
- Pancreas

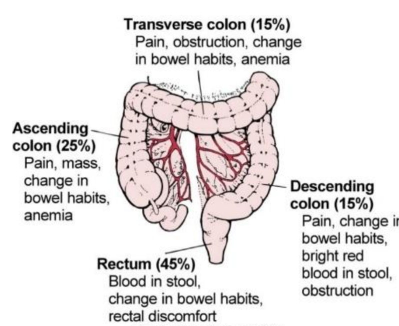


Figure 22: Locations for colon cancer, relative incidence, and associated manifestations

- Colon cancer
- Important for all cancers:
    - Risk factors
    - Screening
    - Precancerous lesions

Selected Digestive Disorders of Children

Gluten-Sensitive Enteropathy (Celiac Disease)

- A type of malabsorption syndrome
- Impaired absorption due to mucosal sensitivity to gluten

Etiology:

- Dietary, genetic, and immunologic factors

Pathophysiology:

- Gluten is the protein component in cereal grains (BROW; Barley, Rye, Oat and Wheat)
- Acts as a toxin that damages villous epithelium in small intestine
- Associated with other autoimmune disorders (IgA deficiency)

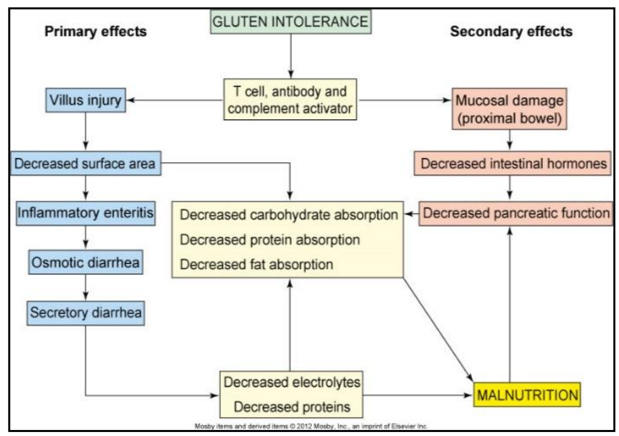


Figure 23: Pathophysiology of gluten-sensitive enteropathy (Celiac disease)

Clinical manifestations:

- Manifestations of malabsorption symptoms
- Failure to grow and thrive
- Confirmation by a tissue biopsy
- **Celiac crisis**:
    - Severe diarrhea, dehydration

Biliary Atresia

- Congenital malformation characterized by <u>absence</u> or <u>obstruction</u> of intrahepatic or extrahepatic bile ducts

Pathophysiology/ Clinical manifestations:

- Block, inflammation, and fibrosis of the bile canaliculi
- Cholestasis
- Jaundice is the primary clinical manifestation
- 80% die before 3 years if untreated
- Liver transplant long-term therapy

Kwashiorkor and Marasmus

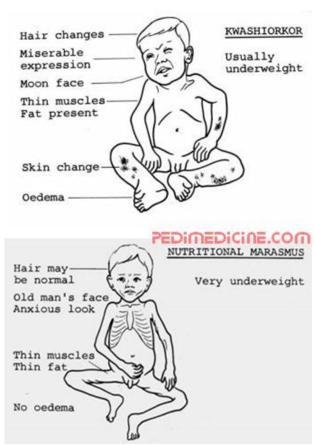
- Both are:
  - Known collectively as protein energy malnutrition (PEM)
  - Types of malnutrition associated with long-term starvation

Pathophysiology:

- **Kwashiorkor**: a severe protein deficiency
- **Marasmus**: a deficiency of all nutrients
- **Impaired liver function**:
  - In kwashiorkor, the lack of proteins causes the liver to swell because of the inability to produce lipoproteins for cholesterol synthesis
  - In marasmus, liver function still continues, but the overall caloric intake is too low to support cellular protein synthesis

Clinical manifestations:

- Manifestations of malnutrition
- Stunted physical and mental development
- The presence of subcutaneous fat, hepatomegaly, and a fatty liver (kwashiorkor) differentiates kwashiorkor from marasmus



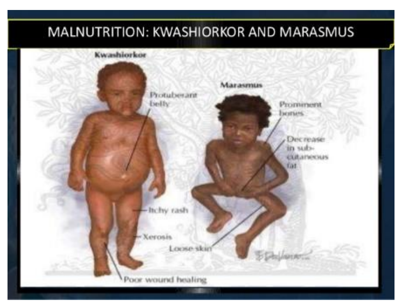


Figure 24: Comparison of kwarshikor and marasmus

12: Alterations of the Musculoskeletal and Integumentary Systems

Learning Objectives

- Review clinical manifestations of musculoskeletal diseases
- Explain the pathophysiology of bone fractures & osteoporosis
- Explain the pathophysiology *osteomyelitis*
- Explain the pathophysiology and types of *osteoarthritis*
- Explain the pathophysiology of *impaired uric acid metabolism* and *gout*
- Explain the pathophysiology of *fibromyalgia*
- Explain the pathophysiology of *various skin lesions*
- Explain the pathophysiology of *pressure ulcers*
- Explain the pathophysiology of *scleroderma*
- Explain the pathophysiology of *burn*
- Explain the pathophysiology of *common inflammatory disorders of the skin* and *psoriasis*
- Explain the pathophysiology of *common skin cancers*

Review: Anatomy and Physiology of the Musculoskeletal and Integumentary Systems

- Bone tissue is a solid connective tissue (review anatomy and physiology of bone tissue)
    - Cartilage is avascular while bone is very vascular
        - This is why intraosseous access can be effectively used for emergency vascular access in emergency and trauma situations where vascular access is not possible
    - The osteocyte is the “Mother Cell” for bone tissue
- Review of anatomy and physiology of the Integumentary System

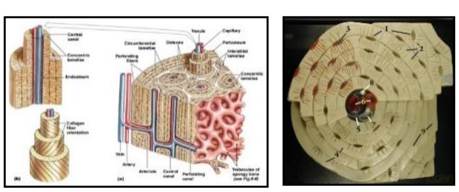


Figure 1: Compact and spongy bone

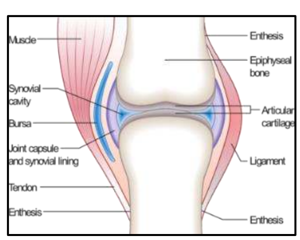


Figure 2: Synovial joint

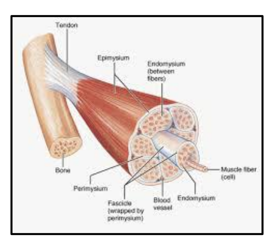


Figure 3: Muscle structure

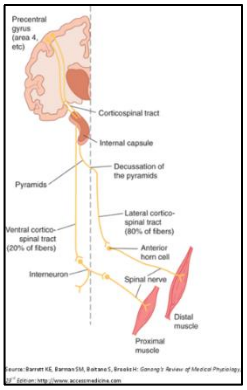


Figure 4: Corticospinal tract

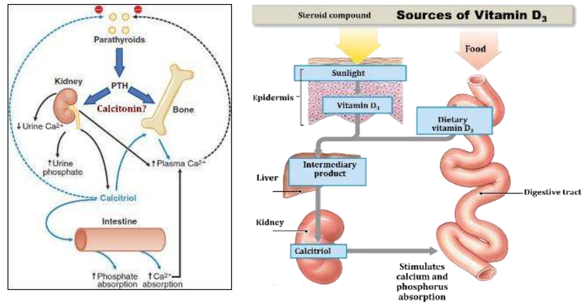


Figure 5: Vitamin D Synthesis

Bone Fractures

- A fracture is a break in the continuity of a bone

Classifications:

- Complete or incomplete
- Closed or open
- Comminuted
- Linear
- Oblique
- Spiral
- Transverse
- Greenstick – children (just a bend)
- Pathologic

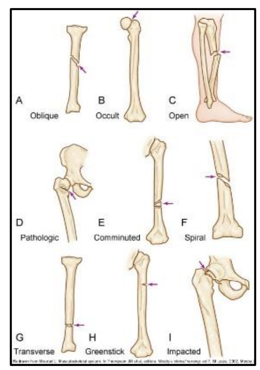


Figure 6: Classifications of bone fractures

Pathophysiology:

- Broken bone causes damage to surrounding tissue, periosteum, and blood vessels with hematoma formation
  - Inflammation process begins
- Bone tissue destruction triggers an inflammatory response and repair (callus formation)

Clinical manifestations:

- Numbness up to 20 minutes following injury
- Swelling, tenderness, pain, impaired sensation, muscle spasms, malalignment

Osteoporosis

- Porous poorly mineralized bone

Etiology:

- Decreased levels of estrogen and testosterone. Common in postmenopausal women (estrogen is an anabolic hormone)
- Inadequate levels of vitamin D (osteomalacia)
- Decreased physical activity level (age-related)
- Excess intake of caffeine, phosphorus, alcohol, nicotine
- Drug-induced eg glucocorticoids

Clinical manifestations:

- Bone aches, weakness & deformities, fractures

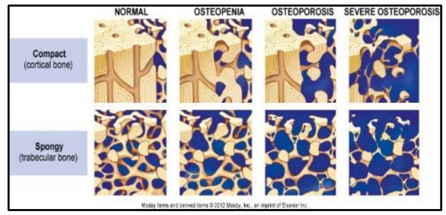


Figure 7: Comparison bone density in stages of osteoporosis to normal bone

Pathophysiology:

- Reduced bone mass/density and an imbalance of bone resorption and formation
  - Normal bone density = >833mg/cm^2
  - Osteopenic bone = 833-648 mg/cm^2
  - *Bone density <648 mg/cm^2 = osteoporosis

Diagnostic Evaluation and Testing

- **Bone histology**:
  - Usually normal but lacks structural integrity
  - Thinning and perforations
- **Bone densitometry** (dual-energy x-ray absorptiometry [DXA, DEXA])
  - Simple imaging test for determining bone density
  - Common in the diagnosis of osteoporosis

Osteomyelitis

- Bone infection with progressive inflammatory destruction, usually from **bacteria**
- Very serious

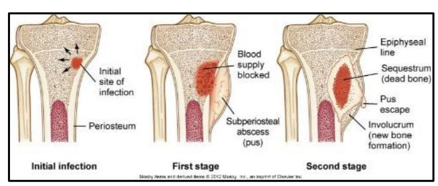


Figure 8: Progression of osteomyelitis

Etiology:

- **Staphylococcal infection**:
  - Open wound (exogenous)
  - Blood-borne (endogenous)

Risk factors:

- Trauma/surgery
- Immunocompromised patients
- Diabetes mellitus
- Poor vascular supply

- Peripheral neuropathy

Pathophysiology and clinical manifestations

- Acute or chronic inflammation
- Necrotic bone
- Pain, erythema, tenderness, swelling± abscess
- Fever
    - Can be a cause for fever of unknown origin (FUO)

Osteoarthritis

- Inflammatory/degenerative joint disease

Etiology:

- Primary disease is idiopathic
- Genetic, biochemical, and biomechanical factors

Risk factors:

- Joint trauma, long-term mechanical stress
- Endocrine disorders (hyperparathyroidism)
- Obesity, aging

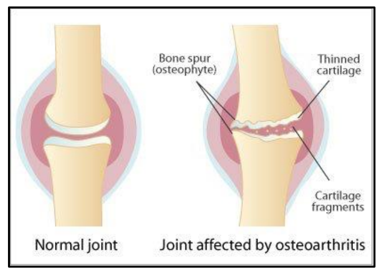


Figure 9: Normal versus osteoarthritic joint

Pathophysiology:

- Local areas of degeneration and loss of articular cartilage
- Increased production of pro-inflammatory cytokines
- Tissue catabolism exceeds repair
- Thickening of the joint capsule
- Formation of bone spurs (**osteophytes**)
- Variable degrees of synovitis (inflammation of the synovial membrane)

Clinical manifestations:

- Pain (worsens with activity)
- Stiffness (diminishes with activity)
- Swelling of the joint
- Tenderness
- Limited mobility
- Deformity

Rheumatoid Arthritis

- Systemic inflammatory/autoimmune disease primarily affecting joints

Etiology:

- Multifactorial with strong genetic predisposition
- Association with **HLA-DR4**
    - **HLA** (Human Leukocyte antigen) OR **MHC** (major histocompatibility complex):
        - Group of proteins coded for by genes on chromosome 6 that makes for our tissue identity.
    - There are 6 sets of genes in HLA forming two classes HLA I and II the pattern of which make us each unique.
        - Class 1 = A, B, C.
        - Class 2 = DR, DQ and DP.
    - HLA typing/matching is required in organ transplantation to avoid/ reduce graft rejection

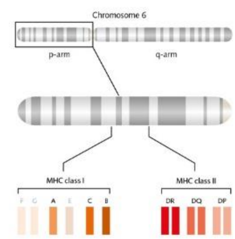


Figure 10: Genetics of rheumatoid arthritis

Pathophysiology:

- Primary site is synovial membrane
- Antibodies formed against joint tissue
- **Type IV hypersensitivity** (T cell mediated)
- Inflammatory cells, exudate & cytokines (TNF-α) leading to destruction of cartilage/bone
- Abnormal layer of granulation tissue (pannus)
    - **Pannus** = granulation tissue that replaces the gaps on destroyed tissue.

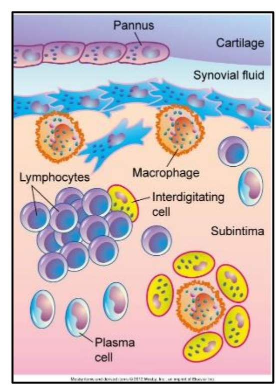


Figure 11: Pathophysiology of rheumatoid arthritis

Clinical manifestations:

- Insidious onset
- **Affected joints**: initially Wrists, MCPs, MTPs
- Symmetric joint involvement
- Morning stiffness
- Constitutional symptoms
- Joint deformities
- Subcutaneous nodules
- Presence of RF (rheumatoid factor) and ANA (antinuclear antibody) in serum

- *Specific criteria for diagnosis, published in the Journal of Arthritis and Rheumatology ("2010 Rheumatoid arthritis classification criteria" can be found at https://onlinelibrary.wiley.com/doi/full/10.1002/art.27584

Ankylosing Spondylitis

- Inflammatory joint disease of the spine or sacroiliac joints causing stiffening and fusion of joints (**LOSE CURVATURE IN SPINE**)

Etiology:

- Unknown
- Strong association with **HLA-B27 antigen**

Pathophysiology:

- Inflammation of fibrocartilage
- Inflammatory cells and fibrocartilage erosion
- Repair and scar tissue
- Calcification leading to joint fusion

Clinical manifestations:

- **Early symptoms:**
    - Low back pain that begins in early 20s and progresses over time
    - Stiffness
    - Pain
    - Restricted motion
- Loss of normal lumbar curvature (lordosis)
- Increased concavity of upper spine (kyphosis)

Gout

- Metabolic disorder that disrupts the body's control of uric acid production or excretion

Etiology and Pathophysiology:

- Accelerated purine breakdown OR poor uric acid secretion in the kidneys
    - Purines (nitrogenous bases) are required in DNA synthesis which is needed for cell division.
    - With cell damage/death, there is breakdown of purines → increase in uric acid production
    - Uric acid buildup is also caused by cancer chemotherapy
    - Uric acid transforms into urea in order to be excreted via the kidney
    - Uric acid buildup → hyperuricemia → deposition in various tissue

- High levels of uric acid in blood (hyperuricemisa)
- Uric acid crystals deposition in connective tissue including joints, also in the kidney causing uric acid stones

Sites:

- MTP (Metatarsophalangeal) joint of big toe (50% of initial attacks)
- Heel, ankle, instep of the foot, knee, wrist, or elbow

Risk factors:

- Male gender
- Increasing age
- High intake of alcohol, red meat, and fructose
- Drugs

Clinical manifestations:

- Recurrent attacks of monoarticular arthritis
- **Tophi**: deposits of urate crystals in and around joints
- **Renal disease**: glomerular, tubular, interstitial, vascular
- Renal stones
- **Acute gouty attack**:
  - Severe pain, especially at night, with hot, red, tender joint & signs of systemic acute inflammation

Figure 12: Tophi deposits of the metacarpal joints

Fibromyalgia

- Chronic widespread diffuse **non-articular** pain associated with fatigue and characteristic tender points. No joint pain.

Pathophysiology/clinical features:

- 80% to 90% are women, ages 25-45 yr, some adolescents
- Genetic predisposition: defective serotonin, catecholamines, and dopamine genes
- CNS sensitization
- Autoimmune disorders often coexist
- Overlaps with chronic fatigue syndrome and myofascial pain syndrome

Clinical manifestations:

- Diffuse, chronic (>3 mo) pain (burning/gnawing)
- Pain often begins in one location, especially neck and shoulders, then becomes more generalized
- Profound fatigue
- Increased sensitivity to touch
- Absence of inflammation
- Sleep disturbances / non-restorative sleep

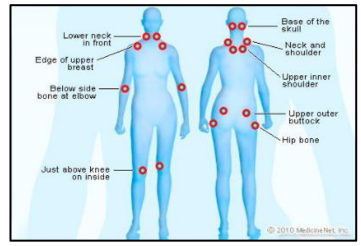


Figure 13: Multiple pain sites of fibromyalgia

Diagnosis

- Tenderness in eleven pairs of tender points along with a history of diffuse pain
- Diagnostic criteria*

Bone Tumors

- Can have different origins:
  - **Osteogenic**: from bone cells
  - **Chondrogenic**: from cartilage (chondroblast)
  - **Collagenic**: from fibrous tissue (fibroblast)
  - **Myelogenic**: from vascular tissue/marrow

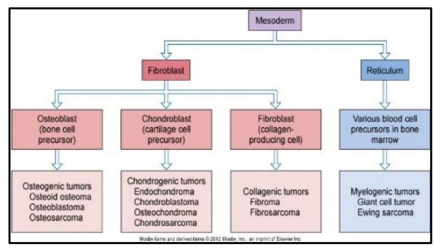


Figure 14: Cellular/ tissue origins of bone tumors

Osteosarcoma

- Adolescents and young adults
- 38% of bone tumors
- Seniors with a history of radiation therapy
- Metaphysis of long bones
- Masses of osteoid

Chondrosarcoma

- Middle-aged and older adults
- Infiltrates trabeculae in spongy bone
- Metaphysis or diaphysis of long bones
- Lobules of hyaline cartilage

Fibrosarcoma

- Firm, fibrous mass of collagen, fibroblasts, and osteoclast-like cells
- Metaphysis of femur or tibia
- Lung metastasis is common

Myelogenic tumors

- Giant cell tumor
  - Extensive bone resorption because of the osteoclastic origin of the giant cells
  - Epiphyses of the femur, tibia, radius, or humerus
  - Slow growth rate

Multiple Myeloma*

Muscle Tumors

Rhabdomyosarcoma

- Malignant tumor of striated muscle
- Usually muscles and CT in the eye, tongue, neck, larynx, nasal cavity, axilla, vulva, and heart
- Highly malignant with rapid metastasis

12: Disorders of the Integument

Clinical Manifestations/ Lesions of Skin Dysfunction

Table: Describing Skin Alterations and Lesions *(Poonawalla & Diven, 2008)*	
Macule	Telangiectasia
Papule	Keloid
Vesicle	Atrophy
Bulla	Sclerosis
Pustule	Scar
Plaque	Erosion
Nodule	Scale
Cyst	Ulcer
Erythema	Vesicle

Pressure Ulcers

- Ulcers result from any unrelieved pressure on the skin, causing underlying tissue damage
  - Pressure
  - Shearing forces

- Friction
- Moisture

Stages:

I. Non-blanchable erythema of intact skin
II. Partial-thickness skin loss (epidermis or dermis)
III. Full-thickness skin loss with damage/loss of SC tissue
IV. Full-thickness skin loss with damage to muscle, bone

Pathophysiology:

- Over bony prominences; 95% on lower body
- **Hyperemia**: early sign
- **Ischemia**: can start 2-6 h of pressure
- **Necrosis**: can start as early as 6 h of constant pressure
- **Ulcer**: necrotic area breaks down (like tip of an iceberg)

Patient care:

- Reposition patient every 2 hours

Keloids

- Elevated, rounded, and firm
- Claw-like margins that extend beyond the original site of injury
- Excessive collagen formation during dermal connective tissue repair
- Common in dark skin types and burn scars
- Type III collagen is increased

Inflammatory Disorders of the Skin

- Most common are:
  - **Dermatitis**; various types
  - Eczema
- The disorders are generally characterized by:
  - Pruritus
  - Distinct lesions
  - Epidermal changes

Allergic contact dermatitis

- Type IV hypersensitivity
- Erythema, swelling, pruritus, vesicular lesions

Atopic dermatitis

- Type I hypersensitivity
- Red, weeping crusts, chronic inflammation, lichenification

Irritant contact dermatitis

- Non-immunologic inflammation of the skin
- Prolonged exposure to irritating substances
- Symptoms similar to allergic contact dermatitis

Seborrheic dermatitis

- Inflammation; scalp, eyebrows, eyelids, nasolabial folds, and ear canals
- Scaly, white, or yellowish plaques

Psoriasis

- A chronic, relapsing, proliferative skin disorder

Etiology:

- Not fully understood, genetic and immunologic factors

Pathophysiology and clinical manifestations:

- Shortened keratinocyte cell cycle
- High rate of mitosis in the basal skin layer. Cells do not have time to mature or adequately keratinize
- T cell immune- mediated inflammatory response
- Scaly, thick, silvery, itchy elevated lesions
- Sites: scalp, elbows, or knees
- Dermal and epidermal thickening

Scleroderma

- Sclerosis (hardening) of skin/ CT, can progress to internal organs

Pathophysiology:

- Increased synthesis, deposition of collagen with inflammation
- Altered connective tissues, damage to small blood vessels
- T lymphocytes activation
- 50% of patients die within 5 years

Clinical manifestations:

- **Skin**: hard, hypopigmented, tightly connected to underlying tissue
  - Facial skin becomes tight, mouth may not open completely
- Fingers become tapered and flexed; nails and fingertips can be lost from atrophy

Burns

Etiology:

- Fire, chemical, electrical
- **Assessment of burn**:
  - Degree (1st, 2nd, 3rd)
  - % body surface area (Rule of 9's)
  - **Parklands formula** (calculation for fluid administration)

Figure 15: Degree of burn (1st, 2nd, 3rd)

Pathophysiology:

- Heat disturbs protein 3D structure → dysfunctional proteins
- Disruption of skin sensation
- Tissue/endothelial injury and inflammation
- Increased capillary permeability → edema
- Loss of water and plasma causing dehydration
- Electrolytes imbalance → lose everything from intravascular space
- Hypovolemia, multiorgan damage

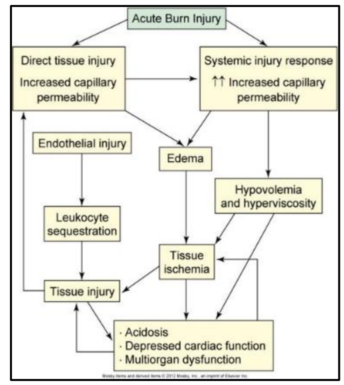


Figure 16: Pathophysiology of burns

Rule of nines

- Method for estimating percentage of body surface area impacted by burn

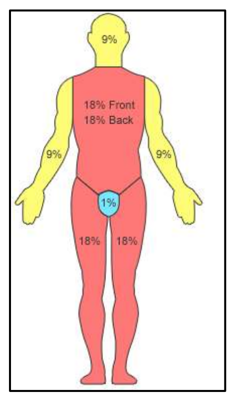


Figure 17: Rule of Nines (Adult)

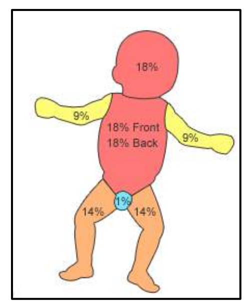


Figure 18: Rule of Nines (Child)

Clinical manifestations:

- Edema
- Manifestations of dehydration
- Manifestations of hypovolemia
- Organ failure
- Death results from dehydration, poor organ perfusion and renal failure

Bibliography

Aleteha, G. et al. (2010). 2010 Rheumatoid arthritis classification criteria: An American College of Rheumatology/European League Against Rheumatism collaborative initiative. *Arthritis & Rheumatism*, 62: 2569-2581. doi:10.1002/art.27584

Huether and McCance. (2012). *Understanding pathophysiology*. (5th ed.). St. Louis, Missouri: Elsevier Inc.

National Institute of Health. (2019, May 29). Childhood Hodgkin Lymphoma Treatment (PDQ®)–Health Professional Version. Retrieved December 2019, from https://www.cancer.gov/types/lymphoma/hp/child-hodgkin-treatment-pdq#link/_557.

Poonawalla, T., & Diven, D. (2008). Learning module: Core concepts of pediatric dermatology. From https://www.utmb.edu/pedi_ed/CoreV2/Dermatology/page_01.htm

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Content Images (in order of appearance):

1) *5: Figure 6: "Life Cycle of an RBC."* Created by: Osborne, Kathryn (2019)
2) *6: Figure 44:* "*Venous Thromboembolism*" Retrieved from: https://www.google.com/url?sa=i&source=images&cd=&ved=2ahUKEwiGvqjnp5fmAhUkhOAKHRkWBXoQjRx6BAgBEAQ&url=https%3A%2F%2Fwww.slideshare.net%2Fnidafkhan1%2Fthromboprophylaxis-in-orthopedic-surgery&psig=AOvVaw21Kyz5D2M6Vm0SSHVP8NKm&ust=1575387918944464
3) *7: Figure 15:* "*Classifying TBI and manifestations."* Created by: Osborne, Kathryn (2018)
4) *10: Figure 12:* "*Anatomy of a kidney glomerulus*." Retrieved from: https://en.wikipedia.org/wiki/Glomerulus_(kidney)

5) *10: Figure 19: "Hemolytic uremic syndrome (overview)."* Retrieved from: https://www.roshreview.com/blog/ep8-fractures-vp-shunts-fight-bites-hemolytic-uremic-syndrome-cyanotic-heart-lesions-anticoagulation-much/
6) *10: Figure 20: "ADAMTS13 and VWF in TTP."* Retrieved from: https://www.sciencedirect.com/science/article/pii/B9780323462020000248

Appendix: Figures, Images, Diagrams

1: Altered Cellular and Tissue Biology

Figure 1: Cellular response to stress and adaptation

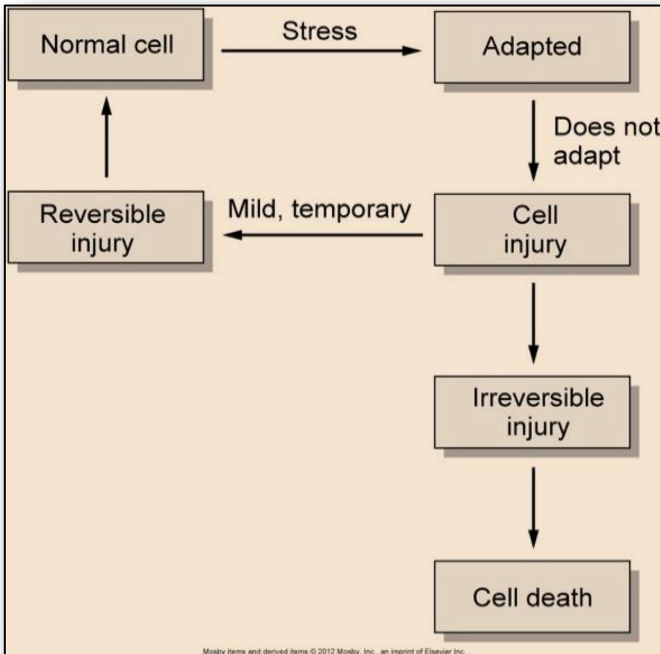


Figure 2: Cellular adaptation

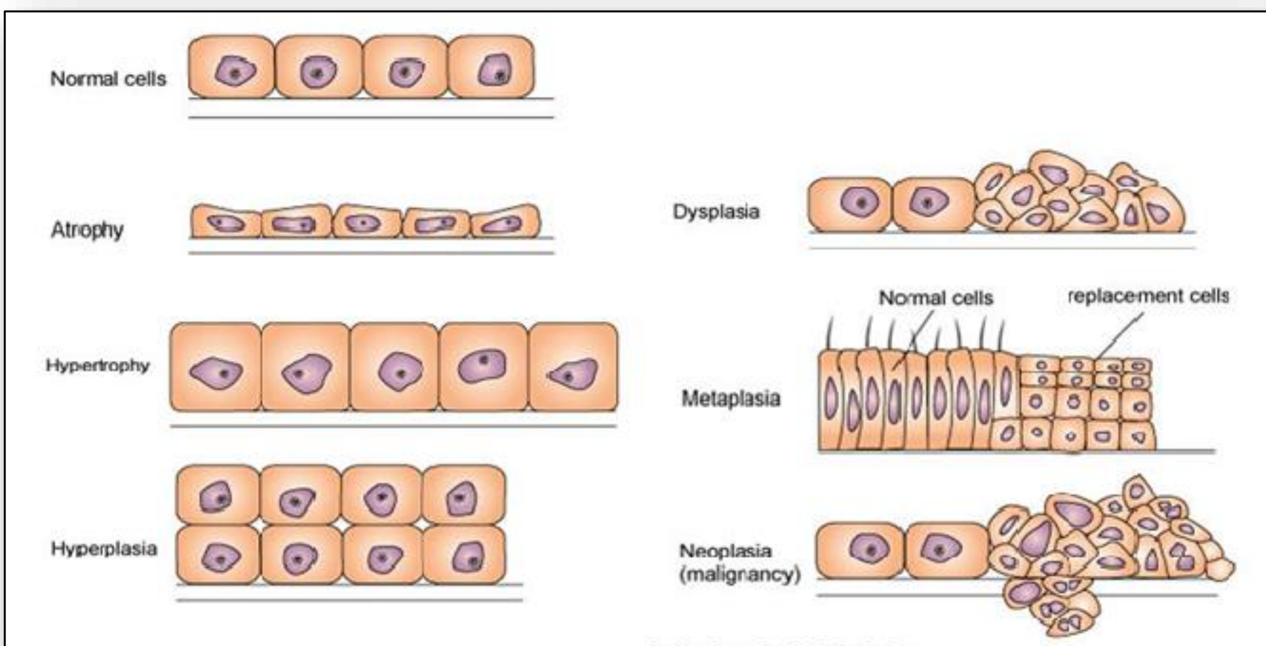


Figure 3: Cellular response in epithelium (e.g. trachea) to chronic, persistent injury or irritation (e.g. smoking)

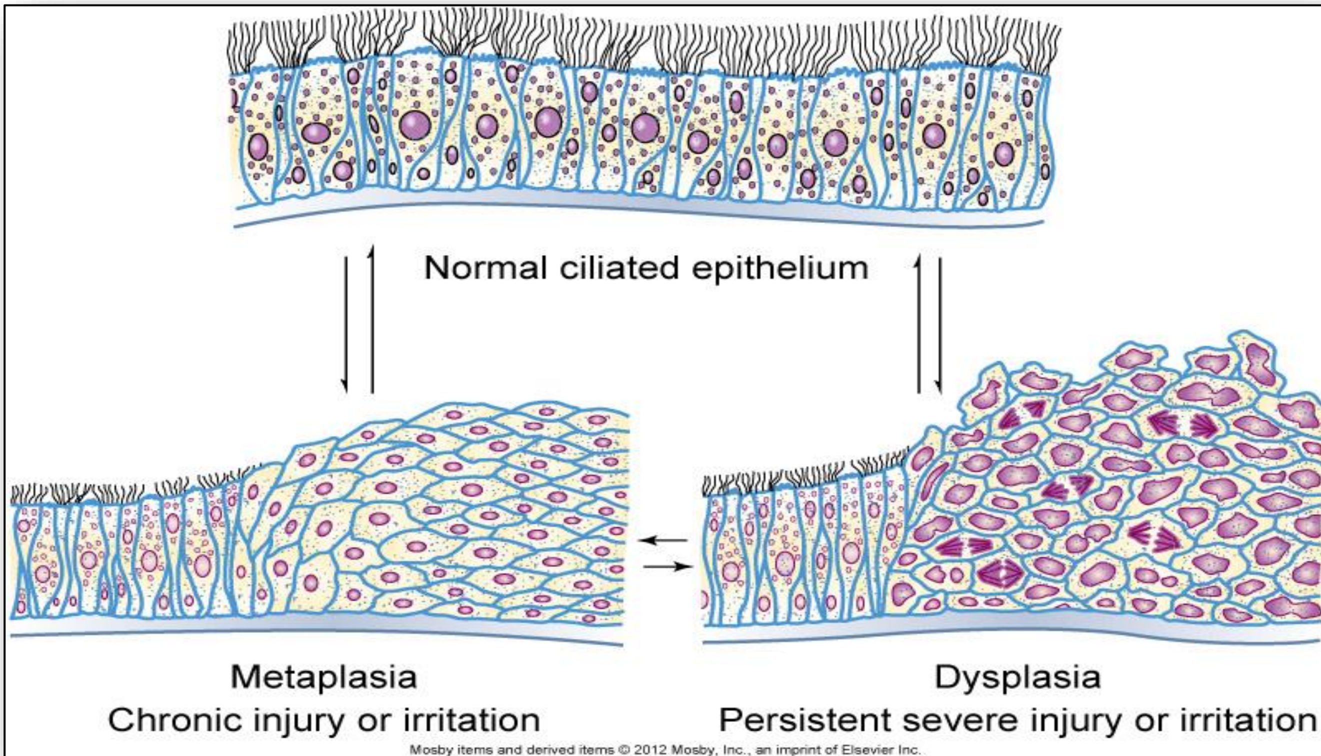


Figure 4: Cellular formation of free radicals

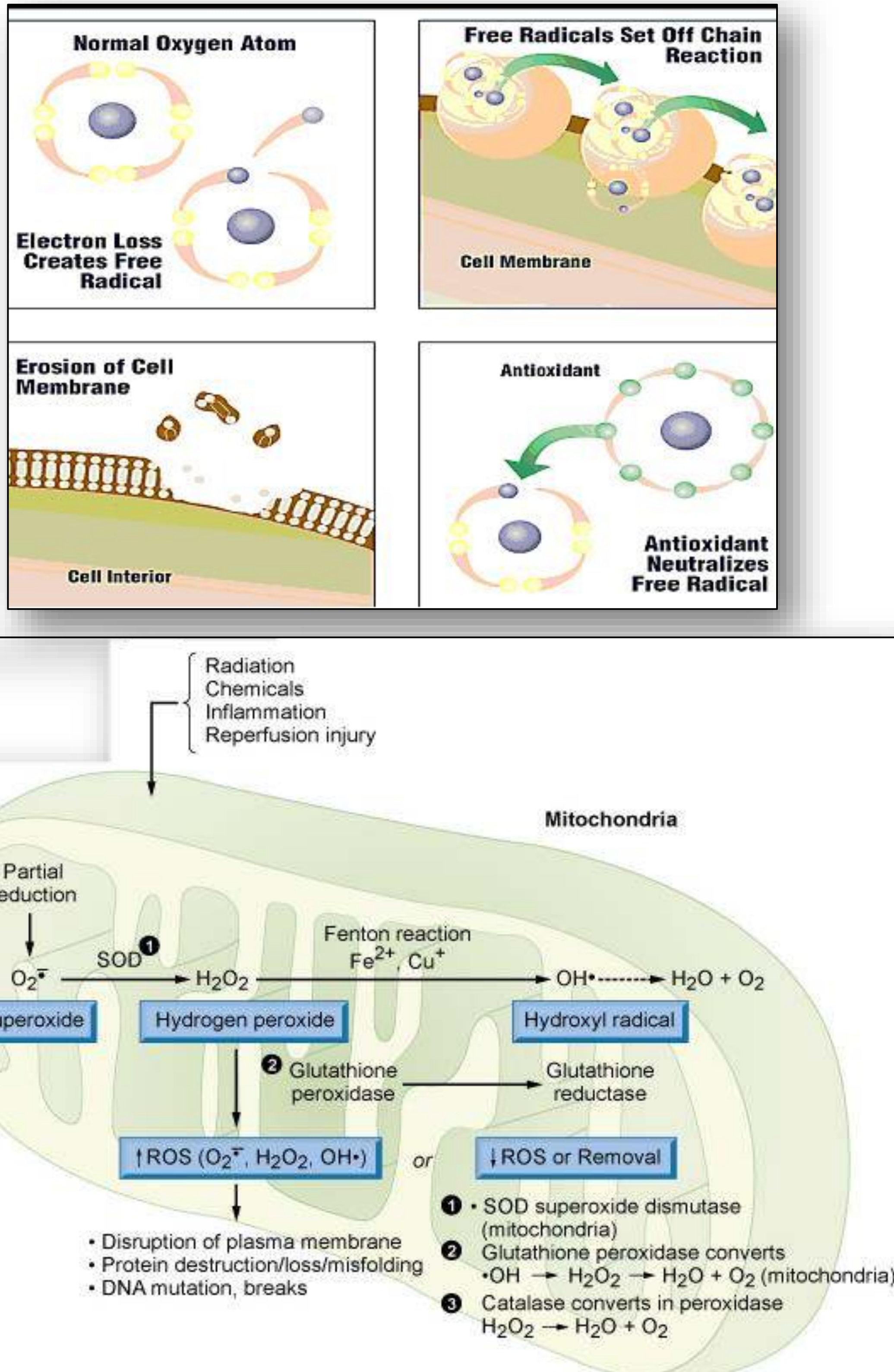


Figure 5: Metabolism and detoxification of ethanol (alcohol)

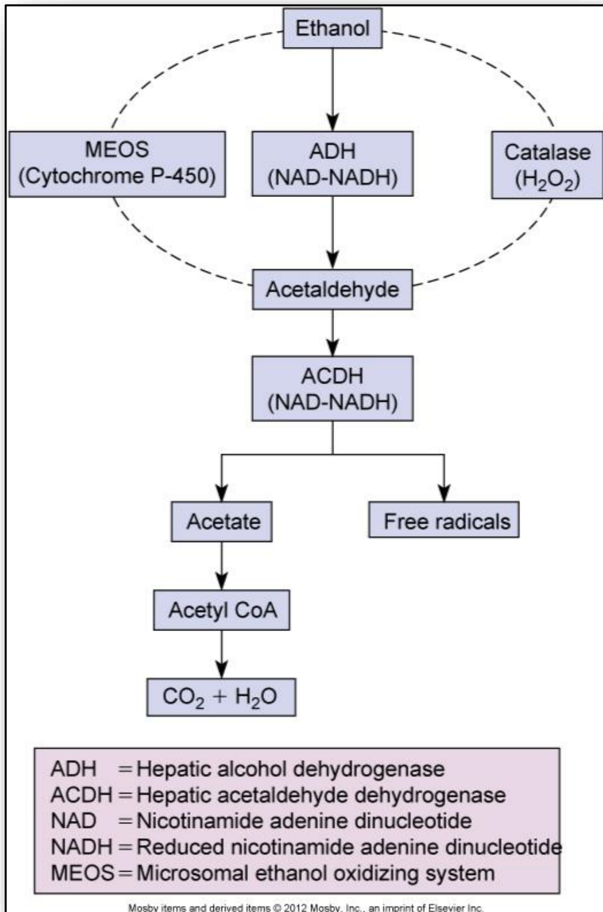


Figure 6: Fetal alcohol syndrome (FAS)

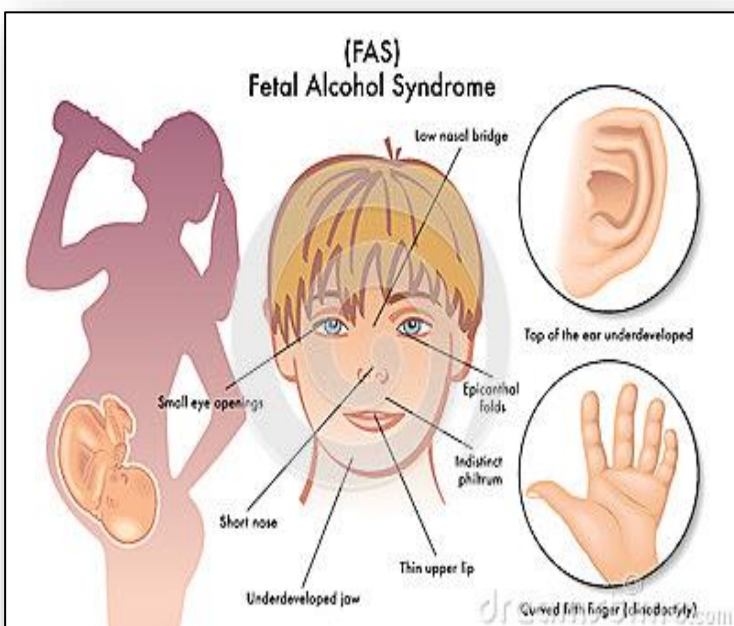


Figure 7: Ethylene glycol toxicity

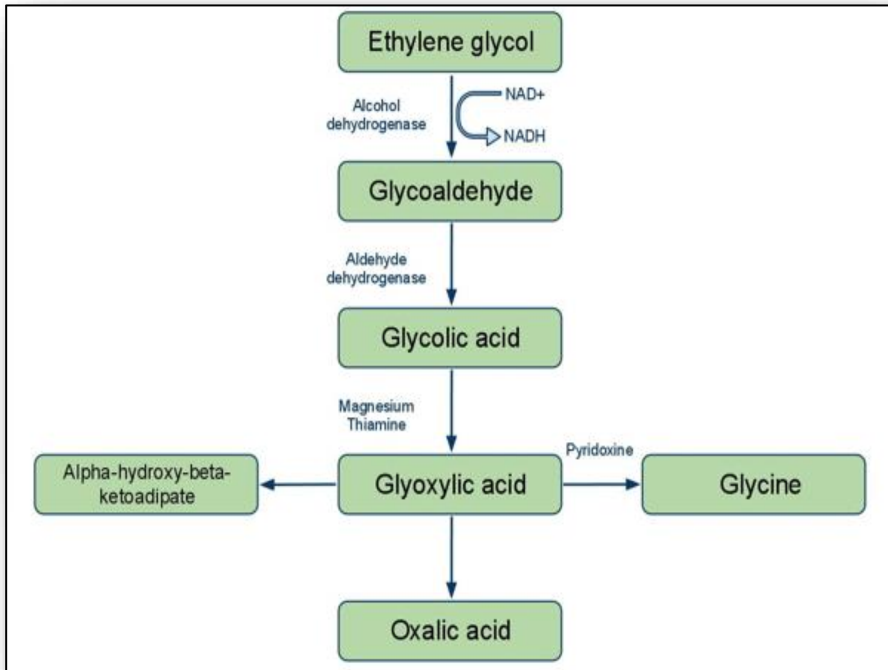


Figure 8: Blood smear of person with lead toxicity

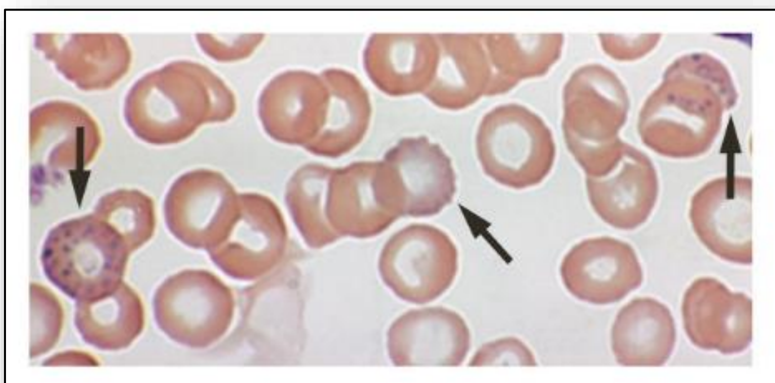


Figure 9: Blue gum line in lead toxicity

Figure 10: Cellular injury from water accumulation (hydropic degeneration)

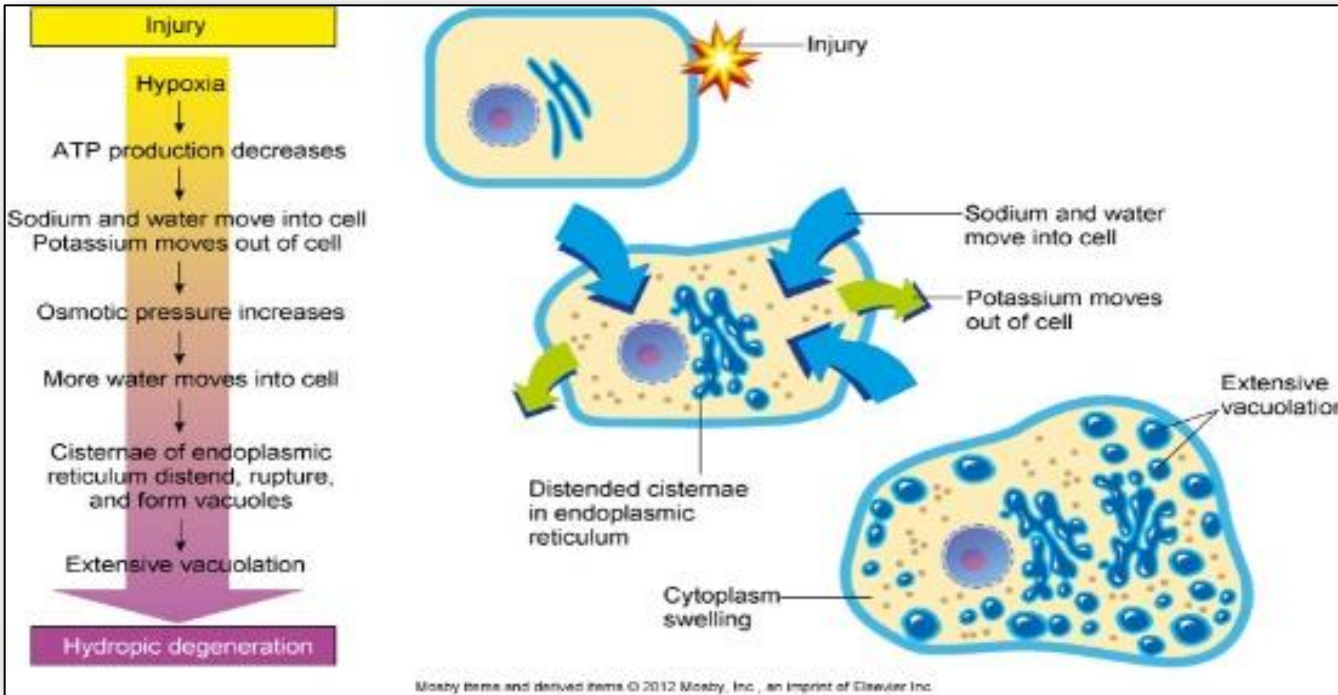


Figure 11: Cellular damage by excess free calcium

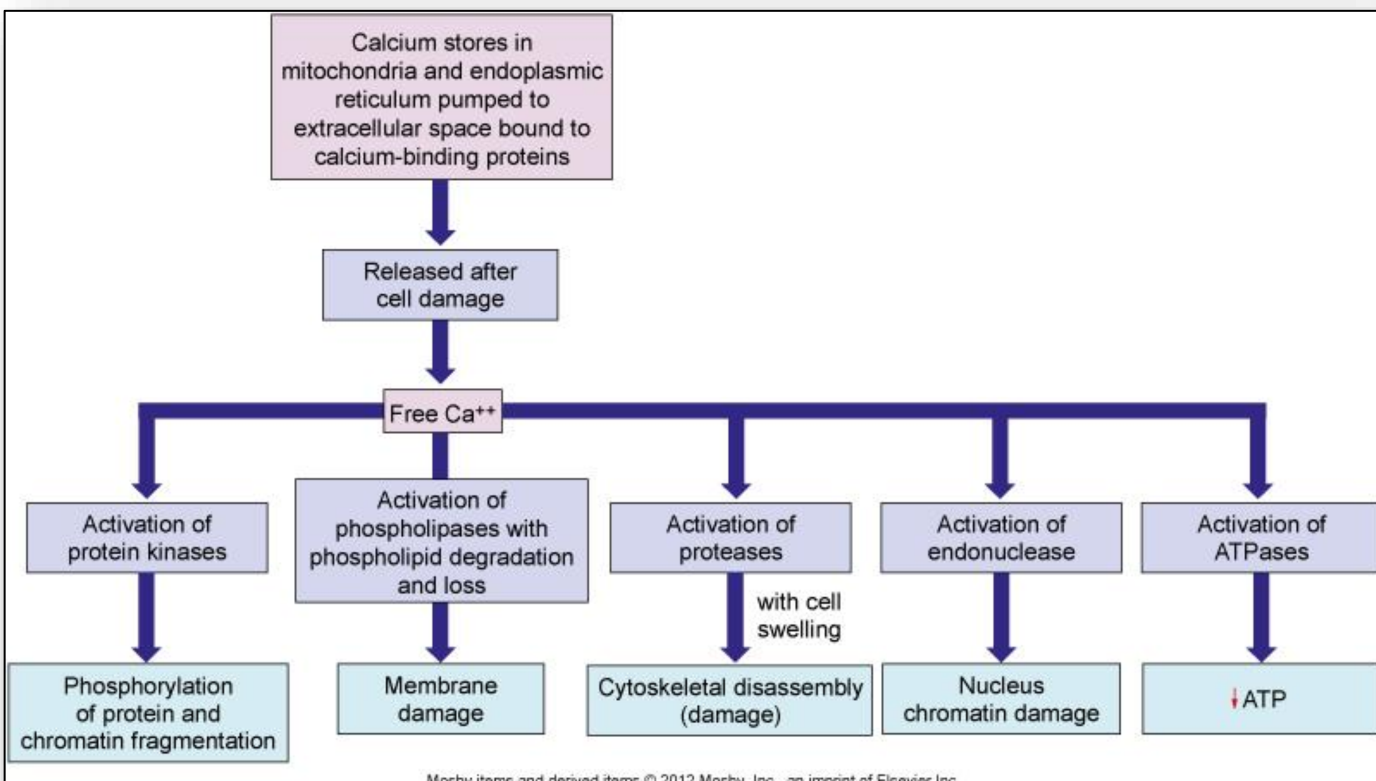


Figure 12: Cell death – Necrosis vs apoptosis

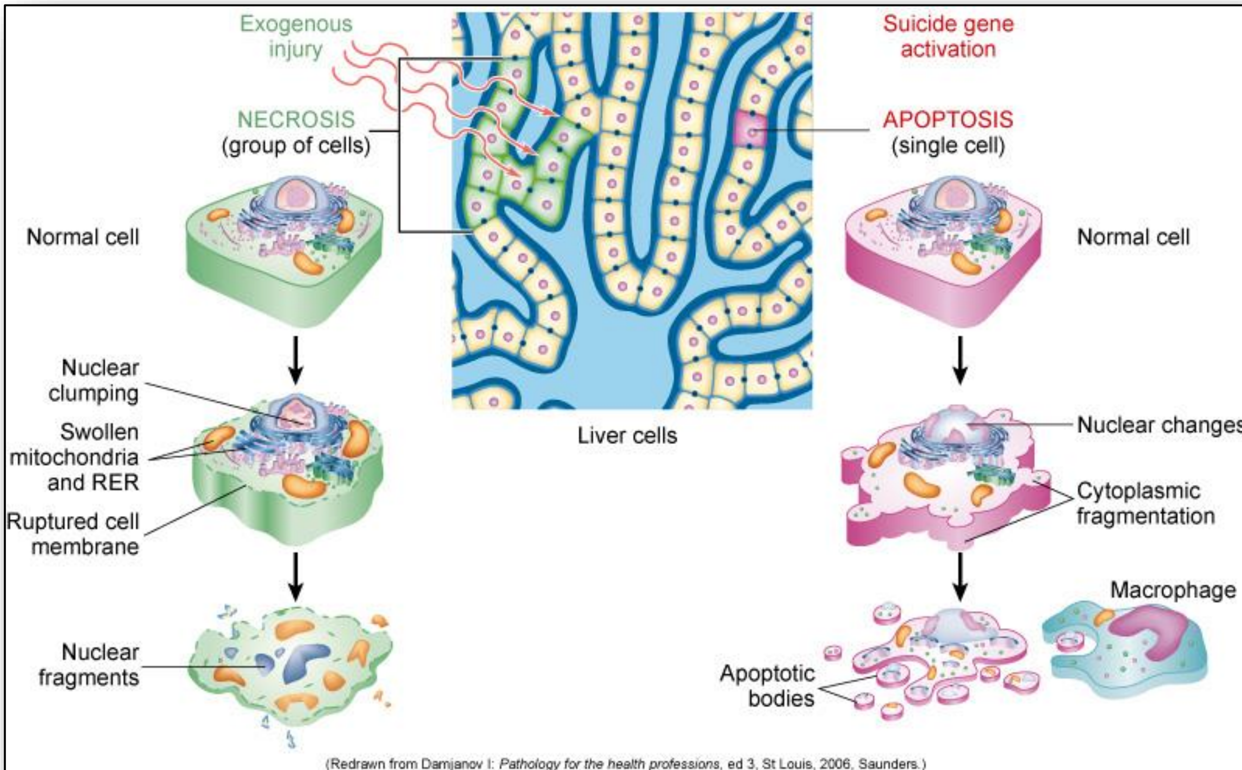


Figure 13: Liquifaction necrosis

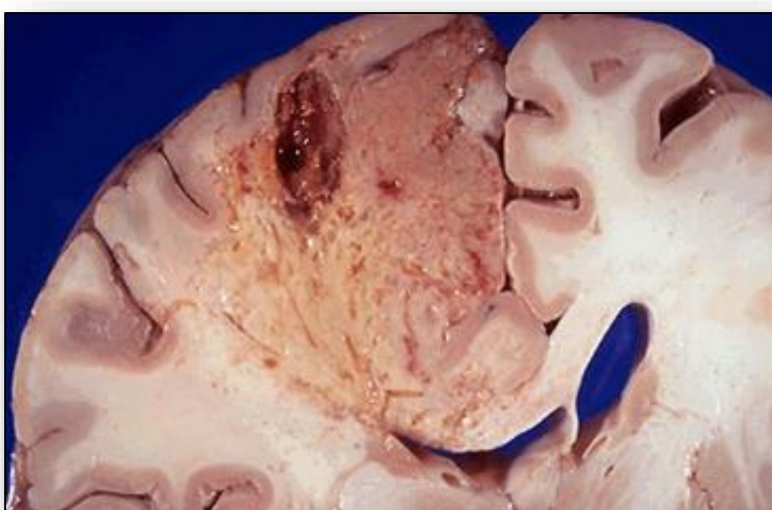


Figure 14: Coagulative necrosis

Figure 15: Fat necrosis

Figure 16: Caseous necrosis

Figure 17: Infarction

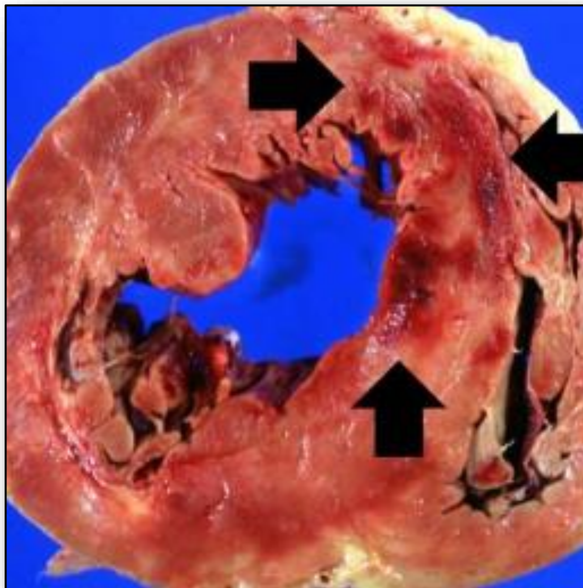
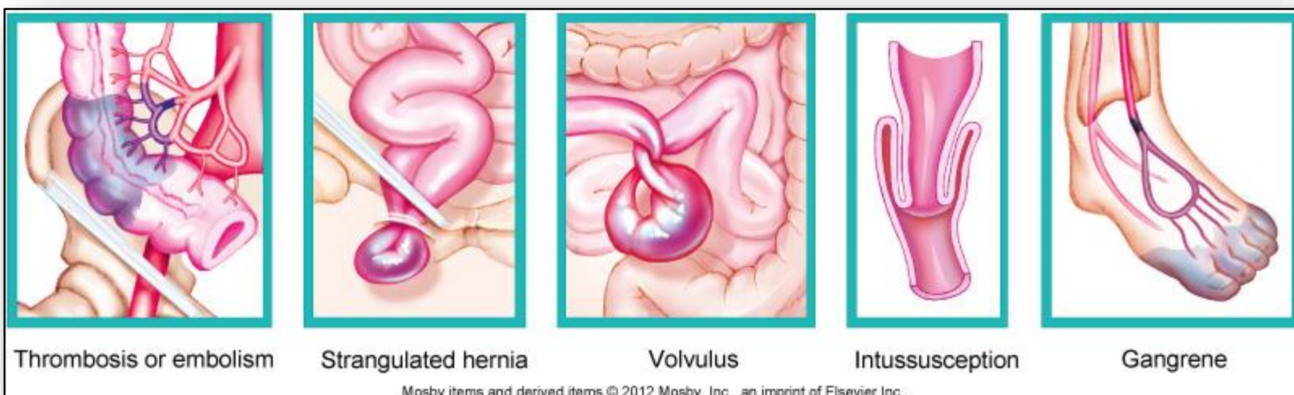


Figure 18: Types of gangrene



2: Mechanisms of Self-Defense

Figure 1: Overview of immunity - Innate vs adaptive

Innate (Nonspecific) Immunity	Natural killer cell	Adaptive (Acquired) Immunity (Chapter 17)
First line of defense	**Second line of defense**	**Third line of defense**
• Intact skin • Mucous membranes and their secretions • Normal microbiota	• Natural killer cells and phagocytic white blood cells • Inflammation • Fever • Antimicrobial substances	• Specialized lymphocytes: T cells and B cells • Antibodies

Figure 2: Processes of white blood response to inflammation (exiting of blood vessel and migration to site by chemotaxis)

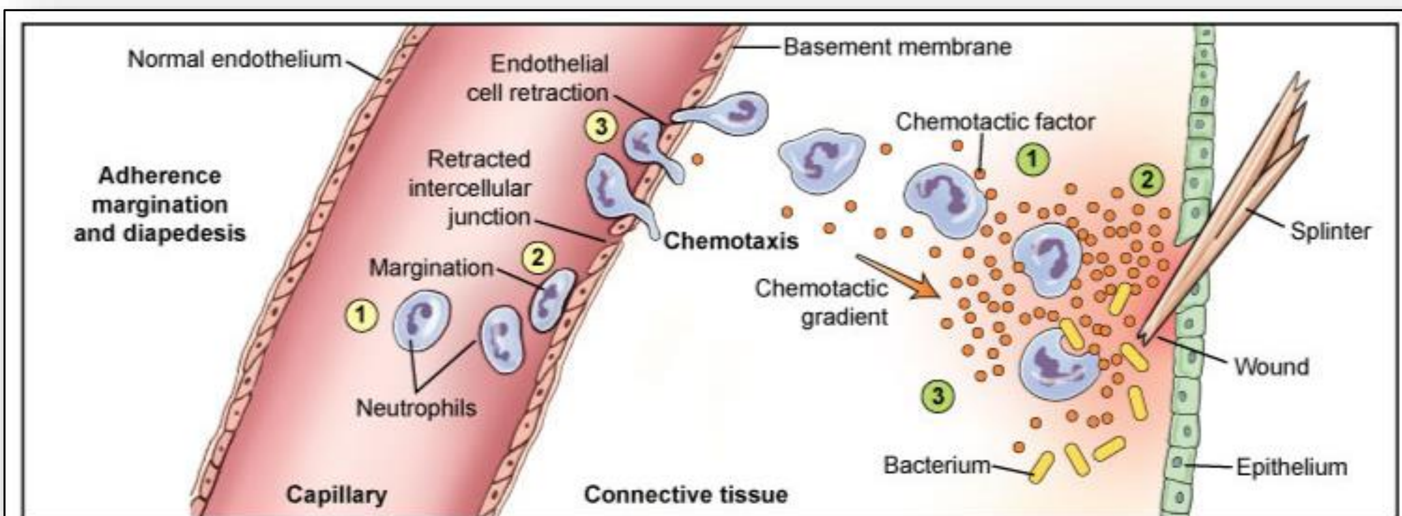


Figure 3: Plasma protein systems (overview)

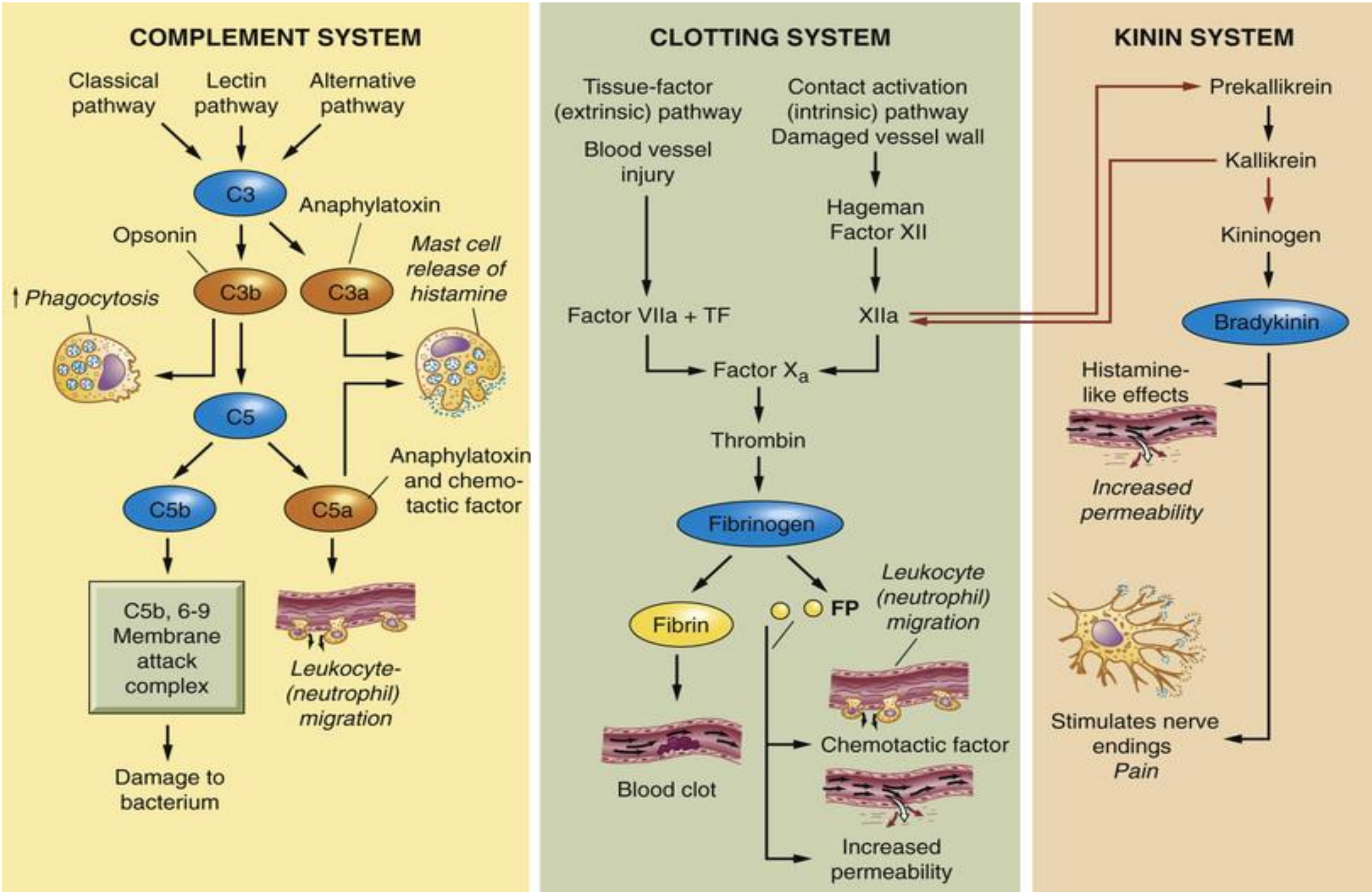


Figure 4: Coagulation (Clotting) System – Intrinsic (left) vs extrinsic (right) pathways

Extrinsic system:
- Activated by TF
- Tested via PT, INR

Intrinsic system:
- Activated by contact
- Tested via PTT

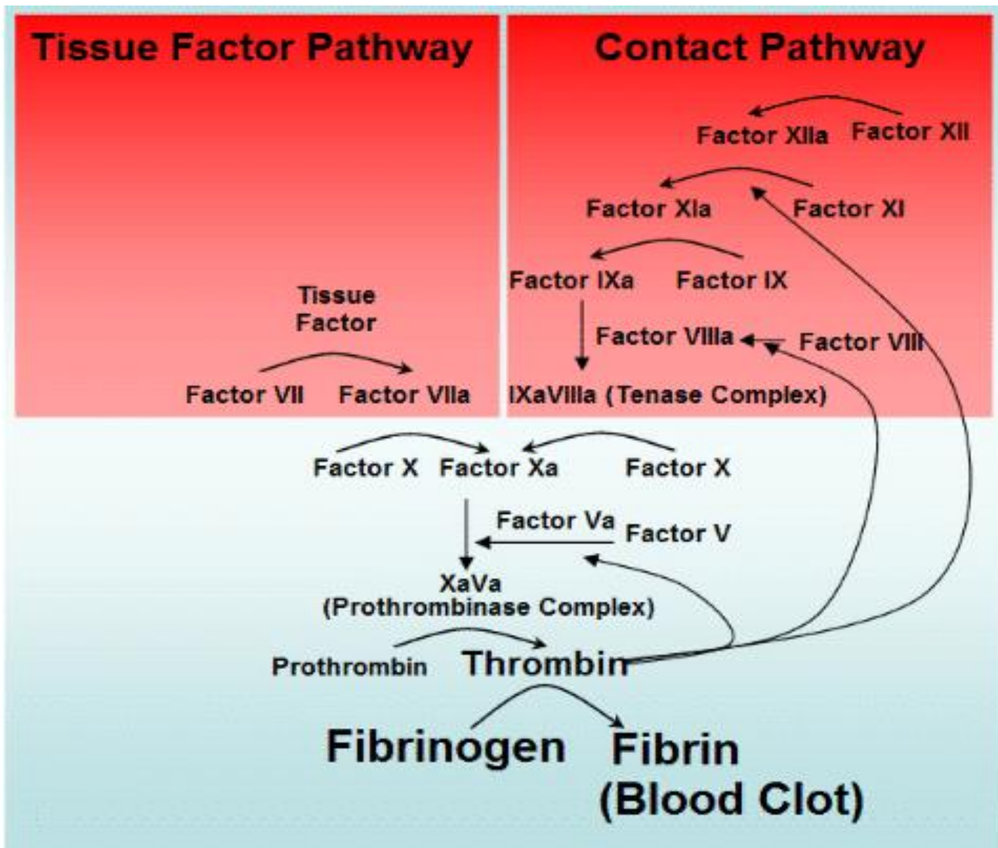


Figure 5: Mast cell processes in inflammation

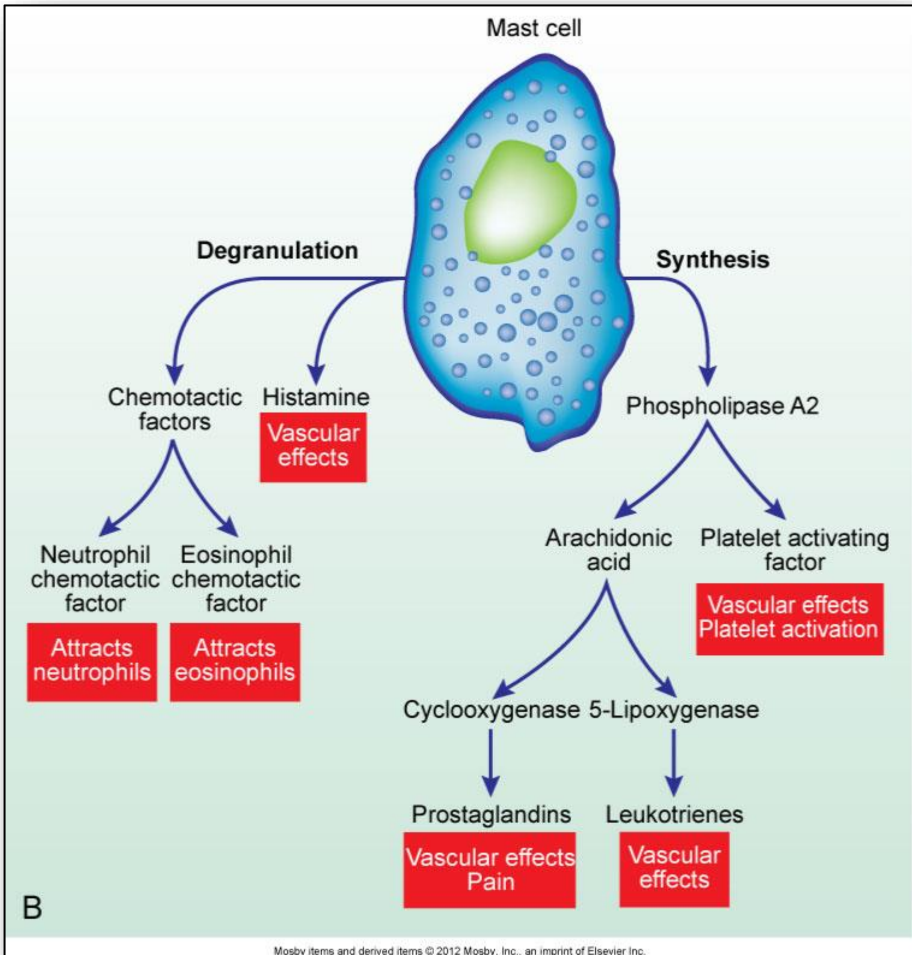


Figure 6: Endothelium

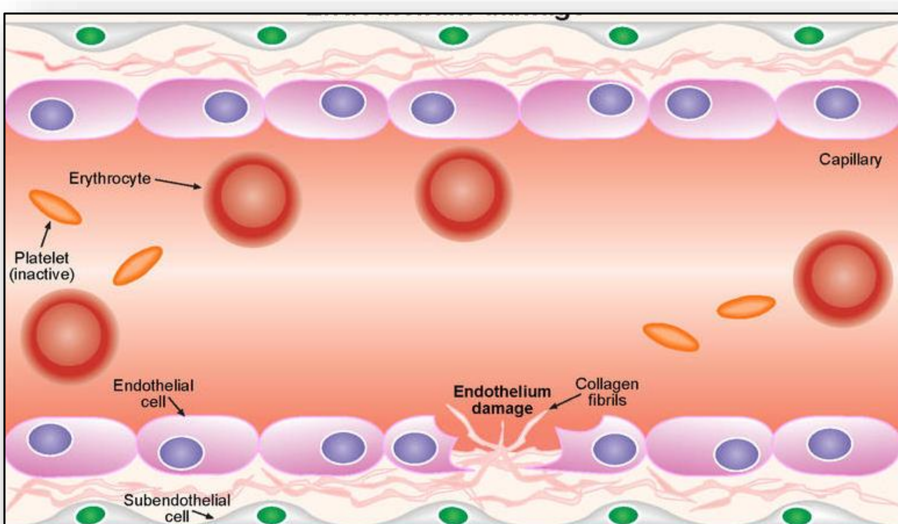


Figure 7: Serous exudate

Figure 8: Fibrinous exudate

Figure 9: Purulent exudate

Figure 10: Hemorrhagic exudate

Figure 11: Langerhans giant cell of chronic inflammation

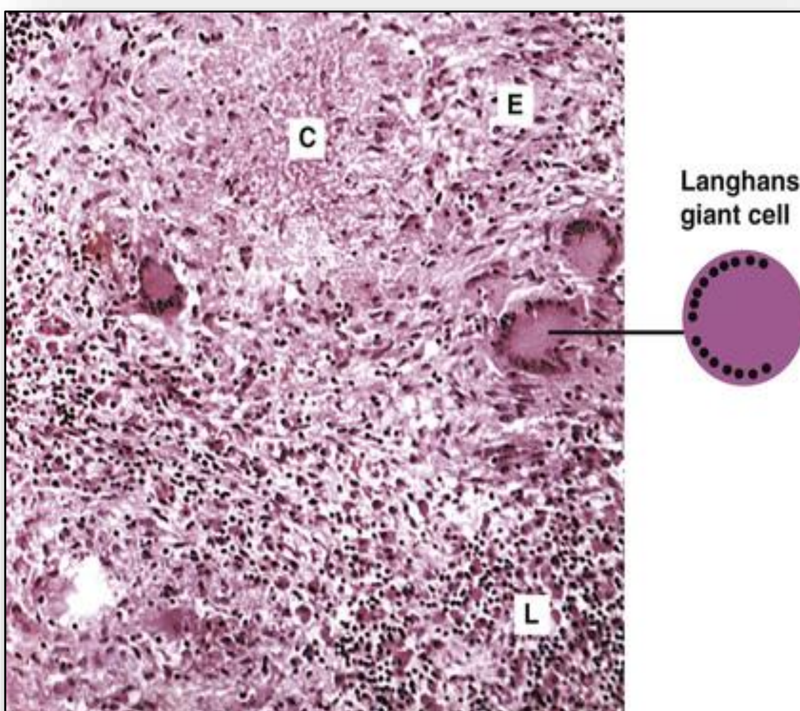


Figure 12: Giant cell and granuloma formation in chronic inflammation

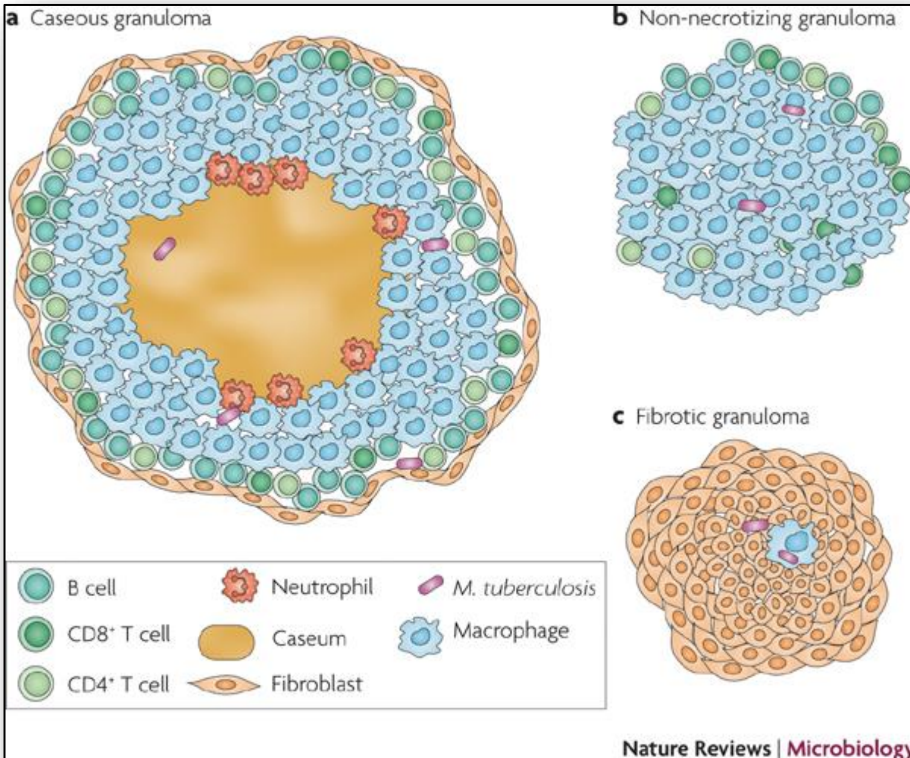


Figure 13: Acute vs chronic inflammation

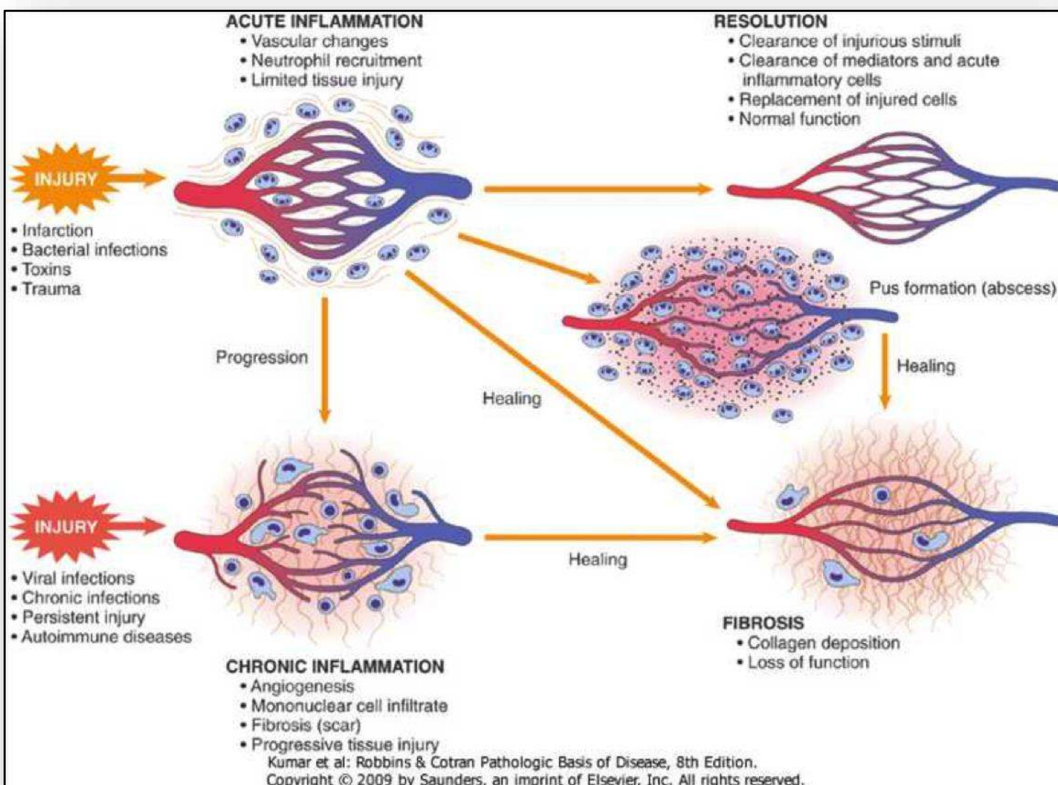


Figure 14: Phases of wound healing

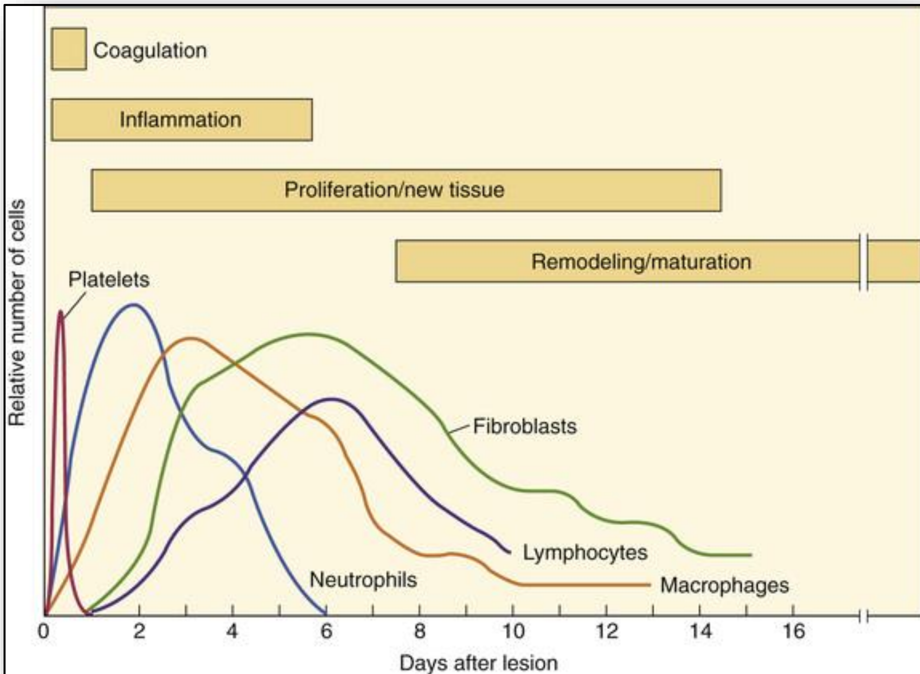


Figure 15: Timescale (minutes-years) for phases of wound healing

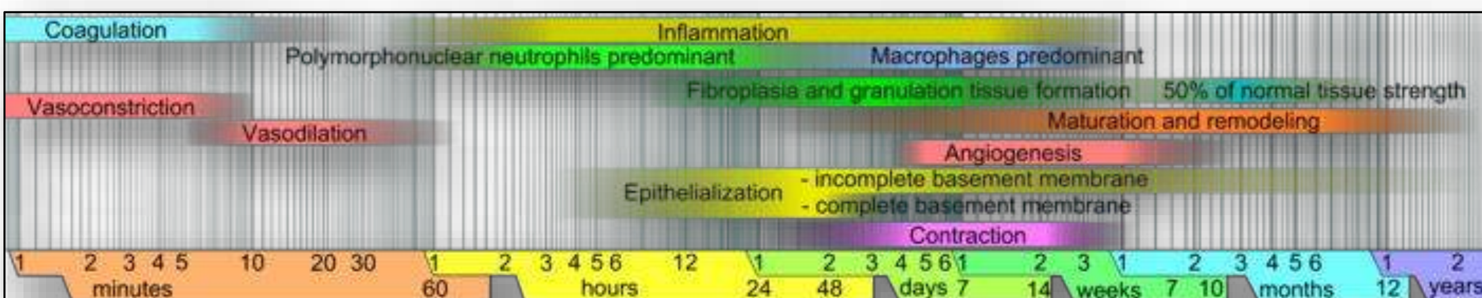


Figure 16: Healing by primary vs secondary intention

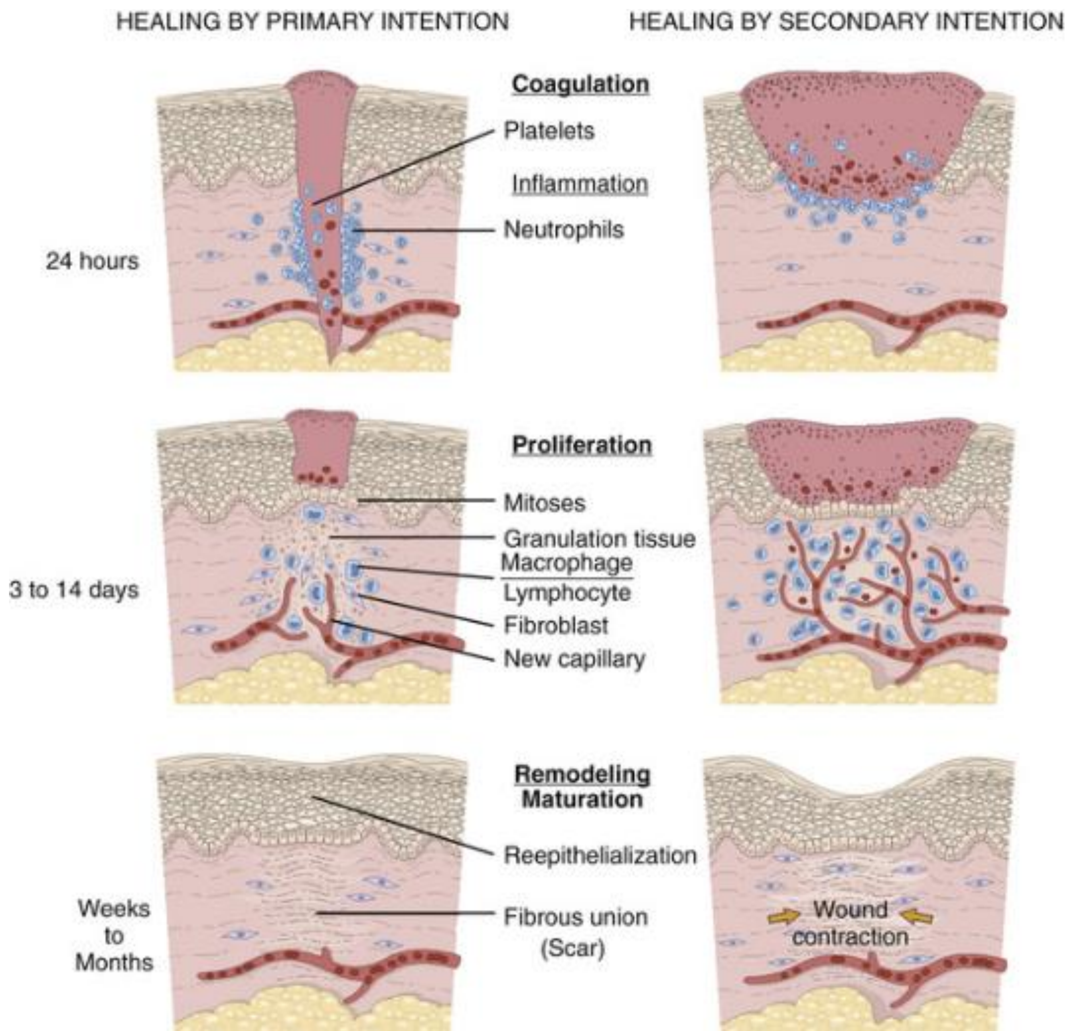


Figure 17: Diabetes and impaired wound healing

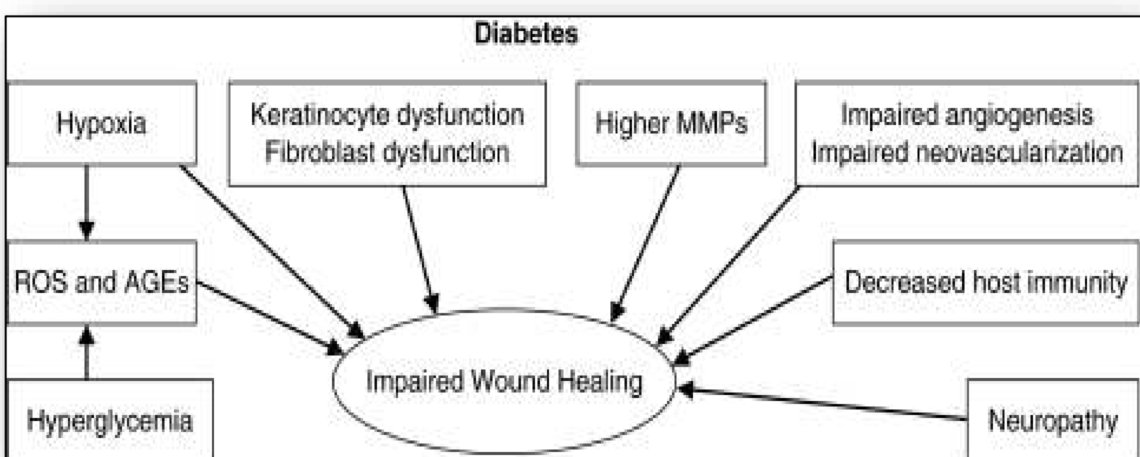


Figure 18: Psychological conditions and impaired wound healing

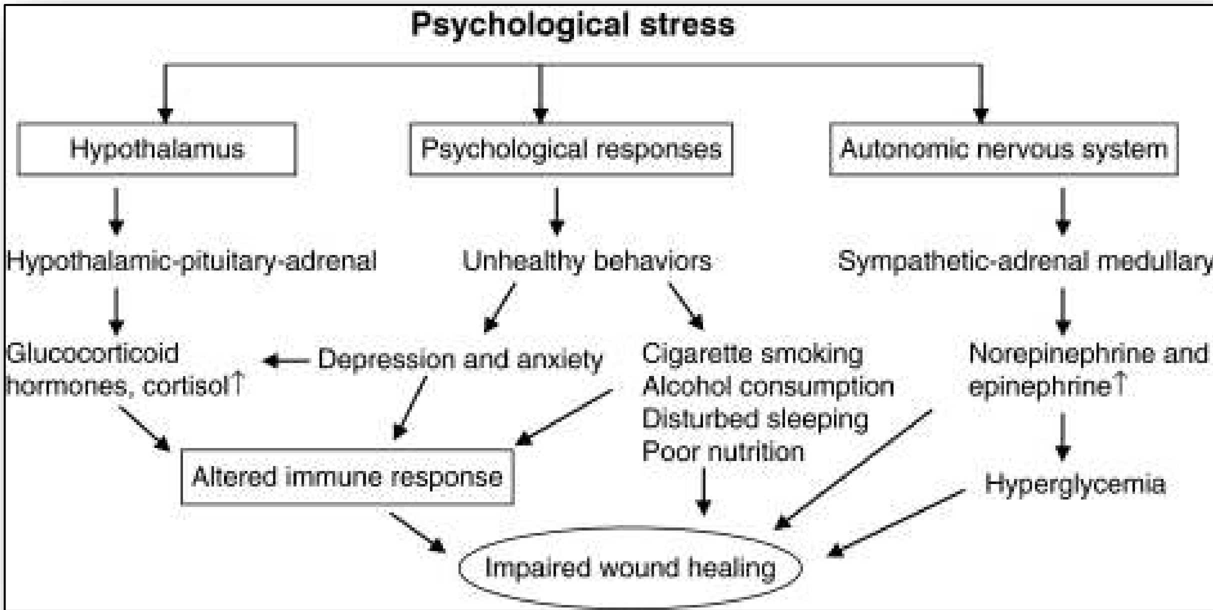


Figure 19: Adaptive immunity (3rd line of defence)

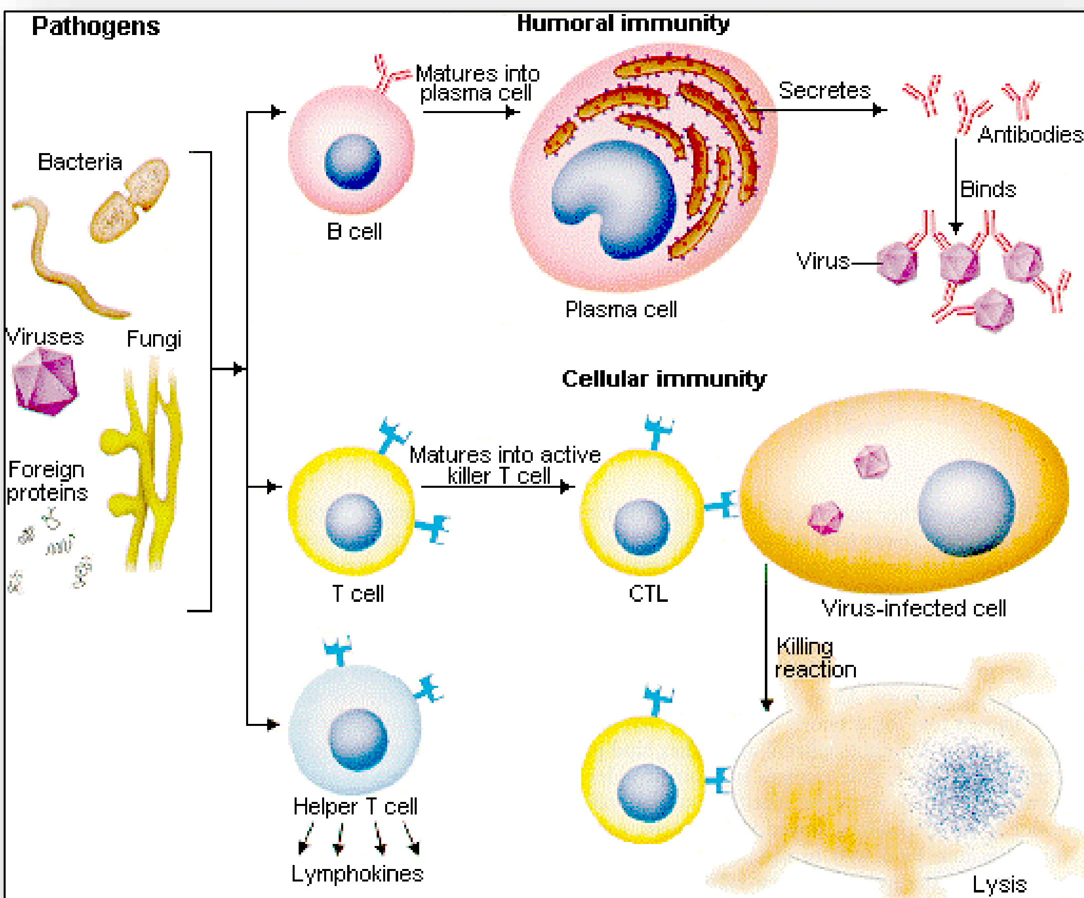


Figure 20: Processes of antigen recognition, processing and presentation

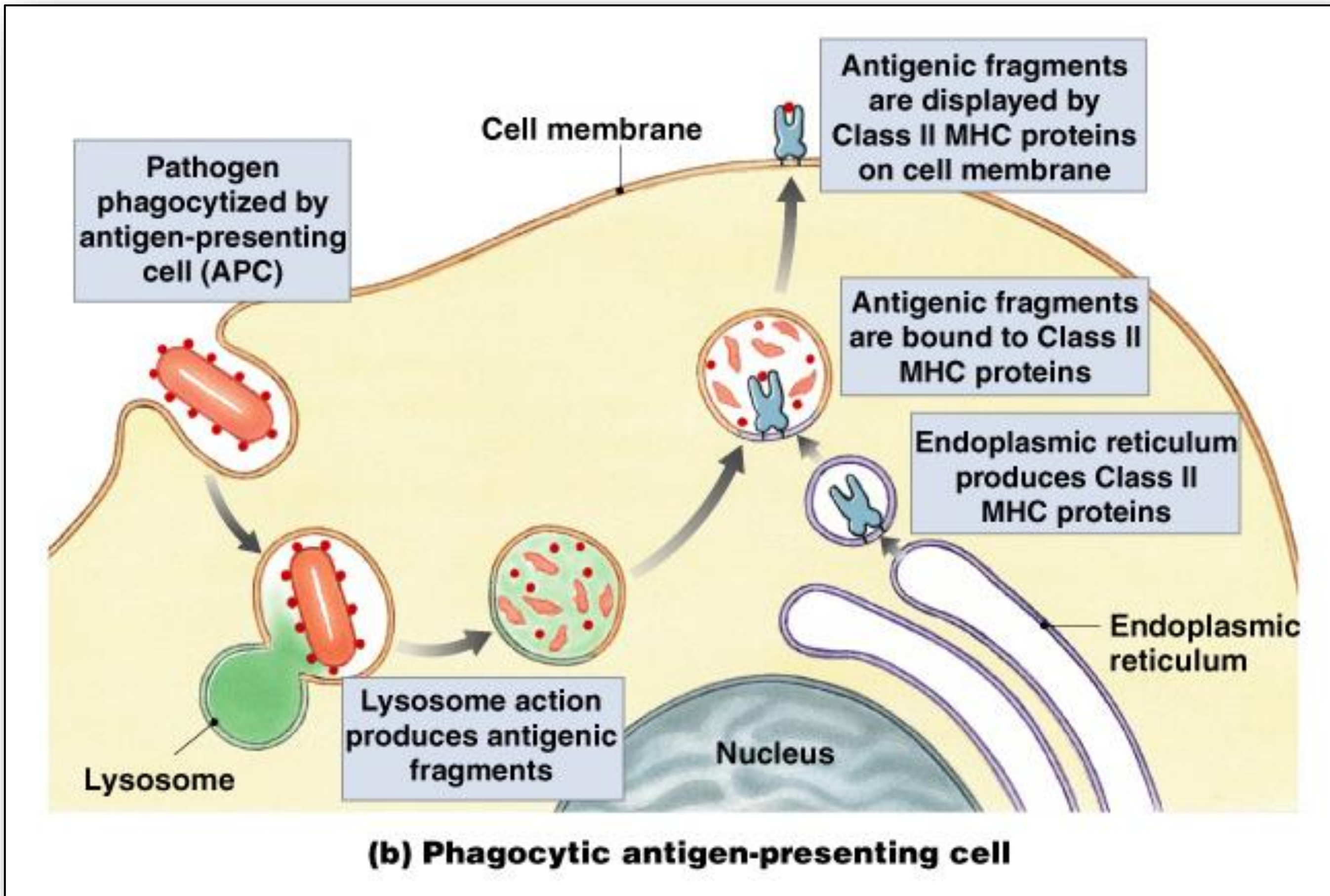


Figure 21: Antibody (Ig) Structure

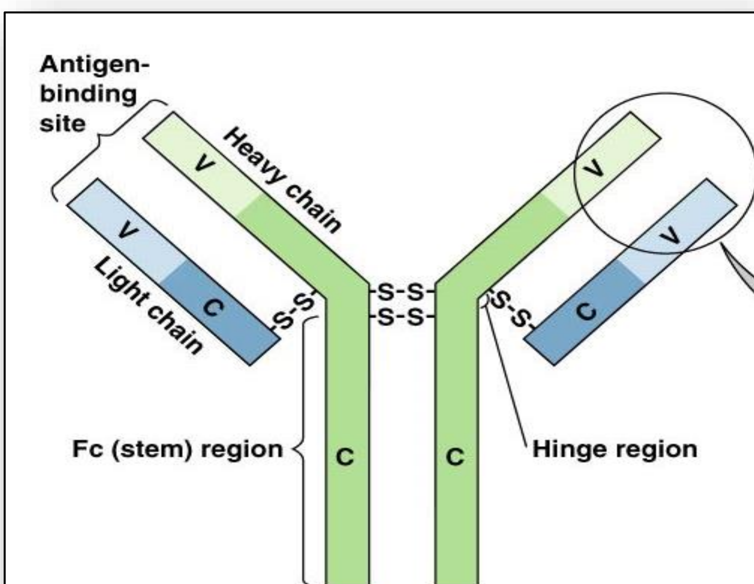


Figure 22: Antibody classes

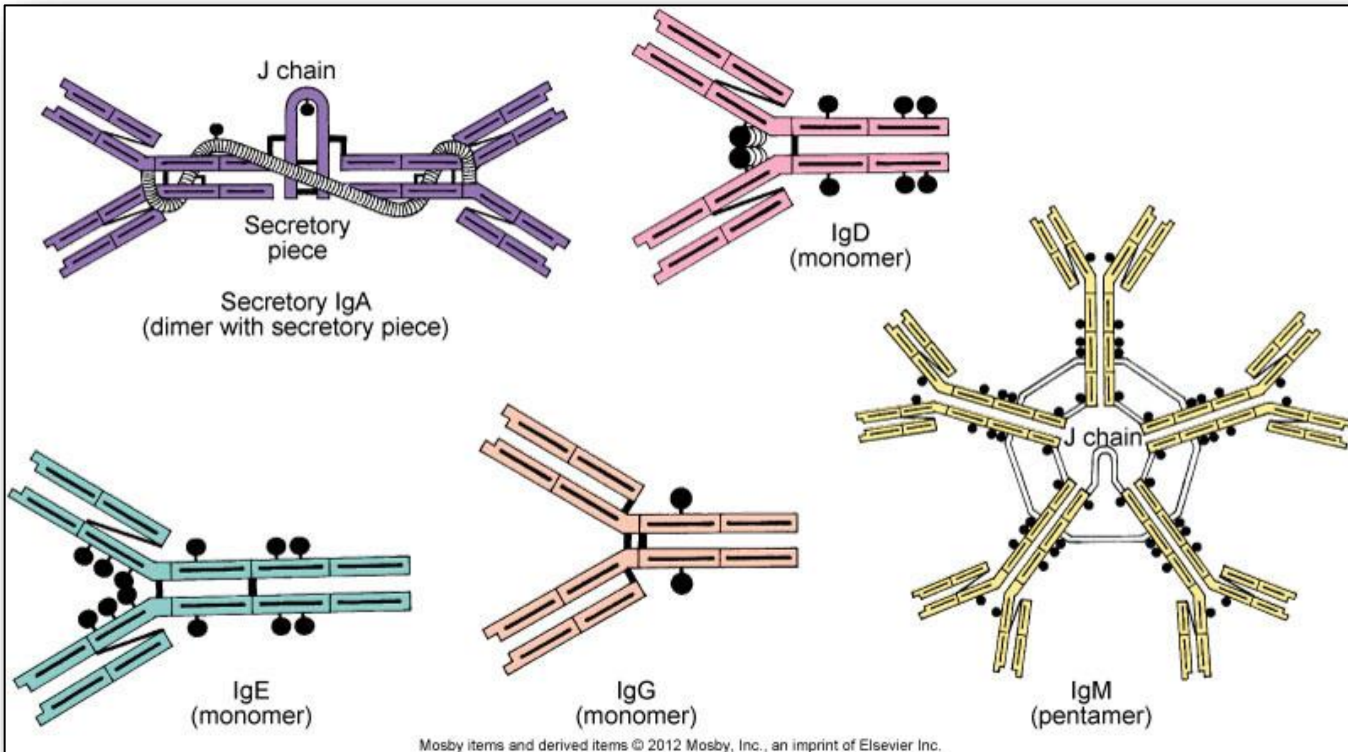


Figure 23: Antibody functions

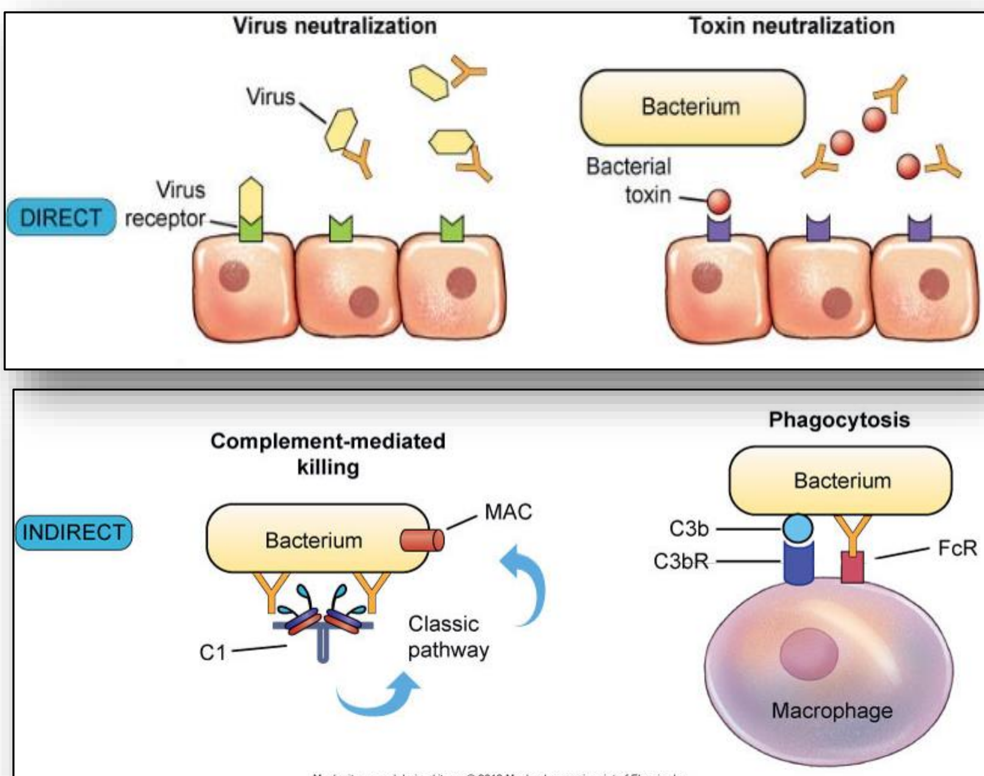


Figure 24: Primary and secondary immune response

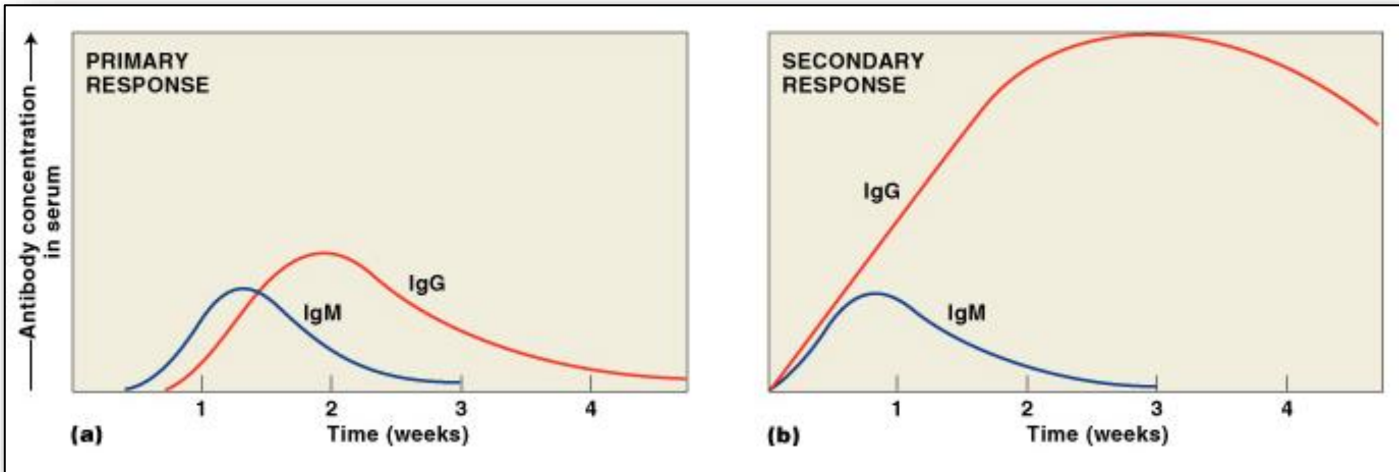


Figure 25: Type-I hypersensitivity

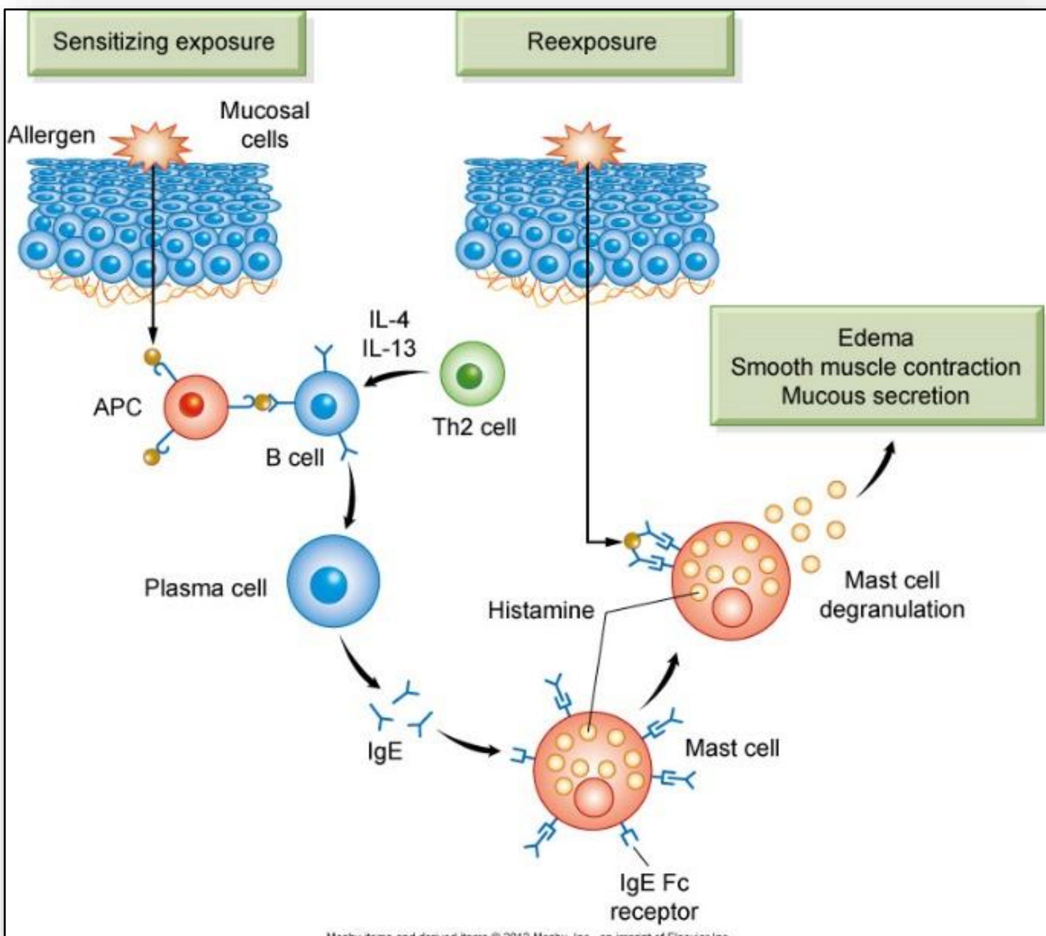


Figure 26: Type-III Hypersensitivity

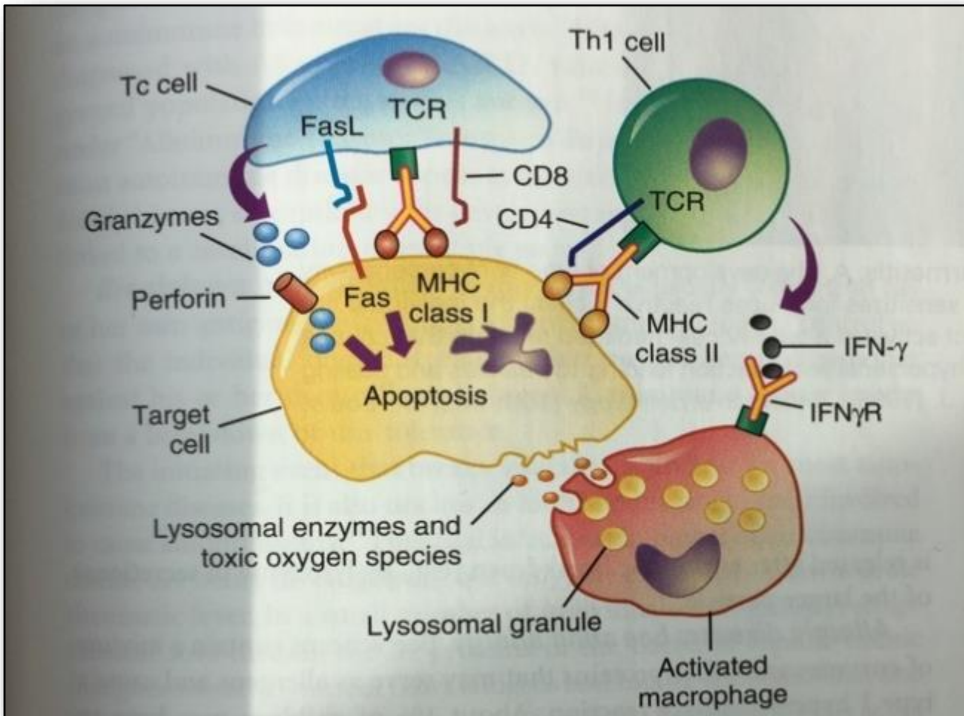


Figure 27: Type-IV cell-mediated immune response

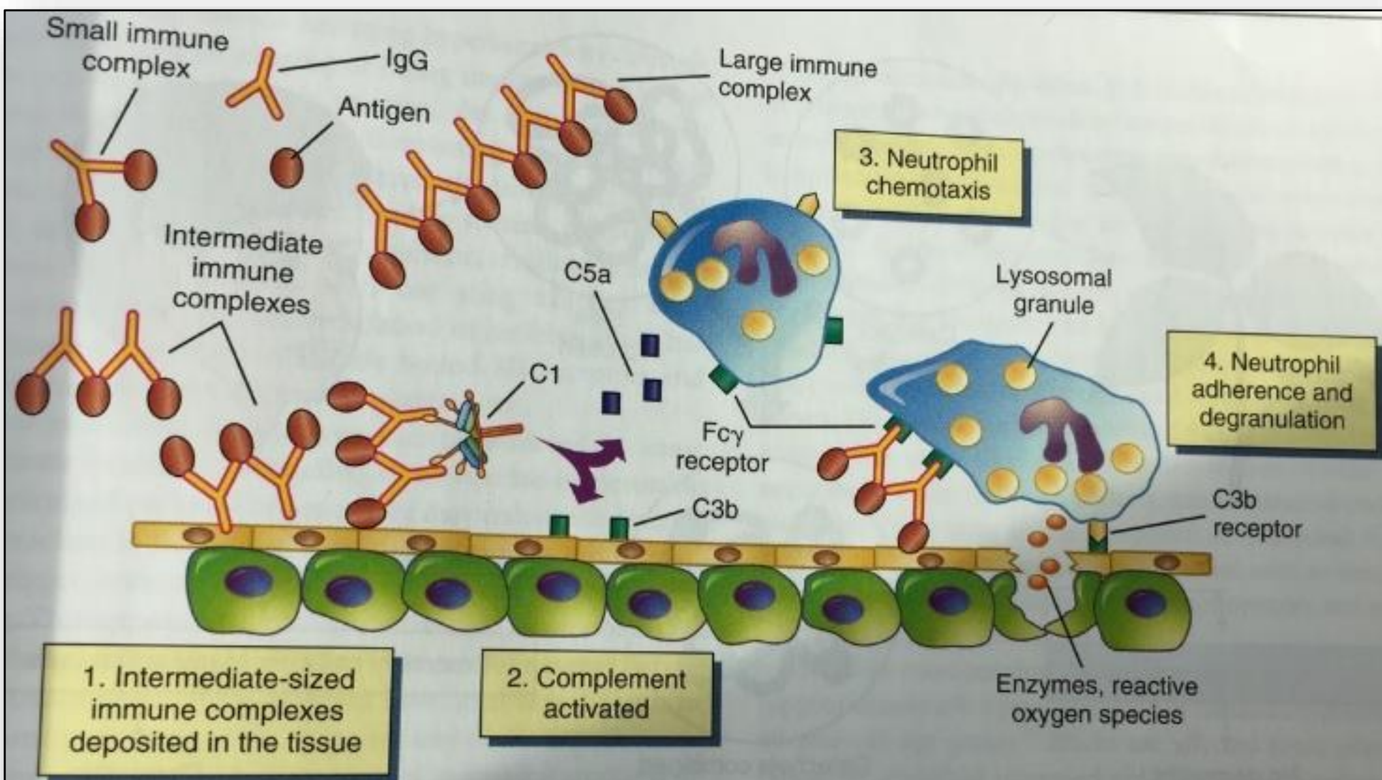


Figure 28: Systemic lupus erythematous (SLE)

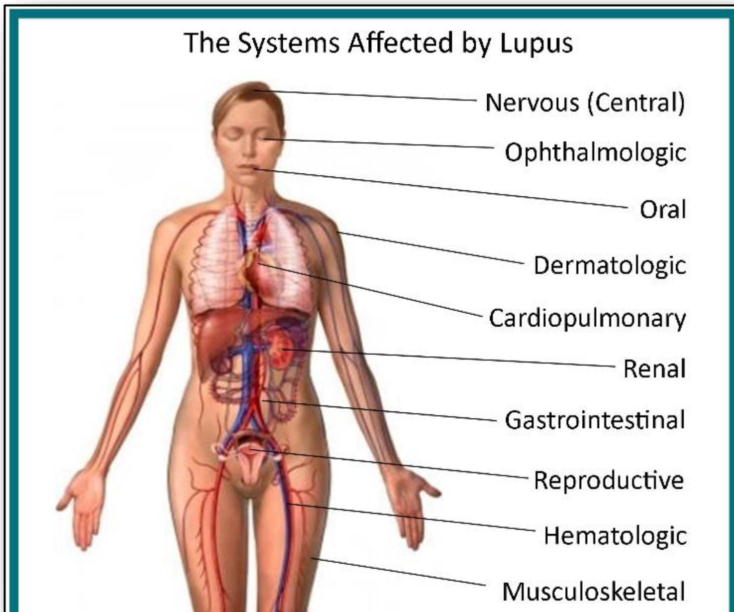


Figure 29: Table – Criteria for the diagnosis of SLE

No	Criteria
1	Malar rash
2	Discoid rash
3	Photosensitivity
4	Oral ulcers
5	Arthritis
6	Serositis
7	Renal disease a. > 0.5 g/d proteinuria, or b. ≥ 3+ dipstick proteinuria, or c. Cellular casts
8	Neurologic Disorder a. Seizures, or b. Psychosis (without other cause)
9	Hematologic disorders a. Hemolytic anemia, or b. Leukopenia (< 4000/mL), or c. Lymphopenia (< 1500/mL), or d. Thrombocytopenia (< 100,000/mL)
10	Immunologic abnormalities a. Positive LE cell preparation, or b. Antibody to native DNA, or c. Antibody to Sm. or d. False-positive serologic test for syphilis
11	Positive antinuclear antibody (ANA)

Figure 30: Malar rash of SLE

Figure 31: Transfusion reaction - ABO blood groups

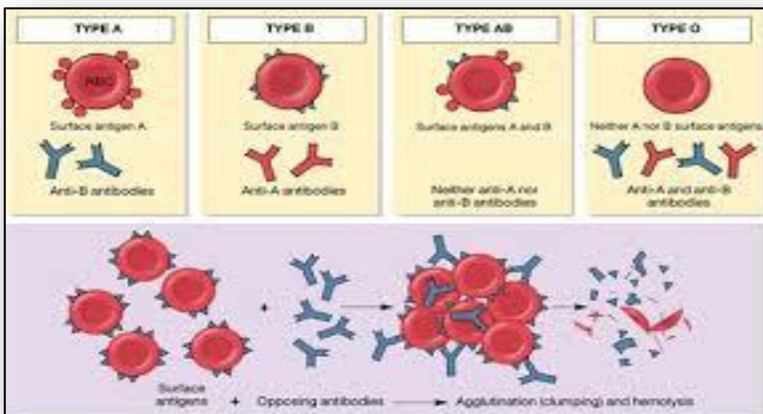
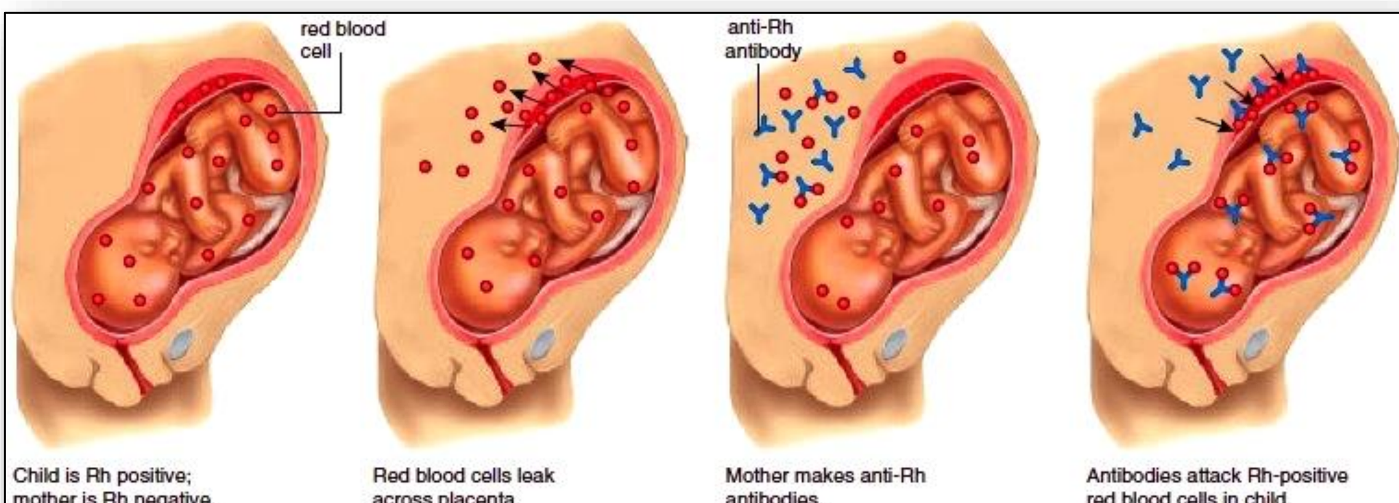


Figure 32: Transfusion reaction - Rh incompatibility in pregnancy



3: Cellular Proliferation and Cancer

Figure 1: Benign vs Malignant Tumors

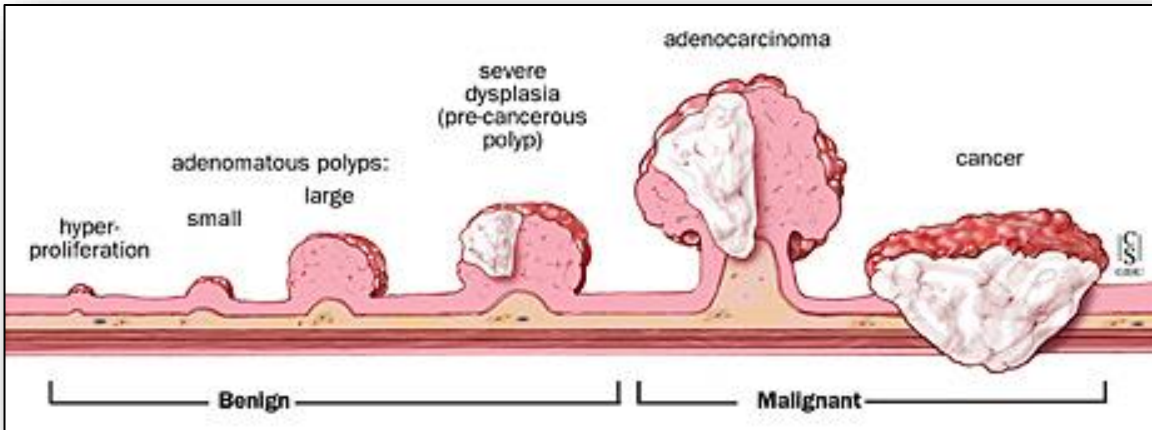


Figure 2: The tumor microenvironment (note significant cell heterogeneity)

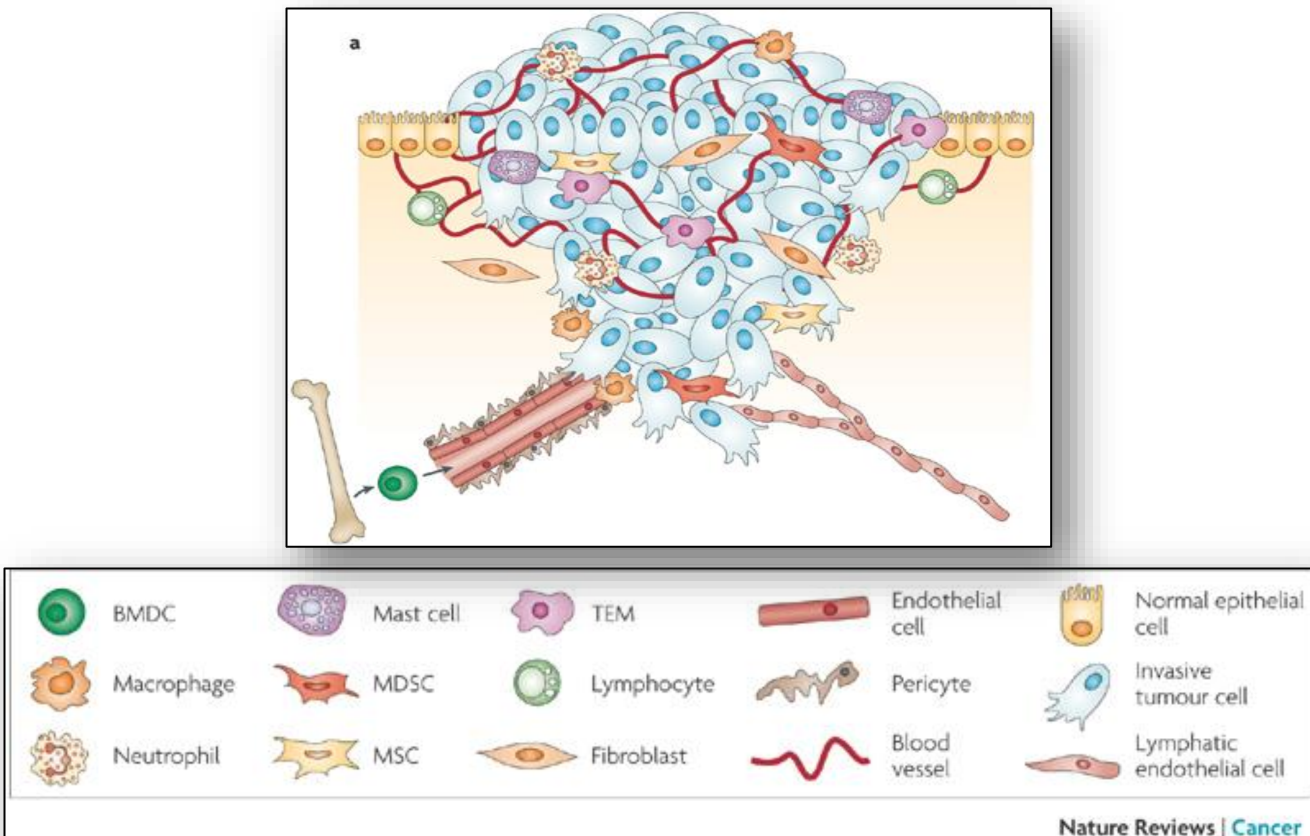


Figure 3: Energy Production in Normal vs. Proliferative (Tumor) Tissue

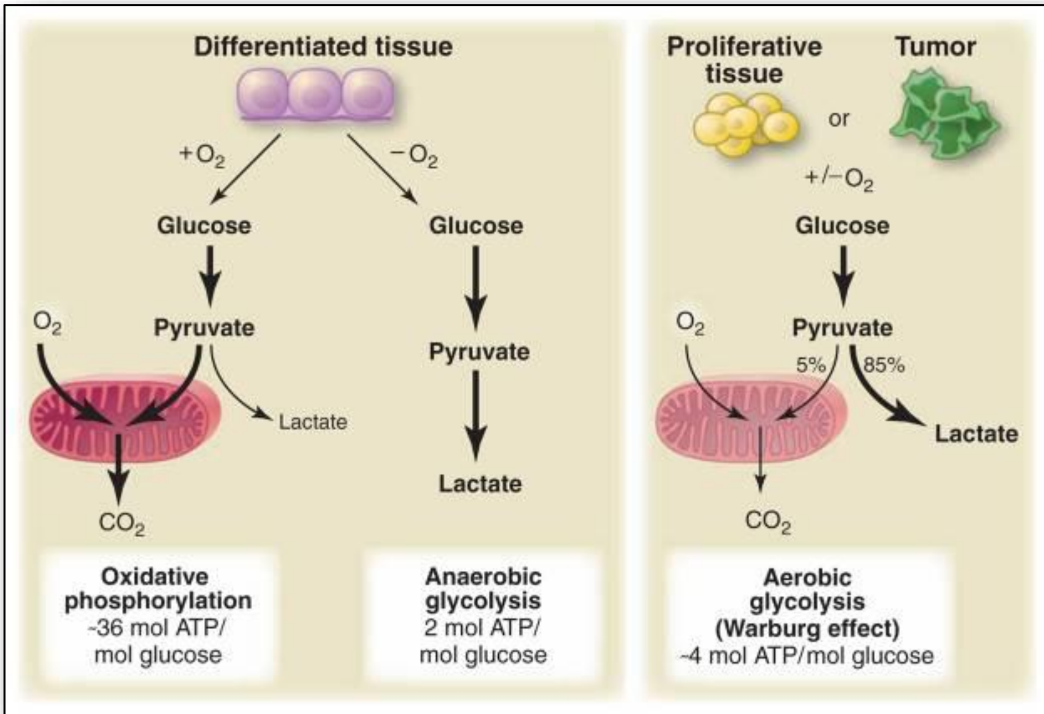
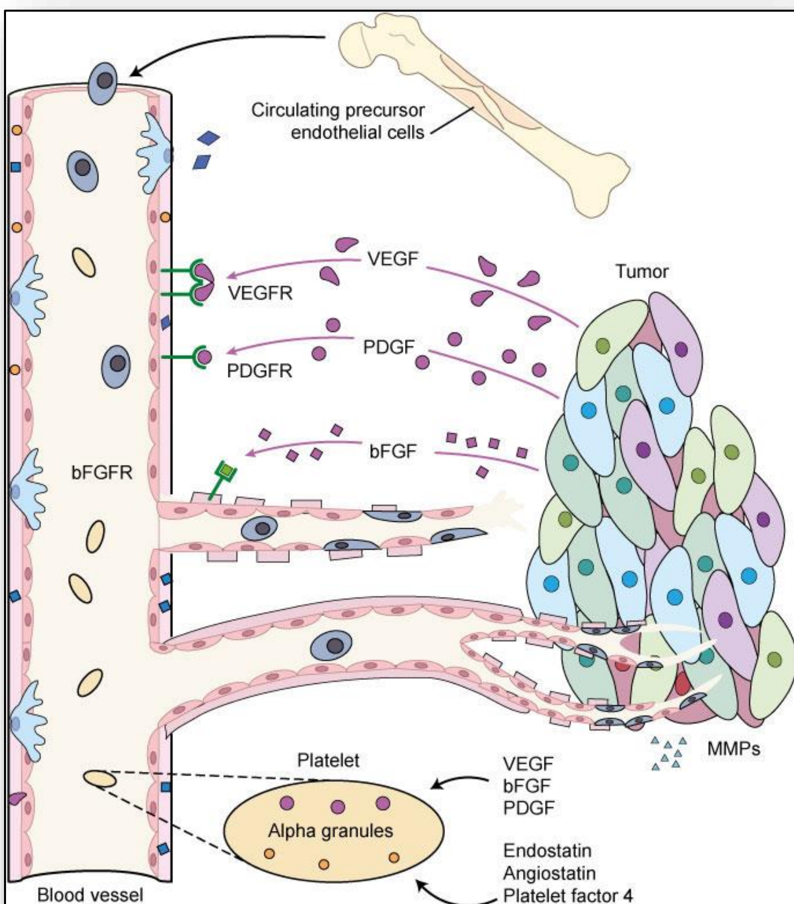
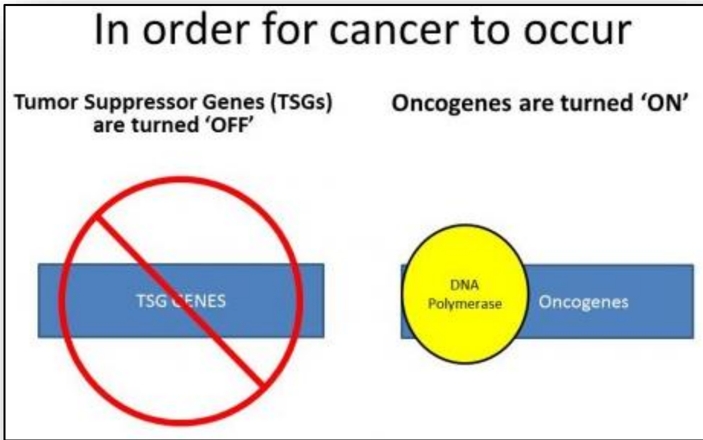


Figure 4: Angiogenesis and the presence of growth factors



(Adapted from Folkman J: Angiogenesis: an organizing principle for drug discovery? *Nat Rev Drug Discov* 6(4):273–286, 2007.)

Figure 5: Oncogenes and tumor-suppressor genes in cancer development



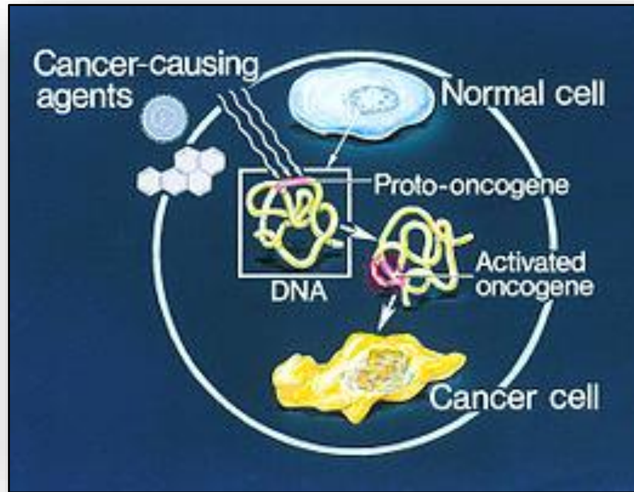


Figure 6: Burkitt lymphoma chromosomal translocation

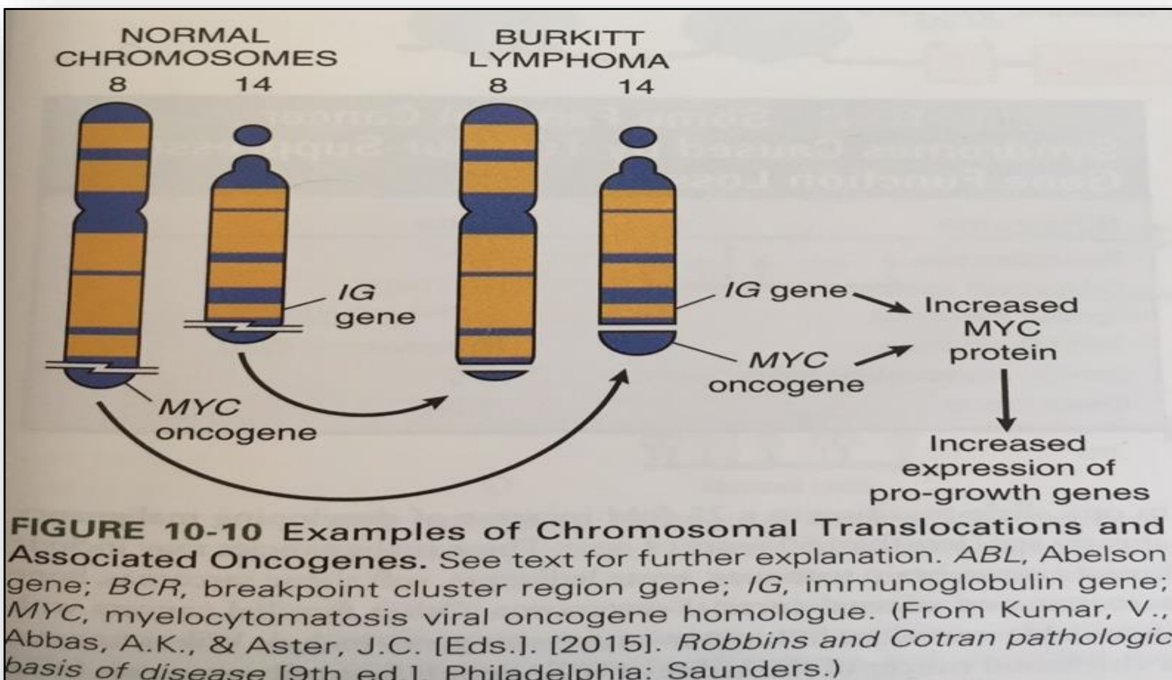


Figure 7: CML Chromosomal Translocation

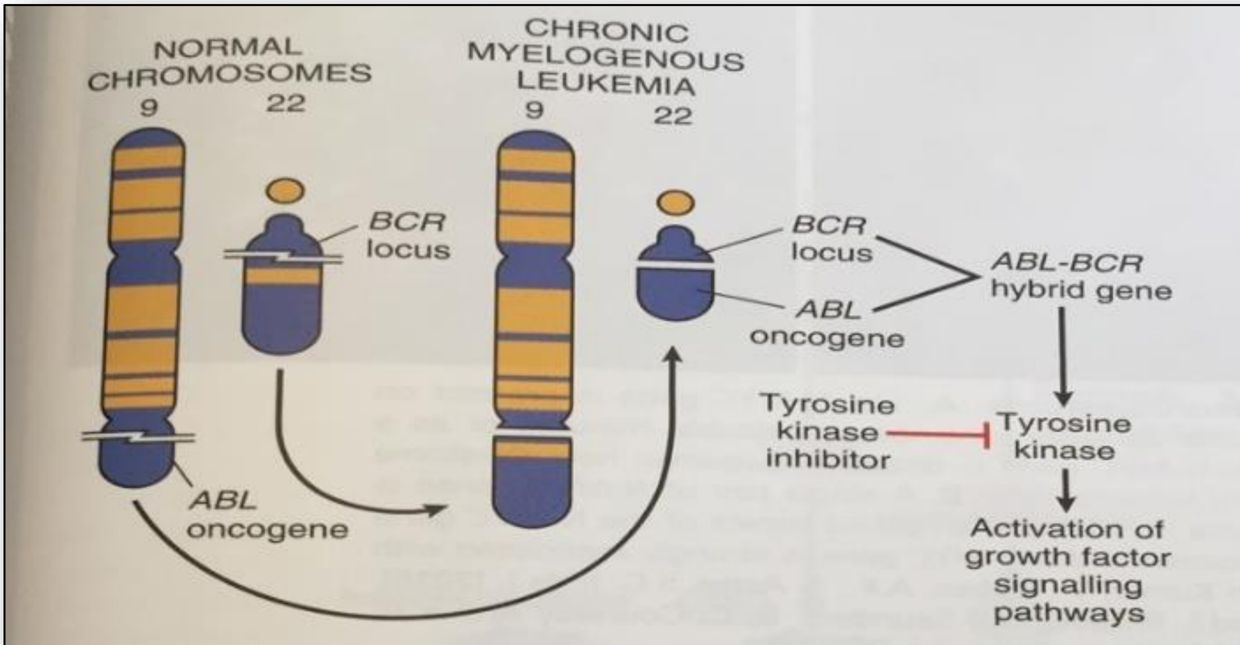


Figure 8: Clinical Tumor Markers

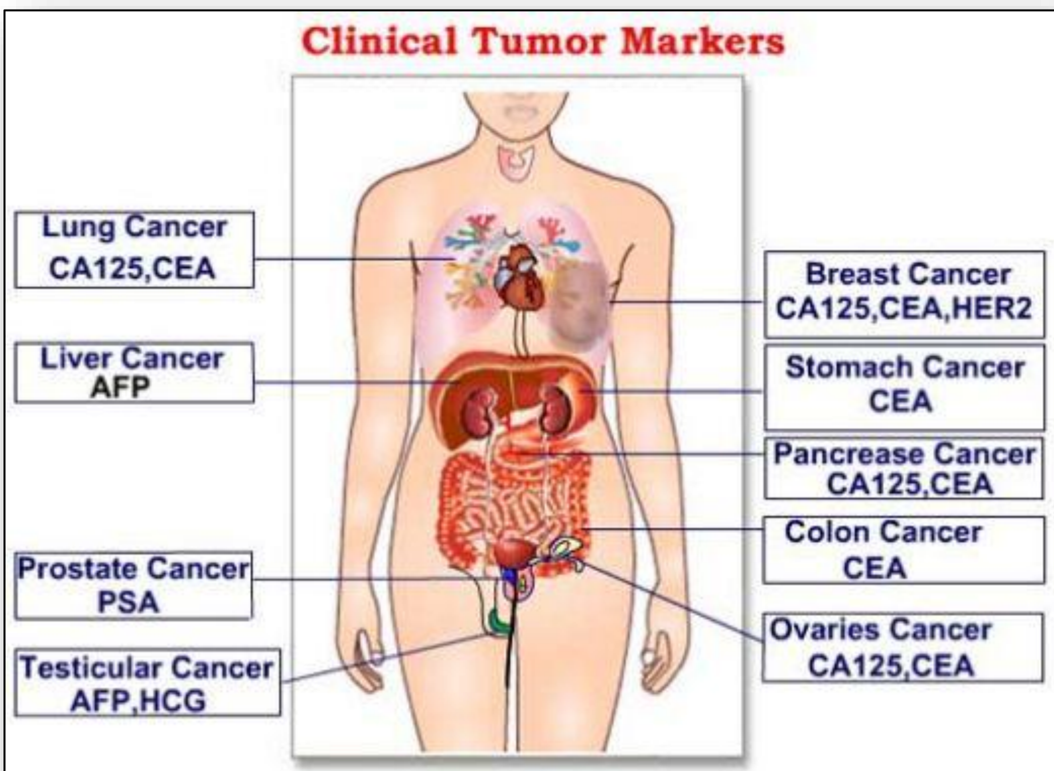


Figure 9: Tumor Markers

Marker name	Type of tumor
Alpha fetoprotein (AFP)	Liver cancer
Prostatic specific antigen (PSA)	Prostate cancer
Bence Jones protein	Multiple myeloma
Homovanilic/ Vanilvlmandelic acid HVA/VMA)	Neuroblastoma Pheochromocvtoma

CA: cancer antigen CEA: cancer embryonic antigen

Figure 10: Stem Cells

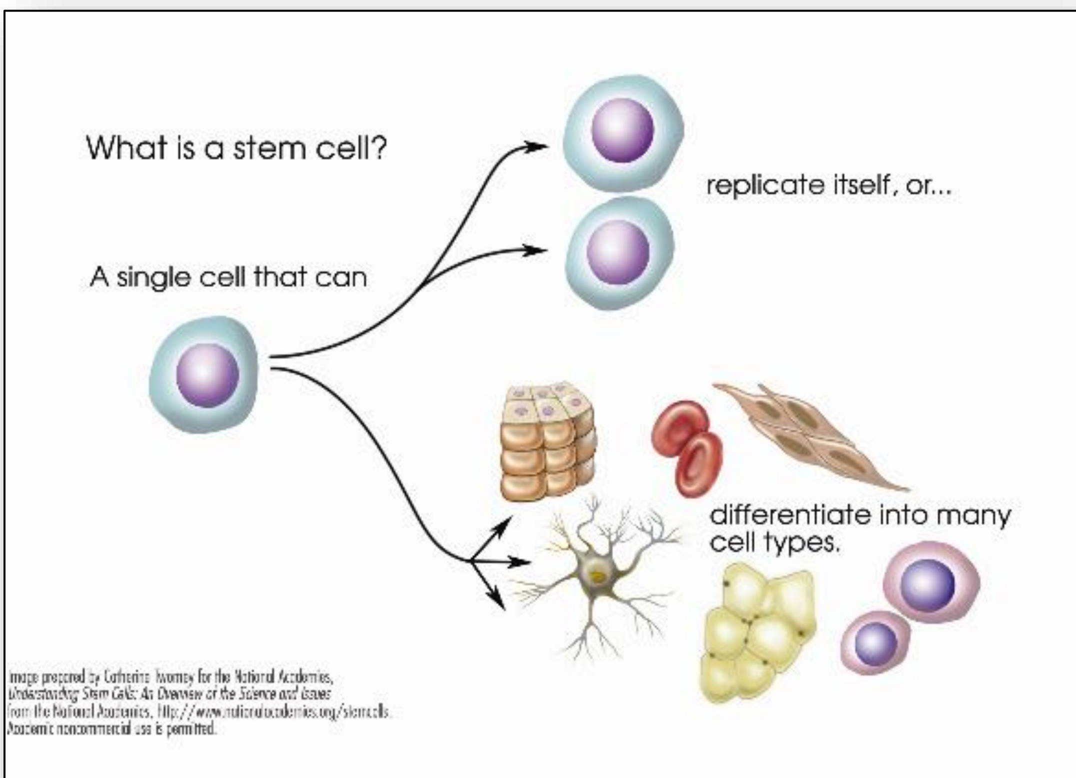


Figure 11: Local vs distant metastasis

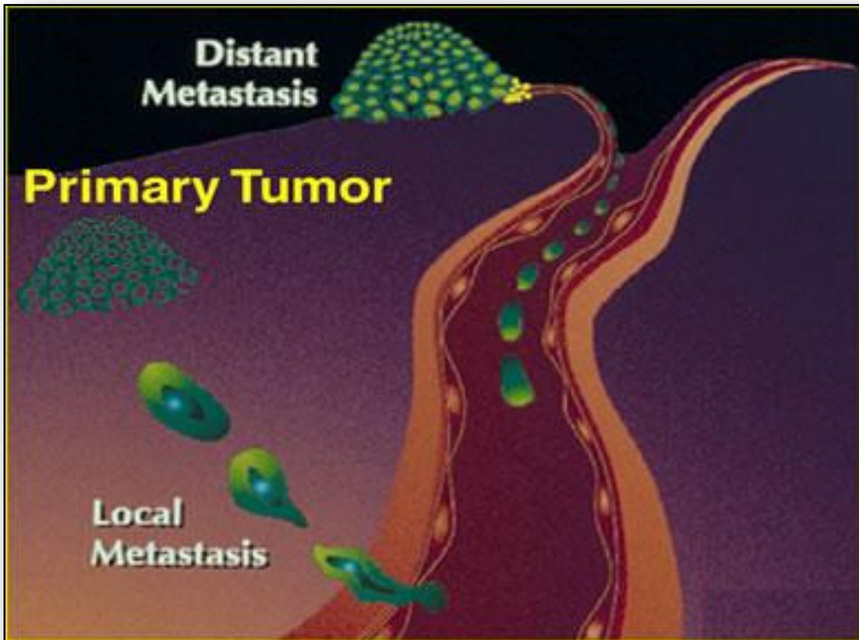


Figure 12: Processes of cancer metastasis

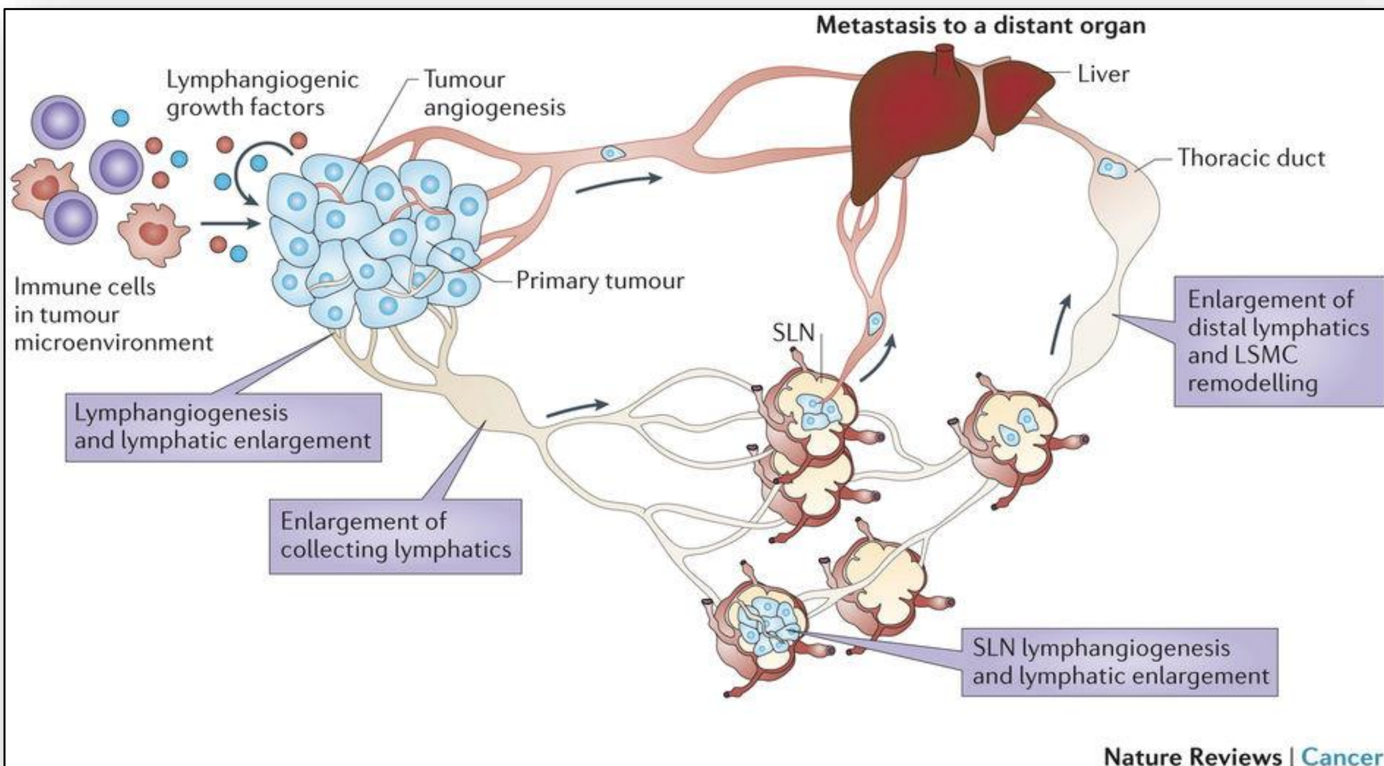


Figure 13: Types of Biopsies

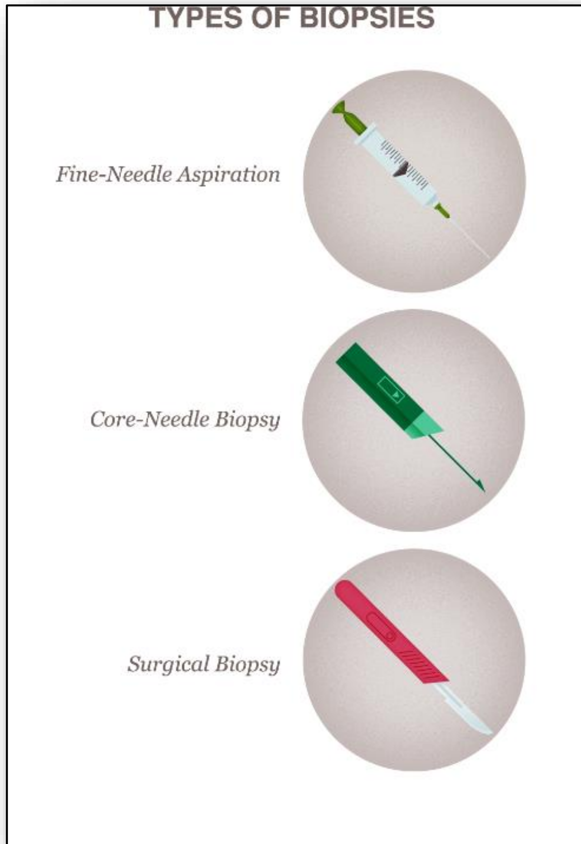


Figure 14: Surgical biopsy of skin lesion

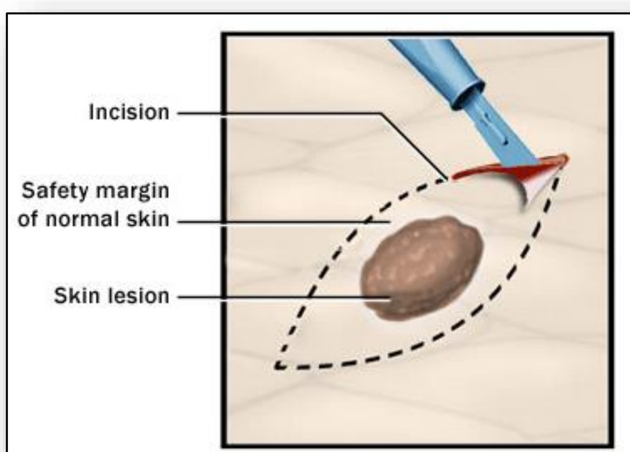


Figure 15: Processing and histopathology of biopsies

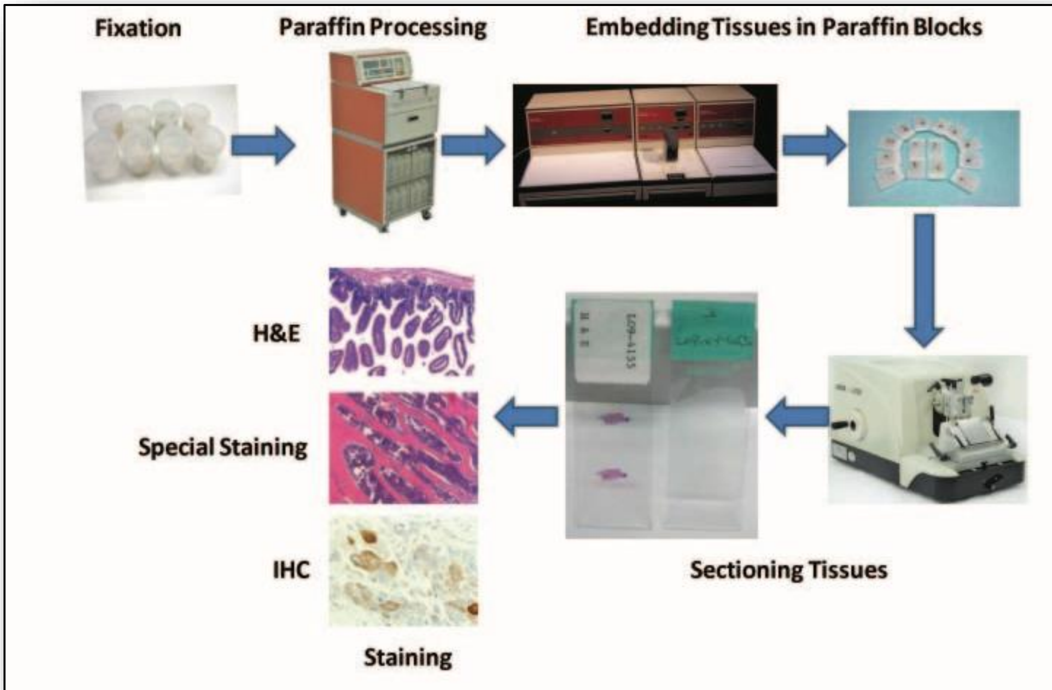


Figure 16: Tumor staging – Microscopic analysis

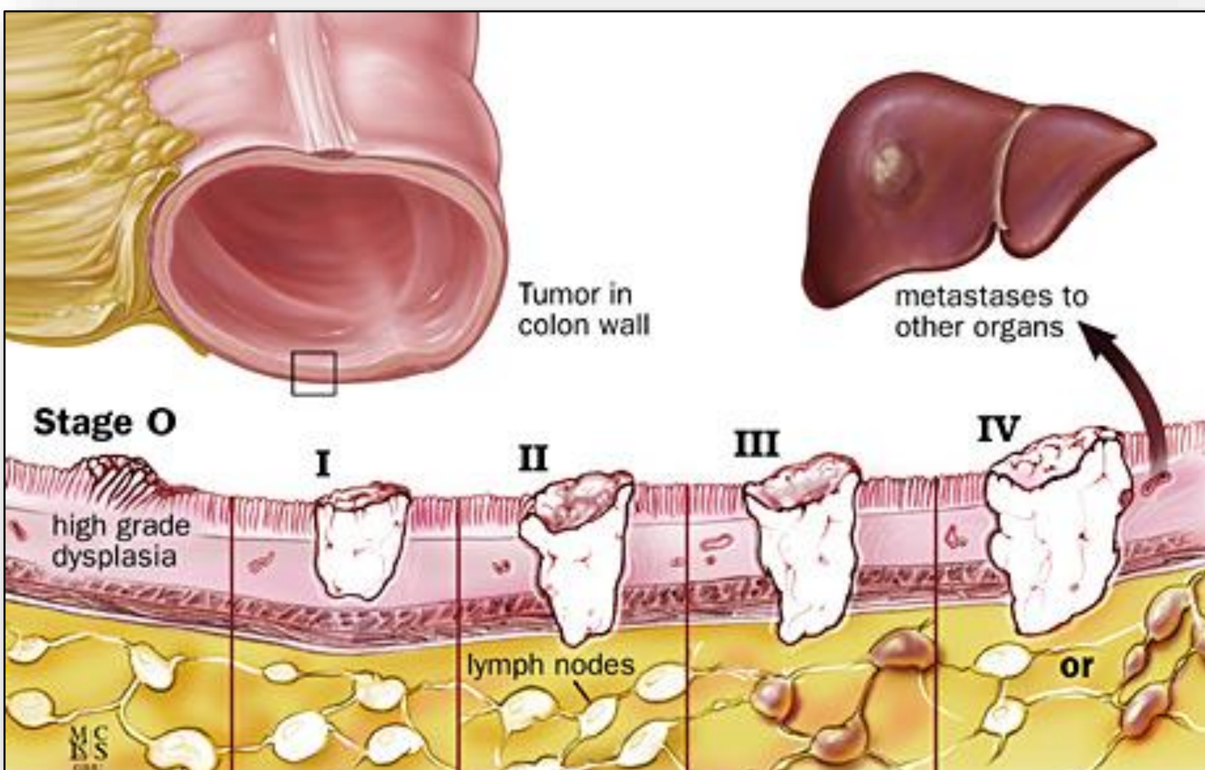


Figure 17: TMN Classification – (example, breast cancer)

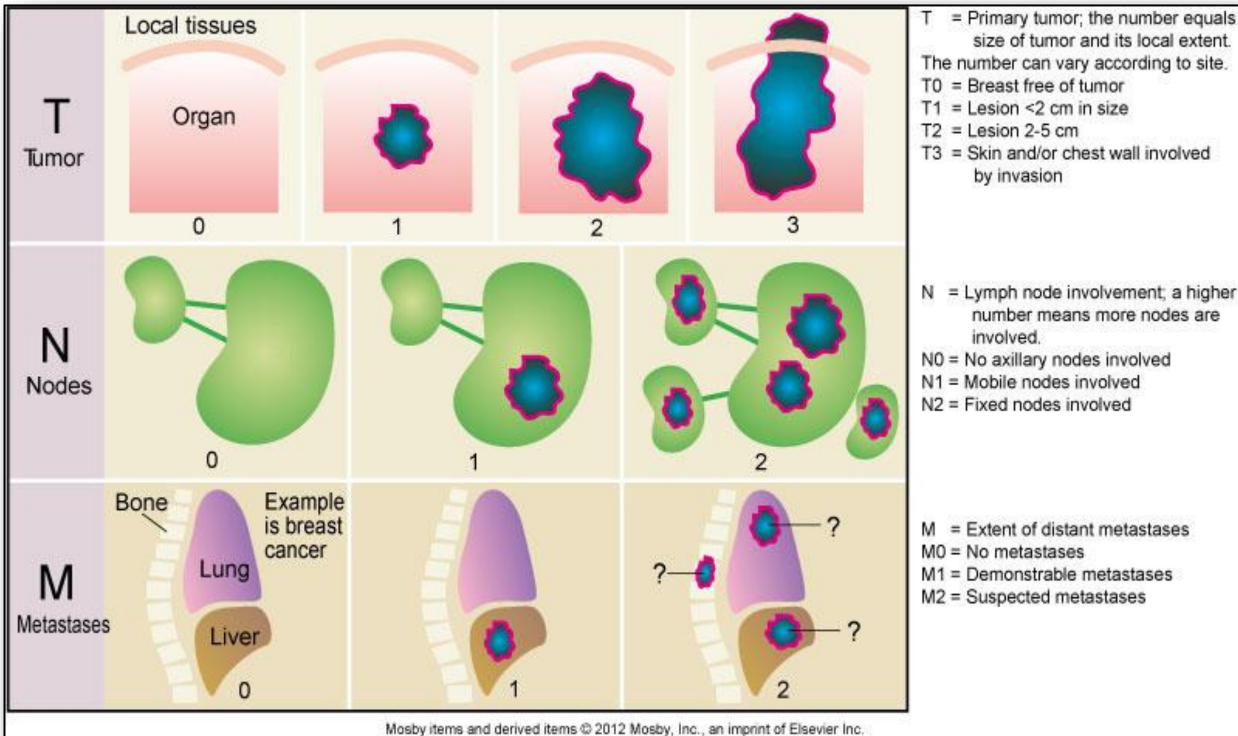


Figure 18: Syndrome of cachexia

Figure 19: Cell types originating from the mesoderm germ layer

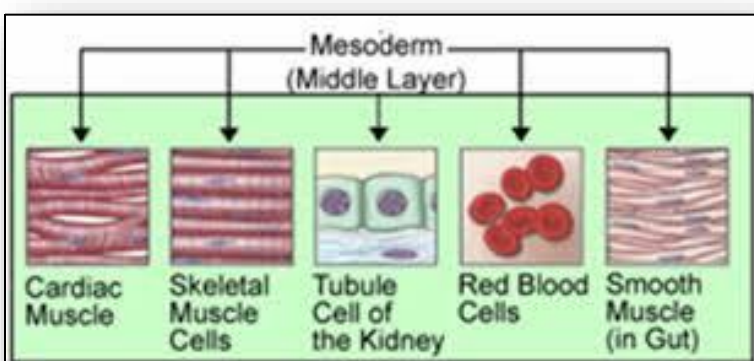


Figure 20: Bone tumors - Osteosarcoma (left) and Ewing sarcoma (right)

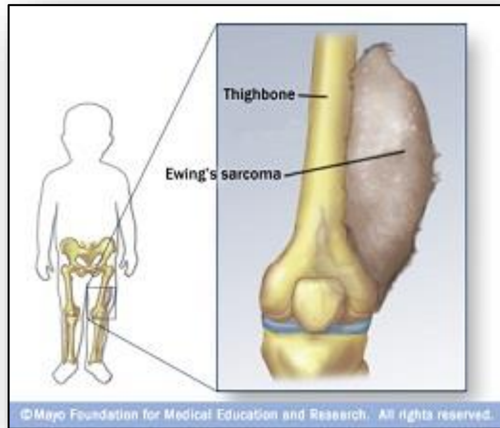


Figure 21: Wilms tumor (two images)

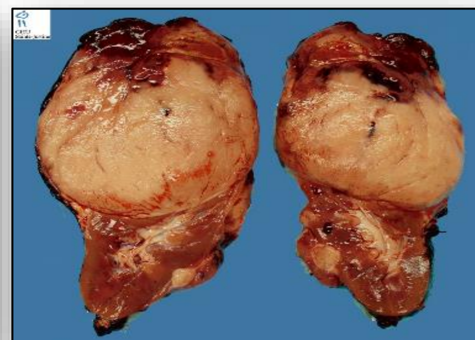


Figure 22: Tobacco and cancer risk

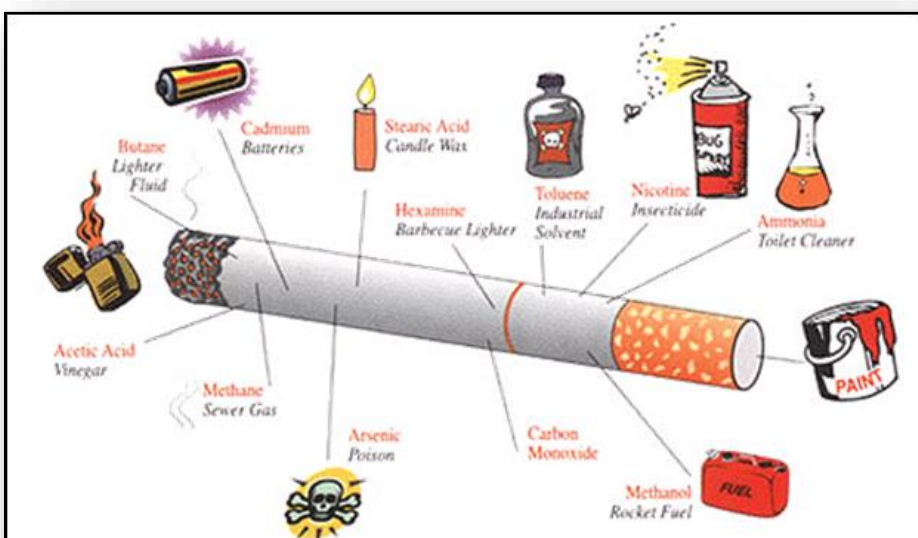


Figure 23: Obesity and cancer risk

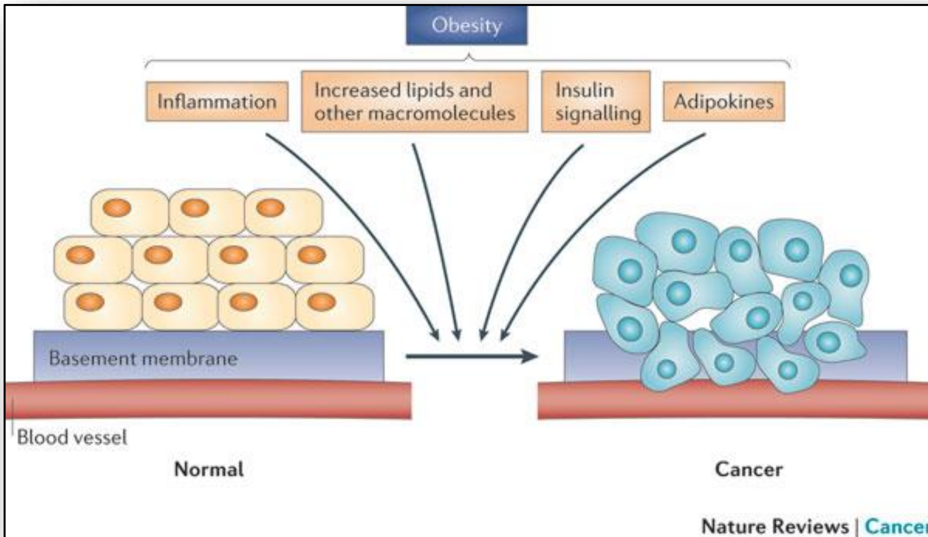
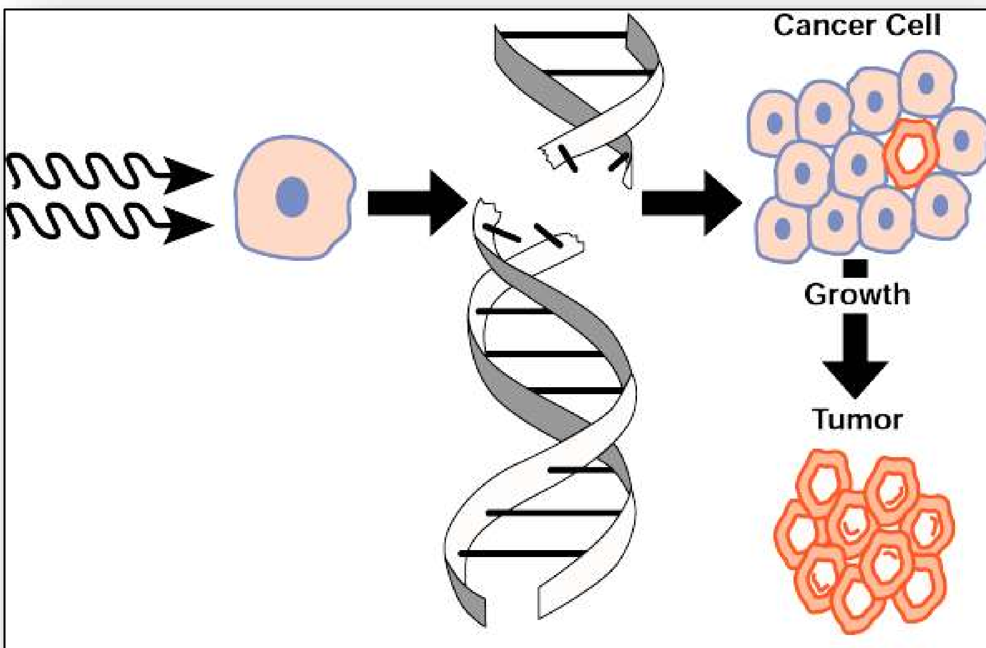


Figure 24: Ionizing radiation and cancer risk



4: Alterations of the Endocrine System

Figure 1: Hypothalamus and pituitary regulation of cortisol secretion in response to stress

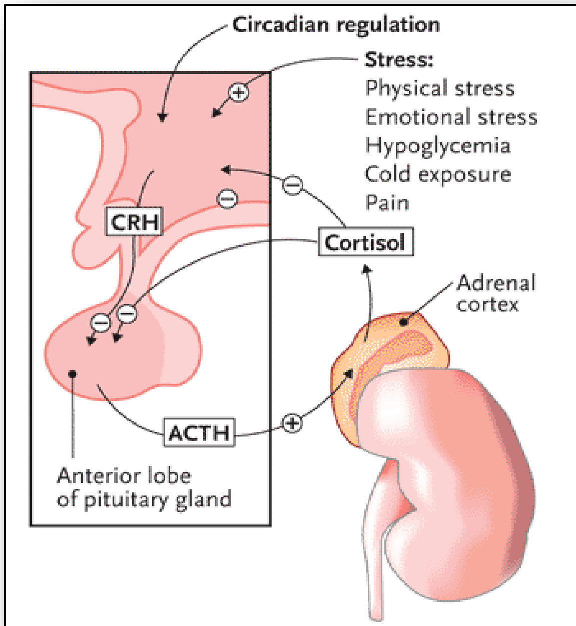


Figure 2: Hypothalamus and Pituitary Hormones, Targets, and Function

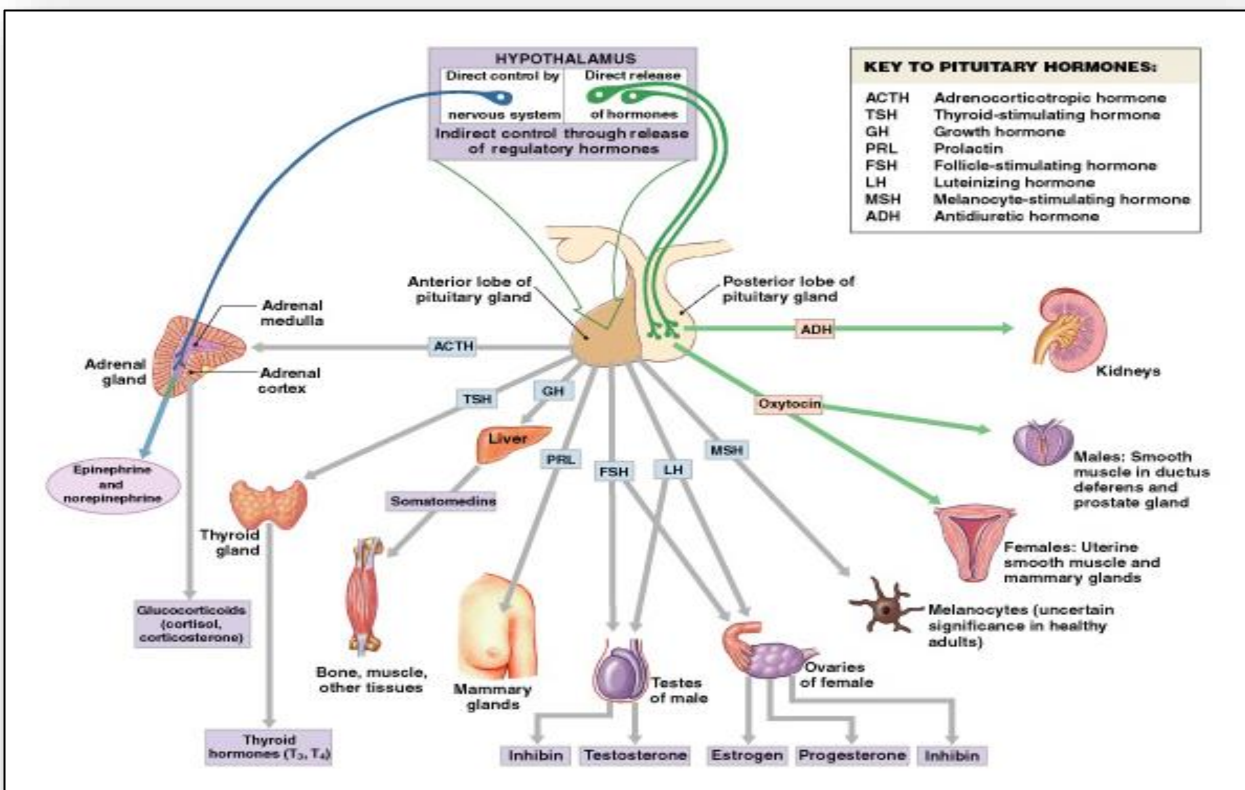


Figure 3: Renin-Angiotensin-Aldosterone-System (RAAS) of kidneys

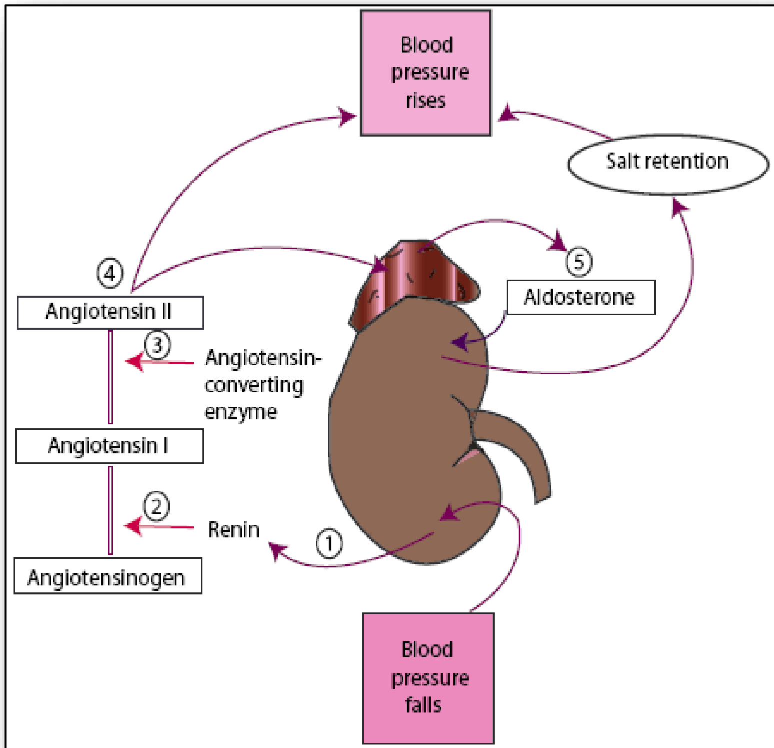


Figure 4: Mechanisms of Calcium Regulation

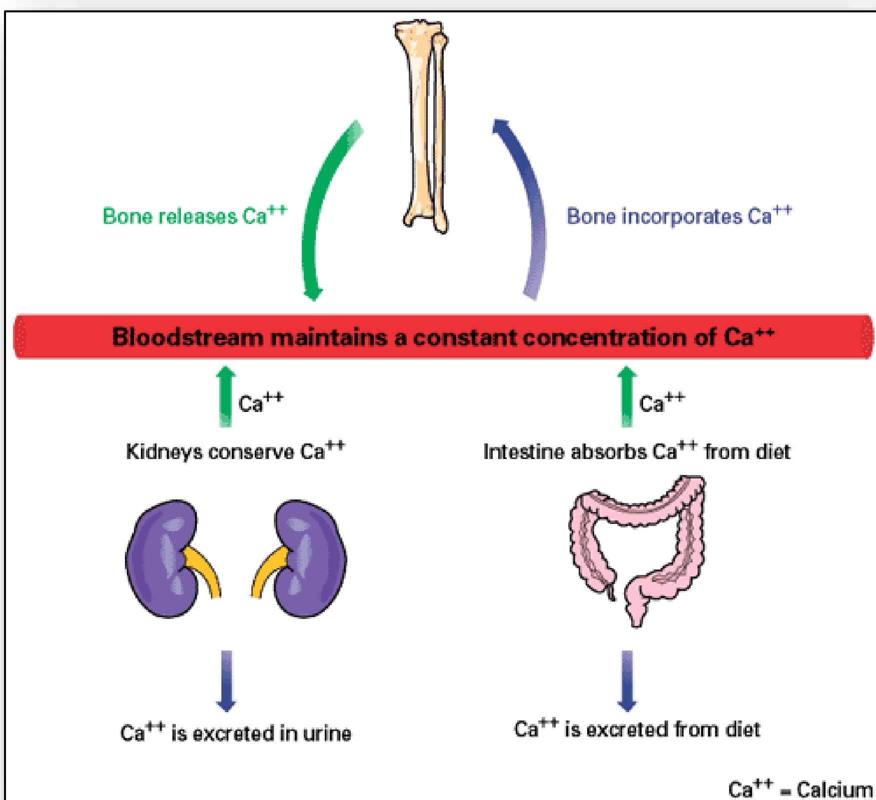


Figure 5: Expected results for normal urine analysis

Normal urine analysis

- Normal values are as follows:
- Color – Yellow (light/pale to dark/deep amber)
- Clarity/turbidity – Clear or cloudy
- pH – 4.5-8
- Specific gravity – 1.005-1.025
- Glucose - ≤130 mg/d
- Crystals – Occasionally
- Bacteria – None
- Yeast - None
- Casts – 0-5 hyaline casts/lpf
- Ketones – None
- Nitrates – Negative
- Leukocyte esterase – Negative
- Bilirubin – Negative
- Urobilirubin – Small amount (0.5-1 mg/dL)
- Blood - ≤3 RBCs
- Protein - ≤150 mg/d
- RBCs - ≤3RBCs/hpf
- WBCs - ≤2-5 WBCs/hpf
- Squamous epithelial cells - ≤15-20 squamous epithelial cells/hpf

Figure 6: Volume status, urine osmolality and sodium

Volume status	Urine osmolality (mOsm/kg)	Urine sodium (mEq/L)	Etiology
Hypovolemic	↑	≤20	Nonrenal sodium loss
		>20	Renal sodium loss
Hypervolemic	↑	≤20	Edematous states
		>20	Kidney failure
Euvolemic	↑		SIADH
	Intermediate	>20	Potomania
	↓		Primary polydipsia
	Variable		Excess fluid intake

↑= >300; **intermediate**: 150-300; ↓: ≤100-150

Figure 7: Hypopituitary-axis and hormonal regulation

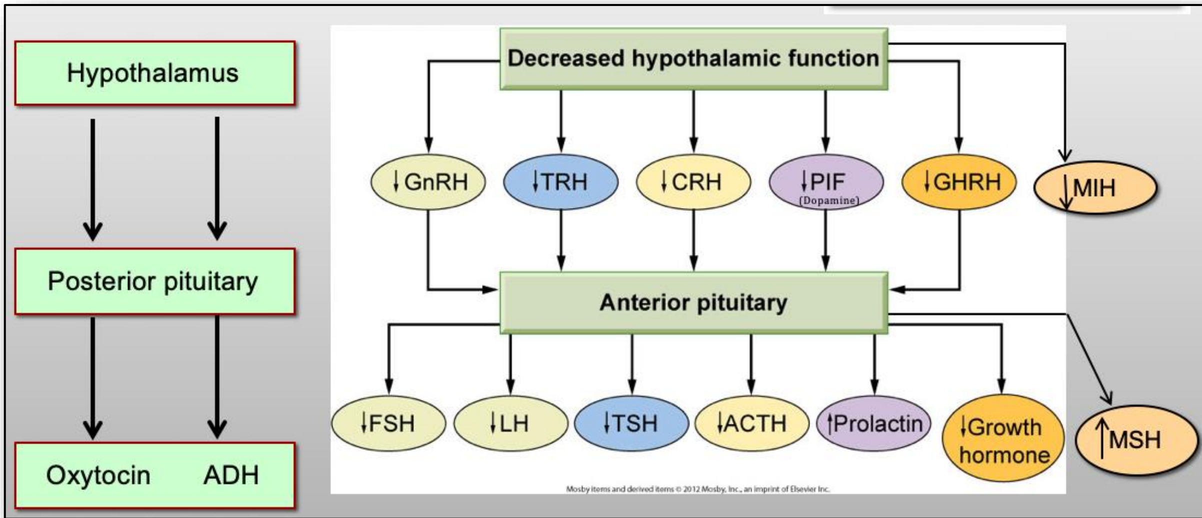


Figure 8: Anatomy of the thyroid gland

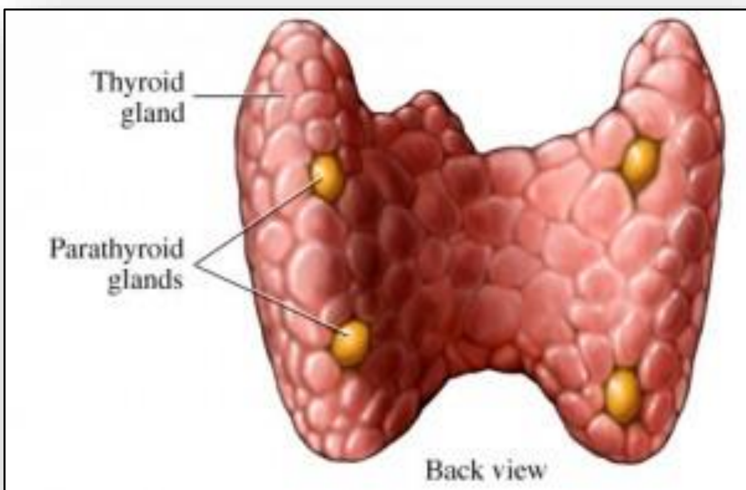


Figure 9: Production and release of thyroid hormones with a follicle

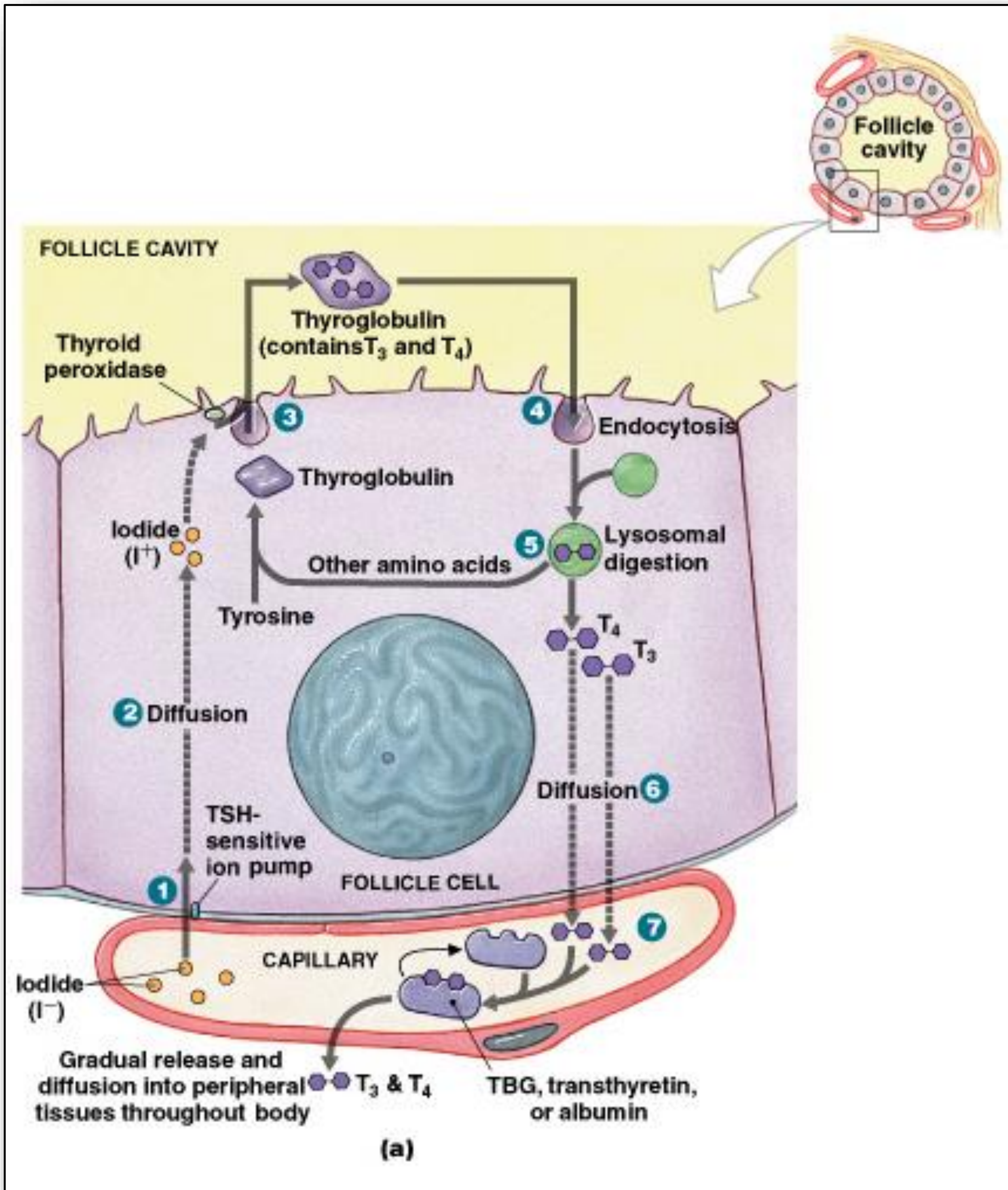


Figure 10: Comparison of hyperfunction and hypofunction of the thyroid gland

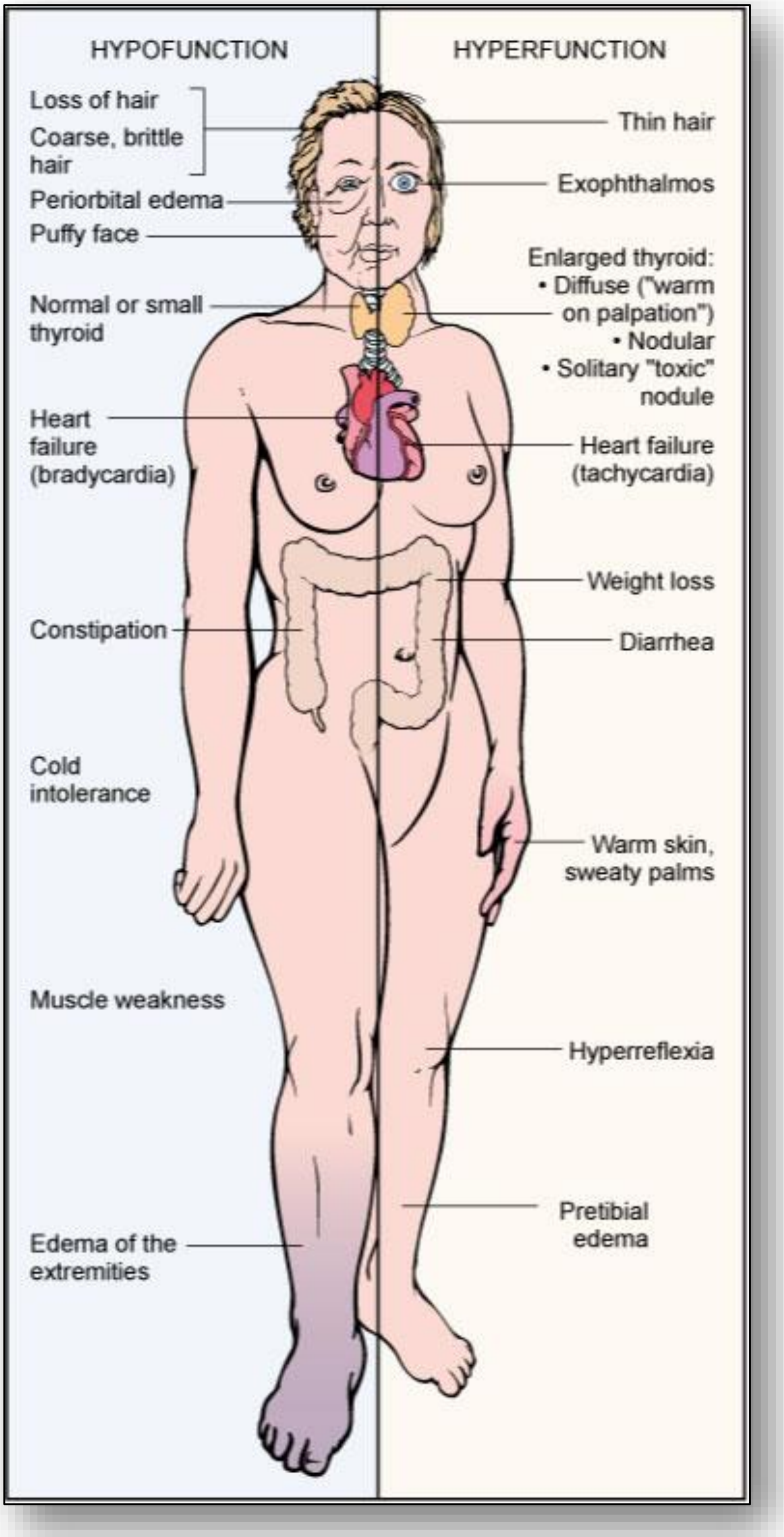


Figure 11: Radioactivity in Graves' versus other multimodule goitre and thyroid cancer (adenoma)

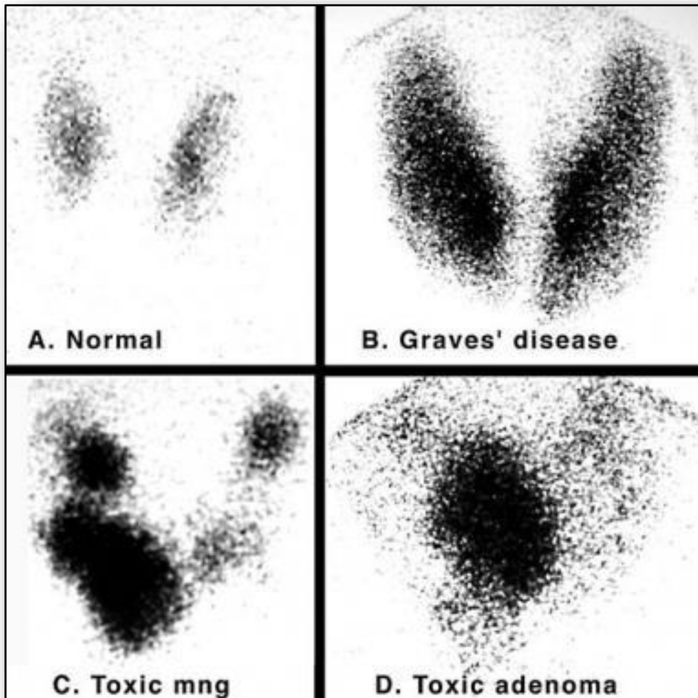


Figure 12: Goitre typically seen in Grave's disease

Figure 13: Example of pretibial myxedema

Figure 14: Example of exopthalmos

Figure 15: Tetany in the hand

Figure 16: Most common, featured symptoms of Diabetes mellitus

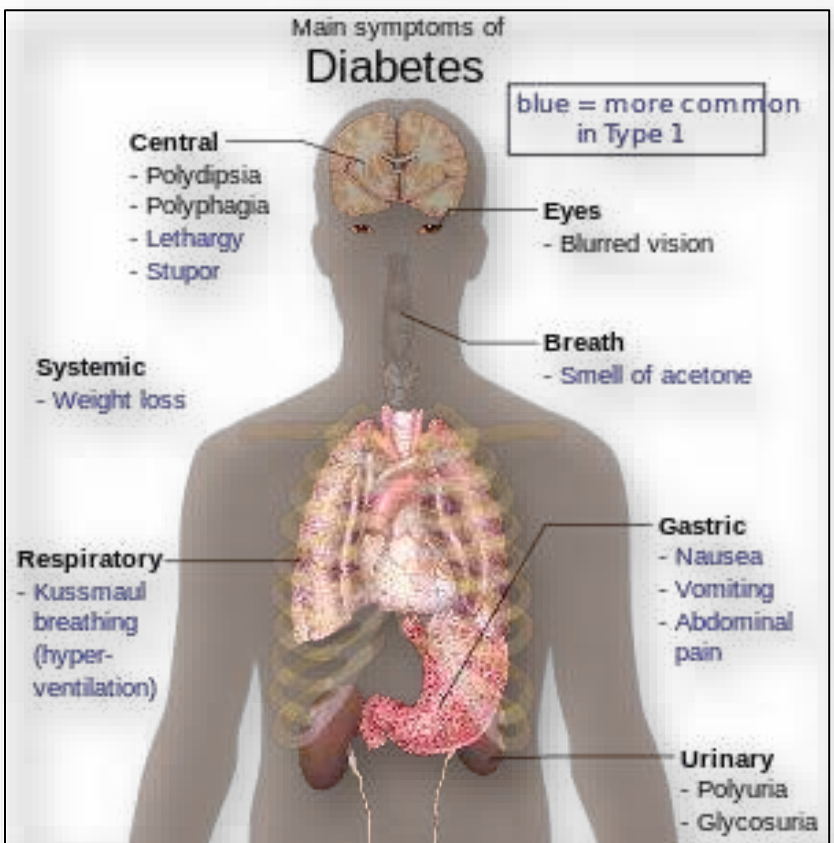


Figure 17: Table – Complications of Diabetes Mellitus

Complications of Diabetes Mellitus	
Acute	**Chronic**
▪ Hypoglycemia - More common in Type I	▪ Hyperglycemia and non-enzymatic glycosylation
▪ Hyperglycemia - Diabetic ketoacidosis (Type I) - Hyperosmolar hyperglycemic nonketotic syndrome (HHNKS) (Type II)	▪ Microangiopathy - Retinopathy - Nephropathy - Neuropathy
	▪ Macroangiopathy - Coronary artery disease (CAD) - Cardiovascular accident (CVA; Stroke) - Peripheral artery disease (PAD)
	▪ Repeated infections
	▪ Skin complications

Figure 18: Complications of Diabetes mellitus (angiopathies)

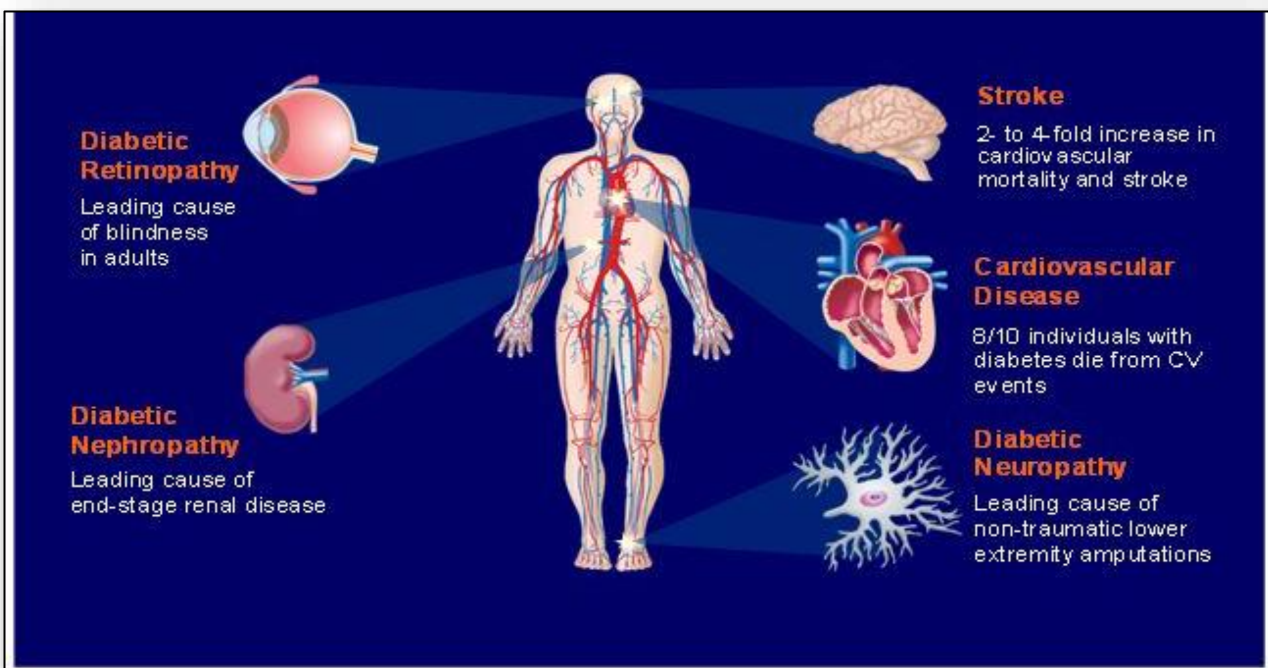


Figure 19: Macrovascular complications of Diabetes mellitus

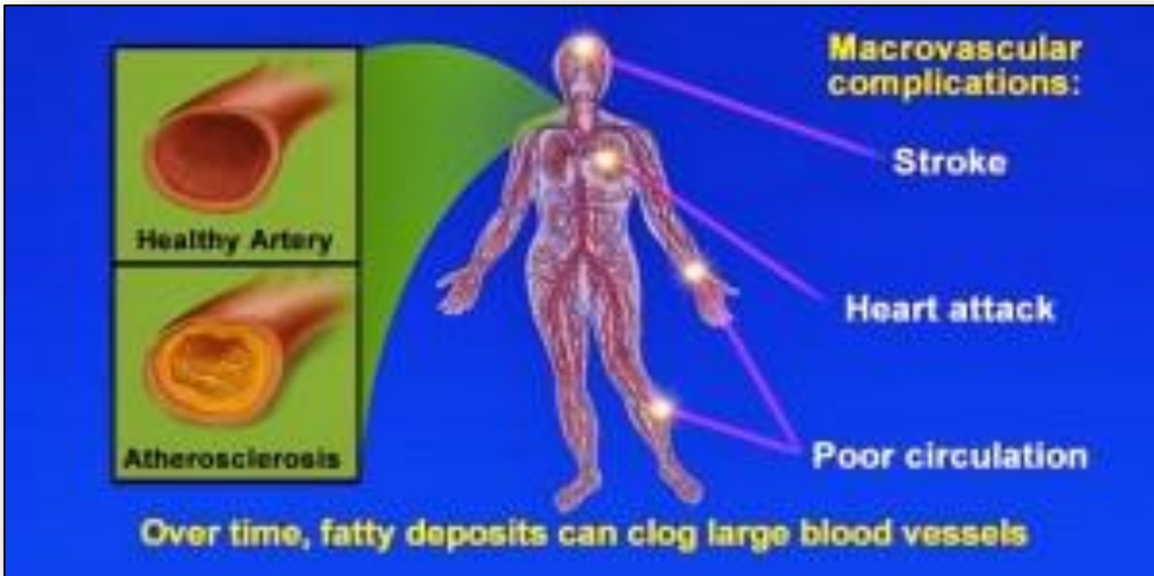


Figure 20: Pathophysiology of DKA and HHNK

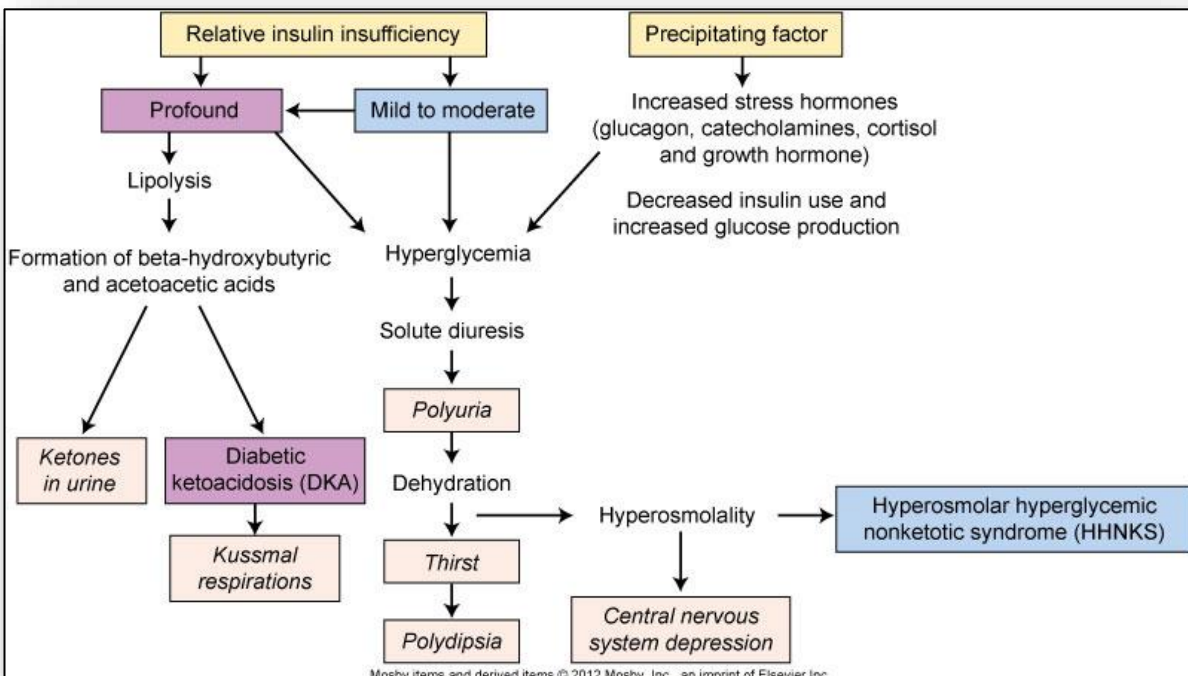


Figure 21: Pathophysiology of diabetic ketoacidosis

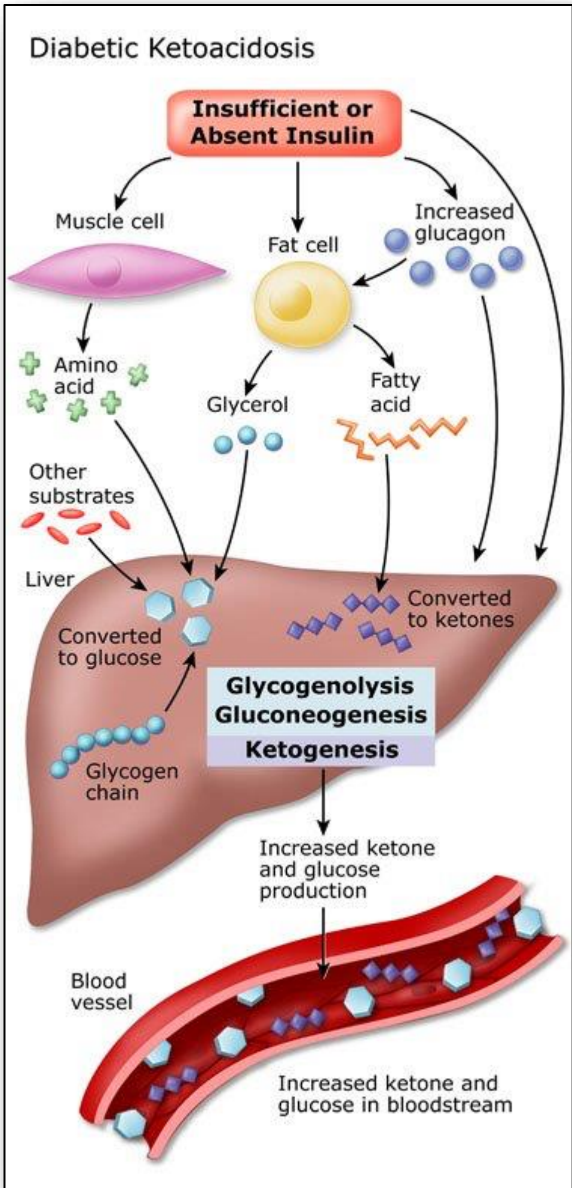


Figure 22: Cellular layers of the adrenal gland

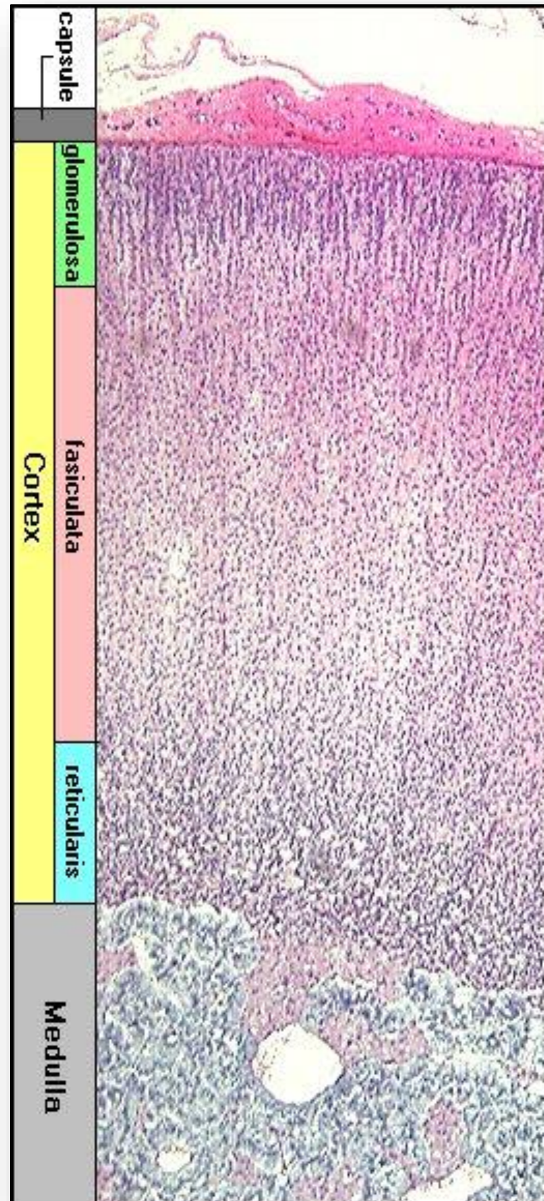


Figure 23: Manifestations of Cushing's syndrome

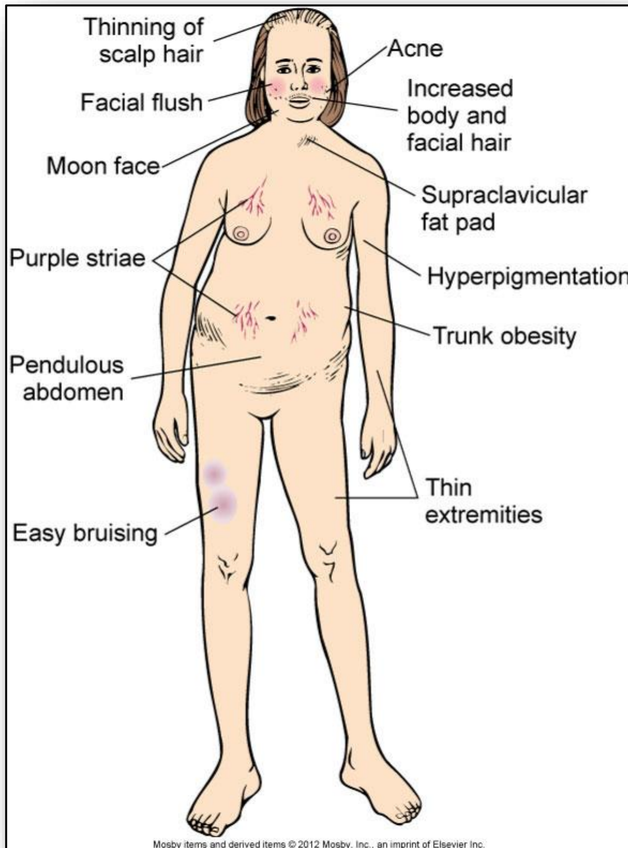


Figure 24: Hyperpigmentation of skin and mucous membranes seen in Addison's disease

4: Alterations of the Reproductive System

Figure 25: Sites of ectopic endometrial cells

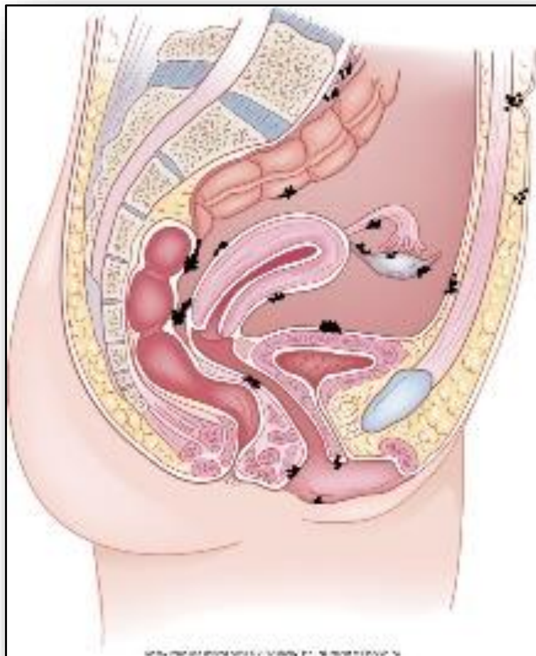


Figure 26: Polycystic ovarian syndrome (comic representation)

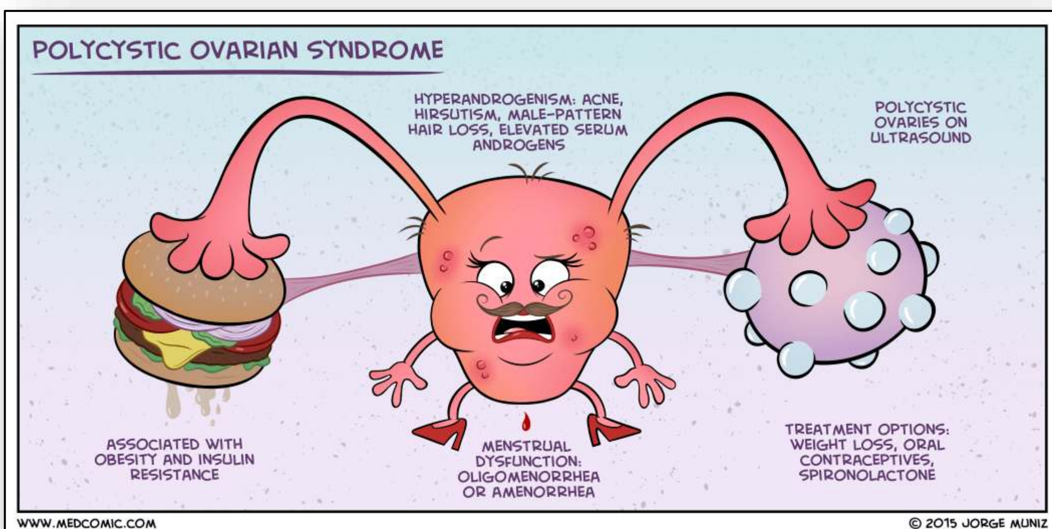
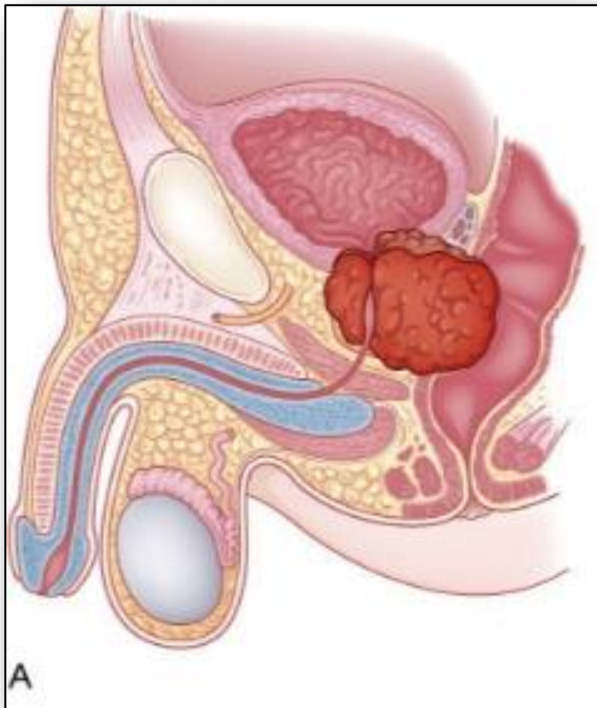


Figure 27: Anatomy of benign prostatic hyperplasia



5: Blood (Disorders of the Hematological System)

Figure 1: Whole Blood Composition

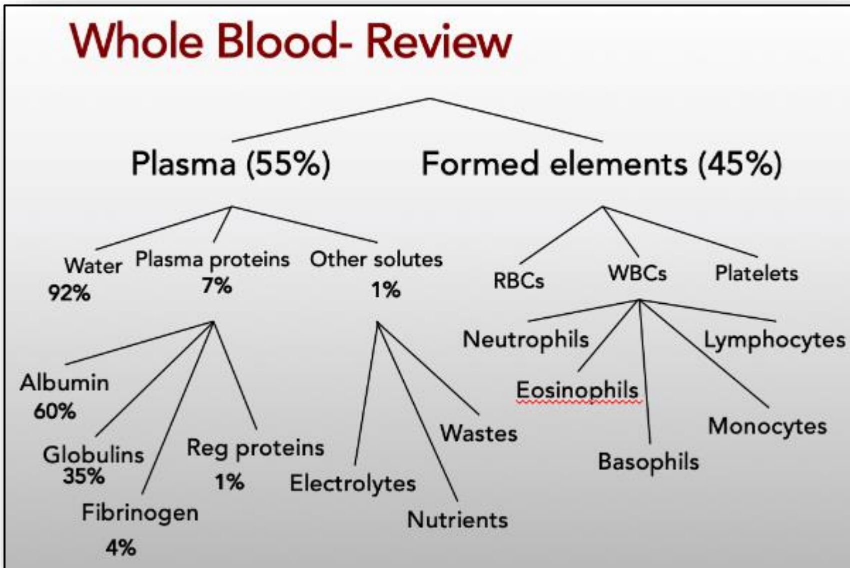


Figure 2: Maturation of Blood Cells

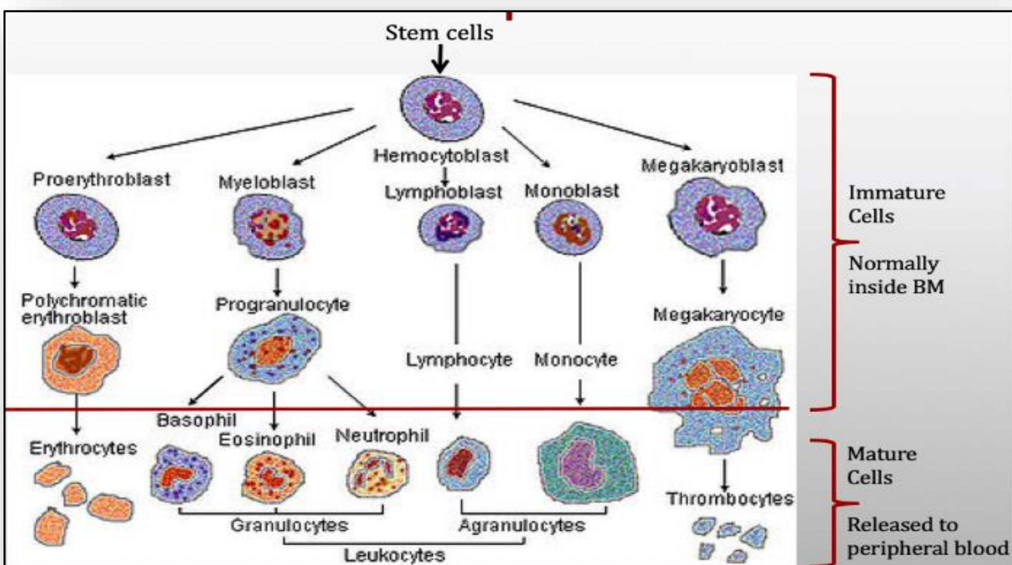


Figure 3: Separated blood components

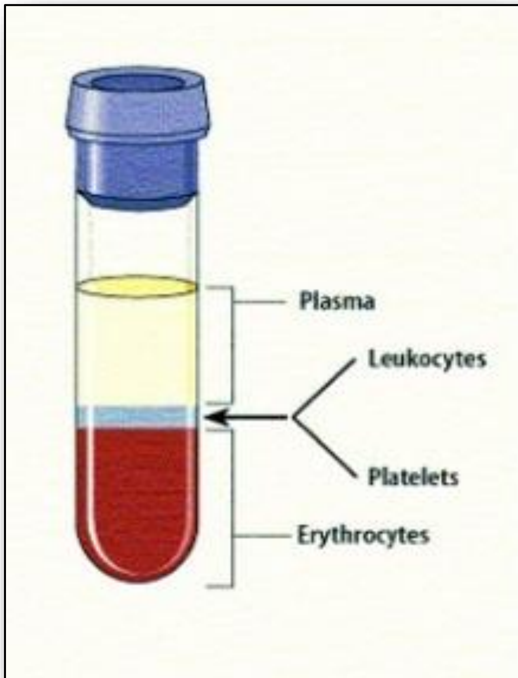


Figure 4: RBC Molecular Structure

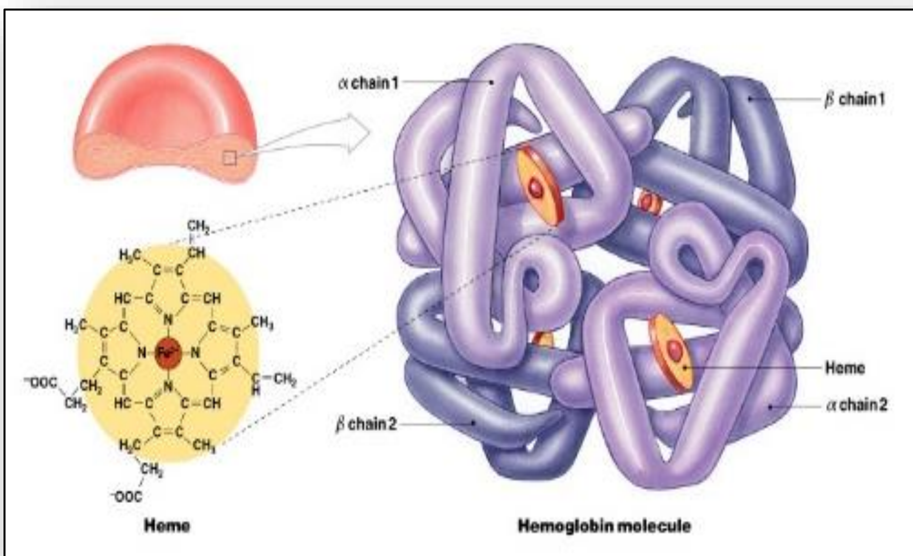


Figure 5: Life Cycle of a Red Blood Cell

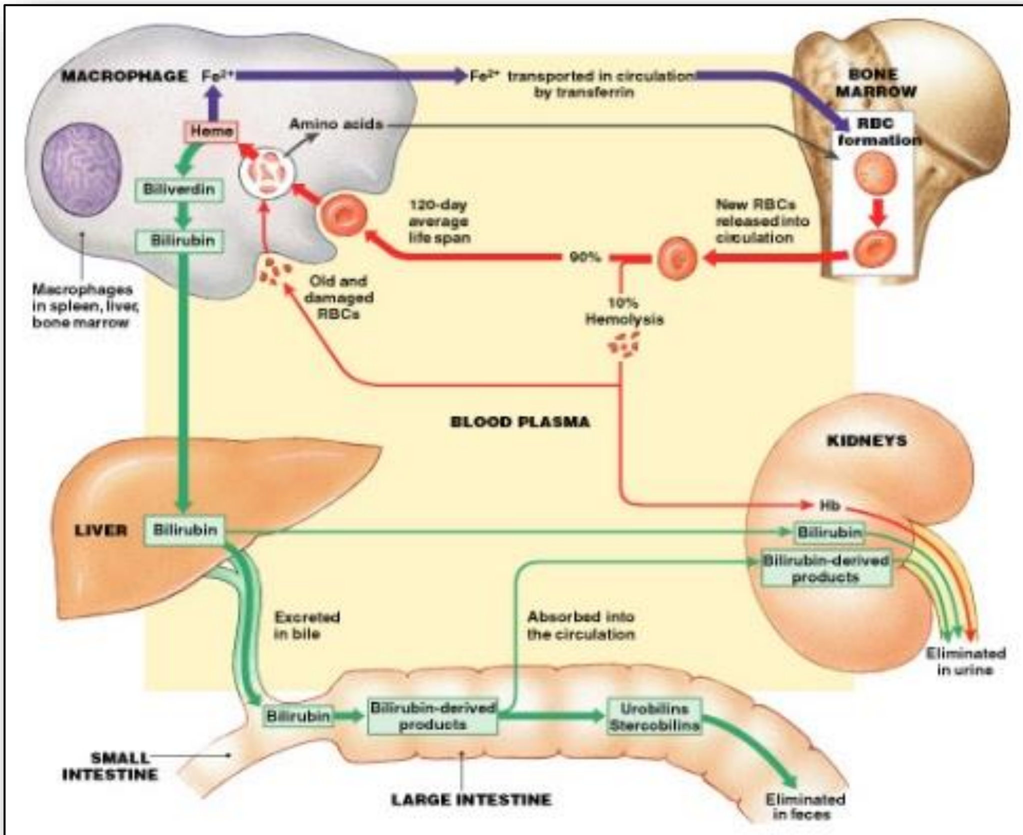


Figure 6: Life Cycle of an RBC

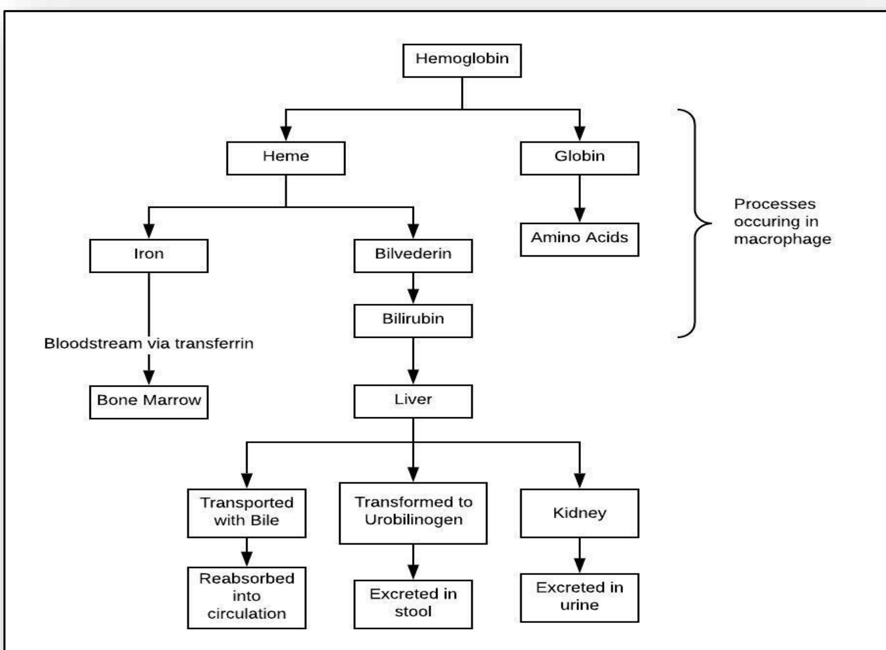


Figure 7: The "pallor sign" of anemia

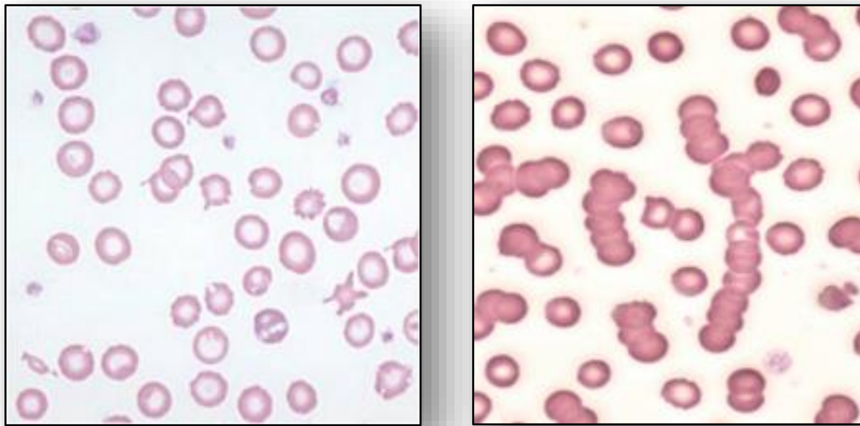


Figure 8: Pallor in the hand (right) compared to normal coloration

Figure 9: Microscopic comparison of normal RBCs (left) to Microcytic-hypochromic RBCs (right)

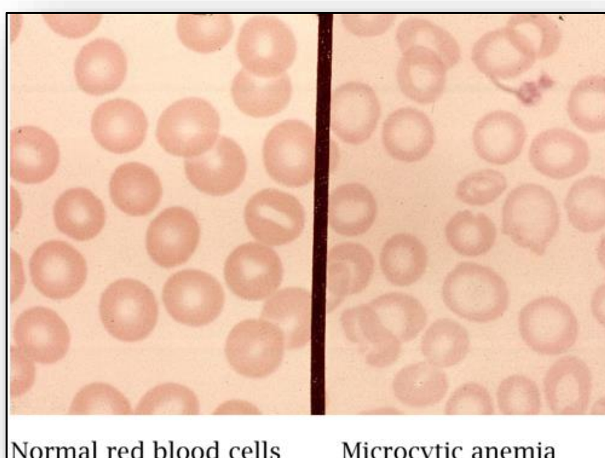


Figure 10: Comparison of healthy liver to hemochromatosis liver

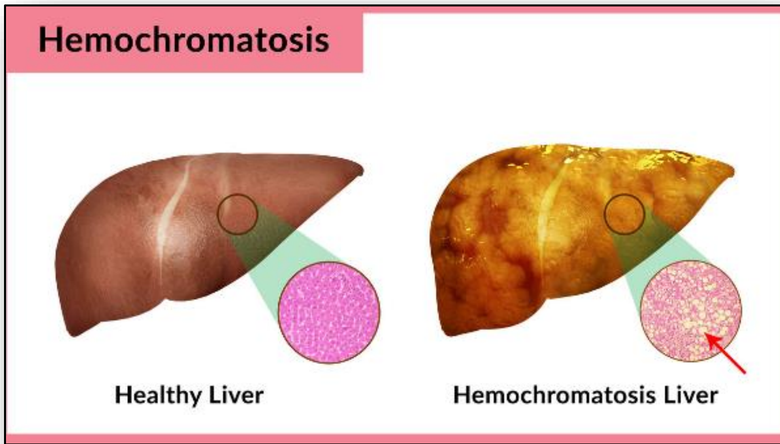


Figure 11: Organs affected in iron overload

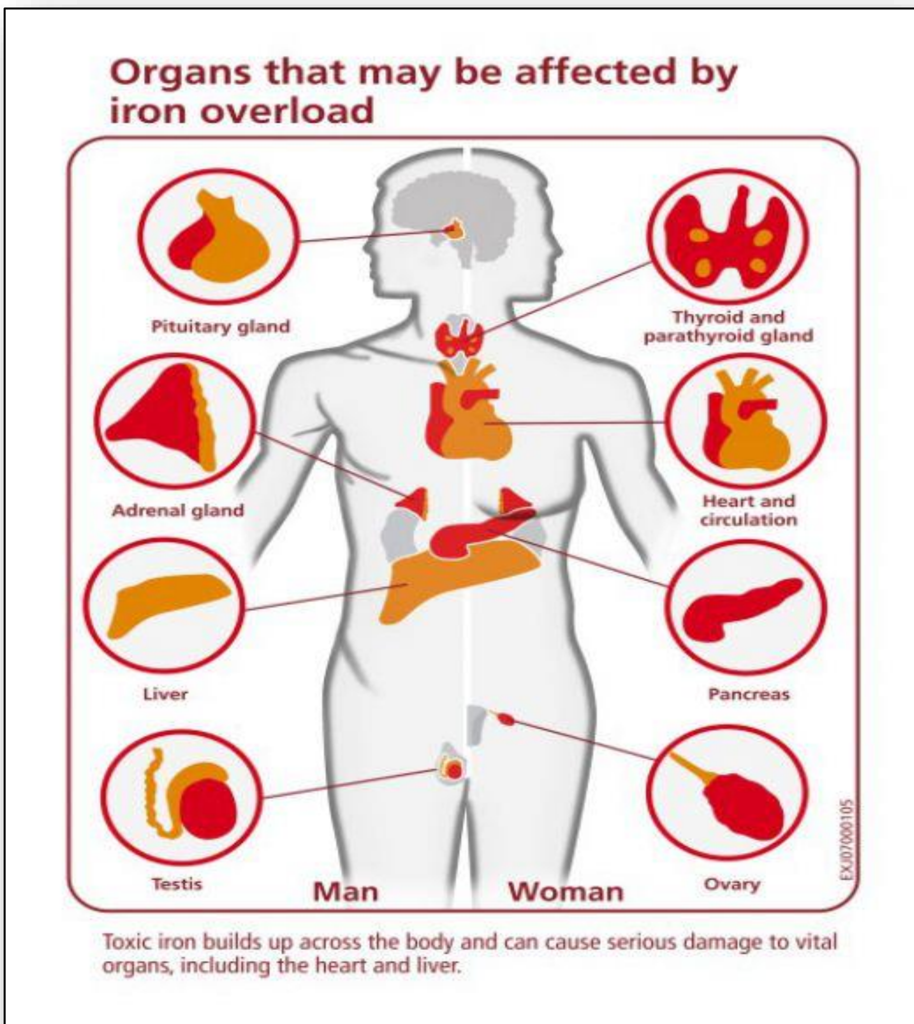


Figure 12: Metabolism of Vitamin B12 (cobalamin)

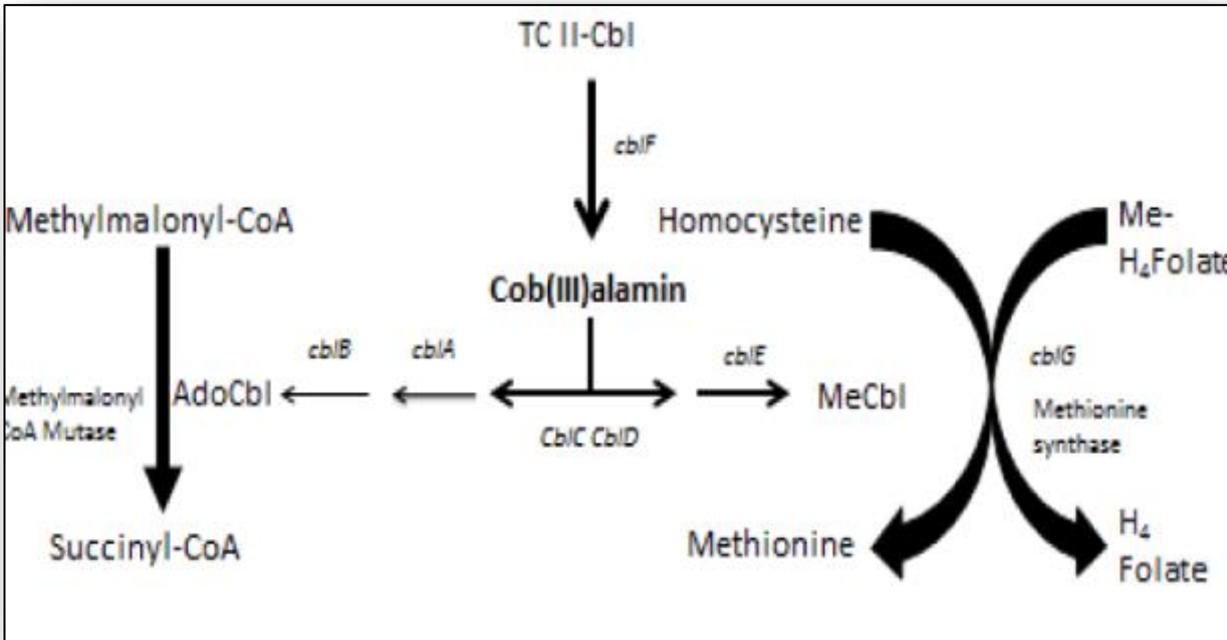


Figure 13: Classic symptoms of leukemia

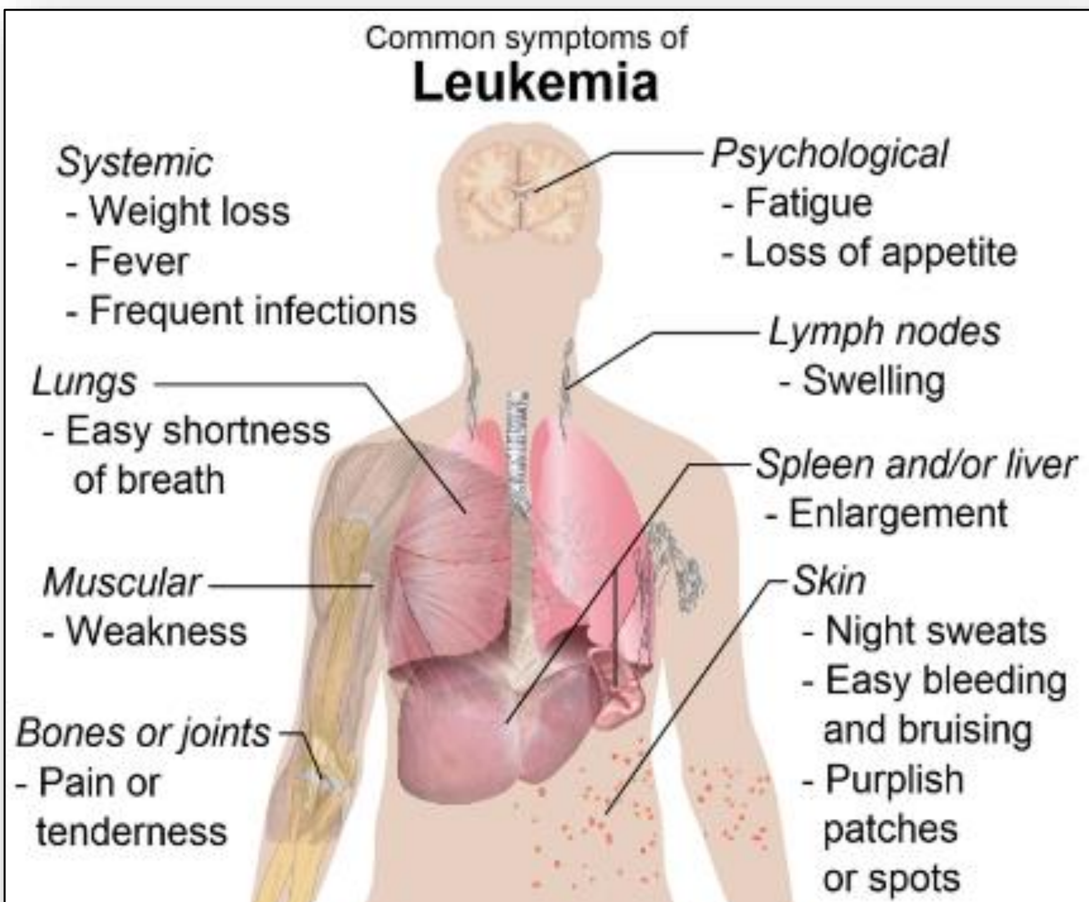


Figure 14: Lymphoid and myeloid stem cell maturation

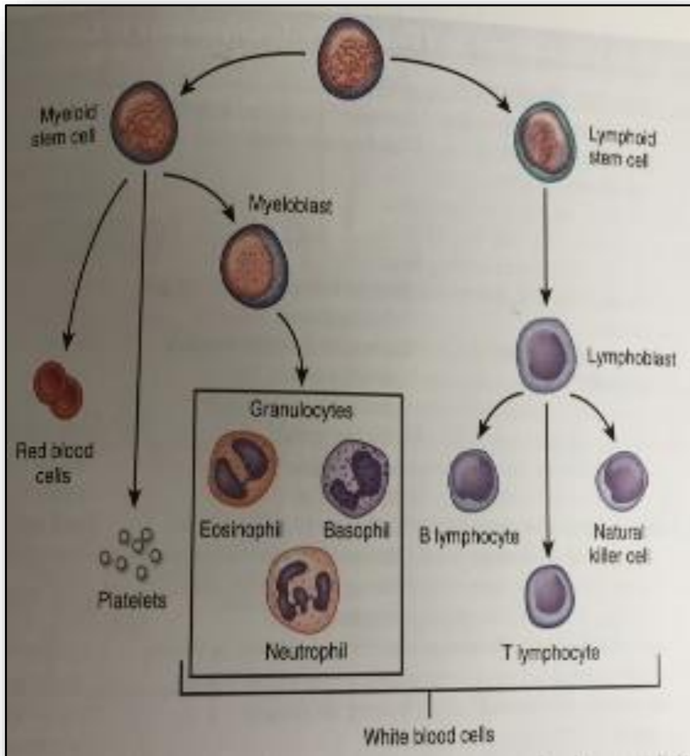


Figure 15: Lymphoblasts in blood smear

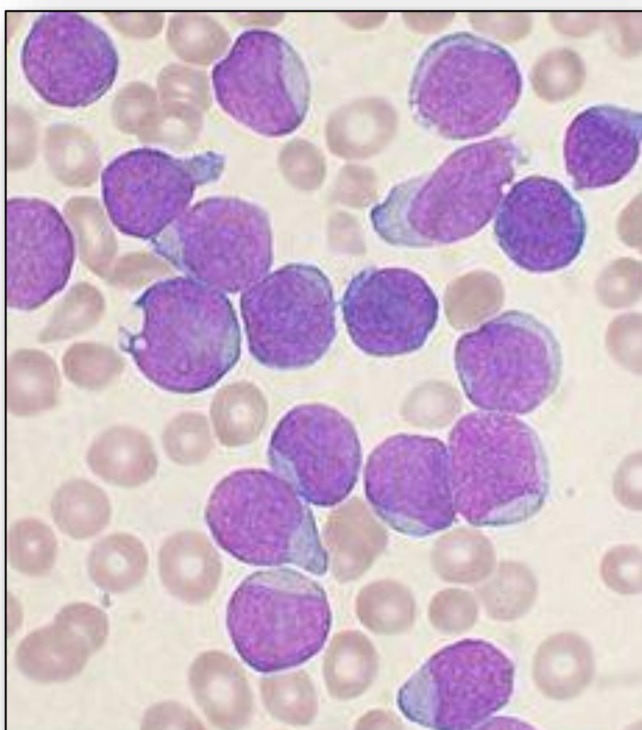


Figure 16: Development of ALL from lymphoid stem cell

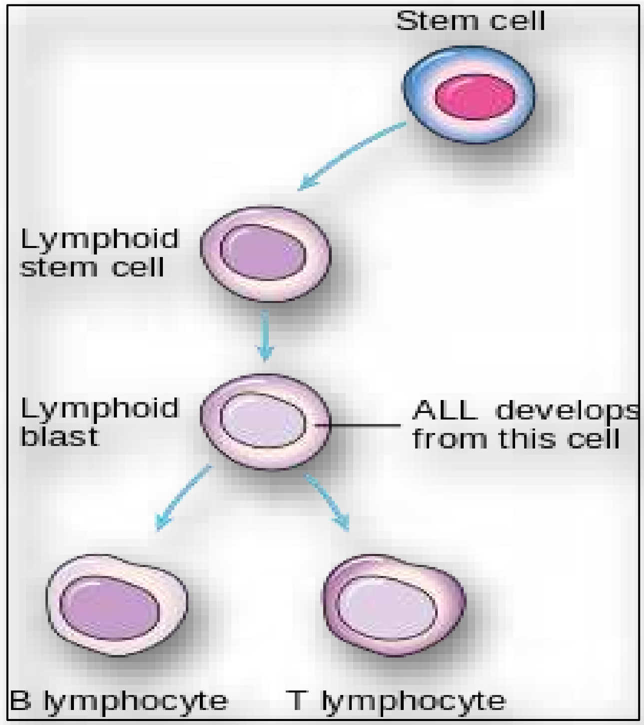


Figure 17: Development of AML from myeloid stem cell

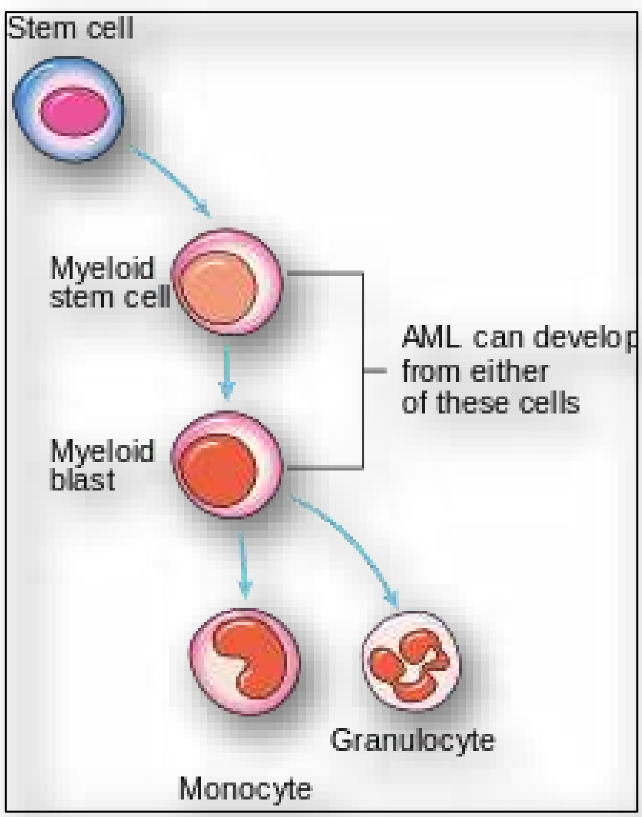


Figure 18: Lymphadenopathy (visible in groin and axilla)

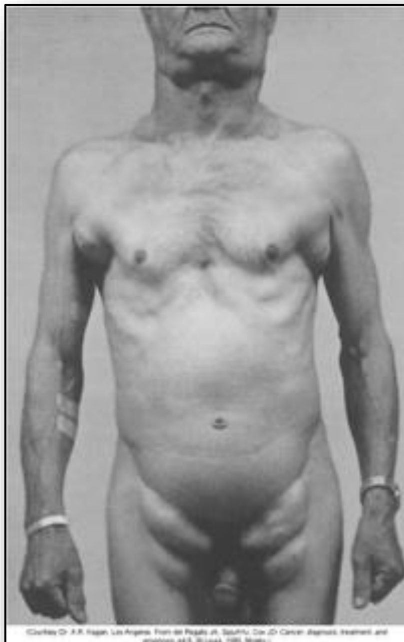


Figure 19: Body lymph node sites

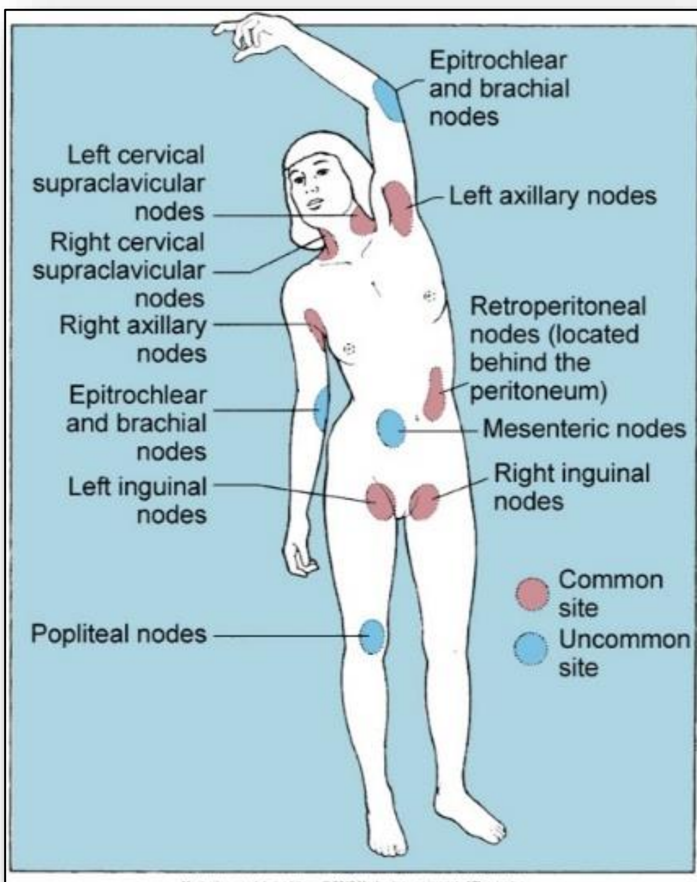
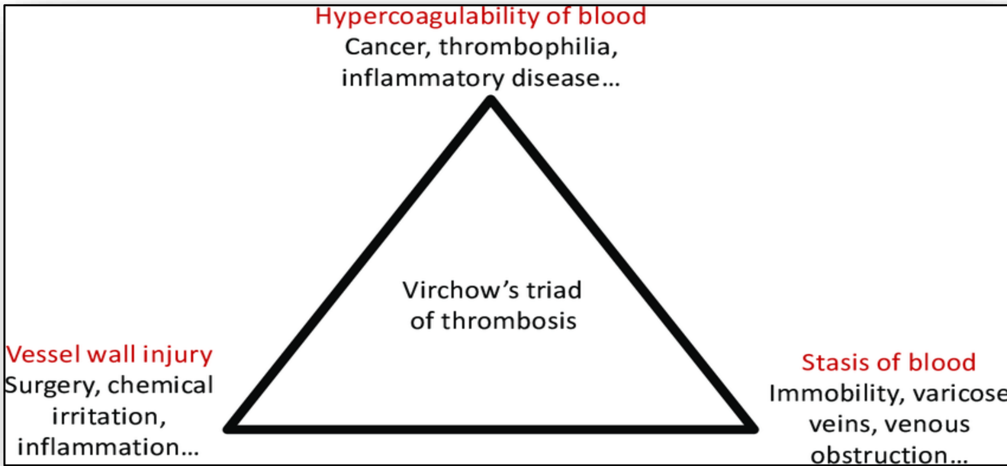


Figure 20: Virchow's Triad of thrombosis



6: Alterations of the Cardiovascular System

Figure 1: Heart anatomy

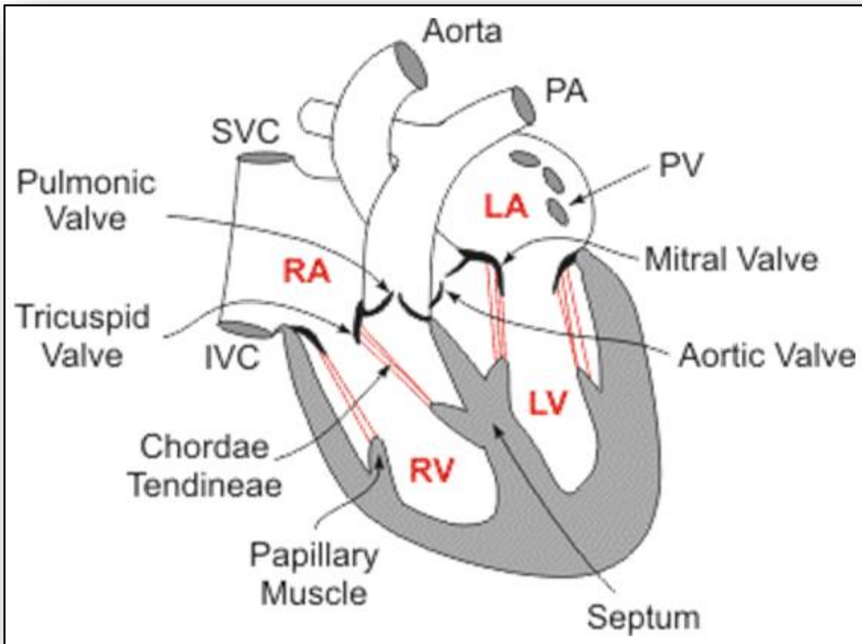


Figure 2: Layers covering heart

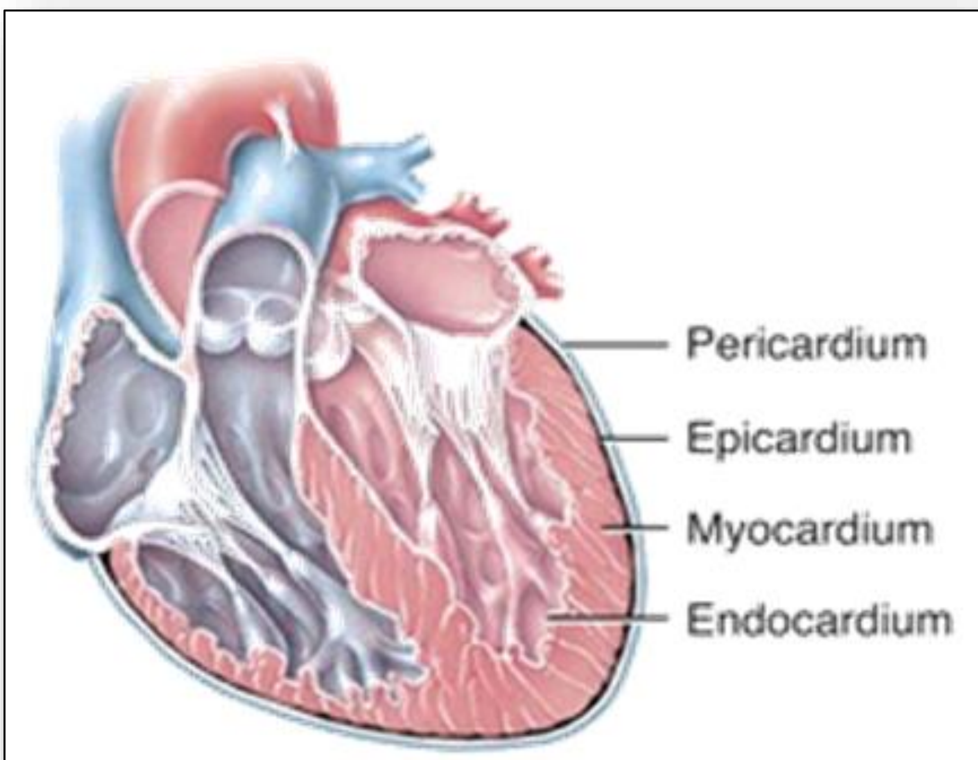


Figure 3: Heart layers showing pericardial cavity

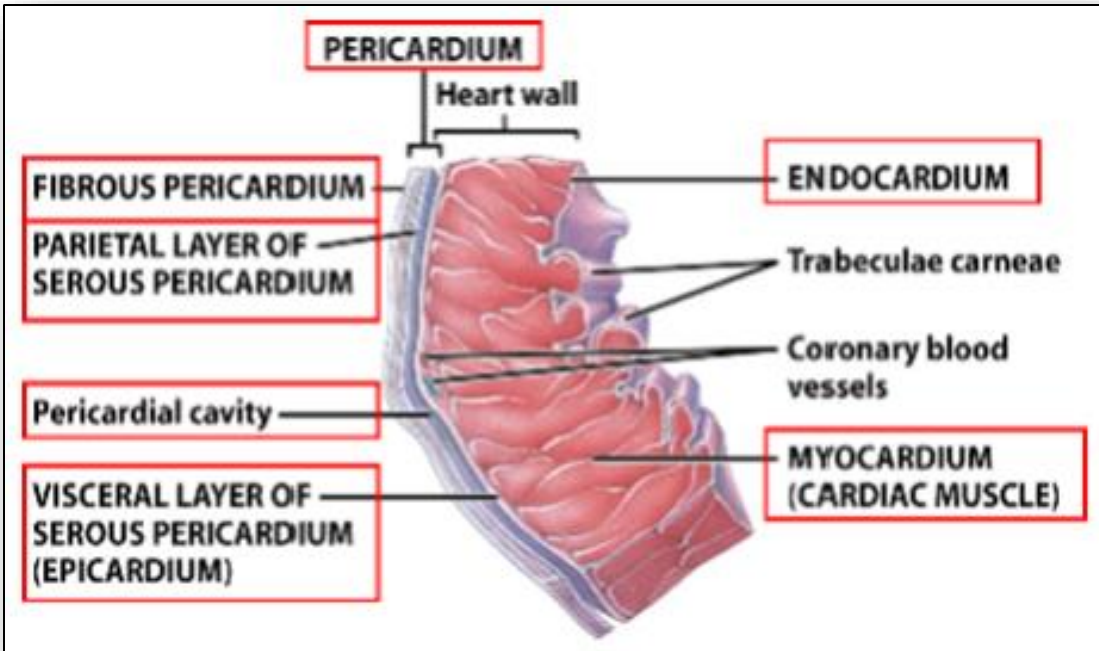


Figure 4: Circulation of blood via the heart

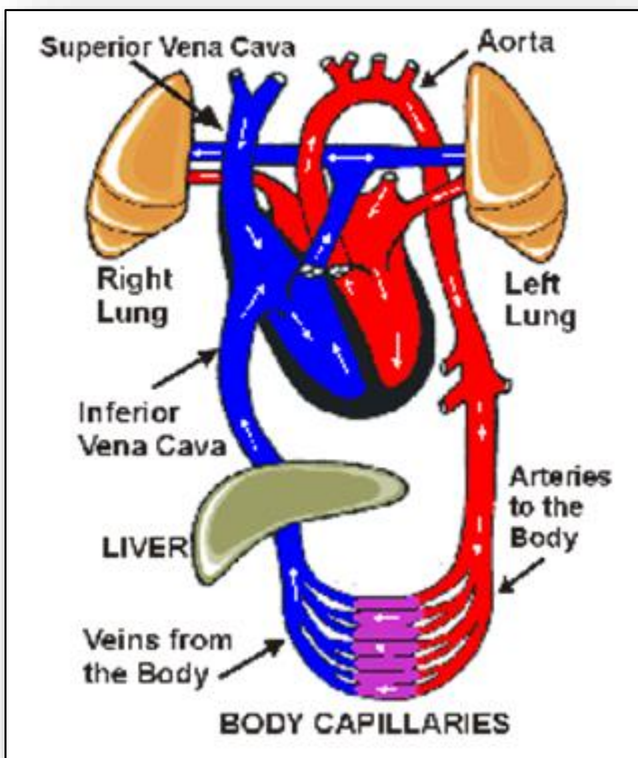


Figure 5: Conduction system of the heart

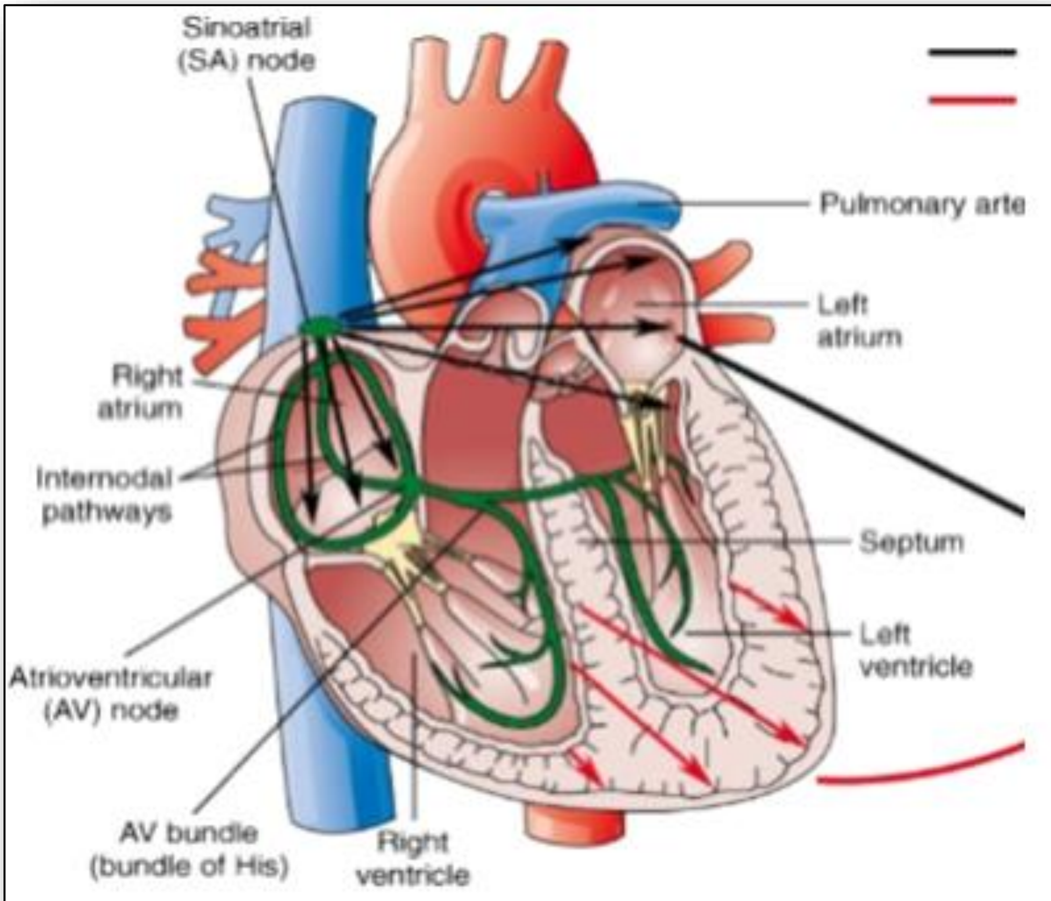


Figure 6: Formation of atheroma within a blood vessel

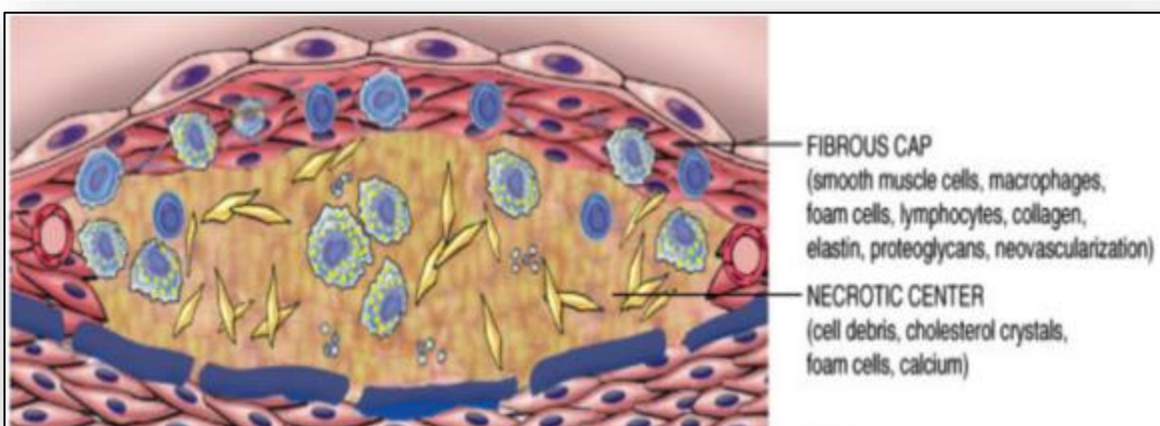


Figure 7: Pathophysiology of atherosclerosis

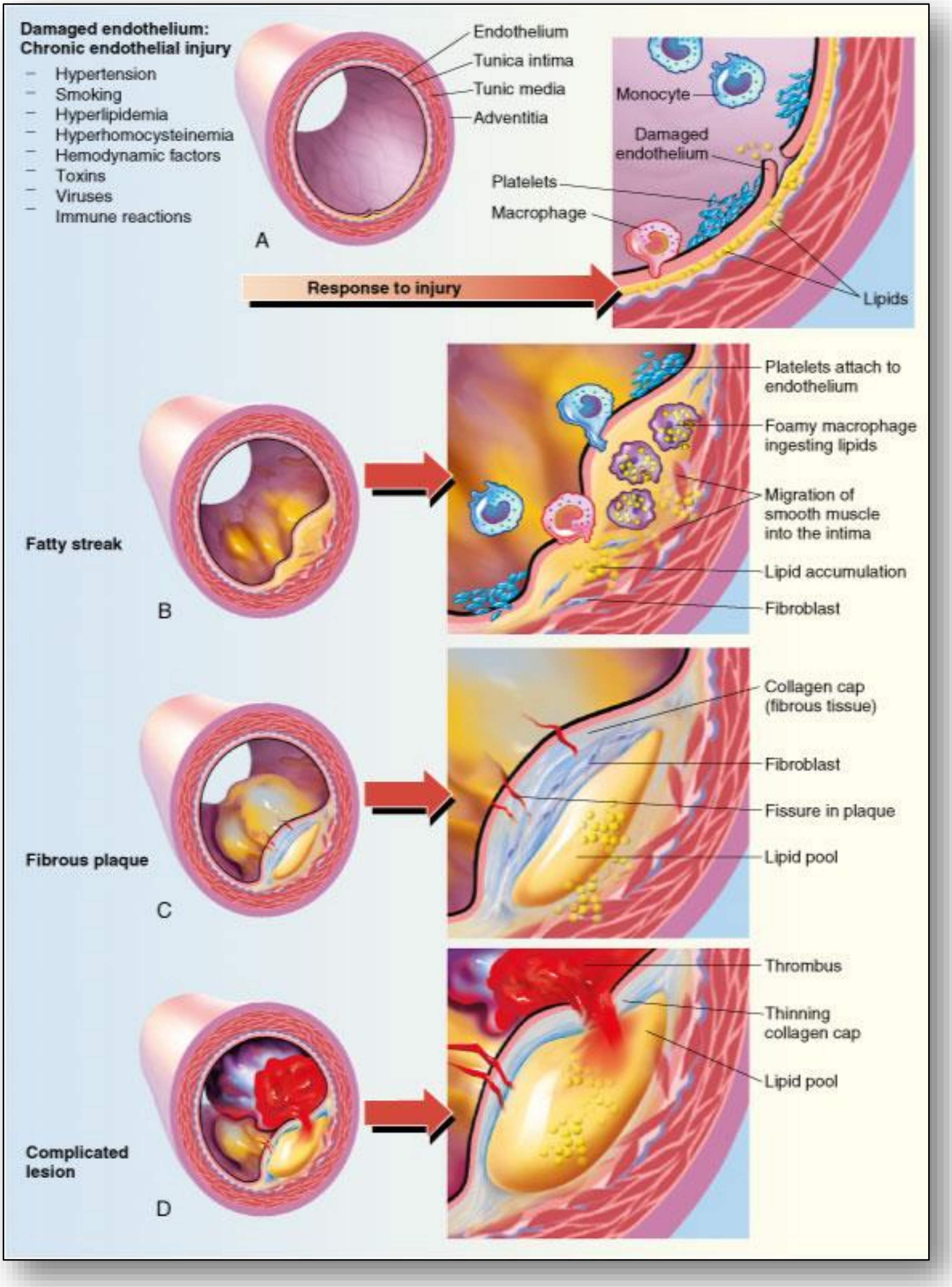


Figure 8: Progression of atherosclerosis within an artery; From normal, healthy artery (left) to occlusion (right)

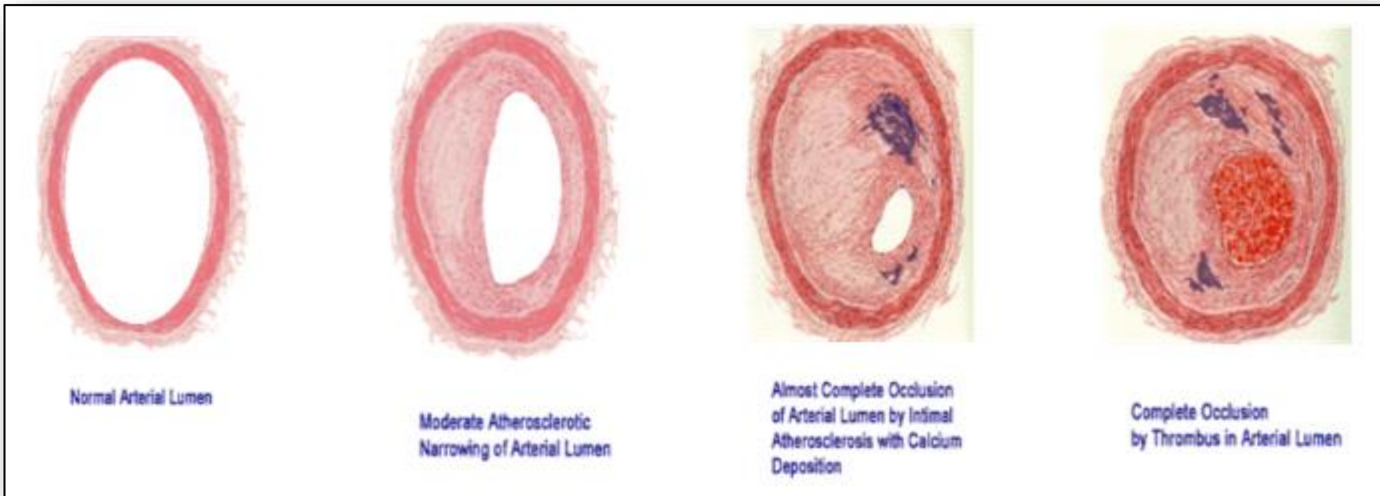


Figure 9: Progression and associated outcomes of coronary artery disease

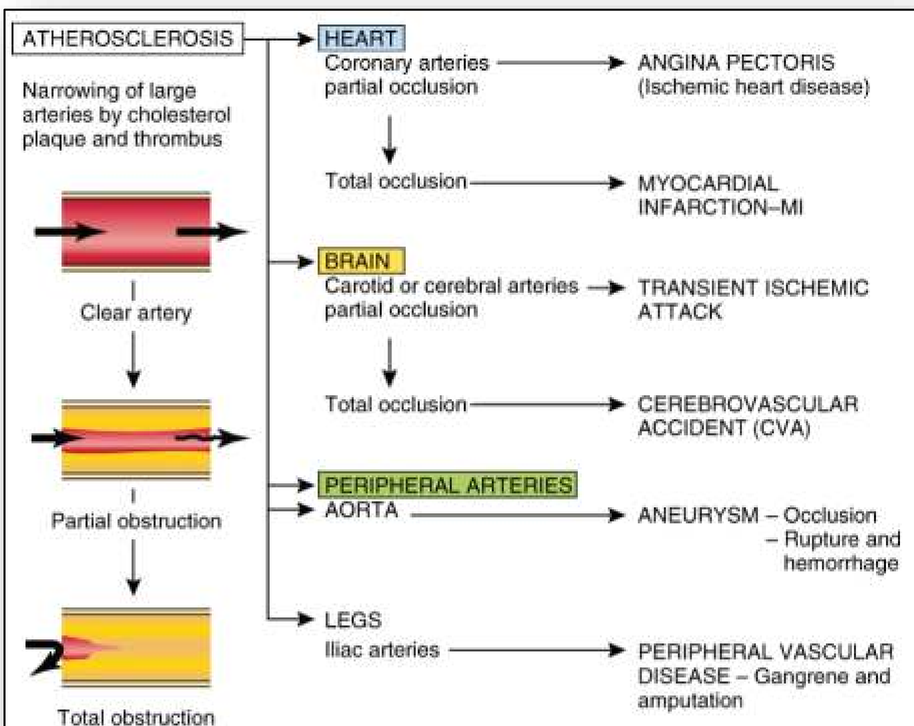


Figure 10: Typical distribution of referred pain in angina

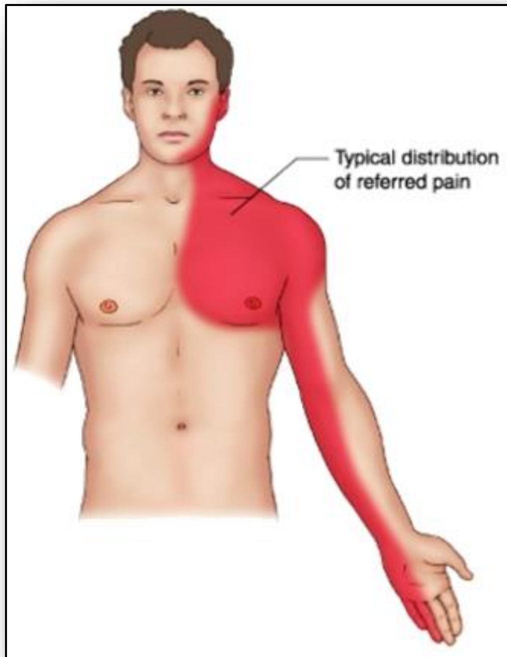


Figure 11: Changes in cardiac enzymes over time following cardiac injury

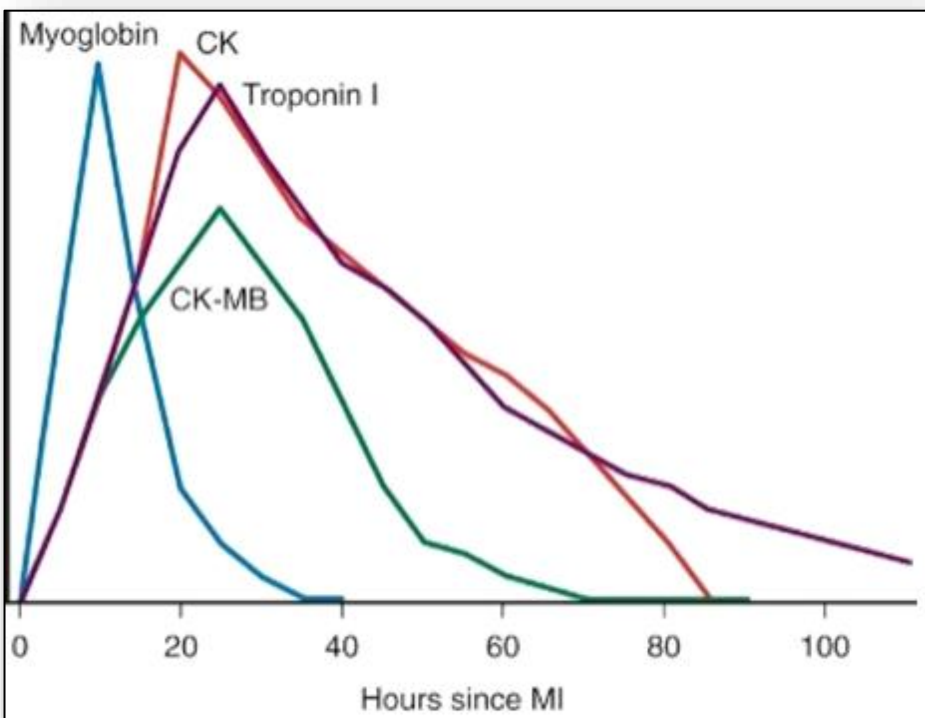


Figure 12: Fibrous tissue scars (arrows) following cardiac injury/infarction

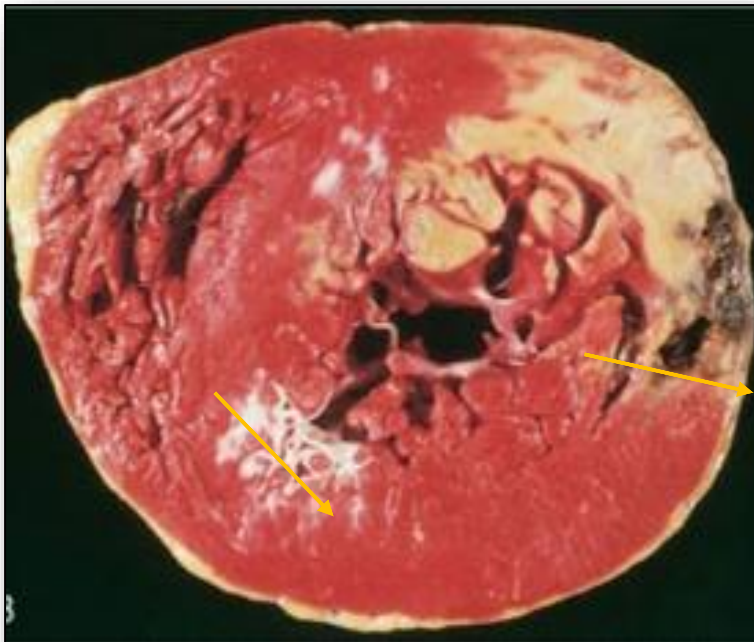


Figure 13: ECG changes related to cardiac damage

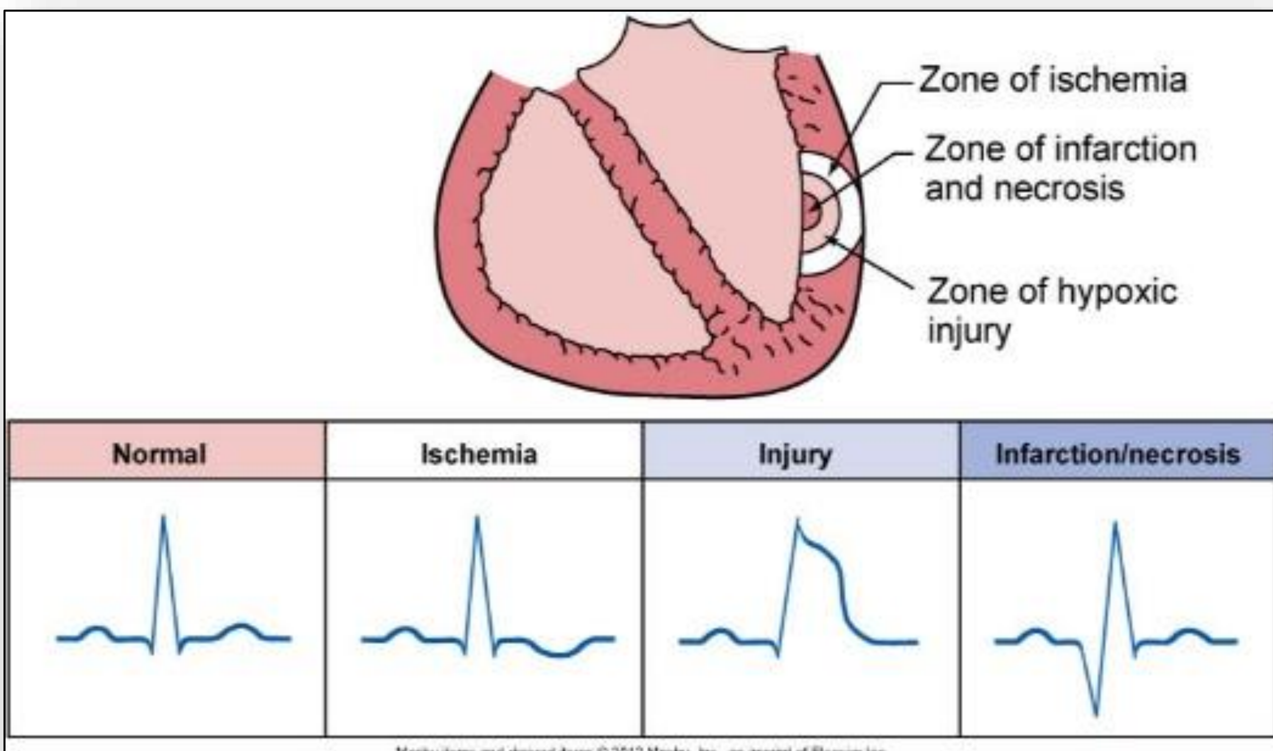


Figure 14: Aschoff body

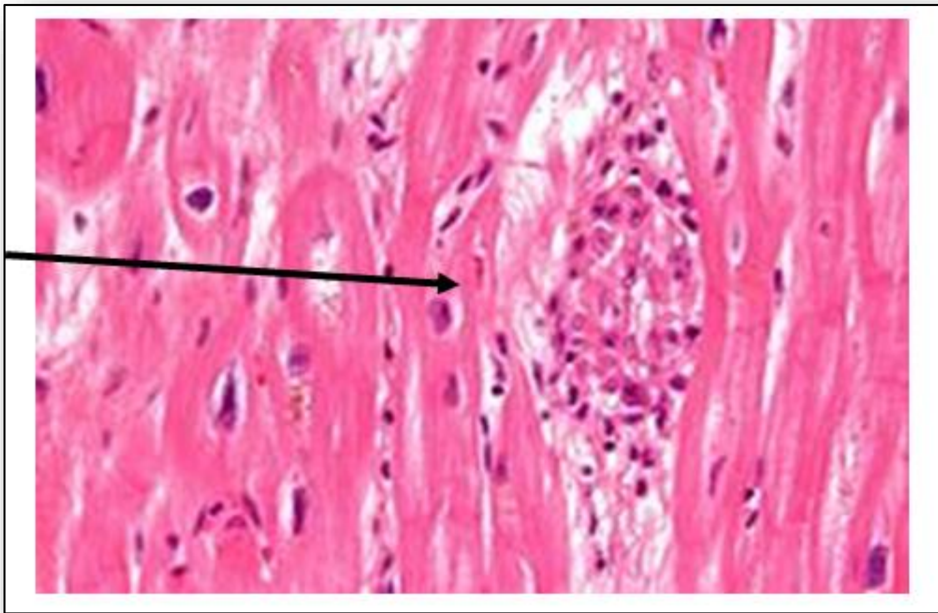


Figure 15: Valvular changes occurring in rheumatic fever

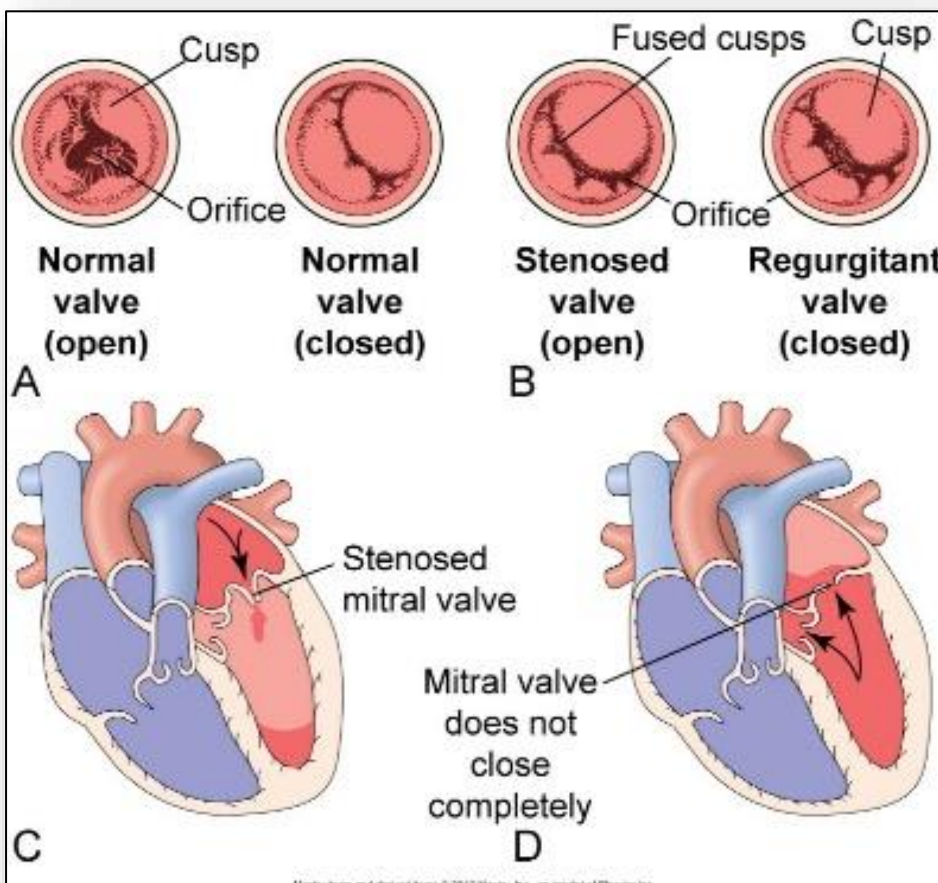


Figure 16: Verrucae along the edge of a valve leaflet; Also thickening of the chordae tendinae visible

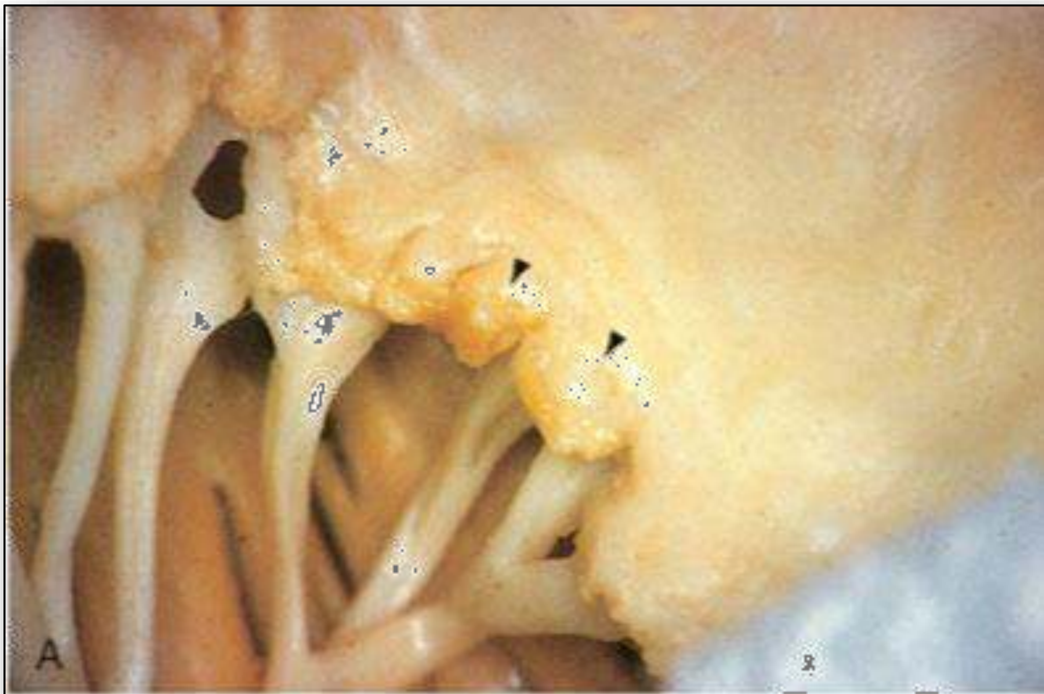


Figure 17: Thickening of valve leaflets and subsequent valve stenosis

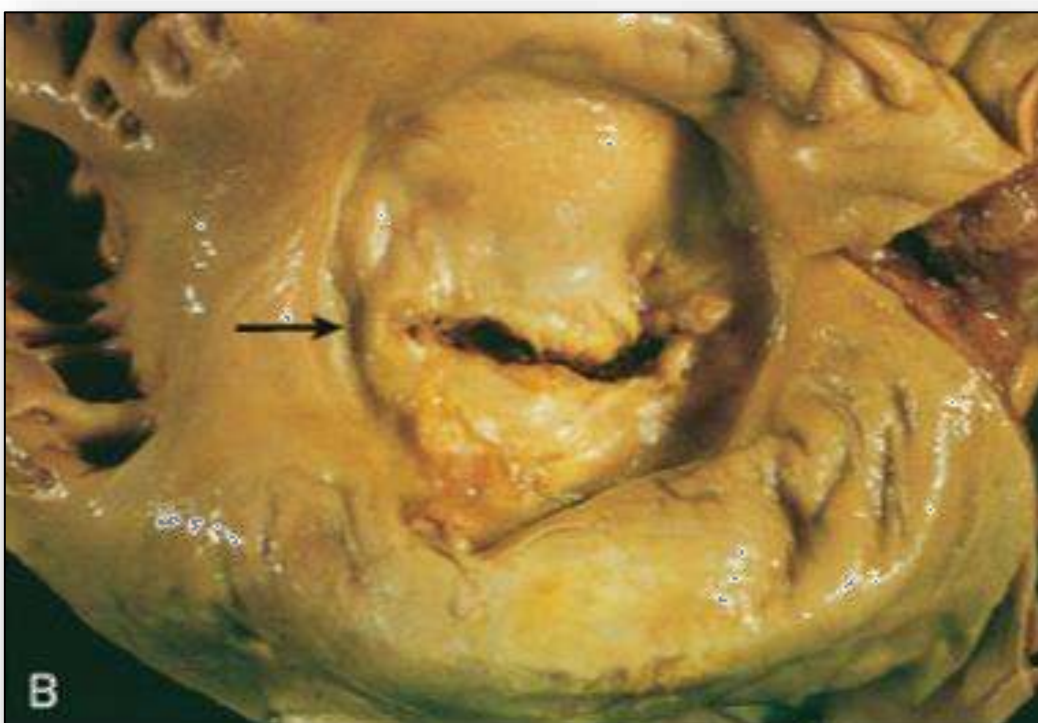


Figure 18: Erythema marginatum

Figure 19: Subcutaneous nodules

Figure 20: Locations of basal nuclei lesions

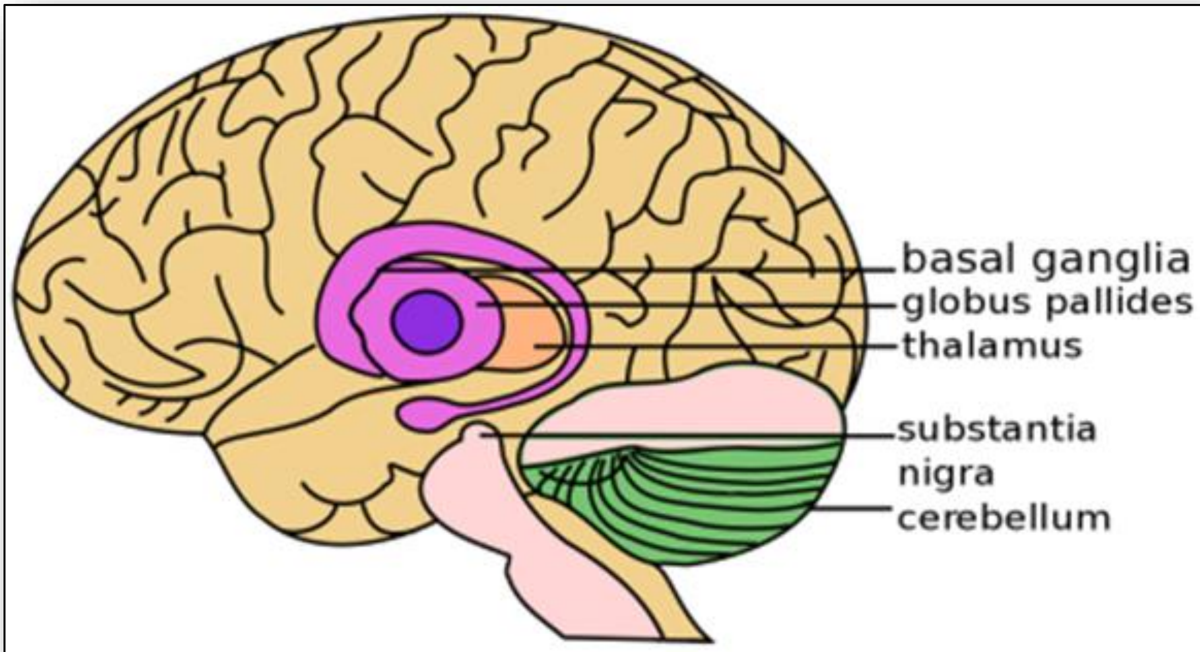


Figure 21: Vegetative growths on endocardium

Figure 22: Osler's nodes

Figure 23: Splinter hemorrhages

Figure 24: Pathological development of pericarditis (acute to constrictive)

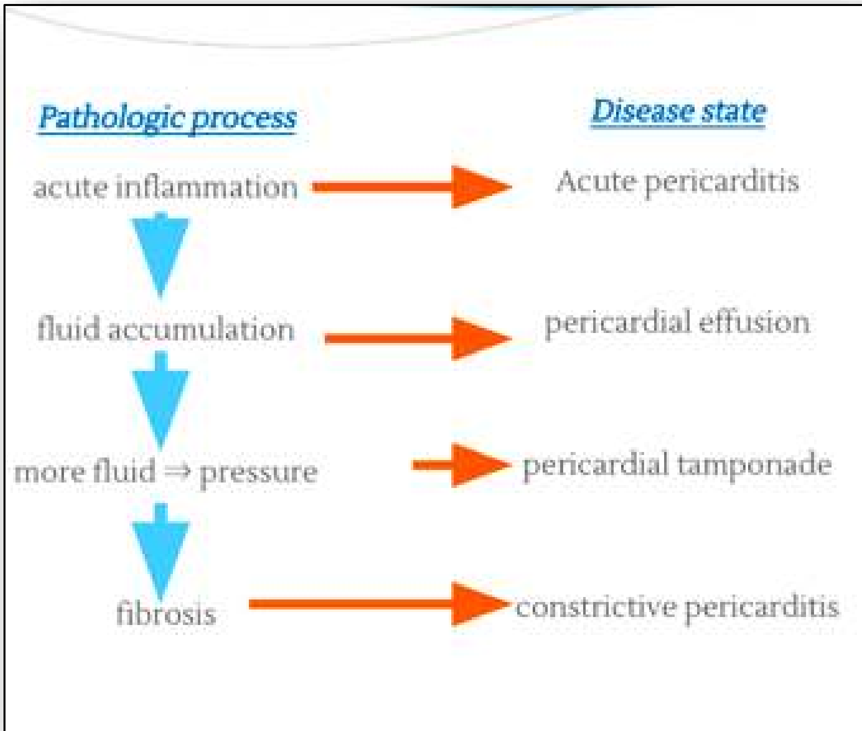


Figure 25: Pericardial effusion

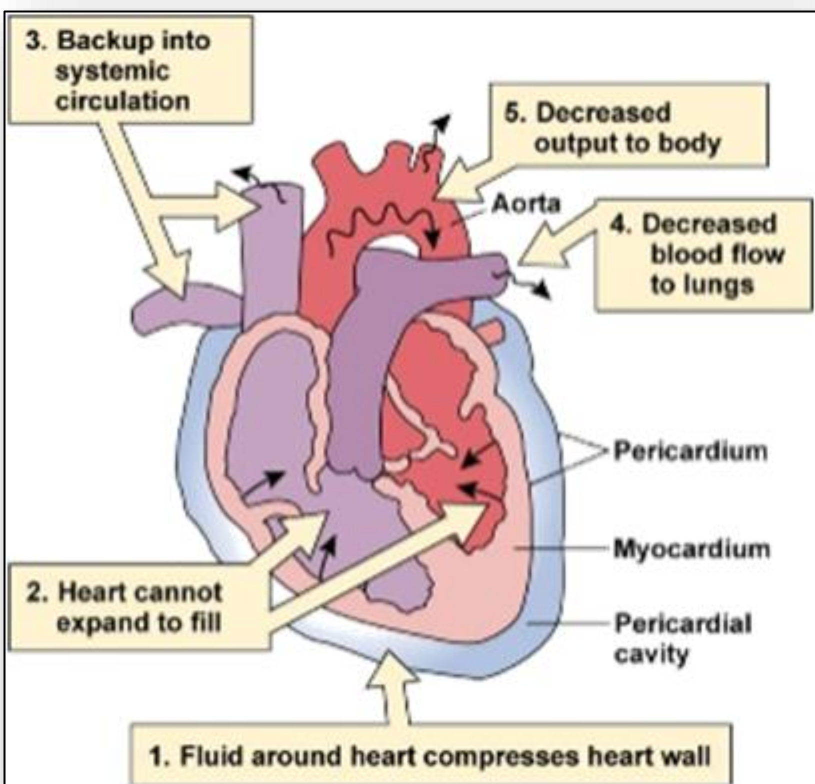


Figure 26: Comparison of pericardial effusion to normal heart on x-ray

Figure 27: Cardiocentesis to remove pericardial fluid

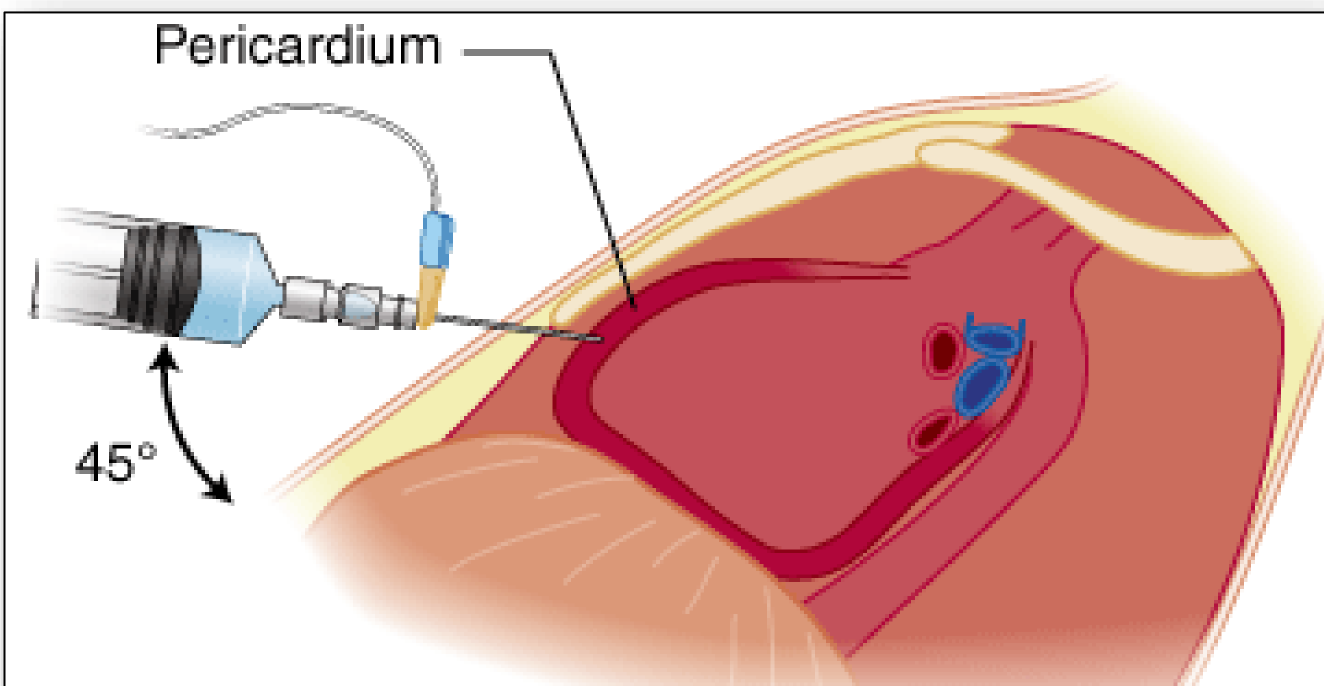


Figure 28: Summary Table Manifestations of Valvular Stenosis and regurgitation (taken from Huether & McCance, 2012)

TABLE 23-6 CLINICAL MANIFESTATIONS OF VALVULAR STENOSIS AND REGURGITATION

MANIFESTATION	AORTIC STENOSIS	MITRAL STENOSIS	AORTIC REGURGITATION	MITRAL REGURGITATION	TRICUSPID REGURGITATION
Most common cause	Congenital bicuspid valve, degenerative (calcific) changes with aging, rheumatic heart disease	Rheumatic heart disease	Infective endocarditis; aortic root disease (connective tissue diseases, Marfan syndrome); dilation of aortic root from hypertension and aging	Myxomatous degeneration (mitral valve prolapse)	Congenital
Cardiovascular outcome (untreated)	Left ventricular hypertrophy followed by left heart failure; decreased coronary blood flow with myocardial ischemia	Left atrial hypertrophy and dilation with fibrillation, followed by right ventricular failure	Left ventricular hypertrophy and dilation, followed by left heart failure	Left atrial hypertrophy and dilation, followed by left heart failure	Right heart failure
Pulmonary effects	Pulmonary edema: dyspnea on exertion	Pulmonary edema: dyspnea on exertion, orthopnea, paroxysmal nocturnal dyspnea, predisposition to respiratory tract infections, hemoptysis, pulmonary hypertension	Pulmonary edema with dyspnea on exertion	Pulmonary edema with dyspnea on exertion	Dyspnea
Central nervous system effects	Syncope, especially on exertion	Neural deficits only associated with emboli (e.g., hemiparesis)	Syncope	None	None
Pain	Angina pectoris	Atypical chest pain	Angina pectoris	Atypical chest pain	Palpitations
Heart sounds	Systolic murmur heard best at right parasternal second intercostal space and radiating to neck	Low rumbling diastolic murmur heard best at apex and radiating to axilla, accentuated first heart sound, opening snap	Diastolic murmur heard best at right parasternal second intercostal space and radiating to neck	Murmur throughout systole heard best at apex and radiating to axilla	Murmur throughout systole heard best at left lower sternal border

With data from Braunwald E, editor: *Heart disease: a textbook of cardiovascular medicine*, ed 5, Philadelphia, 1997, Saunders; Carabello BA, Paulus WJ: Valvular heart disease. In Crawford MH, DiMarco JP, editors: *Cardiology*, London, 2001, Mosby.

Figure 29: Comparison cardiomyopathies (dilated, hypertrophic, restrictive) to normal heart

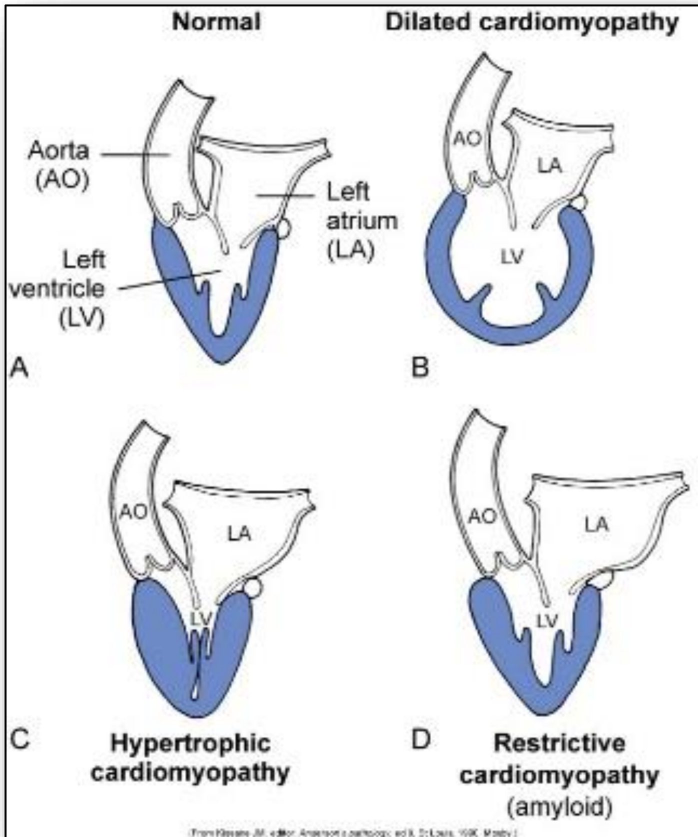


Figure 30: The conduction system of the heart

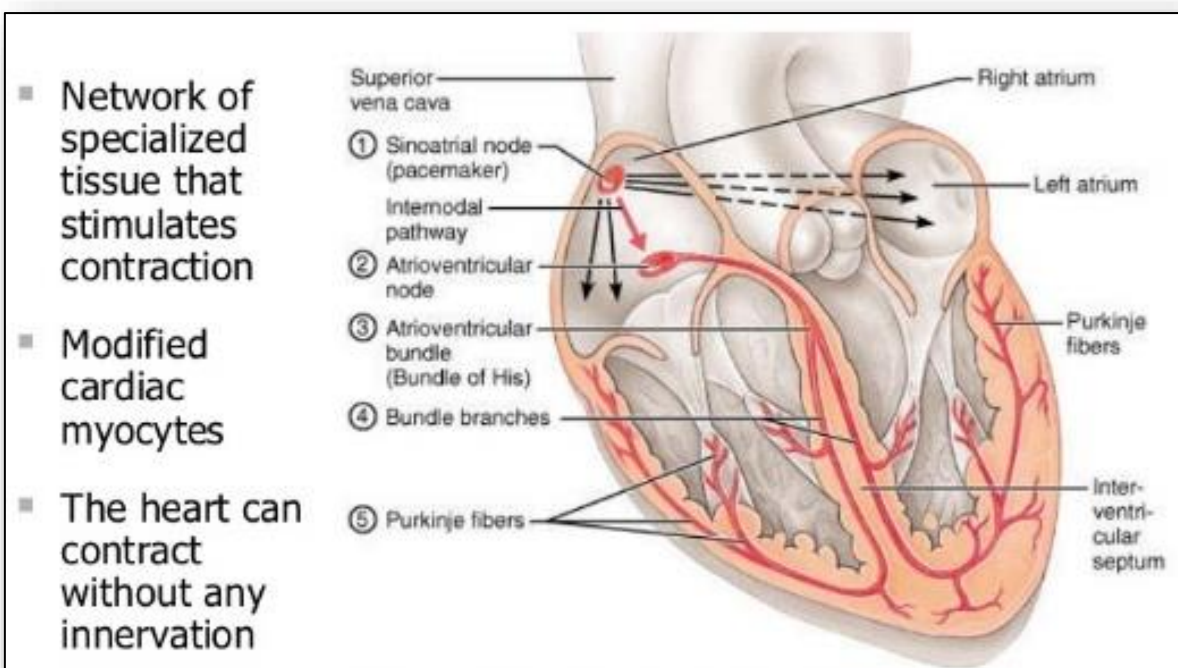


Figure 31: ECG comparison atrial fibrillation to atrial flutter

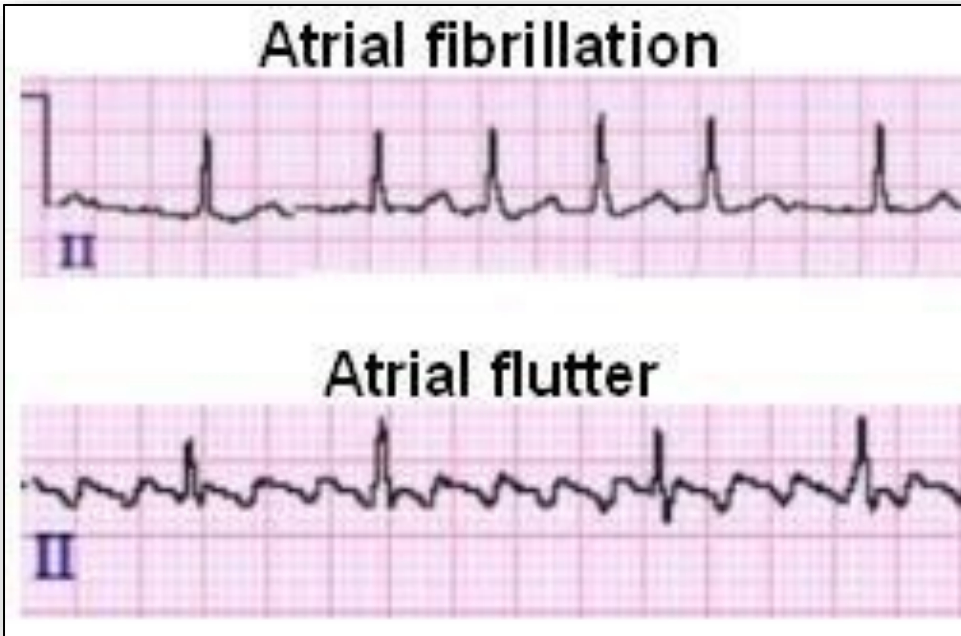


Figure 32: Pathophysiology of heart failure (simplified)

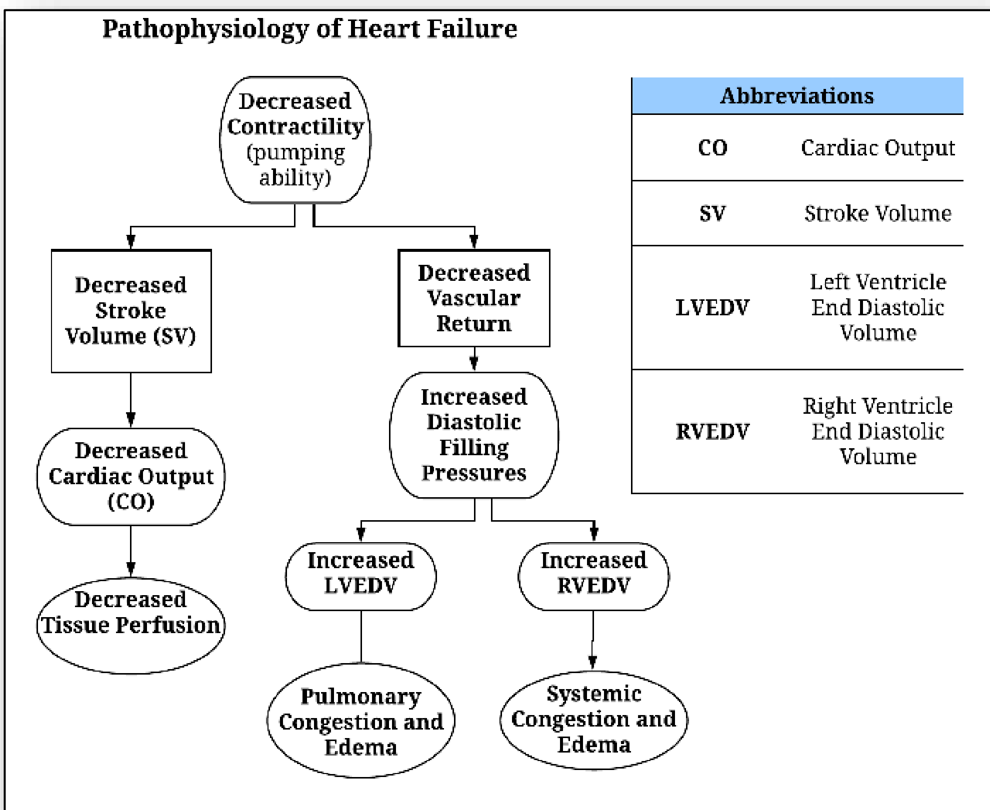


Figure 33: Compensation and impact of afterload on progression of heart failure (from Huether & McCance, 2012)

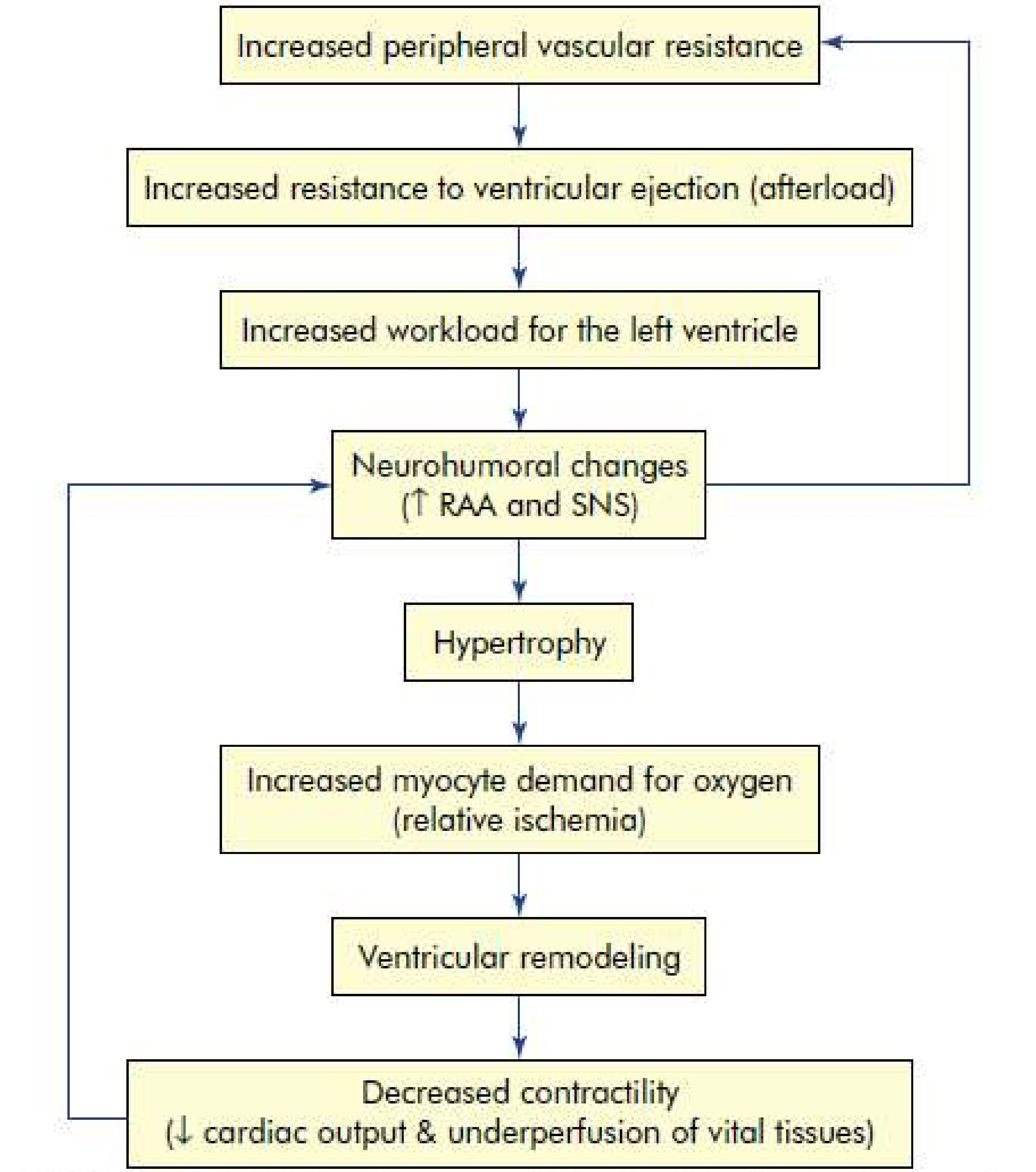


FIGURE 23-37 Role of Increased Afterload in the Pathogenesis of Heart Failure.

Figure 34: Ventricular remodelling and the development of heart failure (Huether & McCance, 2012)

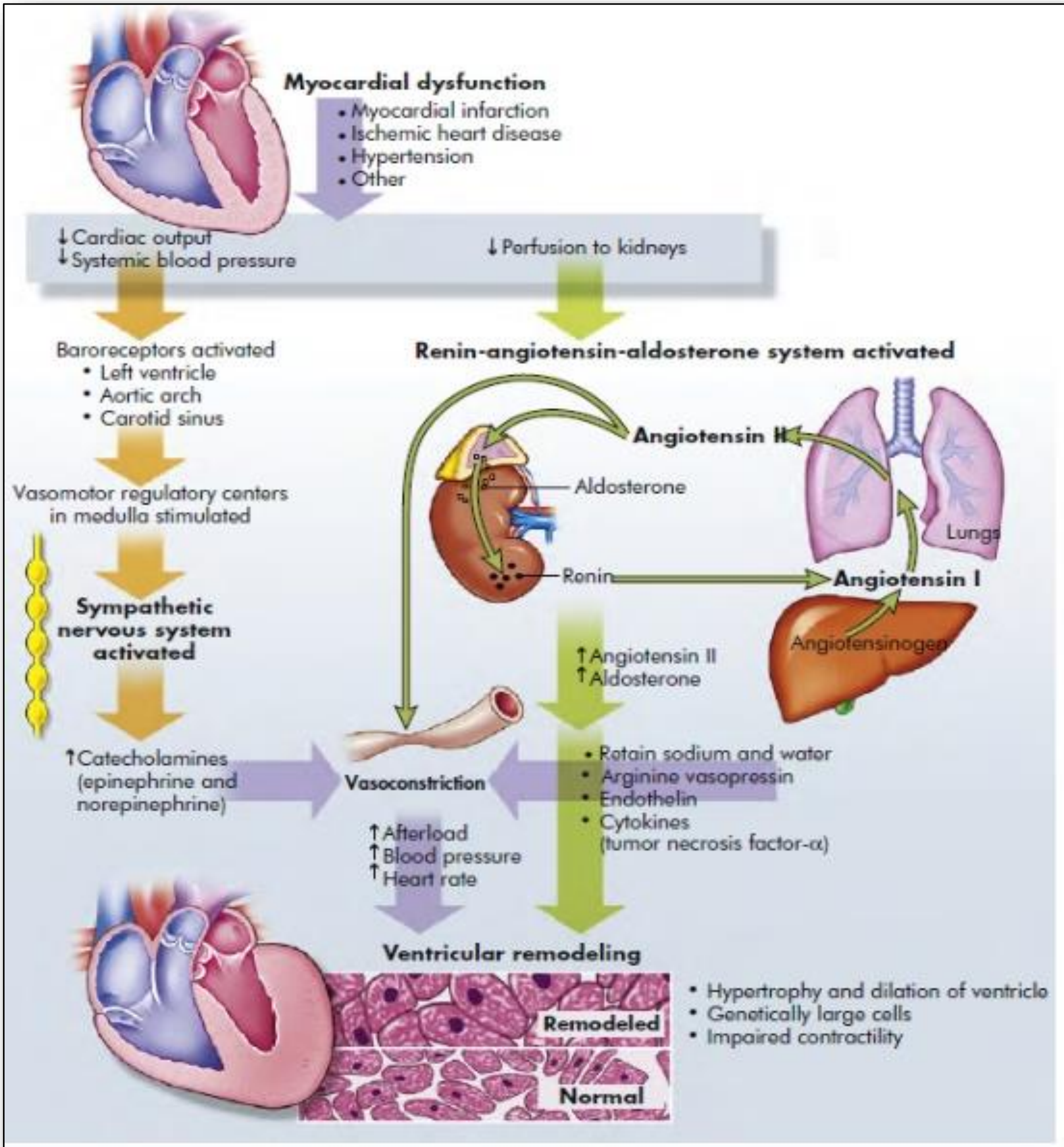


Figure 35: Manifestations of heart failure

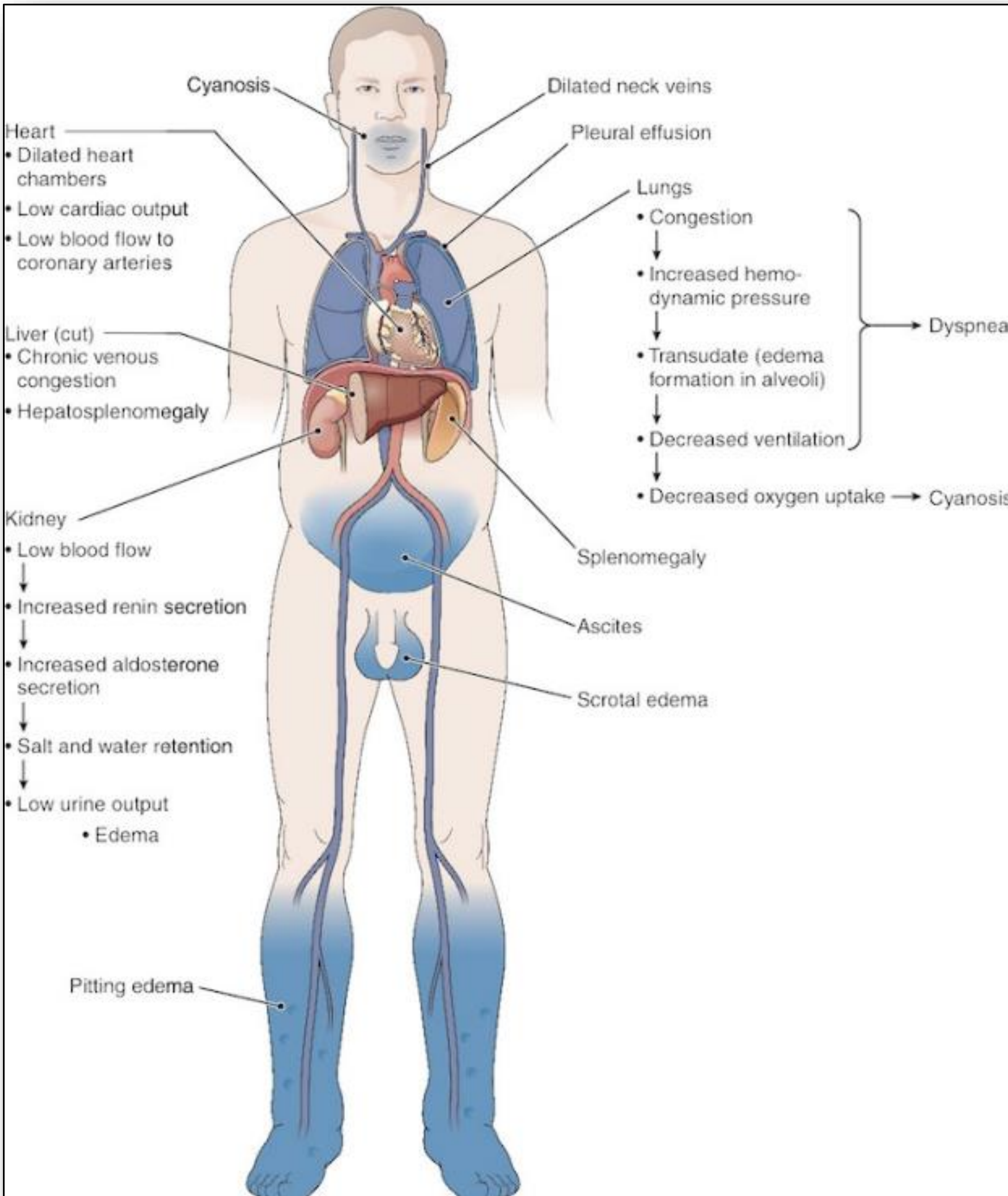


Figure 36: Symptom pathophysiology for L-sided heart failure

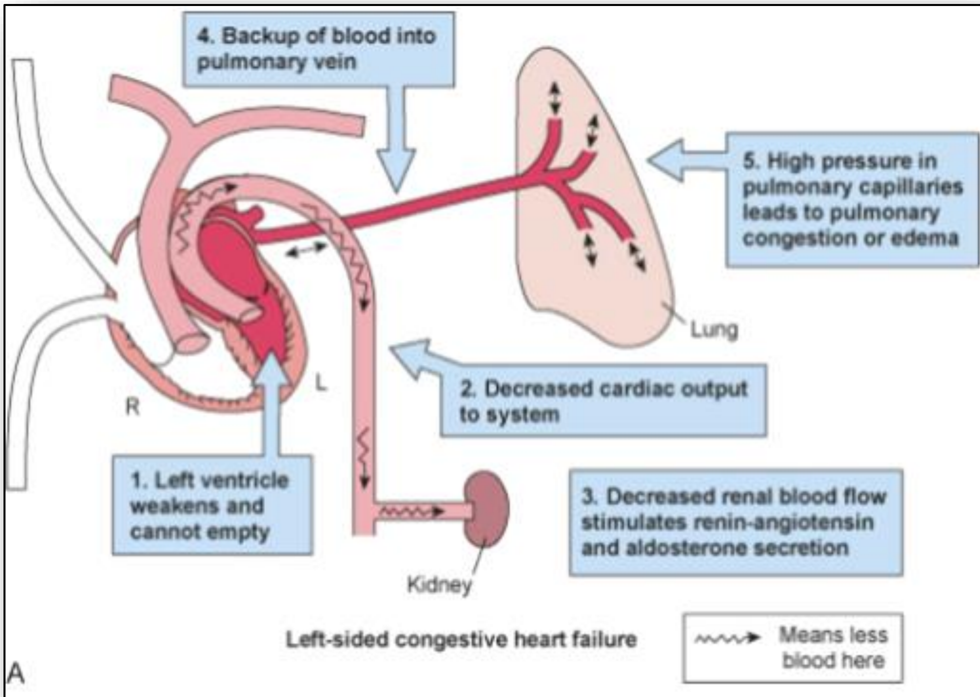


Figure 37: Pathophysiology of manifestations for R-side heart failure

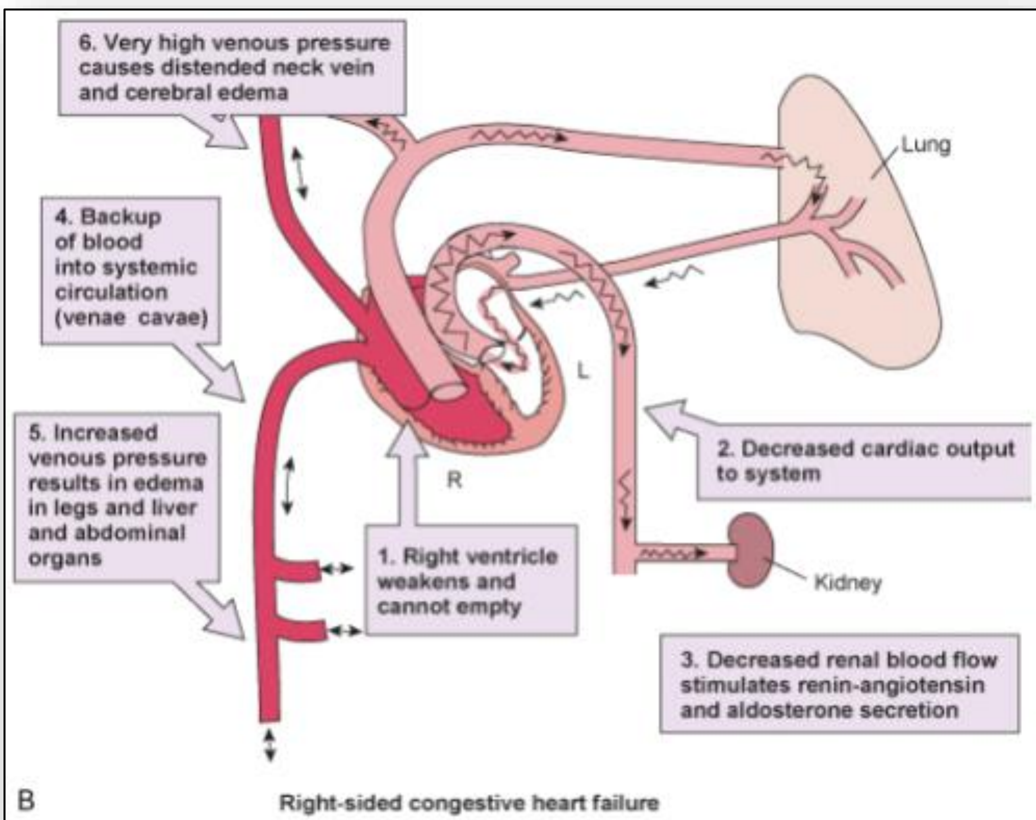


Figure 38: Categories of hypertension

BP	Mild	Moderate	Severe
SBP	140-159	160-179	≥ 180
DBP	90-99	100-109	≥ 110

Figure 39: Narrowing of vascular lumen contributing to hypertension

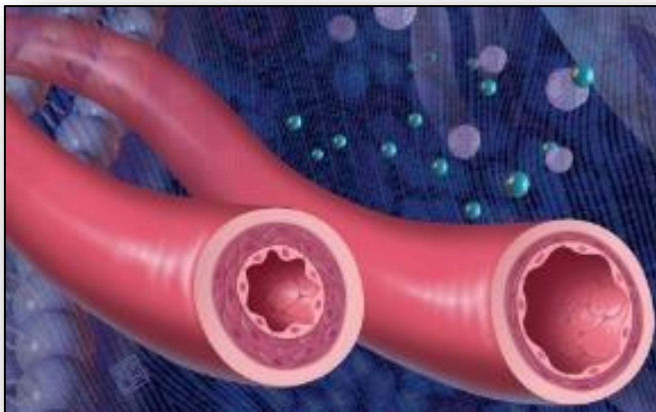


Figure 40: Pathophysiology of hypertension (from Huether & McCance, 2012)

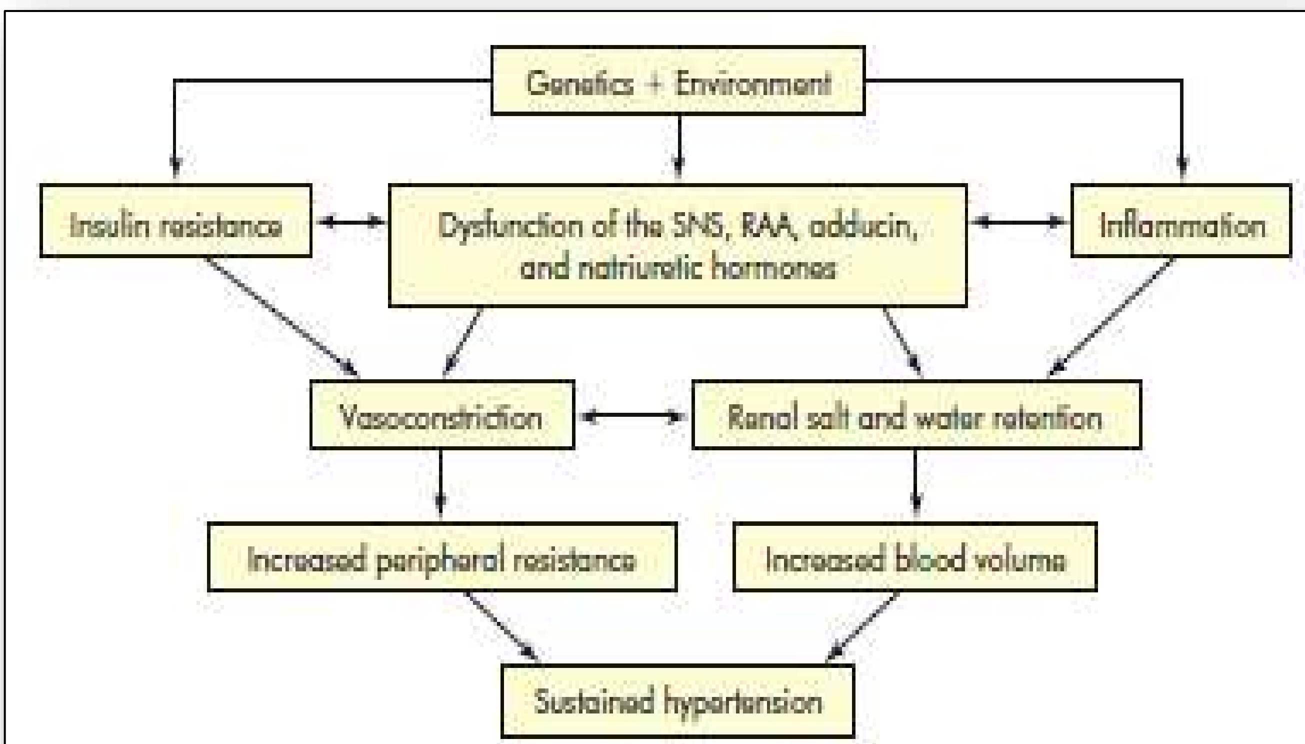


Figure 41: Relationship of arteriolar vasoconstriction on elevation in BP

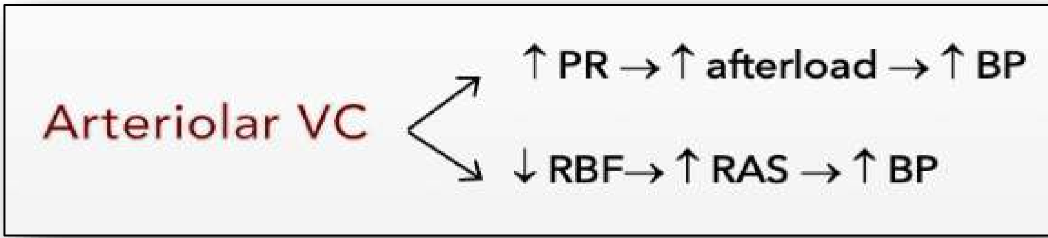


Figure 42: Types of aneurysms (Huether & McCance, 2012)

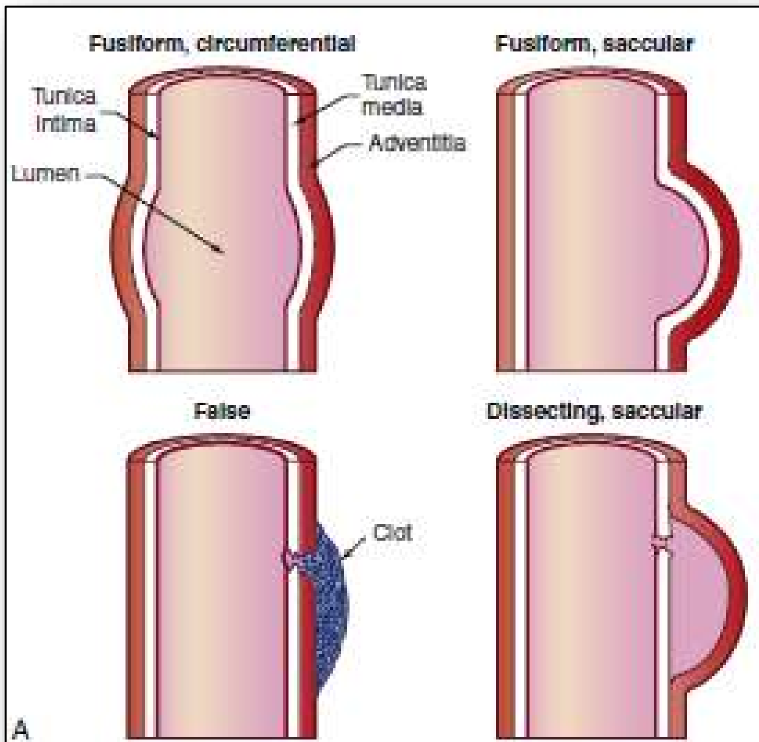


Figure 43: Hypertensive retinopathy; Papilledema; Loss of vision

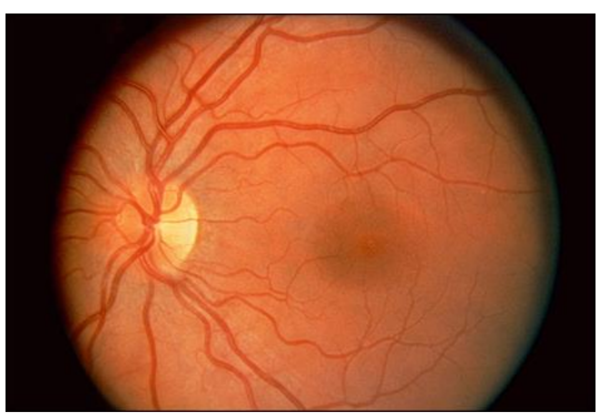


Figure 44: Venous Thromboembolism (VTE)

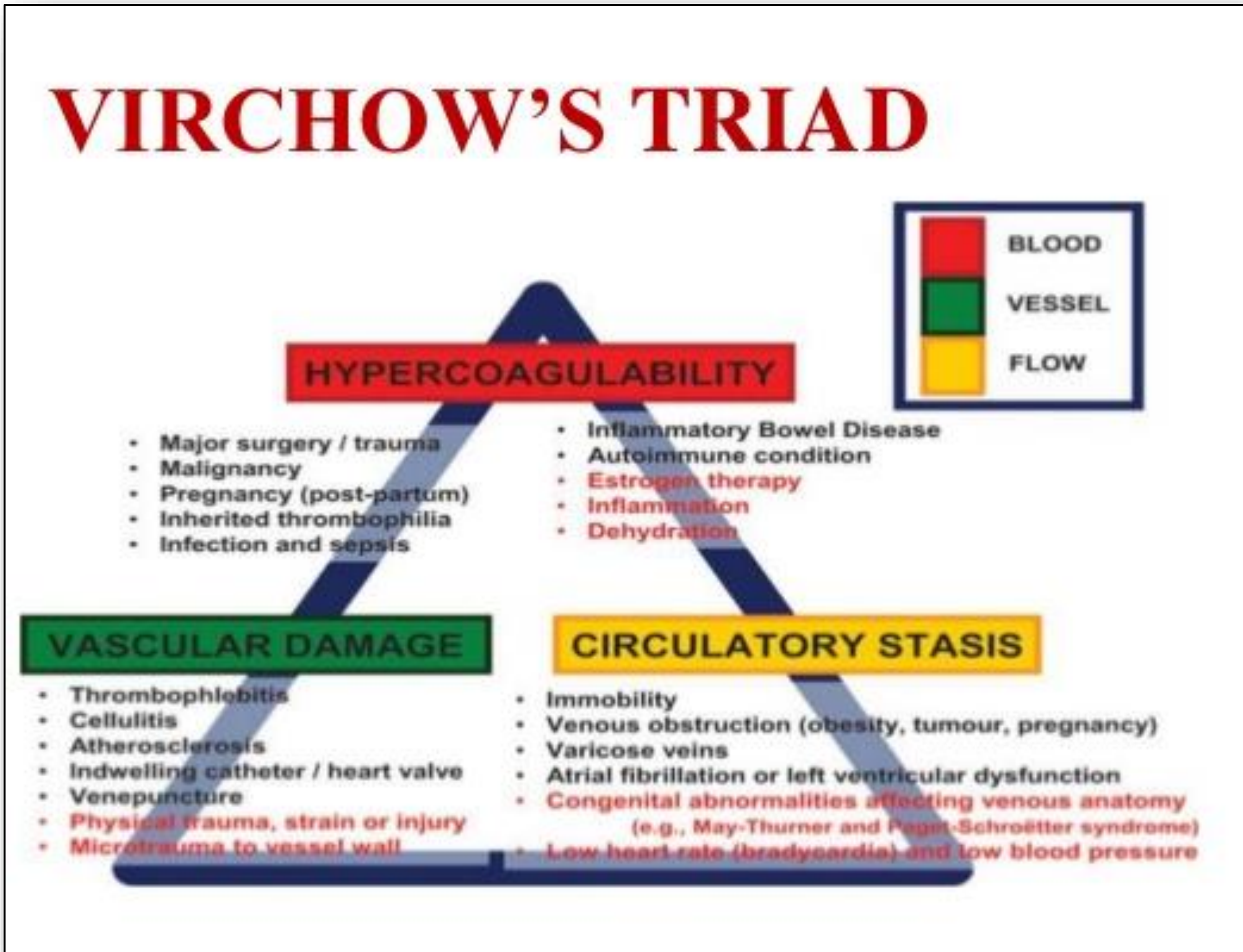


Figure 45: Pathway of pulmonary embolism

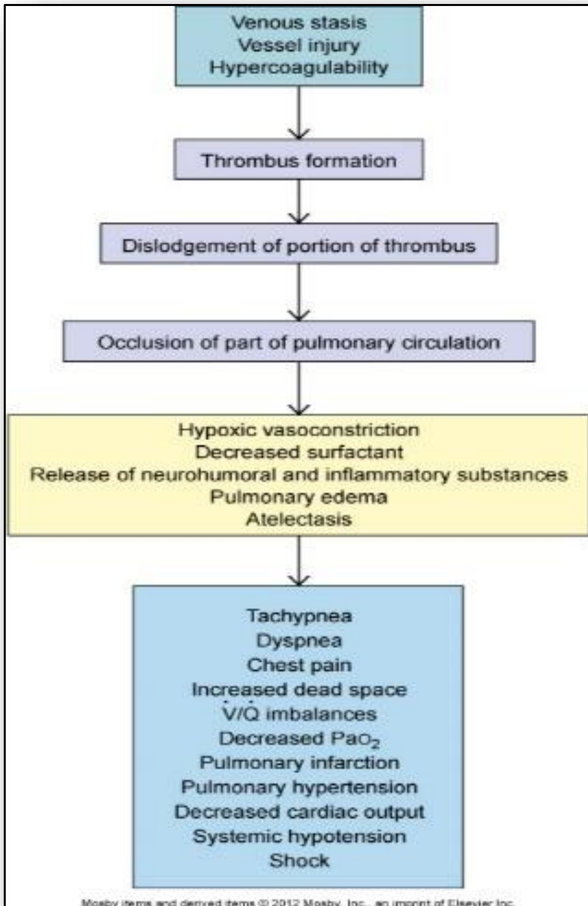
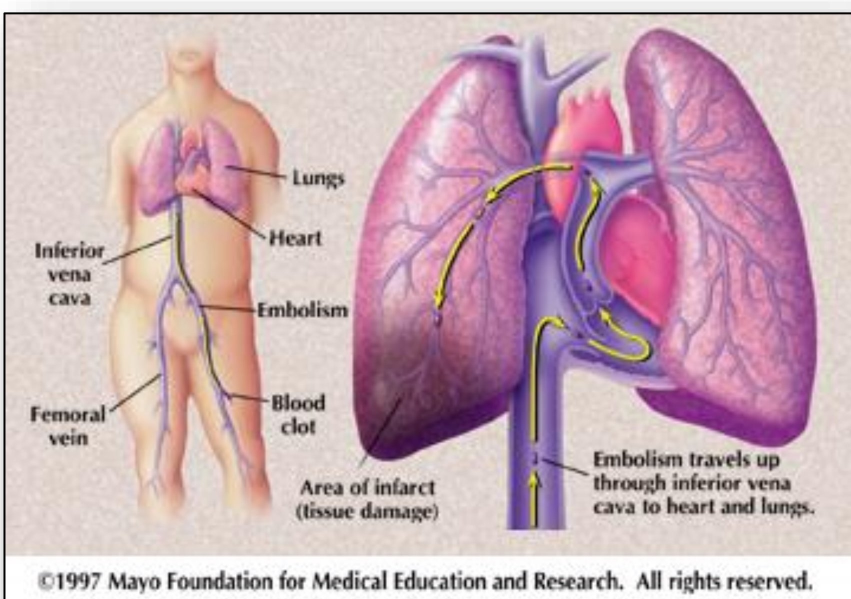


Figure 46: Pulmonary embolism



7: Alterations of the Nervous System

Figure 1: Divisions and structure of the nervous system

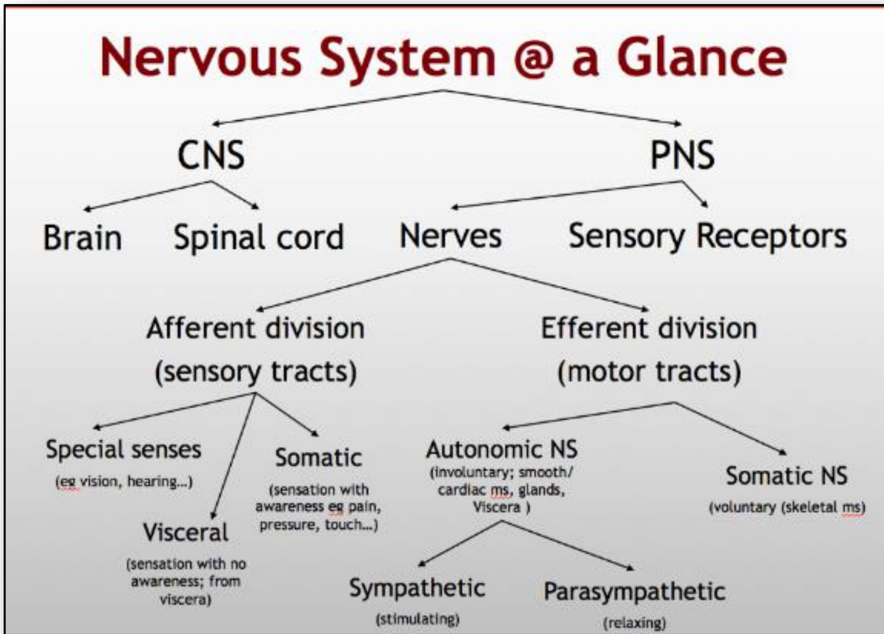


Figure 2: Anatomy of the brain and brainstem

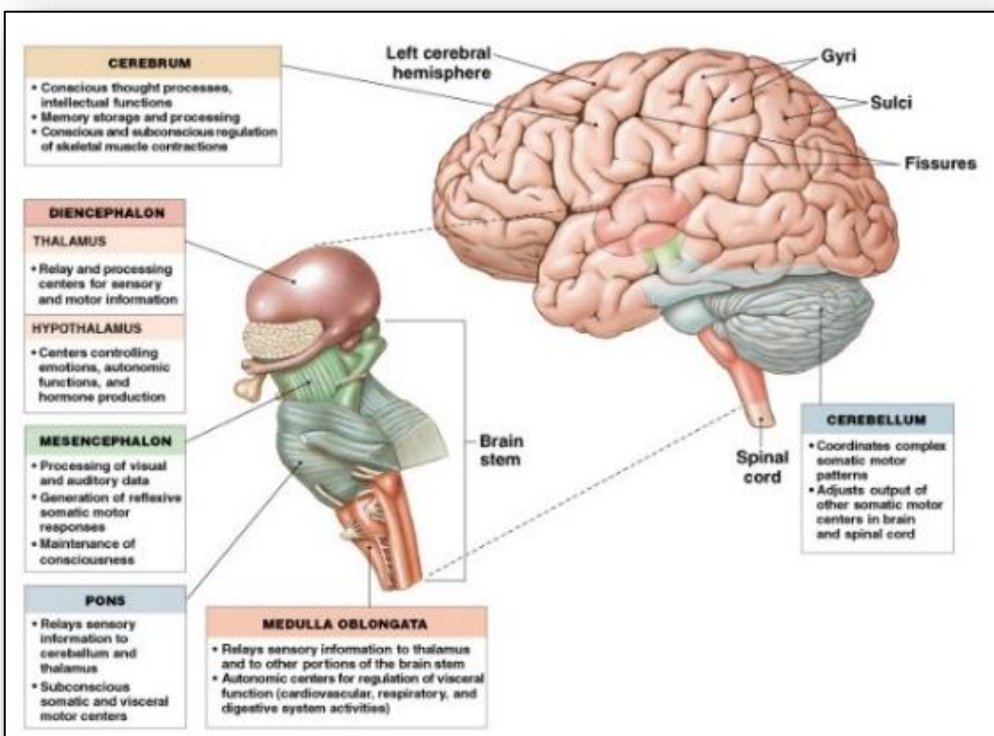


Figure 3: Anatomy of brain and brainstem

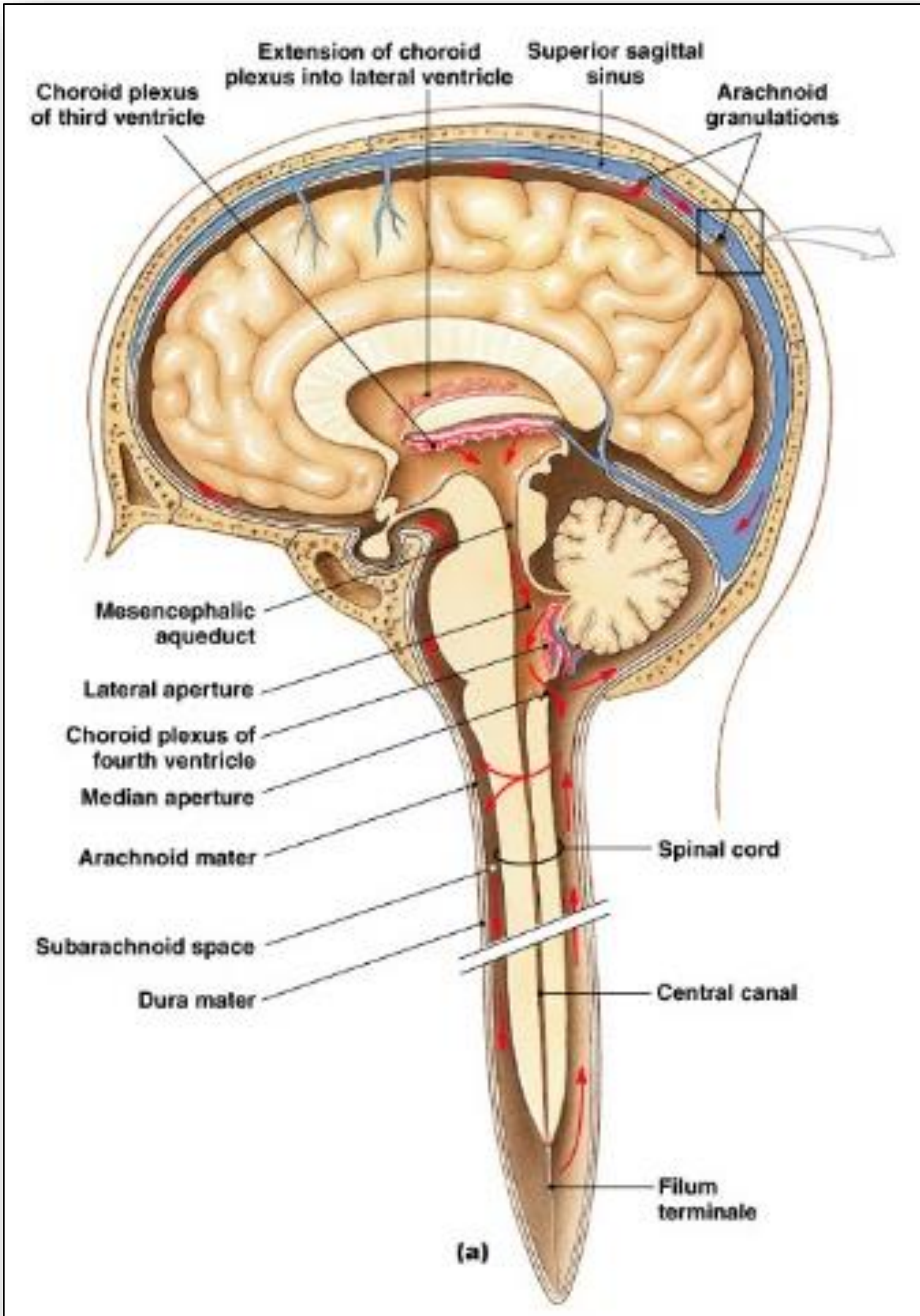


Figure 4: Frontal cross of brain

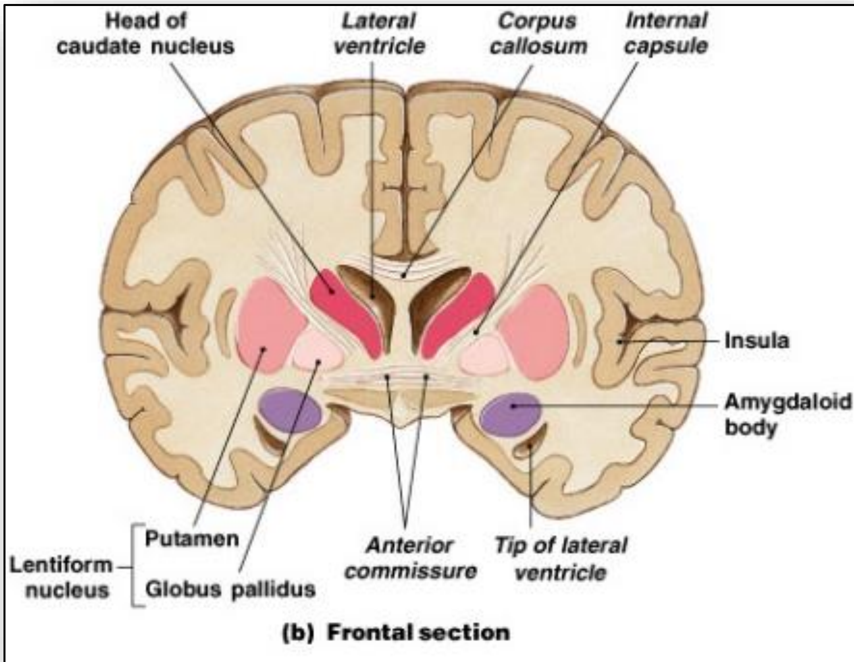


Figure 5: Surface anatomy of the brain

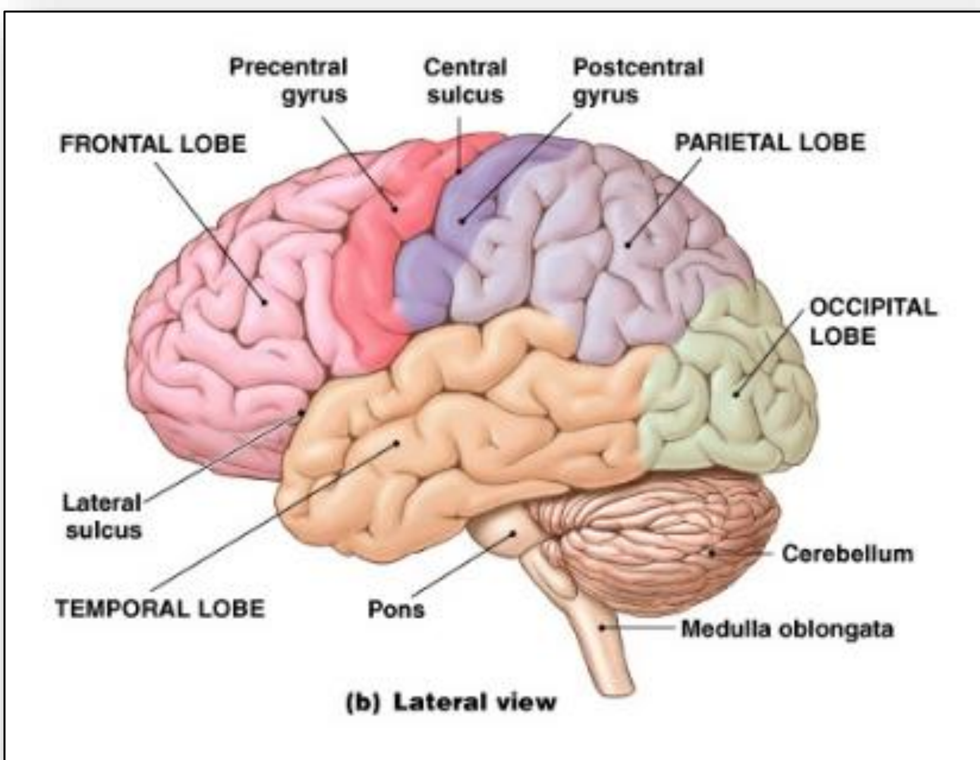


Figure 6: Cells of the nervous system and blood brain barrier

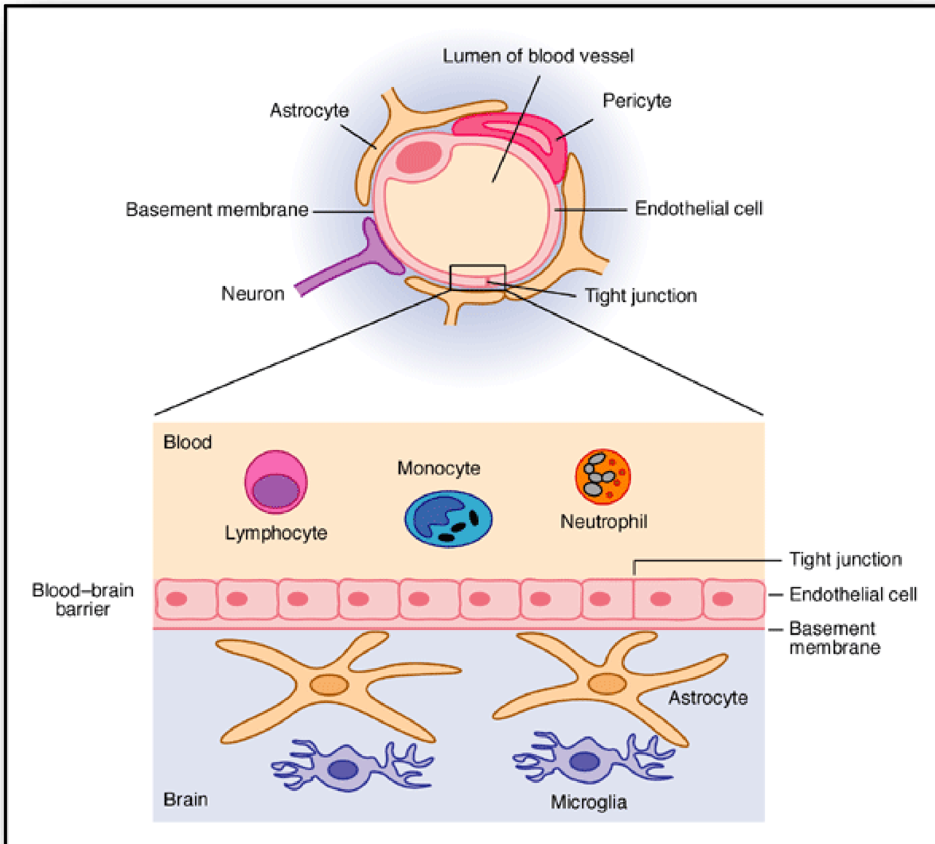


Figure 7: Neuronal synapse

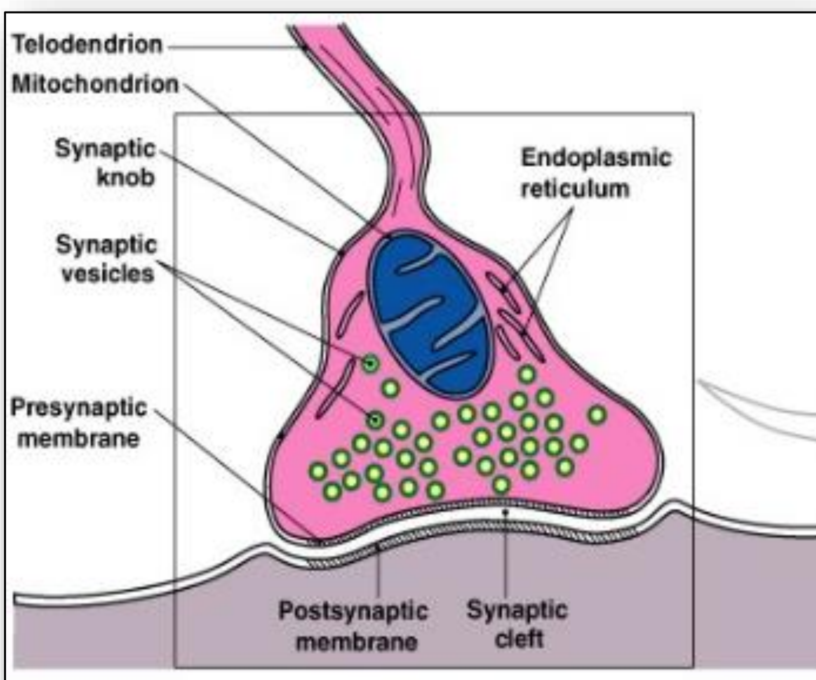


Figure 8: Peripheral nerves

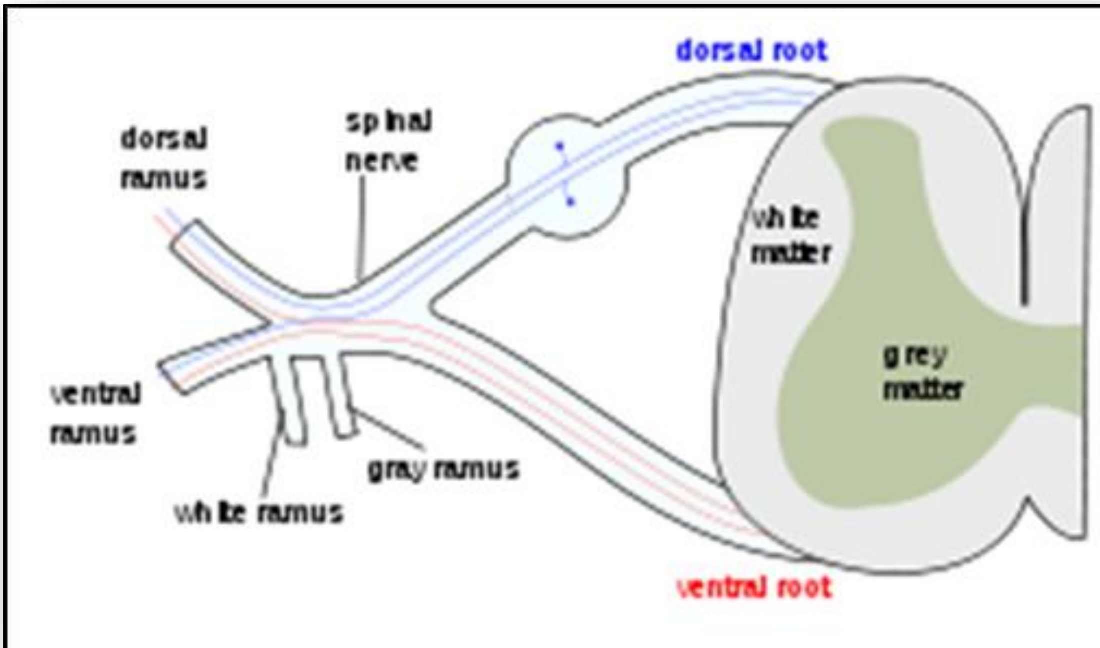


Figure 9: Craniosacral regions of the autonomic nervous system

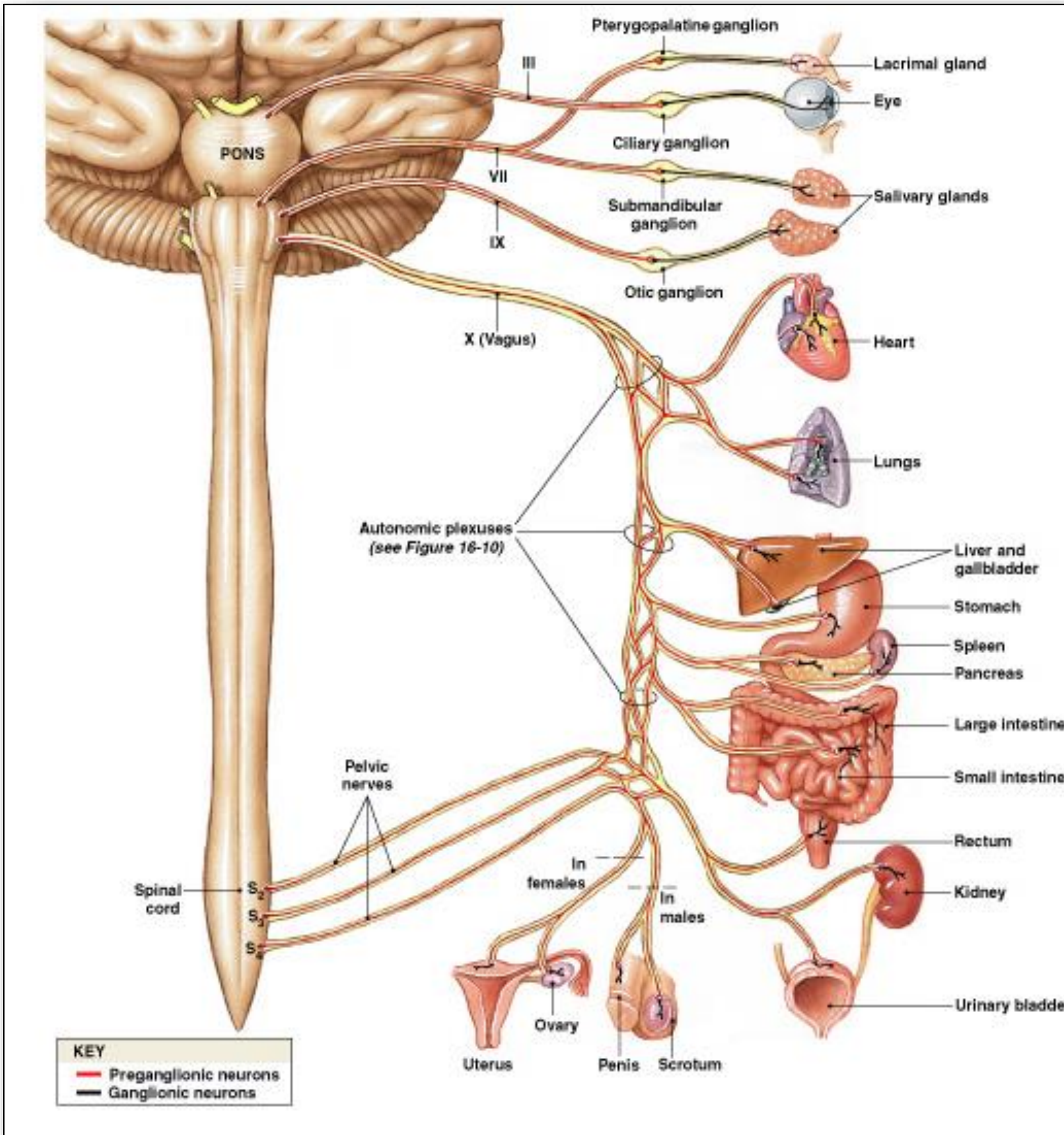


Figure 10: Thoracolumbar region of the autonomic nervous system

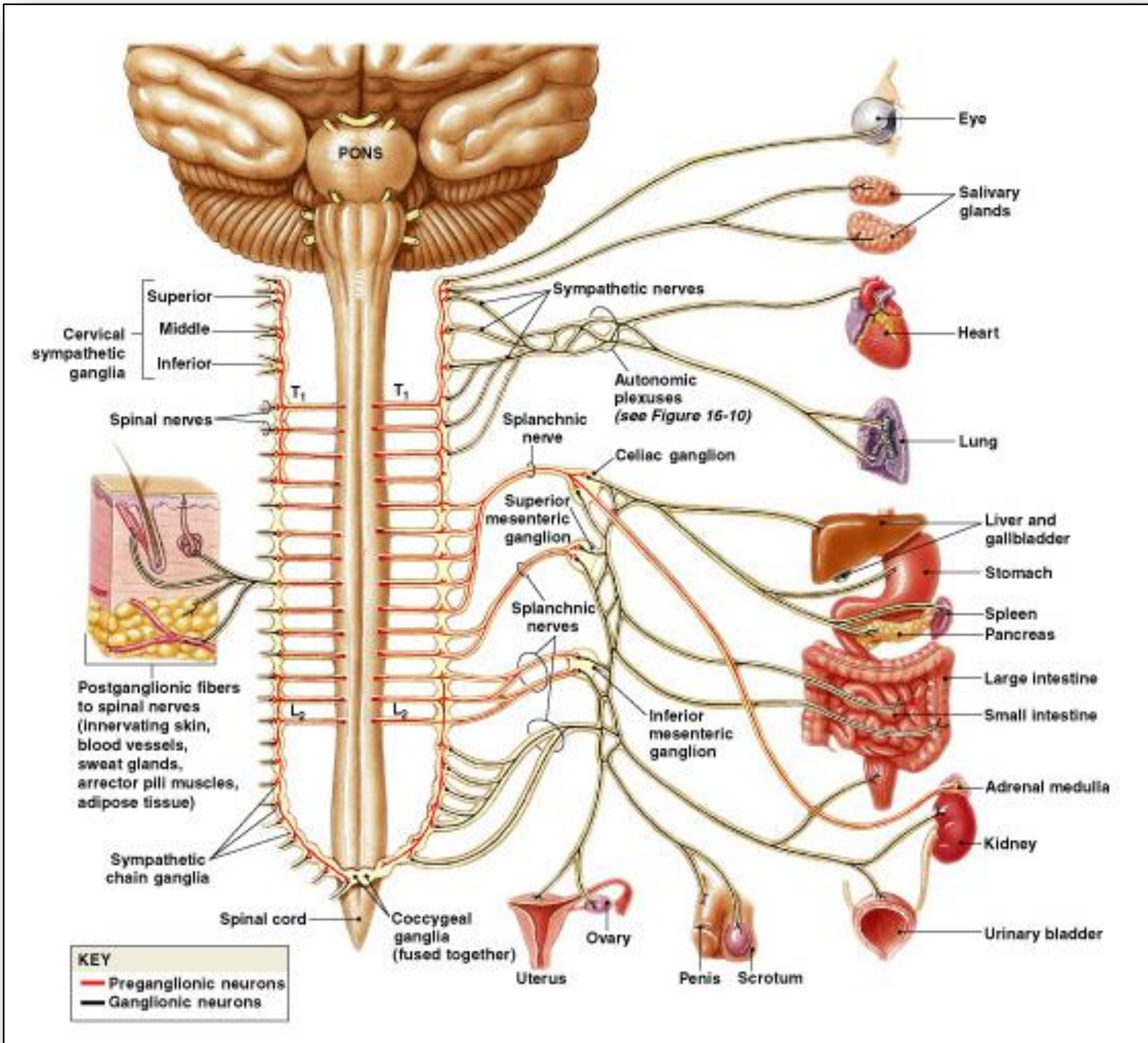


Figure 11: Brain cross section showing white and grey matter

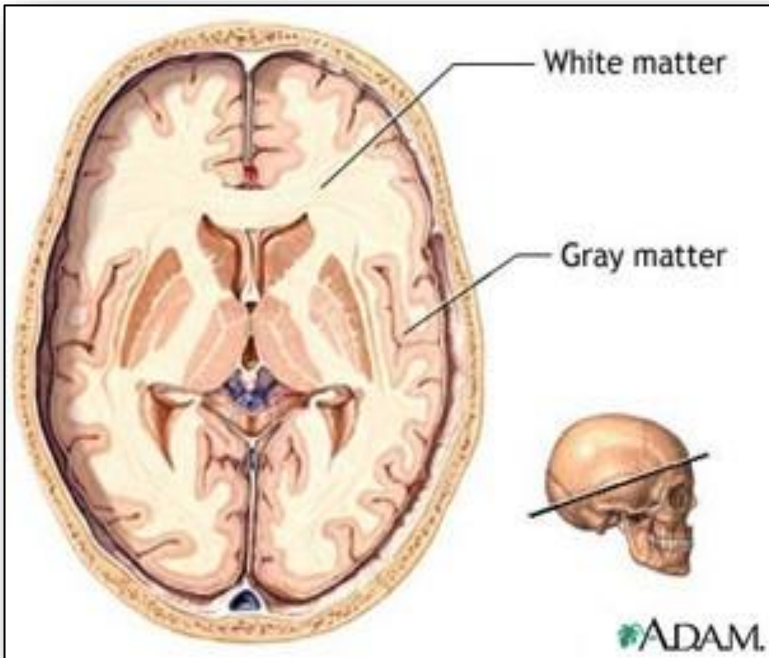


Figure 12: Relationship between central and peripheral nervous system in the spinal cord

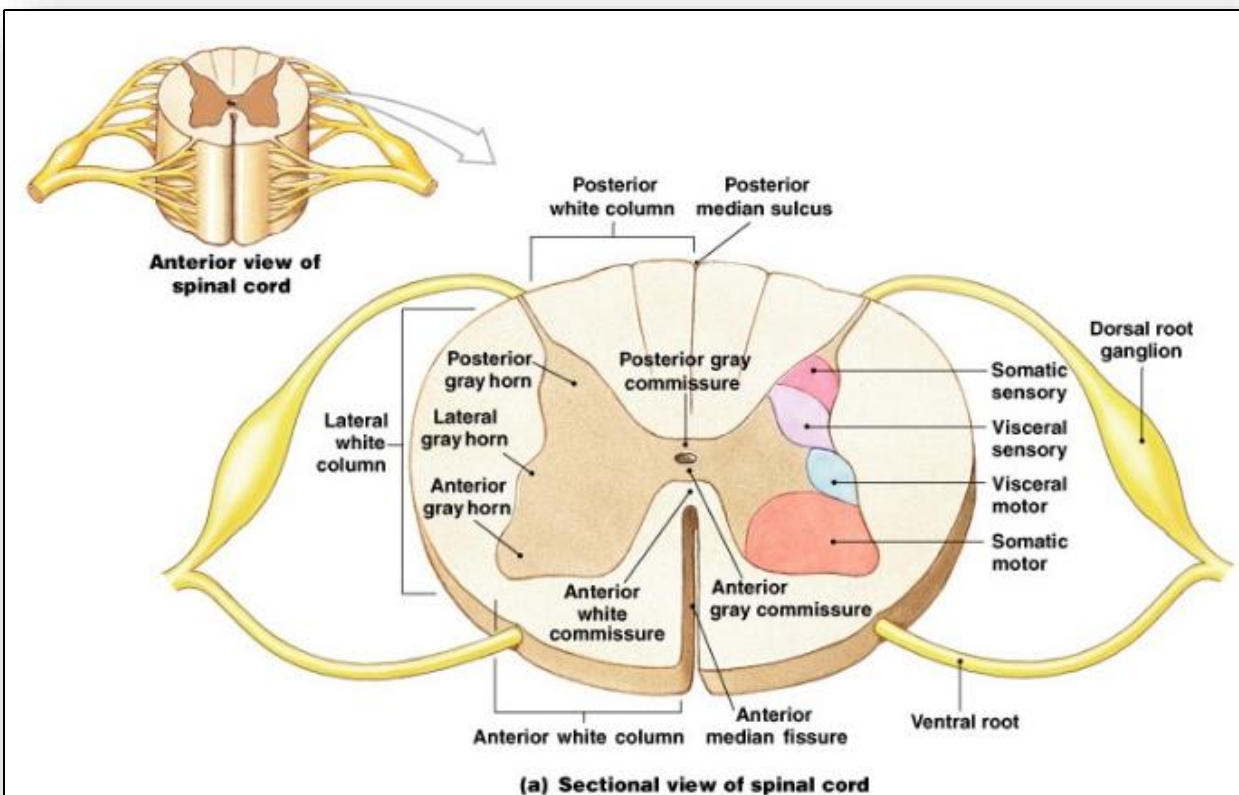


Figure 13: Coup versus contrecoup impact in TBI

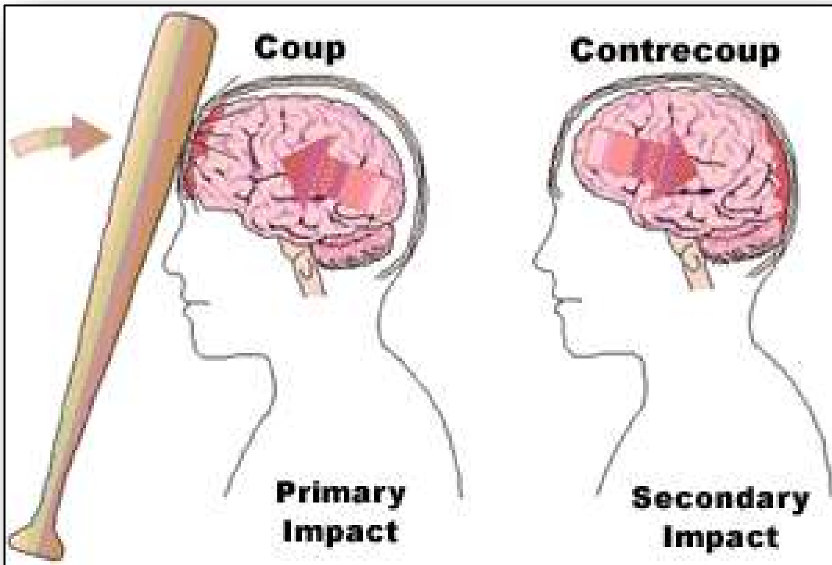


Figure 14: Types of hematoma by location (epidural, subdural, intracerebral)

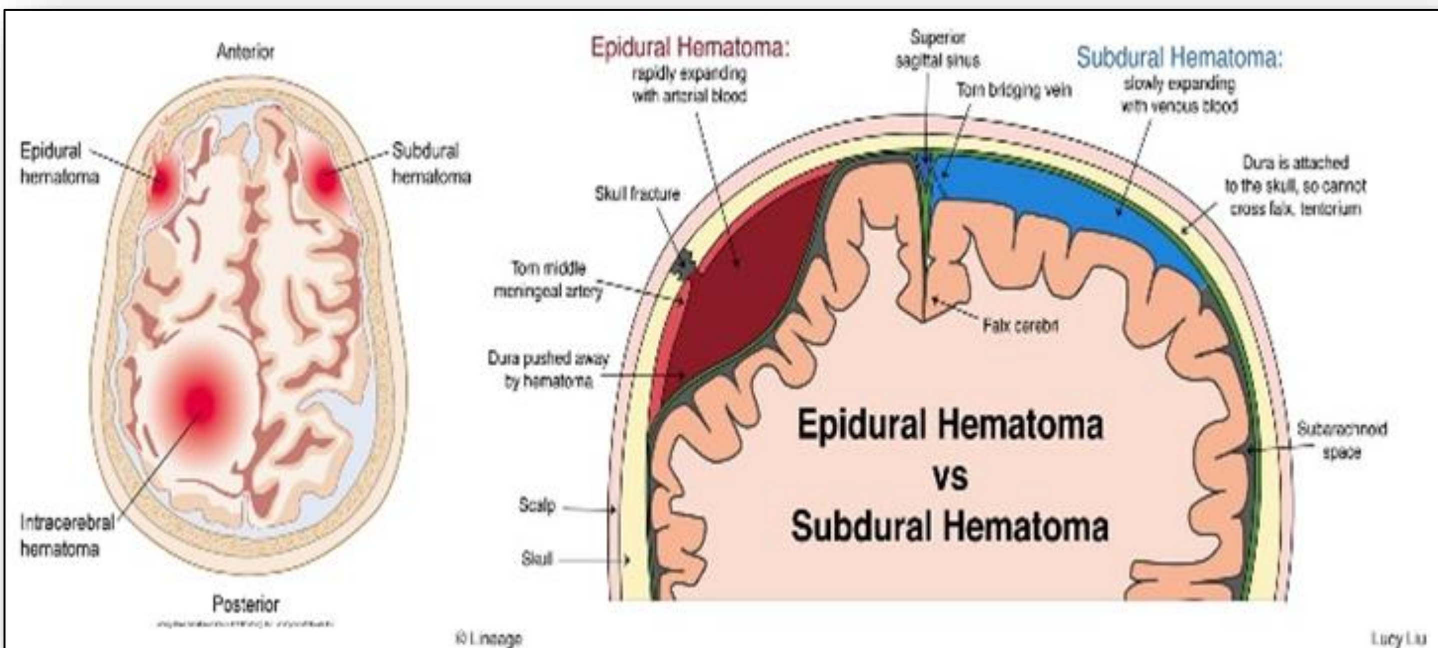


Figure 15: Classifying TBI and manifestations

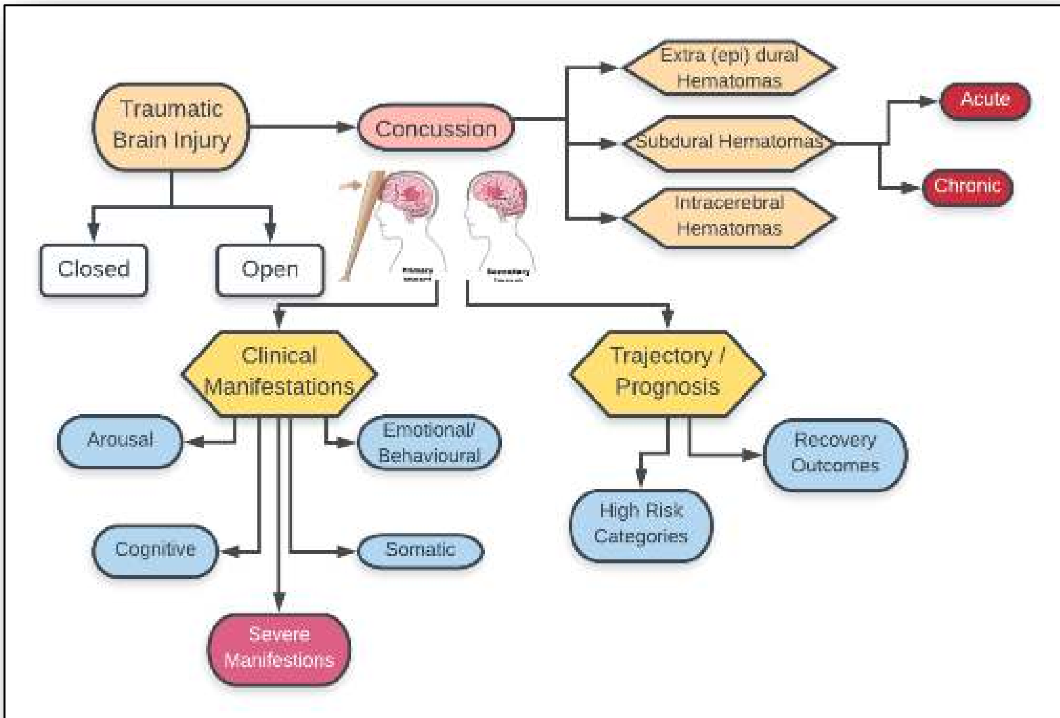


Figure 16: Compression fracture of the spine

Figure 17: Comminuted (burst) fracture on x-ray

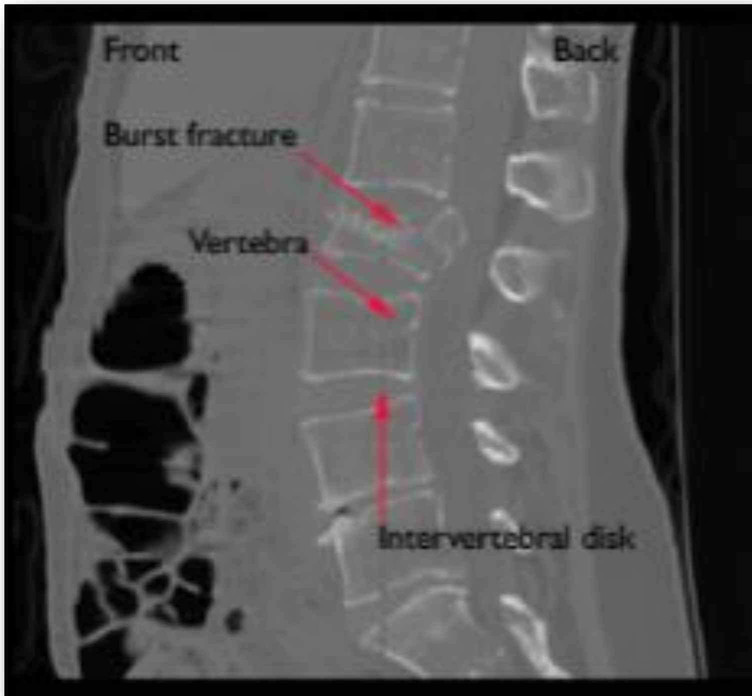


Figure18: Dislocation fracture of the spine

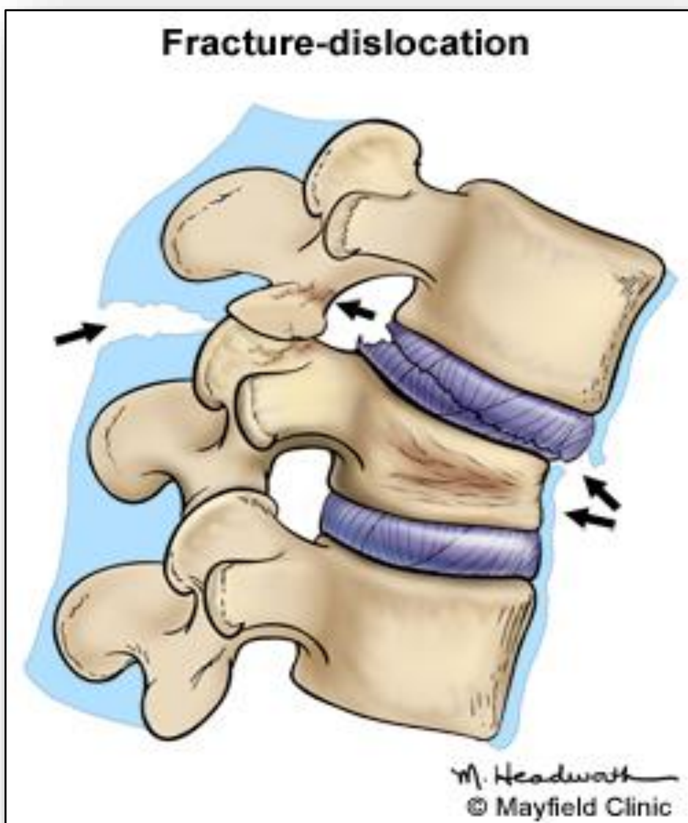


Figure 19: Anatomy of a vertebra

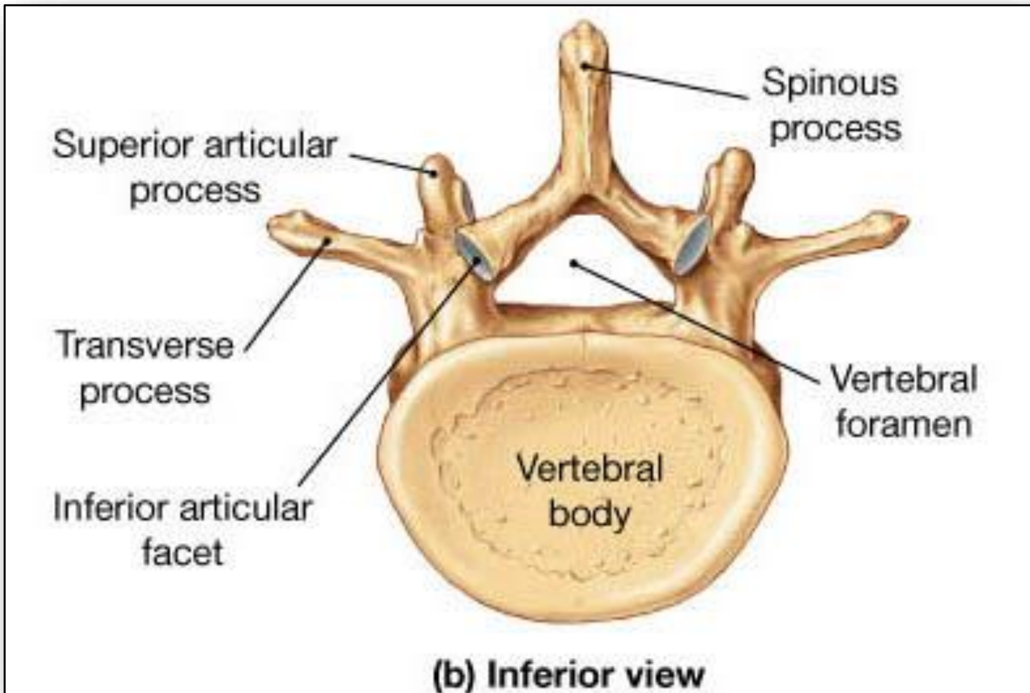


Figure 20: Comparison of normal and herniated intervertebral disc

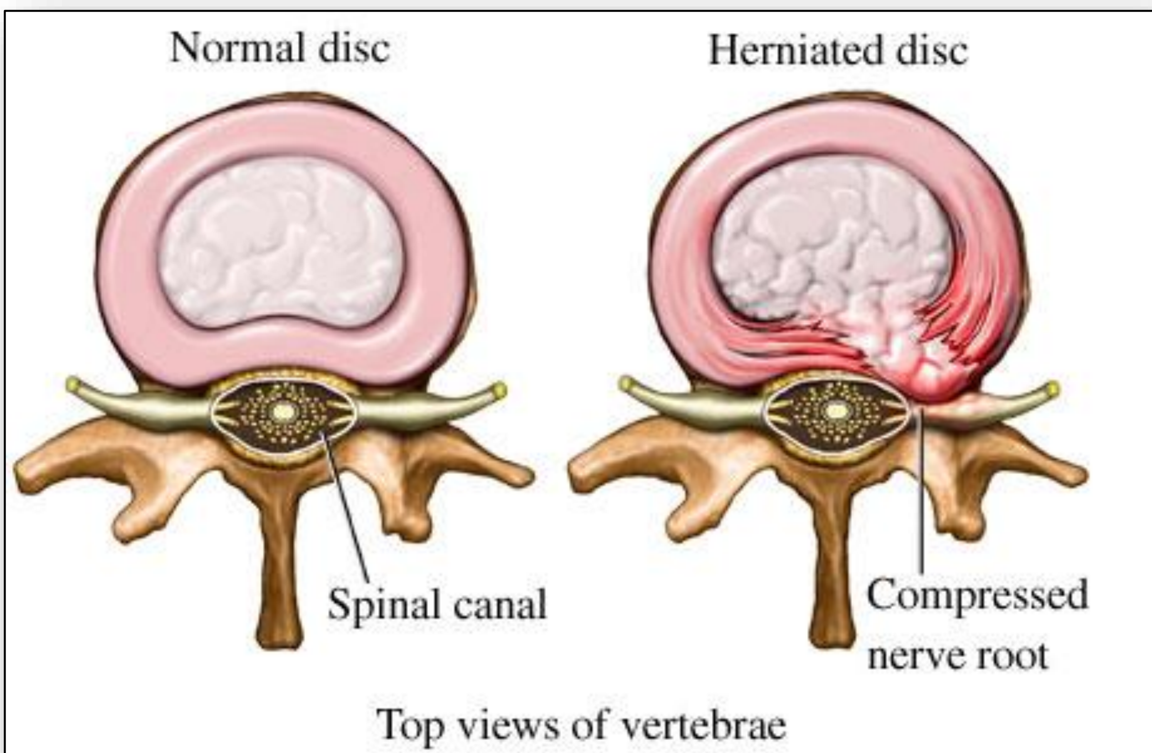


Figure 21: Spondylolysis and spondylolisthesis of the lower spine

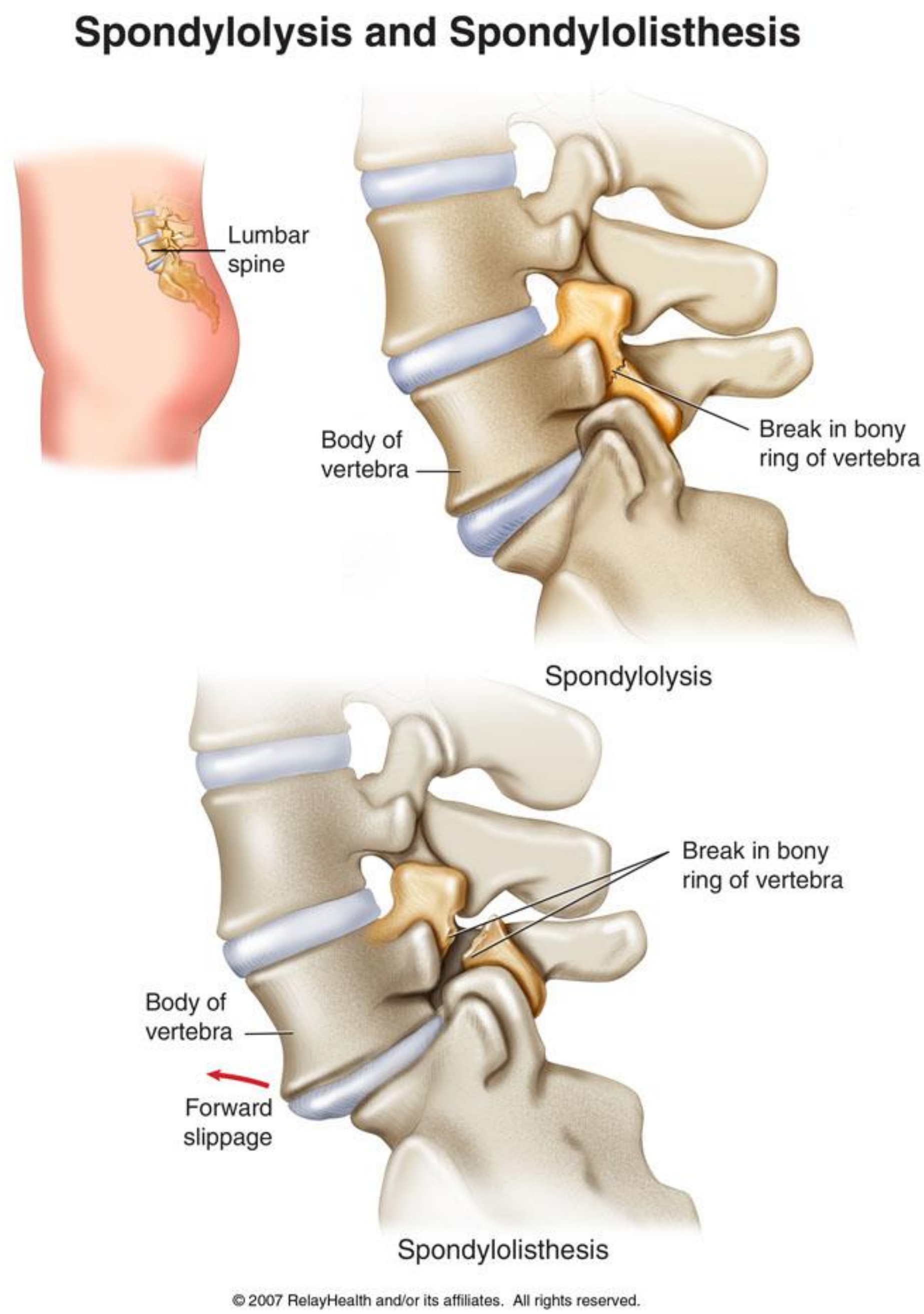


Figure 22: Spinal stenosis

Figure 23: Cerebral vasculature

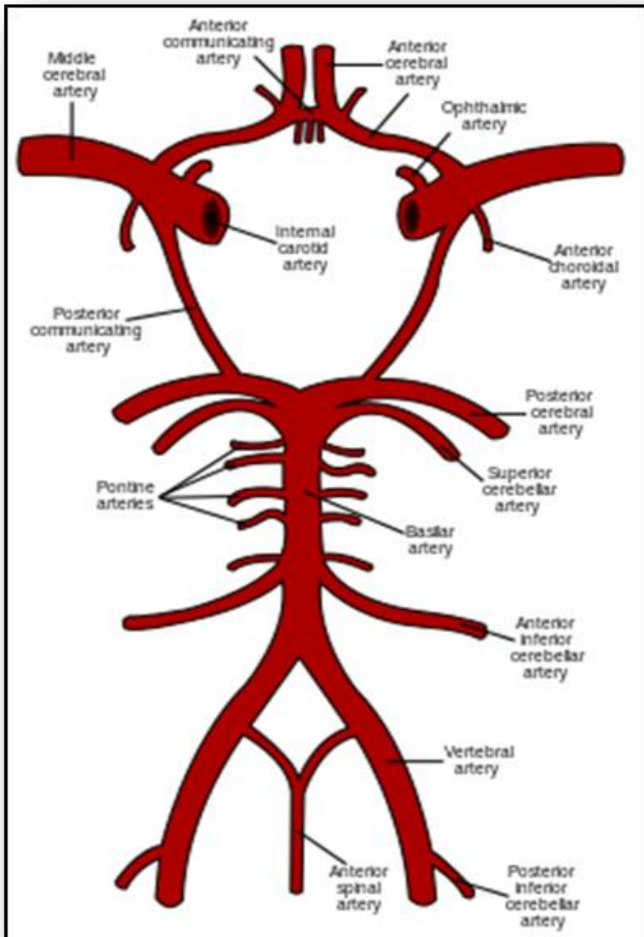


Figure 24: Ischemic stroke

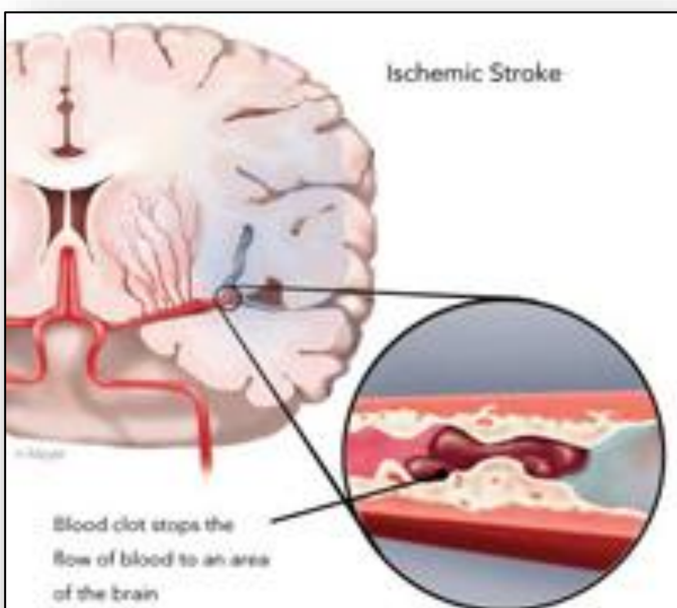


Figure 25: Kernig (A) and Brudinski (B) Signs

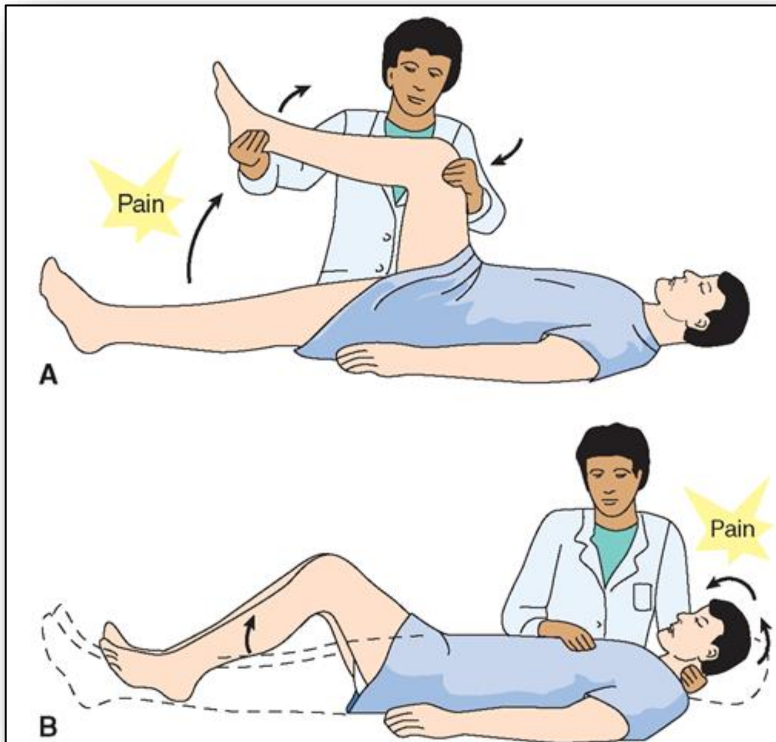


Figure 26: Types of aneurysms

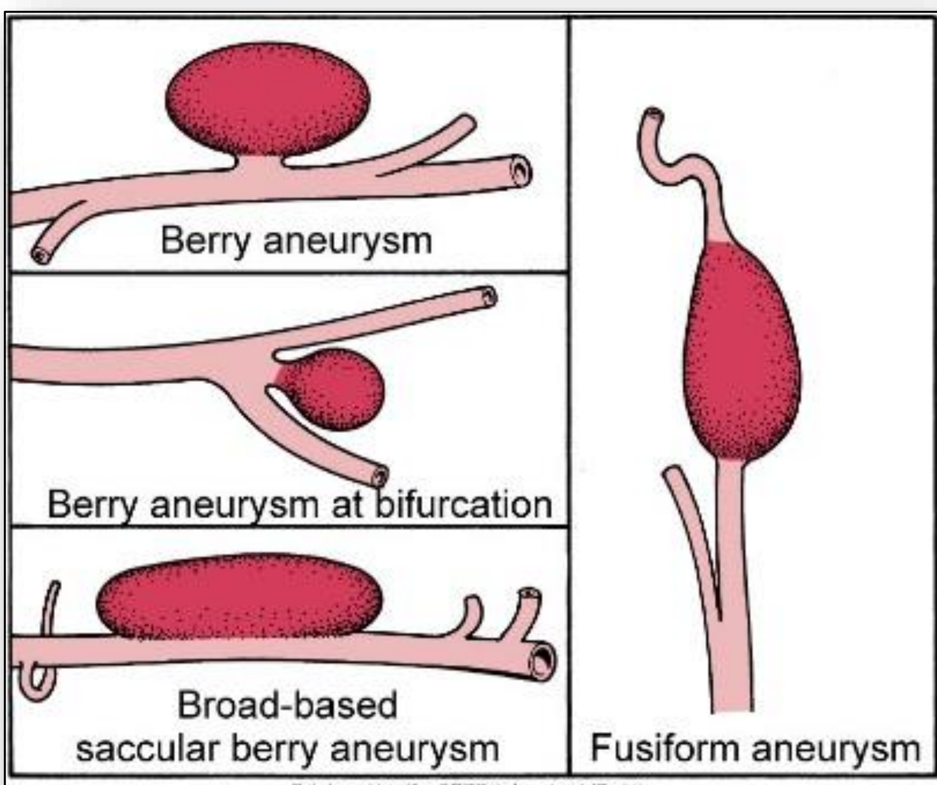


Figure 27: Types of Headache

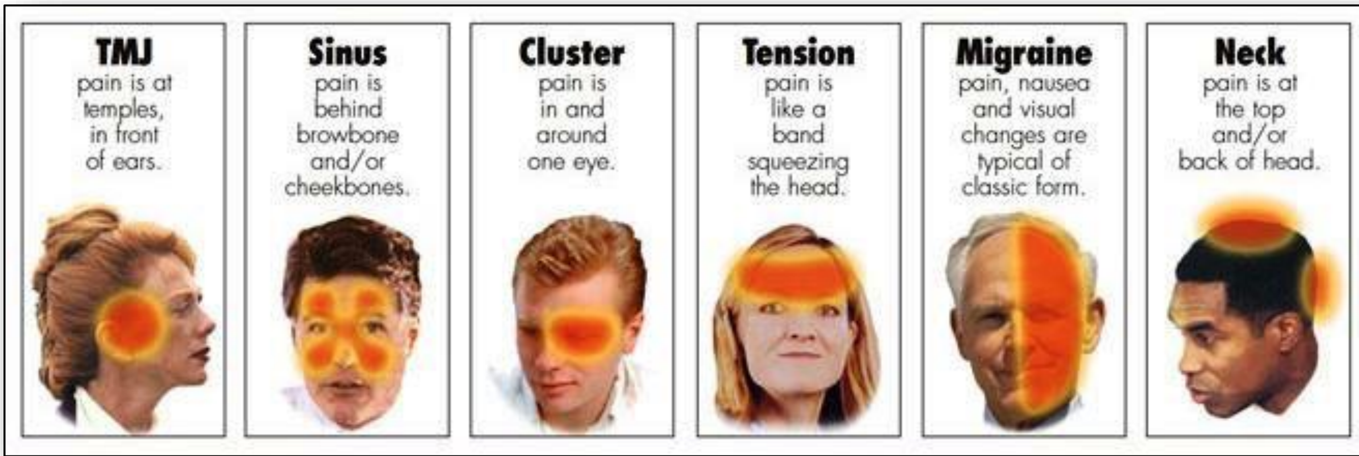


Figure 28: Unilateral location of migraine headache

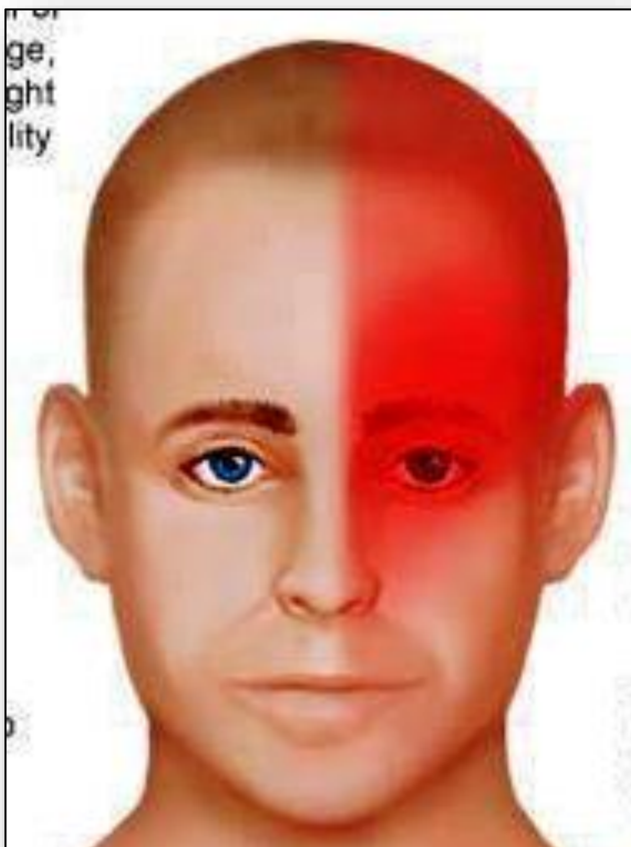


Figure 29: Pathophysiology and manifestations of multiple sclerosis

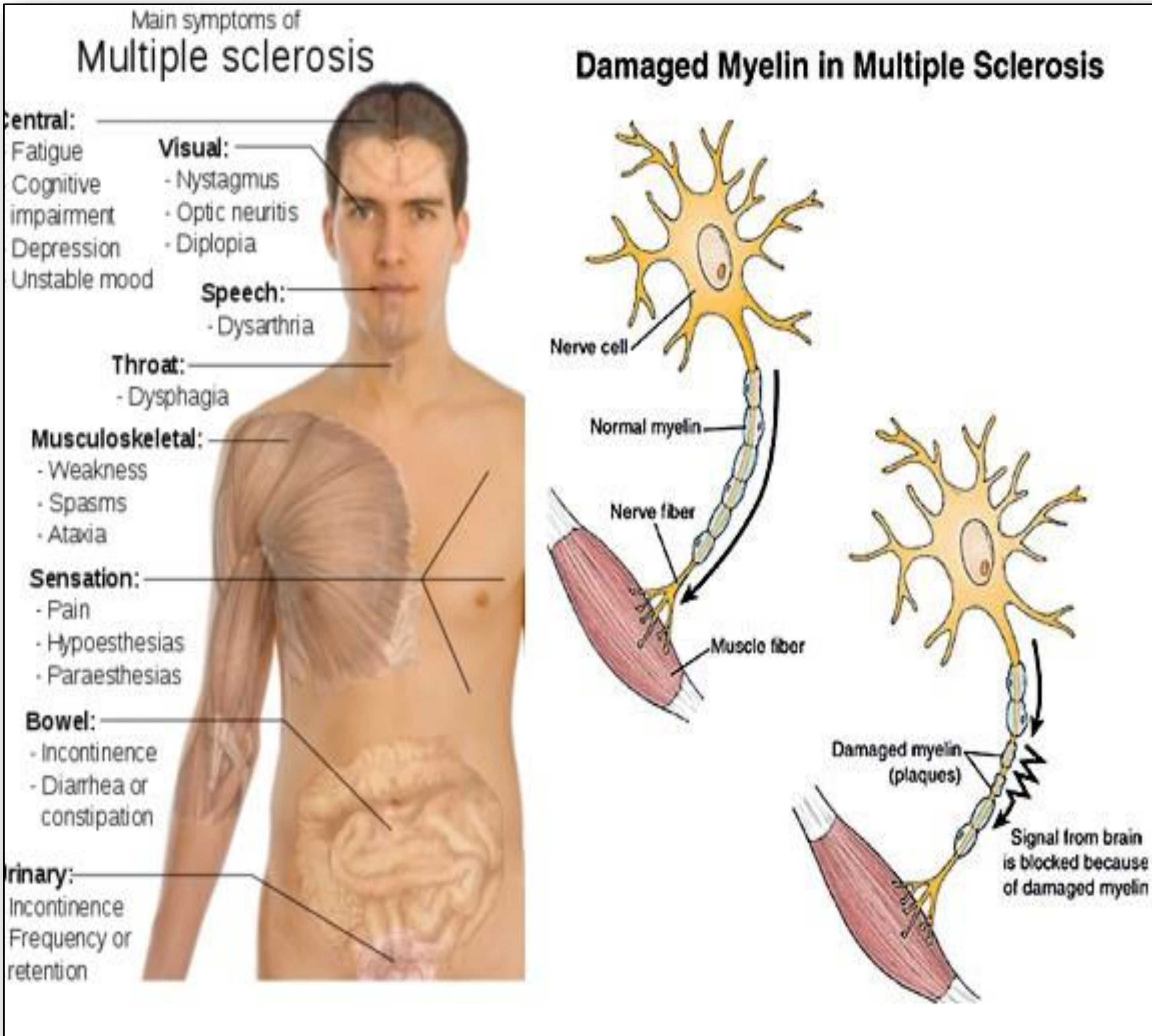
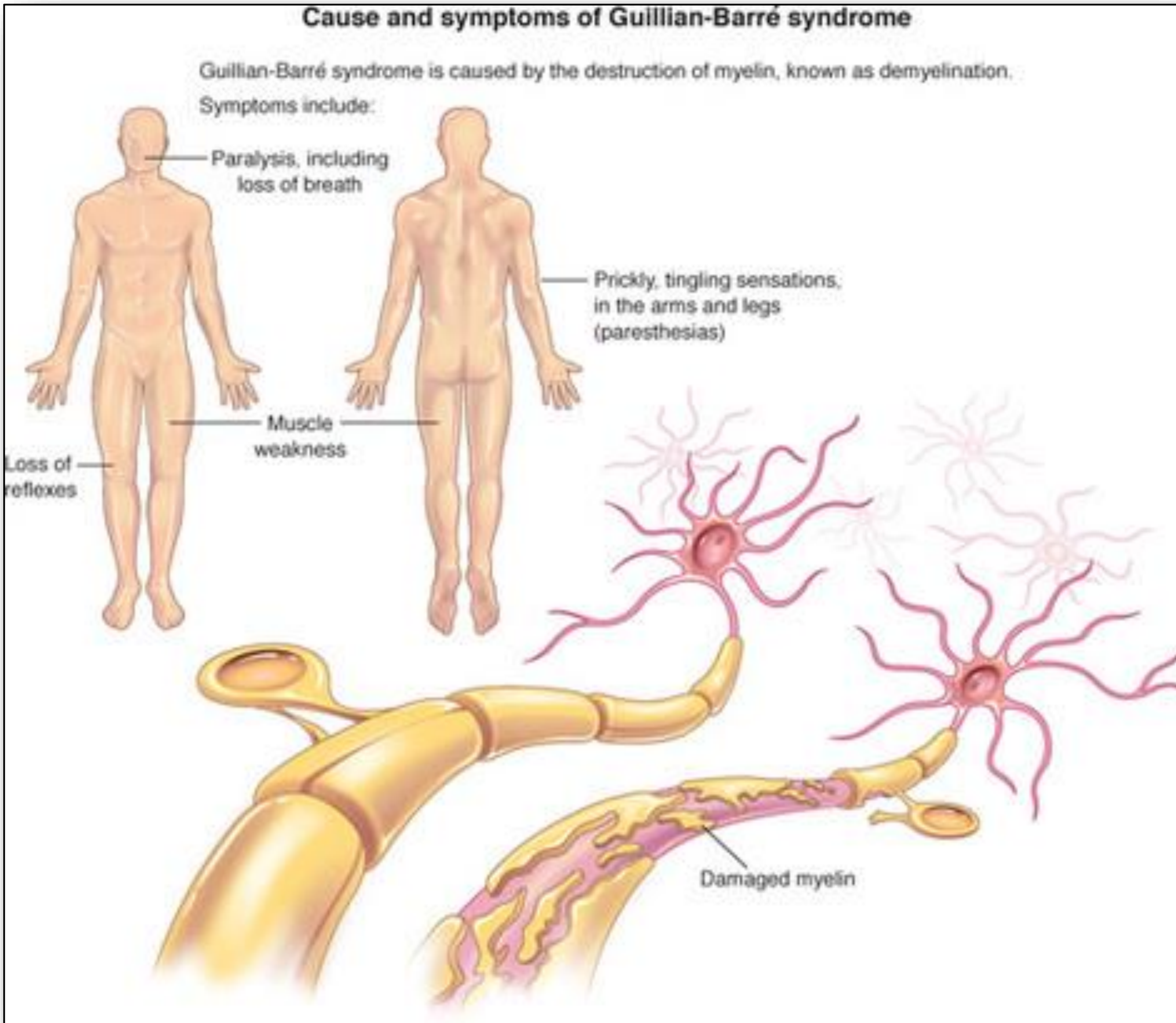


Figure 30: Causes, pathophysiology, and manifestations in Guillain-Barre syndrome



8: Alterations in Cognitive Systems, Cerebral Hemodynamics and Motor Function

Figure 31: Reticular activating system (RAS)

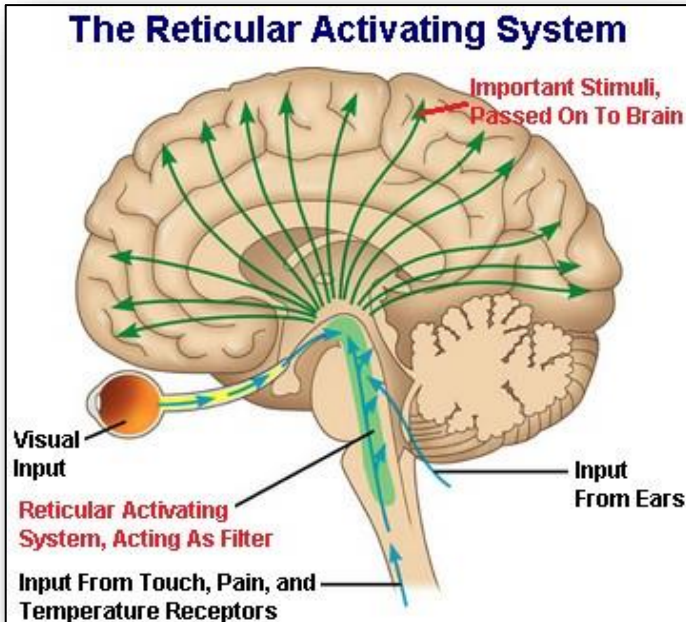


Figure 32: Brodmann's cytoarchitectural map

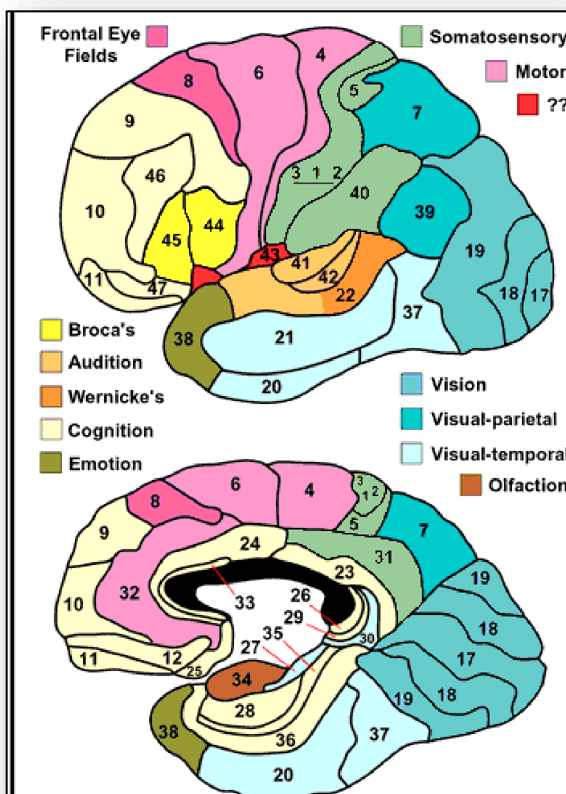


Figure 33: Types of aphasia

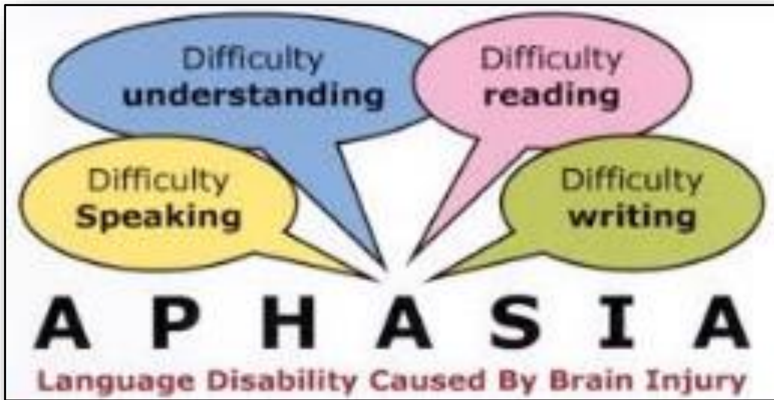


Figure 34: Plaques and tangles implicated in Alzheimer's disease

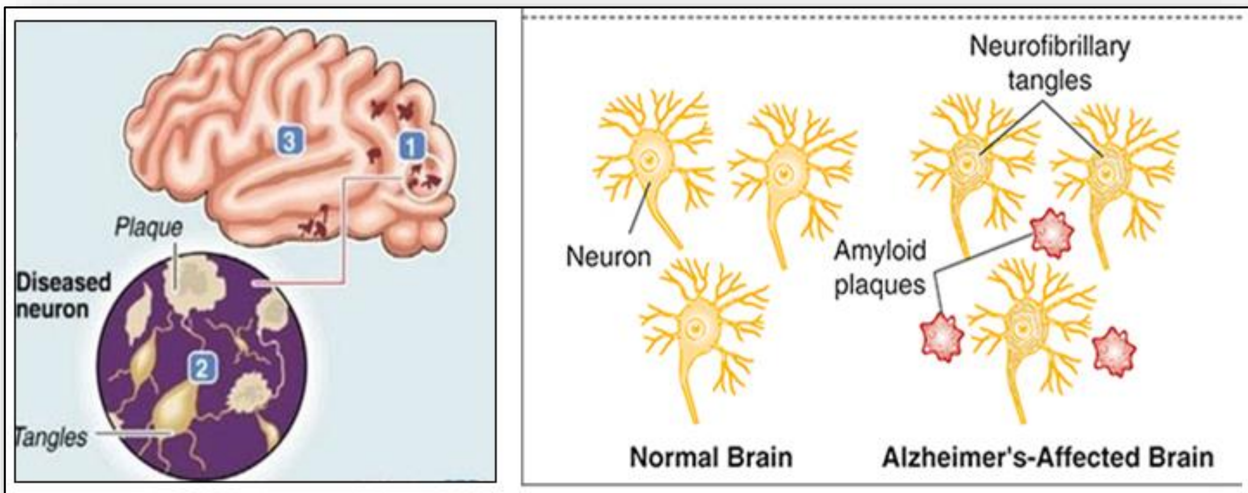


Figure 35: Comparison of normal brain to that of person with severe Alzheimer's

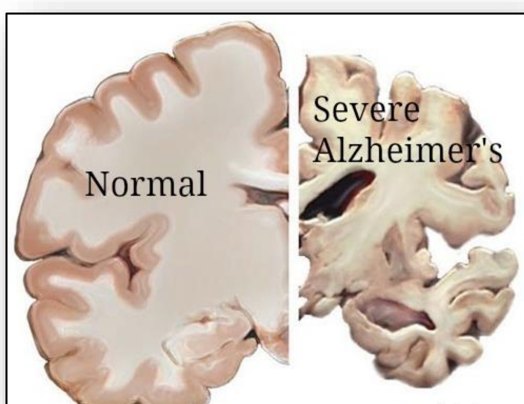


Figure 36: Comparison normal brain (right) to brain with hydrocephalus (left)

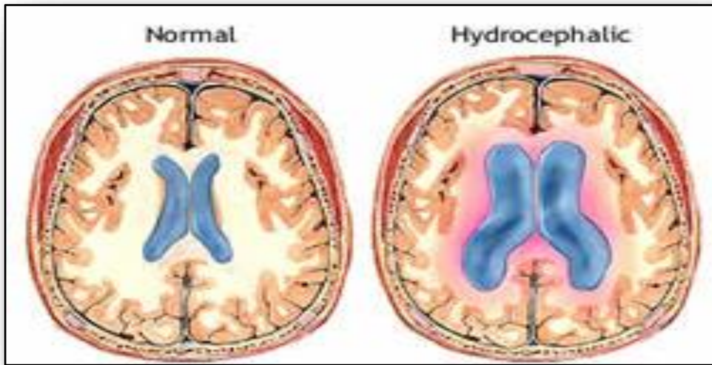


Figure 37: Shunt used to drain fluid in hydrocephalus

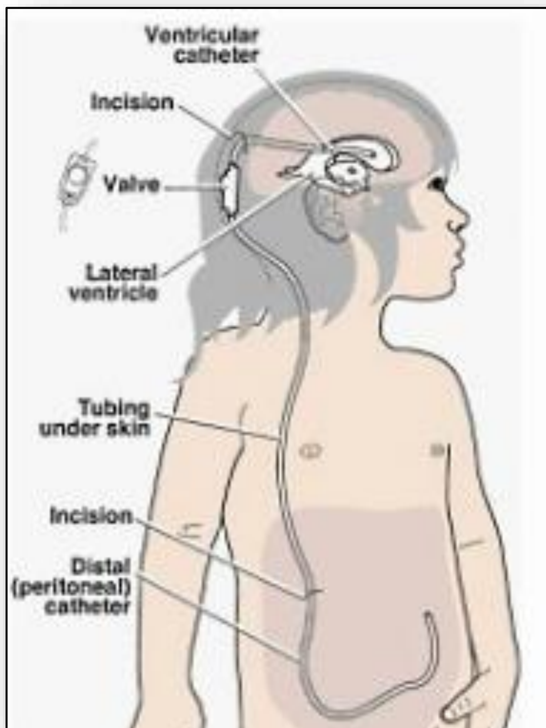


Figure 38: Motor function syndromes

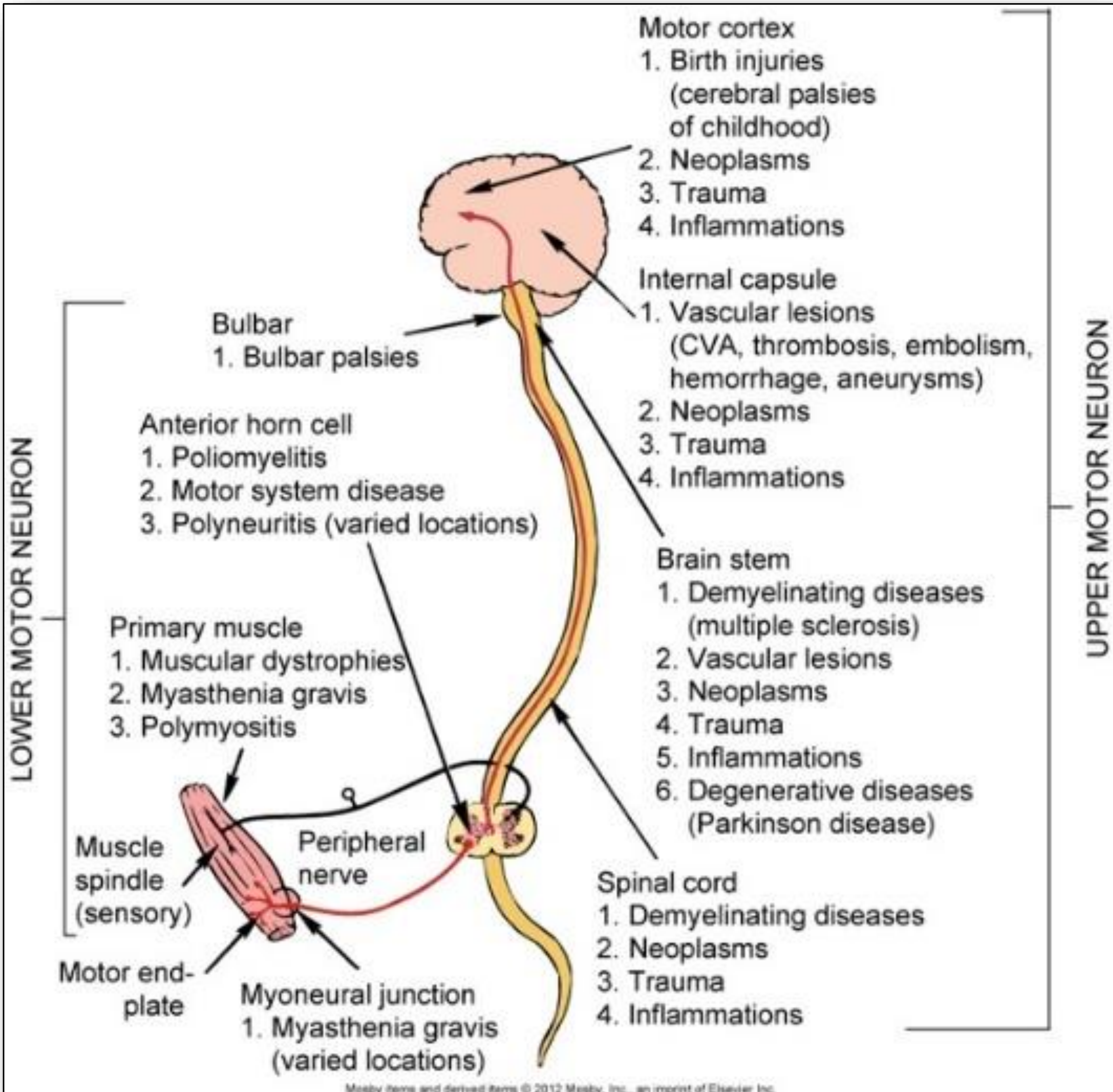


Figure 39: Dopamine levels in normal neurons versus those of one with Parkinson's disease

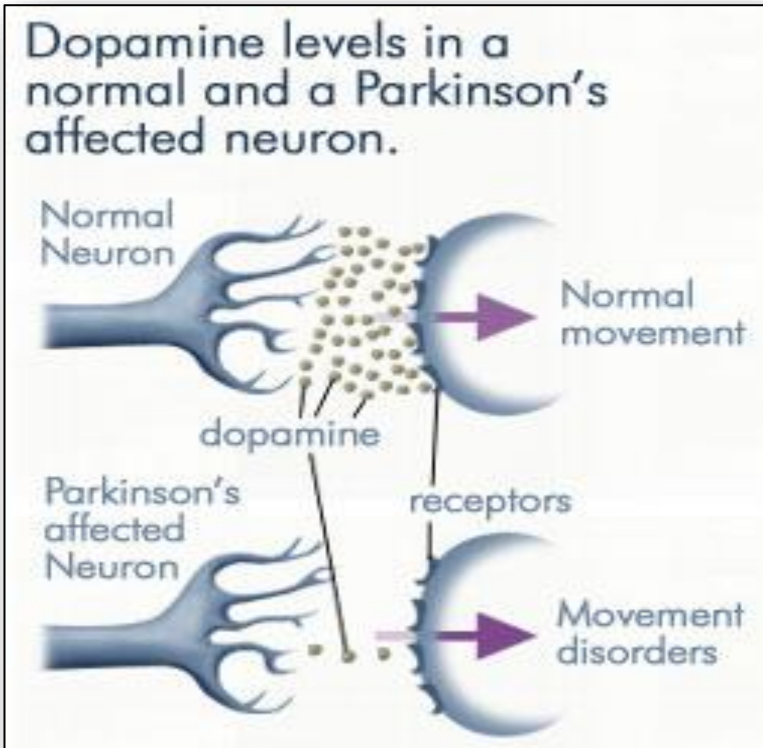


Figure 40: Classic Parkinsonian gait and stance

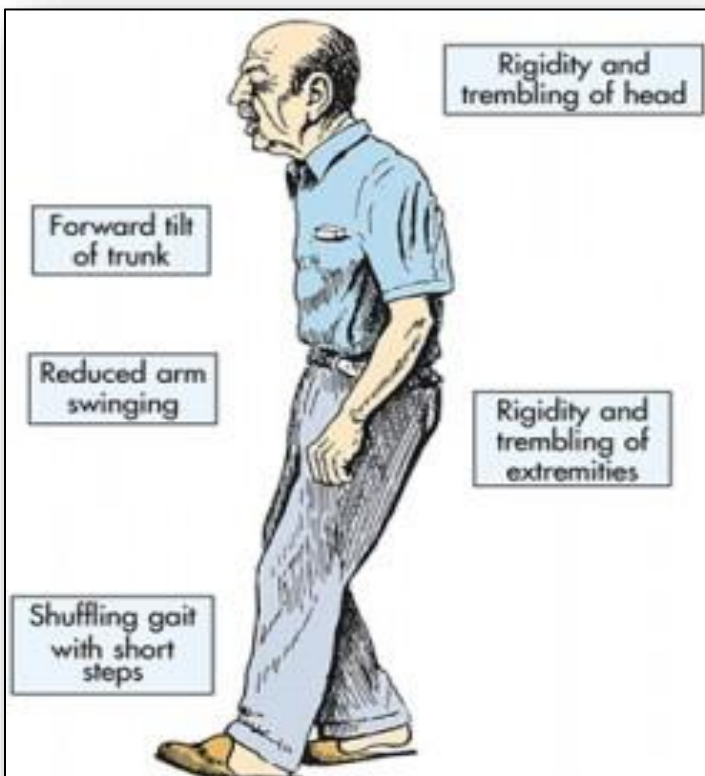


Figure 41: Brain structures impacted in Parkinson Disease

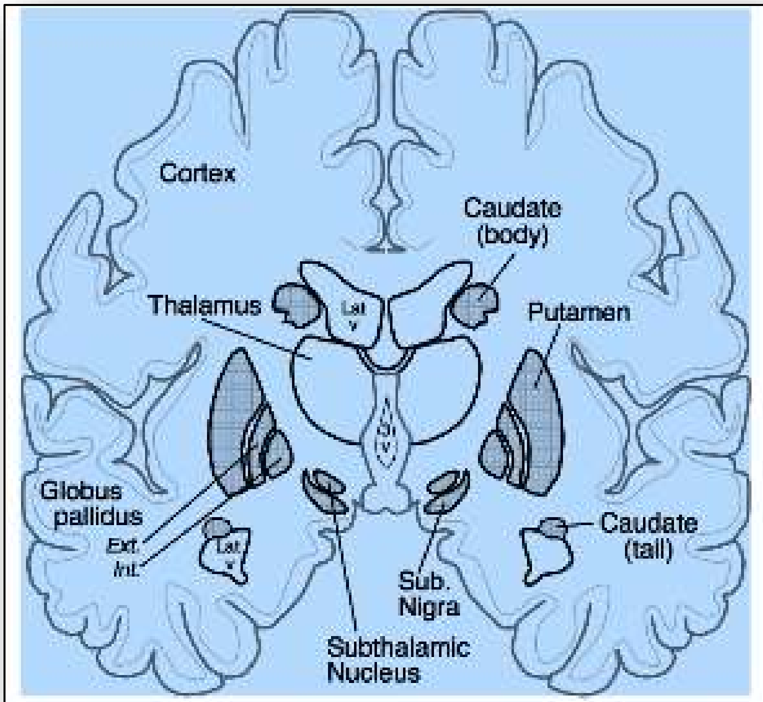


Figure 42: Decorticate and decerebrate posturing

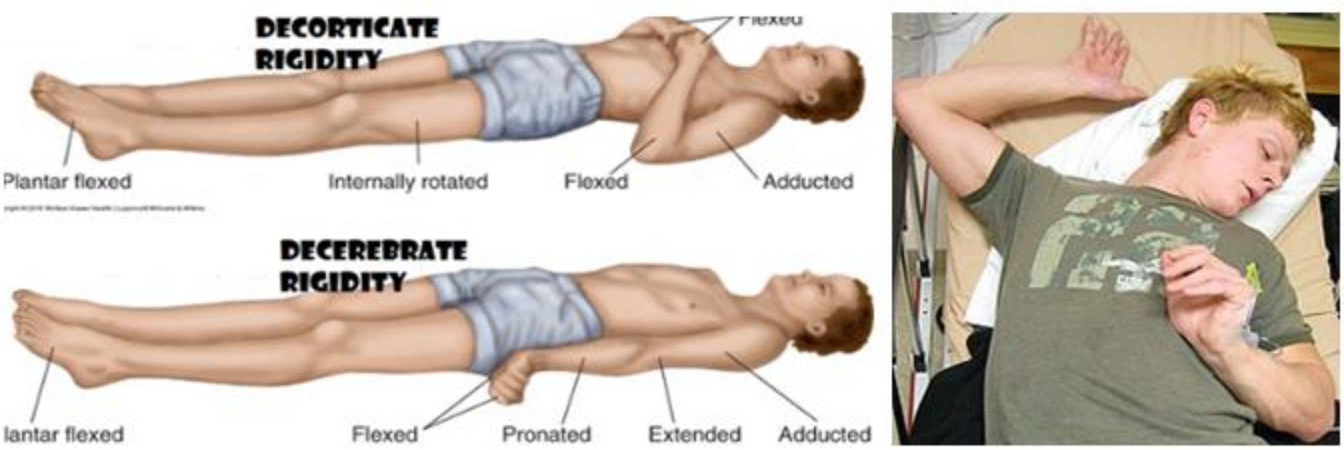


Figure 43: Disorders of gait (spastic, scissors)

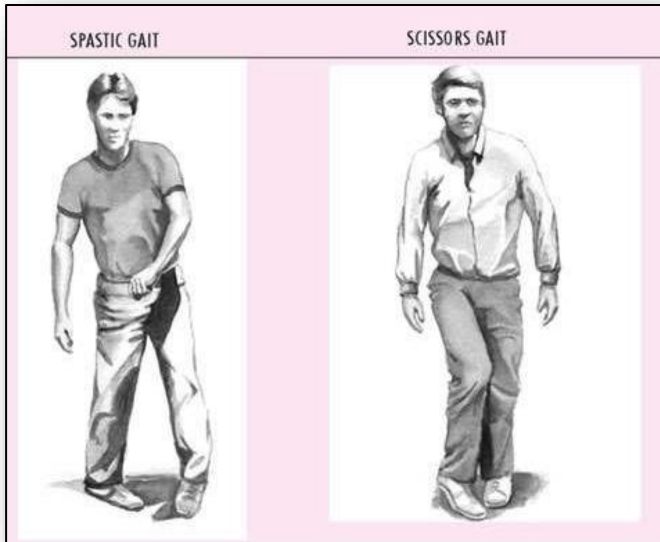


Figure 44: Pathways of nociception in the spinothalamic tract

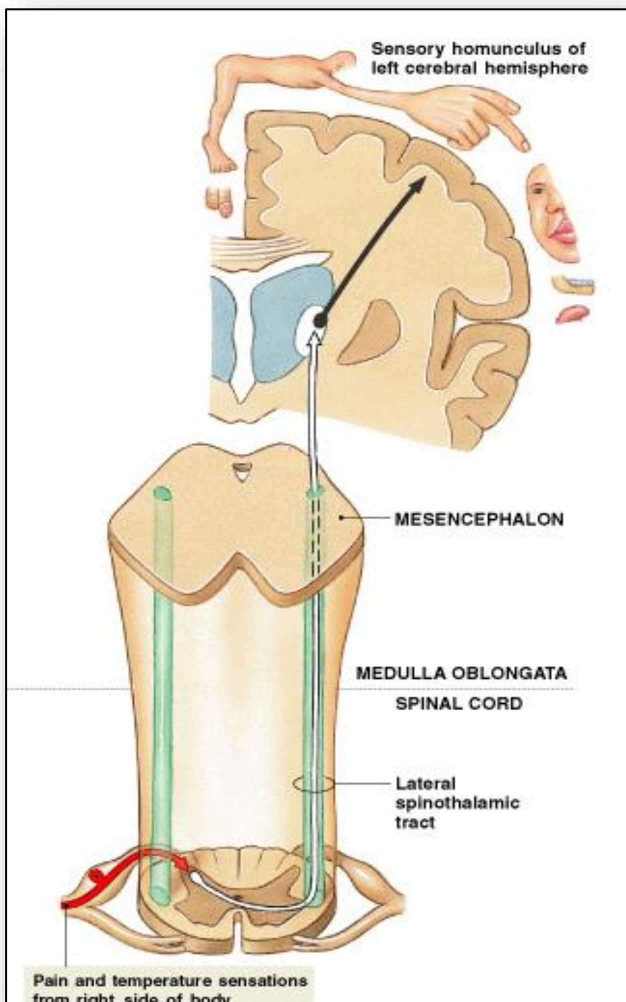


Figure 45: Endogenous opioids in the brain

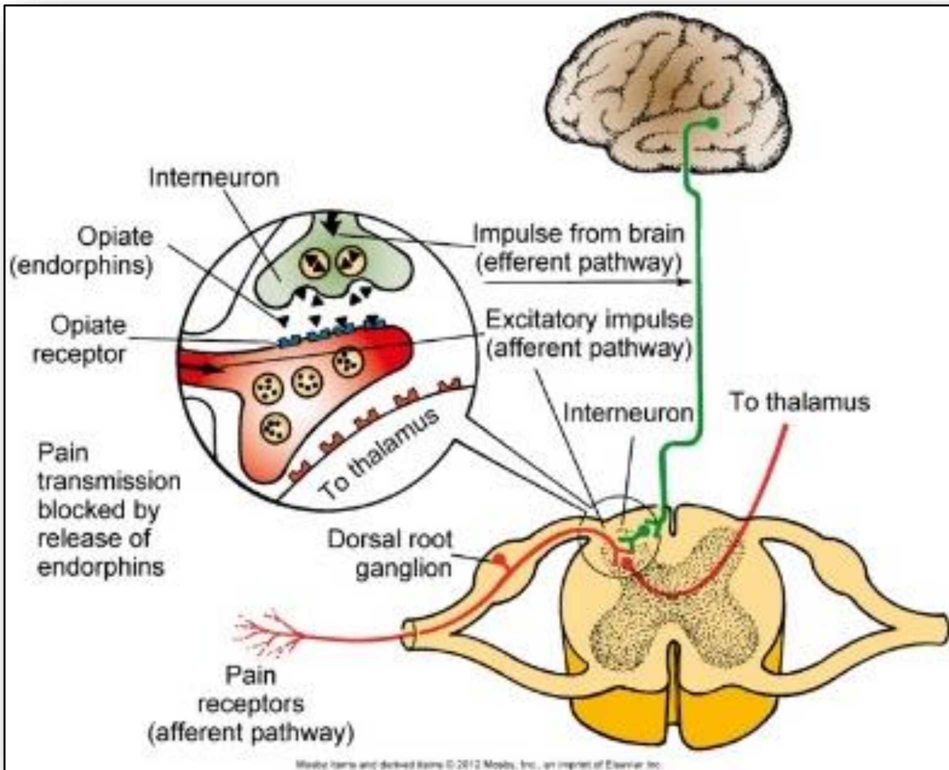


Figure 46: Referred pain locations

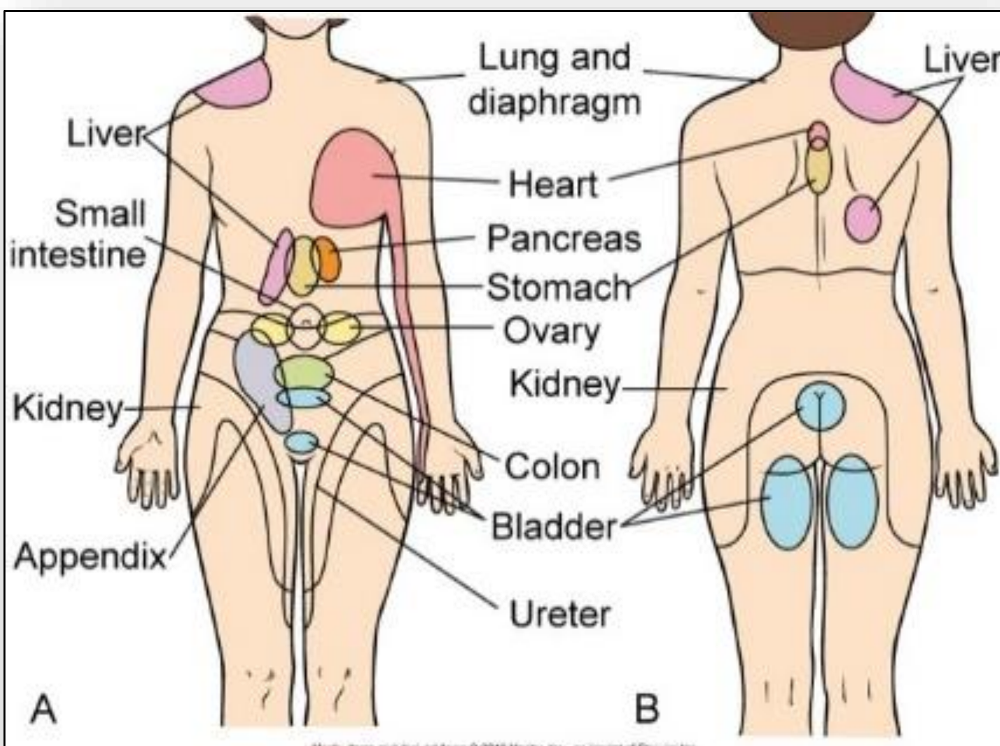


Figure 47: Homeostatic processes in temperature regulation

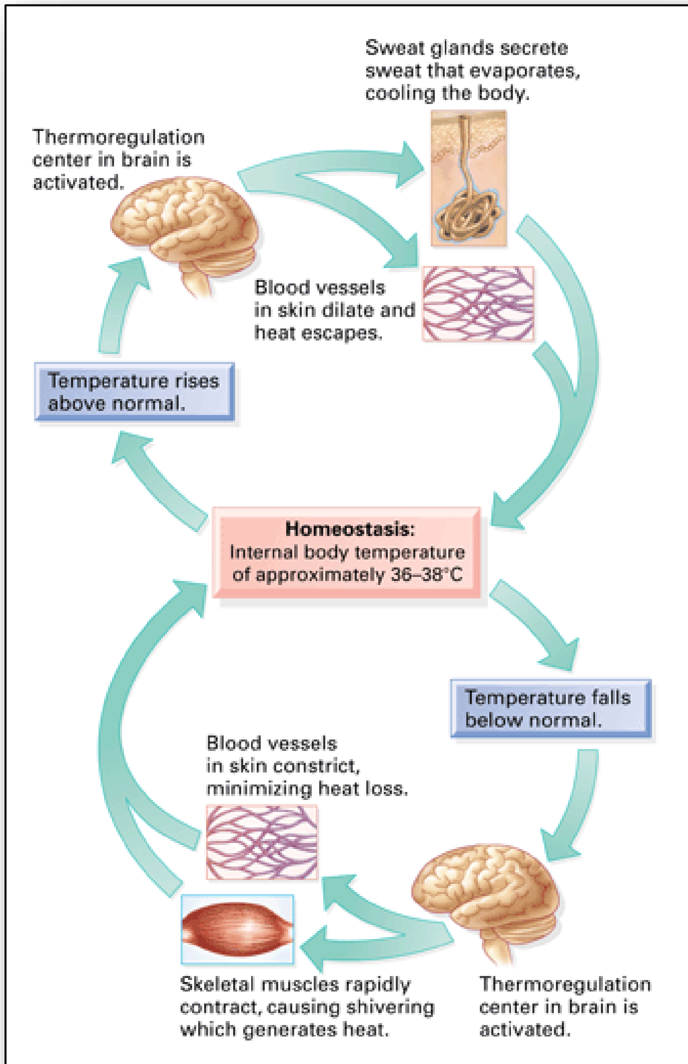
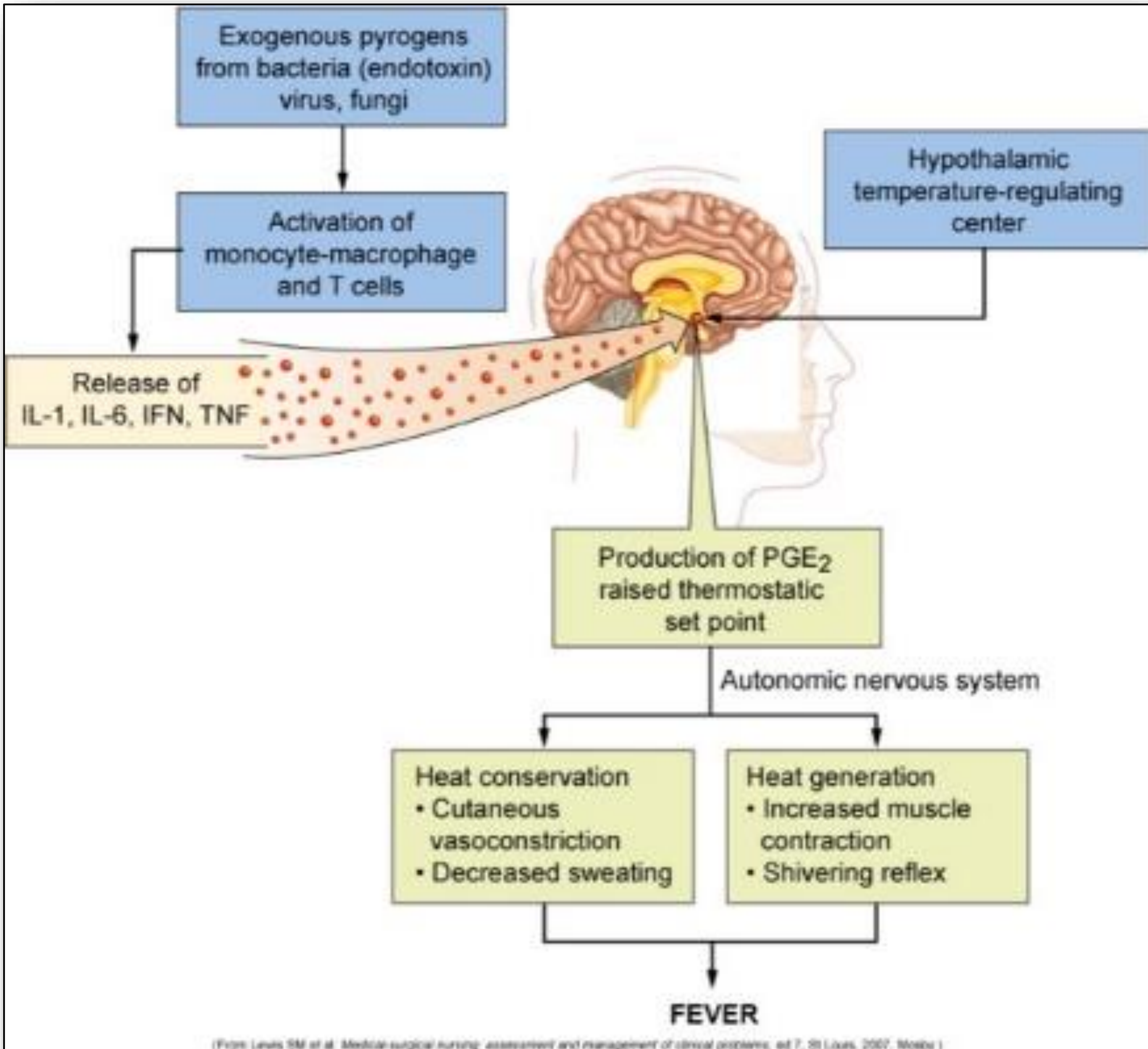


Figure 48: Mechanisms of heat loss and production

TABLE 13-5 MECHANISMS OF HEAT PRODUCTION AND HEAT LOSS

CONDITION	DESCRIPTION
Heat Production	
Chemical reactions of metabolism	Occur during ingestion and metabolism of food and while maintaining body at rest (basal metabolism); occur in body core (e.g., liver)
Skeletal muscle contraction	Gradual increase in muscle tone or rapid muscle oscillations (shivering)
Chemical thermogenesis	Epinephrine is released and produces rapid, transient increase in heat production by raising basal metabolic rate; quick, brief effect that counters heat lost through conduction and convection; involves brown adipose tissue, which decreases markedly in older adults; thyroid hormone increases metabolism
Heat Loss	
Radiation	Heat loss through electromagnetic waves emanating from surfaces with temperature higher than surrounding air
Conduction	Heat loss by direct molecule-to-molecule transfer from one surface to another, so that warmer surface loses heat to cooler surface
Convection	Transfer of heat through currents of gases or liquids; exchanges warmer air at body's surface with cooler air in surrounding space
Vasodilation	Diverts core-warmed blood to surface of body, with heat transferred by conduction to skin surface and from there to surrounding environment; occurs in response to autonomic stimulation under control of hypothalamus
Evaporation	Body water evaporates from surface of skin and linings of mucous membranes; major source of heat reduction connected with increased sweating in warmer surroundings
Decreased muscle tone	Exhausted feeling caused by moderately reduced muscle tone and curtailed voluntary muscle activity
Increased respiration	Air is exchanged with environment through normal process; minimal effect
Voluntary mechanisms	"Stretching out" and "slowing down" in response to high body temperatures; increasing body surface area available for heat loss; dressing in light-colored, loose-fitting garments
Adaptation to warmer climates	Gradual process beginning with lassitude, weakness, and faintness; proceeding through increased sweating, lowered sodium content, decreased heart rate, and increased stroke volume and extracellular fluid volume; and terminating in improved warm weather functioning and decreased symptoms of heat intolerance (work output, endurance, and coordination increase; subjective feelings of discomfort decrease)

Figure 49: Pathophysiology of fever



9: Alterations of Pulmonary Functions

Figure 1: Anatomy of the upper and lower respiratory tracts

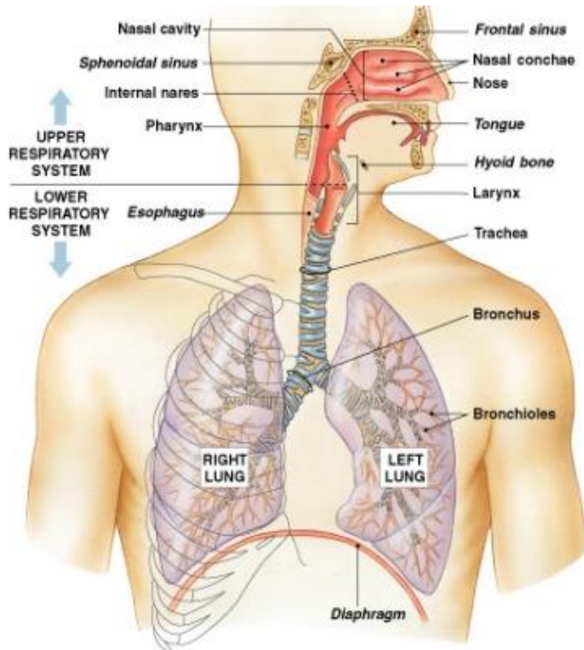


Figure 2: Anatomy of the upper respiratory tract

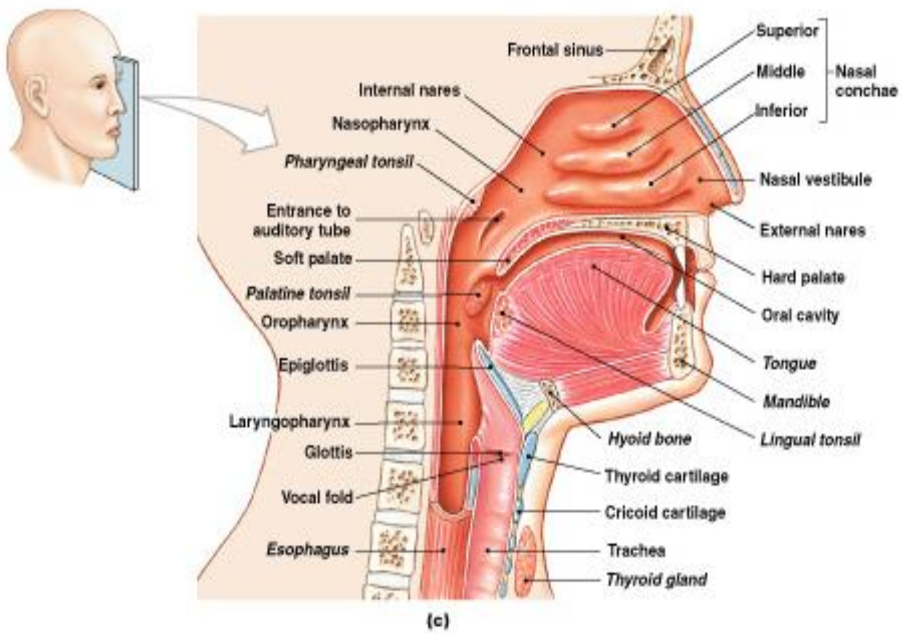


Figure 3: Conducting components of the lower respiratory tract

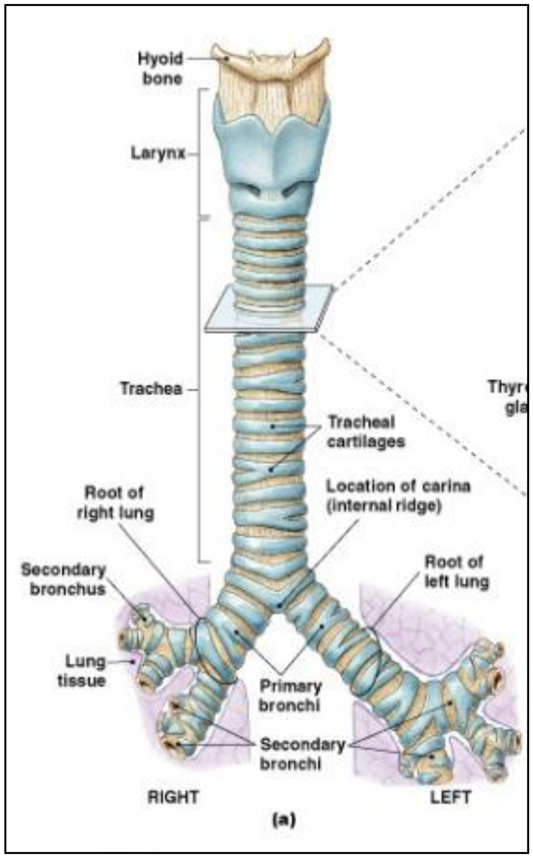


Figure 4: Cross section of alveoli of the lower respiratory tract

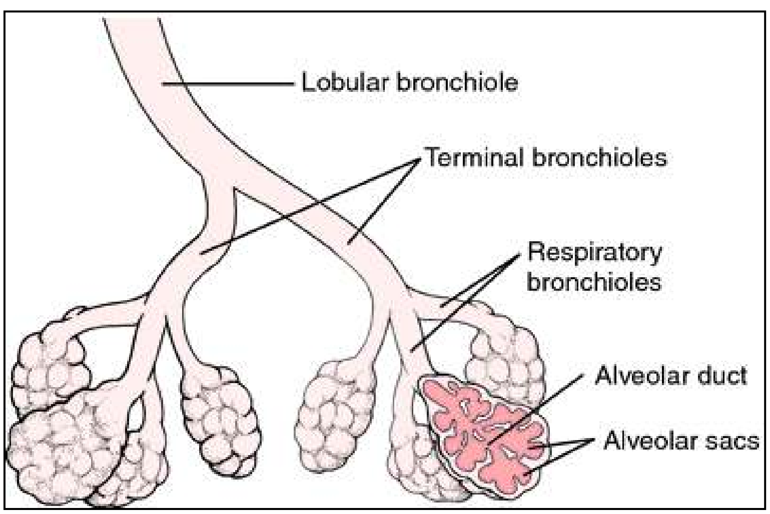


Figure 5: Sites and surfaces of gas exchange

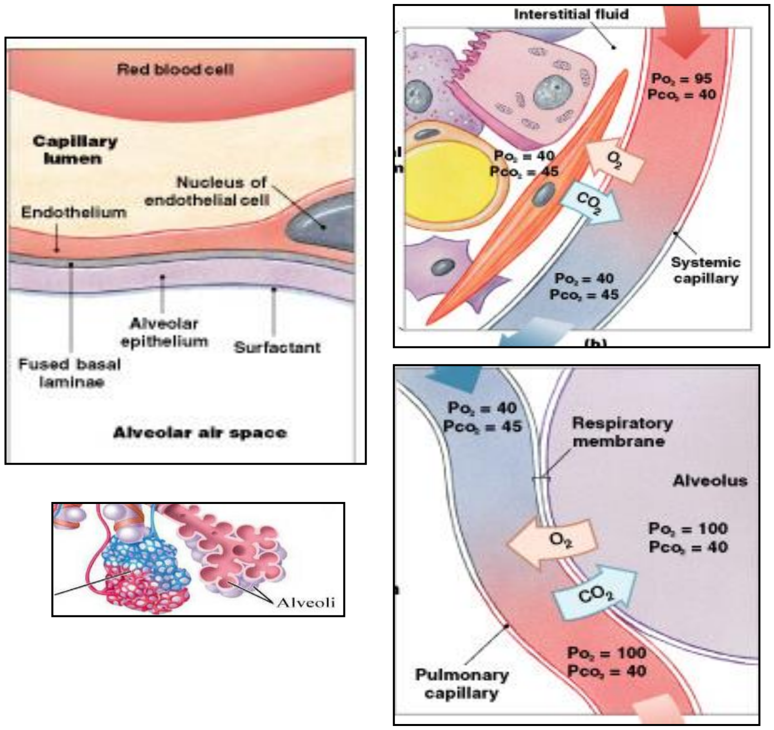


Figure 6: Pulmonary and systemic circuits

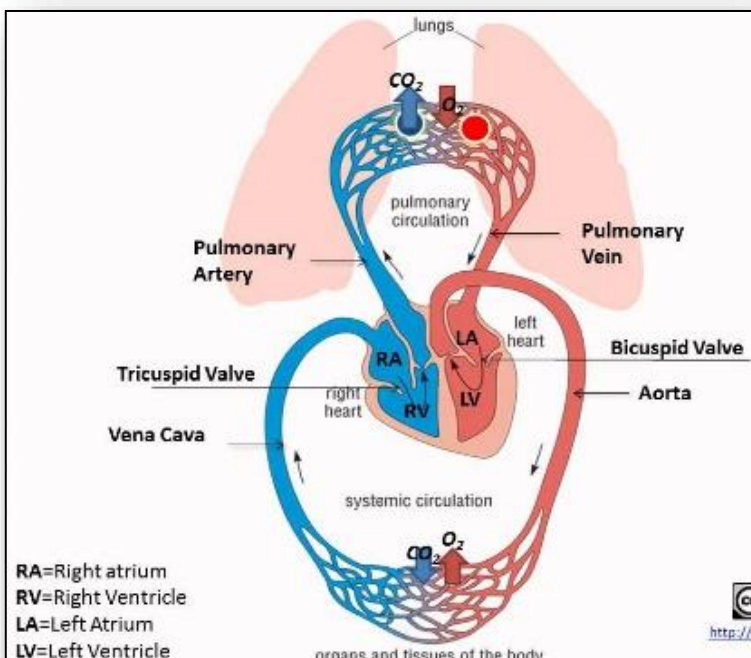


Figure 7: Spirometry measurement of respiratory volumes and capacities

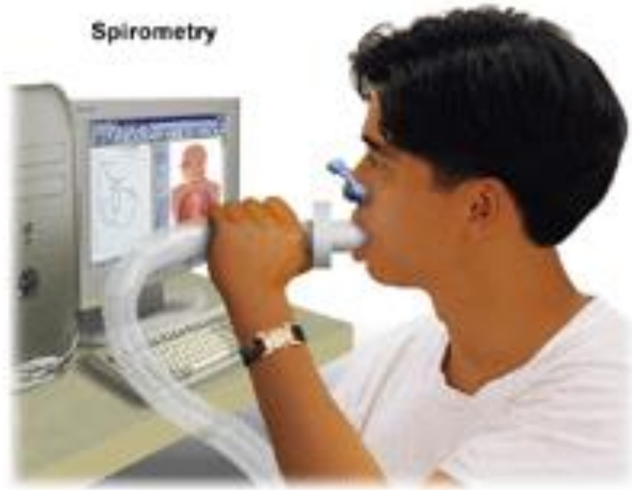


Figure 8: Respiratory volumes (#1-4) and capacities (#5-8)

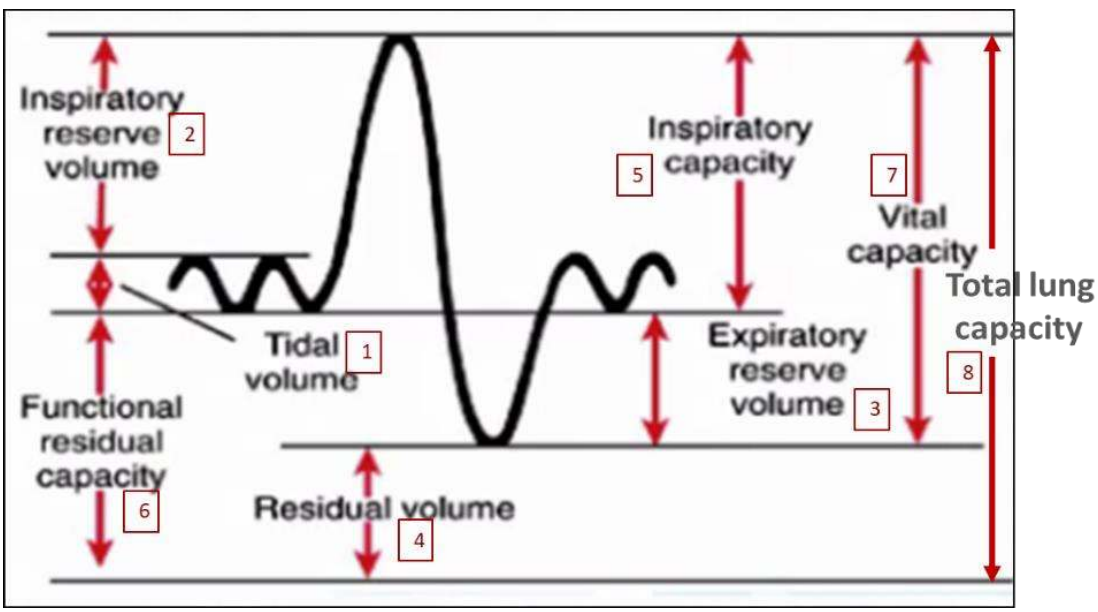


Figure 9: Respiratory rates and patterns

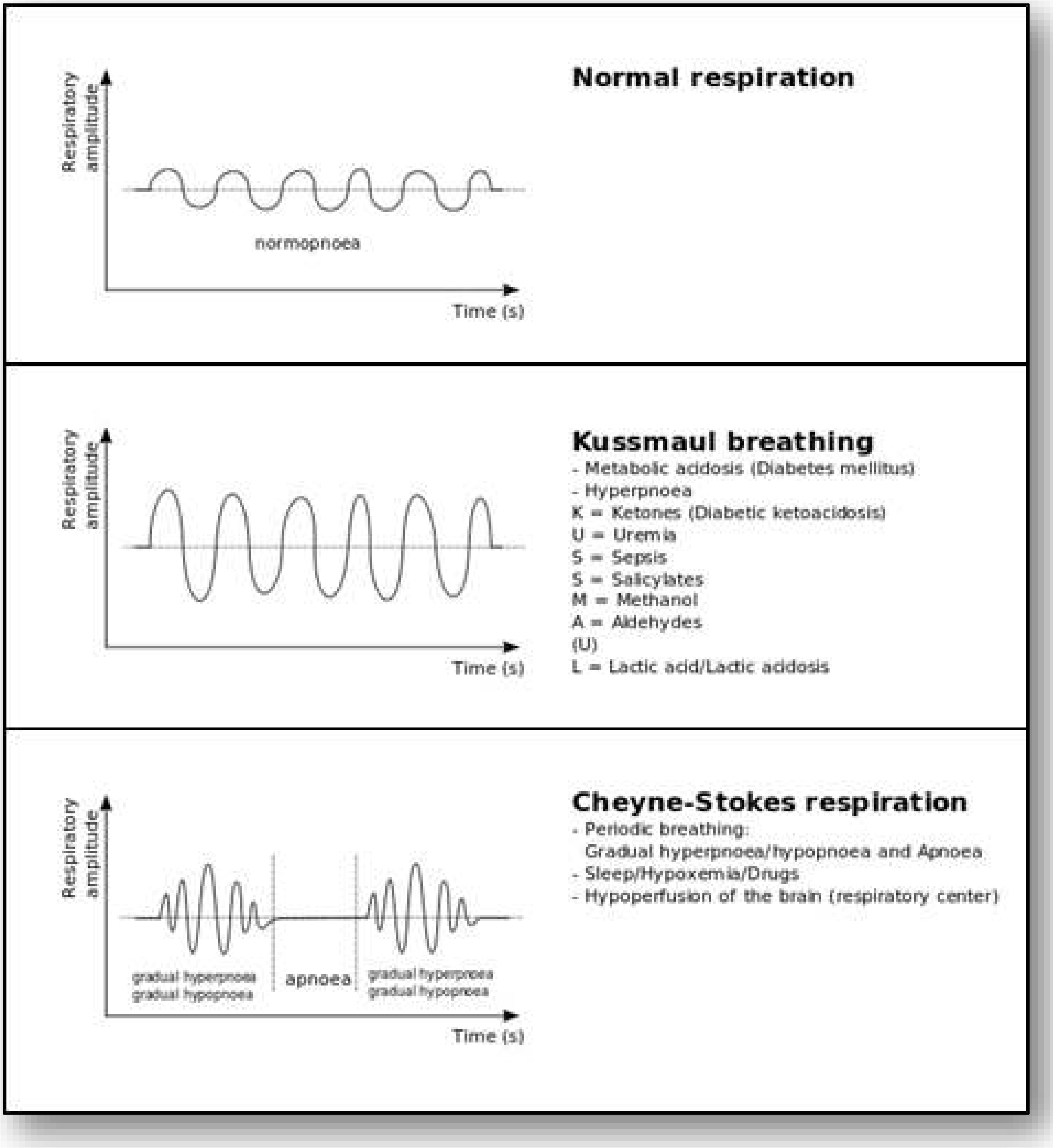


Figure 10: Paradoxical breathing

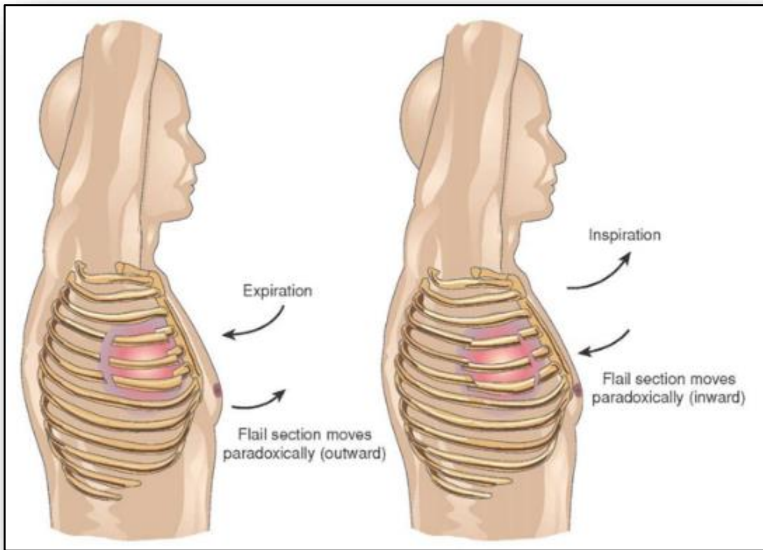


Figure 11: Pneumothorax

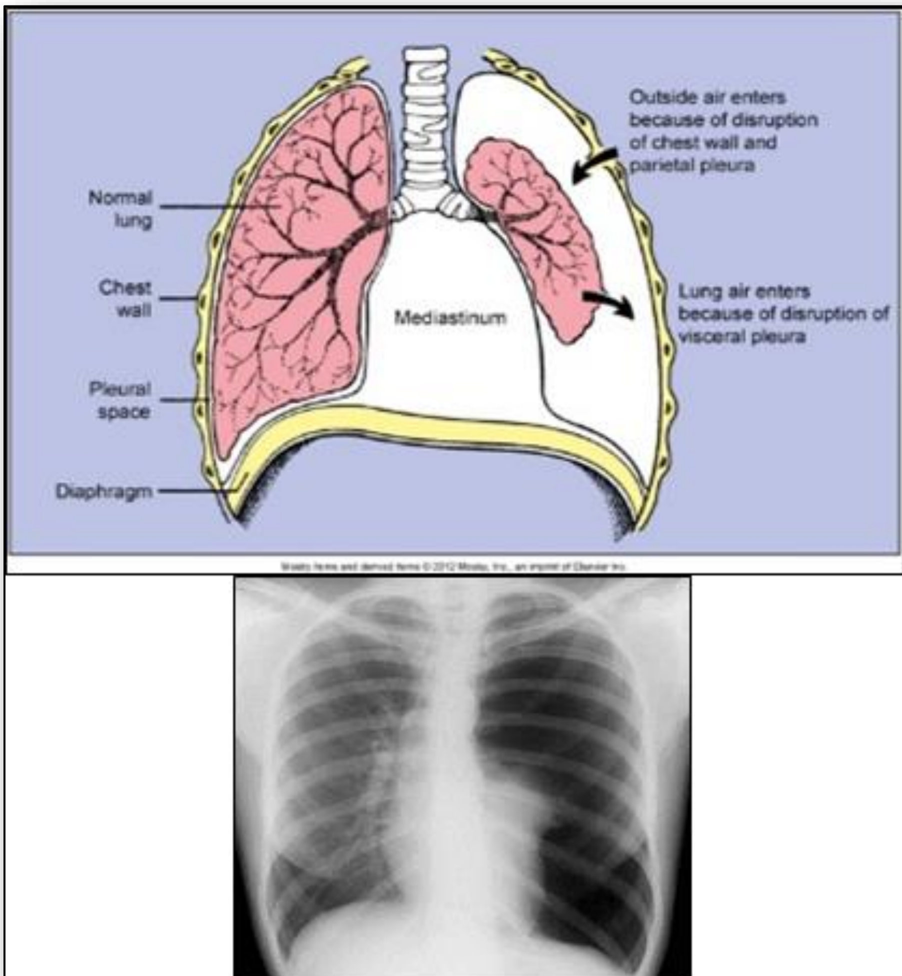


Figure 12: Tension pneumothorax with mediastinal deviation

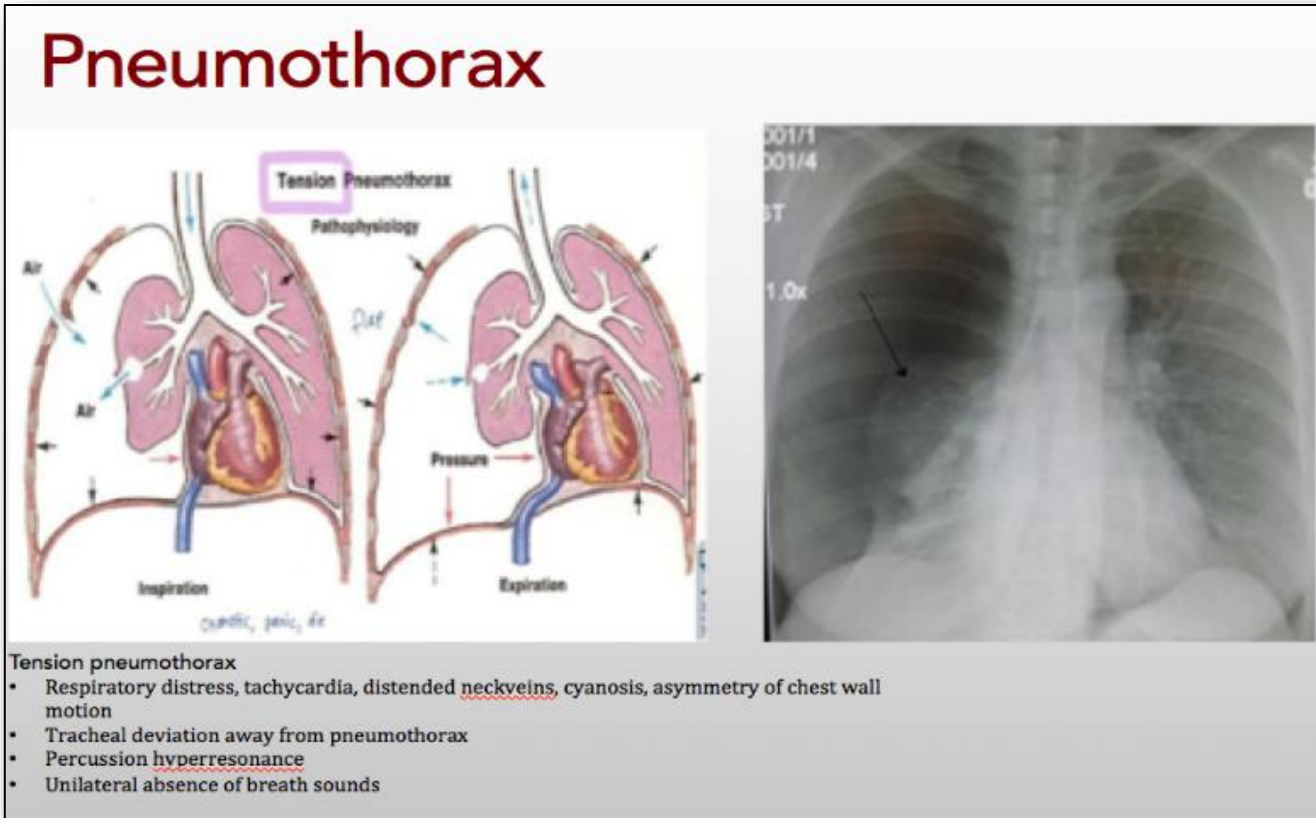


Figure 13: Subpleural bleb

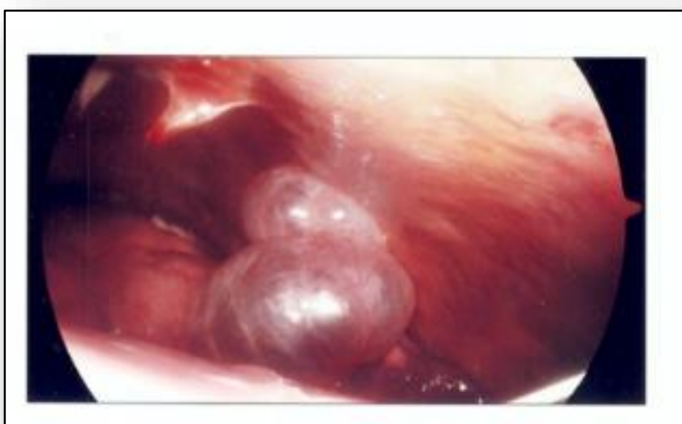


Figure 14: Pleural effusion

Figure 15: Atelectasis

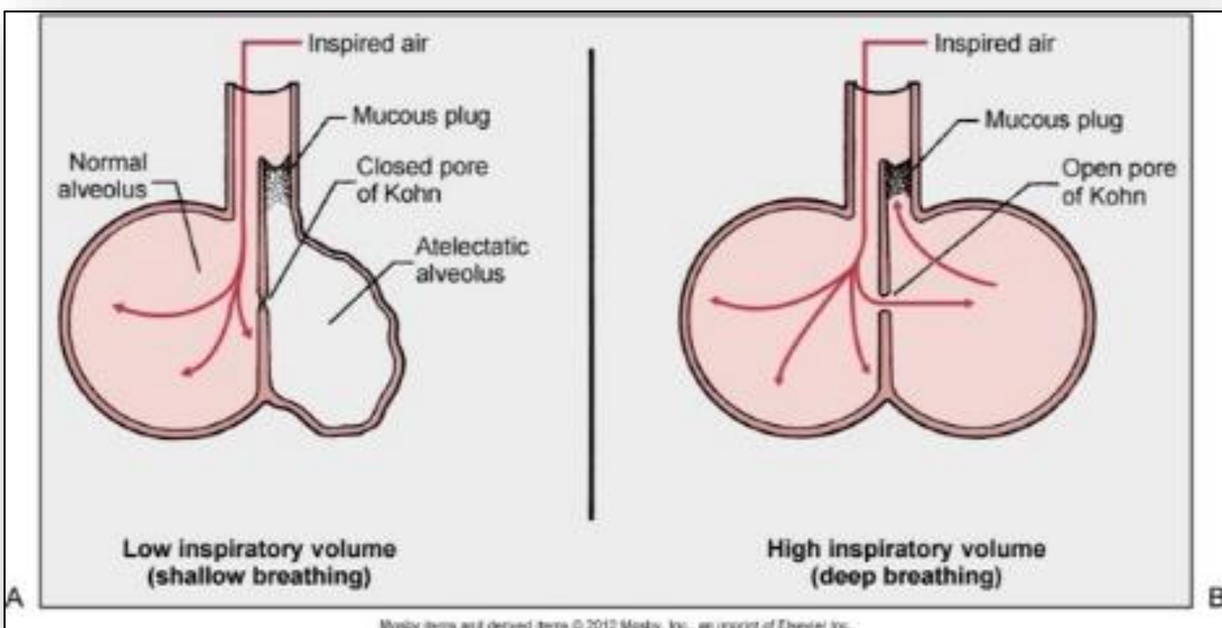


Figure 16: Atelectasis on an x-ray

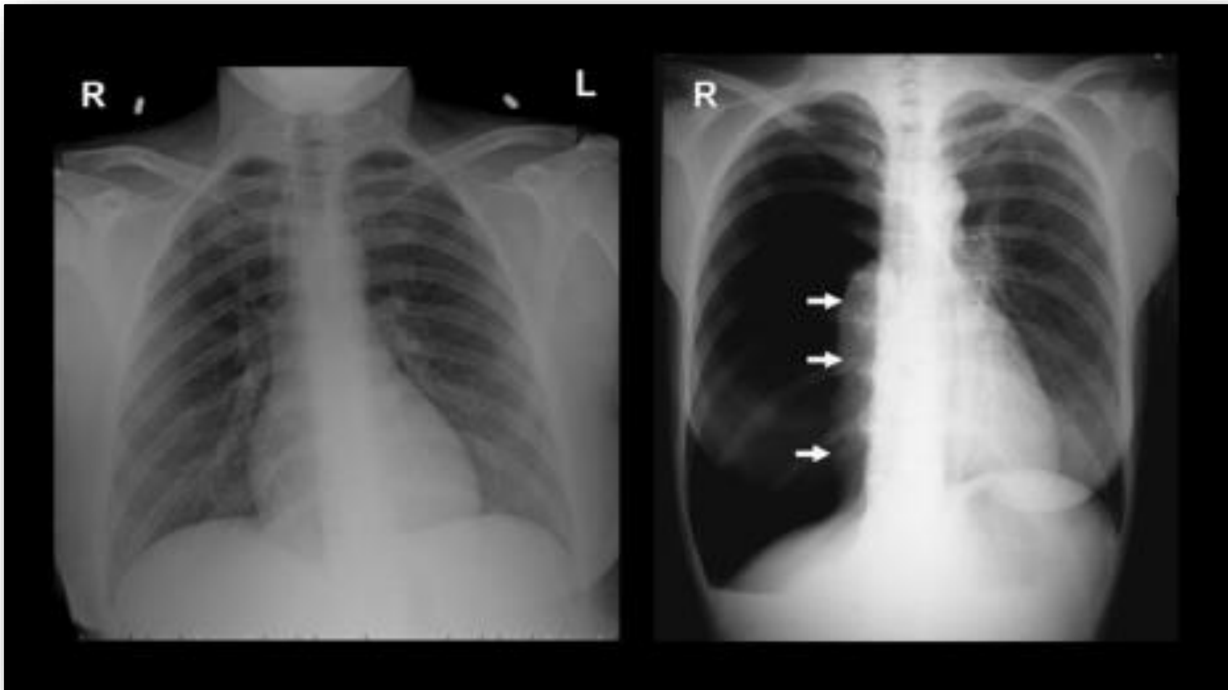


Figure 17: Replacement of normal lung tissue with fibrous tissue

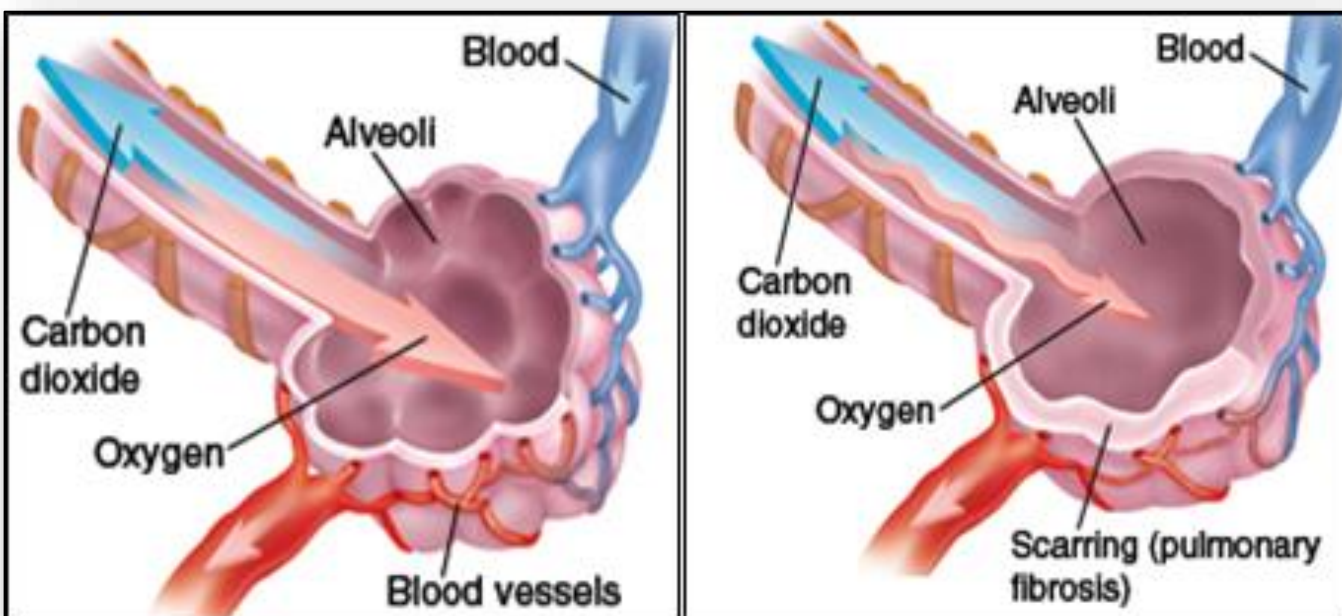


Figure 18: Pathophysiology of Pulmonary Edema

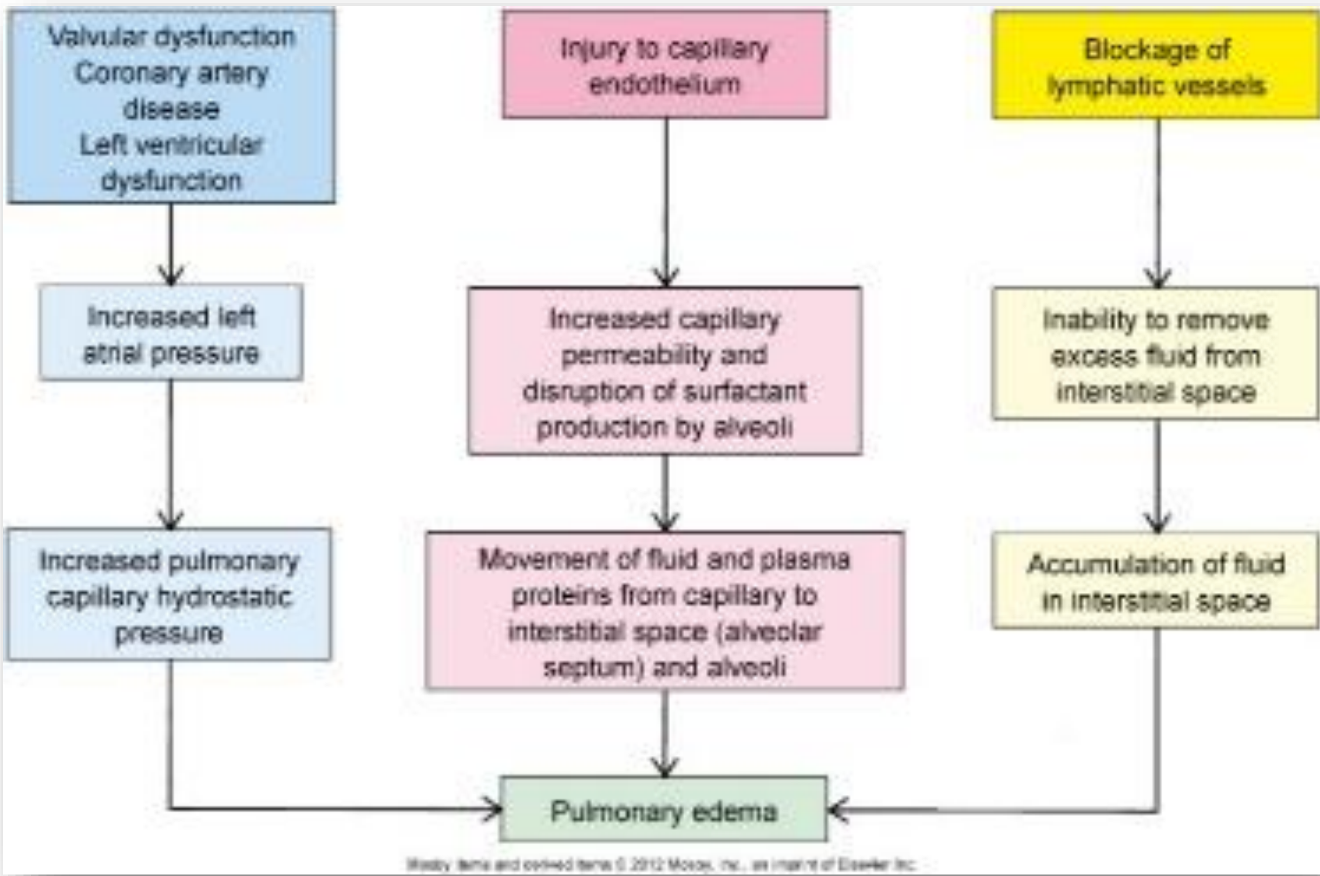


Figure 19: Pathophysiology of ARDS

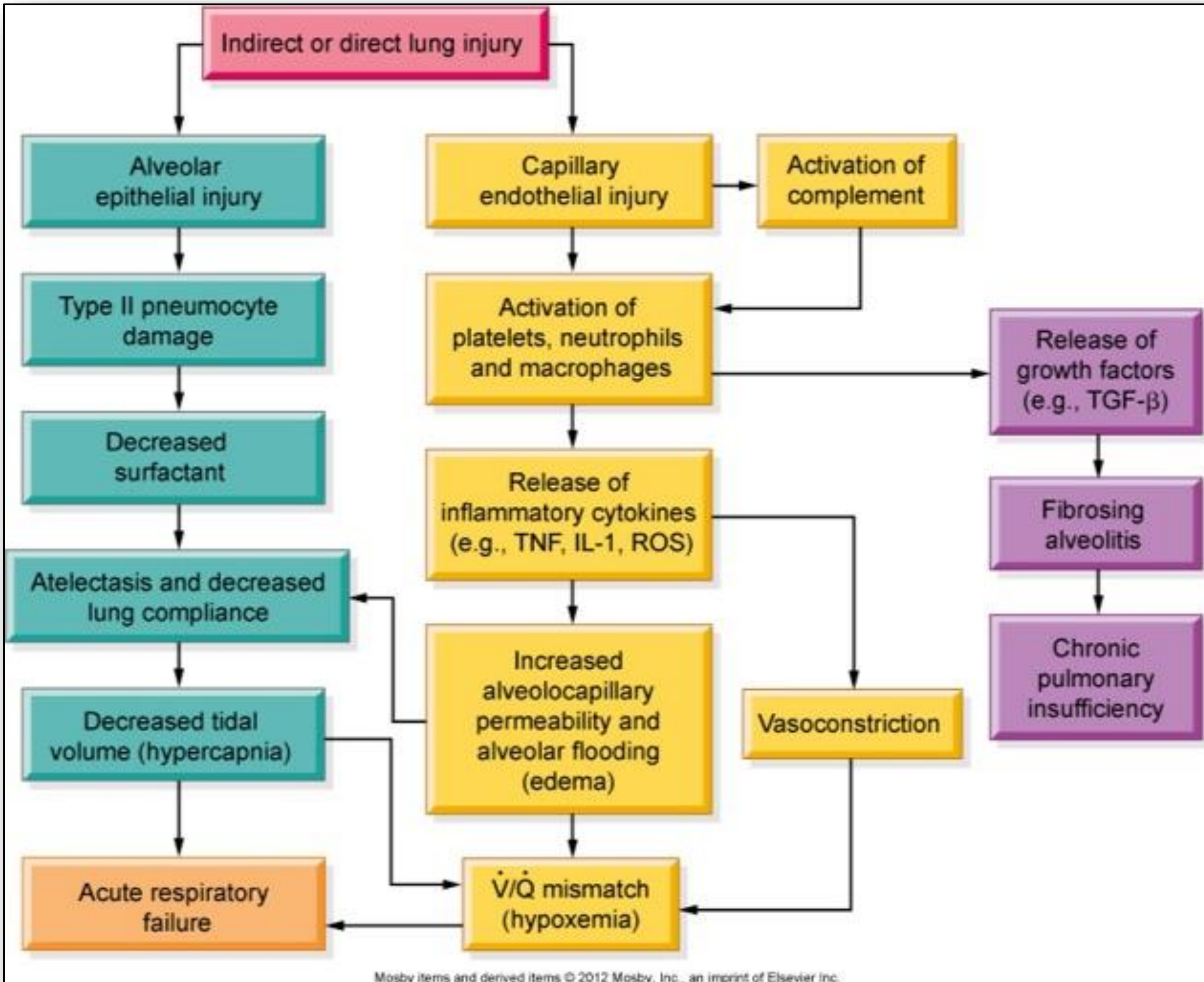


Figure 20: Pathophysiology of bronchiectasis compared to normal bronchus

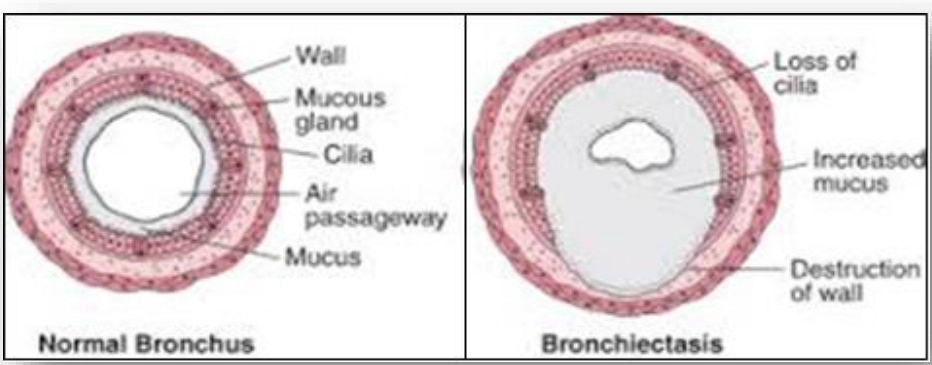


Figure 21(a): Pathophysiology of obstructive pulmonary disease

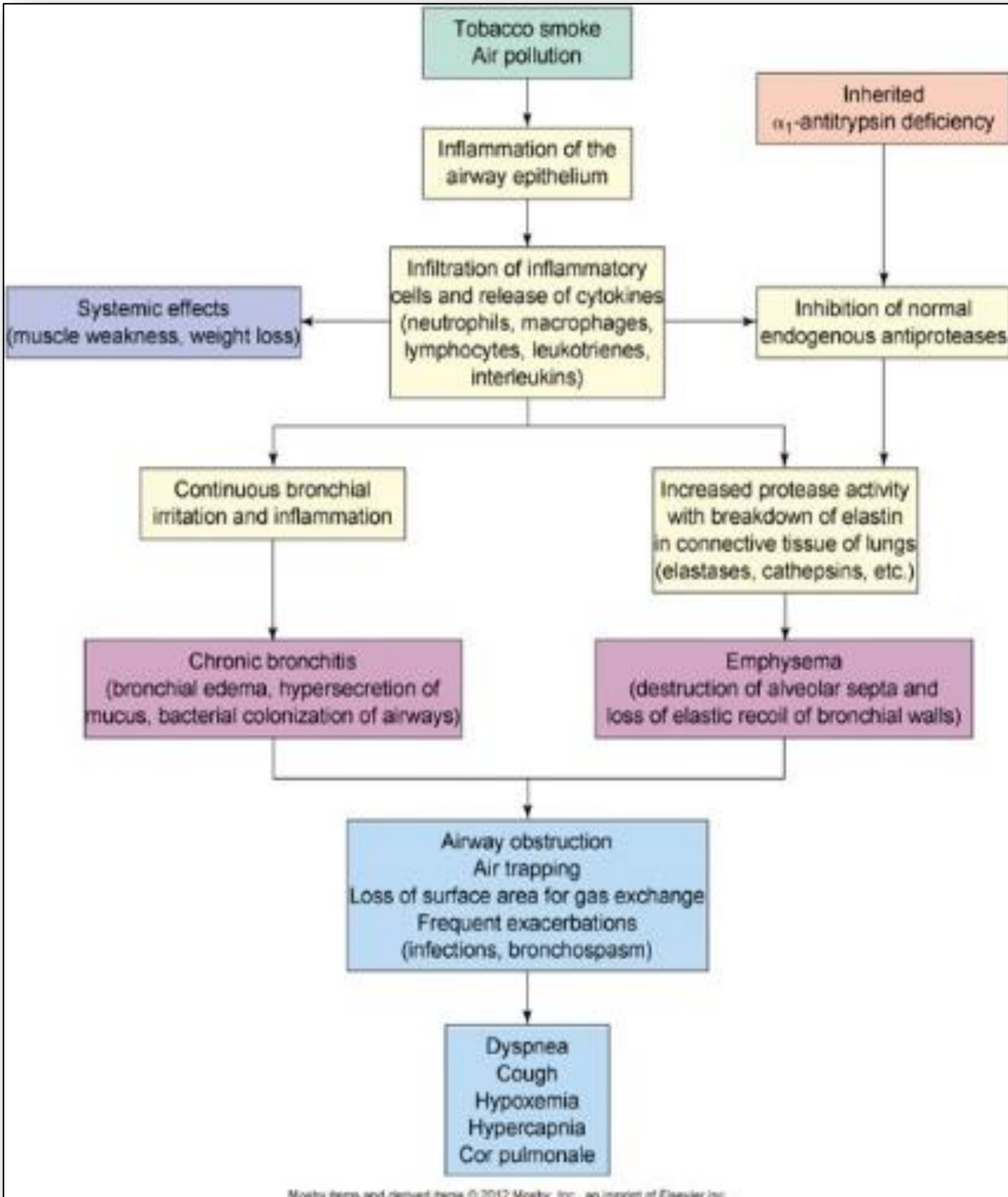


Figure 21(b): Pathophysiology of obstructive pulmonary disease

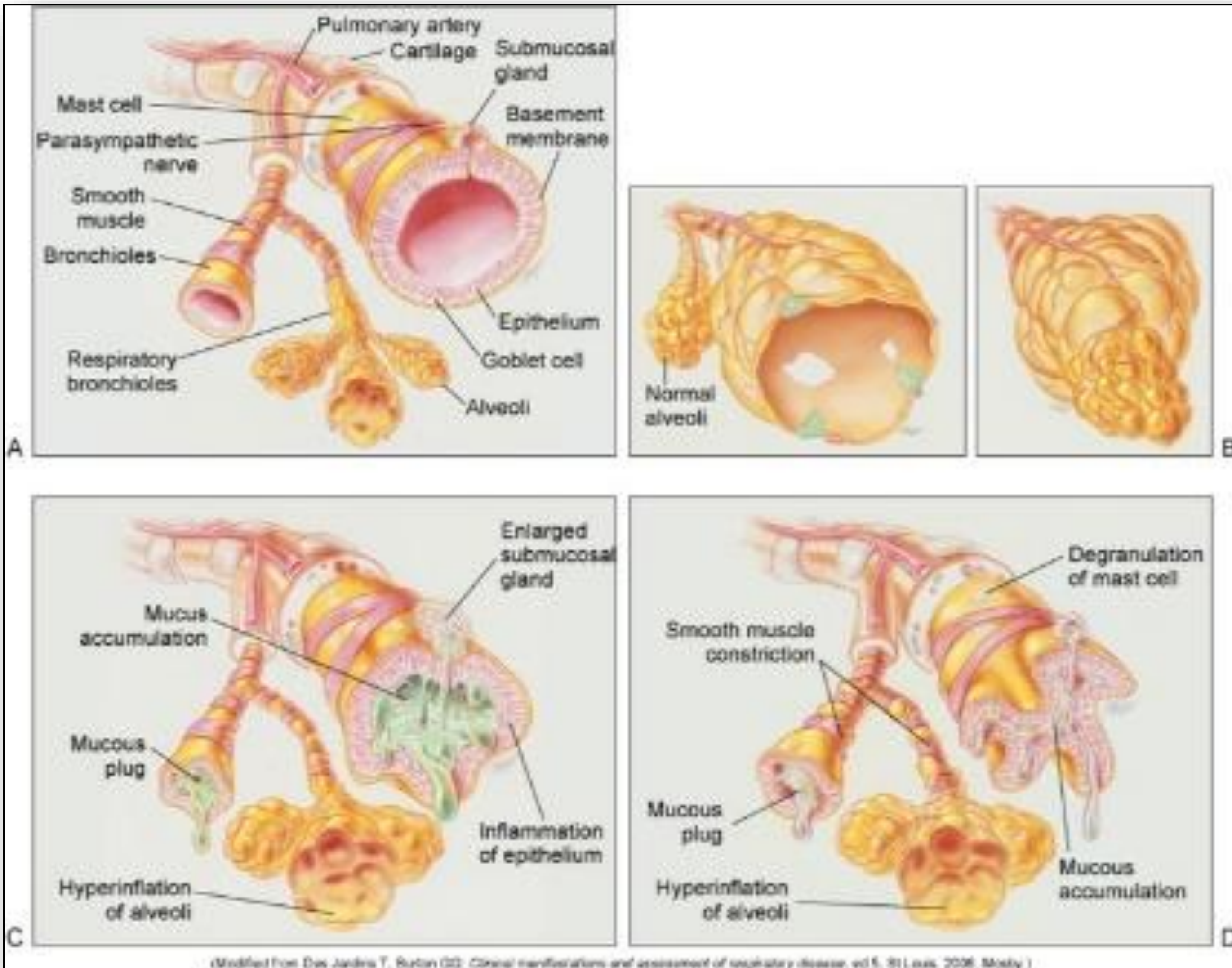


Figure 22: Damaged alveolus of pulmonary disease

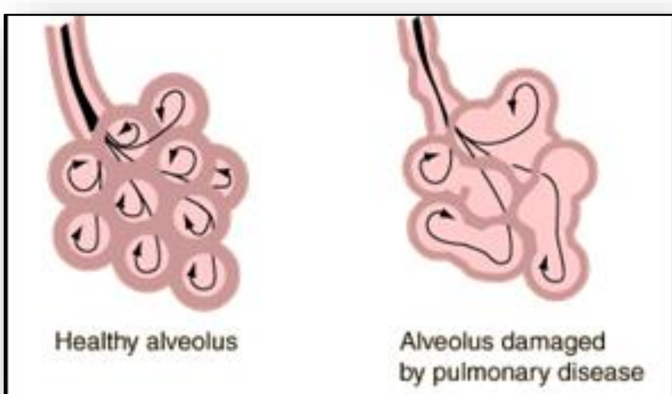


Figure 23: Virchow’s triad

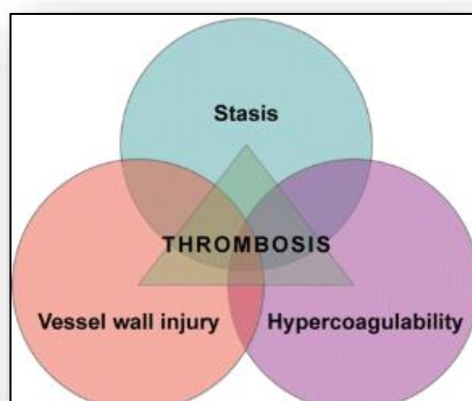


Figure 24: Pathophysiology of pulmonary embolism

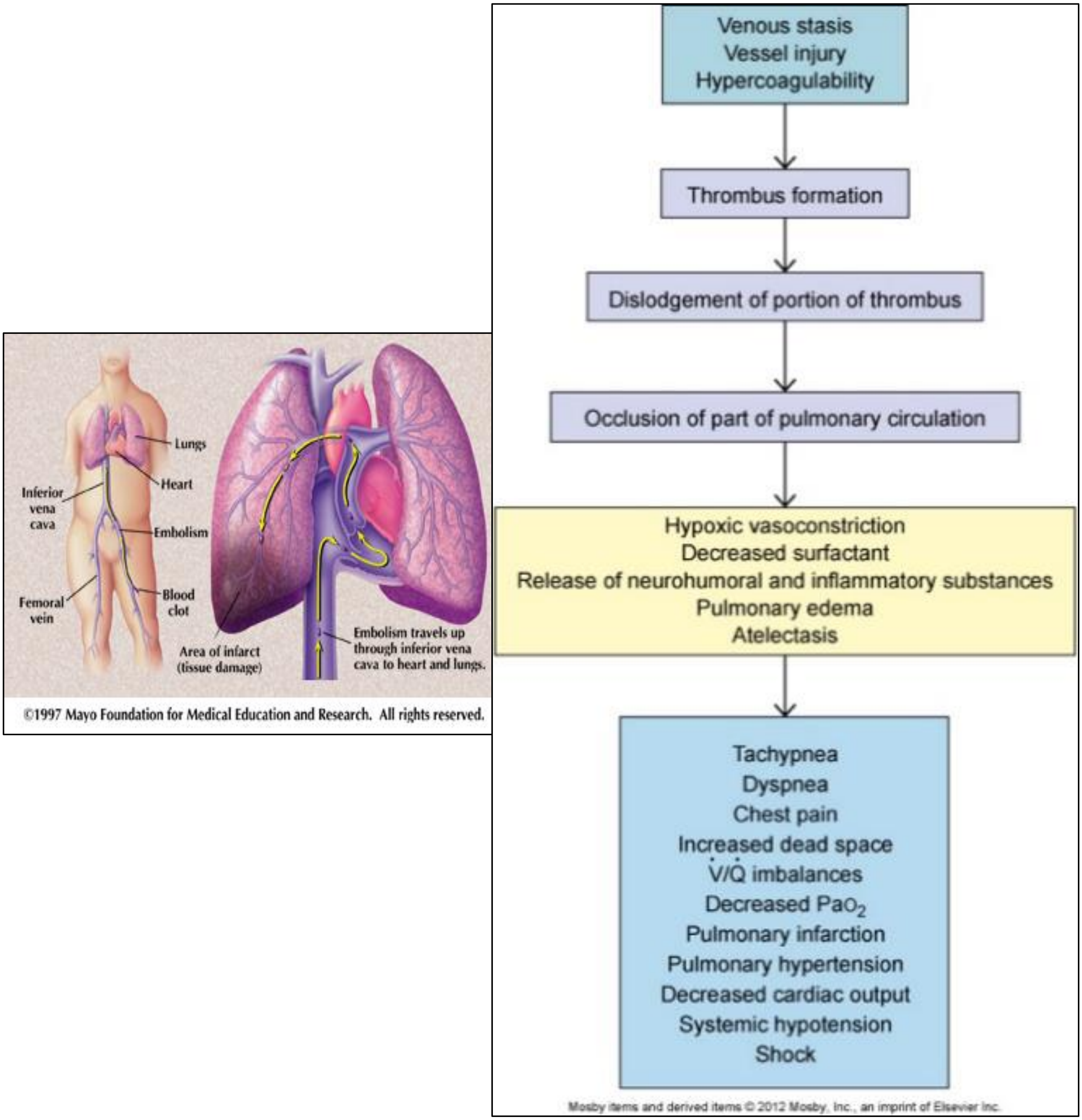


Figure 25: Pathophysiology of pulmonary hypertension

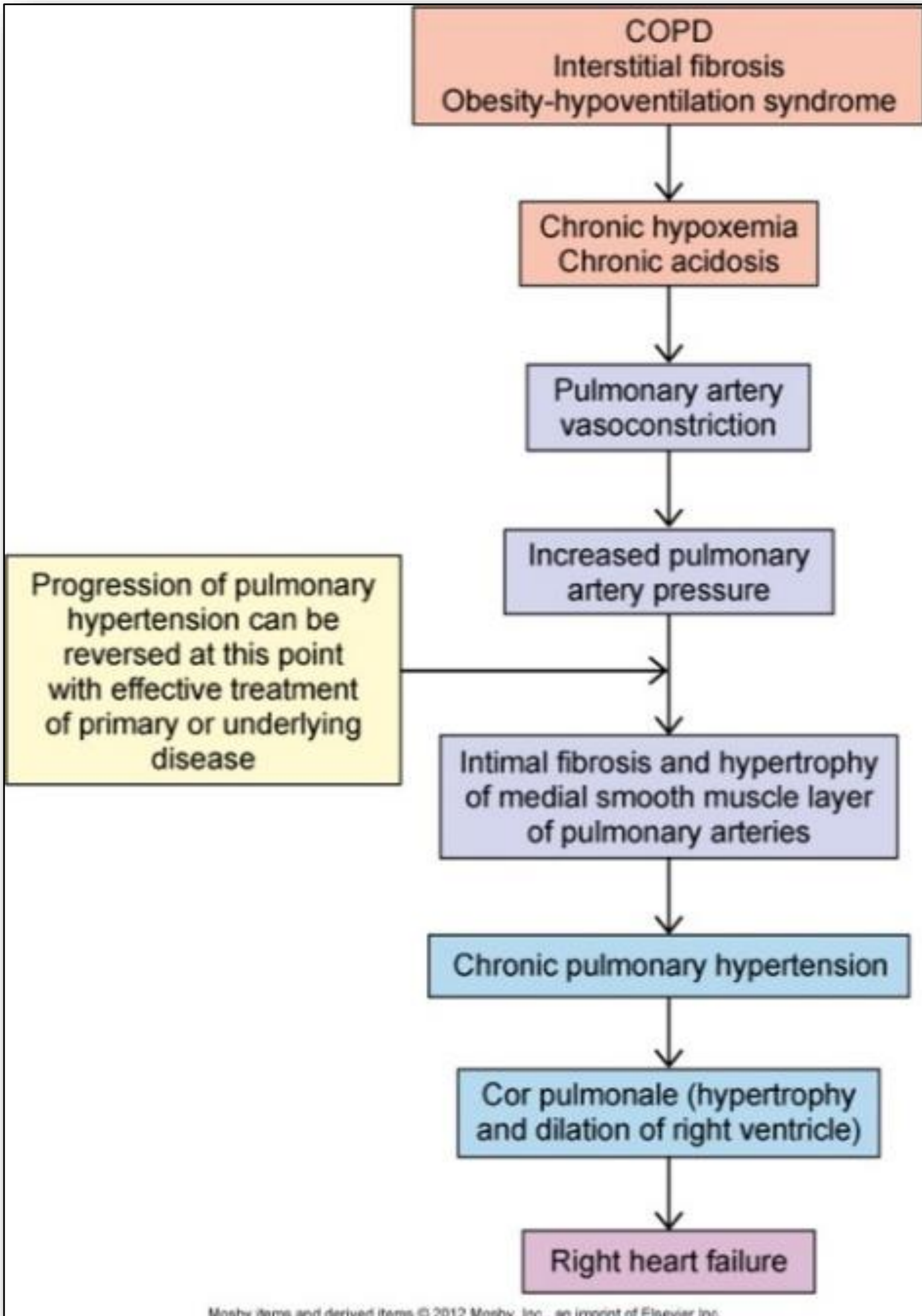


Figure 26: Pathophysiology of Croup

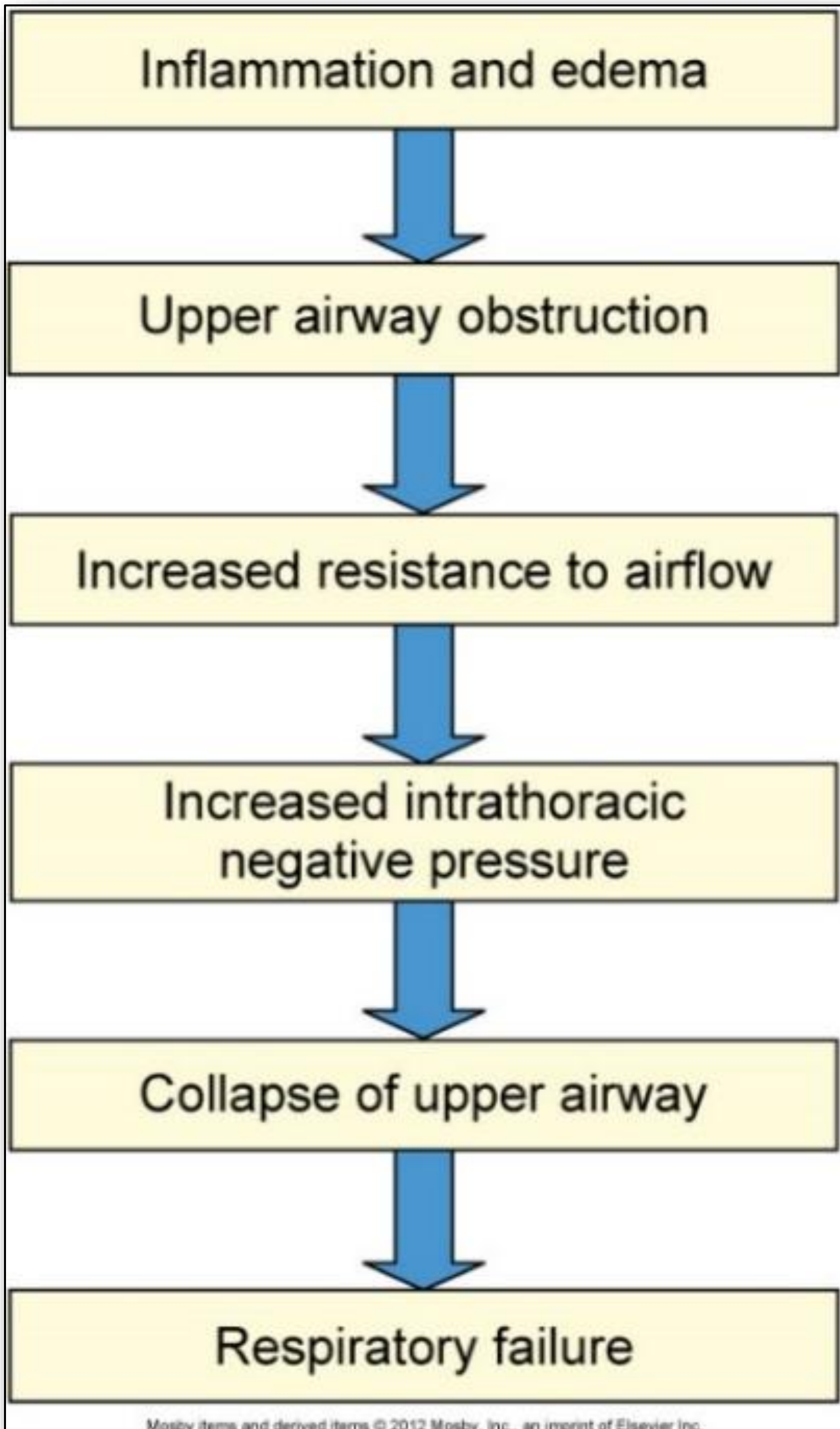
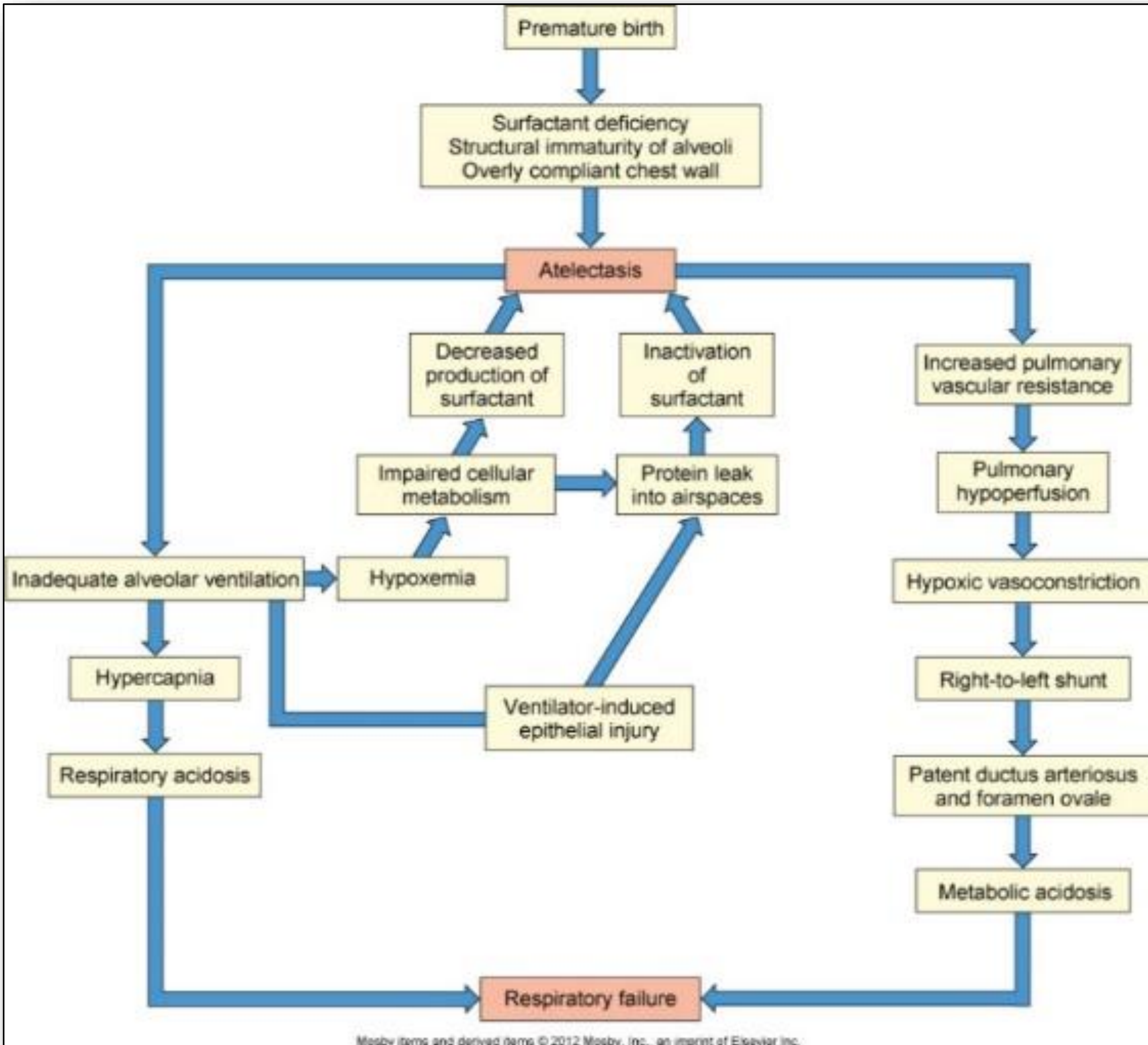


Figure 27: Pathophysiology of RDS of the newborn



10: Alterations in Renal and Urologic Systems / Water, Acid-Base and Electrolyte Imbalance

Figure 1: Urinary tract (anterior view)

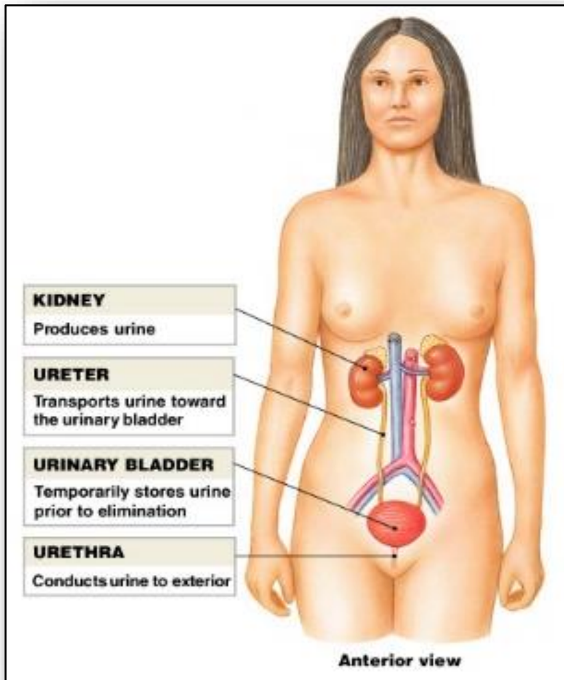


Figure 2: Urinary tract (posterior view)

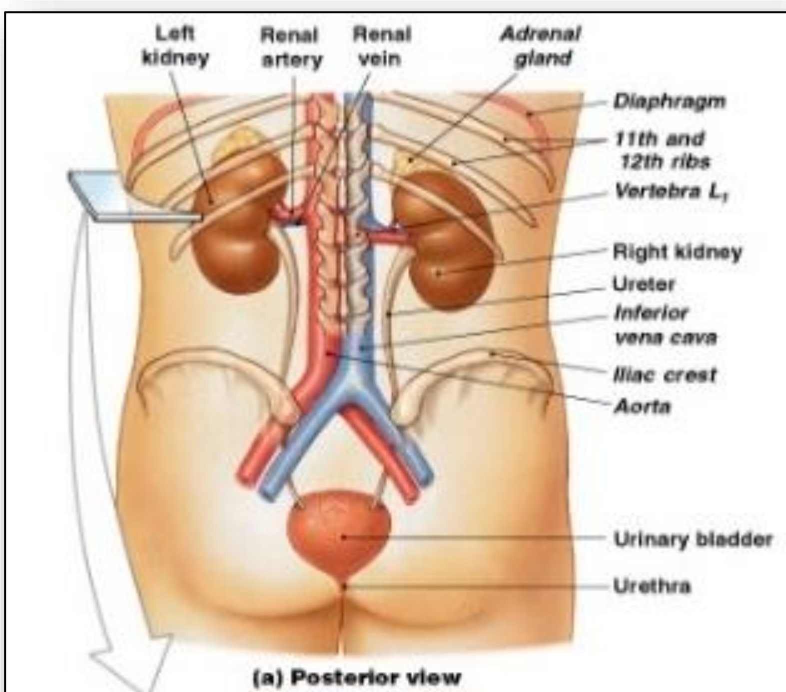


Figure 3: Urinary tract (superior view)

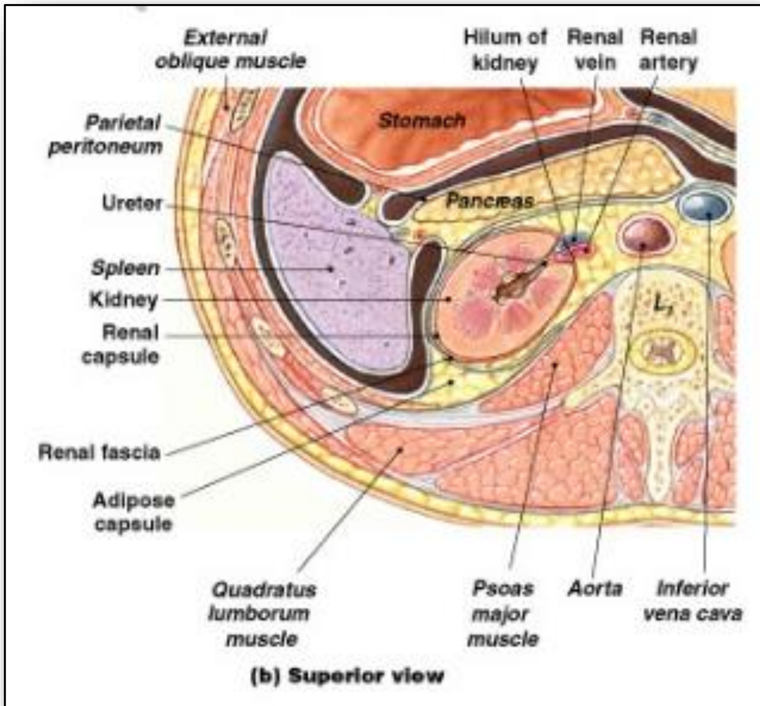


Figure 4: Gross and microscopic anatomy of renal pyramid

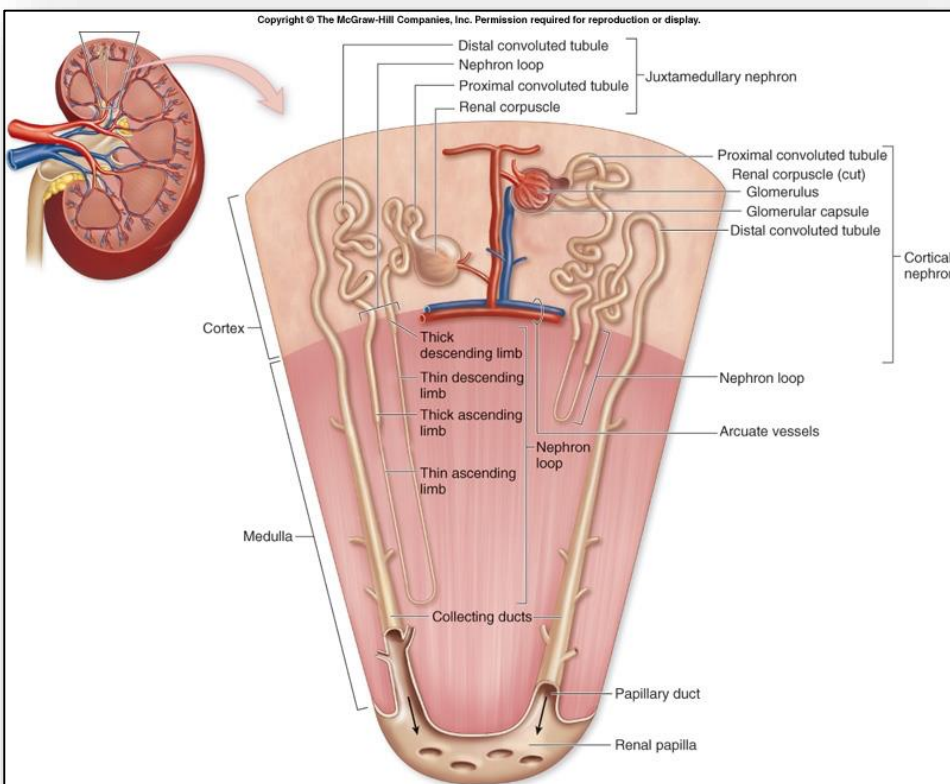


Figure 5: Structure and function of a nephron

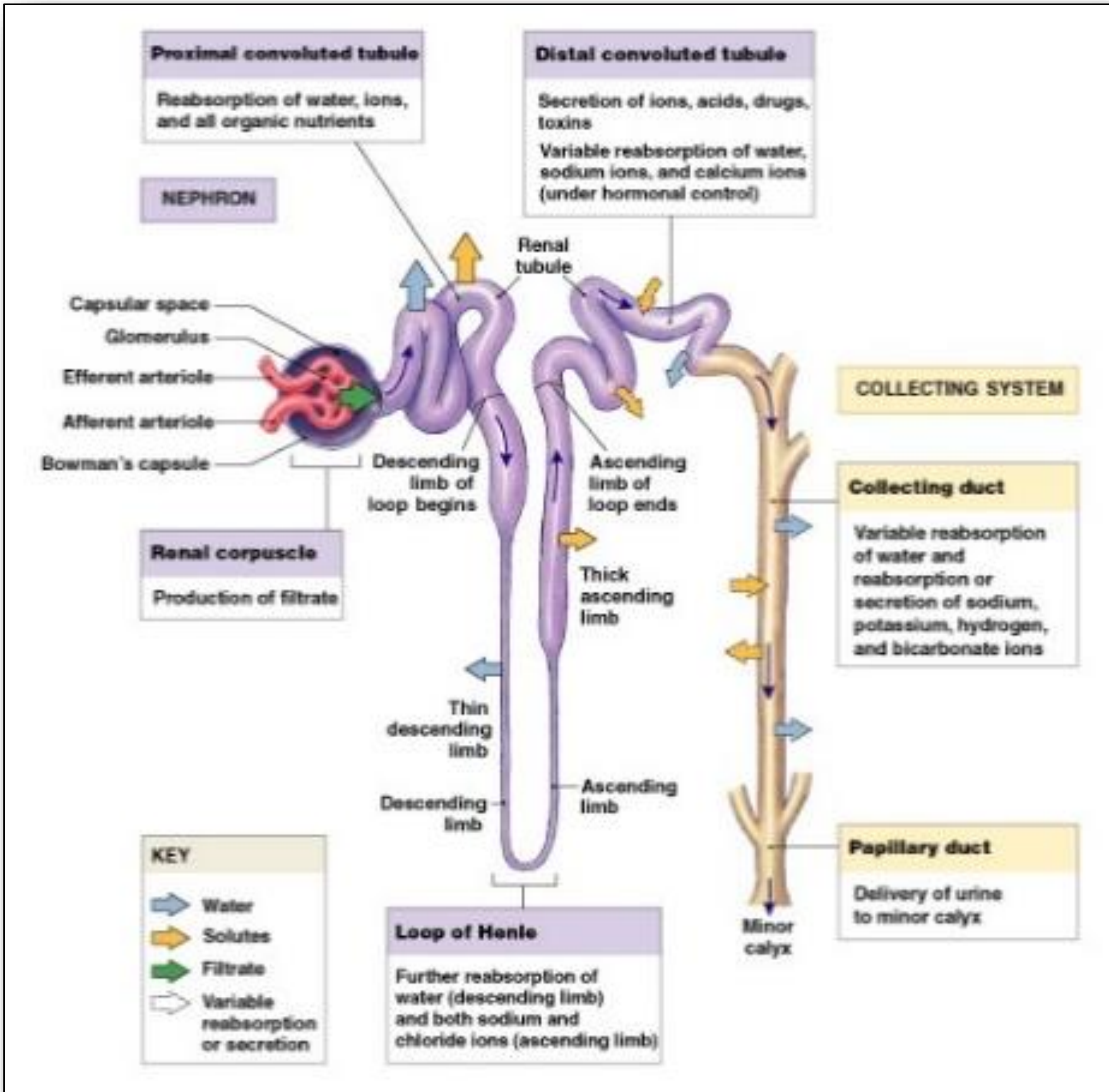


Figure 6: Sectional anatomy of the kidney

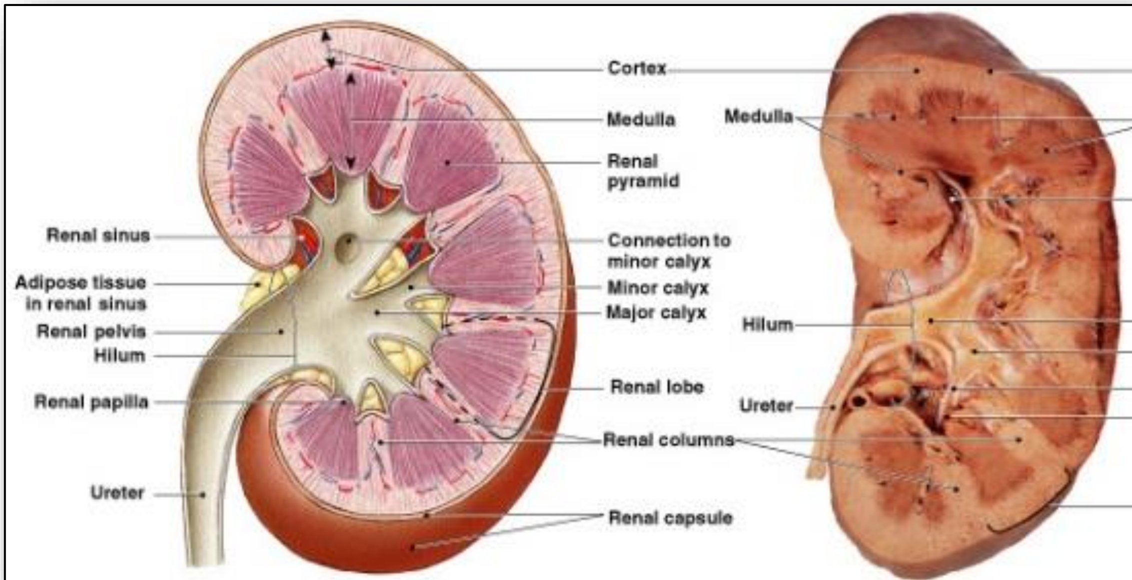


Figure 7: Blood supply of the kidney

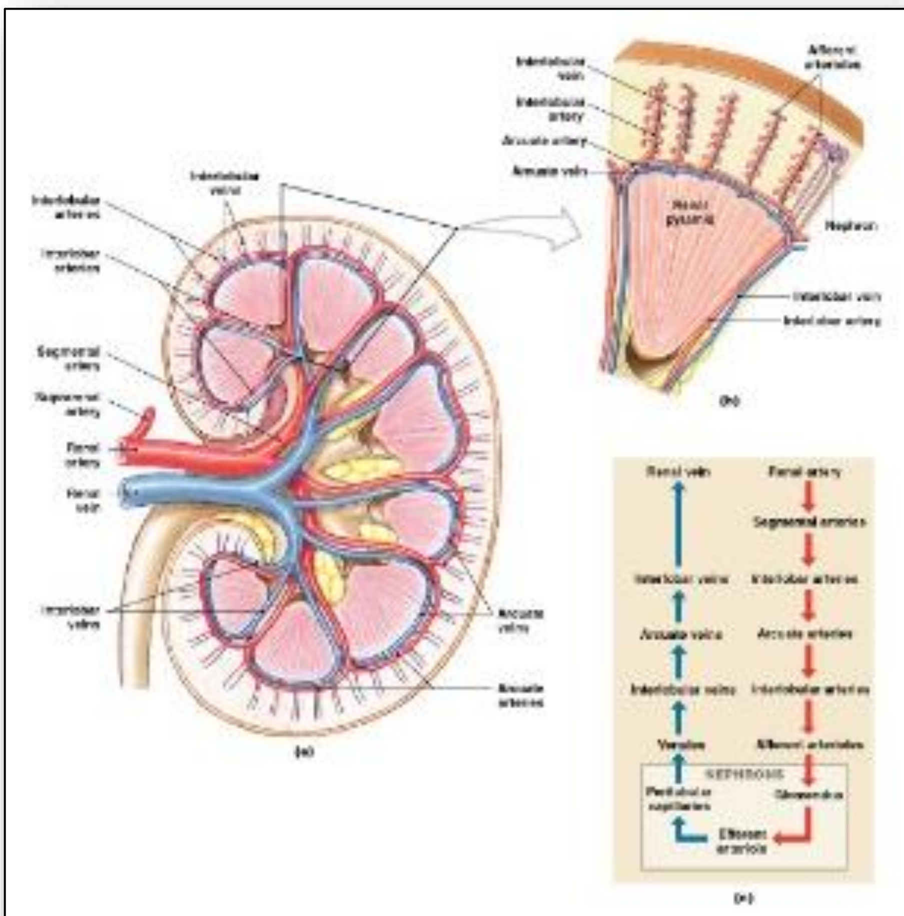


Figure 8: Renin-angiotensin-aldosterone-system (RAAS)

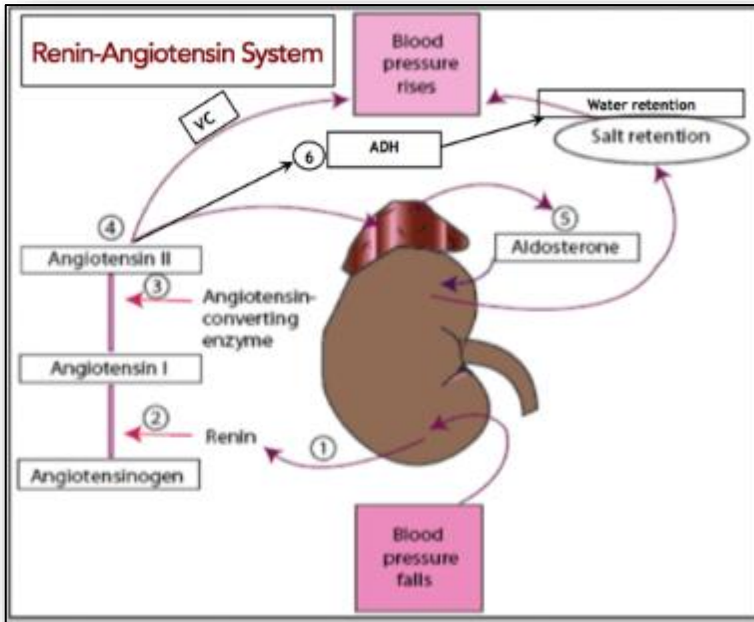


Figure 9: Formation of urine in the nephron

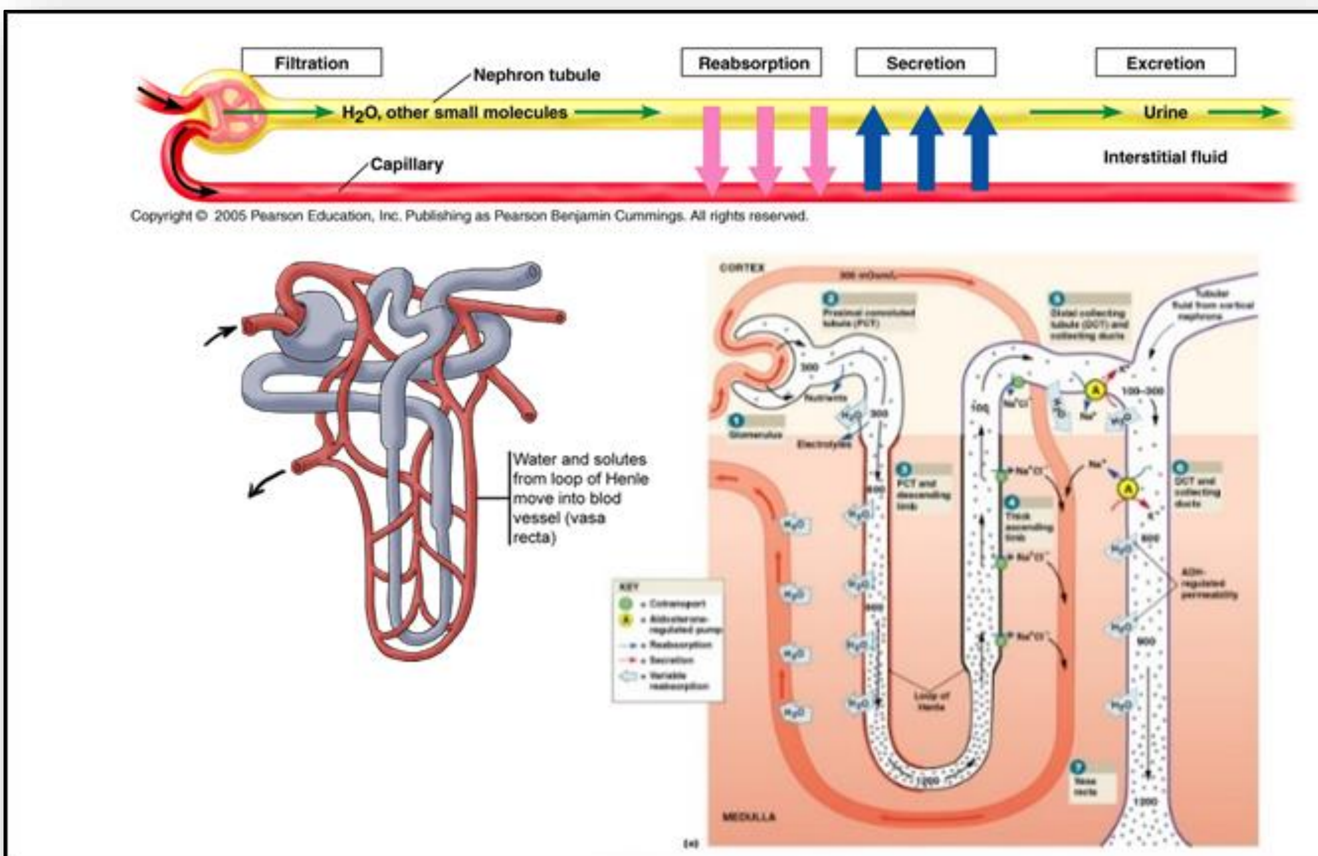


Figure 10: Anatomy of the urologic system

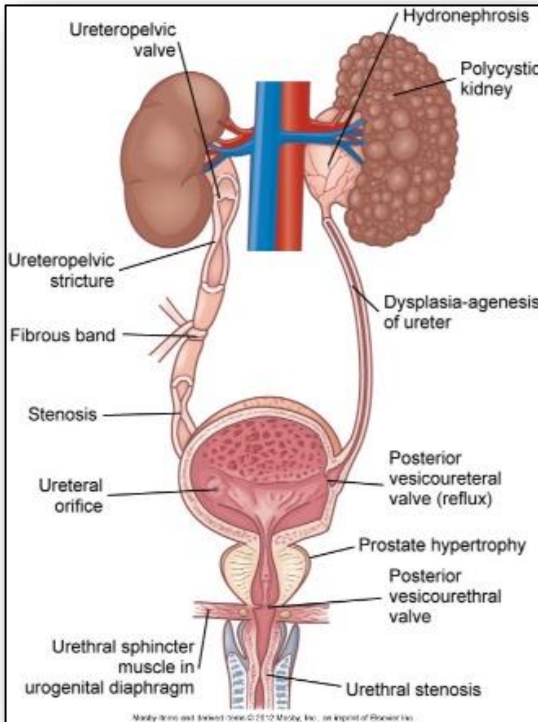


Figure 11: Process of micturition

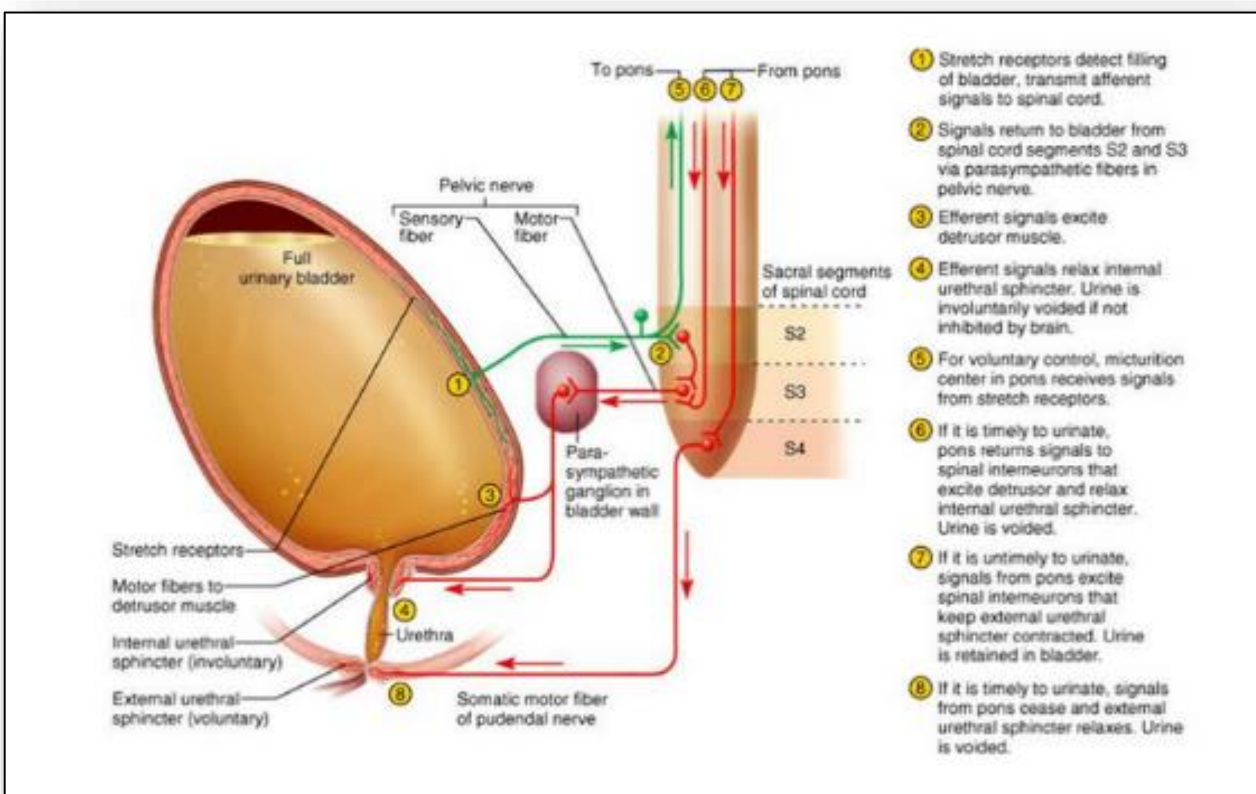


Figure 12: Anatomy of a kidney glomerulus

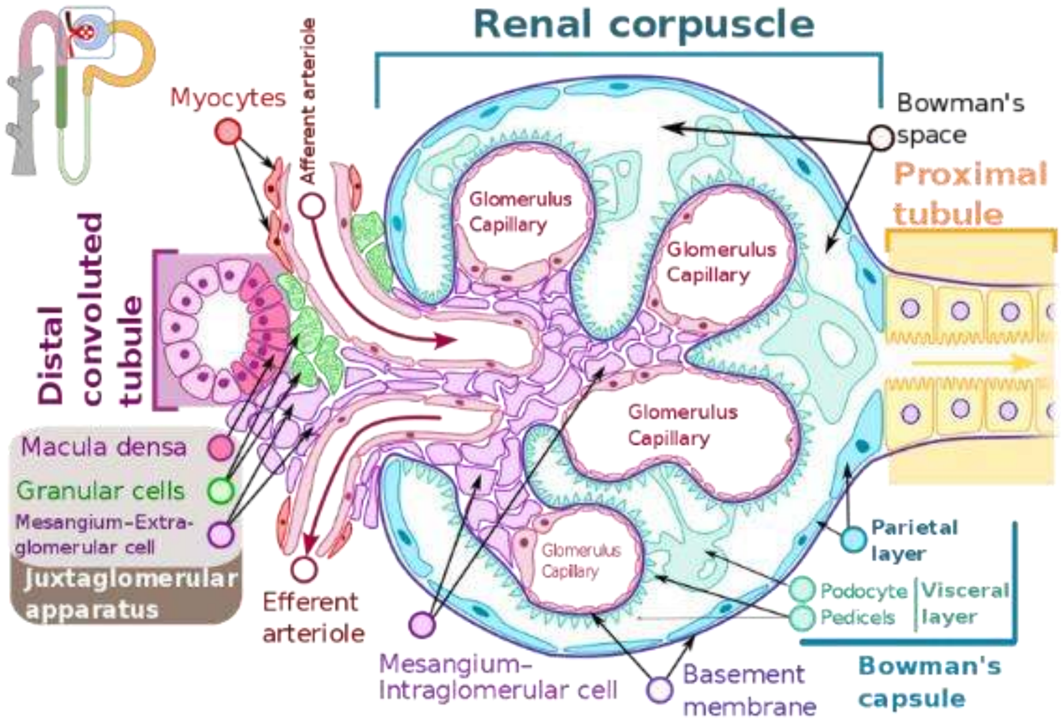


Figure 13: Spectrum of glomerular pathology (4)

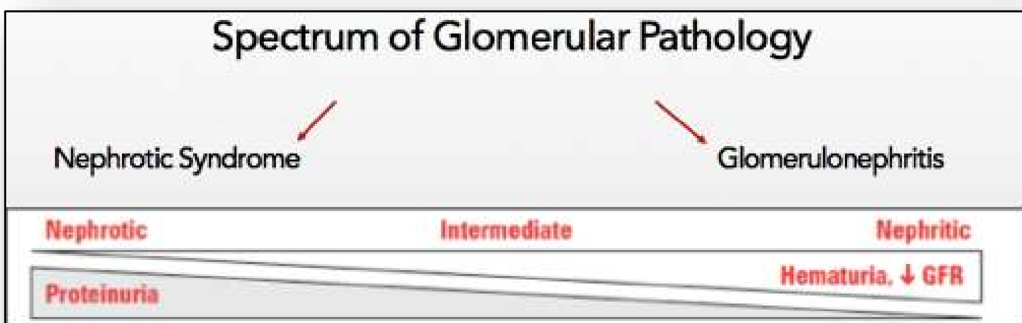


Figure 14: Manifestations of nephrotic syndrome

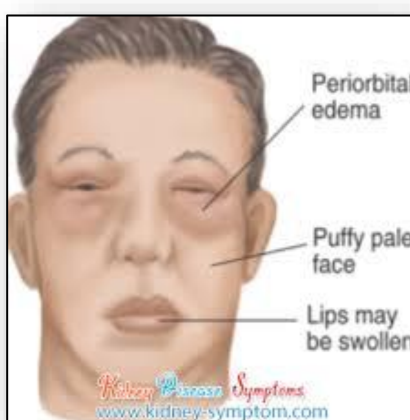


Figure 15: Pathophysiology of nephrotic syndrome

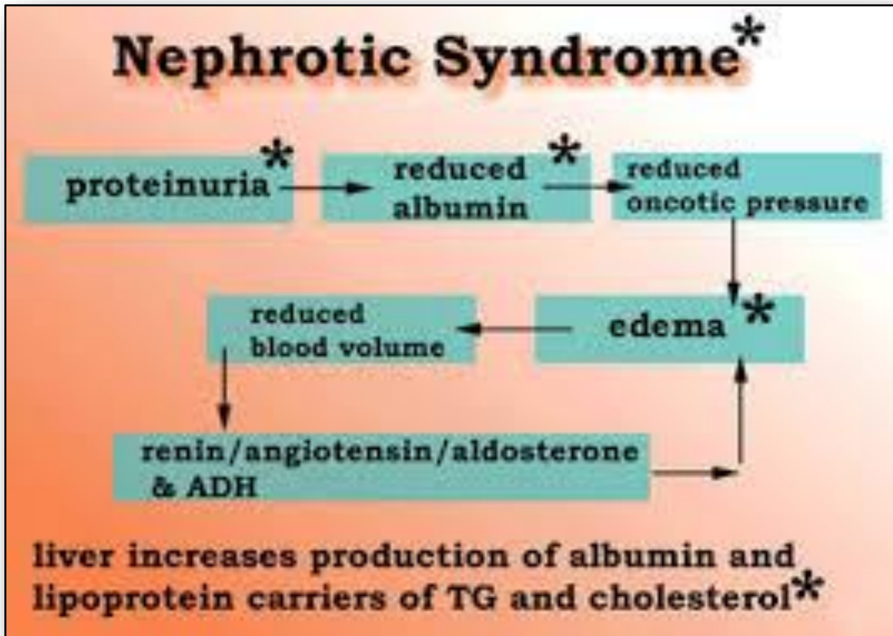


Figure 16: Pathophysiology in acute kidney failure

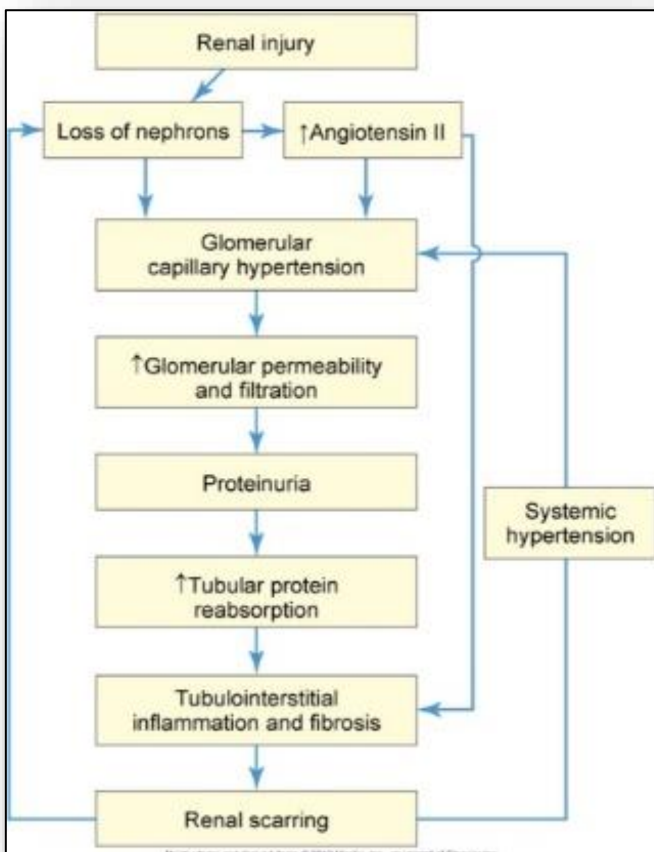


Figure 17: Signs and symptoms of kidney failure

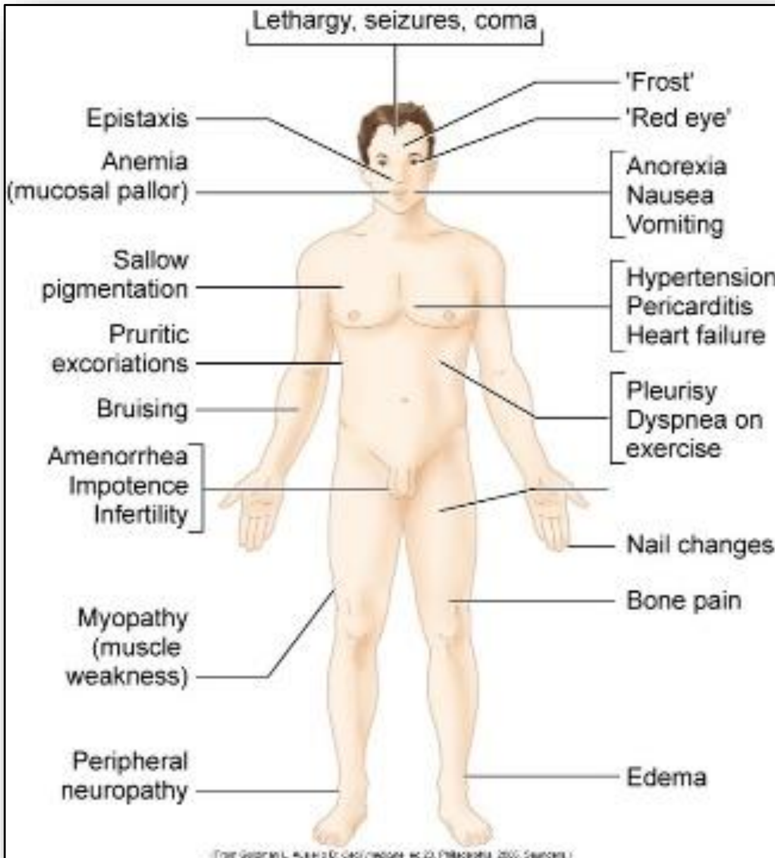


Figure 18: Vessel occlusion related to platelet accumulation and/or damaged red blood cells

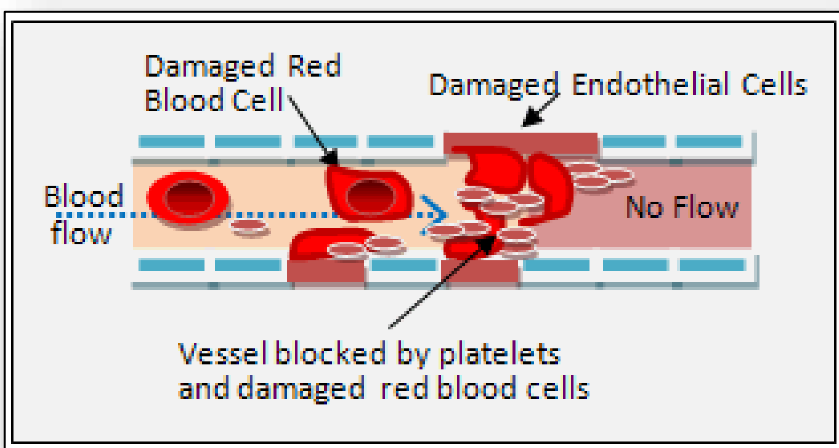


Figure 19: Hemolytic uremic syndrome (overview) (5)

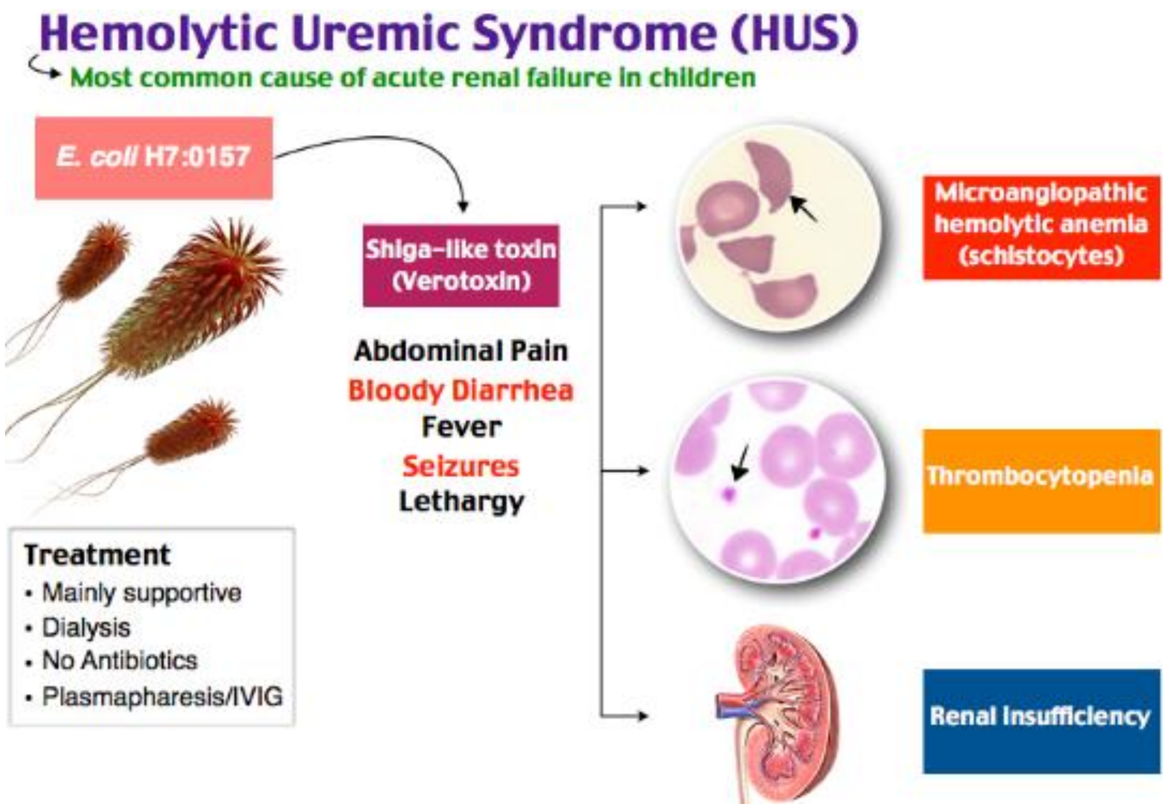


Figure 20: ADAMTS13 and VWF protein in TTP (6)

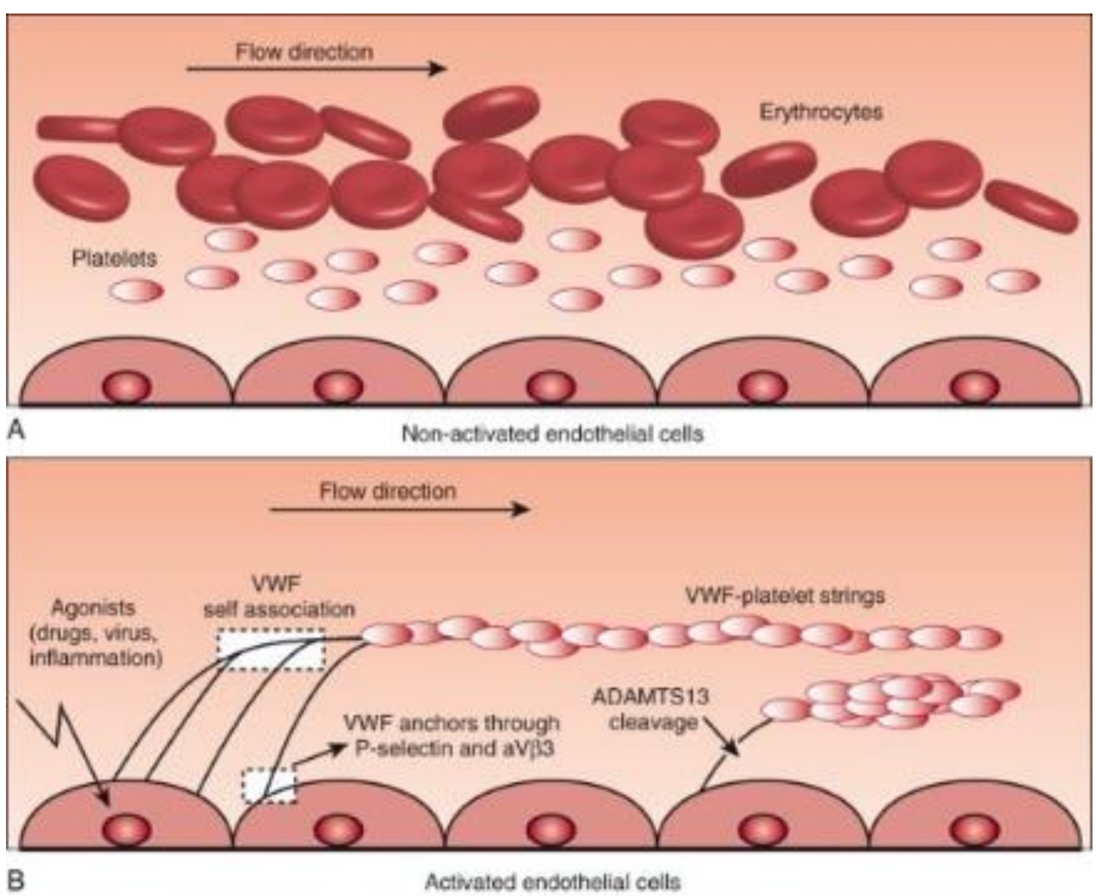


Figure 21: Mechanisms to maintain water balance in the human body

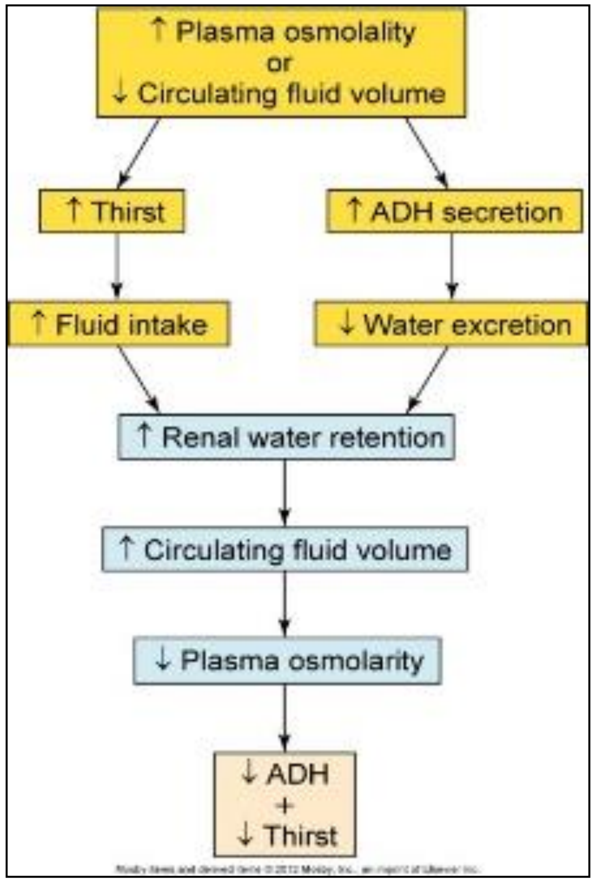


Figure 22: Movement of water between fluid compartments

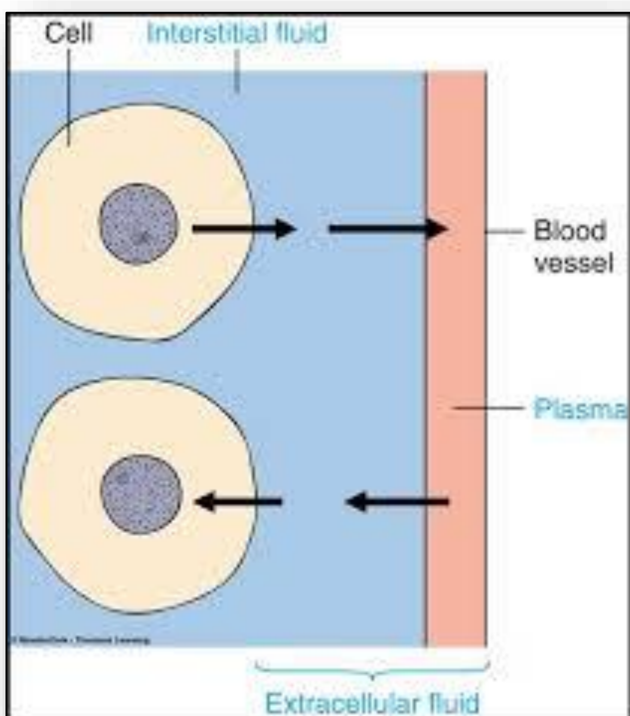


Figure 23: Water movement between fluid compartments

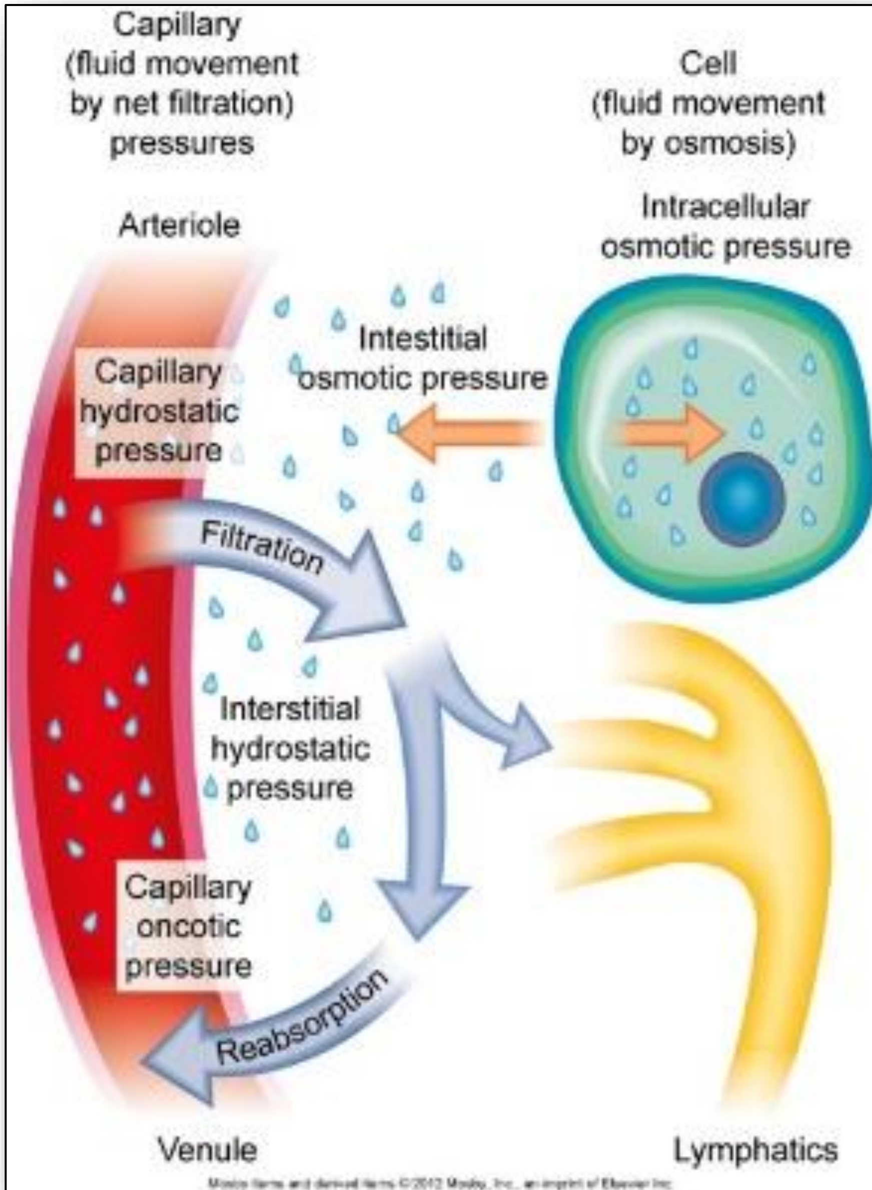


Figure 24: ECG changes related to potassium level

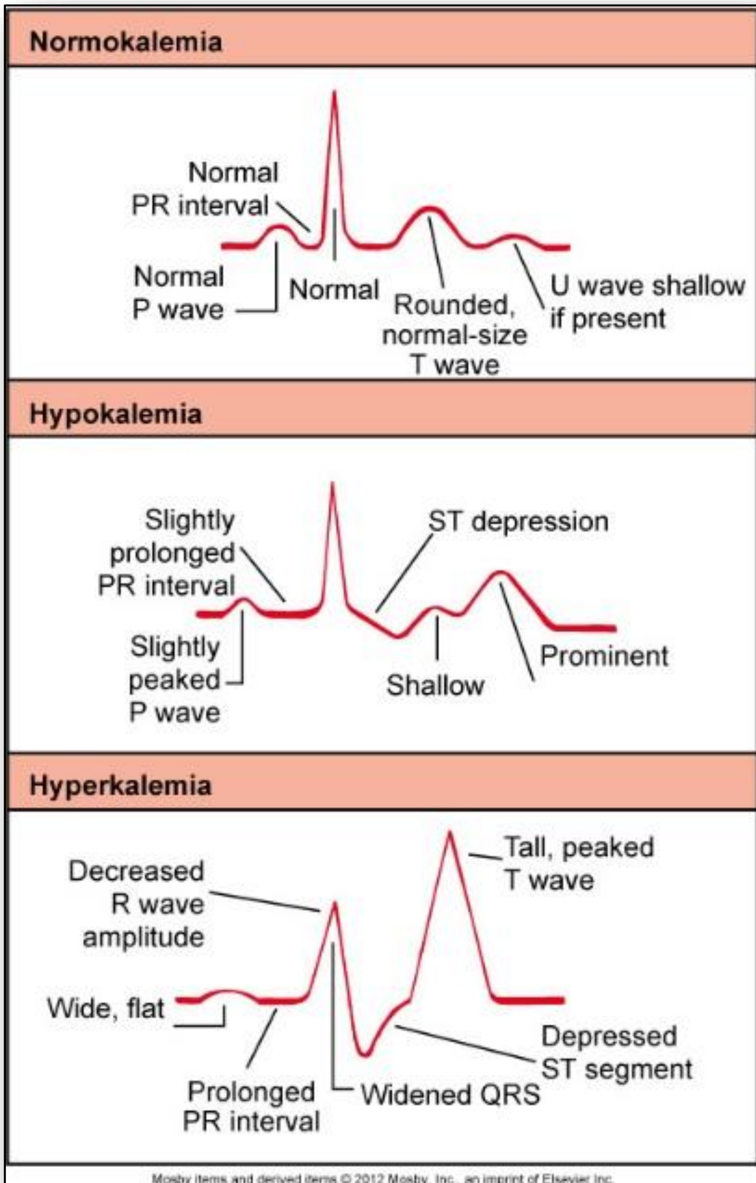


Figure 25: Tetany of the hand

Figure 26: Respiratory (lungs) and metabolic (kidneys) compensation mechanisms in acid-base balance

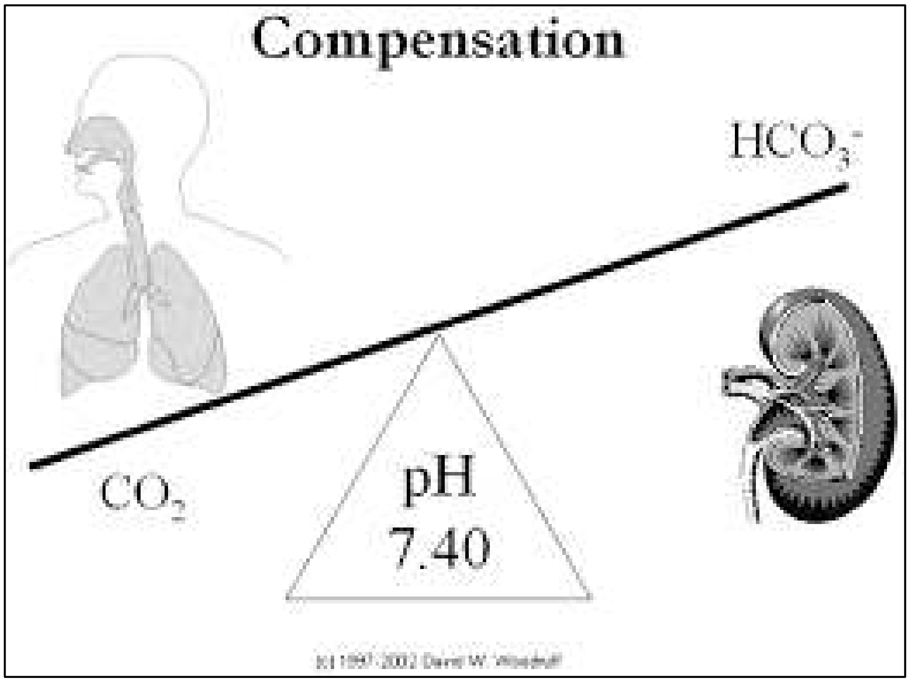


Figure 27: Normal ranges for determining acidosis and alkalosis

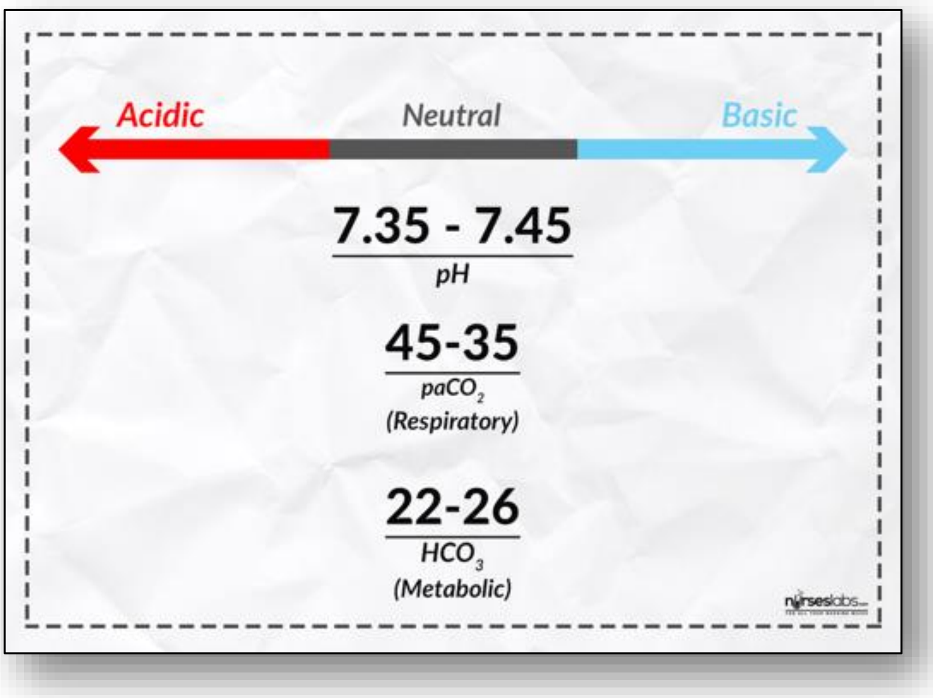


Figure 28: Definition of the anion gap

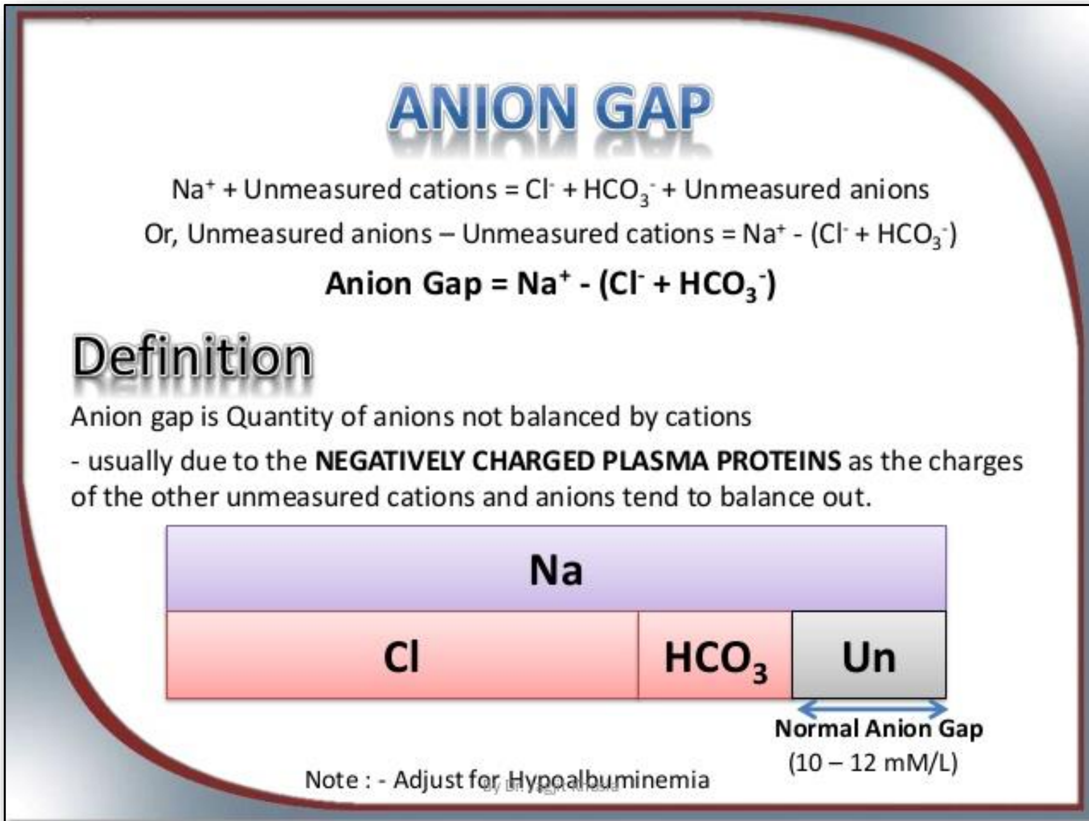


Figure 29: Metabolic acidosis and compensation

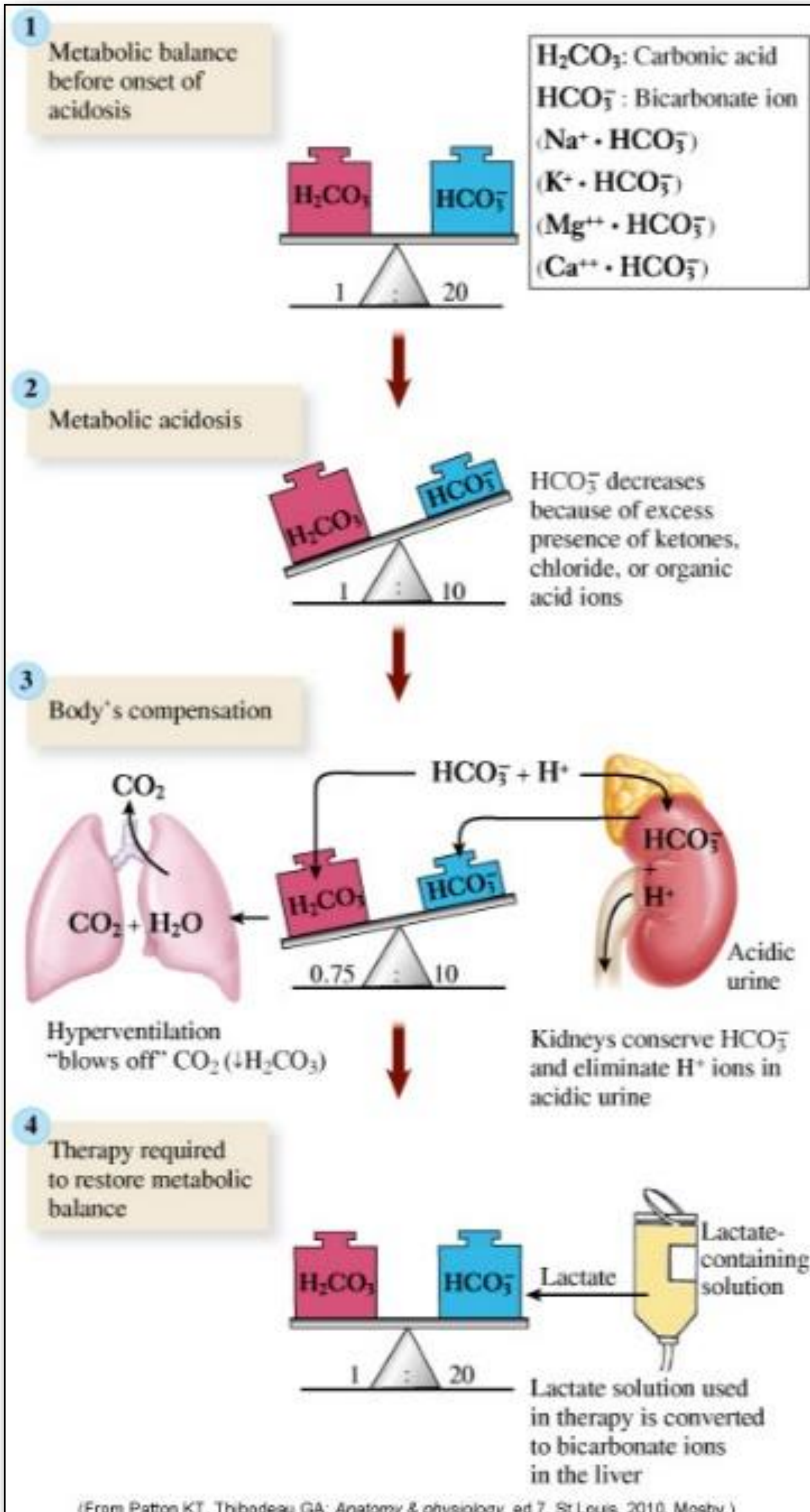


Figure 30: Metabolic alkalosis and compensation

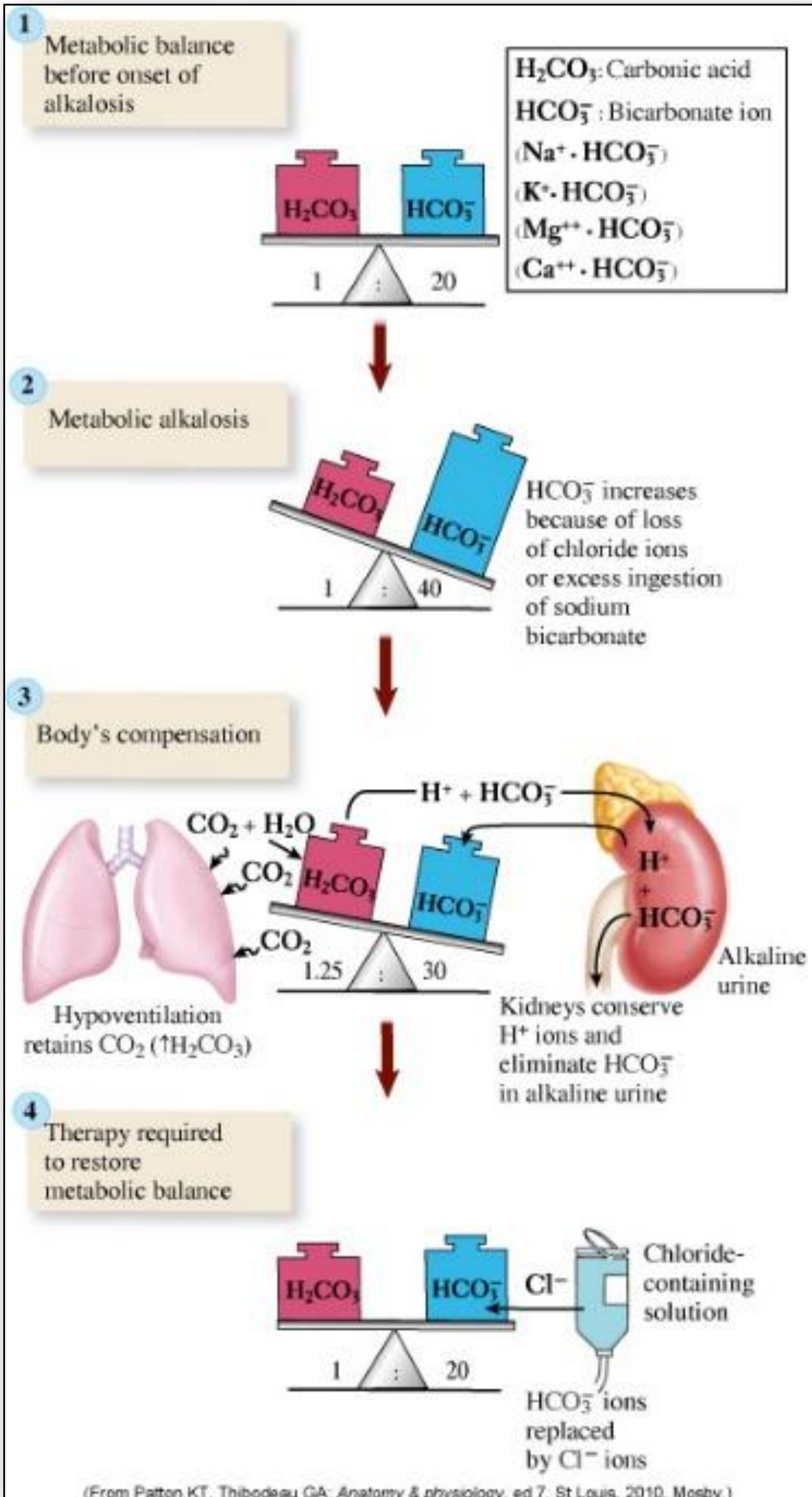
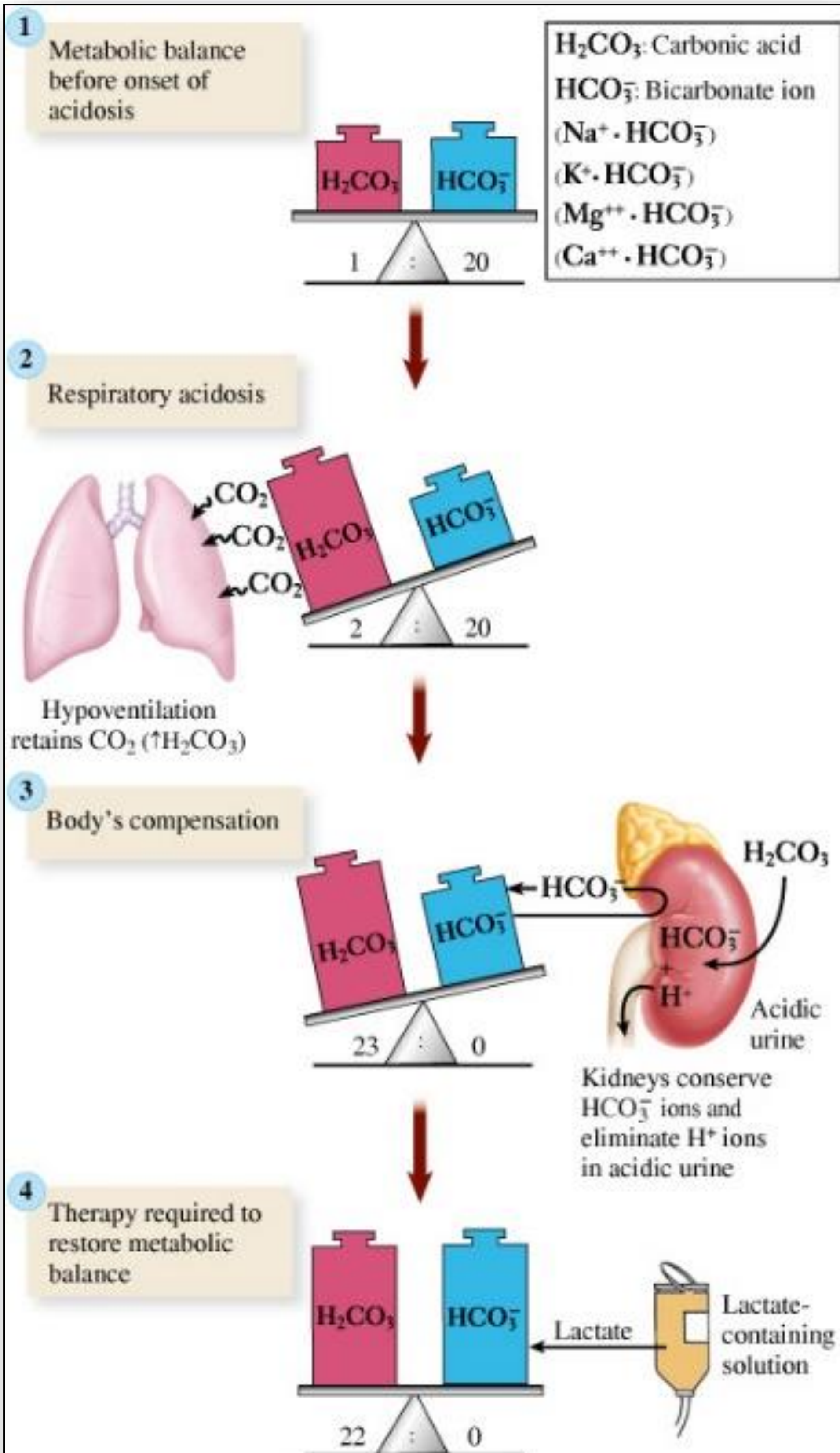
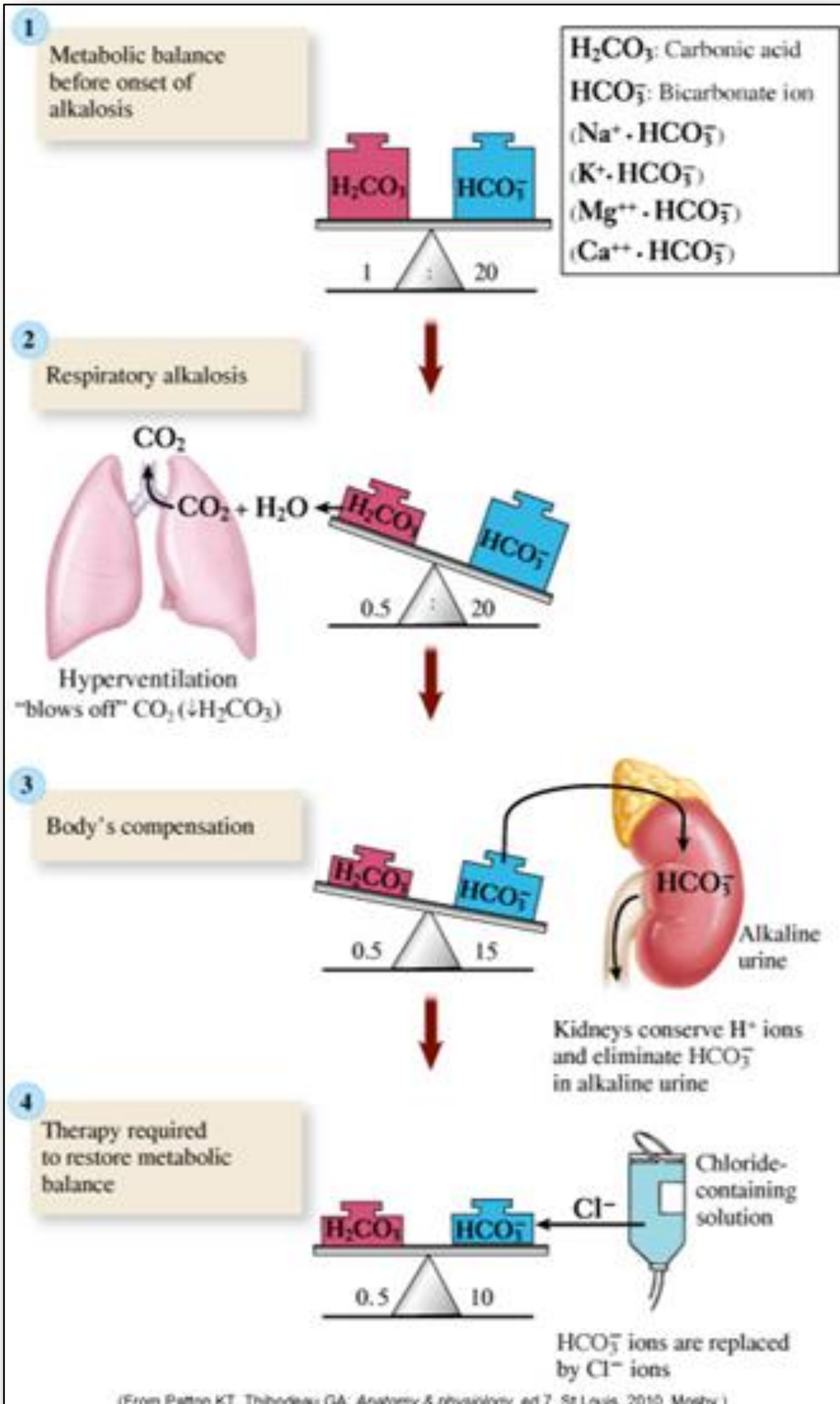


Figure 31: Respiratory acidosis and compensation



(From Patton KT, Thibodeau GA: *Anatomy & physiology*, ed 7, St Louis, 2010, Mosby.)

Figure 32: Respiratory alkalosis and compensation



11: Alterations of the Gastrointestinal Tract

Figure 1: Accessory organs of the gastrointestinal tract

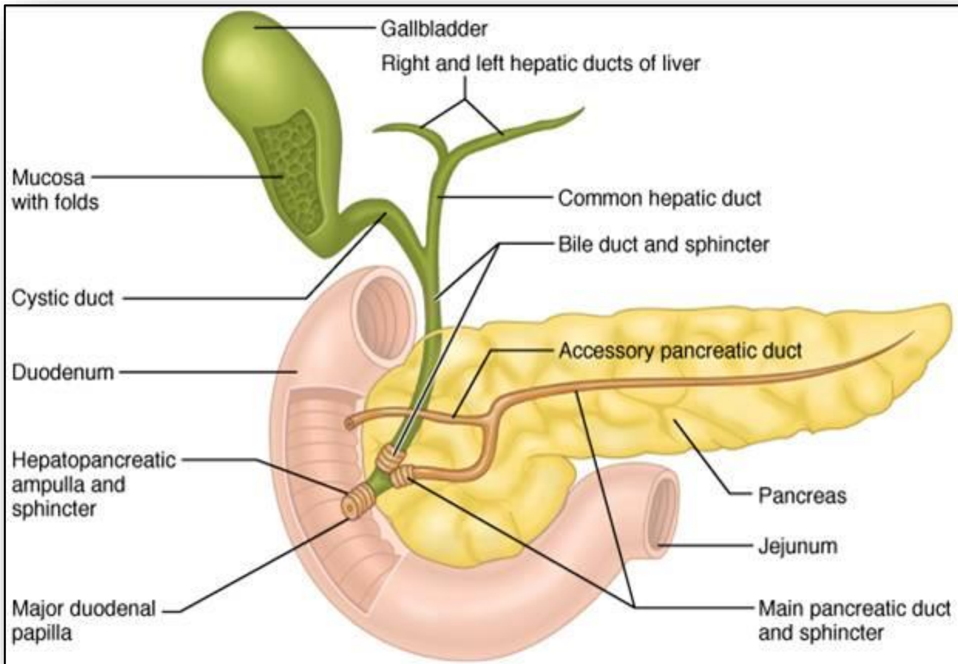


Figure 2: Location of the lower esophageal sphincter implicated in achalasia

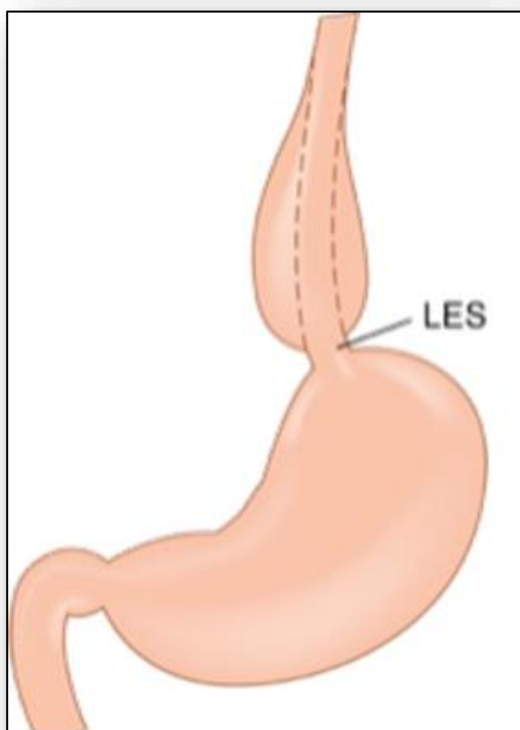


Figure 3: Massively dilated esophagus seen in achalasia

Figure 4: Barrett's esophagus

Figure 5: Sliding hiatal hernia (A) and para-esophageal (rolling) hiatal hernia (B)

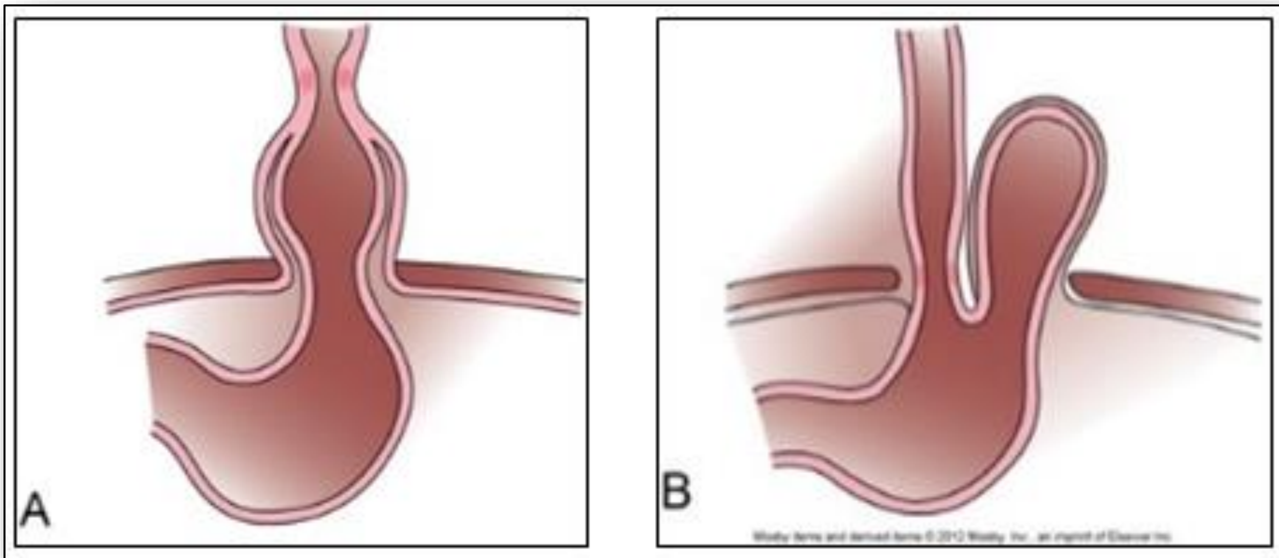


Figure 6: Anatomy of the stomach (focus on the pylorus)

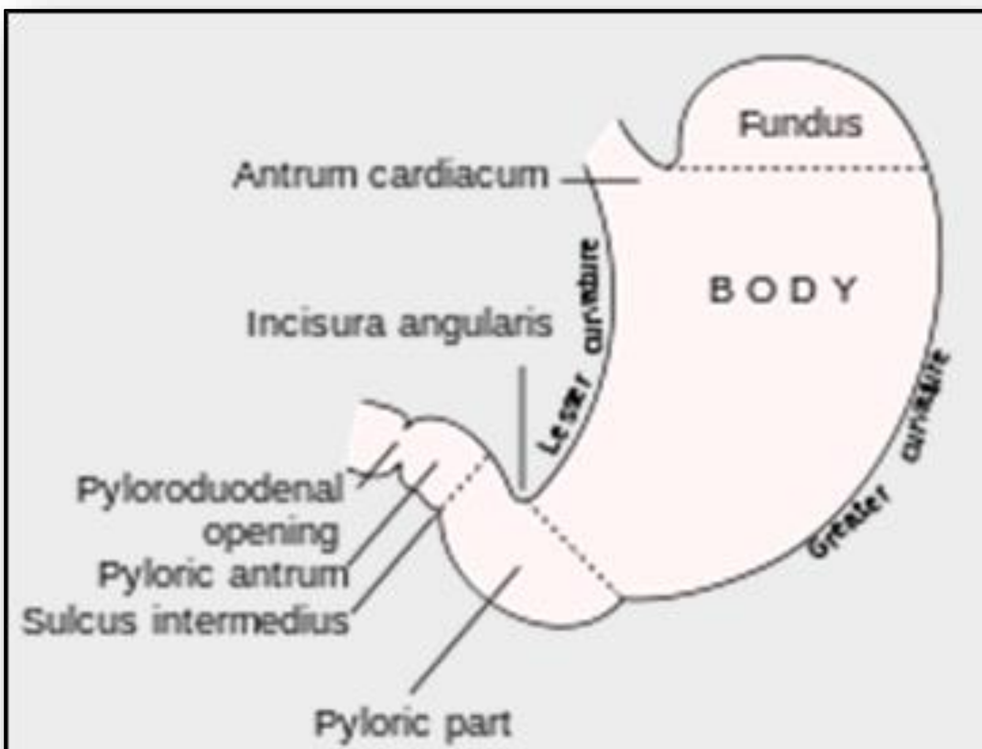


Figure 7: Intestinal obstruction pathophysiology and clinical manifestations

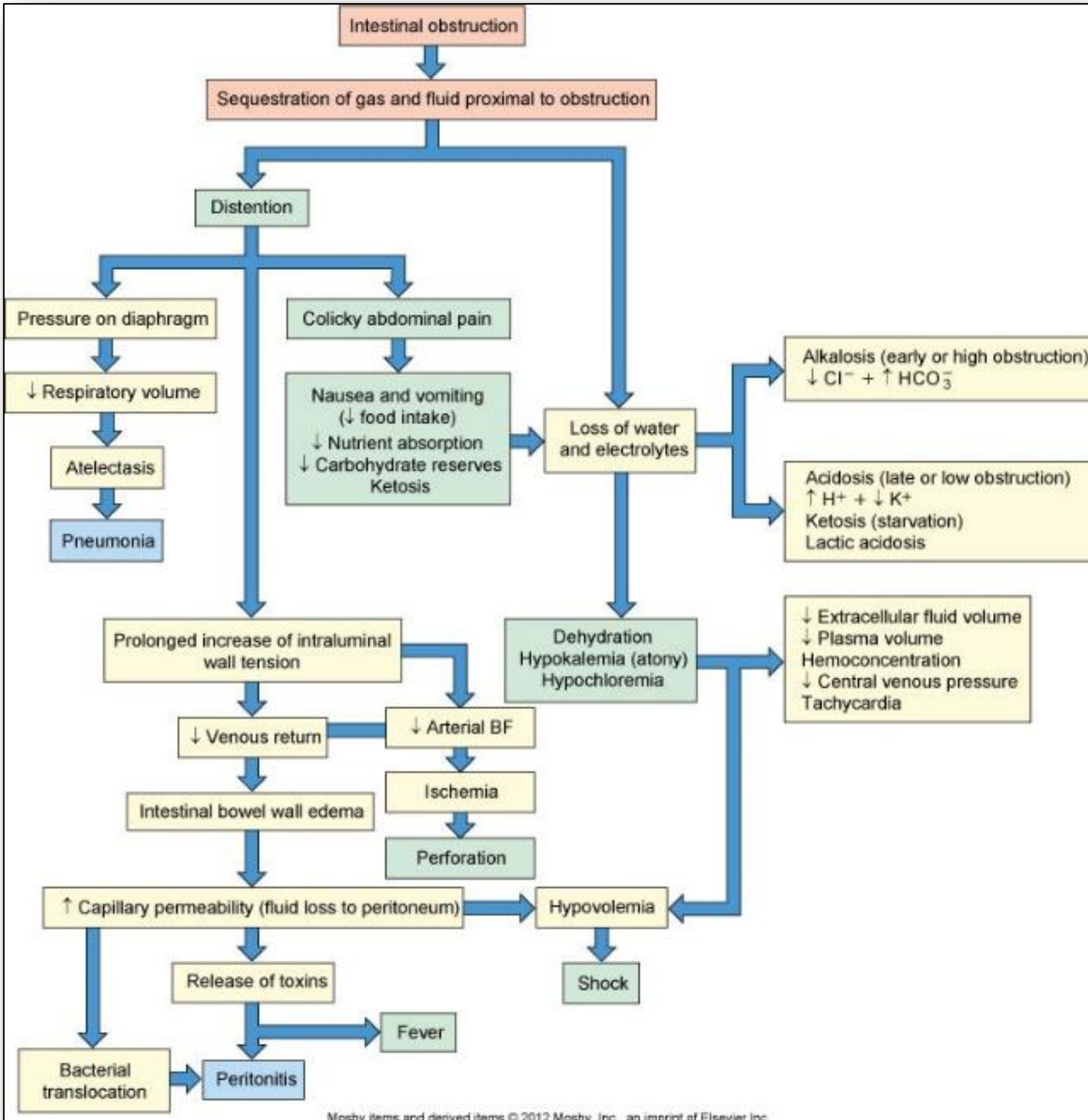


Figure 8: Pathophysiology of gastric ulcers

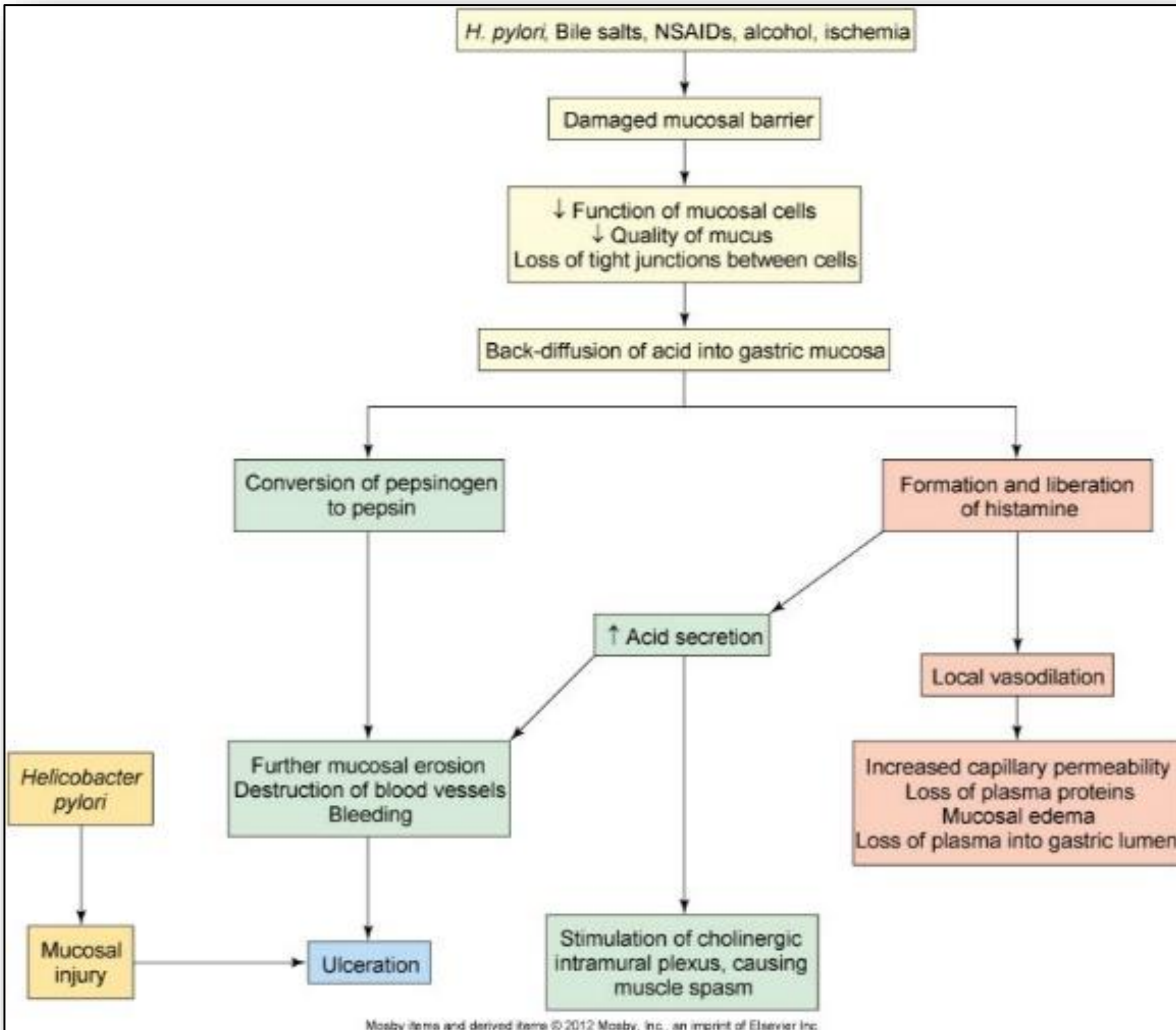


Figure 9: Degrees of erosion in stomach mucosa from superficial erosions to "true ulcer"

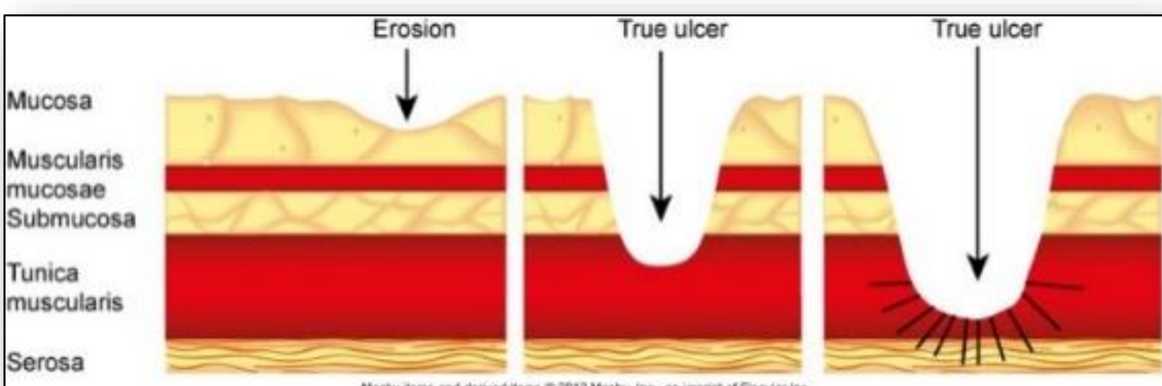


Figure 10: Pathophysiology, clinical manifestations and complications of gastrointestinal bleeding

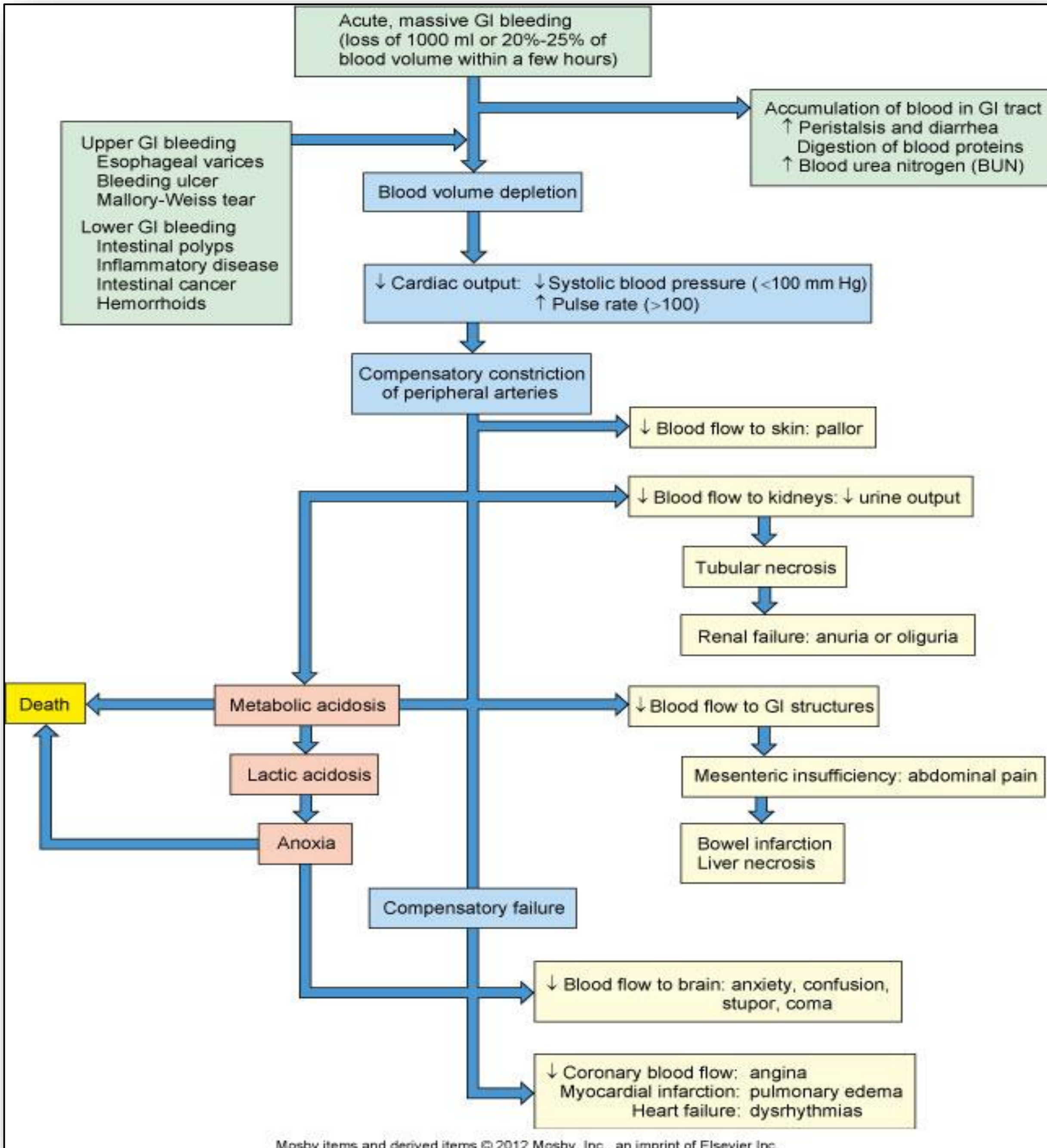


Figure 11: Locations of pain in appendicitis

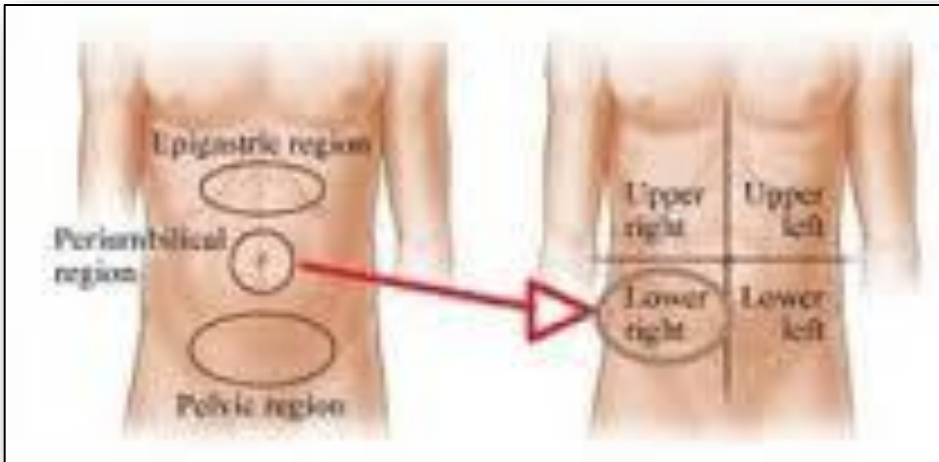


Figure 12: Therapeutic modifications in management of IBD

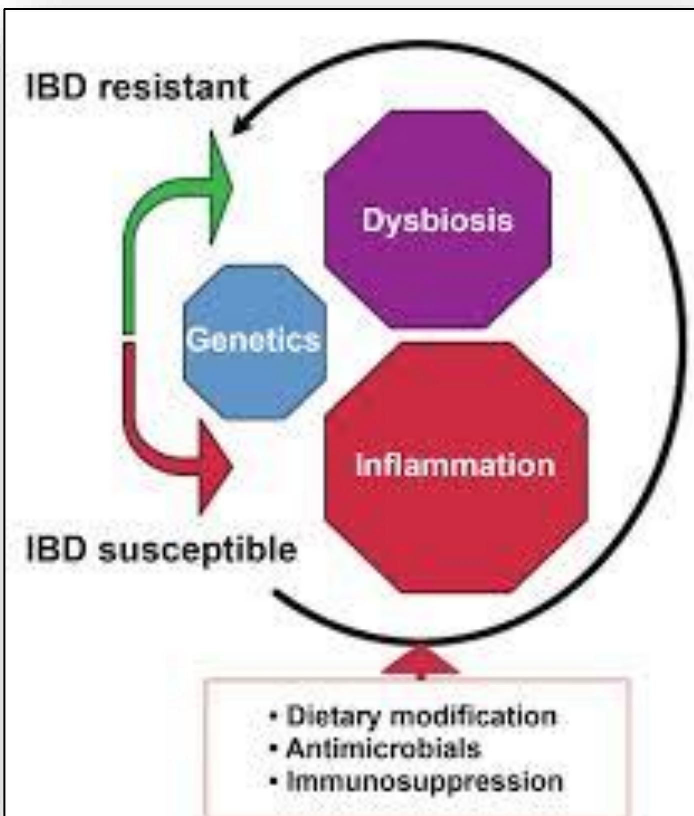


Figure 13: Ulcerative colitis and Crohn's disease

Figure 14: Sites of lesions in UC versus Crohn's disease

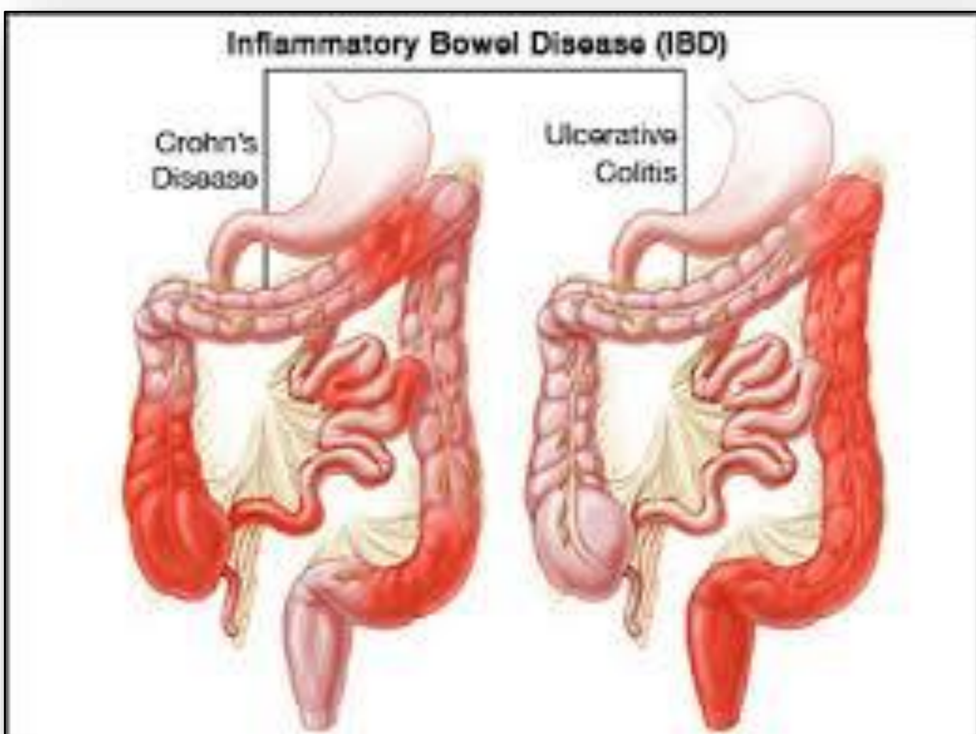


Figure 15: Table - Comparison IBS, Ulcerative Colitis and Crohn's disease (for review)

	Irritable Bowel Syndrome	Ulcerative Colitis	Crohn's Disease
Symptoms	Bloating Cramping Diarrhea Nausea Constipation Abdominal Pain Gas Incomplete Bowel Movements Fatigue	Bloating Cramping Nausea Vomitting Severe Diarrhea Bloodly Stool Abdominal pain Weight Loss Fatigue	Bloating Cramping Nausea Vomitting Severe Diarrhea Bloodly Stool Mucus in Stool Abdominal pain Weight Loss Higher Likelihood of Fistulae Fatigue
Diagnosis	Elimination of other diseases. Colonoscopy & biopsy does not show signs of disease.	Ulcer formation on interior wall of large intestine are visible via colonoscopy. Biopsy will also show signs of disease.	Attacks the interior wall & inner layers of the intestinal wall. Ulceration is less likely, but intestine will show inflammation. Biopsy will show signs of disease.
Classification	Functional Disorder	Inflammatory Bowel Disease	Inflammatory Bowel Disease
Possible Causes	Usually develops after trauma to the gut such as food poisoning or stomach flu. A recent study suggests micro-inflammation causing overproduction of histamines may be the root cause of IBS. Also considered are: bacterial overgrowth, nervous system disorders or psychosomatic causes.	Autoimmune system of the body attacks the gut, causing inflammation & ulcers on interior wall of large intestine. Genetics, bacterial infection or overuse of antibiotics may also play a part.	Autoimmune system of the body attacks the gut, causing deep inflammation in the large intestine and possibly small intestine. Genetics, bacterial infection and/or overuse of antibiotics may also play a part.
Common Treatments	Diet Probiotics Fiber Antispasmodics Laxatives Antidepressants	Diet Antibiotics Anti-inflammatories Corticosteriods Immunosuppressive Drugs Surgery	Diet Antibiotics Anti-inflammatories Corticosteriods Immunosuppressive Drugs Surgery

Figure 16: Diverticular disease

Figure 17: Etiology of hyperbilirubinemia (jaundice; icterus)

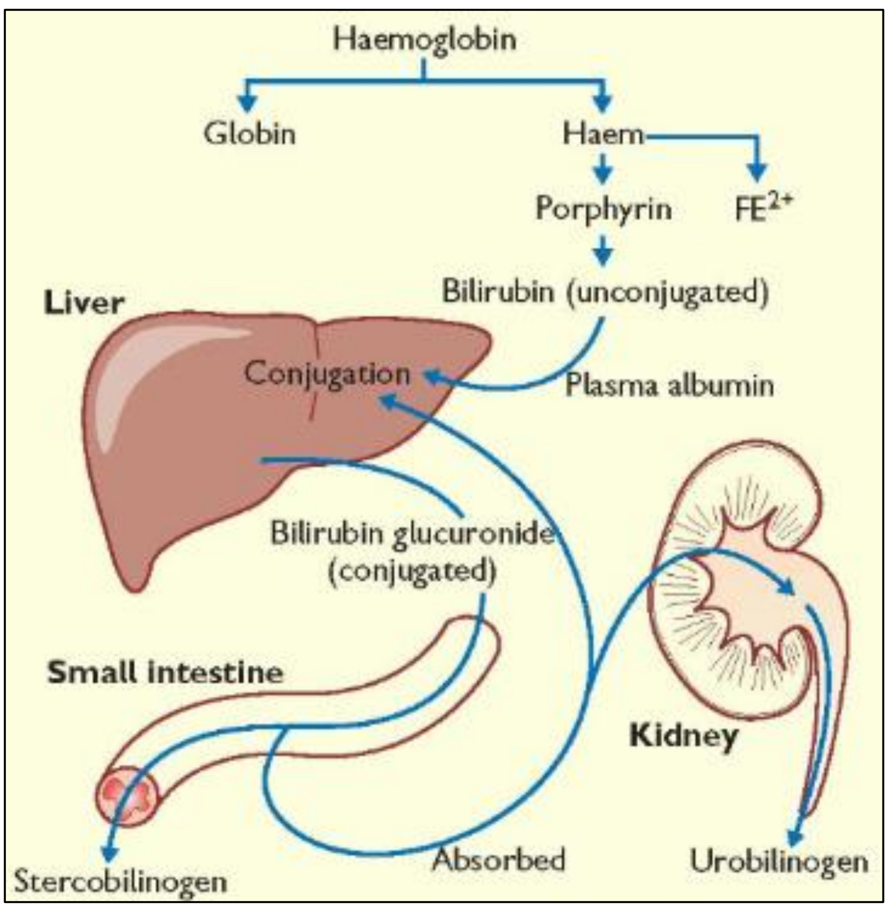


Figure 18: Pathophysiology and clinical manifestations of liver cirrhosis

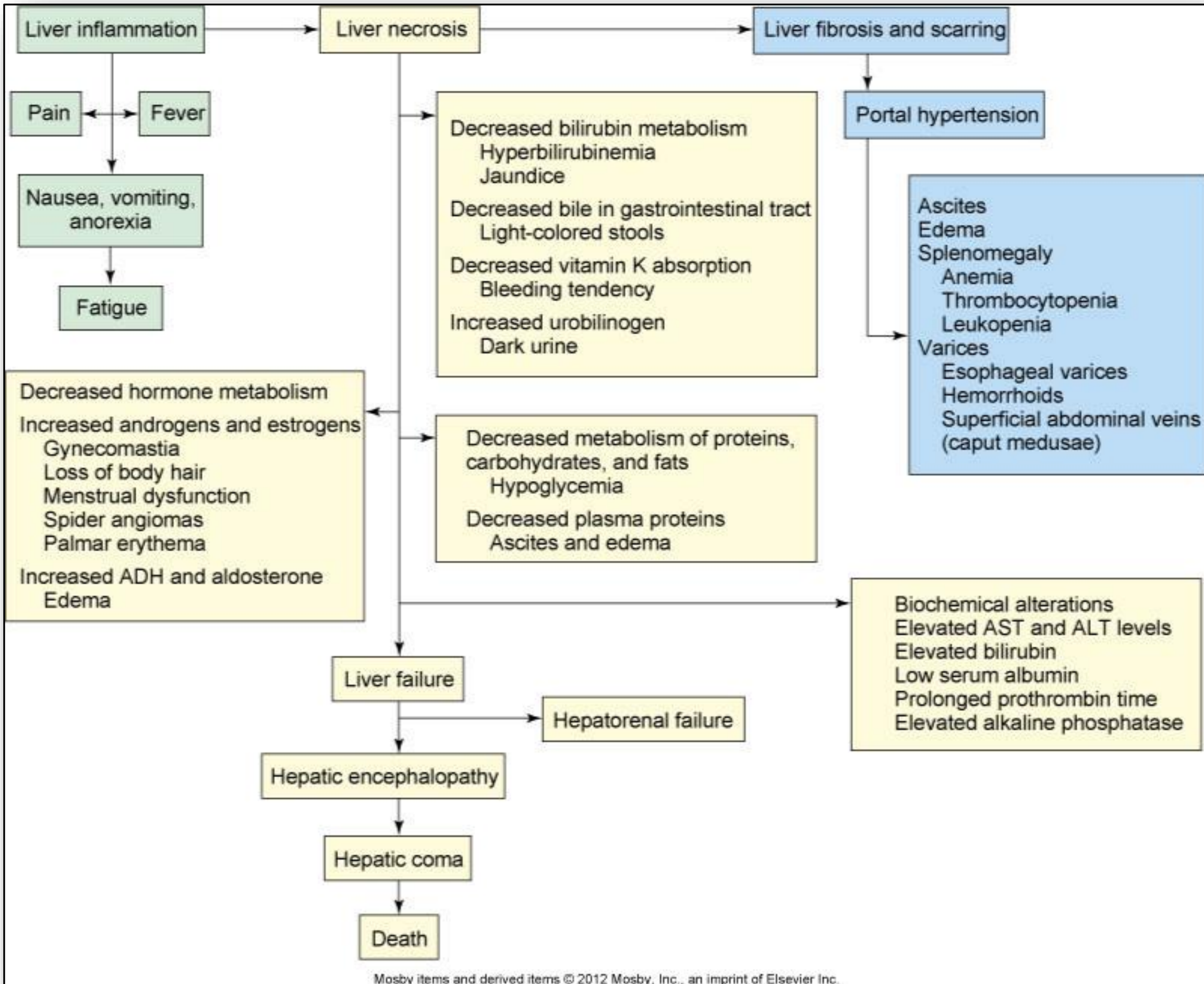


Figure 19: Gallstones

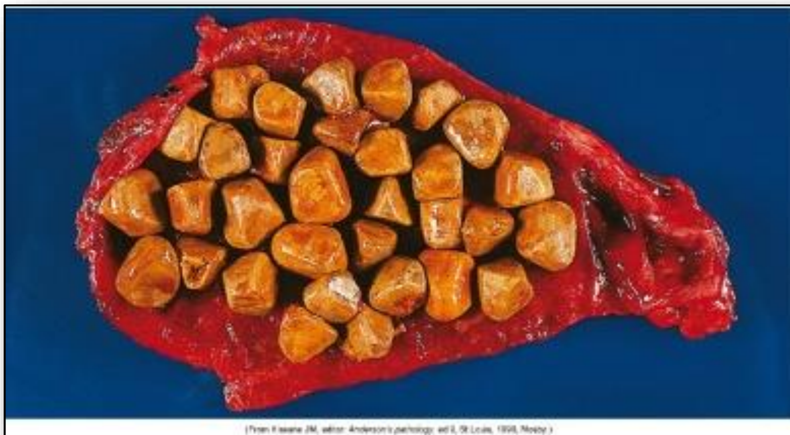


Figure 20: Pathophysiology of cholelithiasis (gallstones)

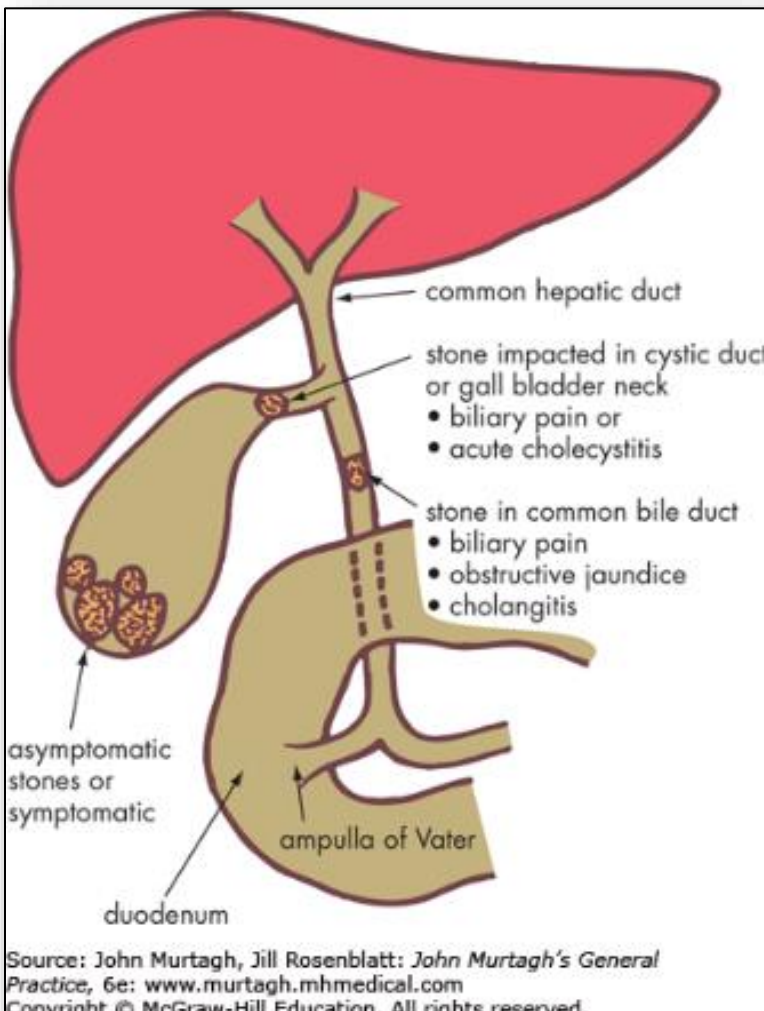


Figure 21: Pathophysiology of obesity

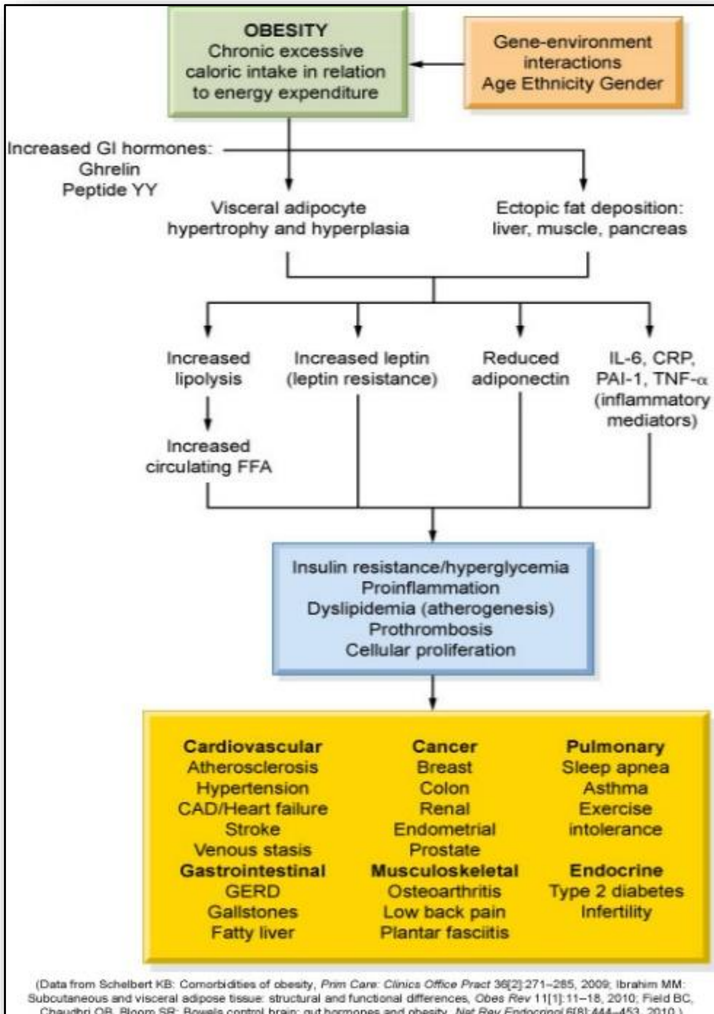


Figure 22: Locations for colon cancer, relative incidence, and associated manifestations

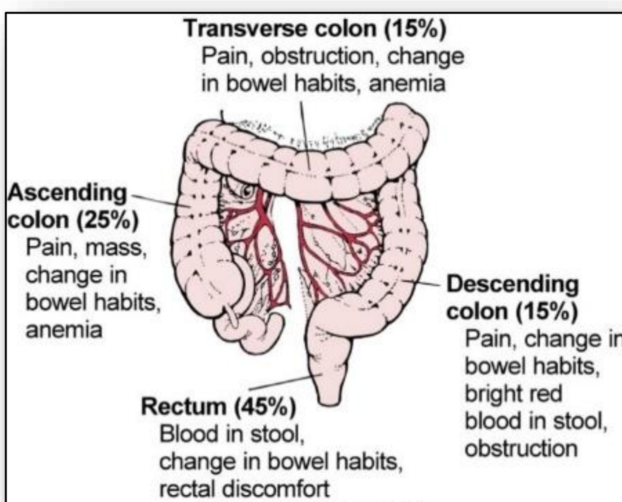


Figure 23: Pathophysiology of gluten-sensitive enteropathy (Celiac disease)

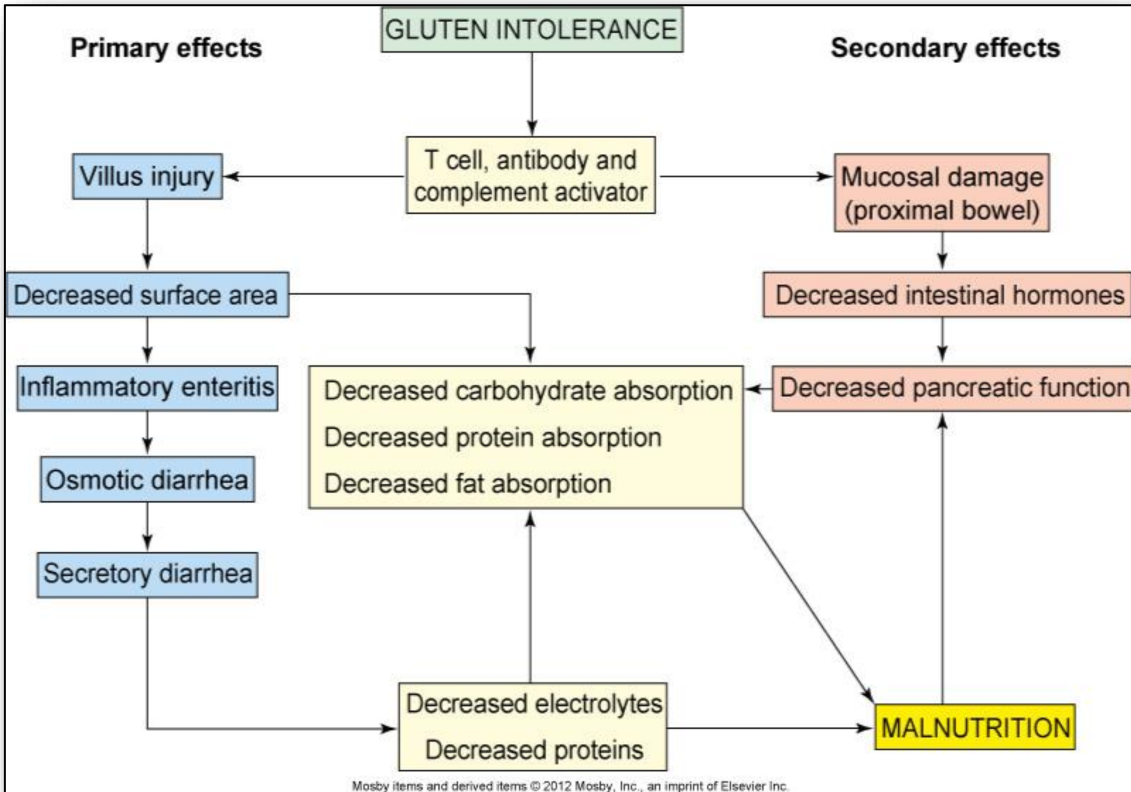


Figure 24: Comparison of kwarshikor and marasmus

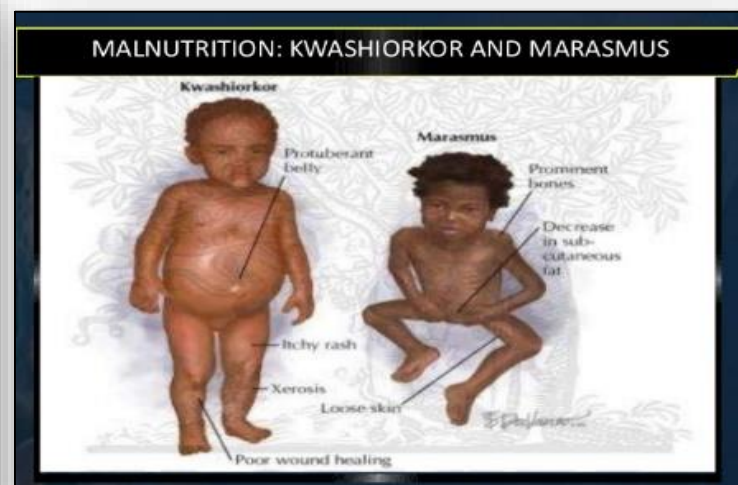


Figure 1: Compact and spongy bone

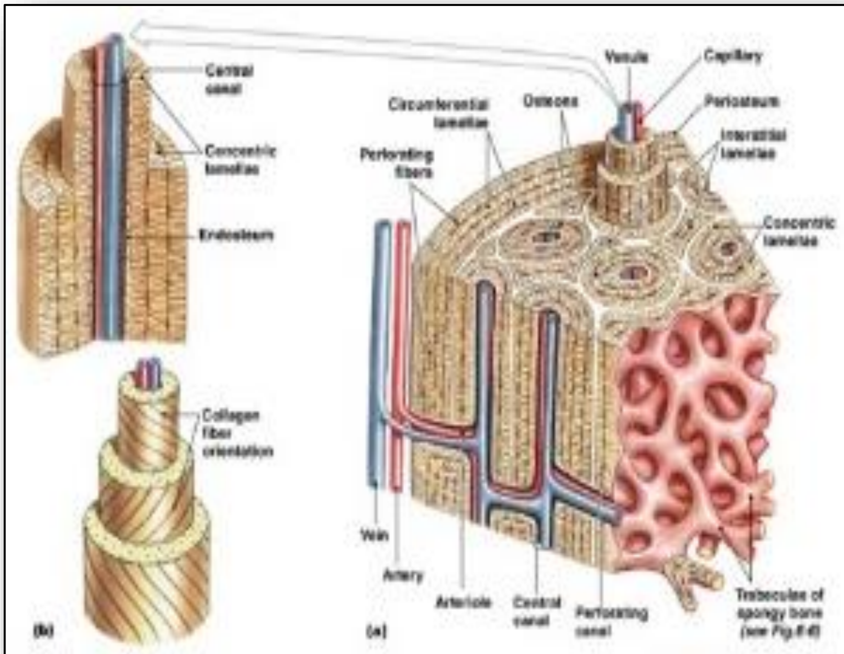


Figure 2: Synovial joint

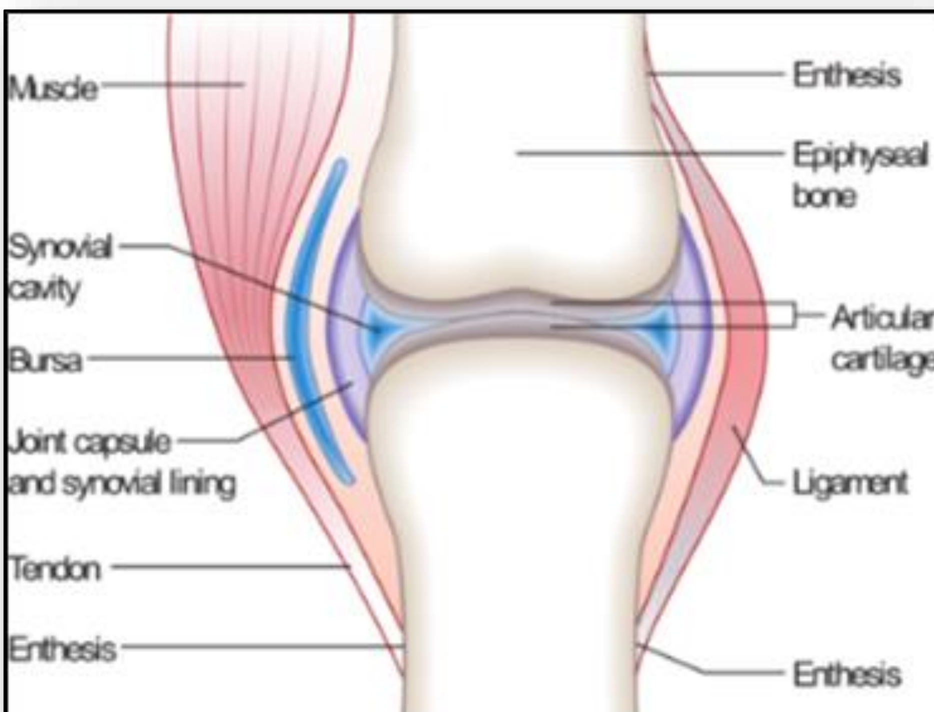


Figure 3: Muscle structure

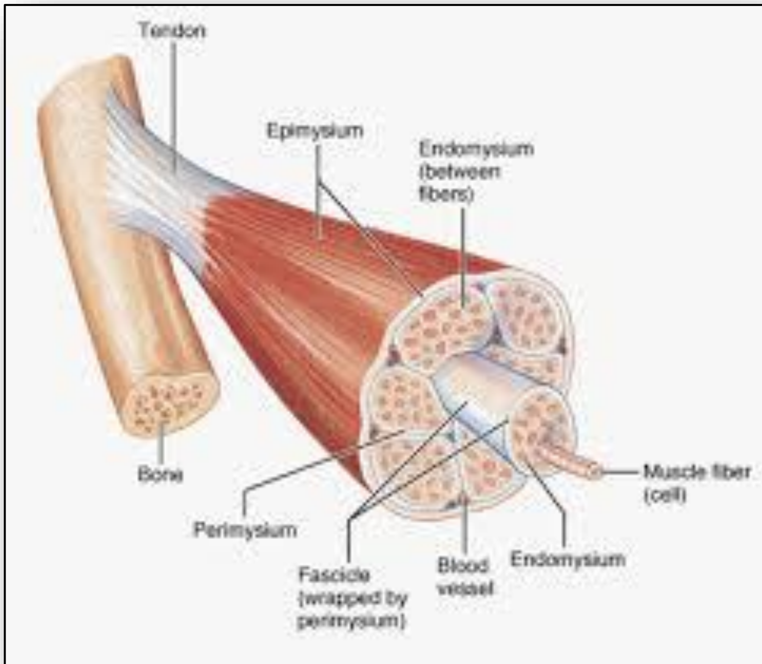
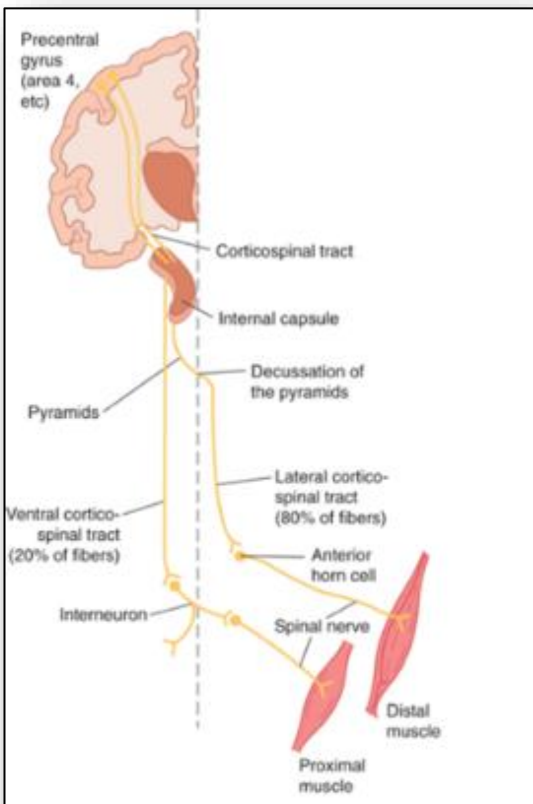


Figure 4: Corticospinal tract



Source: Barrett KE, Barman SM, Boitano S, Brooks H: *Ganong's Review of Medical Physiology*, 23rd Edition: http://www.accessmedicine.com

Figure 5: Vitamin D Synthesis

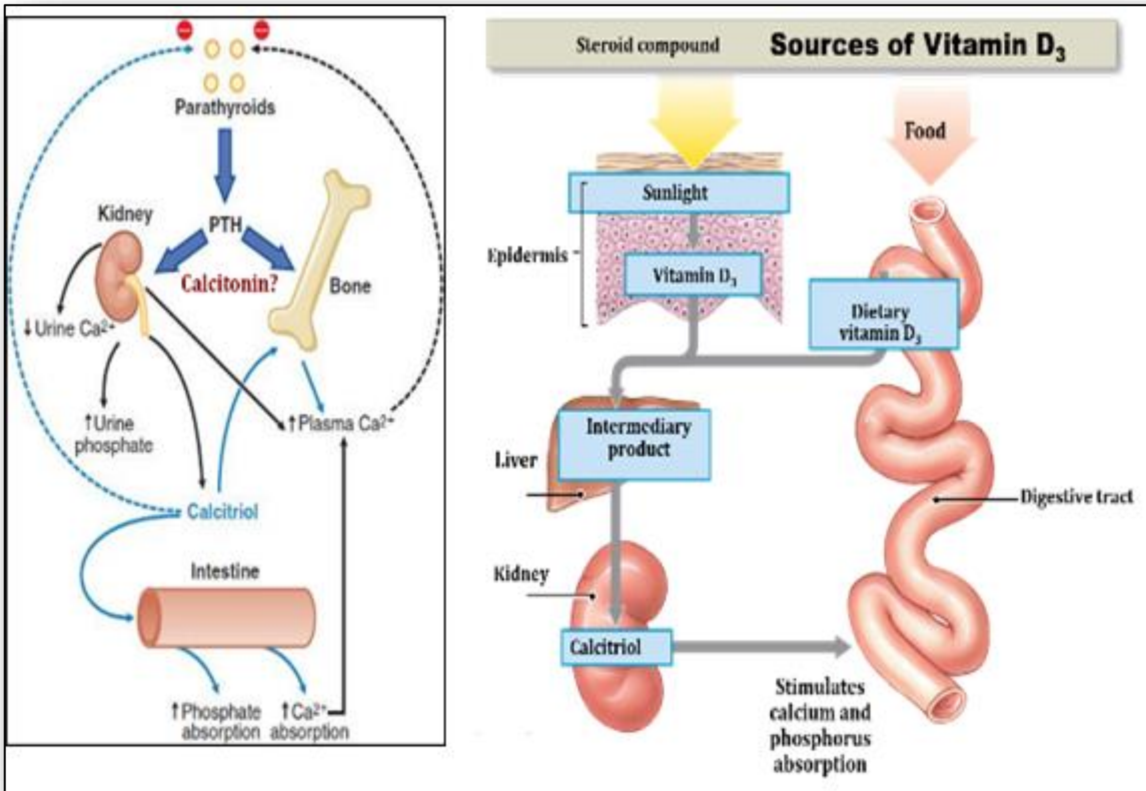


Figure 6: Classifications of bone fractures

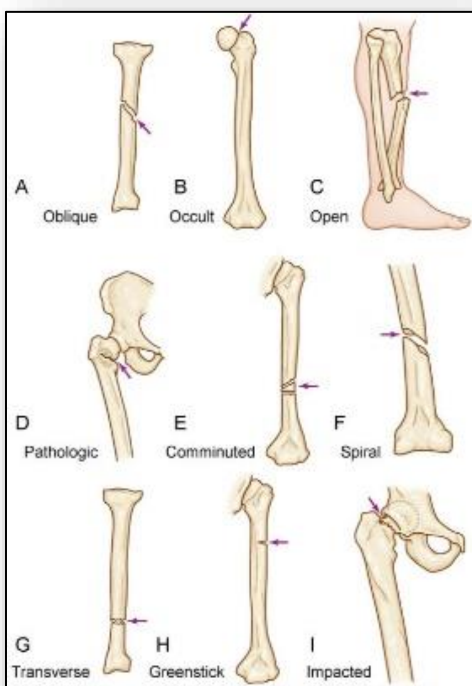


Figure 7: Comparison bone density in stages of osteoporosis to normal bone

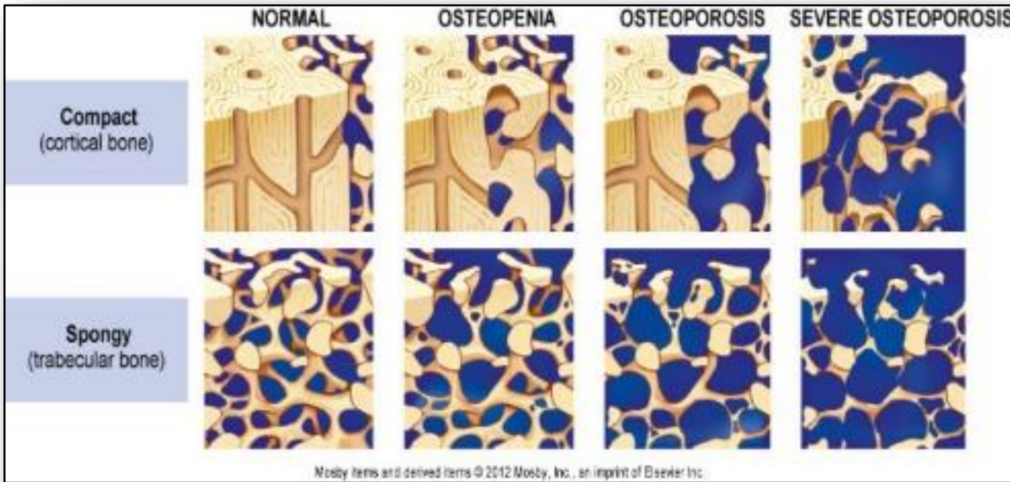


Figure 8: Progression of osteomyelitis

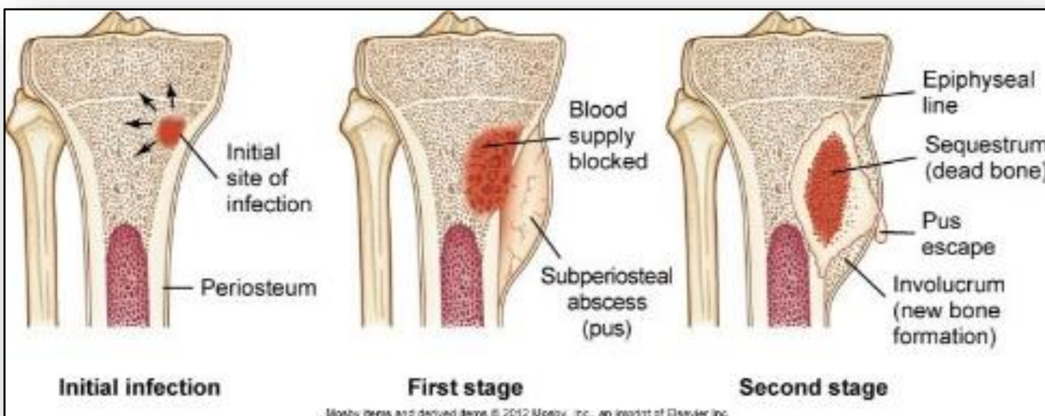


Figure 9: Normal versus osteoarthritic joint

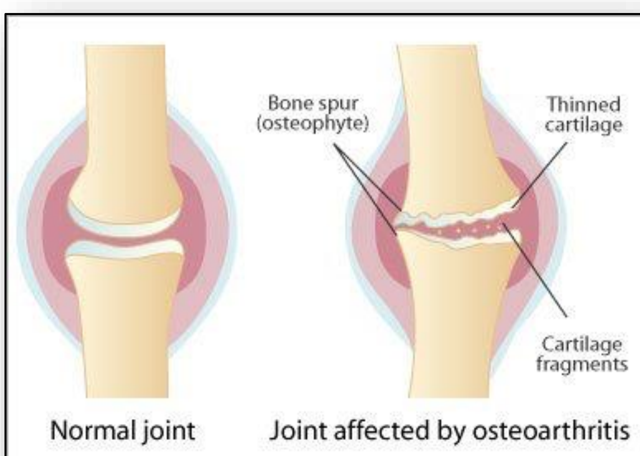


Figure 10: Genetics of rheumatoid arthritis

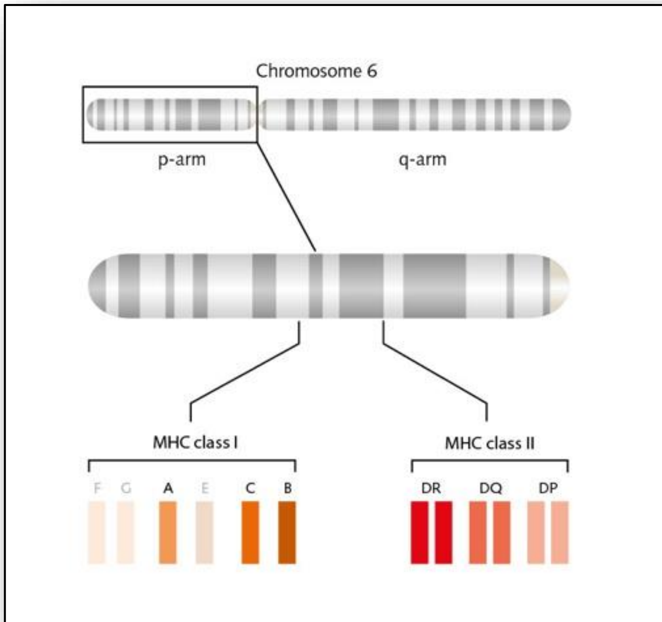


Figure 11: Pathophysiology of rheumatoid arthritis

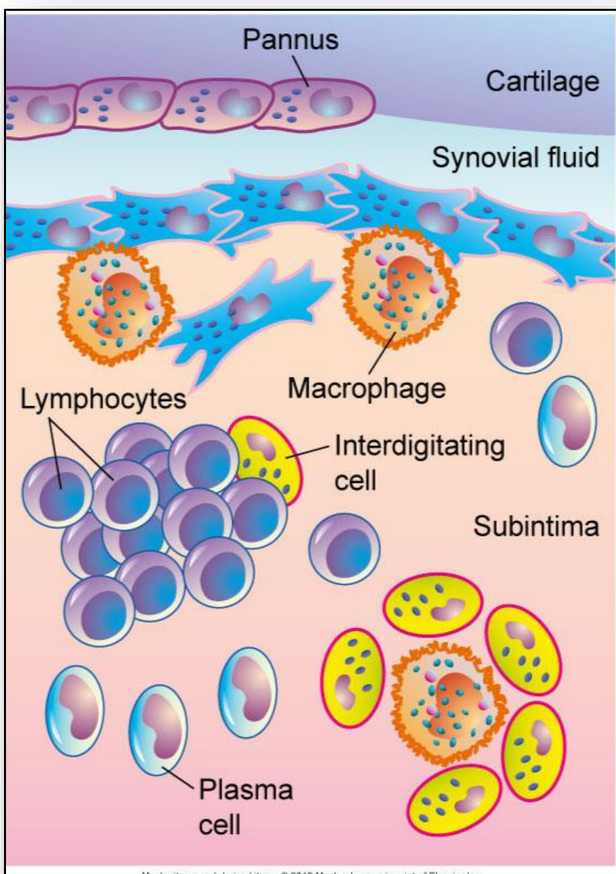


Figure 12: Tophi deposits of the metacarpal joints

Figure 13: Multiple sites of fibromyalgia

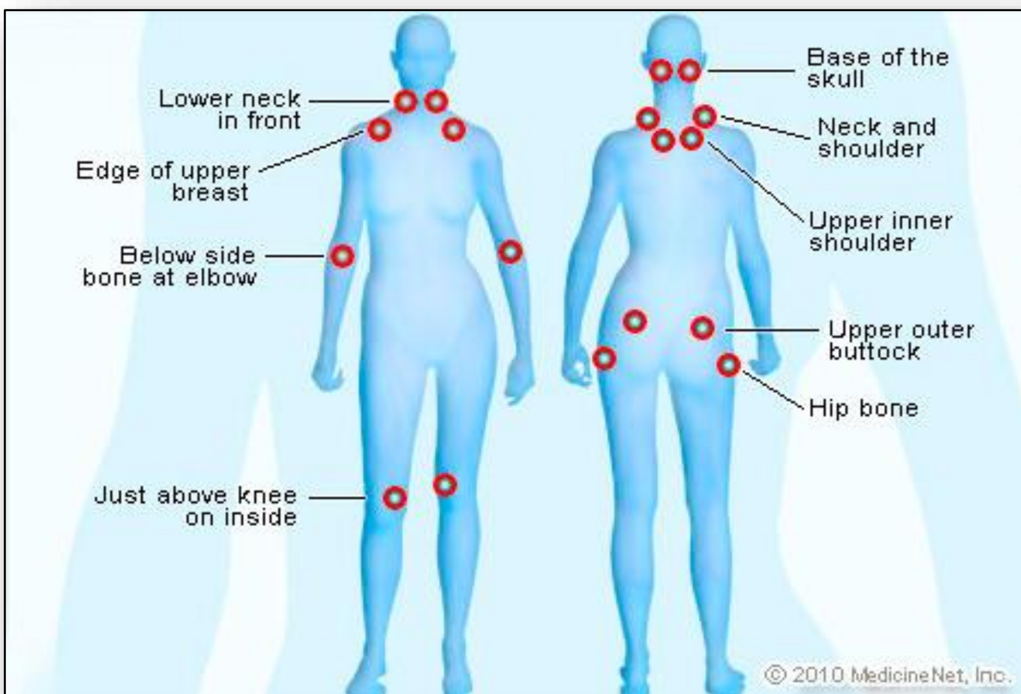


Figure 14: Cellular / tissue origins of bone tumors

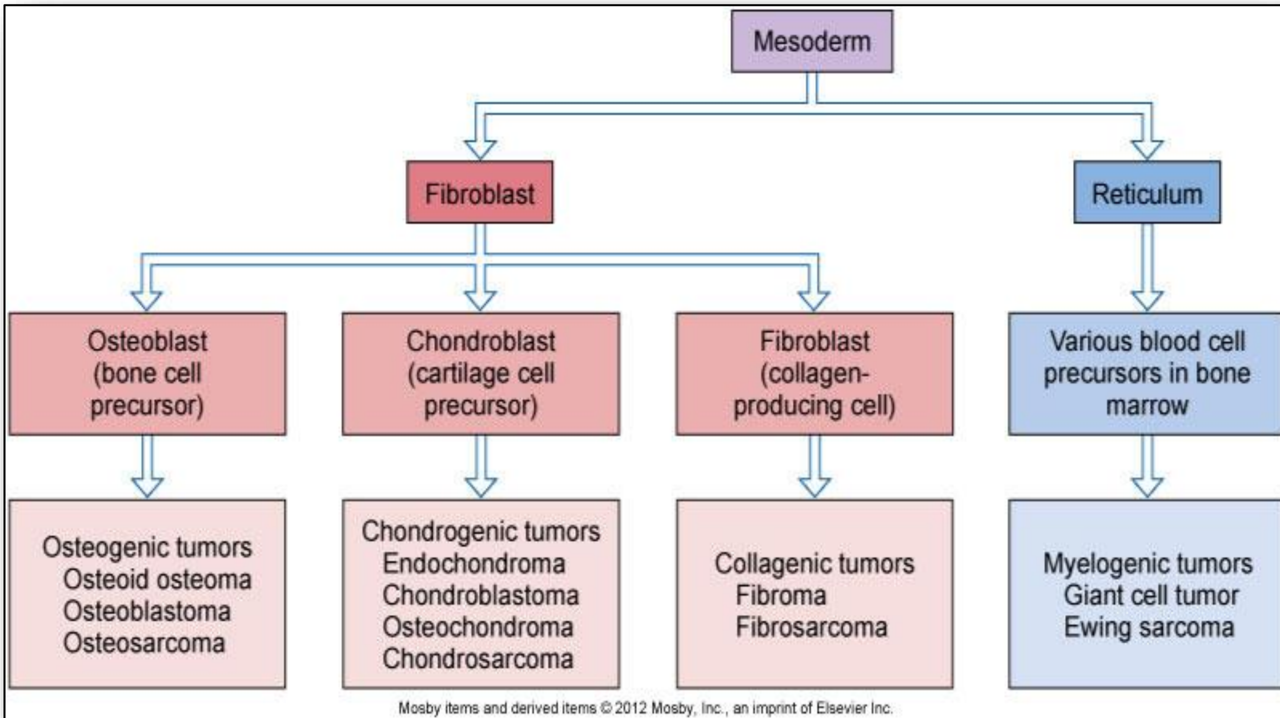


Figure 15: Degree of burns (1^{st}, 2^{nd}, 3^{rd})

Figure 16: Pathophysiology of burns

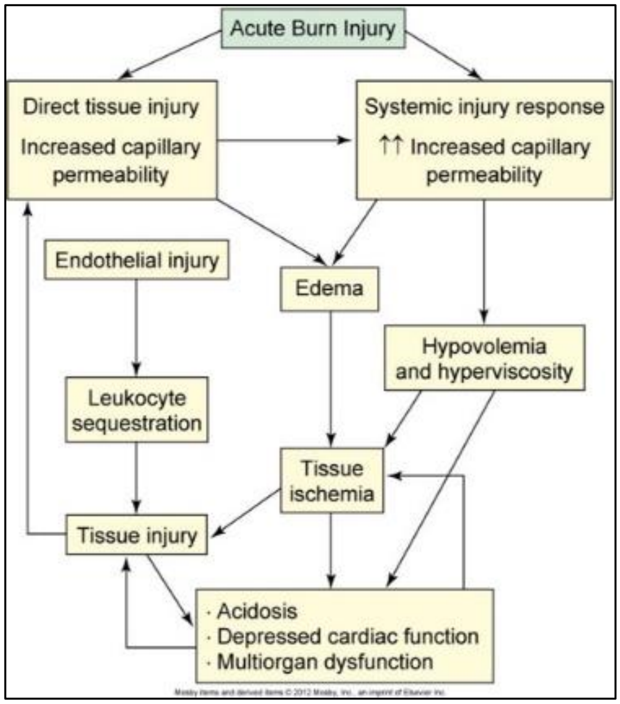


Figure 17: Rule of Nines (Adult)

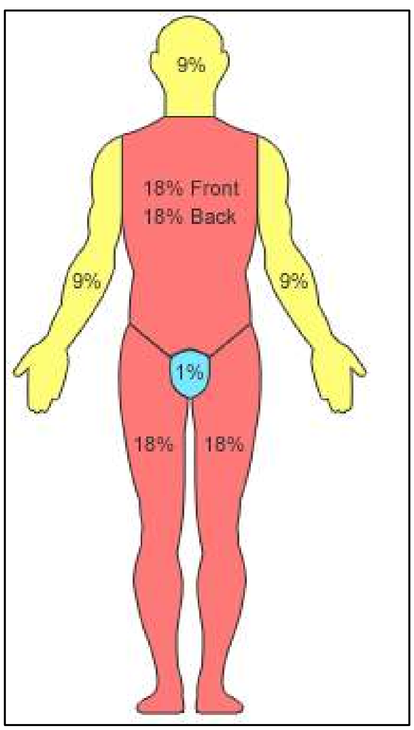


Figure 18: Rule of Nines (Child)